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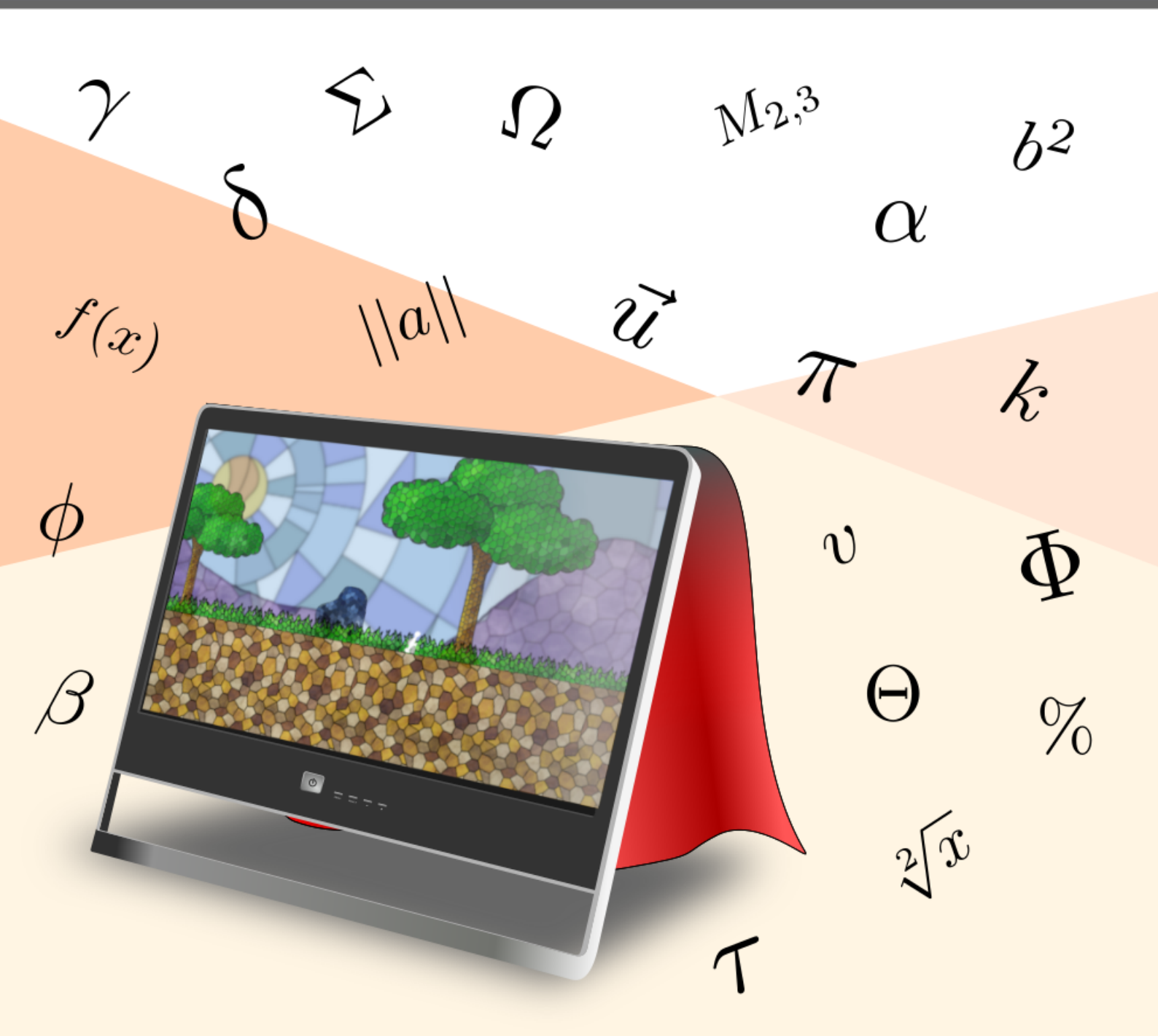

2D Game Development: From Zero to Hero

A compendium of the community knowledge on game design and development

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Perseverance is the backbone of success.

Anonymous

To my family

To my friends, both international and not

To the ones who never give up

Daniele Penazzo

Contents

1 Foreword **1**

2 Introduction **2**

2.1 Why another game development book? . . . 2

2.2 Conventions used in this book . . . 3

2.2.1 Logic Conventions . . . 3

2.2.2 Code Listings . . . 3

2.2.3 Block Quotes . . . 3

2.2.4 Boxes . . . 4

2.2.5 Engine Used . . . 4

2.2.6 About editions . . . 5

2.3 Structure of this Book . . . 5

3 The Maths Behind Game Development **9**

3.1 Some useful symbols . . . 9

3.2 The modulo operator . . . 9

3.3 Powers and Roots . . . 10

3.4 Equations . . . 11

3.4.1 You can add or subtract any number on both sides . . . 11

3.4.2 You can multiple or divide any non-zero number on both sides . . . 11

3.5 Exponentiations and Logarithms . . . 12

3.6 Limits . . . 12

3.7 Derivatives . . . 13

3.8 The Cartesian Plane . . . 14

3.9 Vectors . . . 15

3.9.1 Adding and Subtracting Vectors . . . 15

3.9.2 Scaling Vectors . . . 16

3.9.3 Dot Product . . . 17

3.9.4 Vector Length and Normalization . . . 18

3.9.5 “Clamping” a Vector . . . 18

3.10 Geometry . . . 19

3.10.1 Convex vs Concave polygons . . . 19

3.10.2 Self-intersecting polygons . . . 20

3.10.3 Straight Lines and their equations . . . 20

3.10.3.1 Getting the equation of a straight line, given two points . . . 21

3.10.3.2 Getting the equation, given the slope and a point . . . 22

3.10.4 Projections . . . 22

3.10.4.1 Projecting arbitrary lines on the axes . . . 24

3.11 Matrices . . . 25

3.11.1 What is a matrix . . . 25
3.11.2 Matrix sum and subtraction . . . 26
3.11.3 Multiplication by a scalar . . . 26
3.11.4 Transposition . . . 26
3.11.5 Multiplication between matrices . . . 27
3.11.6 Other uses for matrices . . . 29
3.12 Trigonometry . . . 29
3.12.1 Radians vs Degrees . . . 29
3.12.2 Sine, Cosine and Tangent . . . 30
3.12.3 Pythagorean Trigonometric Identity . . . 31
3.12.4 Reflections . . . 31
3.12.5 Shifts . . . 31
3.12.6 Trigonometric Addition and subtraction . . . 32
3.12.7 Double-Angle Formulae . . . 32
3.12.8 Inverse Formulas . . . 32
3.13 Numerical Analysis . . . 33
3.13.1 Newton-Raphson method . . . 33
3.14 Coordinate Systems on computers . . . 34
3.15 Transformation Matrices . . . 35
3.15.1 Stretching . . . 35
3.15.2 Rotation . . . 36
3.15.2.1 Choosing the direction of the rotation . . . 37
3.15.2.2 Rotating referred to an arbitrary point . . . 38
3.15.3 Shearing . . . 38
3.16 Basics of Probability . . . 39
3.16.1 A simple definition of probability . . . 39
3.16.2 Probability of independent events . . . 40
3.16.3 Probability of mutually exclusive events . . . 41
3.16.4 Probability of non-mutually exclusive events . . . 41
3.16.5 Conditional Probability . . . 42
3.16.6 Uniform Distributions . . . 44
3.16.7 How probability is used in games . . . 44
3.16.8 Tiered Prize Pools . . . 46
3.16.8.1 Introducing a “luck” stat . . . 47

4 Some Computer Science Fundamentals 49

4.1 Number representations . . . 49
4.1.1 The most used representations . . . 49
4.1.1.1 Decimal . . . 49
4.1.1.2 Binary . . . 49

4.1.1.3 Octal . . . 50
4.1.1.4 Hexadecimal . . . 51
4.1.2 Converting between decimal and binary . . . 52
4.1.2.1 Two's complement . . . 53
4.1.2.2 Floating point . . . 54
4.1.3 Converting between binary and octal . . . 54
4.1.4 Gray Code . . . 55
4.2 Basics of Logic . . . 55
4.2.1 Truth tables . . . 56
4.2.2 Common operators . . . 56
4.2.2.1 AND . . . 56
4.2.2.2 OR . . . 56
4.2.2.3 NOT . . . 57
4.2.2.4 XOR . . . 57
4.2.3 Logic operations vs bitwise operations . . . 57
4.2.3.1 Packing more information with less . . . 58
4.2.4 De Morgan's Laws and Conditional Expressions . . . 59
4.3 Algorithms . . . 59
4.4 Recursion . . . 60
4.5 Programming Languages . . . 62
4.5.1 Classifying programming languages . . . 62
4.5.1.1 By how they build . . . 62
4.5.1.2 By Paradigm . . . 64
4.5.1.3 By the way types are determined . . . 64
4.5.1.4 By the "strength" of typing . . . 65
4.5.1.5 By memory management . . . 66
4.5.2 Languages available for this book . . . 67
4.6 Computers are (not) precise . . . 67
4.6.1 Catastrophic cancellation . . . 69
4.7 Random Numbers are not really random . . . 70
4.7.1 How to seed a random number generator . . . 71
4.8 Estimating the complexity of algorithms . . . 73
4.8.1 O(1) . . . 73
4.8.2 O(log(n)) . . . 74
4.8.3 O(n) . . . 74
4.8.4 O(n·log(n)) . . . 75
4.8.5 $O(n^2)$ . . . 75
4.8.6 $O(2^n)$ . . . 76
4.9 A primer on calculating the order of your algorithms . . . 76
4.9.1 Some basics . . . 76

4.9.2 What happens when we have more than one big-O? . . . 77
4.9.3 A problem with asymptotic complexity . . . 78
4.9.4 What do we do with recursive algorithms? . . . 79
4.9.5 How do big-O estimates compare to each other? . . . 79
4.10 Simplifying your conditionals with Karnaugh Maps . . . 80
4.10.1 “Don’t care”s . . . 81
4.10.2 A more complex map . . . 83
4.10.3 Guided Exercise . . . 84
4.11 Object Oriented Programming . . . 86
4.11.1 Introduction . . . 86
4.11.2 Objects . . . 86
4.11.3 Abstraction and Interfaces . . . 87
4.11.4 Inheritance and Polymorphism . . . 87
4.11.5 Mixins . . . 87
4.11.6 The Diamond Problem . . . 88
4.11.7 Composition . . . 88
4.11.8 Composition vs. Inheritance . . . 89
4.11.9 “Composition over Inheritance” design . . . 91
4.11.10 Coupling . . . 92
4.11.11 The DRY Principle . . . 93
4.11.12 SOLID Principles . . . 93
4.12 Designing entities as data . . . 94
4.13 Reading UML diagrams . . . 95
4.13.1 Use Case Diagrams . . . 95
4.13.1.1 Actors . . . 96
4.13.1.2 Use Cases . . . 97
4.13.1.3 Notes . . . 98
4.13.1.4 Sub-Use Cases . . . 98
4.13.2 Class Diagrams . . . 98
4.13.2.1 Classes . . . 98
4.13.2.2 Interfaces . . . 99
4.13.2.3 Relationships between entities of the class diagram . . . 99
4.13.2.4 Notes . . . 102
4.13.3 Activity Diagrams . . . 102
4.13.3.1 Start and End Nodes . . . 103
4.13.3.2 Actions . . . 103
4.13.3.3 Decisions (Conditionals) and loops . . . 103
4.13.3.4 Synchronization . . . 105
4.13.3.5 Swimlanes . . . 105
4.13.3.6 Signals . . . 106

4.13.3.7 Notes . . . 107
4.13.3.8 A note on activity diagrams . . . 107
4.13.4 Sequence Diagrams . . . 108
4.13.4.1 Lifelines . . . 108
4.13.4.2 Messages . . . 109
4.13.4.3 Object Instantiation and Destruction . . . 109
4.13.4.4 Grouping and loops . . . 110
4.13.4.5 Notes . . . 110
4.13.5 Other diagrams . . . 111
4.14 Generic Programming . . . 111
4.15 Basic Data Structures . . . 112
4.15.1 Statically-Allocated Arrays . . . 112
4.15.2 Structs . . . 112
4.15.3 Enums . . . 113
4.16 Advanced Data Structures . . . 114
4.16.1 Graphs . . . 114
4.16.2 Trees . . . 116
4.16.2.1 Depth-first Search . . . 118
4.16.2.2 Breadth-first search . . . 122
4.16.3 Dynamic Arrays . . . 123
4.16.3.1 Performance Analysis . . . 123
4.16.4 Linked Lists . . . 125
4.16.4.1 Performance Analysis . . . 126
4.16.5 Doubly-Linked Lists . . . 127
4.16.6 Hash Tables . . . 128
4.16.7 Binary Search Trees (BST) . . . 129
4.16.8 Heaps . . . 130
4.16.9 Stacks . . . 131
4.16.10 Queues . . . 132
4.16.11 Circular Queues . . . 133
4.17 Equality vs. Identity . . . 134
4.18 Truthiness and “Falsiness” . . . 135
4.19 Operators are functions too . . . 136
4.20 The principle of locality . . . 137
4.21 Treating multidimensional structures like one-dimensional ones . . . 138
4.22 Data Redundancy . . . 139
4.23 Introduction to Multi-Tasking . . . 141
4.23.1 Multi-Threading vs Multi-Processing . . . 142
4.23.2 Coroutines . . . 143
4.24 Introduction to Multi-Threading . . . 143

4.24.1 What is Multi-Threading . . . 143
4.24.2 Why Multi-Threading? . . . 143
4.24.3 Thread Safety . . . 144
4.24.3.1 Race conditions . . . 144
4.24.3.2 Critical Regions . . . 146
4.24.4 Ensuring determinism . . . 146
4.24.4.1 Immutable Objects . . . 146
4.24.4.2 Mutex . . . 146
4.24.4.3 Atomic Operations . . . 149
4.25 Developer Experience (DX/DevEx) . . . 149
4.26 What to do when coding . . . 149
4.26.1 Safe default initialization . . . 150
4.26.1.1 “Smart” default initialization . . . 150
4.27 What to avoid when coding . . . 151
4.27.1 Magic Numbers . . . 151
4.27.2 Code monoliths . . . 152

5 A Game Design Dictionary 154

5.1 Platforms . . . 154
5.1.1 Arcade . . . 154
5.1.2 Console . . . 155
5.1.3 Personal Computer . . . 155
5.1.4 Mobile . . . 157
5.1.5 Web . . . 157
5.1.6 Fantasy consoles . . . 158
5.2 Input Devices . . . 158
5.2.1 Mouse and Keyboard . . . 158
5.2.2 Gamepad . . . 159
5.2.3 Touch Screen . . . 159
5.2.4 Dedicated Hardware . . . 160
5.2.5 Other Input Devices . . . 160
5.3 Game Genres . . . 160
5.3.1 Shooters . . . 160
5.3.2 Strategy . . . 161
5.3.3 Platformer . . . 161
5.3.4 RPG . . . 161
5.3.5 MMO . . . 162
5.3.6 Simulation . . . 162
5.3.7 Rhythm Games . . . 162
5.3.8 Visual novels . . . 162

5.3.9 Puzzle games . . . 162
5.4 Miscellaneous . . . 163
5.4.1 Emergent Gameplay . . . 163

6 Project Management Basics and tips 165
6.1 The figures of game design and development . . . 165
6.1.1 Producer/Project Manager . . . 165
6.1.2 Game Designer . . . 166
6.1.3 Writer . . . 166
6.1.4 Developer . . . 167
6.1.5 Visual Artist . . . 168
6.1.6 Sound Artist . . . 169
6.1.7 Marketing/Public Relations Manager . . . 169
6.1.8 Tester . . . 170
6.2 Some general tips . . . 170
6.2.1 Be careful of feature creep . . . 170
6.2.2 On project duration . . . 171
6.2.3 Brainstorming: the good, the bad and the ugly . . . 171
6.2.4 On Sequels . . . 171
6.3 Common Errors and Pitfalls . . . 172
6.3.1 Losing motivation . . . 172
6.3.2 The “Side Project” pitfall . . . 172
6.3.3 Making a game “in isolation” . . . 172
6.3.4 (Mis)Handling Criticism . . . 173
6.3.4.1 Misusing of the Digital Millennium Copyright Act . . . 173
6.3.5 Not letting others test your game . . . 175
6.3.6 Being perfectionist . . . 175
6.3.7 Using the wrong engine . . . 176
6.4 Software Life Cycle Models . . . 177
6.4.1 Iteration versus Increment . . . 177
6.4.2 Waterfall Model . . . 177
6.4.3 Incremental Model . . . 177
6.4.4 Evolutionary Model . . . 178
6.4.5 Agile Software Development . . . 179
6.4.5.1 User Stories . . . 179
6.4.5.2 Scrum . . . 180
6.4.5.3 Kanban . . . 180
6.4.5.4 ScrumBan . . . 181
6.4.6 Lean Development . . . 181
6.4.7 Where to go from here . . . 182

6.5 Version Control . . . 182
6.6 Metrics and dashboards . . . 182
6.6.1 SLOC . . . 183
6.6.2 Cyclomatic Complexity . . . 183
6.6.2.1 How cyclomatic complexity is calculated . . . 183
6.6.3 Code Coverage . . . 185
6.6.4 Code Smells . . . 186
6.6.5 Coding Style infractions . . . 186
6.6.6 Depth of Inheritance . . . 186
6.6.7 Number of methods / fields / variables . . . 187
6.6.8 Number of parameters . . . 187
6.6.9 Other metrics . . . 187

7 Writing a Game Design Document 188

7.1 What is a Game Design Document . . . 188
7.2 Possible sections of a Game Design Document . . . 188
7.2.1 Project Description . . . 188
7.2.2 Characters . . . 189
7.2.3 Storyline . . . 189
7.2.3.1 The theme . . . 189
7.2.3.2 Progression . . . 190
7.2.4 Levels and Environments . . . 190
7.2.5 Gameplay . . . 190
7.2.5.1 Goals . . . 190
7.2.5.2 Game Mechanics . . . 191
7.2.5.3 Skills . . . 191
7.2.5.4 Items/Powerups . . . 192
7.2.5.5 Difficulty Management and Progression . . . 192
7.2.5.6 Losing Conditions . . . 193
7.2.6 Graphic Style and Art . . . 193
7.2.7 Sound and Music . . . 193
7.2.8 User Interface . . . 194
7.2.9 Game Controls . . . 194
7.2.10 Accessibility Options . . . 195
7.2.11 Tools . . . 195
7.2.12 Marketing . . . 195
7.2.12.1 Target Audience . . . 196
7.2.12.2 Available Platforms . . . 196
7.2.12.3 Monetization . . . 196
7.2.12.4 Internationalization and Localization . . . 197

7.2.13 Other/Random Ideas 197
7.3 Where to go from here 197

8 The Game Loop 200
8.1 The Input-Update-Draw Abstraction 200
8.2 Input 201
8.2.1 Events vs Real Time Input 201
8.3 Timing your loop 202
8.3.1 What is a time step 202
8.3.2 Fixed Time Steps 202
8.3.3 Variable Time Steps 203
8.3.4 Semi-fixed Time Steps 203
8.3.5 Frame Limiting 204
8.3.6 Frame Skipping/Dropping 205
8.3.7 Multi-threaded Loops 205
8.4 Issues and possible solutions 206
8.4.1 Frame/Screen Tearing 206
8.5 Drawing to screen 206
8.5.1 Clearing the screen 207

9 Collision Detection and Reaction 209
9.1 Why Collision Detection is done in multiple passes 209
9.2 Narrow-Phase Collision Detection: did it really collide? 209
9.2.1 Collision Between Two Points 210
9.2.2 Collision Between A Point and a Circle 210
9.2.3 Collision Between Two Circles 212
9.2.4 Collision Between Two Axis-Aligned Rectangles (AABB) 213
9.2.5 Line/Point Collision 215
9.2.6 Line/Circle Collision 217
9.2.7 Point/Rectangle Collision 220
9.2.8 Point/Triangle Collision 220
9.2.9 Circle/Rectangle Collision 222
9.2.10 Line/Line Collision 224
9.2.11 Line/Rectangle Collision 226
9.2.12 Point/Polygon Collision 227
9.2.12.1 Jordan Curve Theorem 227
9.2.12.2 Thinking outside the box: polygon triangulation 229
9.2.12.3 Bounding Boxes 230
9.2.12.4 Point/Polygon collision detection using triangulation 232
9.2.13 Circle/Polygon Collision 235
9.2.14 Line/Polygon Collision 237

9.2.15 Polygon/Polygon Collision 238
9.2.16 Non-convex polygons collision 240
9.2.16.1 Polygon triangulation: the return 241
9.2.17 Pixel-Perfect collision 242
9.3 Broad-phase collision detection: is a collision even possible? 244
9.3.1 The Brute Force Method 245
9.3.2 Building Quad Trees 245
9.3.2.1 A more precise definition 247
9.3.2.2 Querying quad trees 247
9.3.3 Building AABB-Trees 248
9.3.3.1 Querying AABB-trees 249
9.3.4 Collision groups 251
9.4 Other Collision Detection Methods 251
9.4.1 Calculating the position of tiles 251
9.4.2 The “Tile + Offset” Method 254
9.5 Collision Reaction/Correction 256
9.5.1 HitBoxes vs HurtBoxes 256
9.5.2 Collision Reaction Methods 256
9.5.2.1 A naive approach 257
9.5.2.2 Shallow-axis based reaction method 259
9.5.2.3 Interleaving single-axis movement and collision detection 261
9.5.2.4 The “Snapshot” Method 262
9.5.3 When two moving items collide 263
9.6 Common Issues with time-stepping Collision Detection 265
9.6.1 The “Bullet Through Paper” problem 265
9.6.2 Precision Issues 265
9.6.3 One-way obstacles 266
9.7 Separating Axis Theorem 266
9.7.1 Why only convex polygons? 267
9.7.2 How it works 268
9.7.2.1 Finding the axes to analyze 268
9.7.2.2 Projecting the shapes into the axes and exiting the algorithm 270
9.7.2.3 From arbitrary axes to “x and y” 271
9.8 Ray Casting 271
9.8.1 What is Ray Casting? 271
9.8.2 Other uses for ray casting: Pseudo-3D environments 272

10 Scene Trees **273**

10.1 What is a scene 273
10.2 Scene trees and their functionalities 273

10.2.1 How scene trees can make drawing entities easier 273
10.3 Implementing a scene tree 274

11 Cameras **275**
11.1 Screen Space vs. Game Space 275
11.2 Cameras and projections 276
11.3 Most used camera transitions and types 277
11.3.1 Static Camera 277
11.3.2 Grid Camera 277
11.3.3 Position-Tracking Camera 277
11.3.3.1 Horizontal-Tracking Camera 277
11.3.3.2 Full-Tracking Camera 278
11.3.4 Camera Trap 278
11.3.5 Look-Ahead Camera 279
11.3.6 Hybrid Approaches 280
11.4 Clamping your camera position 280

12 Game Design Tips **282**
12.1 Tutorials 282
12.1.1 Do not pad tutorials 282
12.1.2 Integrate tutorials in the lore 282
12.1.3 Let the player explore the controls 282
12.2 Consolidating and refreshing the game mechanics 283
12.2.1 Remind the player about the mechanics they learned 283
12.2.2 Introduce new ways to use old mechanics 283
12.3 Rewarding the player 283
12.3.1 Reward the player for their “lateral thinking” 283
12.3.2 Reward the player for their tenacity 284
12.3.3 Reward the player for exploring 285
12.3.4 Reward the player for not immediately following the given direction 285
12.3.5 Reward the player for not trusting you entirely 285
12.3.6 Reward Backtracking (but don’t make it mandatory!) 286
12.3.7 The “lives” system 286
12.3.7.1 1-UPs 287
12.3.7.2 Other approaches 287
12.4 Loading Screens 288
12.4.1 What to put in a loading screen 288
12.4.2 Letting the player “exit” the loading screen 288
12.4.3 Avoiding a loading screen altogether 289
12.5 Designing the story and gameplay flow 289
12.5.1 Linear Gameplay 289

12.5.2 Branching gameplay . . . 290
12.5.3 Parallel gameplay . . . 291
12.5.4 Threaded gameplay . . . 291
12.5.5 Episodic gameplay . . . 292
12.5.6 Adding parallel paths . . . 292
12.5.7 Looping Gameplay . . . 293
12.5.7.1 Soft-reset mechanics . . . 293
12.6 Some game genres and their characteristics . . . 294
12.6.1 Roguelikes and Rogue-lites . . . 294
12.6.1.1 Use of pseudo-randomness and procedural generation . . . 294
12.6.1.2 Permadeath . . . 294
12.6.1.3 Turn-based Gameplay . . . 295
12.6.1.4 Lack of mode-based gameplay . . . 295
12.6.1.5 Multiple ways to accomplish (or fail!) a task . . . 295
12.6.1.6 Resource Management is key . . . 296
12.6.1.7 Peace was never an option . . . 296
12.6.1.8 Dealing with the unknown . . . 296
12.6.2 Deck builders . . . 297
12.7 Tips and Tricks . . . 297
12.7.1 General Purpose . . . 297
12.7.1.1 Make that last Health Point count . . . 297
12.7.1.2 Avoiding a decision can be a decision itself . . . 297
12.7.1.3 Telegraphing . . . 298
12.7.1.4 Minigames . . . 298
12.7.1.5 When unlockables are involved, be balanced . . . 299
12.7.2 Shooters . . . 299
12.7.2.1 Make the bullets stand out . . . 299
12.7.3 RPGs . . . 300
12.7.3.1 Grinding and farming . . . 300
12.7.3.2 Leveling Curves . . . 301
12.7.3.3 “Mastering” . . . 303
12.8 Perceived Fairness . . . 303
12.8.1 You don’t need precise collision detection . . . 303
12.8.2 Immediate dangers should be well visible . . . 304
12.9 Miscellaneous . . . 306
12.9.1 You cannot use the “Red Cross” in games . . . 306
12.9.2 Auto-saving . . . 306
12.9.3 Feedback is important . . . 307

13 World Building **308**

13.1 Character Design . . . 308
13.1.1 Make your characters suck at something . . . 308
13.2 Level Design . . . 308
13.2.1 Anything can tell a story . . . 309
13.2.2 Details matter . . . 309
13.3 Quest design . . . 310
13.3.1 You can make “busy work” interesting too . . . 310
13.4 Story Design . . . 310
13.4.1 Use the “illusion of choice” to gently push the story forward . . . 310

14 Creating your assets 313
14.1 Assumptions . . . 313
14.2 Graphics . . . 313
14.2.1 Some computer graphics basics . . . 313
14.2.1.1 The “color wheel” . . . 313
14.2.1.2 Color representations . . . 314
14.2.1.3 Primary, Secondary and Tertiary colors . . . 316
14.2.1.4 Analogous Colors . . . 316
14.2.1.5 Complementary Colors . . . 316
14.2.1.6 Color Depth . . . 317
14.2.1.7 Direct Color vs. Indexed Color . . . 320
14.2.1.8 Lossless Formats . . . 320
14.2.1.9 Lossy Formats . . . 320
14.2.1.10 Transparency . . . 321
14.2.1.11 Texture Filtering . . . 321
14.2.1.12 What is a sprite? . . . 322
14.2.2 General Tips . . . 322
14.2.2.1 Practice, Practice, Practice... . . . 322
14.2.2.2 References are your friends . . . 322
14.2.2.3 Don’t compare your style to others’ . . . 323
14.2.2.4 Study other styles . . . 323
14.2.2.5 Learn to deconstruct objects into shapes . . . 323
14.2.2.6 Graphics Smearing . . . 323
14.2.3 Sprite sheets . . . 324
14.2.3.1 Sprite Sheet Gotchas . . . 325
14.2.4 Virtual Resolution . . . 326
14.2.5 Using limited color palettes . . . 327
14.2.5.1 Dithering . . . 327
14.2.5.2 Palette Swapping . . . 329
14.2.6 Layering and graphics . . . 329

14.2.6.1 Detail attracts attention . . . 329
14.2.6.2 Use saturation to separate layers further . . . 330
14.2.6.3 Movement is yet another distraction . . . 330
14.2.6.4 Use contrast to your advantage . . . 330
14.2.6.5 Find exceptions . . . 331
14.2.6.6 Summarizing Layering . . . 331
14.2.7 Pixel Art . . . 333
14.2.7.1 What pixel art is and what it is not . . . 333
14.2.7.2 Tools . . . 333
14.2.7.3 Layers . . . 334
14.2.7.4 Sub-pixel animation with palette cycling . . . 334
14.2.8 Normal Mapping . . . 334
14.2.8.1 A simple example . . . 335
14.2.9 Blending surfaces . . . 337
14.2.9.1 Addition . . . 337
14.2.9.2 Subtraction . . . 338
14.2.9.3 Alpha Blending . . . 339
14.2.10 Tips and Tricks . . . 339
14.2.10.1 Tiles . . . 339
14.2.10.2 Sprites and icons . . . 343
14.3 Sounds And Music . . . 343
14.3.1 Some audio basics . . . 343
14.3.1.1 Sample Rate . . . 343
14.3.1.2 Bit Depth . . . 344
14.3.1.3 Lossless Formats . . . 345
14.3.1.4 Lossy Formats . . . 345
14.3.1.5 Clipping . . . 345
14.3.2 Sound Synthesis . . . 346
14.3.2.1 AM Synthesis . . . 346
14.3.2.2 FM Synthesis . . . 347
14.3.2.3 FM Synthesis vs Sample-based music . . . 347
14.3.3 Basic Wave Forms . . . 348
14.3.3.1 Sine Wave . . . 348
14.3.3.2 Square Wave . . . 349
14.3.3.3 Triangle Wave . . . 349
14.3.3.4 Sawtooth Wave . . . 349
14.3.3.5 Noise . . . 350
14.3.4 ADSR Envelope . . . 350
14.3.4.1 Attack . . . 351
14.3.4.2 Decay . . . 351

14.3.4.3 Sustain 351
14.3.4.4 Release 352
14.3.5 Digital Sound Processing (DSP) 352
14.3.5.1 Reverb 353
14.3.5.2 Pitch Shift 353
14.3.5.3 Filtering 353
14.3.5.4 Doppler Effect 353
14.3.6 Simulating older consoles' audio 354
14.3.6.1 Commodore Vic20 354
14.3.6.2 Commodore 64 354
14.3.6.3 Commodore Amiga 355
14.3.6.4 Sega Master System / GameGear 356
14.3.6.5 Sega Genesis/MegaDrive 356
14.3.6.6 NES 356
14.3.6.7 SNES 357
14.3.6.8 Game Boy 357
14.3.6.9 AdLib / SoundBlaster 358
14.3.7 "Swappable" sound effects 358
14.3.8 Some audio processing tips 358
14.3.8.1 Prefer cutting over boosting 358
14.3.8.2 Spatial sound is not only for 3D Games 359
14.3.9 DAW Basics 360
14.3.9.1 What is a DAW Software? 360
14.3.9.2 The Piano Roll 360
14.3.10 Music Tracker Basics 361
14.3.10.1 What is a Music Tracker Software? 361
14.3.10.2 Samples 363
14.3.10.3 Instruments 363
14.3.10.4 Channels 363
14.3.10.5 Patterns 363
14.3.11 Basic Rhythms 363
14.3.11.1 Four on the floor 364
14.3.11.2 Four on the floor with off-beat hi-hats 364
14.3.11.3 A simple rock beat 364
14.3.12 Adaptive Music 365
14.4 Fonts 365
14.4.1 Font Categories 365
14.4.1.1 Serif and Sans-Serif fonts 365
14.4.1.2 Proportional and Monospaced fonts 366
14.4.2 Using textures to make text 366

14.4.3 Using Fonts to make text . . . 368
14.5 Shaders . . . 368
14.5.1 What are shaders . . . 368
14.5.2 Shader Programming Languages . . . 369
14.5.3 The GLSL Programming Language . . . 370
14.5.3.1 The data types . . . 370
14.5.4 Some GLSL Shaders examples . . . 371

15 Design Patterns **373**
15.1 Creational Design Patterns . . . 373
15.1.1 Singleton Pattern . . . 373
15.1.2 Dependency Injection . . . 375
15.1.3 Prototype . . . 377
15.2 Structural Design Patterns . . . 377
15.2.1 Flyweight . . . 377
15.2.2 Component/Composite Pattern . . . 379
15.2.3 Decorator . . . 381
15.2.4 Adapter . . . 382
15.2.4.1 Object Adapter . . . 383
15.2.4.2 Class Adapter . . . 383
15.2.5 Facade . . . 384
15.2.6 Proxy . . . 385
15.3 Behavioural Design Patterns . . . 387
15.3.1 Command Pattern . . . 387
15.3.2 Observer Pattern . . . 389
15.3.3 Strategy . . . 390
15.3.4 Chain of Responsibility . . . 392
15.3.5 Visitor . . . 393
15.4 Architectural Design Patterns . . . 393
15.4.1 Service Locator . . . 394

16 Useful Containers and Classes **395**
16.1 Resource Manager . . . 395
16.2 Animator . . . 395
16.3 Finite State Machine . . . 396
16.4 Menu Stack . . . 399
16.5 Particle Systems . . . 399
16.5.1 Particles . . . 399
16.5.2 Emitters . . . 402
16.5.3 Force Application . . . 404
16.6 Timers . . . 405

16.6.1 Accounting for “leftover time” 406
16.6.1.1 A naive solution 406
16.6.1.2 A different approach 408
16.7 Inbetweening 409
16.7.1 Bouncing 412
16.8 Chaining 412

17 Artificial Intelligence in Videogames 413

17.1 Path Finding 413
17.1.1 Representing our world 413
17.1.1.1 2D Grids 413
17.1.1.2 Path nodes 414
17.1.1.3 Navigation meshes 414
17.1.2 Heuristics 417
17.1.2.1 Manhattan Distance heuristic 417
17.1.2.2 Euclidean Distance heuristic 418
17.1.3 Algorithms 419
17.1.3.1 A simple “Wandering” Algorithm 419
17.1.3.2 A slightly better “Wandering” algorithm 423
17.1.3.3 The Greedy “Best First” Algorithm 424
17.1.3.4 The Dijkstra Algorithm 426
17.1.3.5 The A* Algorithm 428
17.2 Finite state machines 430
17.3 Decision Trees 430
17.4 Behaviour Trees 431
17.5 Tips and tricks 431
17.5.1 “Jump when the player shoots” 431
17.5.2 Distance-based patterns 433
17.5.2.1 “Ranged pattern” 433
17.5.2.2 “Melee pattern” 434

18 Other Useful Algorithms 436

18.1 World Generation 436
18.1.1 Midpoint Displacement Algorithm 436
18.1.2 Diamond-Square Algorithm 438
18.1.3 Maze Generation 441
18.1.3.1 Randomized Depth-First Search (Recursive Backtracker) 442
18.1.3.2 Randomized Kruskal’s Algorithm 447
18.1.3.3 Recursive Division Algorithm 450
18.1.3.4 Binary Tree Algorithm 452
18.1.3.5 Eller’s Algorithm 455

18.2 Dungeon Generation . . . 456
18.3 Noise Generation . . . 456
18.3.1 Randomized Noise (Static) . . . 456
18.3.2 Perlin Noise . . . 457
18.4 Animation . . . 458
18.4.1 Skeleton animation . . . 458

19 Procedural Content Generation 459
19.1 What is procedural generation (and what it isn't) . . . 459
19.2 Advantages and disadvantages . . . 460
19.2.1 Advantages . . . 460
19.2.1.1 Less disk space needed . . . 460
19.2.1.2 Larger games can be created with less effort . . . 460
19.2.1.3 Lower budgets needed . . . 461
19.2.1.4 More variety and replayability . . . 461
19.2.2 Disadvantages . . . 461
19.2.2.1 Requires more powerful hardware . . . 461
19.2.2.2 Less Quality Control . . . 461
19.2.2.3 Worlds can feel repetitive or "lacking artistic direction" . . . 461
19.2.2.4 You may generate something unusable . . . 462
19.2.2.5 Story and set game events are harder to script . . . 462
19.3 Where it can be used . . . 462
19.4 Procedural Generation and Difficulty Management . . . 463
19.4.1 Static difficulty . . . 463
19.4.2 Adaptive Difficulty . . . 463
19.4.2.1 Rubberbanding . . . 463
19.4.3 Static vs. Adaptive Difficulty . . . 464

20 Developing Game Mechanics 465
20.1 General Purpose . . . 465
20.1.1 I-Frames . . . 465
20.1.2 Tilemaps . . . 466
20.1.2.1 Rectangular Tilemaps . . . 466
20.1.2.2 Hexagonal Tilemaps . . . 467
20.1.2.3 Isometric Tilemaps . . . 469
20.1.3 Scrolling Backgrounds and Parallax Scrolling . . . 470
20.1.3.1 Infinitely Scrolling Backgrounds . . . 470
20.1.3.2 Parallax Scrolling . . . 471
20.1.4 Avoid interactions between different input systems . . . 474
20.1.5 Sprite Stenciling/Masking . . . 474
20.1.6 Loading screens . . . 474

20.1.7 Simulating Inertia . . . 475
20.1.8 Corner correction . . . 480
20.2 2D Platformers . . . 482
20.2.1 Simulating Gravity . . . 482
20.2.2 Avoiding “Floaty Jumps” . . . 482
20.2.3 Making jumps “float differently” . . . 484
20.2.4 Ladders . . . 485
20.2.5 Walking on slanted ground . . . 485
20.2.6 Stairs . . . 485
20.2.7 Ledge Grabbing . . . 485
20.2.8 Jump Buffering . . . 485
20.2.9 Coyote Time . . . 486
20.2.10 Timed Jumps . . . 488
20.2.11 Wall Jumps . . . 489
20.2.12 Screen Wrap . . . 489
20.3 Top-view RPG-Like Games . . . 489
20.3.1 Managing height . . . 489
20.3.1.1 Faking it . . . 489
20.3.1.2 Managing height for real . . . 490
20.4 Rhythm Games . . . 490
20.4.1 The world of lag . . . 490
20.4.1.1 Input Lag . . . 491
20.4.1.2 Video Lag . . . 491
20.4.1.3 Audio Lag . . . 491
20.4.1.4 Lag Tests . . . 491
20.4.2 Synchronizing with the Music . . . 492
20.4.2.1 Time domain vs. Frequency Domain . . . 492
20.4.2.2 The Fast Fourier Transform . . . 493
20.4.2.3 Beat Detection . . . 493
20.5 “Bullet Hell” Style Games . . . 493
20.5.1 Bullets . . . 494
20.5.2 The Ship Hitbox . . . 494
20.5.3 Screen-clearing bombs . . . 495
20.5.4 Clearing bullets on pattern changes . . . 495
20.5.5 Find other chances to clear some bullets . . . 496
20.5.6 Turn enemy bullets into collectibles at the end of a boss fight . . . 496
20.5.7 The “Chain Meter” . . . 496
20.5.8 Managing the player’s death . . . 497
20.5.9 The Enemy AI . . . 497
20.5.10 Be fair to the player, but also to the computer . . . 497

20.5.11 Inertia . . . 498
20.5.12 Some examples . . . 498
20.6 Match-x Games . . . 498
20.6.1 Managing and drawing the grid . . . 498
20.6.2 Finding and removing Matches . . . 499
20.6.2.1 Why don't we delete the matches immediately? . . . 501
20.6.3 Replacing the removed tiles and applying gravity . . . 502
20.7 Cutscenes . . . 504
20.7.1 Videos . . . 504
20.7.2 Scripted Cutscenes . . . 504
20.8 Savefiles . . . 504
20.8.1 Serialization . . . 504
20.8.2 Deserialization . . . 504
20.8.3 Tips and tricks . . . 504
20.8.3.1 Add a version to your savefiles . . . 504

21 Dealing with piracy **505**
21.1 Objectives of anti-piracy measures . . . 505
21.2 Alternatives to anti-piracy measures . . . 505
21.2.1 Regional Pricing . . . 506
21.2.2 Give out free demos . . . 506
21.2.3 No DRM entirely . . . 506
21.2.4 Poisoning the well . . . 507
21.3 Implementing anti-piracy measures . . . 507
21.3.1 Third-party DRM systems . . . 507
21.3.2 Debugger detection . . . 508
21.3.3 Checksums . . . 508
21.3.4 When an anti-piracy flag is triggered, be generic . . . 508
21.3.5 Obfuscation and redundancy of checks . . . 509
21.3.5.1 Play with the pirates . . . 509
21.3.6 Don't stop the pirates immediately . . . 509

22 Balancing Your Game **512**
22.1 Do not annoy the player . . . 512
22.2 Favour the player when possible . . . 512
22.3 Difficulty curves . . . 513
22.3.1 Simple Lines . . . 513
22.3.2 Flat Line . . . 513
22.3.2.1 Linear Increase . . . 514
22.3.2.2 Logarithmic Line . . . 514
22.3.2.3 Exponential Line . . . 515

22.3.3 Wave patterns . . . 516
22.3.3.1 Linearly Increasing wave . . . 516
22.3.3.2 Logarithmically Increasing wave . . . 516
22.3.4 Interval Patterns . . . 517
22.3.4.1 Simple Interval . . . 517
22.3.4.2 Widening Interval . . . 518
22.3.4.3 Widening Interval with Logarithmic trend . . . 519
22.3.5 This is not everything . . . 519
22.3.5.1 Sawtooth pattern . . . 519
22.3.5.2 What not to do . . . 520
22.3.5.3 Beyond difficulty . . . 521
22.4 Economy . . . 521
22.4.1 Supply and Demand . . . 521
22.4.2 Money sources and sinks . . . 522
22.4.3 Inflation . . . 523
22.4.4 Deflation . . . 523
22.5 A primer on Cheating . . . 523
22.5.1 Information-based cheating . . . 524
22.5.2 Mechanics-based cheating . . . 524
22.5.3 Man-in-the-middle . . . 524
22.5.4 Low-level exploits . . . 524
22.6 How cheating influences gameplay and enjoyability . . . 525
22.6.1 Single Player . . . 525
22.6.2 Multiplayer . . . 525
22.6.2.1 P2P . . . 525
22.6.2.2 Dedicated Servers . . . 527
22.7 Cheating protection . . . 528
22.7.1 Debug Mode vs Release Mode . . . 529
22.8 Some common exploits . . . 529
22.8.1 Integer Under/Overflow . . . 529
22.8.1.1 How the attack works . . . 530
22.8.2 Repeat attacks . . . 530

23 Accessibility in video games 532

23.1 What accessibility is and what it is not . . . 532
23.2 UI and HUD Scaling . . . 532
23.3 Subtitles . . . 532
23.4 Mappable Buttons . . . 532
23.5 Button Toggling . . . 533
23.6 Dyslexia . . . 533

23.6.1 Text Spacing . . . 533
23.6.2 Fonts . . . 533
23.7 “Slow Mode” . . . 533
23.8 Colorblind mode . . . 533
23.9 No Flashing Lights . . . 534
23.10 No motion blur . . . 534
23.11 Reduced Motion . . . 534
23.12 Assisted Gameplay . . . 534
23.13 Controller Support . . . 534
23.14 Some special cases . . . 535

24 Testing your game 536
24.1 When to test . . . 536
24.1.1 Testing “as an afterthought” . . . 536
24.1.2 Test-Driven Development . . . 536
24.1.3 The “Design to test” approach . . . 536
24.1.4 You won’t be able to test EVERYTHING . . . 536
24.2 Mocking . . . 537
24.3 Types of testing . . . 537
24.3.1 Automated Testing . . . 537
24.3.2 Manual Testing . . . 538
24.4 Unit Testing . . . 538
24.5 Integration Testing . . . 539
24.6 Regression Testing . . . 539
24.7 Playtesting . . . 539
24.7.1 In-House Testing . . . 539
24.7.2 Closed Beta Testing . . . 540
24.7.3 Open Beta Testing . . . 540
24.7.4 A/B Testing . . . 540

25 Profiling and Optimization 541
25.1 Profiling your game . . . 541
25.1.1 Does your application really need profiling? . . . 541
25.1.1.1 Does your FPS counter roam around a certain “special” value? . . . 541
25.1.1.2 Is the animation of your game stuttering but the FPS counter is fine? . . . 541
25.1.2 First investigations . . . 541
25.1.2.1 Is your game using 100% of the CPU? . . . 541
25.1.2.2 Is your game overloading your GPU? . . . 542
25.1.2.3 Is your game eating up more and more RAM as it’s running? . . . 542
25.2 Optimizing your game . . . 543
25.2.1 Working with references vs. returning values . . . 543

25.2.2 Optimizing Drawing 544
25.2.2.1 Off-screen objects 544
25.2.3 Reduce the calls to the Engine Routines 544
25.2.4 Entity Cleanup and Memory leaks 545
25.2.5 Using analyzers to detect Memory Leaks 546
25.2.5.1 Static Scanners 546
25.2.5.2 Dynamic testing tools 546
25.2.6 Resource Pools 546
25.2.7 Lookup Tables 548
25.2.8 Memoization 548
25.2.9 Approximations 551
25.2.10 Eager vs. Lazy Evaluation 551
25.2.10.1 Eager approach 552
25.2.10.2 Lazy approach 552
25.2.11 Detach your updating from drawing 553
25.2.12 Be mindful of how you query your data structures 553
25.3 Tips and tricks 553
25.3.1 Be mindful of your “updates” 553
25.3.2 Use the right data structures for the job 554
25.3.3 Dirty Bit 555
25.3.4 Far-Away entities (Dirty Rectangles) 555
25.3.5 Tweening is better than animating 556
25.3.6 Remove dead code 556
25.4 Non-Optimizations 557
25.4.1 “Switches are faster than IFs” 557
25.4.2 Blindly Applying Optimizations 559

26 Marketing your game 561

26.1 An Important Note: Keep your feet on the ground 561
26.2 The importance of being consumer-friendly 561
26.3 Pricing 562
26.3.1 Penetrating the market with a low price 562
26.3.2 Giving off a “premium” vibe with a higher price 562
26.3.3 The magic of “9” 563
26.3.4 Launch Prices 563
26.3.5 Bundling 563
26.3.6 Nothing beats the price of “Free” (Kinda) 564
26.4 Managing Hype 564
26.5 Downloadable Content 564
26.5.1 DLC: what to avoid 565

26.5.2 DLC: what to do . . . 565
26.6 Digital Rights Management (DRM) . . . 565
26.6.1 How DRM can break a game down the line . . . 567
26.7 Free-to-Play Economies and LootBoxes . . . 567
26.7.1 Microtransactions . . . 567
26.7.1.1 The human and social consequences of Lootboxes . . . 567
26.7.2 Free-to-Play gaming and Mobile Games . . . 568
26.8 Assets and asset Flips . . . 569
26.9 Crowdfunding . . . 569
26.9.1 Communication Is Key . . . 570
26.9.2 Do not betray your backers . . . 570
26.9.3 Don’t be greedy . . . 570
26.9.4 Stay on the “safe side” of planning . . . 571
26.9.5 Keep your promises, or exceed them . . . 571
26.9.6 A striking case: Mighty No. 9 . . . 571
26.10 Engagement vs Fun . . . 572
26.11 Streamers and Content Creators . . . 573
26.11.1 The game developers’ side . . . 573
26.11.2 The Streamers’ side . . . 573
26.11.3 Other entities and conclusions . . . 574

27 Keeping your players engaged 575
27.1 Communities . . . 575
27.1.1 Forums . . . 575
27.1.2 Wikis . . . 575
27.1.3 Update Previews . . . 575
27.1.4 Speedrunning . . . 576
27.1.5 Streaming . . . 577
27.2 Replayability . . . 577
27.2.1 Modding . . . 577
27.2.2 Fan games . . . 578
27.2.3 Mutators . . . 578
27.2.4 Randomizing . . . 579
27.2.5 New Game+ . . . 579
27.2.6 Transmogrification . . . 579

28 When the time for retirement comes 581
28.1 Remonetization . . . 581
28.2 Free release . . . 582
28.3 Open Sourcing . . . 582
28.3.1 When “Open Sourcing” is not enough . . . 583

28.4 Hybrid Approaches . . . 583
28.5 What not to do . . . 583

29 Dissecting games: three study cases 586
29.1 A bad game: Hoshi wo miru hito . . . 586
29.1.1 Introduction . . . 586
29.1.2 Balance Issues . . . 586
29.1.2.1 You can’t beat starter enemies . . . 586
29.1.2.2 The Damage Sponge . . . 586
29.1.2.3 You can’t run away from battles, but enemies can . . . 587
29.1.2.4 Statistics . . . 587
29.1.3 Bad design choices . . . 587
29.1.3.1 You get dropped in the middle of nowhere . . . 587
29.1.3.2 The starting town is invisible . . . 588
29.1.3.3 The Jump Ability . . . 588
29.1.3.4 Items are invisible . . . 588
29.1.3.5 Item management . . . 588
29.1.3.6 Buying Weapons makes you weaker . . . 588
29.1.3.7 Enemy Abilities . . . 589
29.1.3.8 You can soft lock yourself . . . 589
29.1.4 Confusing Choices . . . 589
29.1.4.1 Starting level for characters . . . 589
29.1.4.2 Slow overworld movement . . . 589
29.1.4.3 Exiting a dungeon or a town . . . 589
29.1.4.4 The Health Points UI . . . 590
29.1.5 Inconveniencing the player . . . 590
29.1.5.1 The battle menu order . . . 590
29.1.5.2 Every menu is a committal . . . 590
29.1.5.3 Password saves . . . 590
29.1.5.4 Each character has their own money stash . . . 591
29.1.6 Bugs and glitches . . . 591
29.1.6.1 Moonwalking and save warping . . . 591
29.1.6.2 The final maze . . . 591
29.1.6.3 The endings . . . 592
29.1.7 Conclusions . . . 592
29.2 The first good game - VVVVVV: Slim story and essential gameplay . . . 592
29.2.1 A Slim story that holds up great . . . 592
29.2.2 Essential gameplay: easy to learn, hard to master . . . 593
29.2.3 Diversified challenges . . . 593
29.2.4 Graphics . . . 593

29.2.5 Amazing soundtrack . . . 594
29.2.6 Accessibility Settings . . . 594
29.2.7 Post-endgame Modes . . . 594
29.2.8 User-generated content . . . 594
29.2.9 “Speedrunnability” . . . 595
29.2.10 Characters are memorable, even if you don’t see them a lot . . . 595
29.2.11 Conclusion . . . 595
29.3 Another good game - Undertale: A masterclass in storytelling . . . 595
29.3.1 The power of choice . . . 595
29.3.2 The game doesn’t take itself very seriously (sometimes) . . . 595
29.3.3 All the major characters are very memorable . . . 596
29.3.4 The game continuously surprises the player . . . 596
29.3.5 Player choices influence the game . . . 596
29.3.6 Great (and extensive!!) soundtrack . . . 597
29.3.7 Conclusion . . . 597

30 Project Ideas **598**
30.1 Life . . . 598
30.1.1 Basic Level . . . 598
30.1.2 Advanced Level . . . 599
30.1.3 Master Level . . . 599
30.2 Tic-Tac-Toe . . . 599
30.2.1 Basic Level . . . 600
30.2.2 Advanced Level . . . 600
30.2.3 Master Level . . . 600
30.3 Space Invaders . . . 601
30.3.1 Basic Level . . . 601
30.3.2 Advanced Level . . . 602
30.3.3 Master level . . . 602
30.4 Breakout . . . 603
30.4.1 Basic Level . . . 603
30.4.2 Advanced Level . . . 604
30.4.3 Master Level . . . 604
30.5 Shooter Arena . . . 605
30.5.1 Basic Level . . . 606
30.5.2 Advanced level . . . 606
30.5.3 Master Level . . . 607

31 Game Jams **608**
31.1 Have Fun . . . 608
31.2 Stay Healthy . . . 608

31.3 Stick to what you know . . . 608
31.4 Hacking is better than planning (But still plan ahead!) . . . 608
31.5 Graphics? Sounds? Music? FOCUS! . . . 609
31.6 Find creativity in limitations . . . 609
31.7 Involve Your Friends! . . . 610
31.8 Write a Post-Mortem (and read some too!) . . . 610
31.9 Most common pitfalls in Game Jams . . . 610

32 Where To Go From Here 612
32.1 Collections of different topics . . . 612
32.1.1 Books . . . 612
32.1.2 Videos . . . 612
32.1.3 Multiple formats . . . 612
32.2 Pixel Art . . . 613
32.2.1 Multiple Formats . . . 613
32.3 Sound Design . . . 613
32.3.1 Multiple Formats . . . 613
32.4 Game Design . . . 613
32.4.1 Books . . . 613
32.5 Game Development . . . 613
32.5.1 Web Resources . . . 613
32.5.2 Videos . . . 613
32.6 References and Cheat Sheets . . . 613

A Glossary 615

B Engines, Libraries And Frameworks 619

C Some other useful tools 631

D Free assets and resources 637

E Contributors 640

1 Foreword

Every time we start a new learning experience, we may be showered by an immeasurable amount of doubts and fears. The task, however small, may seem daunting. And considering how large the field of Game Development can be, these fears are easily understandable.

This book is meant to be a reference for game developers, oriented at 2D, as well as being a collection of “best practices” that you should follow when developing a game by yourself (or with some friends).

But you shouldn’t let these “best practices” jail you into a way of thinking that is not yours, actually my first tip in this book is **do not follow this book. Yet.**

Do it wrong.

Learn why these best practices exist by experience, make code so convoluted that you cannot maintain it anymore, don’t check for overflows in your numbers, **allow yourself to do it wrong**.

Your toolbox is there to aid you, your tools don’t have feelings that can be hurt (although they will grumble at you many times) in the same way that you cannot hurt a hammer when missing the nail head. You cannot break a computer by getting things wrong (at least 99.9999% of the time). Breaking the rules will just help you understand them better.

Write your own code, keep it as simple as you can, and practice.

Don’t let people around you tell you that “you shouldn’t do it that way”, if you allow that to happen you’re depriving yourself of a great opportunity to learn. Don’t let others’ “lion tamer syndrome” get to you, avoid complex structures as much as possible; cutting and pasting code will get you nowhere.

But most of all, never give up and try to keep it fun.

There will be times where you feel like giving up, because something doesn’t work exactly as you want it to, or because you feel you’re not ready to put out some code. When you don’t feel ready, just try making something simple, something that will teach you how to manipulate data structures and that gives you a result in just a couple days of work. Just having a rectangle moving on the screen, reacting to your key presses can be that small confidence boost that can get you farther and farther into this world.

And when all else fails, take a pen, some paper and your favorite Rubber Duck (make sure it is impact-proof) and **think**.

Coding is hard, but at the same time, it can give you lots of satisfaction.

I really hope that this book will give you tips, tricks and structures that one day will make you say “Oh yeah, I can use that!”. So that one day you are able to craft an experience that someone else will enjoy, while you enjoy the journey that brings to such experience.

2 Introduction

> A journey of a thousand miles begins with a single step
>
> *Laozi - Tao Te Ching*

Welcome to the book! This book aims to be an organized collection of the community's knowledge on game development techniques, algorithms and experience with the objective of being as comprehensive as possible.

2.1 Why another game development book?

It's really common in today's game development scene to approach game development through tools that abstract and guide our efforts, without exposing us to the nitty-gritty details of how things work on low-level and speeding up and easing our development process. This approach is great when things work well, but it can be seriously detrimental when we are facing against issues: we are tied to what the library/framework creators decided was the best (read "applicable in the widest range of problems") approach to solving a problem.

Games normally run at 30fps, more modern games run at 60fps, some even more, leaving us with between 33ms to 16ms or less to process a frame, which includes:

- Process the user input;
- Update the player movement according to the input;
- Update the state of any AI that is used in the level;
- Move the NPCs according to their AI;
- Identify Collisions between all game objects;
- React to said Collisions;
- Update the Camera (if present);
- Update the HUD (if present);
- Draw the scene to the screen.

These are only some basic things that can be subject to change in a game, **every single frame**.

When things don't go well, the game lags, slows down or even locks up. In that case we will be forced to take the matter in our hands and get dirty handling things exactly as we want them (instead of trying to solve a generic problem).

When you are coding a game for any device that doesn't really have "infinite memory", like a mobile phone, consoles or older computers, this "technical low-level know-how" becomes all the more important.

This book wants to open the box that contains everything related to 2D game development, plus some small tips and tricks to make your game more enjoyable. This way, if your game encounters some issues, you won't fear diving into low-level details and fix it yourself.

Or why not, make everything from scratch using some pure-multimedia interfaces (like SDL or SFML) instead of fully fledged game engines (like Unity).

This book aims to be a free (as in price) teaching and reference resource for anyone who wants to learn 2D game development, including the nitty-gritty details.

Enjoy!

2.2 Conventions used in this book

2.2.1 Logic Conventions

When talking about logic theory, the variables will be represented with a single uppercase letter, written in math mode: A

The following symbol will be used to represent a logical "AND": $\wedge$

The following symbol will be used to represent a logical "OR": $\vee$

The logical negation of a variable will be represented with a straight line on top of the variable, so the negation of the variable A will be $\bar{A}$

2.2.2 Code Listings

Listings, algorithms and anything that is code will be shown in monotype fonts, using syntax highlighting where possible, inside of a dedicated frame:

Listing 1: Example code listing

```
function Example(phrase){
    // This is a simple example function
    console.log(phrase);
}

class ExampleClass{
    // This is a simple example class
    constructor(){
        // This is an example constructor
    }
}
```

2.2.3 Block Quotes

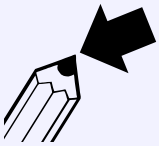
There will be times when it's needed to write down something from another source verbatim, for that we will use block quotes, which are styled as follows:

> Hi, I'm a block quote! You will see me when something is... quoted!
>
> I am another row of the block quote! Have a nice day!

2.2.4 Boxes

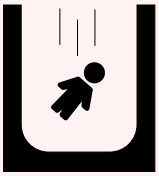
In your journey through this book, you may find some boxes, let's see which ones you may come across.

Tip!



This is a tip box, here you will find tips that are loosely related to the chapter at hand. These small tips will help you make a better game, or wiggle your way through something difficult.

Pitfall Warning!



This is a pitfall box, it will warn you of traps behind the corner, as well as possible shortcomings of a certain solution.

Random Trivia!

This is a trivia box, it will give out some small facts that can help you understand things better, or just give you a small break from all the learning.

Note!

This is just a note box, it's not a pitfall, a tip or a trivia. This is used for reminders and just as a general purpose note

Advanced Wizardry!

This will warn you of complex sections, or sections treating advanced topics that have limited game development usefulness. Such sections may be skimmed over.

2.2.5 Engine Used

Most editions of this book does not use any engine. All algorithms will be presented pretending there is some "generic engine" behind the scenes that handles sprites, vectors and the like. The objective of this book is teaching algorithms, tips and tricks and game design in the most engine-agnostic (and language-agnostic, if you're looking at the "pseudocode edition") way possible.

If instead you're reading a version that features "language extensions", all algorithms will be in your favourite language, using your favourite engine.

2.2.6 About editions

This book comes in various editions, and they come with some caveats.

- **Pseudocode Edition:** This is the standard edition, using a C-like syntax that tries to be as readable as possible and abstracts itself from any kind of engine.
- **Python Edition:** Python is considered one of the easiest language to start coding on. Many tend to complain about its performance, but its similarity to Godot Engine's GDScript and its flexibility make it a good candidate for starters.
- **C++ Edition:** C++ is probably the most used language in game development (along with C#) but it can be really difficult to manage. It has no garbage collection, forcing you to manage the memory manually, and pointers can prove to be a difficult concept for many.
- **JavaScript Edition:** Javascript is the de-facto "internet language" and its influence is spreading to desktop applications and video games too. Many games now can be played on the browser thanks to it and the HTML5 canvas elements. This is a language that can be very forgiving and frustrating at the same time.
- **Lua Edition:** Lua is one of the most spread scripting languages in the world of video games. Since it has a very small interpreter, it can be added to a lot of code bases without weighing them down much. It is not a proper object-oriented language, but it has very strong metaprogramming capabilities (where you can "program the programming language"). There are also some libraries that allow for classes and object-oriented concepts to fit in Lua.
- **Rust Edition:** Rust is the most admired language since it was release. Rust has gained a lot of traction as an alternative to C++ thanks for its performance and focus on memory safety.

2.3 Structure of this Book

This book is structured in many chapters, here you will find a small description of each and every one of them.

- **Foreword:** You didn't skip it, right?
- **Introduction:** Here we present the structure of the book and the reasons why it came to exist. You are reading it now, hold tight, you're almost there!
- **The Maths Behind Game Development:** Here we will learn the basic maths that are behind any game, like vectors, matrices and screen coordinates.
- **Some Computer Science Fundamentals:** Here we will learn (or revise) some known computer science fundamentals (and some less-known too!) and rules that will help us managing the development of our game.
- **A game design dictionary:** Here we will introduce some basic concepts that will help us in understanding game design: platforms, input devices and genres.
- **Project Management Basics and Tips:** Project management is hard! Here we will take a look at some common pitfalls and tips that will help us deliver our own project and deliver it in time.
- **Writing a Game Design Document:** In this section we will take a look at one of the first documents that comes to exist when we want to make a game, and how to write one,
- **The Game Loop:** Here we will learn the basics of the "game loop", the very base of any video game.
- **Collision Detection and Reaction:** In this section we will talk about one of the most complex and compu-

tationally expensive operations in a video game: collision detection.

- **Scene Trees:** Here we will briefly talk about probably the most important structure in games and game engines: the scene tree.
- **Cameras:** In this section we will talk about the different types of cameras you can implement in a 2D game, with in-depth analysis and explanation.
- **Game Design Tips:** In this chapter we will talk about level design and how to walk your player through the learning and reinforcement of game mechanics, dipping our toes into the huge topic that is game design.
- **World Building:** A section completely dedicated to the art of building a world that will make sense in the player's mind, with tips and tricks on how to make it more engaging and fun.
- **Creating your own assets:** Small or solo game developers may need to create their own assets, in this section we will take a look at how to create our own graphics, sounds and music.
- **Design Patterns:** A head-first dive into the software engineering side of game development, in this section we will check many software design patterns used in many games.
- **Useful Containers and Classes:** A series of useful classes and containers used to make your game more maintainable and better performing.
- **Artificial Intelligence in Video games:** In this section we will talk about algorithms that will help you coding your enemy AI, as well as anything that must have a "semblance of intelligence" in your video game.
- **Other Useful Algorithms:** In this section we will see some algorithms that are commonly used in game, including path finding, world generation and more.
- **Procedural Content Generation:** In this chapters we will see the difference between procedural and random content generation and how procedural generation can apply to more things than we think.
- **Developing Game Mechanics:** Here we will dive into the game development's darkest and dirtiest secrets, how games fool us into strong emotions but also how some of the most used mechanics are implemented.
- **Balancing Your Game:** A very idealistic vision on game balance, in this chapter we will take a look inside the player's mind and look at how something that may seem "a nice challenge" to us can translate into a "terrible balance issue" to our players.
- **Accessibility in video games:** Here we will learn the concept of "accessibility" and see what options we can give to our players to make our game more accessible (as well as more enjoyable to use).
- **Testing your game:** This section is all about hunting bugs, without a can of bug spray. A deep dive into the world of testing, both automated and manual.
- **Profiling and Optimization:** When things don't go right, like the game is stuttering or too slow, we have to rely on profiling and optimization. In this section we will learn tips and tricks and procedures to see how to make our games perform better.
- **Marketing Your Game:** Here we will take a look at mistakes the industry has done when marketing and maintaining their own products, from the point of view of a small indie developer. We will also check some of the more controversial topics like loot boxes, micro transactions and season passes.
- **Keeping your players engaged:** a lot of a game's power comes from its community, in this section we will take a look at some suggestion you can implement in your game (and out-of-game too) to further engage your loyal fans.
- **Dissecting Games:** A small section dedicated to dissecting the characteristics of one (very) bad game, and

two (very) good games, to give us more perspective on what makes a good game “good” and what instead makes a bad one.

- **Project Ideas:** In this section we take a look at some projects you can try and make by yourself, each project is divided into 3 levels and each level will list the skills you need to master in order to be able to take on such level.
- **Game Jams:** A small section dedicated on Game Jams and how to participate to one without losing your mind in the process, and still deliver a prototype.
- **Where to go from here:** We’re at the home stretch, you learned a lot so far, here you will find pointers to other resources that may be useful to learn even more.
- **Glossary:** Any world that has a $_{g}$ symbol will find a definition here.
- **Engines and Frameworks:** A collection of frameworks and engines you can choose from to begin your game development.
- **Tools:** Some software and tool kits you can use to create your own resources, maps and overall make your development process easier and more manageable.
- **Premade Assets and resources:** In this appendix we will find links to many websites and resource for graphics, sounds, music or learning.
- **Contributors:** Last but not least, the names of the people who contributed in making this book.

Have a nice stay and let’s go!

Part 1: The basics

3 The Maths Behind Game Development

> Do not worry about your difficulties in Mathematics. I can assure you mine are still greater.
>
> *Albert Einstein*

This book assumes you already have some knowledge of maths, but we will also try to keep the bar of entry as low as possible.

Also we will represent derivatives with the $f'(x)$ symbol, instead of the more verbose $\frac{\partial f}{\partial x}$.

In this chapter we'll take a quick look (or if you already know them, a refresher) on the basic maths needed to make a 2D game.

3.1 Some useful symbols

While reading this book, we may need to delve into some mathematical lingo that not everyone may understand immediately, so here's a small glossary of some of mathematical the symbols we may use.

- $x \in S$ Denotes a "set membership", so the object to the left of the symbol is an element of the set at the right: x is an element inside the set S;
- $A \subset B$ Denotes a "subset relationship": A is a subset of B;
- $A \subseteq B$ Denotes a "subset relationship" where equality is possible: A is a subset of B, but also it may happen that A equals B;
- $A \cup B$ Denotes "set union", the result is composed by all elements of A and B, combined;
- $A \cap B$ Denotes "set intersection", the result is composed by all elements of A that are also found in B;
- $\forall$ Means "for all";
- $\exists$ Means "exists";
- $\exists!$ Means "exists only one";
- $P \rightarrow Q$ Means "implies", so you can read this as "P implies Q" or "if P is true then Q is true";
- $P \leftrightarrow Q$ Logical equivalence: means "if and only if" or "is equivalent", so you can read this as "P is equivalent to Q" or "P if and only if Q";

3.2 The modulo operator

Very basic, but sometimes overlooked, function in mathematics is the "modulo" function (or "modulo operator"). Modulo is a function that takes 2 arguments, let's call them "a" and "b", and returns the remainder of the division represented by a/b.

So we have examples like $mod(3, 2) = 1$ or $mod(4, 5) = 4$ and $mod(8, 4) = 0$.

In most programming languages the modulo function is hidden behind the operator "%", which means that the function $mod(3, 2)$ is represented with $3\%2$.

The modulo operator is very useful when we need to loop an ever-growing value between two values (as will be

shown in infinitely scrolling backgrounds).

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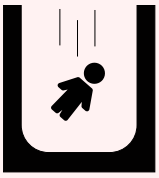

Be careful when using the modulo operator with negative arguments: it may lead to unexpected results, which may depend on the programming language you are using.

3.3 Powers and Roots

We start our revision of maths by remembering powers and roots. A power is just a short way to multiply a number by itself a certain amount of times.

For example: $2^3 = 2 \cdot 2 \cdot 2 = 8$, 2 is multiplied by itself 3 times, giving 8 as a result.

Some other examples can be $4^4 = 4 \cdot 4 \cdot 4 \cdot 4 = 256$ and $0^{9532} = 0 \cdot 0 \cdot ... \cdot 0 = 0$

One rule that we need to remember is that any number, when elevated to the zero-th power is always 1, so $256^0 = 1$ as well as $2^0 = 1$.

Note!

Technically 0^0 might be considered "undefined", but in most non-rigorous mathematical environments $0^0 = 1$ is accepted.

Roots are the inverse operation of powers, which means that if $4^2 = 16$ then $\sqrt[2]{16} = 4$

So we can say that

> The nth root of a number x is a number r so that $r^n = x$

Taking the examples of earlier, we have that $\sqrt[3]{8} = 2$, $\sqrt[4]{256} = 4$ and $\sqrt[9532]{0} = 0$. Omitting the index n on the root is a short way to write the "square root", which is the root with index 2. That means:

$\sqrt[2]{4} = \sqrt{4} = 2$

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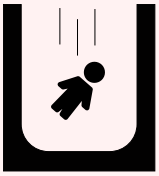

When talking "real numbers", there is no $\sqrt{-1}$: that would fall into the "complex numbers" category, which are a matter outside the scope of this revision. That's because there is no real number that multiplied by itself an even amount of times would give a negative number. To make things more complex, roots with odd indices of negative numbers are part of the real numbers instead: $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$

3.4 Equations

Equations are a way to express equality between two expressions, we've seen equations all our lives, just "hidden". Every operation is an equation.

In their more known form, equations can have one or more "unknowns", usually represented with letters (the most used are, in order x, y and z) and "solving an equation" means finding values for the unknowns that make the equation true.

Here's a simple equation:

$$2 \cdot x = 10$$

Which can be read as "x is the number, that multiplied by 2, gives 10", the solution of this equation is $x = 5$, because $2 \cdot 5 = 10$.

There are some basic rules, here's a quick rundown.

3.4.1 You can add or subtract any number on both sides

This is one of the rules that will help us making things a bit easier. Let's take the following example:

$$15 + 2x = 45$$

We can subtract 15 on both sides to make our life easier:

$$-15 + 15 + 2x = 45 - 15$$

$$2x = 30$$

3.4.2 You can multiple or divide any non-zero number on both sides

This is another one of those rules that makes things a lot easier, taking the previous example:

$$2x = 30$$

We can divide each side by 2 (or multiply it by $\frac{1}{2}$) to get to the final result:

$$\frac{1}{2} \cdot 2x = 30 \cdot \frac{1}{2}$$

$$\frac{2x}{2} = \frac{30}{2}$$

$$x = 15$$

3.5 Exponentiations and Logarithms

Similarly to powers involving simple numbers, we can involve letters in powers too, making them "exponentiations".

$$2^x = 32$$

In this case x is the number that makes the previous equation true (by the way, the result is $x = 5$).

Its inverse is called a "logarithm", and it's represented as follows:

$$\log_2 32 = x$$

In the previous example "2" is called the "base" of the logarithm. So this formula is read as "x is the base 2 logarithm of 32" (the result is still 5, by the way).

Here is a quick table of rules that can be used to make logarithms easier to calculate.

Table 1: Some rules that would help us calculating logarithms

Rule	Formula
Product	$\log_b(x \cdot y) = \log_b x + \log_b y$
Quotient	$\log_b(\frac{x}{y}) = \log_b x - \log_b y$
Power	$\log_b(x^p) = p \cdot \log_b x$
Root	$\log_b(\sqrt[p]{x}) = \frac{log_b x}{p}$

3.6 Limits

Advanced Wizardry!

We're entering some complex math territory here, so I will give you an "intuitive" definition of a limit. Having an idea of what it is will suffice for the needs of this book

Limits are an interesting beast: we can see them as the value a function approaches as the input approaches some value. Limits are written as follows:

$$\lim_{x \to a} f(x) = y$$

In this case it can be read as "y is the limit, for x approaching a, of $f(x)$". Limits can be seen as, "the more x gets closer to a, the more $f(x)$ gets closer to y".

y and a can be any value, including infinity. In fact the following statement is true:

$$\lim_{x \to +\infty} x = +\infty$$

The further we count down the line of numbers the closer we get to infinity. Which also means that:

$$\lim_{x \to +\infty} \frac{1}{x} = 0$$

Because as we are counting up with x, we are dividing 1 by bigger and bigger numbers until (at the limit) we reach 0.

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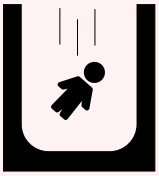


There are some situations where a limit cannot be determined immediately (or sometimes at all). Some of these are $+\infty - \infty$, $0 \cdot \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$, ∞^0, 0^0 and 1^∞.

3.7 Derivatives

Note!

This is not a complete guide to derivatives, there is so much more to it than written in here. This is mostly for informational purposes when the term "derivative" will be used in this book.

Derivatives are technically just a limit, to be precise they are the following limit:

$$\lim_{h \to 0} \frac{f(x+h) - f(h)}{h}$$

They also have a nifty property that is used extensively in calculus: if $f'(x) > 0$ then $f(x)$ is increasing, while if $f'(x) < 0$ then $f(x)$ is decreasing. This means that the equation $f'(x) = 0$ can be used to find local extrema: also known as "local minimums" and "local maximums".

There are some rules to quickly derivative functions, here we list some of the most basic.

Table 2: Some simple derivation rules (k is any constant number and e is Euler's number)

Function	Derivative
k	0
x^k	$k \cdot x^{k-1}$
e^x	e^x

Then there are rules for sums, multiplications and divisions.

Table 3: Some derivation rules for combined functions (a and b are constants)

Functions	Derivative
$af(x) + bg(x)$	$af'(x) + bg'(x)$
$f(x)g(x)$	$f'(x)g(x) + f(x)g'(x)$
$\frac{f(x)}{g(x)}$	$\frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$
$f(g(x))$	$f'(g(x)) \cdot g'(x)$

3.8 The Cartesian Plane

The Cartesian plane is a plane that features a 2-dimensional coordinate system. This way we can represent points with a pair of coordinates (x, y).

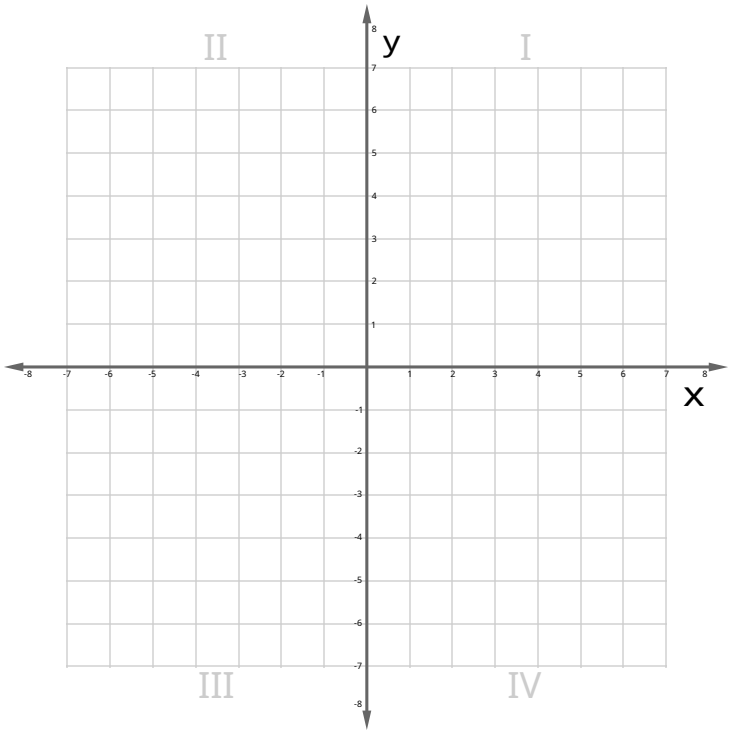


Figure 1: Example of a Cartesian plane

Using a Cartesian plane we can represent the position of items, as well as their shape, space occupation, as well as vectors that represent forces, velocity and direction.

It is an essential tool for 2D game development, and it will be one of the abstractions we will use to represent items in a 2-Dimensional plane.

3.9 Vectors

For our objective, we will simplify the complex matter that is vectors as much as possible.

In the case of 2D game development, a vector is just a pair of values `(x,y)`.

Vectors usually represent a force applied to a body, its velocity or acceleration and are graphically represented with an arrow.

On a Cartesian plane it can be seen as "the x and y quantities you need to move to get from a point to another".

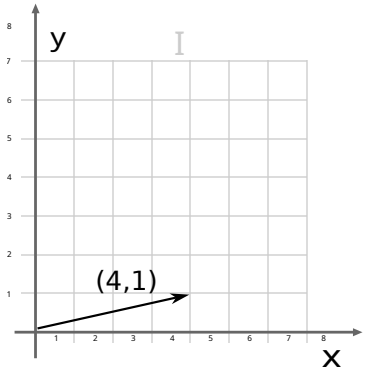


Figure 2: Image of a vector

From the previous example, the vector $v = (4, 1)$ can be thought of as the following:

> you need to move 4 units on the x axis and 1 on the y axis to go from the origin to the point $P(4, 1)$

The pain of learning about vectors is paid off by their capacity of being added and subtracted among themselves, as well as being multiplied by a number (called a "scalar") and between themselves.

3.9.1 Adding and Subtracting Vectors

Adding vectors is as easy as adding its "members". Let's consider the following vectors:

$v = (4, 1)$

$u = (1, 4)$

The sum vector s will then be:

$s = v + u = (4 + 1, 1 + 4) = (5, 5)$

Graphically it can be represented by placing the tail of the arrow v on the head of the arrow u, or vice-versa:

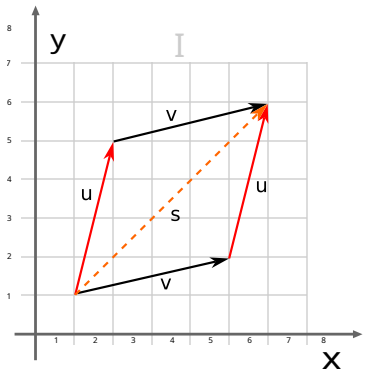


Figure 3: Graphical representation of a sum of vectors

A different example could be the following:

$v = (2, 4)$

$u = (1, 5)$

The sum vector s will be:

$s = u + v = (2 + 1, 4 + 5) = (3, 9)$

3.9.2 Scaling Vectors

There may be situations where you need to make a vector x times longer. This operation is called "scalar multiplication" and it is performed as follows:

$v = (1, 2)$

$3 \cdot v = (1 \cdot 3, 2 \cdot 3) = (3, 6)$

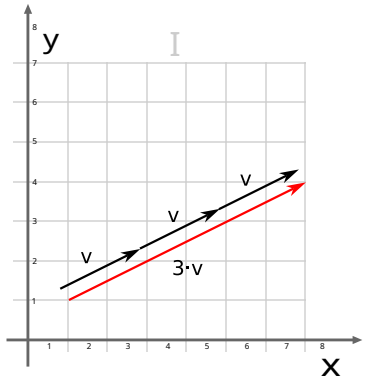


Figure 4: Example of a vector multiplied by a value of 3

Obviously this works with scalars with values between 0 and 1:

$v = (4, 2)$

$\frac{1}{2} \cdot v = (\frac{1}{2} \cdot 4, \frac{1}{2} \cdot 2) = (2, 1)$

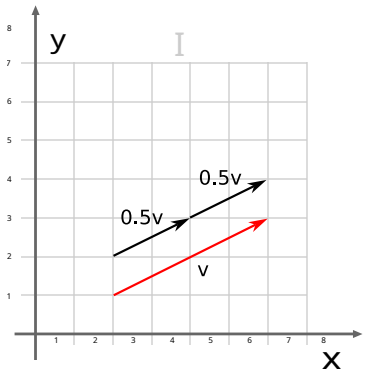


Figure 5: Example of a vector multiplied by a value of 0.5

When you multiply the vector by a value less than 0, the vector will rotate by $180°$.

$v = (1, 2)$

$-2 \cdot v = (-2 \cdot 1, -2 \cdot 2) = (-2, -4)$

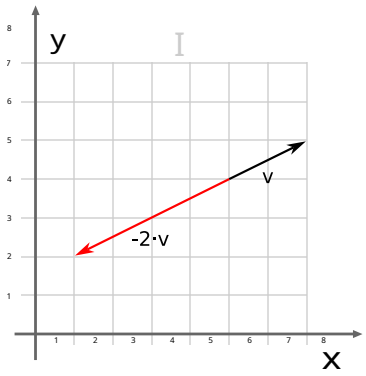


Figure 6: Example of a vector multiplied by a value of -2

3.9.3 Dot Product

The dot product (or scalar product, projection product or inner product) is defined as follows:

Given two n-dimensional vectors $v = [v_1, v_2, ...v_n]$ and $u = [u_1, u_2, ..., u_n]$ the dot product is defined as:

$$v \cdot u = \sum_{i=1}^{n} (v_i \cdot u_i) = (v_1 \cdot u_1) + ... + (v_n \cdot u_n)$$

So in our case, we can easily calculate the dot product of two two-dimensional vectors $v = [v_1, v_2]$ and $u = [u_1, u_2]$ as:

$$v \cdot u = (v_1 \cdot u_1) + (v_2 \cdot u_2)$$

Let's make an example:

Given the vectors $v = [1, 2]$ and $u = [4, 3]$, the dot vector is:

$$v \cdot u = (1 \cdot 4) + (2 \cdot 3) = 4 + 6 = 10$$

3.9.4 Vector Length and Normalization

Given a vector $a = [a_1, a_2, ..., a_n]$, you can define the length of the vector as:

$$||a|| = \sqrt{a_1^2 + a_2^2 + ... + a_n^2}$$

Or alternatively

$$||a|| = \sqrt{a \cdot a}$$

We can get a 1-unit long vector by "normalizing" it, getting a vector that is useful to affect (or indicate) direction without affecting magnitude. A normalized vector is usually indicated with a "hat", so the normalized vector of $a = [a_1, a_2, ..., a_n]$ is

$$\hat{a} = \frac{a}{||a||}$$

Knowing that the length of a vector is a scalar (a number, not a vector), normal scalar multiplication rules apply. (See Scaling Vectors)

3.9.5 "Clamping" a Vector

This is not an operation "per se", but there are occasions where we need to limit the length of a vector: this usually happens when we are working with velocity, as not limiting it would allow an object to change position faster and faster, making the game less playable and even breaking time-stepping collision detection algorithms.

To clamp a vector, we need to find its magnitude and direction first, which is the "normalized vector". Let's think about the vector v, its magnitude and direction are:

$$||v|| = \sqrt{v \cdot v}$$

$$\hat{v} = \frac{v}{||v||}$$

After that, we can build a new vector using the "clamped magnitude" (which we'll call $||v||_{clamp}$), calculated as such:

$$||v||_{clamp} = \begin{cases} ||v|| \; when \; ||v|| < ||v||_{max} \\ ||v||_{max} \; otherwise \end{cases}$$

To build the new vector, we just need to multiply $||v||_{clamp}$ by $\hat{v}$:

$$v_{clamp} = ||v||_{clamp} \cdot \hat{v}$$

The new vector will have the same direction as the old one, but its magnitude will be clamped, just like we wanted.

3.10 Geometry

Among all the maths we found so far (and the maths we will explain later), we cannot avoid talking a bit about geometry: in this book we will talk about the minimal amount of geometry necessary to understand the underlying concepts of what's coming up.

3.10.1 Convex vs Concave polygons

A polygon is considered convex essentially when **any line** (not tangent to an edge or corner) drawn through the shape crosses the shape itself only twice (at its ends).

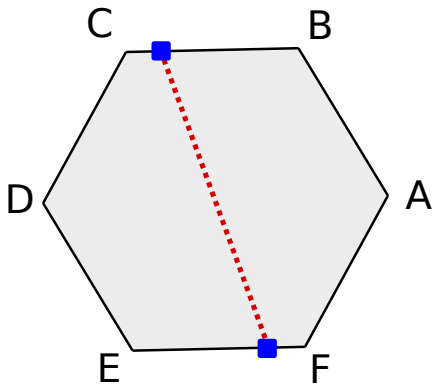


Figure 7: Example of a convex shape

Any shape where you can find at least one line that crosses the shape more than twice is considered "non-convex" (commonly referred as "concave").

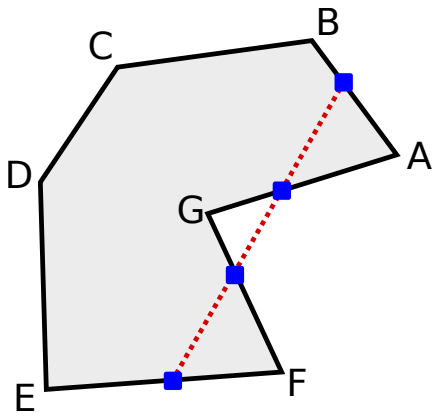


Figure 8: Example of a concave shape

Note!

Not all non-convex shapes are technically called “concave” (they should be called “non-convex”), but for the sake of simplicity we’ll use the term “non-convex” and “concave” interchangeably in this book.

3.10.2 Self-intersecting polygons

Contrary to what many think, polygons can self-intersect too, which can make calculations a lot harder.

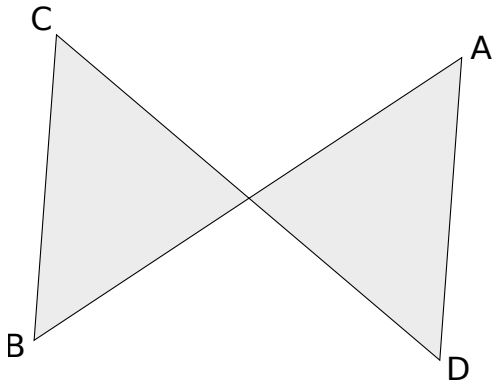


Figure 9: Example of a self-intersecting polygon

For the sake of game development, we will usually talk about simple polygons which are polygons that don’t self-intersect and have no holes in them. More strictly we will (for 99.9% of the time) talk about **convex simple polygons**.

3.10.3 Straight Lines and their equations

One of the main topics we will encounter over and over in our game development adventure will be “straight lines”. We will need to draw them, see if two straight lines collide, project stuff onto them, and much more. So it’s important that we know them well.

Here's a straight line:

$$ax + by + c = 0$$

That's not what you expected, right? What you've seen is the "general form" of a straight line's equation, because you can represent lines using equations (also circles, and other stuff). This is not a much-used form, though, probably the most used form is called the "slope-intercept form":

$$y = mx + q$$

Random Trivia!

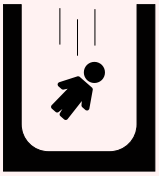
To transform a "general form" equation into the relative "slope-intercept from" just remember the following formulas:

$$m = -\frac{a}{b} \quad q = -\frac{c}{b}$$

This doesn't work well when $b = 0$, which will be subject of the next "pitfall".

Where in this case m is the *slope* of our straight line, and q represents the so-called *y-intercept* (the value of y when $x = 0$). If $q = 0$ the line goes through the origin of the Cartesian coordinate system, if $m = 0$ the line is horizontal.

Pitfall Warning!



"Vertical straight lines" is where the slope-intercept form fails, in fact vertical straight lines have an equation in the form of $x = k$, which would mean that $b = 0$ which is problematic (see previous trivia).

3.10.3.1 Getting the equation of a straight line, given two points

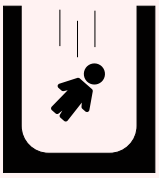
We all know that given two points we can strike one and only one line. How many times did you measure two points (maybe while doing some D.I.Y.) and stroke a line between them?

It will be useful in our adventure to be able to get the equation of a straight line starting from two points, so let's call our two points $P(x_1, y_1)$ and $Q(x_2, y_2)$, then the straight line that crosses both those points will have equation:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

This may seem really complicated, but with some small calculations we can reach a formula for our straight line in any form (generic or "slope-intercept").

Pitfall Warning!



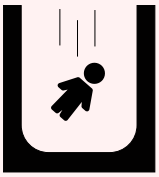
Again, this formula fails when we are dealing with "vertical lines", because the denominator at the right side of the equation will be zero. But in that case we'll already know the formula: it will be $x = x_1$ (which in turn will be equal to x_2)

3.10.3.2 Getting the equation, given the slope and a point

If we have a point $P(x_p, y_p)$ and the slope m (for instance if we need to find a line perpendicular to another line), in that case we can use the following formula:

$$y - y_p = m(x - x_p)$$

Pitfall Warning!



Guess what? This (again) doesn't allow us to create "vertical lines", because we need a slope value, which we don't have when it comes to vertical lines. You can see (non rigorously) a vertical line as a line with "infinite slope".

3.10.4 Projections

In some situations (as you will see in the SAT), we may need to get to project polygons onto a line, this usually involves projecting **points** to a line.

Given the formulas we've seen earlier, and doing some thinking, we can easily project a point onto any straight line. Let's see how to do it.

First of all, the line we will be projecting onto will have equation $y = mx + q$, just as in the slope-intercept formula.

We will assume that we have a point $P(x_p, y_p)$ that we want to project onto a line r with equation $y = mx + q$, with $m \neq 0$ (thus excluding horizontal lines). We will call the projected point "P onto r" with the name $P_r(x_r, y_r)$.

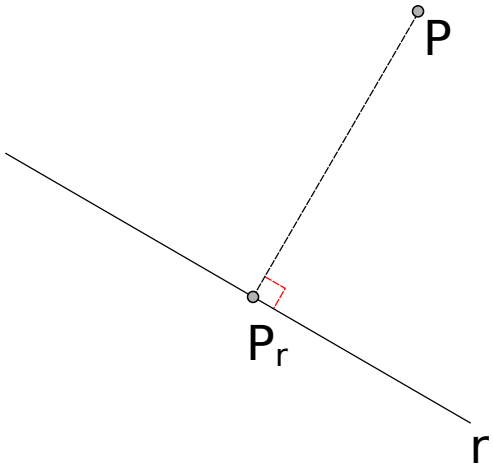
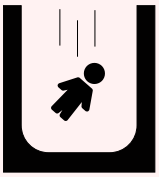


Figure 10: Projecting the point P onto the line r

First, we need to find the line that goes through P and is perpendicular to r, this is really easy. To find a slope m_1 of a line perpendicular to another line with slope m we use the formula

$$m_1 = -\frac{1}{m}$$

Pitfall Warning!



This is why we excluded the case $m = 0$ (horizontal lines), if we didn't we would have the chance of having $m_1 = \frac{1}{0}$ which doesn't make sense.
In this case we can easily conclude that if $m = 0$, the projection of the point P onto the line r has coordinates (x_P, y) (with y taken from the line we're projecting onto).

Now we have a point and a slope, so we can use one of the formulas we've already seen to find the line with that slope that crosses P:

$$y - y_p = m_1(x - x_p) \Leftrightarrow y - y_p = -\frac{1}{m}(x - x_p)$$

To find P_r we just need to find the point where the two lines collide, which is the solution to the equation system:

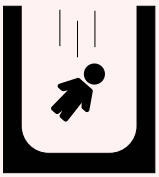
$$\begin{cases} y = mx + q \\ y - y_p = -\frac{1}{m}(x - x_p) \end{cases}$$

Which finds solution in:

$$\begin{cases} x = \frac{x_p + m y_p - m q}{m^2 + 1} \\ y = \frac{m x_p + m^2 y_p + q}{m^2 + 1} \end{cases}$$

The coordinates x and y we just found are actually the coordinates x_r and y_r of our projected point P_r.

Pitfall Warning!



Due to the fact that we used $m_1 = -\frac{1}{m}$ the previous results are not valid for $m = 0$. The denominator of the results gives no issue, since $m^2 + 1 = 0$ does not have a solution in real numbers (and we won't need to delve into the Complex number territory).

3.10.4.1 Projecting arbitrary lines on the axes

Similarly to what we've done with points, we can project arbitrary lines (or, to be precise, **the ends** of such lines) onto the axes. This will help us in doing some calculations later (when we'll talk about SAT).

To project any line r to the x-axis we can just "pass all the line's points through" the following function:

$$proj_x(P_r(x, y)) = (x, 0)$$

for each point P_r in the line r.

If we want to project such line on the y-axis, we can just use this other function:

$$proj_y(P_r(x, y)) = (0, y)$$

for each point P_r in the line r.

We can see an intuitive representation of projecting a line onto the axes below:

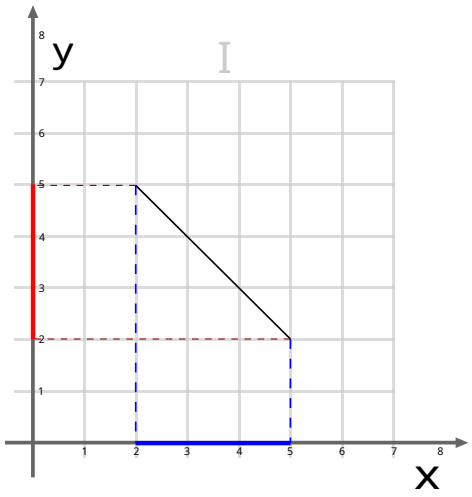


Figure 11: Projecting a line onto the axes

3.10.4.1.1 How does it work?

Let's take the point $P(2,5)$ from the previous figure. We want to project it on the x axis: that means we need to find a line that is 90 degrees with the x axis and passes through P.

Such line is the line with equation $x = 2$, now to find the projection of P onto the x axis, we will just need to solve a simple equation system.

$$\begin{cases} x = 2 \\ y = 0 \end{cases}$$

Where $y = 0$ is the equation of the x axis. So our projected point is $P_x(2,0)$.

Similar thing goes for projecting the point on the y axis: the line that is 90 degrees with the y axis and goes through P has equation $y = 5$, the y axis has equation $x = 0$, thus the system of equation is solved with $P_y(0,5)$.

3.11 Matrices

3.11.1 What is a matrix

Matrices are essentially an $m \times n$ array of numbers, which are used to represent linear transformations.

Here is an example of a 2×3 matrix.

$$A_{2,3} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 0 \end{bmatrix}$$

3.11.2 Matrix sum and subtraction

Summing and subtracting $m \times n$ matrices is done by summing or subtracting each element, here is a simple example. Given the following matrices:

$$A_{2,3} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 0 \end{bmatrix} B_{2,3} = \begin{bmatrix} 1 & 3 & 0 \\ 4 & 2 & 4 \end{bmatrix}$$

We have that:

$$A_{2,3} + B_{2,3} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 0 \\ 4 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2+1 & 1+3 & 4+0 \\ 3+4 & 2+2 & 0+4 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 4 \\ 7 & 4 & 4 \end{bmatrix}$$

3.11.3 Multiplication by a scalar

Multiplication by a scalar works in a similar fashion to vectors, given the matrix:

$$A_{2,3} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 0 \end{bmatrix}$$

Multiplication by a scalar is performed by multiplying each member of the matrix by the scalar, like the following example:

$$3 \cdot A_{2,3} = 3 \cdot \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 \cdot 2 & 3 \cdot 1 & 3 \cdot 4 \\ 3 \cdot 3 & 3 \cdot 2 & 3 \cdot 0 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 12 \\ 9 & 6 & 0 \end{bmatrix}$$

3.11.4 Transposition

Given an $m \times n$ matrix A, its transposition is an $n \times m$ matrix A^T constructed by turning rows into columns and columns into rows.

Given the matrix:

$$A_{2,3} = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 0 \end{bmatrix}$$

The transpose matrix is:

$$A^T_{2,3} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix}$$

3.11.5 Multiplication between matrices

Given 2 matrices with sizes $m \times n$ and $n \times p$ (mind how the number of columns of the first matrix is the same as the number of rows of the second matrix):

$$A_{3,2} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} B_{2,4} = \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

We can calculate the multiplication between these two matrices, in the following way.

First of all let's get the size of the resulting matrix, which will be always $m \times p$.

Now we have the following situation:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

Matrix multiplication is called a "rows by columns" multiplication, so to calculate the first row - first column value we'll need the first row of one matrix and the first column of the other.

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

The values in the example will be combined as follows:

$$2 \cdot 2 + 3 \cdot 0 = 4$$

Obtaining the following:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

Notice how it was required to have the same number of elements in the first matrix's row (which is determined by the number of columns) and the second matrix's column (which is determined by the number of rows) to be able to combine them in a valid way.

Let's try the next value:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

The values will be combined as follows:

$$2 \cdot 3 + 3 \cdot 1 = 9$$

Obtaining:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 9 & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

Same goes for the last 2 values. When we are done with the first row, we keep going similarly with the second row:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 9 & 8 & 5 \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

Which leads to the following calculation:

$$1 \cdot 2 + 2 \cdot 0 = 2$$

Which we will insert in the result matrix:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 9 & 8 & 5 \\ 2 & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

You can try completing this calculation yourself, the final result is as follows:

$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 9 & 8 & 5 \\ 2 & 5 & 4 & 3 \\ 8 & 12 & 16 & 4 \end{bmatrix}$$

Note!

Multiplication between matrices is **non commutative**, which means that the result of $A \times B$ is not equal to the result of $B \times A$: actually one of the results may not even be possible to calculate.

3.11.6 Other uses for matrices

Matrices can be used to quickly represent equation systems, with equation that depend on each other. For instance:

$$\begin{bmatrix} 2 & 3 & 6 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Can be used to represent the following system of equations:

$$\begin{cases} 2x + 3y + 6z = 4 \\ 1x + 4y + 9z = 5 \end{cases}$$

Or, as we'll see, matrices can be used to represent transformations in the world of game development.

3.12 Trigonometry

When you want to develop a game, you will probably find yourself needing to rotate items relative to a certain point or relative to each other. To do so, you need to know a bit of trigonometry, so here we go!

3.12.1 Radians vs Degrees

In everyday life, angles are measured in degrees, from 0 to 360 degrees. In some situations in maths, it is more comfortable to measure angles using radians, from 0 to 2π.

You can convert back and forth between radians and degrees with the following formulas:

$$angle\ in\ degrees = angle\ in\ radians \cdot \frac{180}{\pi}$$

$$angle\ in\ radians = angle\ in\ degrees \cdot \frac{\pi}{180}$$

This book will always refer to angles in radians, so here are some useful conversions, ready for use:

Table 4: Conversion between degrees and Radians

Degrees	Radians
0°	0
30°	$\frac{\pi}{6}$
45°	$\frac{\pi}{4}$
60°	$\frac{\pi}{3}$
90°	$\frac{\pi}{2}$
180°	π
360°	2π

3.12.2 Sine, Cosine and Tangent

The most important trigonometric functions are sine and cosine. They are usually defined in reference to a "unit circle" (a circle with radius 1).

Given the unit circle, let a line through the origin with an angle θ with the positive side of the x-axis intersect such unit circle. The x coordinate of the intersection point is defined by the measure $cos(\theta)$, while the y coordinate is defined by the measure $sin(\theta)$.

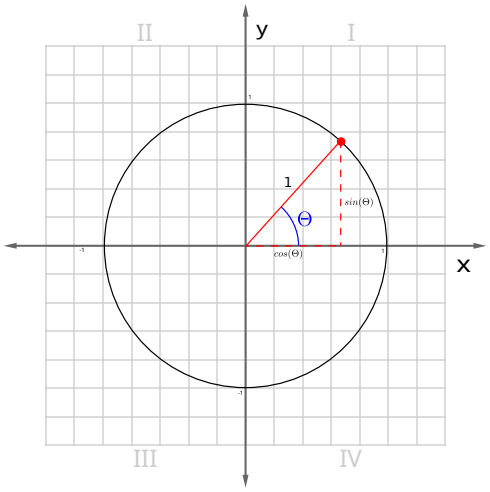


Figure 12: Unit Circle definition of sine and cosine

For the purposes of this book, we will just avoid the complete definition of the tangent function, and just leave it as a formula of sine and cosine:

$$tan(\theta) = \frac{sin(\theta)}{cos(\theta)}$$

3.12.3 Pythagorean Trigonometric Identity

One of the most important identities in Trigonometry is the "Pythagorean Trigonometric Identity", which is expressed as follows, valid for each angle θ:

$$sin^2(\theta) + cos^2(\theta) = 1$$

Using this identity, you can express functions in different ways:

$$cos^2(\theta) = 1 - sin^2(\theta)$$

$$sin^2(\theta) = 1 - cos^2(\theta)$$

Also remember that $sin^2(\theta) = (sin(\theta))^2$ and $cos^2(\theta) = (cos(\theta))^2$.

3.12.4 Reflections

Sometimes we may need to reflect an angle to express it in an easier way, and their trigonometric formulas will be affected, so the following formulas may come of use:

Table 5: Some reflection formulas for trigonometry

Reflection Formulas
$sin(-\theta) = -sin(\theta)$
$cos(-\theta) = cos(\theta)$
$sin(\frac{\pi}{2} - \theta) = cos(\theta)$
$cos(\frac{\pi}{2} - \theta) = sin(\theta)$
$sin(\pi - \theta) = sin(\theta)$
$cos(\pi - \theta) = -cos(\theta)$
$sin(2\pi - \theta) = -sin(\theta) = sin(-\theta)$
$cos(2\pi - \theta) = cos(\theta) = cos(-\theta)$

3.12.5 Shifts

Trigonometric functions are periodic, so you may have an easier time calculating them when their arguments are shifted by a certain amount. Here we can see some of the shift formulas:

Table 6: Some Shift Formulas for Trigonometry

Shift Formulas
$sin(\theta \pm \frac{\pi}{2}) = \pm cos(\theta)$
$cos(\theta \pm \frac{\pi}{2}) = \mp sin(\theta)$
$sin(\theta + \pi) = -sin(\theta)$
$cos(\theta + \pi) = -cos(\theta)$
$sin(\theta + k \cdot 2\pi) = sin(\theta)$
$cos(\theta + k \cdot 2\pi) = cos(\theta)$

3.12.6 Trigonometric Addition and subtraction

Sometimes you may need to express a trigonometric formula with a complex argument by splitting such argument into different trigonometric formulas. If such argument is a sum or subtraction of angles, you can use the following formulas:

Table 7: Some addition and difference identities in trigonometry

Addition/Difference Identities
$sin(\alpha \pm \beta) = sin(\alpha)cos(\beta) \pm cos(\alpha)sin(\beta)$
$cos(\alpha \pm \beta) = cos(\alpha)cos(\beta) \mp sin(\alpha)sin(\beta)$

3.12.7 Double-Angle Formulae

Other times (mostly on paper) you may have an argument that is a multiple of a known angle, in that case you can use double-angle formulae to calculate them.

Table 8: Some double-angle formulae used in trigonometry

Double-Angle Formulae
$sin(2\theta) = 2sin(\theta)cos(\theta)$
$cos(2\theta) = cos^2(\theta) - sin^2(\theta)$

3.12.8 Inverse Formulas

As with practically all maths formulas, there are inverse formulas for sine and cosine, called $arcsin$ and $arccos$, which allow to find an angle, given its sine and cosine.

In this book we won't specify more, besides what could be the most useful: the 2-argument arctangent.

This formula allows you to find the angle of a vector, relative to the coordinate system, given the x and y coordinates of its "tip", such angle θ is defined as:

$$\theta = arctan(\frac{y}{x})$$

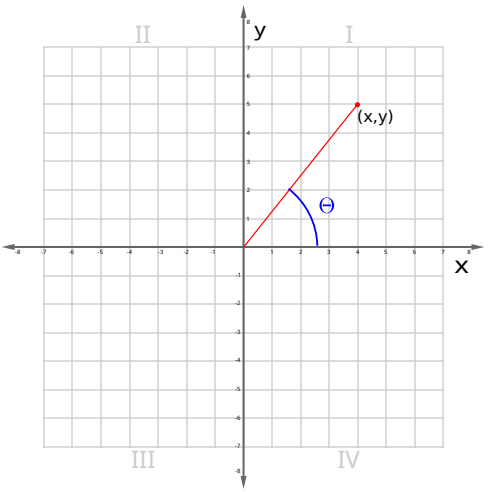


Figure 13: Graphical plotting of the angle of a vector

3.13 Numerical Analysis

Here we will give some pointers over some algorithms and methods that may be useful to better explain some topics treated in this book. Feel free to skip or quickly read this section if you don't want to dive into too much detail over this kind of maths.

3.13.1 Newton-Raphson method

Advanced Wizardry!

This section treats of how to approximate a function value in an iterative way. This will be useful to know what the "Fast Inverse Square Root" algorithm uses. Feel free to skim through this section.

Also known as Newton's method, this is an iterative algorithm that is used to get progressively better approximations to the roots of a function.

The algorithm starts with a "guess", called x_0, and produces the first approximation using the formula:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Each subsequent guess (and thus iteration) can be obtained similarly by using the formula:

$$x_{n+1} = x_n - \frac{f(n)}{f'(n)}$$

And such guess will be more precise than the previous one (if we don't consider some situations where approaching the root can be problematic or not possible). The algorithm will stop when you reach an approximation that is "good enough".

Obviously all limitations of standard functions apply, such as domain and trouble with divisions by zero.

3.14 Coordinate Systems on computers

When it comes to 2D graphics on computers, our world gets quite literally turned upside down.

In our maths courses we learned about the Coordinate Plane, with an origin and an `x` axis going from left to right and a `y` axis going from bottom to top, where said axis cross it's called the "Origin".

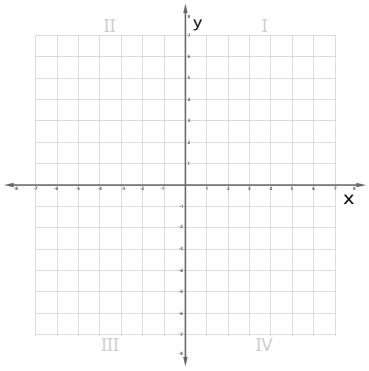


Figure 14: Image of a coordinate plane

When it comes to 2D graphics on computers and game development, the coordinate plane looks like this:

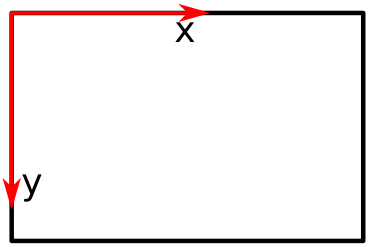


Figure 15: Image of a screen coordinate plane

The origin is placed on the top left of the screen (at coordinates `(0,0)`) and the `y` axis is going from top to bottom. It's a little weird at the beginning, but it's not hard to get used to it.

3.15 Transformation Matrices

There will be a time, in our game development journey where we need to rotate an object, and that's bound to be pretty easy because rotation is something that practically all engines and tool kits do natively. But also there will be times where we need to do transformations by hand.

An instance where it may happen is rotating an item relative to a certain point or another item: imagine a squadron of war planes flying in formation, where all the planes will move (and thus rotate) relative to the "team leader".

In this chapter we'll talk about the 3 most used transformations:

- Stretching/Squeezing/Scaling;
- Rotation;
- Shearing.

And to do so, we will use the following reference image, complete with a quadrant of the Cartesian plane.

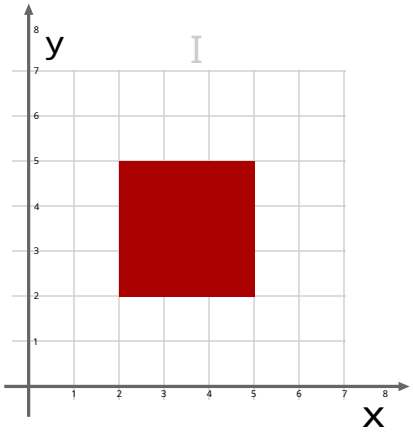


Figure 16: Reference image for transformation matrices

3.15.1 Stretching

Stretching is a transformation that enlarges all distances in a certain direction by a defined constant factor. In 2D graphics you can stretch (or squeeze) along the x-axis, the y-axis or both.

If you want to stretch something along the x-axis by a factor of k, you will have the following system of equations:

$$\begin{cases} x' = k \cdot x \\ y' = y \end{cases}$$

which is translated in the following matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Likewise, you can stretch something along the y-axis by a factor of k by using the following matrices:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Stretching our reference image along the x and y axes respectively would look something like this:

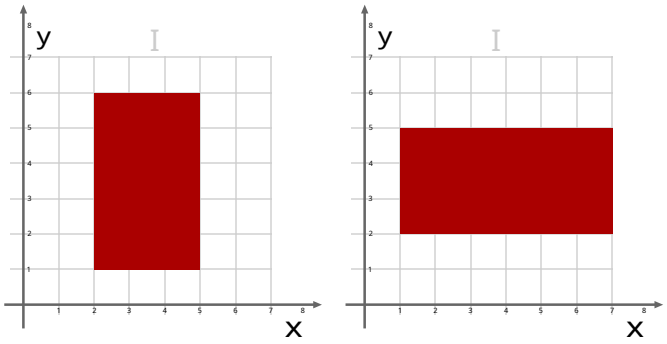


Figure 17: Stretching along the x and y axes

You can mix and match the factors and obtain different kinds of stretching, if the same factor k is used both on the x and y-axis, we are performing a *scaling* operation, like follows:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

In instead of stretching you want to squeeze something by a factor of k, you just need to use the following matrices for the x-axis:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{1}{k} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

and respectively, the y-axis:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{k} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

3.15.2 Rotation

If you want to rotate an object by a certain angle θ, you need to decide upon two things (besides the angle of rotation):

- Direction of rotation (clockwise or counterclockwise);
- The point of reference for the rotation.

3.15.2.1 Choosing the direction of the rotation

We will call T_R the transformation matrix for the "rotation" functionality.

Similarly to stretching, rotating something of a certain angle θ leads to the following matrix form:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = T_R \begin{bmatrix} x \\ y \end{bmatrix}$$

If we want to rotate something **clockwise**, relative to its reference point, we will have the following transformation matrix:

$$T_R = \begin{bmatrix} cos(\theta) & sin(\theta) \\ -sin(\theta) & cos(\theta) \end{bmatrix}$$

This could how our square could look, after a rotation:

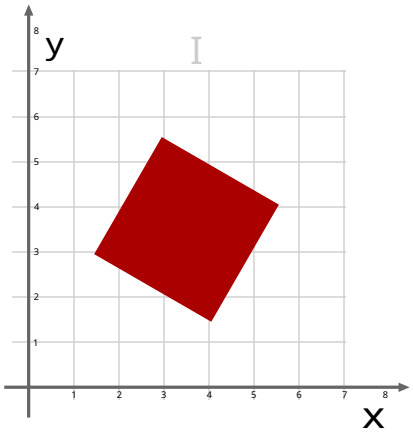
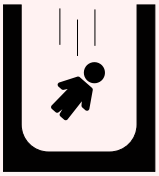


Figure 18: The result of applying a rotation matrix

If instead we want our rotation to be **counterclockwise**, we will instead use the following matrix:

$$T_R = \begin{bmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{bmatrix}$$

Pitfall Warning!



These formulas **assume that the x-axis points right and the y-axis points up**, if the y-axis points down in your implementation, you need to swap the matrices.

3.15.2.2 Rotating referred to an arbitrary point

The biggest problem in rotation is rotating an object relative to a certain point: you need to know the point of rotation (x_p, y_p) in relation to the origin of the coordinate system you're using, and modify the matrices as follows:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = T_R \begin{bmatrix} x - x_p \\ y - y_p \end{bmatrix} + \begin{bmatrix} x_p \\ y_p \end{bmatrix}$$

In short, you need to rotate the item by first "bringing it centered to the origin", rotate it, and then bring it back into its original position.

3.15.3 Shearing

During stretching, we used the elements that are in the "main diagonal" to stretch our objects. If we modify the elements in the "anti-diagonal", we will obtain shear mapping (or shearing).

Shearing will move points along a certain axis with a "strength" defined by the distance along the other axis: if we shear a rectangle, we will obtain a parallelogram.

A shear parallel to the x-axis will have the following matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

While a shear parallel to the y-axis will instead have the following matrix:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Here is how shearing would look, as an example:

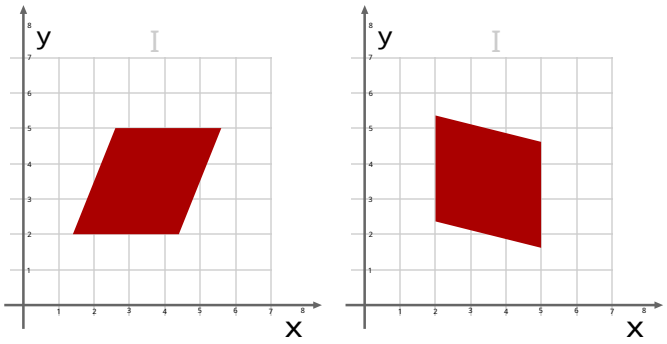


Figure 19: Shearing along the x and y axes

3.16 Basics of Probability

Games can make heavy use of probability: for instance when spawning items and treasures. Having a basic grasp of how probability works can make things a lot easier.

3.16.1 A simple definition of probability

We will define the probability of an event A with a fraction:

$$P(A) = \frac{The\ outcome\ is\ A}{All\ Outcomes}$$

The numerator is called "event space", while the denominator is called "sample space".

For instance: let's take a coin. We want to calculate the probability that a coin toss ends with "head": first we count how many outcomes are possible. Since a coin can land on "tails" or "heads", we have 2 possible outcomes, and head is only one of them.

For practicality, we will call "Heads" H and "Tails" T.

Thus:

$$P(H) = \frac{1}{2} = 0.5$$

This result can be converted to a percentage, by multiplying it by 100. That means that there's a 50% chance that a coin toss ends in "heads", shocking, I know.

What if we wanted to know the probability of a coin "not landing on heads"?

Here's a useful formula:

$$P(\overline{A}) = 1 - P(A)$$

Thus, by applying such formula on our coin example we have:

$$P(\overline{H}) = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2} = P(T)$$

Perfect. Everything as expected.

3.16.2 Probability of independent events

But what if we wanted to calculate the probability of more than one event?

If our events are independent (that means that the result of one doesn't affect the result of others), we can use the following formula:

$$P(A\ and\ B) = P(A \cap B) = P(A) \cdot P(B)$$

Let's return to our coin example.

If we wanted to know the probability of two coin tosses landing both on heads, we would have:

$$P(H\ and\ H) = P(H) \cdot P(H) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Let's demonstrate that intuitively: since the example is simple, we can literally count the possible outcomes:

Table 9: Counting the possible outcomes of two coin tosses

First Toss	Second Toss
Heads	Heads
Heads	Tails
Tails	Heads
Tails	Tails

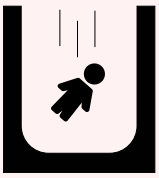
Now we know that there are 4 possible outcomes, and the "Heads + Heads" is only one of them. This confirms our formula.

In the exact same way, we can calculate the probability of a "Heads + Tails" result:

$$P(H\ and\ T) = P(H) \cdot P(T) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

And the previous table confirms our calculations.

Pitfall Warning!



Someone may argue that the probability of "Heads + Tails" is $\frac{1}{2}$, but that would not be correct. We are still strictly tied to the events, that means that "Heads + Tails" (First toss is heads, second toss is tails), is different from "Tails + Heads" (first toss is tails, second is heads).

3.16.3 Probability of mutually exclusive events

In case the events are mutually exclusive (that means, if one event happens, none of the others can happen), the following formula may be helpful in some occasions:

$$P(A\ or\ B) = P(A \cup B) = P(A) + P(B)$$

Going back to our coin example: the probability of a coin toss being "either heads or tails" is $\frac{1}{2} + \frac{1}{2} = 1$.

Another example could be done using a 6-sided dice.

Each face can be on top with a probability of $\frac{1}{6}$. Let's calculate the probability of either 1 or 6 being face up:

$$P(1\ or\ 6) = P(1) + P(6) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

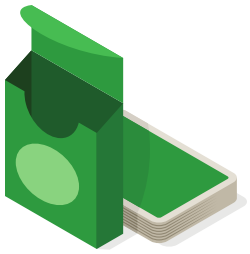
Note!

Considering the latest "tossing two coins" example, we can calculate the probability of "one coin lands on heads and the other lands on tails" with the previous formulas, since coin tosses tick both the "independence" and "mutual exclusivity" boxes.

$$P((H\ and\ T)\ or\ (T\ and\ H)) = P(H\ and\ T) + P(T\ and\ H) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

3.16.4 Probability of non-mutually exclusive events

Not all events are mutually exclusive. Let's think, for instance, about a deck of cards: what if you wanted to know the probability of drawing either a card of hearts or a face card (Jack, Queen or King)?



We need to use a different formula in that case, which is the following one:

$$P(A\ or\ B) = P(A \cup B) = P(A) + P(B) - P(A\ and\ B)$$

Note!

Why are we subtracting $P(A\ and\ B)$? Because if we didn't, we would be counting the face cards of hearts twice: once when we count the card of hearts, and once when we count all the face cards.

Let's continue with our example.

A standard deck has 52 cards, 13 for each suit. This means we would have 13 cards of hearts: $P(A) = \frac{13}{52}$.

The same deck of cards also has 3 "face cards" for each suit, totalling 12: $P(B) = \frac{12}{52}$.

Since there are face cards of the hearts suit, we need to account for those too, totalling 3: $P(A\ and\ B) = \frac{3}{52}$.

This means that the probability we're looking for is calculated as follows:

$$P(A\ orB) = P(A) + P(B) - P(A\ and\ B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26}$$

3.16.5 Conditional Probability

Advanced Wizardry!

Conditional probability doesn't have a lot of uses in game development, but it's worth mentioning it if you want to have a probabilistic approach to AI. Feel free to quickly skim through this section.

Sometimes you may need to consider the probability of a certain event, given that another event happens. This is called "conditional probability", and it can be calculated as follows:

$$P(A|B) = \frac{P(A\ and\ B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$

Conditional probability can be used to enrich the decision making used in enemy AI, for instance.

Let's take a concrete example, taken straight from the famous tabletop RPG Dungeons&Dragons, and see how probability can be applied to decision making.

> You're fighting against an enemy. Both you and the enemy are close to fatal damage: you have 1HP, while the enemy has 3HP left.
> To attack an enemy you need to roll a 20-sided dice (called a d20): if the number rolled is 13 or higher you will hit, else you will miss.
> If you hit, you will roll a 6-sided dice (called a d6): the number rolled will decide how much damage you will deal, so you need 3 or more.
> We need to find the probability of killing the enemy within the next turn to decide our next move.

First of all, let's name the events:

- **H** Will be the event "hit", which means that the d20 rolled a number that is 13 or higher.
- **F** Will be the event "fatal damage", which means that the d6 rolled a number that is 3 or higher.

Now we will calculate the probabilities we need for our calculation:

$$P(H) = \frac{8}{20} = \frac{2}{5}$$
$$P(F) = \frac{4}{6} = \frac{2}{3}$$

Our objective is calculating "the probability of doing at least 3HP of damage, given that we hit the enemy". This is represented as:

$$P(F|H) = \frac{P(F \cap H)}{P(H)}$$

This means we will have to calculate another probability, which is quite easy:

$$P(F \cap H) = P(F) \cdot P(H) = \frac{2}{3} \cdot \frac{2}{5} = \frac{4}{15}$$

Now we are ready to calculate everything we need:

$$P(F|H) = \frac{P(F \cap H)}{P(H)} = \frac{\frac{4}{15}}{\frac{2}{5}} = \frac{4}{15} \cdot \frac{5}{2} = \frac{2}{3}$$

Given a 66% chance of success, you may decide that attacking is worth the risk. Such decision may be hard-coded into an AI, for instance if the probability is higher than 50% the AI may choose to attack instead of retreating and call for backup.

3.16.6 Uniform Distributions

In most cases, we will speak in terms of “uniform distributions”, that means that we will be operating on a system where all outcomes have the same probability of happening.

That means that all dices are “fair”, all coins are “fair” and all our “bingo bags” have only one instance of a certain number, all of the same size, shape and feel (thus making it impossible for a number to appear more often than any other).

In the grand scheme of things, we are assuming that the `random()` function of our programming language is a uniform distribution, where any number may come out with the same probability of any other.

3.16.7 How probability is used in games

You can use probability to govern how items spawn: surely you want more precious items to spawn more rarely (with less probability), while more common items should spawn more often.

Let’s say we want an item to spawn with 20% probability: how can we do it?

20% probability can be rewritten as the decimal 0.2, such decimal can be obtained with the fraction $\frac{1}{5}$. We have practically solved the problem: we decide on one number between 1 and 5 (inclusive) and we will know that such number will be “extracted” 20% of the time.

Listing 2: 1 (out of 5) will be extracted with about 20% probability

```
function main(){
    let happened = 0;
    // Monte Carlo Method we do 10000 "extractions"
    for (let i = 0; i < 1000; i++){
        // Get a random number between 1 and 5
        let n = Math.random() * 5 + 1;
        if (n == 1){
            // If it's 1, we have a match!
            happened++;
        }
    }
    // We print the result
    console.log(happened / 10000)
}
```

We will obtain the following result.

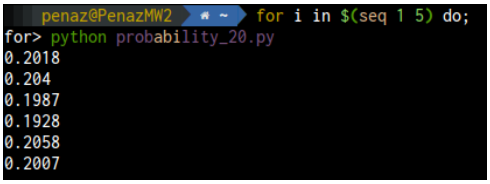


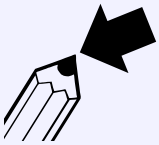
Figure 20: Running the probability_20 example shows the probability floating around 20%

But what if we wanted to be a lot more precise? Let's say we want to spawn an item with 13% probability, how would we go at it?

It's actually pretty simple: out 13% probability can be represented by the fraction $\frac{13}{100}$. Each number between 1 and 100 (inclusive) has a $\frac{1}{100}$ chance of being extracted. Since extracting one number bars any other number to appear in that extraction we can use the "mutually exclusive events" formula.

$$P(1\ or\ 2\ or\ ...\ or\ 13) = P(1) + P(2) + ... + P(13) = \frac{1}{100} + \frac{1}{100} + ... + \frac{1}{100} = \frac{13}{100}$$

Tip!



If the example is not 100% clear yet, try reading the previous formula right-to-left. That may help.

This means that the event "a number between 1 and 13 appears" has a 13% probability of appearing. We can simplify that statement with "a number less or equal than 13". We can experiment that easily with the following code:

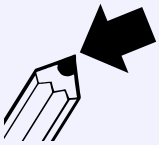
Listing 3: A number less or equal than 13 (out of 100) has 13% probability of appearing

```
function main(){
    let happened = 0;
    // Monte Carlo Method we do 10000 "extractions"
    for (let i = 0; i < 1000; i++){
        // Get a random number between 1 and 100
        let n = Math.random() * 100 + 1;
        if (n <= 13){
            // If it's less or equal than 13, we got a match
            happened++;
        }
    }
    // We print the result
    console.log(happened / 10000)
}
```

```
penaz@PenazMW2 # ~ for i in $(seq 1 5) do;
python probability_le_13.py
0.1273
0.1283
0.1323
0.1268
0.1314
0.1305
```

Figure 21: Running the probability_le_13 example shows the probability floating around 13%

Tip!



You can extend the example above to fractions of a percentage by using bigger numbers: if you wanted a 13.5% probability, you would use all numbers less than or equal to 135, out of 1000.

3.16.8 Tiered Prize Pools

We can use what we learned with probability to create a tiered prize pool. For instance we decide that killing a certain enemy will always drop something, the tier of such item is according to the follow probability list:

- 50% probability for a common item to drop (for instance a scrap of leather);
- 30% probability for an uncommon item to drop (like a lower-grade potion);
- 15% probability for a rare item to drop (a good sword, for instance);
- 5% probability for an epic item to drop (a unique armor, for example);

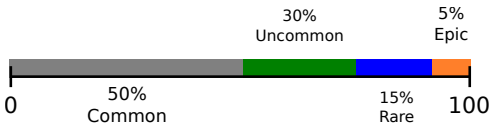


Figure 22: Intuitive representation of our prize pool

In that case we can chain ifs to bring our tiered prize pool to life:

Listing 4: How to implement a tiered prize pool selector

```
function get_tiered_drop(){
    // 1 = Common, 2 = Uncommon, 3 = Rare, 4 = Epic
    let n = Math.random() * 100 + 1;
    if (n <= 50){
        // Common Tier
        return 1;
    }
    if (n <= 80){
        // Uncommon Tier
        // Since n <=50 has already returned false, we know this
        // branch will only happen if 50<n<=80
        return 2;
    }
    if (n <= 95){
        // Rare Tier
        // Since both n<=50 and n<=80 both returned false, we know
        // this branch will only happen if 80<n<=95
        return 3;
    }
    // Epic Tier
    // All other branches failed, so we'll get here only if
    // 95<n<=100
```

```
    return 4;
}
```

3.16.8.1 Introducing a “luck” stat

In many RPGs there is a “luck” statistic that affects how item drops happen, in that case we will need to change how tiered prize pools are given out. Things can get complicated quite quickly.

Let’s imagine a simple situation: one point of “luck” gives a 1% probability of getting an item of each tier higher than “Common”, while at the same time reducing the probability of finding a “common” item.

At a first glance, it seems simple: take each “non-common” class and “add 1”, then take the “common” class and “remove 1 for each point given”. But what would happen if the luck stat is higher than the probability of a “common” item? It should probably start taking away probability from “uncommon” items to give out “rare” and “epic” items.

Let’s see a possible implementation:

Listing 5: A possible implementation of a luck stat

```
// Our probabilities, from least to most common
let pool = [
    ["EPIC", 5],
    ["RARE", 15],
    ["UNCOMMON", 30],
    ["COMMON", 50],
];

// Our "luck stat": each point gives 1% more chance to get a higher-tier item
let luck = 25;

// We cap the Luck stat at 100, the limit is 100% epic items
luck = Math.min(luck, 100);

// We "overload" the prize pool, making the sum go over 100%
let overloadedPool = [];
for (let i = 0; i < pool.length; i++){
    let newProb = pool[i][1] + luck;
    overloadedPool.push([pool[i][0], newProb]);
}
// We calculate how much we "overloaded" the prize pool
let overloadFactor = luck * pool.length;

// We rebalance the prizes to a total of 100, from most to least common
let rebalancedPool = [];
// We need to start from the most common, which means we will iterate backwards
for (let i = overloadedPool.length - 1; i >= 0; i--){
    const item = overloadedPool[i][0];
    const probability = overloadedPool[i][1];
    // This will be modified later, if the pool is "overloaded"
```

```
31      let newProbability = probability;
32      // If the prize pool is still "overloaded"
33      if (overloadFactor > 0){
34          // We calculate a "discharge factor" of sorts
35          let valueToRemove = Math.min(probability, overloadFactor);
36          // We reduce our "overload"
37          overloadFactor -= valueToRemove;
38          // And put the new probability for the class
39          newProbability = probability - valueToRemove;
40      }
41      // We append the new pool item
42      rebalancedPool.push([item, newProbability]);
43  }
```

4 Some Computer Science Fundamentals

> The computing scientist's main challenge is not to get confused by the complexities of his own making.
>
> *Edsger W. Dijkstra*

In order to understand some of the language that is coming up, it is necessary to learn a bit of the computer science language and fundamentals.

This chapter will briefly explain some of the language and terms used, their meaning and how they contribute to your activity of developing games.

4.1 Number representations

When you work with computers, it's impossible to avoid learning a bit of number representations. Computers work with a different logic than humans do: humans have complex minds and thoughts, while most of the time computers work in ones and zeroes. Most of what we see on a screen can be reduced to electrons going through a semiconductor in kind of an orderly fashion: changing from 0 volts (ground) to 5 volts.

4.1.1 The most used representations

Here we will take a quick look at the most used representations. Some are more fundamental than others, but they are all useful in their own way.

Each representation will use a subscript to represent its representation. If no subscript is present it means the standard decimal representation is used.

4.1.1.1 Decimal

This is the standard decimal notation everyone is used to, we have 10 digits at our disposal:

$$0\,1\,2\,3\,4\,5\,6\,7\,8\,9$$

And we place them in certain positions (units, tens, hundreds, thousands, etc...) to represent a certain quantity. We will use this as a basis for all other representations.

So if you want to represent $9 + 1$ you will use the 1 digit, followed by the 0 digit to make 10.

4.1.1.2 Binary

This is the most used representation in computer science, we have only two digits at our disposal: $0\,1$.

Thus if you want to make $1_{bin} + 1_{bin}$, you will have to use the 1 digit, followed by the 0 digit, thus making 10_{bin}, which is the binary representation of 2.

Here are the first 10 numbers for comparison purposes:

Table 10: Comparison between decimal and binary representations

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010

Binary numbers can be used at low level to represent any kind of "binary condition" too: yes/no, true/false are usually mapped to 1 and 0 respectively. This will probe useful in some cases where we will use "binary numbers" to represent groups of "binary conditions" in a compact way, but that's an advanced thing we'll see later.

4.1.1.3 Octal

In the octal representation we have 8 digits at our disposal:

$$0\,1\,2\,3\,4\,5\,6\,7$$

Thus the representation of the decimal number 8 in the octal system is 10_{oct}.

The octal number system doesn't find much use in computer science besides being a quicker way to represent binary numbers. The conversion is quite easy and will be explained in a bit.

Here's a quick comparison between decimal and octal representations:

Table 11: Comparison between decimal and octal representations

Decimal	Octal
0	0
1	1
2	2
3	3

Decimal	Octal
4	4
5	5
6	6
7	7
8	10
9	11
10	12

4.1.1.4 Hexadecimal

Hexadecimal is definitely the second most used representation in computer science, due to how easy it is to represent 4 bytes in a very compact notation.

In the hexadecimal notation, we have sixteen digits at our disposal:

$$0\,1\,2\,3\,4\,5\,6\,7\,8\,9\,A\,B\,C\,D\,E\,F$$

Here's a table of the first 20 numbers to clarify a bit how things work:

Table 12: Comparison between decimal and hexadecimal representations

Decimal	Hex
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D

Decimal	Hex
14	E
15	F
16	10
17	11
18	12
19	13
20	14

4.1.2 Converting between decimal and binary

The algorithm to convert between decimal and binary is quite simple. It is an iterative algorithm that consists in integer dividing the number by 2, until the result of the division is 1. The modulo of such divisions will make up our binary number.

An example is worth a thousand words: let's convert the number 38 to binary.

First of all, we integer divide 38 by two: the result is 19, there is no remainder, so we'll use zero.

Dividend	Remainder
38	0
19	

Let's continue: we integer divide 19 by two: the result is 9, with 1 as remainder.

Dividend	Remainder
38	0
19	1
9	

We iterate some more, by integer dividing until we get 1 as a dividend, at that point we make the last division, which will have remainder 1:

Dividend	Remainder
38	0
19	1
9	1

Dividend	Remainder
4	0
2	0
1	1

Now we just need to read our remainders from bottom to top. So the binary representation of 38_{dec} is 100110_{bin}.

Note!

This is actually a much more generic algorithm: you can convert from decimal to octal and hexadecimal for instance, just by dividing by 8 and 16 respectively. You can convert 38 to octal and hexadecimal as an exercise: the results are 46_{oct} and 26_{hex}.

4.1.2.1 Two's complement

So far we've seen how to convert positive integers from decimal to binary, but how do we represent negative integers?

That's where "two's complement" representation comes into play: there is a bunch of theory behind why it's called this way, and how it works, but what we need to know will be how to represent a negative number.

Let's start with a simple example with 3 binary digits (this means we're pretending our computer can process only up to 3 bits):

Decimal	Binary
-4	100
-3	101
-2	110
-1	111
0	000
1	001
2	010
3	011

As you can see, the most significant bit being set (that means having value of 1) is a telltale sign that a number is negative. But there are some interesting features about two's complement that make it a very nice method of representing integers.

This is because it makes easier to implement hardware that does operations on such numbers. If we sum 3 and -3 in two's complement we will obtain the following:

$$\begin{array}{rr} & 011 \\ + & 101 \\ \hline & 1000 \end{array}$$

This may look completely wrong, but since our "computer" can only process up to 3 bits, the left-most bit will be discarded, giving us the right result: 000_{bin}.

Now let's see how to represent negative numbers in two's complement.

> To represent a negative binary number in two's complement you flip all the bits of such number, then add 1.

As usual, an example is worth a thousand words. We want to convert the number -38 into binary.

First of all we need to define what our range of numbers will be, so that we know how many bits we will use. This is done because this range will be equally split between positive and negative numbers. In this example I will choose a normal 8-bit representation, which can represent numbers spacing from -128 to 127.

The first step is to convert 38 to binary, which as we saw is 100110_{bin}. We will pad this binary number to 8 bits, obtaining 00100110_{bin} as a result.

Now we just need to invert the all the bits in the number, obtaining 11011001_{bin} as a result.

Last step is adding 1 to what we got in the previous step, thus the final result is 11011010_{bin}.

Note!

The more perceptive of you may have noticed a problem: what if we tried to represent the number 128 with 8 bits?
We would obtain 1000000_{bin} which is actually the representation of -128 in two's complement. This is called an "integer overflow", so be careful when mixing unsigned and signed integers.

4.1.2.2 Floating point

[This section is a work in progress and it will be completed as soon as possible]

4.1.3 Converting between binary and octal

As mentioned before, octal can be used as a "shorthand way" to represent binary. The conversion is pretty simple.

> To convert from binary to octal, take the binary digits in groups of 3 (with the necessary padding) and convert them in octal. Then just "stick them together".

Let's take our number 38, it has the following representation:

$$100\ 110_{bin}$$

100_{bin} converts to 4_{oct}, while 110_{bin} converts to 6_{oct}. If we stick them together we obtain the final result: 46_{oct}.

4.1.4 Gray Code

Advanced Wizardry!

Gray code isn't really used in game development, but it will be briefly explained here since it will be used in Karnaugh Maps

Gray code (sometimes known as "reflected binary code") is a particular ordering of the binary system where two successive values differ by only one bit.

Gray code is used in many fields, from Digital (and cable) TV (for error-correction) to analog to digital conversion. In this book we will use it as a representation inside Karnaugh Maps.

Here is a simple representation of the first 10 numbers in decimal, binary and gray code:

Decimal	Binary	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111

4.2 Basics of Logic

If we want our algorithms to be smart enough to be useful, we have to deal with conditionals. That's where logic comes in. In this section we will take a quick look at truth tables as well as logic operations.

4.2.1 Truth tables

Truth tables are used to represent the output of a logic operation. It represents the inputs on the left side, while on the right side the result is shown.

Truth tables in this book will have the following look:

A	B	f
0	0	**0**
0	1	**0**
1	0	**1**
1	1	**0**

4.2.2 Common operators

After we distinguish "true" (1) from "false" (0), we will need to start mixing and matching them (similarly to what we do with numbers). That's where operators come into play: they are a bit different than what we're used to in arithmetics, but they are quite intuitive.

4.2.2.1 AND

The "AND" operator is a binary operator that outputs 1 when both inputs are 1. Here is its truth table:

A	B	**AND**
0	0	**0**
0	1	**0**
1	0	**0**
1	1	**1**

This operator is used to express conditionals where you want two conditions to be true at the same time.

4.2.2.2 OR

The "OR" operator (sometimes called "inclusive or", as opposed to the XOR operator) is a binary operator that outputs 1 when either of the inputs is 1, including the case when both are 1. Here is its truth table:

A	B	**OR**
0	0	**0**
0	1	**1**
1	0	**1**

A	B	**OR**
1	1	**1**

This operator is used to express conditionals where you want at least one condition to be true.

4.2.2.3 NOT

The “NOT” operator is a unary operator that takes a single input and “inverts” it. That means that if the input is 1, the “NOT” operator will output 0, if the input is 0 the “NOT” operator will output 1 instead.

Here is its truth table:

A	**NOT**
0	**1**
1	**0**

4.2.2.4 XOR

The “XOR” operator (called “exclusive or”) is an operator that takes two input and outputs 1 when only one of the two inputs is 1. If both inputs have the value 1, the “XOR” operator will output 0.

Here is its truth table:

A	B	**XOR**
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**0**

This operator is used when you want to express conditionals where only one of the two inputs is true.

4.2.3 Logic operations vs bitwise operations

Advanced Wizardry!

The confines between logic operations and bitwise operations can get blurry. This section introduces bitwise operations and alternative representations of data as a way to fit more data in less space. Feel free to skim over this section.

So far we’ve seen operations that work on single binary digits, which can be seen as the numeric representation of logical statements (0 meaning “false” and 1 meaning “true”). These are logic operations.

Such operations can be applied on a bit-by-bit basis to groups of bits, that's when we talk about about "bitwise operations".

$$\begin{array}{ll} 0110\,0010 & AND \\ 0101\,1010 & \\ \hline 0100\,0010 & \end{array}$$

As you can see the bitwise AND operation takes each bit of the two bytes and does an "AND" operation on each one of them.

4.2.3.1 Packing more information with less

Let's imagine the following situation: we have a structure that represents a tile in a maze. We want to efficiently store whether each side of a certain tile has a wall.

This can be solved by using a 4-bit positive integer and having each bit represent a side of the tile: if that bit is 1, there is a wall, 0 otherwise.

After creating a convention, we can start storing data. For instance we can have the bits representing walls starting from top, going clockwise.

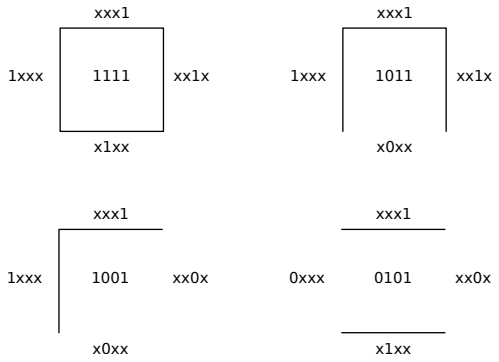


Figure 23: How we can pack wall information with a 4-bit integer

This convention could be summarized as follows:

- 0001 represents a "top wall";
- 0010 represents a "right wall";
- 0100 represents a "bottom wall";
- 1000 represents a "left wall".

This means that 0110 represents a tile having two walls: on the right and bottom side (this is the integer number 6, by the way). If we wanted to check if a certain tile has a wall, we would just need to AND it (bitwise) with the number that represents such wall.

If the result of such operation is not zero, the wall we searched for is in our tile. Continuing with our example, if we test for a right wall we will obtain zero:

$$\begin{array}{ll} 0110 & AND \\ 0001 & \\ \hline 0000 & \end{array}$$

But if we test for a bottom wall, we will obtain something that is not zero:

$$\begin{array}{ll} 0110 & AND \\ 0100 & \\ \hline 0100 & \end{array}$$

4.2.4 De Morgan's Laws and Conditional Expressions

De Morgan's laws are fundamental in computer science as well as in any subject that involves propositional logic. We will take a quick look at the strictly coding-related meaning.

De Morgan's laws can be written as:

> not (A and B) = not A or not B
> not (A or B) = not A and not B

In symbols:

$$\overline{(A \wedge B)} = \bar{A} \vee \bar{B}$$

$$\overline{(A \vee B)} = \bar{A} \wedge \bar{B}$$

These laws allow us to express our own conditionals in different ways, allowing for more readability and maybe avoid some boolean manipulation that can hinder the performance of our game.

4.3 Algorithms

When you talk about computer science, you always hear about algorithms: what is an algorithm?

An algorithm can be informally defined as a finite sequence (as in "not infinite") of instructions that are followed to solve certain problems.

There are numerous examples of algorithms, among them we can find:

- Finding the Greatest Common Divisor (GCD) of two numbers;
- Finding the largest number in a list;

- Calculating the nth Fibonacci number;
- ...

Algorithms are usually represented in flow charts, or it's more modern counterpart: the UML activity diagram. Sometimes algorithms can be represented in "plain language" (in that case we may end up talking about "pseudocode") or in a programming language.

4.4 Recursion

Starting from the (arguably hard) theme of recursion may seem weird, but it is important to understand recursion as soon as possible so we can make the best use of it.

There is a joke I like telling around about recursion:

> To understand recursion, you must first understand recursion.

What is recursion? Recursion is the usage of a function that calls itself.

Your first question will probably be: wouldn't that make the program lock up forever in some kind of loop? It may. But if you're careful, recursion is an amazing tool that allows you to earn a lot of clarity and brevity.

Let's imagine a simple algorithm: we want to make our program count backwards from a number `n` to 0. In a simple "loop" fashion, we may write the following:

Listing 6: Counting from n to 0 using a loop

```
function count_backwards(n){
    // Condition for the loop
    while (n != 0){
        // The function body
        console.log(n);
        // We update the condition to count down
        n = n - 1;
    }
}
```

Pretty simple, right? A real-world example would be counting back from 10 to 0: we print 10, we subtract 1 to get 9, we print 9, subtract 1 to get 8, ...

Let's turn our thinking around for a second. We can see counting back from to 10 to 0 like this: we print 10 and then we count backwards from 9. Counting backwards from 9 would just mean printing 9 and then counting back from 8, etc...

We just turned our simple loop into a recursive function:

Listing 7: Counting from n to 0 using recursion

```
function count_backwards(n){
```

```
    // Stop condition
    if (n == 0){
        // If we don't do this, we won't print 0
        console.log(n);
        return;
    }
    // Procedure
    console.log(n);
    // Recursive call
    count_backwards(n-1);
}
```

Recursive functions have three main components:

- A **base case** (sometimes called a "stop condition"): this allows the function to stop calling itself when a certain condition is reached;
- A **procedure** that elaborates on data or simply does something (in our example, it just prints the number);
- A **recursive call** to the same function we are writing, the call is done in a way that every call gets closer to the "stop condition". It can be done by calling the function on a subset of its argument (if it is a list), until the list has only 1 item or on a smaller number (if the function argument is a number instead).

Recursion can be classified in many ways:

- **By the number of recursive calls:** single vs multiple recursion;
- **By how the recursive call is made:** direct (a function calls itself directly) vs indirect (a function A is called by another function B, which in turn is called by function A)
- **By the position of the recursive call:** head vs tail recursion.

I want to underline the last distinction: what we've seen in the previous listing is called "tail recursion": the recursive call is done **after** everything else (the procedure).

Head recursion is instead done when the recursive call is done **before** the procedure starts, so we can transform our "count down" function to a "count up" just by switching from "tail" to "head" recursion and adding a print statement.

Listing 8: Counting from 0 to n using head recursion

```
function count_forwards(n){
    // Stop condition
    if (n == 0){
        // If we don't do this, we won't print 0
        console.log(n);
        return;
    }
    // Recursive call
    count_forwards(n-1);
    // Procedure
    console.log(n);
}
```

4.5 Programming Languages

Programming languages are a programmer's way to talk to a computer (or a console): they are a way to make an electronic apparatus do something (without involving analogue electronics).

4.5.1 Classifying programming languages

Programming languages can be distinguished by many traits, it is important to know such differences, even though you may have already chosen your programming language.

4.5.1.1 By how they build

The way that a programming language gets you from code to "working product" can heavily influence both the final product as well as the speed of development.

4.5.1.1.1 Compiled Languages

Compiled languages need to go through a building process before it is possible for the product to be run anywhere. This has some advantages, as well as some disadvantages.

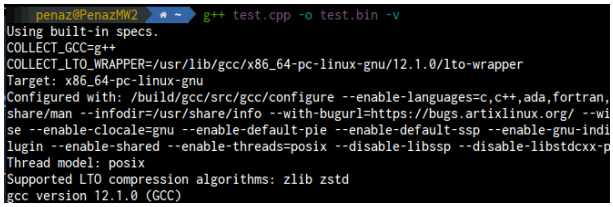


Figure 24: Example of a compiler output (G++)

Among the disadvantages we have that the final product is usually non-portable, that means it cannot be run anywhere besides the machine it was compiled for. This means that you will have to create separate builds for each console, as well as different builds for each operating system.

Another disadvantage can be development speed: before you can test anything your game needs to be rebuilt. Sometimes the rebuild process can be quick (thanks to some techniques that avoid building things that didn't change), sometimes it can be long.

A very strong advantage of compiled languages is speed. Being essentially compiled to machine code, compiled languages have an easier time squeezing every last drop of performance from the platform you're building for. In addition, some languages can use features to physically remove unused code from the build: this way release builds can be much faster than debug ones, because the debug code is physically removed.

Among compiled languages we can find C and C++, as well as Rust and Go.

4.5.1.1.2 Interpreted Languages

Interpreted languages, in their strictest sense, are at the other side of the spectrum: the program is not compiled ahead of time but instead the source code is fed into an interpreter, which executes each row of instructions, one after the other.

Most interpreted languages feature an interactive *REPL*[g] (read-eval-print loop) which allows to test code in real time.

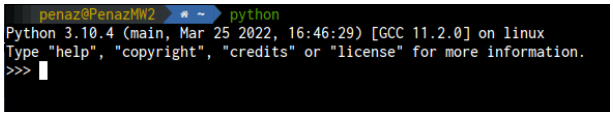


Figure 25: Python's REPL Shell

They have the disadvantage of being usually slower than compiled languages and it's not easy to create builds that physically remove unused (debug) code without having to modify the sources manually. Also each console or operating system will need to have the interpreter installed, which may be an issue.

The advantage is in development speed: you can edit the source code and immediately run the interpreter to see the result, without having to wait for a new build to complete. Another advantage is portability: you don't need to create a new build for every system you want to run your game in, as long as an interpreter is available your game will run.

An example of a purely interpreted language is BASIC.

4.5.1.1.3 Hybrid Approaches

In any project, the ability to code quickly is as important as the performance of the final product: there is a thin balance to strike between "having a product with good performance" and "having a product that is released when needed". If your product releases too late, it doesn't matter how performing it is, the market will have chosen another product. If your product releases early but it underperforms, it will be replaced by better products.

Thus some hybrid approaches have been invented: one of these is, for instance, bytecode-compiled languages.

Bytecode-compiled languages (sometimes called "Languages with intermediate representation") are something that is not quite compiled, but it's not precisely interpreted either: the code is converted into bytecode, which is then fed to the interpreter (or "virtual machine") to run.

Being a representation that is "closer to the hardware" than the original source code, there is a gain in performance, while keeping the flexibility of interpreted code.

Random Trivia!

Some programming languages, like Haskell and Vala use the C programming language as an intermediate language, since C was meant to be an abstraction of the assembly language.

Other approaches include Just-In-Time compiling, which trades off some longer starting times (sometimes called "warm-up times") for better overall performance.

Among the bytecode-compiled languages we can find Java and Python, while Lua can be considered a Just-In-Time Compiled language (thanks to LuaJIT).

4.5.1.2 By Paradigm

A programming paradigm is how the programming language lets you program. There is not a single, definitive way to code, thus programming languages can be distinguished by their paradigm.

4.5.1.2.1 Imperative Languages

Imperative languages are probably the most spread in modern programming: they make use of "orders" (called "statements") to change the status of the program.

This paradigm makes use of variables, statements, subroutines to make the program look like a set of instructions, a recipe, to make the program do what it needs to do (an algorithm).

Imperative languages include C, COBOL, Basic and Fortran.

4.5.1.2.2 Functional Languages

Functional languages make programs work by applying and composing functions (in the mathematical sense). Functions can be bound to variables and chained together (composed) to reach the result.

Functional languages include Haskell, Common Lisp and Scheme.

4.5.1.2.3 Multi-paradigm Languages

Many programming languages tend to "meld together" many programming paradigms, allowing (for instance) for functional style programming in imperative languages.

This means that functions can be bound to variables and passed around as any other object, they can be composed to reach the result if the programmer decides to do so (for instance for readability).

Multi-paradigm languages include Python, Lua and Go.

4.5.1.3 By the way types are determined

Sometimes underrated, how types are evaluated can completely change the way you program your game. Not knowing precisely how your language of choice treats types can lead to hard-to-debug issues.

4.5.1.3.1 Static Typing

Statically typed languages have their types decided ahead of time (usually when the program is compiled) and usually they cannot be changed.

This means that you have to have full awareness of which types will be used while writing your game. Which can be difficult at times.

Statically-typed languages include C, C++ and C#, as well as Java.

4.5.1.3.2 Dynamic Typing

Dynamically typed languages have their types decided at runtime. This allows for simpler syntax, but at the cost of lower performance, due to the fact that types are determined and verified at runtime.

Dynamically-typed languages include JavaScript and Ruby.

4.5.1.3.3 Duck Typing

Duck typing is probably the most misunderstood typing system. It can be described by the following sentence:

> If it walks like a duck and it quacks like a duck, then it must be a duck.

This means that types are inferred by their behaviour (their capabilities), thus creating a series of `-like` objects that behave more or less the same. This means that types can make use of the iteration capabilities of the language as long as they implement some basic methods that allow iteration (like `nextElement()` and `length()`).

This means that we have “file-like” objects, which behave like files, are used like files, but not necessarily have a counterpart in mass storage (they could be in-memory files), or “iterables” (sometimes called “list-like”) which behave like lists of items, but may actually be something else (for instance strings could be seen as a “list of letters”).

In the end, in duck typing, interfaces are treated as some kind of “informal protocol” that tells the language how to use an object. The “protocol” doesn’t even need to be implemented fully: if you have a “file-like” object that implements only the reading method, you can still use it in the same way you’d use a file, as long as you don’t try to write to it.

Duck Typing is used in the Python programming language.

4.5.1.4 By the “strength” of typing

How types are treated after each variable is instantiated can be the source of a lot of headaches while coding, thus it is paramount to be aware of how strong your preferred language’s typing system is.

4.5.1.4.1 Strong Typing

Strongly typed languages don't allow one type to be treated like it was another type without an explicit conversion (usually called "cast"). This prevents unforeseen automatic type conversions that may lead to bugs and faults being undetected at compile time or runtime.

Some examples of strongly typed languages are C++, C#, Python and Java.

4.5.1.4.2 Weak Typing

Weak typed languages allow one type to be treated like another without explicit conversion. This may make the syntax simpler, but may be source of unforeseen bugs.

For instance a string may be treated as it was a number, this means that in some languages (where the operator `+` means both "addition between numbers" and "joining strings together") you may find that a result is a sum of numbers instead of two strings joined together.

An example of a weakly typed language is JavaScript.

Random Trivia!

What about the good old C language? C has strong typing for the great majority of the time, unless we consider the `void*` generic pointer. This kind of pointer can be used in other pointer variables without an explicit cast.

4.5.1.5 By memory management

Another way to classify programming languages is how you can (or have to) manage your memory.

4.5.1.5.1 Languages without Garbage Collection

Some programming languages allow you to play with your system's memory as you wish: they give you all the tools (pointers, references, ...) to manually allocate and free memory.

This comes with its advantages and drawbacks: higher performance is surely a big advantage. A huge disadvantage is the fact that memory management is completely manual: dangling pointers and unreachable memory are commonplace, because there is nothing to clear after you.

Non Garbage-collected languages include C and C++.

4.5.1.5.2 Garbage-collected Languages

Some other languages prefer taking away part of the control on memory to help avoiding the problems that non Garbage-collected languages bring: there is something that cleans after you, which is the Garbage Collector.

The big disadvantage of this approach is that the garbage collector needs reference counting, CPU cycles to run, which means that the whole program runs slower.

Garbage collected languages include Java and Python.

4.5.2 Languages available for this book

Here is a quick rundown of how the languages used in the various editions of this book (excluding "pseudocode", which is not really a programming language) are classified.

- **C++**, a compiled programming language with strong static typing. It is multi-paradigm (although it was born as an imperative language) and has no garbage collector.
- **JavaScript**, an interpreted language (although some engines support Just-In-Time compiling), with weak dynamic typing that supports some duck typing principles. It is multi-paradigm and features a garbage collector.
- **Lua**, a bytecode-compiled (or Just-In-Time compiled) language, with strong dynamic typing that supports some duck typing principles. It is multi-paradigm and garbage-collected.
- **Python**, a bytecode-compiled language, with strong duck typing. It is multi-paradigm and garbage-collected.

4.6 Computers are (not) precise

There are many differences between humans and computers, among those there is one that will keep haunting us in our journey: humans make calculations in "base 10" (decimal), computers make calculation in "base 2" (binary).

This requires computers to represent numbers differently, usually with the exponent+fraction representation (IEEE 754). Also computers have limited resources, thus have no concept of "infinity" (and conversely of "infinitesimal").

Let's assume a computer with a fixed (and reduced) precision and we execute the following C++ program (you can just copy it verbatim):

Listing 9: A simple float precision test

```cpp
#include <iostream>
#include <iomanip>

int main ()
{
    // This will reduce and fix the computer's precision for this execution
    std::cout << std::setprecision(20);

    float d1(1.0);
    std::cout << "This should be 1.0: " << d1 << std::endl;

    float d2(0.1);
    std::cout << "This should be 0.1: " << d2 << std::endl;

    float d3(0.1*0.1);
    std::cout << "This should be 0.01:" << d3 << std::endl;

    bool x (0.1 + 0.1 + 0.1 == 0.3);
    std::cout << "This should be true (1): " << x << std::endl;

```

```
21      return 0;
22  }
```

We save it as “precision_test.cpp” and compile it with the following command line (on Linux):

```
1 g++ -Wall -Wextra -Werror -O0 precision_test.cpp -o precision_test.bin
```

This program will temporarily set a reduced precision in our number representation, and try to output the values of the numbers 1, 0.1 and $0.1^2 = 0.01$, let’s see the results:

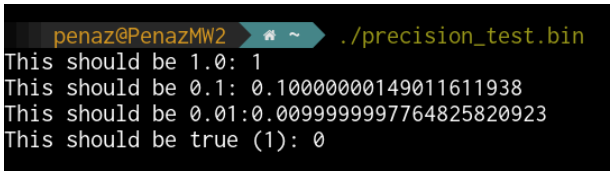


Figure 26: Results of the simple float precision test

With the number 1 it’s all good, but... what is going on with 0.1? What is all that garbage? The number 0.01 is even worse! That’s not even close! Why $0.1 + 0.1 + 0.1$ comes out as not 0.3! **What is maths anymore?**

We have just met one of the (many) limitations of computers: computers cannot represent certain numbers without “approximating”. Compilers and libraries exist to work around these issues, but we need to be ready to avoid surprises.

Just to reiterate: this is not a problem of the single programming language, we can see that C++ is affected, but also Python has the same issue:

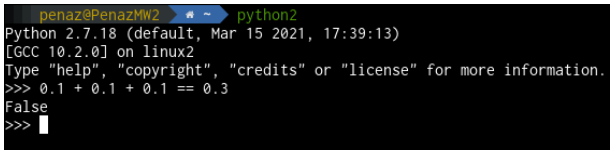


Figure 27: Python 2 has the same issues with precision as C++

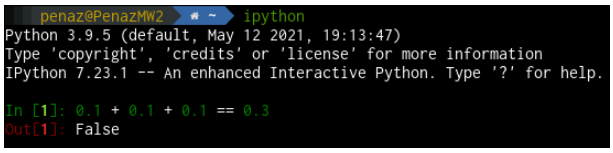


Figure 28: Python 3 doesn’t fare much better when it comes to precision

This is a computer issue in general: this may not be a huge problem for general use but, if we try to be too precise with our calculations, this may come back to bite us.

4.6.1 Catastrophic cancellation

Advanced Wizardry!

Catastrophic cancellation is one of the many pitfalls that you may encounter when dealing with very small numbers. This doesn't happen really often in the world of game development, feel free to just skim through this mostly informative section.

With a name as dangerous-sounding as "catastrophic cancellation", this sure looks like a dangerous phenomenon, but it's only dangerous if we don't know what it is.

Catastrophic Cancellation (sometimes called "cancellation error") is an event that may happen when subtracting two (usually large) numbers that are close to each other in value.

Warning: from here on, in this section, there will be some technical language. I will try to make it as simple and understandable as possible.

Let's imagine a computer, such computer's memory can handle at most 8 decimals while its *A.L.U.*$_{[g]}$ (the unit that takes case of "doing maths") can handle at most 16 decimal places.

Now let's take two numbers:

$$x = 0.5654328749846 \; y = 0.5654328510104$$

When we transfer such numbers in our memory, the computer will approximate such numbers to fit in its memory constraints. We'll represent that by applying to each number a function $fl()$ that we can read as "float representation of this number". So we'll end up having:

$$fl(x) = 0.56543287 \; fl(y) = 0.56543285$$

This is generally called an "assignment error", where during the assignment to a variable, a number loses part of its information.

Let's try an calculate how off those approximations are (by calculating the percent "relative error"), just to get an idea of what we lost by just loading the numbers on our "fake computer":

$$\delta_x = \frac{|x - fl(x)|}{x} = 0.00000088\%$$
$$\delta_y = \frac{|y - fl(y)|}{y} = 0.00000017\%$$

We can see that our approximations are **very close** to the numbers we want to calculate, now let's calculate $x - y$. Making things by hand we would have:

$$x - y = 0.239772 \times 10^{-7}$$

That's a tiny number right there. Now let's calculate $fl(x) - fl(y)$, remembering that the A.L.U. will fill up to 16 decimals:

$$fl(x) - fl(y) = 0.5654328700000000 - 0.5654328500000000 = 0.0000000200000000 = 0.2 \times 10^{-7}$$

That doesn't look so bad, unless we look at the "relative error":

$$\delta = \frac{|0.239772 \times 10^{-7} - 0.2 \times 10^{-7}|}{0.239772 \times 10^{-7}} = 16.6\%$$

We are off by 16% of the total result, this is actually really bad.

What happened? If you look closely, the numbers are really close and even have 7 decimal digits in common, since our computer can memorize only 8 digits, the 9th to 13th decimal digits that looked so unimportant suddenly become a huge part of the result (due to the subtraction) but are already lost.

4.7 Random Numbers are not really random

Computers are deterministic machines, given the same set of instructions and inputs, they will **always** return the same output. Someone may think about "random number generators" and sure, those programs look like they spit random numbers on your screen, but they actually don't.

The most important number when generating random numbers is called *seed* and it's the number used by the *generator* to produce random numbers.

Let's see an example of a random number generator in C++ (you can copy this program verbatim to try it):

Listing 10: A simple random number generation program

```
#include<iostream>
int main(){
    // First of all we get the seed
    unsigned int seed;
    std::cout << "Type the seed: " ;
    std::cin >> seed;
    // Now we seed the randomizer
    srand(seed);
    // Small presentation
    std::cout << "This generator will now generate 10 random numbers" << std::endl;
    // Output 10 random numbers
    for (int i = 0; i < 10; ++i) {
        std::cout << rand() << std::endl;
    }
```

```
15      // Finish the program
16      return 0;
17  }
```

We can save this program as `random_seed.cpp` compile this program with the following command:

```
1  g++ -Wall -Wextra -Werror -O0 random_seed.cpp -o random_seed.bin
```

When we run the program, it will ask us to input a seed (which in our case is a number), after that it will just print 10 random numbers based on that seed. What would happen if we ran the program twice and use the same seed?

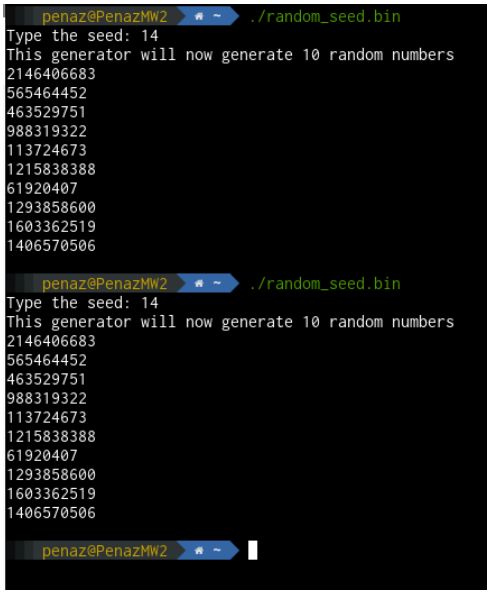


Figure 29: Running a random number generator with the same seed will always output the same numbers

Random numbers generated by computers are never truly random, that's why they are more properly called "pseudo-random numbers".

4.7.1 How to seed a random number generator

From what we have seen earlier, the seed of our random number generator is something we need to be mindful about.

Choosing a static seed will make our game completely deterministic (if played in the same conditions), like we didn't use random numbers at all.

Some games use internal timers to see the random number generator, be it the time that the game has been running, the time that has passed from the beginning of the mission or something similar. This allows you to have some kind of "controlled RNG" that still has a bit of reproducibility.

Some choices expose the game to the possibility of RNG manipulation: where the player has partial or total control over the random number generator, by performing specific actions at specific times, for instance.

A very easy way to seed a generator is using the system time. Here's a more advanced random number generator that uses system time as its seed: if you run the program reasonably slow (not quicker than once a second) you will see the numbers changing.

Listing 11: A random number generation program that uses system time as seed

```
#include<iostream>
#include<time.h>
int main(){
    // First of all we get the seed
    unsigned int seed = time(nullptr);
    // We print it for reference
    std::cout << "The current seed is:" << seed << std::endl;
    // Now we seed the randomizer
    srand(seed);
    // Small presentation
    std::cout << "This generator will now generate 10 random numbers" << std::endl;
    // Output 10 random numbers
    for (int i = 0; i < 10; ++i) {
        std::cout << rand() << std::endl;
    }
    // Finish the program
    return 0;
}
```

We can save this program as `rand.cpp` compile this program with the following command:

```
g++ -Wall -Wextra -Werror -O0 rand.cpp -o rand.bin
```

This is the result of the program being run twice, one second apart:

```
penaz@PenazMW2 ~ ./rand.bin
The current seed is:1667851777
This generator will now generate 10 random numbers
1613303786
780154366
2080970745
2014960335
232397101
2044510406
1960776734
947950517
691212607
706142778

penaz@PenazMW2 ~ ./rand.bin
The current seed is:1667851778
This generator will now generate 10 random numbers
236875944
597010600
1660602459
1630365049
1168494461
1449261863
1664405749
1586092769
278797166
890542202
```

Figure 30: Using the system time as RNG seed guarantees a degree of randomness

4.8 Estimating the complexity of algorithms

Now more than ever, you need to be able to be efficient. How do you know how "efficient" some piece of algorithm is?

Seeing how much time it takes is not an option, computer specifications change from system to system, so we need something that could be considered "cross-platform".

This is where notations come into play.

There are 3 types of Asymptotic notation you should know: Ω, Θ and O.

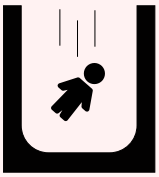
Ω() represents **a lower bound**: this means that the algorithm will take **at least** as many cycles as specified.

O() represents **an upper bound**: it's the most used notation and means that the algorithm will take **at most** as many cycles as specified.

Θ() is a **tight bound**, used when the big-O notation and the big-Ω notation have the same value, which can help define the behavior of the algorithm better.

We will now talk about the most common Big-O notations, from "most efficient" to "least efficient".

Pitfall Warning!



Be mindful of one specific thing: these notations simply tie how the algorithm performs in relation to how a certain variable grows (usually a dataset). If you know for certain that a dataset stays relatively small, an algorithm with a "worse" **O()** may not make a huge difference or may even be more efficient.

4.8.1 O(1)

An algorithm that executes in **O(1)** is said to execute "in constant time", which means that no matter how much data is input in the algorithm, said algorithm will execute in the same time.

An example of a simple O(1) algorithm is an algorithm that, given a list of elements (with at least one element), returns `True` if the first element is `null`.

Listing 12: Example of an O(1) algorithm

```
function isFirstElementNull(elements){
    if (elements[0] == null){
        return true;
    }else{
        return false;
    }
}
```

To be precise, this algorithm will perform both in O(1) and $\Omega(1)$, so it will perform in $\Theta(1)$.

4.8.2 O(log(n))

An algorithm that executes in O(log(n)) is said to execute in "logarithmic time", which means that given an input of **n** items, the algorithm will execute **log(n)** cycles at most.

An example of a O(log(n)) algorithm is the so-called "binary search" on a ordered list of items.

Listing 13: Example of an O(log(n)) algorithm (Binary Search)

```
function binarySearch(elements, element_to_find, begin, end){
    // We need some indexes to make this work
    if (begin === undefined){
        begin = 0;
    }
    if (end === undefined){
        end = elements.length - 1;
    }
    // Get the middle element
    let middle_element_index = Math.floor((end - begin) / 2) + begin;
    let middle_element = elements[middle_element_index];
    if (end <= begin && element_to_find != middle_element){
        // We found nothing, if we don't have this base case the algorithm will keep going
     forever
        return null;
    }
    if (element_to_find == middle_element){
        return middle_element_index;
    }
    if (element_to_find > middle_element){
        return binarySearch(elements, element_to_find, middle_element_index + 1, end);
    }else{
        return binarySearch(elements, element_to_find, begin, middle_element_index - 1);
    }
}
```

The best case is the time when you get the element to find to be the "middle element" of the list, in that case the algorithm will execute in linear time: $\Theta(1)$ - You need **at least one lookup** ($\Omega(1)$) and **at most one lookup** ($O(1)$).

In the worst case, the element is not present in the list, so you have to split the list and find the middle element until you realize that you don't have any more elements to iterate - this translates into a **tight bound** of $\Theta(log_2 n)$

4.8.3 O(n)

An algorithm that executes in O(n) is said to execute in "linear time", which means that given an input of **n** items, the algorithm will execute at most **n** cycles.

An example of a simple O(n) algorithm is the one that prints a list, element by element.

Listing 14: Example of an O(n) algorithm (printing of a list)

```
function printList(list){
    for (const element in list){
        console.log(element);
    }
}
```

It's evident that this algorithm will call the `print` function `n` times, where `n` is the size of the list. This translates in a $\Theta(n)$ complexity, which is both $O(n)$ and $\Omega(n)$.

There is no "best" or "worst" case here, the algorithm prints `n` elements, no matter their order, the alignment of planets and stars or the permission of its parents.

4.8.4 O(n·log(n))

An algorithm that executes in O(n·log(n)) executes in a time slightly longer than a linear algorithm, but it's still considered "ideal". These algorithms are said to execute in "quasi-linear", "log-linear", "super-linear" or "linearithmic" time.

Given an input of **n** elements, these algorithms execute **n·log(n)** steps, or cycles.

Some algorithms that run in O(n·log(n)) are:

- Quick Sort
- Heap Sort
- Fast Fourier Transforms (F.F.T.)

These algorithms are more complex than a simple example and would require a chapter on their own, so we'll leave examples aside for now.

4.8.5 O(n^2)

Quadratic algorithms, as the algorithms that execute in O(n^2) are called, are the door to the "danger zone".

These algorithms can eat your CPU time quite quickly, although they can still be used for small computations somewhat efficiently.

Given an input of **n** elements, these algorithms execute **n^2** cycles, which means that given an input of **20** elements, we'd find ourselves executing **400** cycles.

A simple example of a quadratic algorithm is "bubble sort". A pseudo-code implementation is written here.

Listing 15: Example of an O(n²) algorithm (bubble sort)

```
function bubbleSort(A){
    let n = length(A);
    let swapped = false;
    do{
        swapped = false;
```

```
        for (let i = 1; i < n; i++){
            if (A[i-1] > A[i]){
                // Swap the items
                let tmp = A[i-1];
                A[i-1] = A[i];
                A[i] = tmp;
                swapped = true;
            }
        }
    }while(swapped);
}
```

Anything with complexity higher than $O(n^2)$ is usually considered unusable.

4.8.6 $O(2^n)$

Algorithms that execute in exponential time are considered a major code red, an will usually be replaced with heuristic algorithms (which trade some precision for a lower complexity).

Given an input of `20` elements, an algorithm that executes in $O(2^n)$ will execute 2^{20} = 1 048 576 cycles!

4.9 A primer on calculating the order of your algorithms

4.9.1 Some basics

When you estimate an algorithm, you usually want to calculate how it functions "in the worst case", which usually means that all loops get to their end (of the list or the counter) and everything takes the longest time possible.

Let's start with an example:

Listing 16: A simple O(1) algorithm

```
// A simple O(1) algorithm: assigning to a variable
let my_variable = 1;
```

This is a simple assignment operation, we are considering this instantaneous. So its complexity is $O(1)$.

Now let's see another algorithm:

Listing 17: A simple o(n) algorithm

```
// A simple O(n) algirithm: iterating through a list
for (const item in my_list){
    console.log(item);
}
```

In this case we are iterating through a list, we can see that as the list grows, the number of times we print an element on our screen grows too. So if the list is n items long, we will have n calls to the output statement. This is an $O(n)$ complexity algorithm.

Now let's take something we already saw and analyze it: the bubble sort algorithm:

Listing 18: The bubble sort algorithm, an O(n²) algorithm

```
function bubbleSort(A){
    let n = length(A);
    let swapped = false;
    do{
        swapped = false;
        for (let i = 1; i < n; i++){
            if (A[i-1] > A[i]){
                // Swap the items
                let tmp = A[i-1];
                A[i-1] = A[i];
                A[i] = tmp;
                swapped = true;
            }
        }
    }while(swapped);
}
```

This will require a small effort on our part: we can see that there are 2 nested loops in this code. What's our worst case? The answer is "The items are in the reverse order".

When the items are in the reverse order, we will need to loop through the whole list to get the biggest item at the end of the list, then another time to get the second-biggest item on the second-to-last place on the list... and so on.

So every time we bring an item to its place, we iterate through all the list once. This happens for each item.

So, in a list of length "n", we bring the biggest item to its place "n times" and each "time" requires scanning "n" elements: the result is $n \cdot n = n^2$.

The algorithm has time complexity of $O(n^2)$.

4.9.2 What happens when we have more than one big-O?

There are times when we have code that looks like the following:

Listing 19: A more complex algorithm to estimate

```
// -------------------------------------
let n = A.length;
let swapped = false;
do{
    swapped = false;
    for (let i = 1; i < n; i++){
        if (A[i-1] > A[i]){
            // Swap the items
            let tmp = A[i-1];
            A[i-1] = A[i];
```

```
            A[i] = tmp;
            swapped = true;
        }
    }
}while(swapped);

// -------------------------------------

for (const item in A){
    console.log(item);
}
```

As we can see the first part is the bubble sort algorithm, followed by iterating through the (now ordered) list, to print its values.

We can calculate the total estimate as $O(n^2) + O(n)$ and that would be absolutely correct, but as the list grows, the growth rate of $O(n)$ is very minor if compared to $O(n^2)$, as can be seen from the following figure:

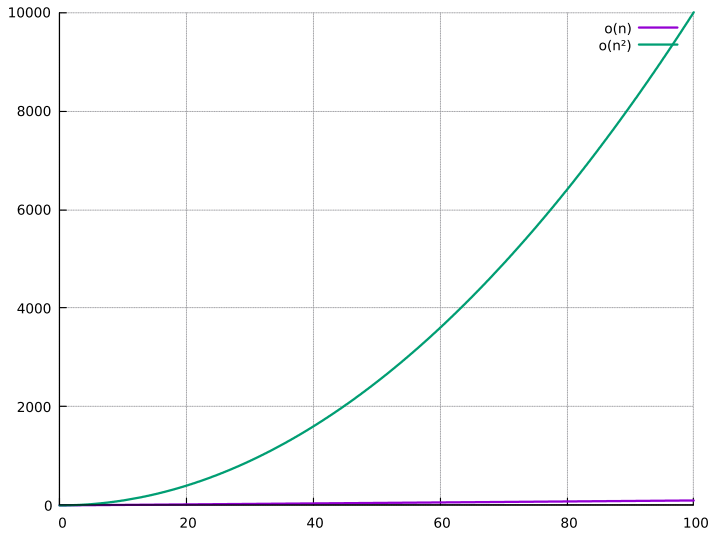


Figure 31: O(n) growth rate, compared to O(n²)

So we can drop the $O(n)$ and consider the entire algorithm as an $O(n^2)$ algorithm in its entirety: this means that when dealing with complexity estimates, you always keep the terms that have the largest "growth rate" (check the Big-O estimates comparison section for more details).

4.9.3 A problem with asymptotic complexity

An important problem with asymptotic complexity is that it tends to hide coefficients and smaller terms, no matter how important they may be.

Let's take an example: we need to order a list of 500000 elements and we found two algorithms:

- Algorithm 1 works in $O(n^2)$

- Algorithm 2 works in $O(n)$, but its "non-simplified" complexity is $O(1000000n)$

Which one would be more efficient? From a first inspection it may seem surprising that until we reach 1 million elements, algorithm 1 is better performing.

If we plot how the CPU cycles behave for each algorithm, we can see how the reality is different.

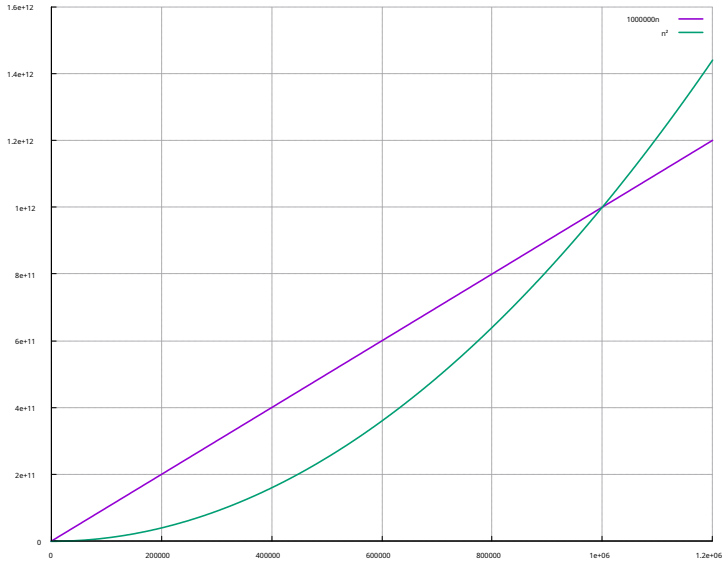


Figure 32: When coefficients have important values, asymptotic complexity may trick us

4.9.4 What do we do with recursive algorithms?

When recursive algorithms are involved, things get a lot more complex, and they involve building recursion trees and sometimes you'll have to use the so-called "master theorem for divide-and-conquer recurrences".

Such methods are outside the scope of this book as of now.

4.9.5 How do big-O estimates compare to each other?

Here we can see how big-O estimates compare to each other, graphically and how important it is to write not-inefficient algorithms.

If we had to write it as an inequality, from more to least efficient, we would have something like this (only considering Big-O notation):

$$O(1) < O(\log n) < O(n) < O(n \cdot \log n) < O(n^2) < O(2^n)$$

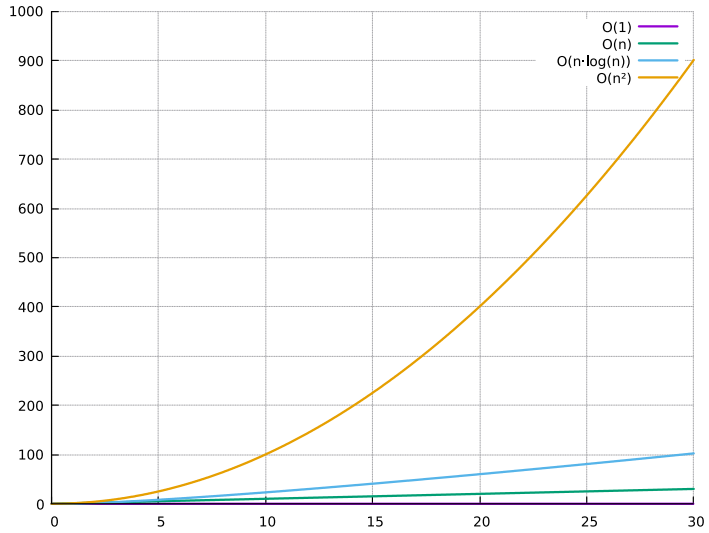


Figure 33: Big-O Estimates, plotted

There is a very specific reason why the $O(2^n)$ estimate is missing from the previous plot: we wouldn't be able to see anything worthwhile if it was included, as seen from the following plot:

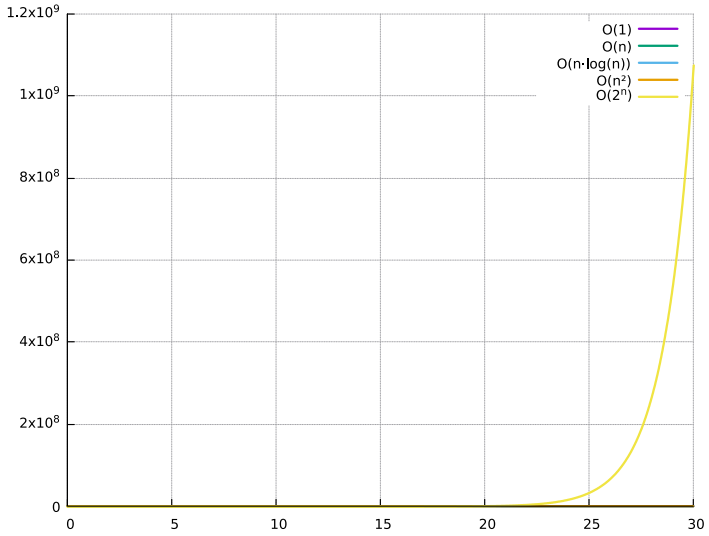


Figure 34: How $O(2^n)$ overpowers lower complexities

[This section is a work in progress and it will be completed as soon as possible]

4.10 Simplifying your conditionals with Karnaugh Maps

Karnaugh maps are a useful tool to simplify boolean algebra expressions, as well as identifying and potentially solving race conditions.

The output of a Karnaugh Map will always be an "OR of ANDs".

The best way to explain them is to give an example.

Let's take the following truth table:

Table 23: The first truth table we'll simplify with Karnaugh Maps

A	B	f
0	0	0
0	1	1
1	0	1
1	1	0

Said table can contain any number of variables (we'll see how to implement those). To be precise, this table represents the formula `A XOR B` (XOR means 'exclusive or').

Let's arrange it into a double-entry table, like this (Values of A are on top, values of B are on the left):

		A	
		0	**1**
B	**0**	0	1
	1	1	0

Figure 35: Karnaugh Map for A XOR B

Now we have to identify the biggest squares or rectangles that contain 2^n elements equal to 1 so that we can cover all the "1" values we have (they can overlap). In this case we're unlucky as we have only two small rectangles that contain one element each:

		A	
		0	**1**
B	**0**	0	1
	1	1	0

Figure 36: Karnaugh Map where the elements of the two "rectangles" have been marked green and red

In this case, we have the result we want with the following formula: $f = (A \wedge \bar{B}) \vee (\bar{A} \wedge B)$

Not an improvement at all, but that's because the example is a really simple one.

4.10.1 "Don't care"s

Karnaugh Maps show more usefulness when we have the so-called "don't care"s, situations where we don't care (wow!) about the result. Here's an example.

Table 24: Truth table with a “don’t care” value

A	B	f
0	0	0
0	1	1
1	0	1
1	1	x

Putting this truth table into a Karnaugh map we get something a bit more interesting:

		A	
		0	**1**
B	**0**	0	1
	1	1	x

Figure 37: Karnaugh Map with a “don’t care” value

Now we have a value that behaves a bit like a “wild card”, that means we can pretend it’s either a 0 or 1, depending on the situation. In this example we’ll pretend it’s a 1, because it’s the value that will give us the biggest “rectangles”.

		A	
		0	**1**
B	**0**	0	1
	1	1	1

Figure 38: Karnaugh Map where we pretend the “don’t care” value is equal to 1

Now we can find two two-elements rectangles in this map.

The first is the following one:

		A	
		0	**1**
B	**0**	0	1
	1	1	1

Figure 39: First Rectangle in the Karnaugh map

In this case, we can see that the result is 1 when $B = 1$, no matter the value of A. We’ll keep this in mind.

The second rectangle is:

		A	
		0	**1**
B	**0**	0	1
	1	1	1

Figure 40: Second Rectangle in the Karnaugh map

In this case, we can see that the result is 1 when $A = 1$, no matter the value of B.

This translates into a formula of: $f = (A) \vee (B)$, considering that we don't care about the result that comes out when $A = 1$ and $B = 1$.

Note!

If instead of 1, we ended up choosing 0 for our "don't care", we would have obtained $f = (A \wedge \bar{B}) \vee (\bar{A} \wedge B)$ (the extended form of $A\ XOR\ B$, which we saw earlier). For our needs, this would have been a good solution too.

4.10.2 A more complex map

When we have more variables, like the following truth table:

A	B	C	D	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	x

Now we'll have to group up our variables and put them in a Karnaugh Map using Gray Code, practically each row or column differs from the adjacent ones by only one bit.

The resulting Karnaugh map is the following (AB on columns, CD on rows):

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	x	1
	10	0	1	1	1

Figure 41: A more complex Karnaugh map

We can see two rectangles that contain 2^n items, one with 2 items, the other with 8, considering the only "don't care" value as 1.

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	x	1
	10	0	1	1	1

Figure 42: First rectangle of the more complex Karnaugh map

In this first rectangle, we can see that the values of C and D don't matter towards the result, as well as the value of B. The only variable that gives the result on this rectangle is $A = 1$. We'll keep that in mind

Let's see the second rectangle:

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	x	1
	10	0	1	1	1

Figure 43: Second rectangle of the more complex Karnaugh map

In this case A doesn't give any contribution to the result, but at the same time we need $B = 1$, $C = 1$ and $D = 0$ to get the wanted result.

$D = 0$ translates into $\bar{D} = 1$, which brings the formula to: $f = A \vee (B \wedge C \wedge \bar{D})$.

If we didn't have that "don't care" value, everything would have been more complex.

4.10.3 Guided Exercise

Let's remove the "don't care" value and have the following truth table:

A	B	C	D	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Let's put it into a Karnaugh Map:

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	0	1
	10	0	1	1	1

Figure 44: Guided Exercise: Karnaugh Map (1/4)

Find the biggest rectangles:

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	0	1
	10	0	1	1	1

Figure 45: Guided Exercise: Karnaugh Map (2/4)

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	0	1
	10	0	1	1	1

Figure 46: Guided Exercise: Karnaugh Map (3/4)

		AB			
		00	**01**	**11**	**10**
CD	**00**	0	0	1	1
	01	0	0	1	1
	11	0	0	0	1
	10	0	1	1	1

Figure 47: Guided Exercise: Karnaugh Map (4/4)

Extract the result: $f = (A \wedge \bar{C}) \vee (A \wedge \bar{B}) \vee (B \wedge C \wedge \bar{D})$

4.11 Object Oriented Programming

4.11.1 Introduction

One of the biggest programming paradigms in use is surely the "Object Oriented Programming" (from now on: "O.O.P.") paradigm. The fundamental unit of a program, in this paradigm is the *Object*. This paradigm allows to structure your code in a more modular and re-usable way, as well as implementing abstractions, allowing for more solid code and making it possible for other code to make use of your own code without needing to know any details besides its *Interface*.

4.11.2 Objects

Objects are the fundamental unit in O.O.P., objects are essentially a collection of data and functions. Objects are actually the physical instantiation of what is called a "Class".

To simplify the concept: a "Class" is a house blueprint, an "Object" is the house itself.

Objects contain data and functions, for the sake of precision, we will use their technical names:

- Functions that are part of an object are called **methods** and they can be classified as:
  - *Instance Methods* when they act on a single object instance;
  - *Static Methods* when they don't (usually they're utility functions), that also means that these methods belong to the Class itself and not to its instance.
- Each piece of data contained in the class is called a **Field** and they can be classified as:
  - *Instance Fields* when they're part of the instance and can change from instance to instance;
  - *Static Fields* when they're part of the class but don't change between instances (**Caution:** it does not mean they cannot change, in that case the change will snowball into all the instances).

4.11.3 Abstraction and Interfaces

Abstraction is a fundamental point in O.O.P., and it is usually taken care of via so-called **Interfaces**.

Interfaces are the front-end that an object offers to other objects so they can interact.

As an example: the interface to your PC is given by Keyboard, Mouse and Screen - you don't need to know how the single electron travels through the circuits to be able to use a computer; same goes for a class that offers a well-abstracted interface.

Being able to abstract a concept, removing the necessity to know the internal workings of the code that is used, is fundamental to be able to write solid and maintainable code, because implementations change, but interfaces rarely do.

Making classes work together with interfaces allows you to modify and optimize your code without having each edit snowball into a flurry of compiler (or interpreter) errors. For instance: a rectangle class exposes in its interface a method `getArea()` - you don't need to know how to calculate the area, you just call that method and know it will return the area of the rectangle.

The concept of keeping the internal workings of a class is called *Information Hiding*.

4.11.4 Inheritance and Polymorphism

One of the biggest aspects of O.O.P. is **inheritance**: you can create other classes based on a so-called "base class", allowing for extensibility of your software.

You can create a "Person" class, with a name, surname and age as fields, and by inheriting from the "Person" class you can create a "Student" class, which has all the fields from Person, plus the "clubs" and "grade" fields.

This allows to create a "tree of classes" that represents part of your software.

From inheritance, O.O.P. presents a concept called **Polymorphism** (From "Poly" - Many, "Morph" - Shape), where you can use the base class to represent the entire class tree, allowing for substitution.

In our "Person-Student" example, you could use a pointer to either a Person or a Student for the sake of getting their first name.

In some languages it is possible for an object to inherit from multiple other objects, this is called "Multiple Inheritance"

4.11.5 Mixins

Mixins are classes that contain certain methods that are made to be used by other classes. We can see mixins as some kind of interface with methods already implemented.

Mixins encourage the reuse of code (since the common functionalities get separated into their own classes), allowing for some interesting mechanisms and enforcing the Dependency Inversion principle.

Many times, Mixins are described as "included" rather than "inherited", due to their nature.

Random Trivia!

The python web framework Django makes heavy use of mixins in its class-based views: you can create a standard "View" (representing a web page, for instance), and then add login protection (via LoginRequiredMixin) or permissions (via PermissionRequiredMixin). This is all done using Python's multiple inheritance.

A code example of mixins is beyond the scope of this book, since each language has its own way of implementing mixins, some easy (like Python), other a bit more complex (like C++, see "Curiously Recurring Template Patterns", or C.R.T.P.).

4.11.6 The Diamond Problem

Usually when you call a method that is not present in the object itself, the program will look through the object's parents for the method to execute. This usually works well when there is no ambiguity. What if there is ambiguity instead?

When multiple inheritance is involved, there is a serious possibility of a situation similar to the following

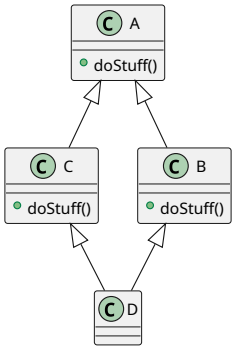


Figure 48: Example of a diamond problem

In this example, class A implements a method `dostuff()` that is overrode by both classes B and C (which inherit from A): now class D inherits from both B and C but does not override `dostuff()`, which one will be chosen?

This is the reason many languages do not implement multiple inheritance between classes (like Java, which allows multiple inheritance only between interfaces), other implement the so-called "virtual inheritance" (C++) and others again use an ordered list to solve the problem (Python).

This is not something you usually need to worry about, but you may want to be careful when you structure your classes to avoid "diamond problems", so to avoid headaches.

4.11.7 Composition

As opposed to inheritance's "IS-A" relationship, composition makes use of a "HAS-A" type of relationship.

Composition allows to define objects by declaring which properties they have: a player character can be a sprite with a “Movable” component, or a box could have a “RigidBody” component.

This way we can create new objects by reusing basic components, making maintenance easier as well as saving lines of code, avoiding “the diamond problem” and reducing coupling.

4.11.8 Composition vs. Inheritance

Let’s be clear right from the get go: there is no “silver bullet” here. Composition and inheritance target different problems (IS-A vs. HAS-A relationships). Inheritance binds classes closely, while composition tends to induce less coupling (we’ll talk about coupling in a second).

Let’s make an example of inheritance: we have a “Shape” base class, from where we create two new classes: Rectangle and Circle. For the purposes of our usage (which will be getting perimeter and area), Rectangle IS-A Shape, as well as Circle IS-A Shape.

Listing 20: An example of inheritance: Shapes

```
class Shape{
    // An abstract shape class
    area(){
        // An abstract function that will be overridden by subclasses
        throw "Not Implemented";
    }
    perimeter(){
        // An abstract function that will be overridden by subclasses
        throw "Not Implemented";
    }
};

class Rectangle extends Shape{
    // A simple rectangle class

    constructor(w, h){
        this.width = w;
        this.height = h;
    }

    area() {
        // Returns the Area of the rectangle
        return this.width * this.height;
    }

    perimeter() {
        // Returns the Perimeter of the rectangle
        return 2 * (this.width + this.height);
    }
}

class Circle extends Shape{
```

```
33      // A simple circle class
34
35      constructor(r){
36          this.radius = r;
37      }
38
39      area() {
40          // Returns the Area of the circle
41          return this.radius * Math.PI ** 2;
42      }
43
44      perimeter() {
45          // Returns the circumference of the circle
46          return 2 * Math.PI * this.radius;
47      }
48  }
```

Let's continue with another example: we have a Coffee machine, such coffee machine HAS-A grinder, as well as brewing unit. We can express such relationships with composition and build our coffee machine from our components.

Listing 21: An example of inheritance: A coffee machine

```
1  class Grinder{
2      // A simple coffee grinder component
3      grind(){
4          // Pretend to grind some coffee
5          console.log("Grinding coffee");
6      }
7  }
8
9  class BrewingUnit{
10     // A simple brewing unit component
11     brew(){
12         // Pretend to brew a good coffee
13         console.log("Brewing your coffee");
14     }
15 }
16
17 class CoffeeMachine{
18     // A simple coffee machine, has a grinder and a brewing unit
19     constructor(){
20         this.grinder = new Grinder();
21         this.brewer = new BrewingUnit();
22     }
23
24     make_coffee(){
25         // Uses the brewing component and the grinder to make some fresh coffee
26         this.grinder.grind();
27         this.brewer.brew();
```

```
28          console.log("Here's your fresh coffee!");
29      }
30  }
```

4.11.9 "Composition over Inheritance" design

Often cited in programming, the "composition over inheritance" design states that code reuse and polymorphism should be achieved using composition as the preferred method, while leaving subclassing (inheritance) alone as much as possible.

This allows for easier code reuse, as well as more flexibility and less coupling. Let's make a simple example.

Note!

If you're having trouble understanding the diagrams that follow, head to the Reading UML Diagrams section for a full explanation on UML.

Let's imagine a physical object that can be Visible/Invisible, Solid/NonSolid and Movable/Immovable. In UML, an inheritance hierarchy would look something like this:

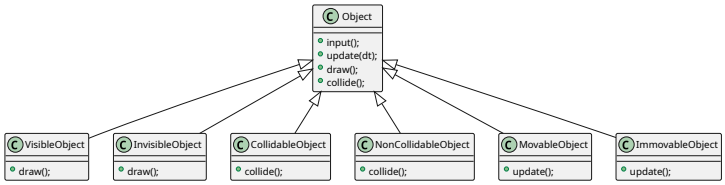


Figure 49: How an object may look using inheritance

If we think of some objects, like a playable character, or a building, things get a lot more complicated: a playable character is movable, solid and visible, while the building is not movable. Things can get ugly really fast.

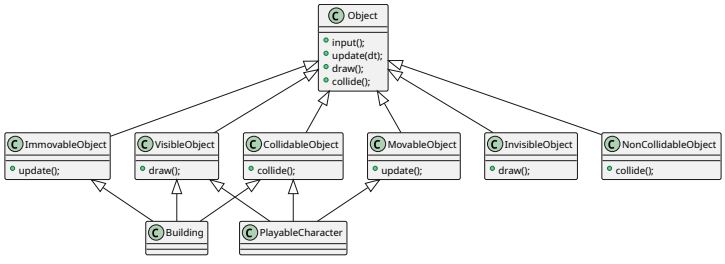


Figure 50: How inheritance can get complicated quickly

This can lead to confusion, as well as the diamond problem.

Using composition, we can separate the behaviours into the "Visible", "Updatable" and "Movable" components and then use those as "reusable puzzle pieces" for our objects.

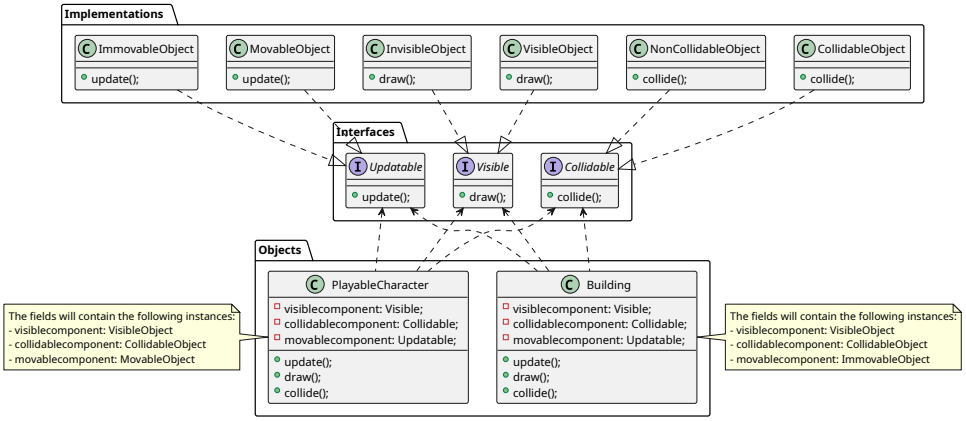


Figure 51: How components make things a bit simpler

4.11.10 Coupling

Coupling is a term used to define the phenomenon where an edit to some part of a software snowballs into a bunch of edits in parts of the software that depend on the modified part, and the part that depend on the previously edited dependency, etc...

The parts involved are defined as “coupled” because, even though they are separated, their maintenance very much behaves like they were a single entity. This also means that the elements that are coupled are harder to reuse, since they are so tightly related that they end up serving each other and nothing else.

Introducing unnecessary coupling in our software will come back to bite us in the future, affecting maintainability in a very negative way, since any edit we make (for instance, to fix a bug) can potentially lead to the need to edit the rest of the software (or game) we are writing.

Reducing coupling is done by reducing interdependence, coordination and information flow between elements of a program.

Examples of coupling include:

- A module uses code of another module (this breaks the principle of *information hiding*[g]);
- Many modules access the same global data;
- A module controls the flow of another module (like passing a parameter that decides “what to do”);
- Subclassing.

This means that it’s in our best interest to reduce code coupling as much as possible, following the good principles of “nutshell programming” and following the SOLID principles, shown next.

Note!

We may be tempted to try and "remove coupling completely", but that's usually a wasted effort. We want to reduce coupling as much as possible and instead improve "cohesion". Sometimes coupling is unavoidable (as in case of subclassing). Balance is the key.

4.11.11 The DRY Principle

DRY is a mnemonic acronym that stands for "**D**on't **R**epeat **Y**ourself" and condenses in itself the principle of reducing repetition inside a software, replacing it with *abstractions* and by *normalizing data*.

This allows for each piece of code (and knowledge, since the DRY principle applies to documentation too) to be unambiguous, centralizing its responsibilities and avoiding repetition.

Violations of the DRY principle are called "WET" (**W**rite **E**verything **T**wice) solutions, which base themselves on repetition and give higher chances of mistakes and inconsistency.

4.11.12 SOLID Principles

SOLID is a mnemonic acronym that condenses five principles of good design, to make code and software that is understandable, flexible and maintainable.

- **Single Responsibility**: Each class should have a single responsibility, it should take care of one part of the software specification and each change to said specification should affect only said class. This means you should avoid the so-called "God Classes", classes that take care of too much, know too much about the system and in a nutshell: have too much responsibility in your software.
- **Open-closed Principle**: Each software entity should be open to extension, but closed for modification. This means that each class (for instance) should be extensible, either via inheritance or composition, but it should not be possible to modify the class's code. This is practically enforcing *Information Hiding*.
- **Liskov Substitution Principle**: Objects in a program should be replaceable with instances of their subtypes and the correctness of the program should not be affected. This is the base of inheritance and polimorphism, if by substituting a base class with one of its children (which should have a Child-is-a-Base relationship, for instance "Circle is a shape") the program is not correct anymore, either something is wrong with the program, or the classes should not be in a "IS-A" relationship.
- **Interface Segregation**: Classes should provide many specific interfaces instead of one general-purpose interface, this means that no client should depend on methods that it doesn't use. This makes the software easier to refactor and maintain, and reduces coupling.
- **Dependency Inversion**: Software components should depend on abstractions and not concretions. This is another staple of nutshell programming and O.O.P. - Each class should make use of some other class's interface, not its inner workings. This allows for maintainability and easier update and change of code, without having the changes snowball into an Armageddon of errors.

[This section is a work in progress and it will be completed as soon as possible]

4.12 Designing entities as data

Sometimes it can be useful to design your entities as data, instead of making them into static objects that possibly require a new release of your product.

Designing your objects as data allows you to use configuration files to create, configure, tinker and extend your product, as well as allow for modifications by people who are fans of your game.

For instance, in a fantasy RPG you could have 3 types of enemies all defined as classes:

- Skeleton
- Zombie
- Ghost Swordsman

Which all have the same behavior but different animations and sprites.

These classes can inherit from an “entity” abstract class which defines the base behavior and then can be extended to create each unique enemy.

Another idea could be designing an “entity” class that can be instantiated, and have a configuration file that defines what each entity is and what its properties are.

An idea could be the following, using YAML:

Listing 22: Example of an entity declared as YAML data

```yaml
entity:
  name: skeleton
  health: 10
  damage_on_hit: 2.5
  spritesheet: "./skelly.png"
  animations:
    walking:
      start_sprite: 4
      frame_no: 4
      duration: 0.2
    attacking:
      start_sprite: 9
      frame_no: 2
      duration: 0.1
```

Another often used alternative is JSON, which would look like this:

Listing 23: Example of an entity declared as JSON data

```json
{
    "entity": {
        "name": "skeleton",
        "health": 10,
        "damage_on_hit": 2.5,
```

```
        "spritesheet": "./skelly.png",
        "animations":{
            "walking":{
                "start_sprite": 4,
                "frame_no": 4,
                "duration": 0.2
            },
            "attacking":{
                "start_sprite": 9,
                "frame_no": 2,
                "duration": 0.1
            }
        }
    }
}
```

With more complex building algorithms, it is possible to change behaviors and much more with just a configuration file, and this gives itself well to rogue-like games, which random selection of enemies can benefit from an extension of the enemy pool. In fact, it's really easy to configure a new type of enemy and have it work inside the game without recompiling anything.

This allows for more readable code and a higher extensibility.

4.13 Reading UML diagrams

UML (Universal Modeling Language) is a set of graphical tools that allow a team to better organize and plan a software product. Diagrams are drawn in such a way to give the reader an overall assessment of the situation described while being easy to read and understand.

In this chapter we will take a look at 4 diagrams used in UML:

- Use Case Diagrams
- Class Diagrams
- Activity Diagrams
- Sequence Diagrams

4.13.1 Use Case Diagrams

Use Case Diagrams are usually used in software engineering to gather requirements for the software that will come to exist. In the world of game development, use case diagrams can prove useful to have an "outside view" of our game, and understand how an user can interact with our game.

Here is an example of a use case diagram for a game:

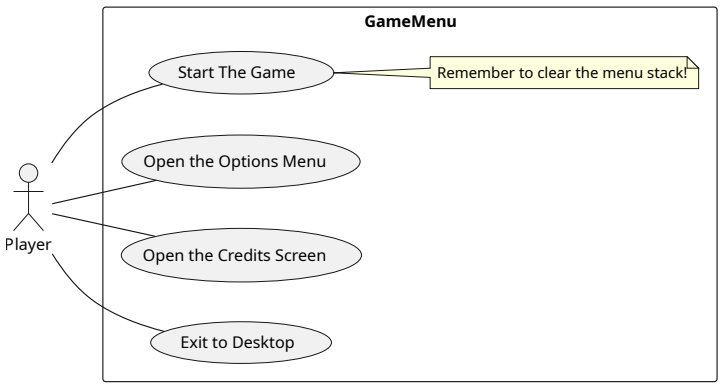


Figure 52: Example of a use case diagram

4.13.1.1 Actors

Actors are any entity that can interface with our system (in this case, our game) without being part of it. Actors can both be human, machines or even other systems.

Actors are represented with a stick figure and can inherit from each other: this will create an “IS-A” relationship between actors.

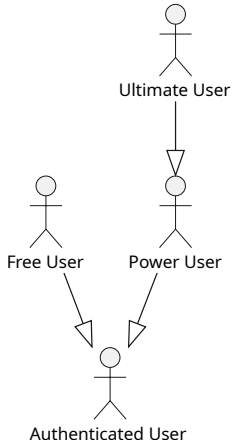


Figure 53: Example of an actor hierarchy

In the previous example, we can see that a “Free User” is an “Authenticated User”, as well as a “Power User” (which could be a paying user) is itself an “Authenticated User” while an “Ultimate User” (which could be a higher tier of paying user) is a “Power User” (thus has all the “Power User” capabilities, plus some unique) and by transitive property an “Authenticated User”.

As seen, inheritance between actors is represented with a solid line with a hollow closed arrow. Such arrow points towards the “super-type” or “parent” from which the subject (or “sub-type”, or “child”) inherits.

This representation will come back in the UML language for other diagrams too.

4.13.1.2 Use Cases

Use cases represent the functionalities that our system offers, and the relationships between them.

Use cases are represented with an ellipse with the name of the use case inside. Choosing the right name for a use case is extremely important, since they will represent the functionality that will be developed in our game.

Figure 54: Example of a use case

4.13.1.2.1 Inheritance

As with many other elements used in UML, use cases can inherit from each other. Inheritance (also called "Generalization") is represented with a closed hollow arrow that points towards the parent use case.

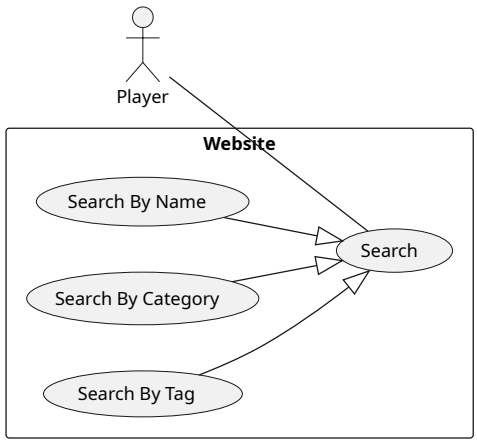


Figure 55: Example of a use case hierarchy

4.13.1.2.2 Extensions

Use case extensions specify how and when optional behavior takes place. Extended use cases are meaningful on their own and are independent from the extending use case, while the extending use case define the optional behavior that may not have much sense by itself.

Extensions are represented via a dashed line with an open arrow on the end, labeled with the `<<extend>>` keyword, pointing towards the extending use case.

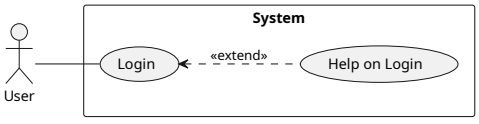


Figure 56: Example of a use case extension

4.13.1.2.3 Inclusions

Inclusions specify how the behavior of the included use case is inserted in the behavior of the including use case. Inclusions are usually used to simplify large use cases by splitting them or extract common behaviors of two or more use cases.

In this situation, the including use case **is not** complete by itself.

Inclusions are represented via a dashed line with an open arrow on the end, labeled with the `<<include>>` pointing towards the included use case.

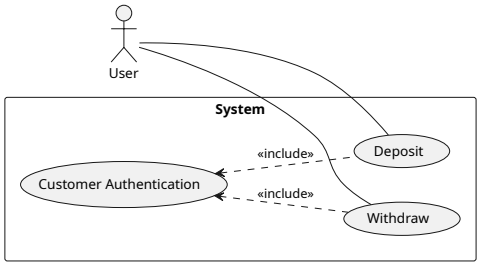


Figure 57: Example of a use case inclusion

4.13.1.3 Notes

In use case diagrams, as well as in many other UML diagrams, notes are used to jot down conditions, comments and everything useful to better understanding the diagram that cannot be conveyed through a well definite structure inside of UML.

Notes are shaped like a sheet of paper with a folded corner and are usually connected to the diagram with a dashed line. Each note can be connected to more than one piece of the diagram.

You can see a note at the beginning of this chapter, in the use case diagram explanation.

4.13.1.4 Sub-Use Cases

Use cases can be further detailed by creating sub-use cases, like the following example.

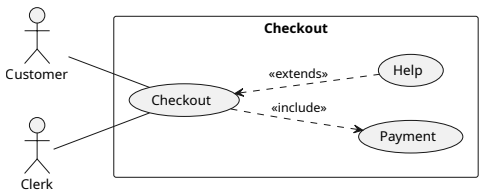


Figure 58: Example of a sub-use case

4.13.2 Class Diagrams

4.13.2.1 Classes

Class diagrams are used a step after analyzing your game, since they are used for planning classes. The central element of a class diagram is the “class”, which is represented as follows:

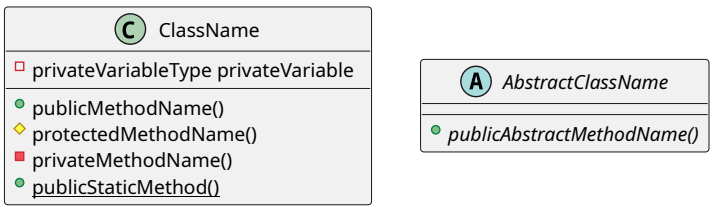


Figure 59: Example of classes in UML

Classes are made up by a class name, which is shown on top of the class; abstract classes are shown with a name in *italics*.

Public members are highlighted by a “+” symbol (or in our case, a green symbol) before their name, protected members use a “#” symbol (or a yellow symbol) and private members use a “-” symbol.

Static members are shown with an underlined name, while abstract members are shown in *italics*.

4.13.2.2 Interfaces

Sometimes there is a need to convey the concept of “interface” inside a UML class diagram, that can easily be done in 2 ways:

- By using the class construct, with the keyword (called “stereotype”) `<<interface>>` written on top of it;
- By using the “lollipop notation” (also called “interface realization”).

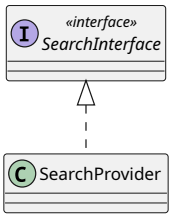


Figure 60: Defining an interface in UML

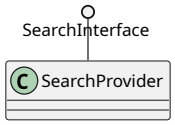


Figure 61: Interface Realization in UML

4.13.2.3 Relationships between entities of the class diagram

Expressing only single classes on their own doesn’t give UML a lot of expressive power when it comes to planning your games. Here we’ll take a quick look at the most used relationships between classes.

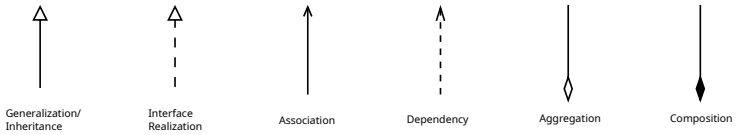


Figure 62: Relationships between classes in an UML Diagram

4.13.2.3.1 Inheritance

Inheritance is represented via a hollow closed arrow head that points towards the base class (exactly like in Actor inheritance), this means that the classes are in a “super-type and sub-type” relationship.

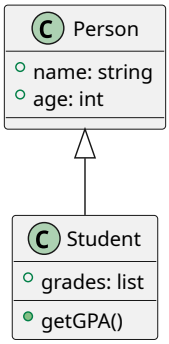


Figure 63: Example of inheritance in UML class diagrams

In this example we say that “Student IS-A Person” and inherits all Person’s methods and fields.

4.13.2.3.2 Interface realization

Interface realization can be complex to understand at first, given its formal definition:

> The interface realization relationship specifies that the realizing class must conform to the contract that the provided interface specifies.

In short, it means that the class is implementing all the methods specified by the interface (thus “realizing” it, as in making it real).

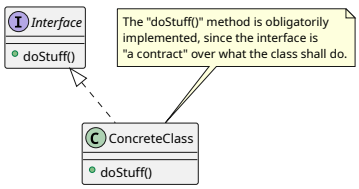


Figure 64: Example of interface realization in UML class diagram

4.13.2.3.3 Association

Association represents a static relationship between two classes. This is usually represented with a solid line with an arrow. The arrow usually shows the reading order of the association, so if you see an “Author” class and a “Book” class, the “wrote” association will be pointing from the “Author” to the “Book” class.

In case the relationship is bi-directional, the arrow points are omitted, leaving only a solid line between the two classes.

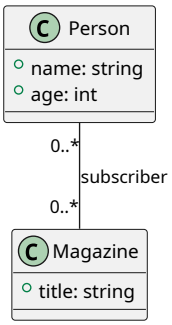


Figure 65: Example of association in UML class diagrams

An example of an association is the relationship between a “Person” and a “Magazine”, such relationship is the “Subscription”. In this case the relationship is bi-directional, since a “Magazine” can be subscribed by many people, but a single “Person” can subscribe to many “Magazine”s.

4.13.2.3.4 Aggregation and Composition

Aggregation is a special case of the association relationship, and represents a more specific case of it. Aggregation is a variant of a “has-a” relationship and represents a part-whole relationship.

Aggregation is represented with a hollow diamond and a line that points to the *contained* class, classes involved in an aggregation relationships *do not* have their life cycles dependent one another, that means that if the container is destroyed, the contained objects will keep on living. An example could be a teacher and their students, if the teacher dies the students will keep on living.

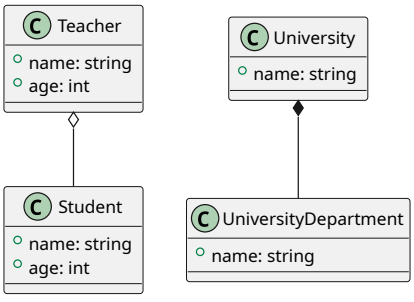


Figure 66: Example of aggregation and composition in UML class diagrams

Composition is represented with a filled diamond instead than a hollow one, in this case there is a life cycle dependency, so when the container is destroyed the contents are destroyed too. Like when a university is dissolved,

its departments will cease to exist. Conversely, a teacher may have some students under their wing, but when a teacher remains without students they won't magically disappear: the teacher's life cycle is independent from their students'.

4.13.2.3.5 Dependency

The dependency relationship is the one that gives us the least amount of coupling, it represents a "supplier-client" relationships, where the supplier supplies its functions (methods) to the client. The association is represented in a dashed line with an open arrow that points towards the supplier.

This means that the client class requires, needs or depends on the supplier.

There are many categories of dependency, like `<<create>` or `<<call>>` that explain further the type of dependency exists between the supplier and the client.

An example could be between a "Car Factory" and a class "Car": the "CarFactory" class depends on the "Car" class, and such dependency is an instantiation dependency.

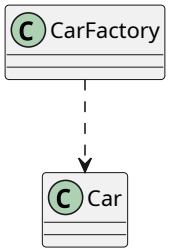


Figure 67: Example of dependency in UML class diagrams

4.13.2.4 Notes

As with Use Case diagrams, class diagrams can make use of notes too, and the graphical language used to represent them is exactly the same one used in the Use Case Diagrams.

4.13.3 Activity Diagrams

Activity diagrams are the more powerful version of flow charts: they represent the flux of an activity in detail, allowing to have a better understanding of a process or algorithm.

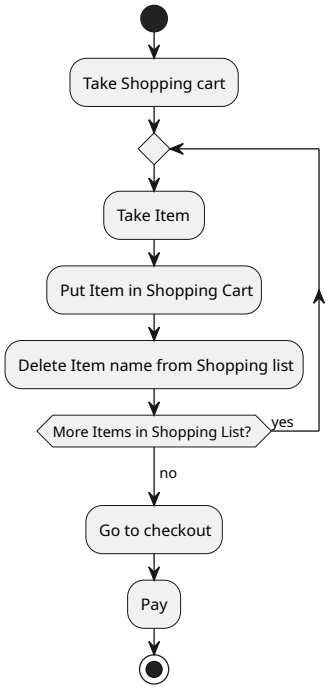


Figure 68: Example of an activity diagram

4.13.3.1 Start and End Nodes

Each diagram begins what a “start node”, represented with a filled black circle, and they end with an “end node” which is represented with a filled black circle inside of a hollow circle.

Figure 69: Example of activity diagrams start and end nodes

4.13.3.2 Actions

Each action taken by the software is represented in the diagram via a rounded rectangle, a very short description of the action is written inside the rounded rectangle space.

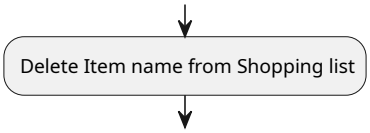


Figure 70: Example of Action in activity diagrams

4.13.3.3 Decisions (Conditionals) and loops

Decisions and loops are enclosed in diamonds. If a condition needs to be written, the diamond can become an hexagon, to make space for the condition to be written or guards can be used to express the condition.

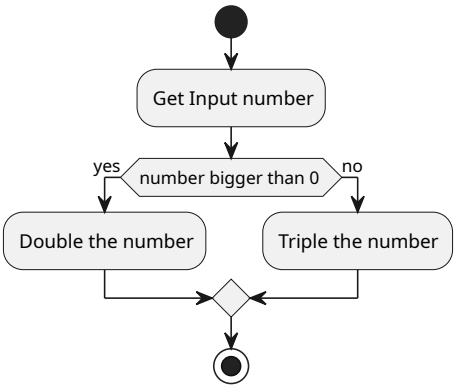


Figure 71: Example of decision, using hexagons to represent the condition

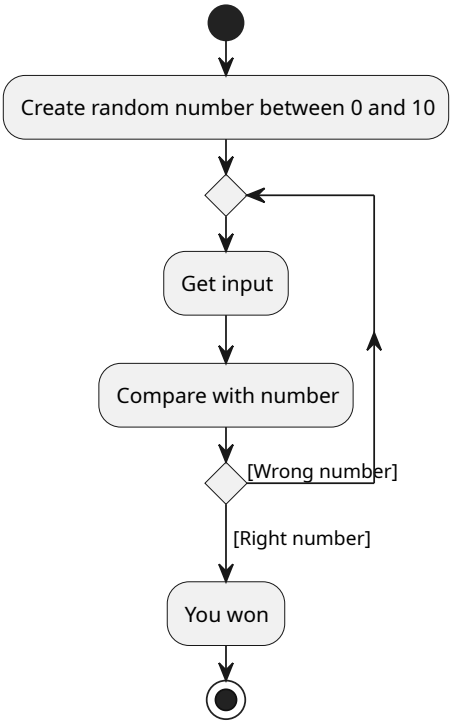


Figure 72: Example of loops, using guards to represent the condition

All the branches that depend on a condition start on the condition itself and end on a diamond, as shown below.

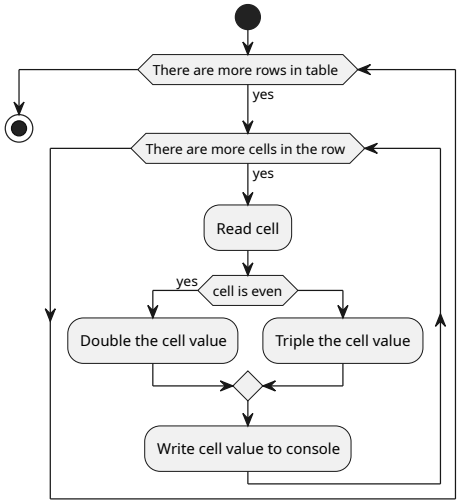


Figure 73: Example of how nested loops and conditions are performed

Note!

Sometimes loops can make use of empty diamonds (called “merges”) to make the diagram clearer.

4.13.3.4 Synchronization

Synchronization (or parallel processing) is represented in activity diagrams by using filled black bars that enclose the concurrent processes: the bars are called “synchronization points” or “forks” and “joins”

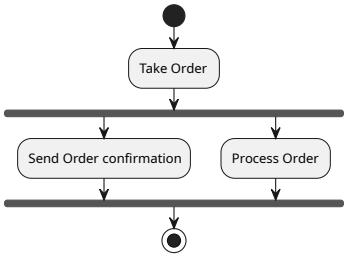


Figure 74: Example of concurrent processes in activity diagrams

In the previous example, the activities “Send Order Confirmation” and “Process Order” are processed in parallel, independently from each other, the first activity that finishes will wait until the other activity finishes before entering the end node.

4.13.3.5 Swimlanes

Swimlanes are a way to organize and group related activities in columns. For instance a shopping activity diagram

can have the “Customer”, “Order”, “Accounting” and “Shipping” swimlanes, each of which contains activities related to their own categories.

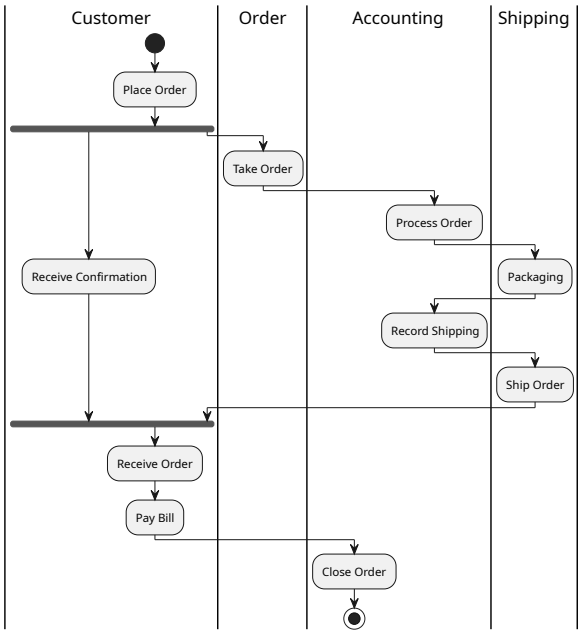


Figure 75: Example of swimlanes in activity diagrams

4.13.3.6 Signals

Signals are used to represent how activities can be influenced or modified from outside the system. There are two symbols used to represent signals.

The “Sent Signal” symbol is represented with a convex pentagon (which reminds an arrow going away from our system), while the “Received Signal” is represented by a concave pentagon (which reminds a “slot” where the “sent signal” symbol can connect to).

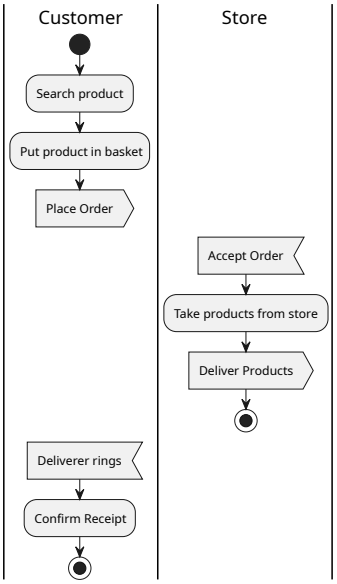


Figure 76: Example of signals in activity diagrams

4.13.3.7 Notes

As with Use Case and Class diagrams, Activity Diagrams can make use of notes, in the same way as the other two diagrams we presented in this book do.

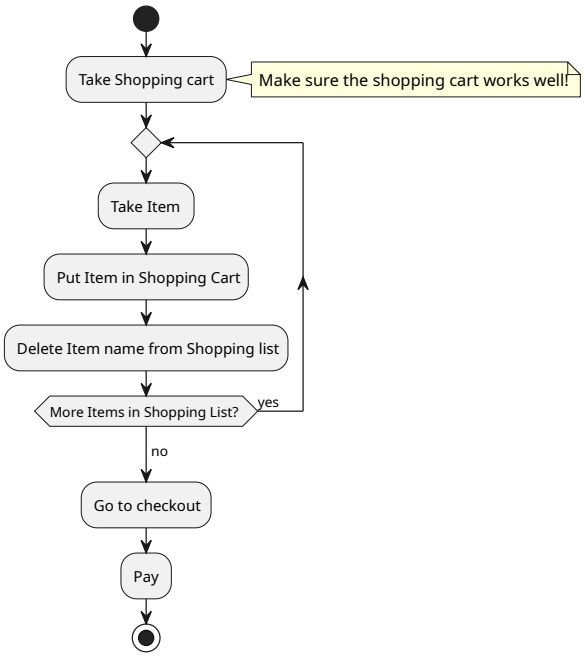


Figure 77: Example of a note inside of an activity diagram

4.13.3.8 A note on activity diagrams

The components of activity diagrams shown here are just a small part of the used components, but they should be enough to get you started designing and reading most of the activity diagrams that exist.

4.13.4 Sequence Diagrams

Sequence diagrams are used to represents how objects (called “participants”) interact with each other and such interactions are represented in a time sequence.

4.13.4.1 Lifelines

The central concept of sequence diagrams are lifelines: the represent the time a participant is “alive” and when it is doing something.

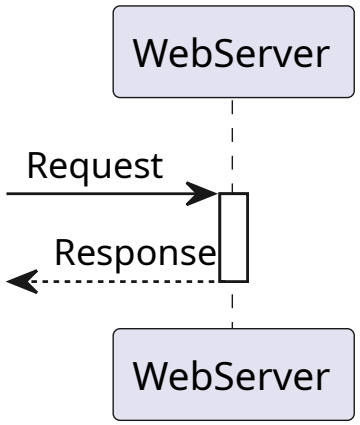


Figure 78: Example of a sequence diagram lifeline

The time flows from top to bottom, a dashed line represents the participant exists (for instance an object is instantiated in memory), while the rectangle that replaces the dotted line represents the participant being “active” (for instance one of the object’s functions is called).

4.13.4.1.1 Participants

The participants don’t have to be actual classes, since sequence diagrams represent interactions at a “higher level” than mere code-bound planning.

Some UML drawing software allows for custom shapes for each participant, like the following:

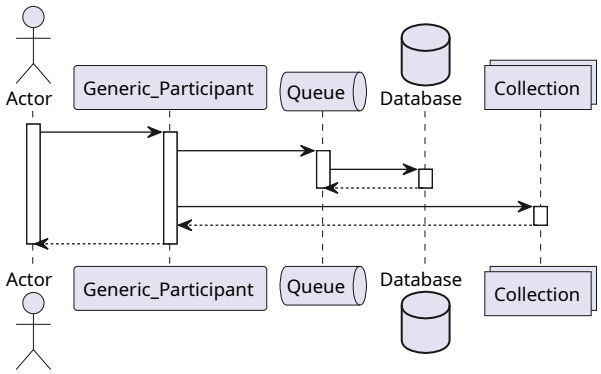


Figure 79: Some alternative shapes for participants

4.13.4.2 Messages

Each object (represented by a lifeline) communicates with other objects (and the "outside") through "messages".

Messages are represented by arrows and an example can be seen here:

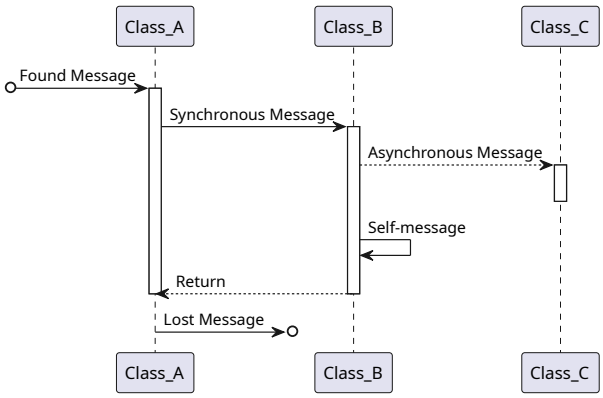


Figure 80: Messages in a sequence diagram

Let's analyze them one by one:

- **Found Messages** are messages that come from "outside", from the perspective of the part of the system we are analyzing, they may come from another system or even the user.
- **Synchronous Messages and returns** are messages that activate a class and wait for a "return message". These usually represent a synchronous function call (but it can represent a more abstract concept).
- **Asynchronous Messages** are messages that activate a class but don't wait for a return value. These usually represent asynchronous functions calls.
- **Self-messages** are messages from an object to itself, they usually represent an internal function call.
- **Lost Messages** are messages sent towards the "outside", from the perspective of the part of the system we are analyzing.

4.13.4.3 Object Instantiation and Destruction

Sometimes it may be useful to represent the instantiation and destruction of objects in a sequence diagram. UML provides such facilities via the `<<instantiate>>`, `<<create>>` and `<<destroy>>` keywords, as well as a symbol for the destruction of an object.

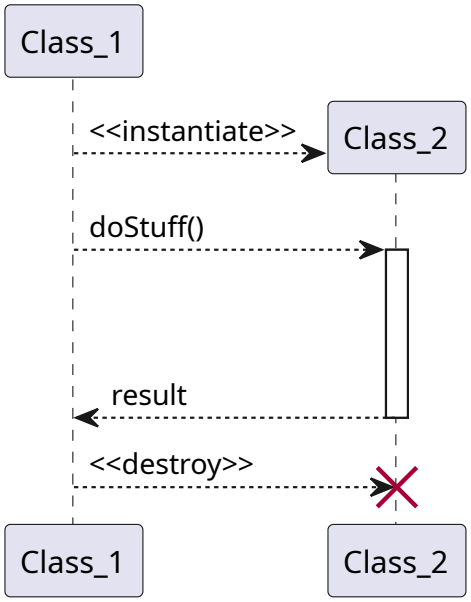


Figure 81: Object instantiation and destruction in a sequence diagram

4.13.4.4 Grouping and loops

From time to time, we may need to represent a series of messages being sent in parallel, a loop, or just group some messages to represent them in a clearer manner. This is where grouping comes of use: it has a representation based on “boxes”, like the following:

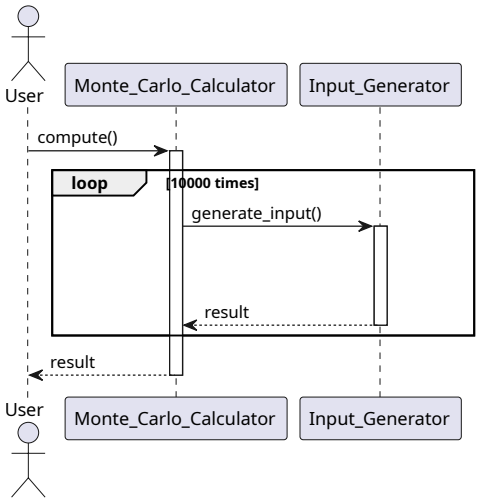


Figure 82: A loop grouping in a sequence diagram

4.13.4.5 Notes

Like all UML diagrams, it is possible to use notes to add some comments that may be useful for the interpretation of our diagrams, like follows.

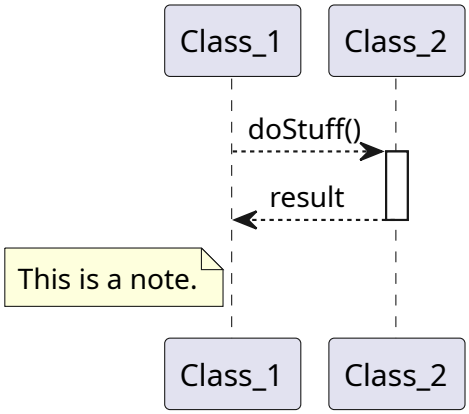


Figure 83: Example of notes in a sequence diagram

4.13.5 Other diagrams

UML is composed by a ton of diagrams that can be used to communicate with your teammates and organize your work, among them we find:

- Component Diagrams;
- Communication Diagrams;
- Composite Structure Diagrams;
- Deployment Diagrams;
- Package Diagrams;
- Profile Diagrams;
- State Diagrams;
- Timing Diagrams.

In this chapter we just saw the ones that will help you the most when reading the rest of this book, as well as effectively planning any project you have in mind.

4.14 Generic Programming

Sometimes it may be necessary (mostly in the case of containers) to have the same kind of code to work on different data types, which means that we need to **abstract types into variables** and be able to code accounting for such types.

Generic Programming is a blanket-term that defines a style of computer programming where algorithms are written in terms of "to be specified later" data types, this usually applies to languages that make use of *static typing*[g].

4.15 Basic Data Structures

Before diving into the more complicated and customizable structures, we should familiarize ourselves with those basic data structures that are offered by essentially all programming languages in one way or another.

These will be the foundational blocks upon which we'll build more advanced structures.

4.15.1 Statically-Allocated Arrays

Statically-Allocated arrays are a data structure that is equivalent to a list of variables of the same type, and such list is referred by the variable name.

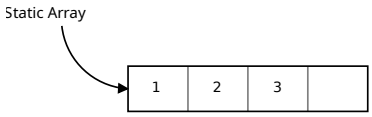


Figure 84: Example of a 4-element integer array

At a lower level a certain amount of memory is allocated, depending on the type of items contained in the array (ints, strings, classes, ...) and how many items such array should contain at most. You can think of it as some machine code that allocates a defined number of "cells" that are contiguous in memory, dedicated to your array.

Since the program needs to allocate memory and the amount of memory allocated depends on how many items are contained in the array, such number of items is defined while coding and changing size is not really possible (there will be solutions in the advanced data structures section).

Given the structure of "adjacent cells", arrays are efficient at being iterated and directly indexed (so you can browse through all the items quickly, or find an item quickly if you know its position), but they are not as efficient in searching its contents (there will be solutions in the advanced data structures section).

4.15.2 Structs

Structs are composite data types that are composed of other, simpler, data types.

Sometimes structs may be mixed up with classes, but the difference usually is that structs have data storage as their objective, while classes may have functions (called "methods") to work on themselves or the data they contain (usually called "fields").

Structs can be used to group together a bunch of variables that are related to each other, without making a class.

Structs is mainly a C/C++ term, the most similar types in other languages are:

- Dicts in Python;
- Tables in Lua (which is also the only complex data type available);
- Objects in JavaScript;

4.15.3 Enums

Enums (or “Enumerated Types”) are types of variable which can take a limited set of values, which are decided by the programmer while writing code.

For instance we can have a “VehicleFuel” Enum that can only take the following values:

- Gasoline
- Diesel
- LPG

Usually Enums are used as constants to make the code more readable and the programmer doesn’t really care how these types are represented in memory (they could be just numbers or something more complex).

Let’s take a very common usage for Enums: removing magic numbers:

Listing 24: A code block featuring some magic numbers

```
if (car.fuel_type === 1) {
    // Gasoline
    gasolinepump.use();
} else if (car.fuel_type === 2) {
    // Diesel
    dieselpump.use();
} else if (car.fuel_type === 3) {
    // LPG
    lpgpump.use();
}
```

If those comments weren’t there, the code would be much harder to understand, and every time we need to work with “fuel types”, we need to remember which “magic number” is which fuel type.

Now let’s see the same code, using Enums.

Listing 25: A code block featuring enums for readability

```
// As of now, Javascript doesn't have native enums, so we use Object freezing to simulate it
const FUEL_TYPES = Object.freeze({
    GASOLINE: 1,
    DIESEL: 2,
    LPG: 3,
});

if (car.fuel_type === FUEL_TYPES.GASOLINE) {
    gasolinepump.use();
} else if (car.fuel_type === FUEL_TYPES.DIESEL) {
    dieselpump.use();
} else if (car.fuel_type === FUEL_TYPES.LPG) {
    lpgpump.use();
}
```

A bit longer, but much easier to read, right? Notice the absence of comments to explain the code, the code explains itself.

4.16 Advanced Data Structures

This section is dedicated to give some basic explanation of some advanced containers and data structures that are used in computer science, allowing us to make an informed choice when we want to implement some even more advanced containers in the future.

Where possible, we will include big-O performance counters for the basic functions of: adding/removing and item at the beginning, adding/removing an item at the end, adding/removing an item in an arbitrary position and indexing at a certain position.

This section is in no way exhaustive, but should be enough to make an informed decision on what containers to use for our components, according to necessities.

Note!

This section will be purely theoretical and many data structures will have no code blocks, this is because implementations vary wildly between programming languages and some of these structures are integrated in such languages.

4.16.1 Graphs

A graph is a data structure that contains a set of vertices (or nodes) which may be connected by a set of edges (or links).

Graphs can be represented in code in two common ways (there are surely other ways to do so): using *adjacency lists* or using *adjacency matrices*.

To explain the two main ways to represent graphs, we will use the following reference image:

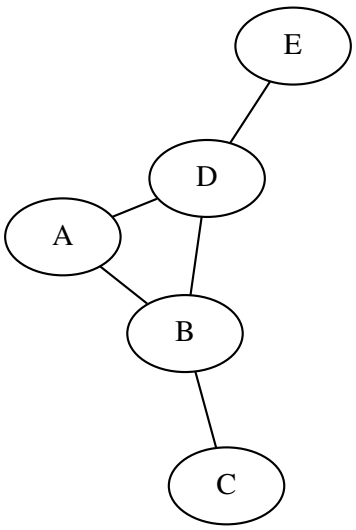


Figure 85: Graphical representation of a simple graph

4.16.1.0.1 Adjacency Lists

Adjacency lists are very simple: they just list the "neighbours" inside a list-like container, every time the graph gets changed, so will the adjacency lists. This method is really flexible and easy to implement. In fact it can be represented in a simple table:

Table 27: A simple adjacency list for our reference image

Node	Adjacency List
A	[B, D]
B	[A, C, D]
C	[B]
D	[A, B, E]
E	[D]

[This section is a work in progress and it will be completed as soon as possible]

4.16.1.0.2 Adjacency Matrices

Another method is to use matrices as a way to store relations between nodes. We have an $n \times n$ matrix (where n is the number of nodes involved) filled with zeros; we put a 1 for every connection that the nodes have (in many conventions, self-loops use the value 2).

Here is an example:

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

If we label the matrix, things are a little bit easier to read:

Table 28: How to read an adjacency matrix

	A	**B**	**C**	**D**	**E**
A	0	1	0	1	0
B	1	0	1	1	0
C	0	1	0	0	0
D	1	1	0	0	1
E	0	0	0	1	0

Note!

For non-directed graphs (like the one in the reference image), adjacency matrices are mirrored on the main diagonal. This may be useful information if you really want to squeeze the last bit of space out of your implementation.

Using the table, we can see that we have a 1 in "row A, column B", which means there is a link "A to B", since there is a 1 in "row B, column A", it means that there is a link "B to A" too. This makes it easy to store single-direction relationships (for Directed Graphs) in a compact way.

[This section is a work in progress and it will be completed as soon as possible]

4.16.2 Trees

When you are a programmer, sooner or later you will have to deal with trees: they are a data structure that represents a hierarchy, using a set of nodes.

Trees can be defined as a "recursive data structure", made up of a node and a bunch of sub-trees connected to it.

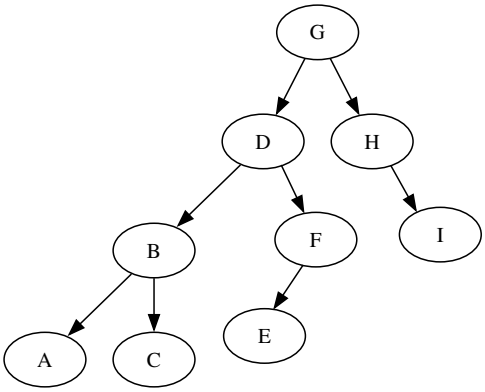


Figure 86: Example of a tree structure

The fact that we can define trees recursively also means that they're a good candidate for all kinds of recursive algorithms, which can help simplifying the code quite a bit.

Trees are the base structure for a lot of other data structures, like heaps and binary search trees.

In this book we will focus mostly on binary trees: trees where each node has at most 2 children.

A possible implementation of a tree could be the following:

Listing 26: A possible implementation of a tree class

```
class Node{
    /*
     * This is an example of a simple node structure for a tree.
     * It can be used as root or any other node
     */

    constructor(String value){
        this.content = value;
        this.left = null;
        this.right = null;
    }
}


function build_example_tree(){
    // Let's build the example tree; starting with the nodes
    let A = Node("A");
    let B = Node("B");
    let C = Node("C");
    let D = Node("D");
    let E = Node("E");
    let F = Node("F");
    let G = Node("G");
    let H = Node("H");
    let I = Node("I");
```

```
26      // Now we connect the various components (the edges)
27      B.left = A;
28      B.right = C;
29      F.left = E;
30      D.left = B;
31      D.right = F;
32      H.right = I;
33      G.left = D;
34      G.right = H;
35      // The tree is ready to be used, let's return the root (G)
36      return G;
37  }
```

4.16.2.1 Depth-first Search

The Depth-first search is a so-called "tree traversal algorithm", which means that it's essentially a way to explore a tree structure. In this case, the algorithm will try to reach the nodes farthest from the root first, before "backtracking" (that means before going "back towards the root").

As said earlier, we will focus on binary trees.

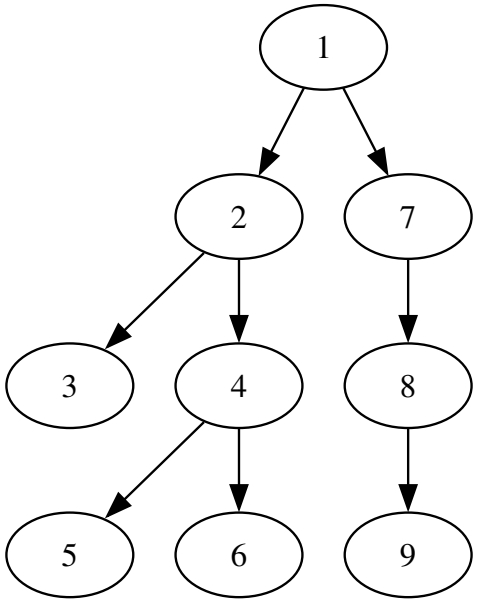


Figure 87: Order in which the nodes are visited during DFS

Depth-first search can be useful in the following situations (as well as others):

- Sorting;
- Maze generation (see the Randomized DFS Method in the Maze generation section);
- Maze solving (which may be useful for Path finding);

The DFS algorithm hides some subtleties, though: the algorithm will "traverse" the tree in the same order, but different implementations will "visit the tree nodes" differently. We will take a look at how nodes are visited now.

In explaining the DFS algorithm, we will refer to the example tree we saw earlier, here it is again:

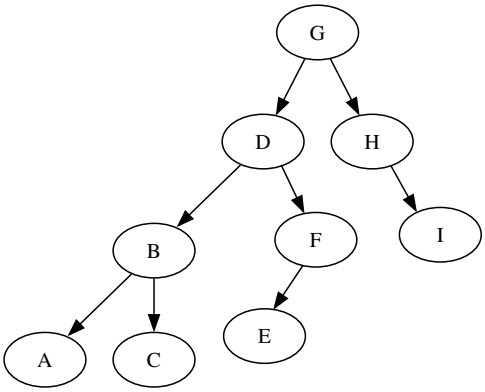


Figure 88: Example tree that will be traversed by DFS

4.16.2.1.1 Pre-order Traversal

The pre-order traversal visits the current node before visiting its children. That means that the algorithm performs the following operations, in order:

1. Visit the current node
2. Recursively traverse the current node's left subtree
3. Recursively traverse the current node's right subtree

If we traverse the example tree with pre-order traversal, and print the visited node, the output will be: `GDBACFEHI`

Here is how an example implementation of a pre-order traversal of a binary tree using DFS would look like:

Listing 27: Pre-order traversal of a tree using DFS

```
function dfs_traverse_preorder(Node n){
    // Step 1: Visit the node, in this case we print its value
    print(n.content);
    // Step 2: We traverse the left subtree, using recursion;
    if (n.left != null){
        dfs_traverse_preorder(n.left);
    }
    // Step 3: We traverse the right subtre, using recursion;
    if (n.right != null){
        dfs_traverse_preorder(n.right);
    }
}

function main(){
    let root = build_example_tree();
    dfs_traverse_preorder(root);
```

```
17 }
```

4.16.2.1.2 In-order Traversal

The in-order traversal visits the tree "from left to right", by prioritizing the traversal of the left subtrees before visiting the current node. That means that the algorithm performs the following operations, in order:

1. Recursively traverse the current node's left subtree
2. Visit the current node
3. Recursively traverse the current node's right subtree

If we traverse the example tree with in-order traversal, and print the visited node, the output will be: `ABCDEFGHI`

Notice how in this case, the output is ordered. This is because the example tree is a special kind of tree, called a "binary search tree". We will see more in the dedicated paragraph.

Here is how an example implementation of a in-order traversal of a binary tree using DFS would look like:

Listing 28: In-order traversal of a tree using DFS

```
1  function dfs_traverse_inorder(n){
2      // Step 1: We traverse the left subtree, using recursion;
3      if (n.left != null){
4          dfs_traverse_inorder(n.left);
5      }
6      // Step 2: Visit the node, in this case we print its value
7      console.log(n.content);
8      // Step 3: We traverse the right subtre, using recursion;
9      if (n.right != null){
10         dfs_traverse_inorder(n.right);
11     }
12 }
13
14 function main(){
15     let root = build_example_tree();
16     dfs_traverse_inorder(root);
17 }
```

4.16.2.1.3 Post-order Traversal

The post-order traveral method prioritizes traversing both the children to visiting the current node, thus it will perform the following operations:

1. Recursively traverse the current node's left subtree
2. Recursively traverse the current node's right subtree
3. Visit the current node

If we traverse the example tree with post-order traversal, and print the visited node, the output will be: `ACBEFDIHG`

Here is how an example implementation of a post-order traversal of a binary tree using DFS would look like:

Listing 29: Post-order traversal of a tree using DFS

```
function dfs_traverse_postorder(Node n){
    // Step 1: We traverse the left subtree, using recursion;
    if (n.left != null){
        dfs_traverse_postorder(n.left);
    }
    // Step 2: We traverse the right subtre, using recursion;
    if (n.right != null){
        dfs_traverse_postorder(n.right);
    }
    // Step 3: Visit the node, in this case we print its value
    print(n.content);
}

function main(){
    let root = build_example_tree();
    dfs_traverse_postorder(root);
}
```

4.16.2.1.4 Reverse Traversals

These kinds of traversals are essentially the same of the ones we've already seen, but the right subtree is given priority over the left. Here are the operations, listed for reference.

Reverse Pre-Order:

1. Visit the current node
2. Recursively traverse the current node's right subtree
3. Recursively traverse the current node's left subtree

Reverse In-Order:

1. Recursively traverse the current node's right subtree
2. Visit the current node
3. Recursively traverse the current node's left subtree

Reverse Post-Order:

1. Recursively traverse the current node's right subtree
2. Recursively traverse the current node's left subtree
3. Visit the current node

The code will be omitted, since it is easy to infer how the code would look, given the previous examples.

4.16.2.2 Breadth-first search

Breadth-first search, or BFS, uses a concept that is opposite of the one in DFS (Depth-first search): instead of going as deep as possible inside the tree, this algorithm prefers exploring "in layers".

The root will be visited first, then all its children, after that all its nephews, etc...

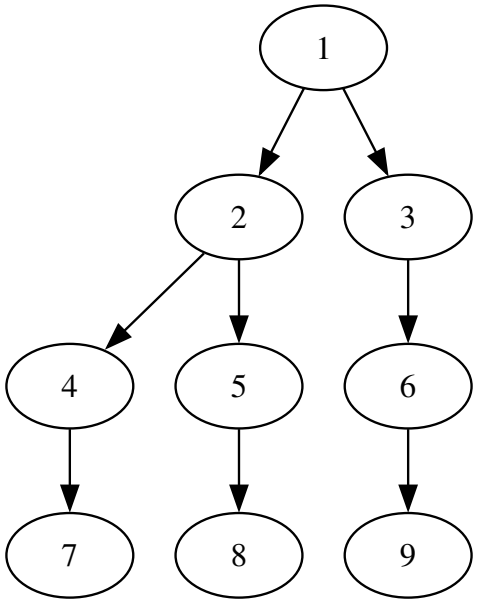


Figure 89: Order in which the nodes are visited during BFS

In the implementation shown here, the steps are the following:

1. Make a queue and enqueue the root
2. If the queue is not empty, take the first node, if it is empty, just stop
3. If such node has any children, enqueue them, in order
4. Visit the node
5. Go back to point 2

A possible implementation of a BFS algorithm could be the following:

Listing 30: Traversal of a tree using BFS

```
function traverse_bfs(root){
    // We will use a queue for this algorithm
    let q = [];
    // First thing, we enqueue the root
    q.push(root);
    // Now comes the iterative part. This will keep going until
    // the tree is completely explored.
    while (q.length != 0){
        // We take the first node in the queue
        let n = q.shift();
```

```
11          // We enqueue its children, if they exist
12          if (n.left != null){
13              q.push(n.left);
14          }
15          if (n.right != null){
16              q.push(n.right);
17          }
18          // Now we visit the current node
19          console.log(n.content);
20          // The loop will continue with the next node in the layer,
21          // automatically start the next layer, or stop because there
22          // are no more nodes to visit.
23      }
24  }
```

BFS is a great algorithm to solve mazes and find the shortest path between two nodes, making it a good choice for Path finding.

4.16.3 Dynamic Arrays

In many languages, arrays are sized statically, with a size decided at compile time. This severely limits the array's usefulness.

Dynamic Arrays are a wrapper around arrays, allowing it to extend its size when needed. This usually entails some additional operations when inserting or deleting an item.

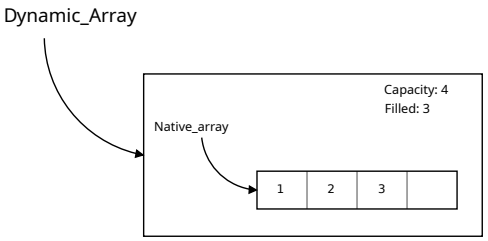


Figure 90: Dynamic Arrays Reference Image

4.16.3.1 Performance Analysis

Indexing an item is immediate, since arrays allow to natively index themselves.

Inserting an item at the beginning is a heavy task, since it requires either moving all the present items or rebuilding the internal native array. Such operations require copying or moving each element, giving us a time complexity averaging on `O(n)`.

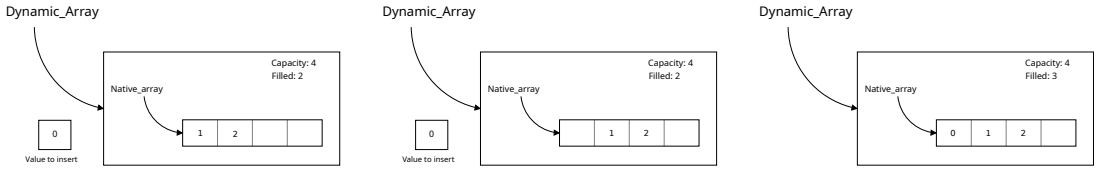


Figure 91: Adding an element at the beginning of a Dynamic Array

Inserting an item at the end, if we keep a pointer to the last item inserted, is an operation that usually happens immediately (time complexity `O(1)`), but when the array is full, we need to instantiate a new native array (usually double the size of the current one) and copy all elements inside the new array (operation that has time complexity of `O(n)`). Since the number of `O(1)` operations outweighs by a long shot the number of `O(n)` operations, it's possible to demonstrate that in the long run appending an item at the end of a dynamic array has a time complexity averaging around `O(1)`.

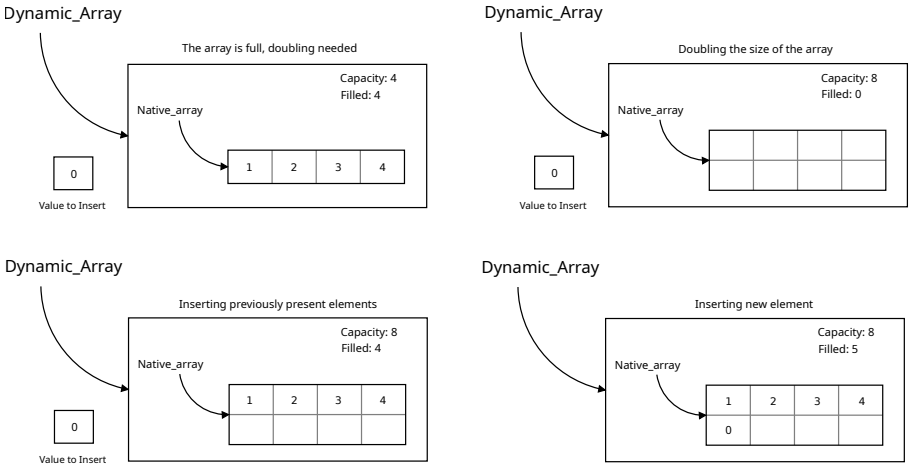


Figure 92: Adding an element at the end of a Dynamic Array

Inserting an item in an arbitrary position, much like inserting an item at the beginning requires moving some items further into the array, potentially all of them (when the arbitrary position is the beginning of the array), thus giving us a time complexity of `O(n)`. Such operation could trigger an array resize, which has no real influence on the estimate.

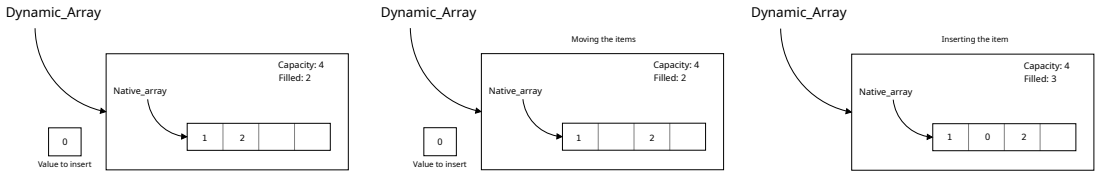


Figure 93: Adding an element at an arbitrary position of a Dynamic Array

Some implementations of the Dynamic Arrays try to save space when the number of items goes lower than $\frac{1}{4}$ of the array capacity during a deletion, the internal array is rebuilt with half the size. Such operation has a time complexity of `O(n)`.

Some other implementations use a $\frac{1}{4}/\frac{3}{4}$ rule, halving the array capacity when the item deletion brings the number of items lower than $\frac{1}{4}$ of the array and doubling it when an insertion makes the number of elements higher than $\frac{3}{4}$ of the array capacity.

Note!

Not all programming languages have native support for arrays, for instance Python normally uses lists (although it supports arrays via the `array` standard library).

Table 29: Performance table for Dynamic Arrays

Operation	Average Cost
Indexing	O(1)
Insert/Delete At Beginning	O(n)
Insert/Delete At End	O(1) amortized
Insert/Delete at position	O(n)

Table 30: Summary Table for Dynamic Arrays

Container Name	Dynamic Array
When To Use it	All situations that require direct indexing of a container, but insertions and removals are not extremely common, and usually take the form of "push back" (insertion at the end)
Advantages	Direct Indexing, Fast iteration through all the elements, given by the fact that arrays are stored compact in memory, fast appending.
Disadvantages	Slow insertions in arbitrary positions and at the head of the array.

4.16.4 Linked Lists

Linked Lists are a data structure composed by "nodes", each node contains data and a reference to the next node in the linked list. Differently from arrays, nodes may not be contiguous in memory, which makes indexing problematic.

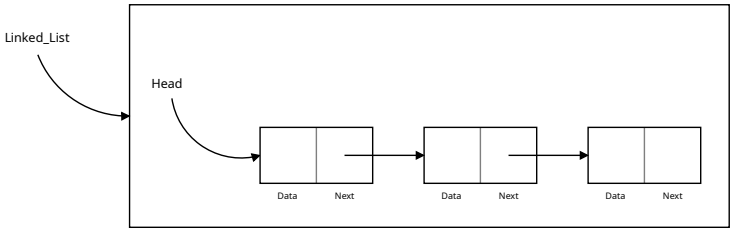


Figure 94: Linked List Reference Image

Some implementations feature a pointer to the last element of the list, to make appending items at the end easier and quicker.

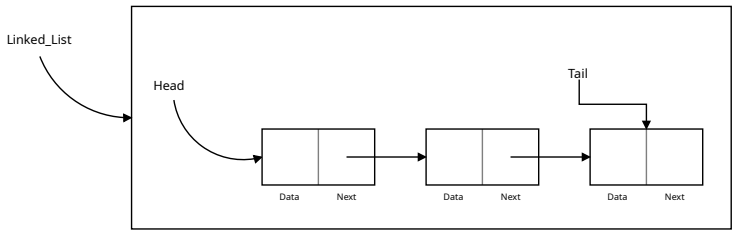


Figure 95: Double-Ended Linked List Reference Image

4.16.4.1 Performance Analysis

Since we only have a handler on the first node, indexing requires us to scan all the elements until we reach the one that was asked for. This operation has a potential time complexity of `O(n)`.

Inserting an item at the beginning is immediate, we just need to create a new node, make it point at the current head of the list and then update our "handle" to point at the newly created node. The number of operations is independent of how many data we already have, so the time complexity is `O(1)`.

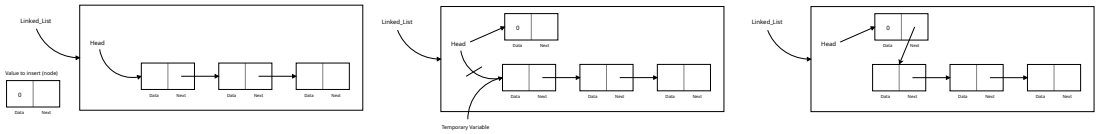


Figure 96: Inserting a new node at the beginning of a linked list

Appending an item at the end has a time complexity that varies depending on the chosen implementation: if the list has a reference to the final node, we just need to create a new node, update the final node's reference (usually called "next") to point at the new node and then update the reference to the final node to point at the newly created node (time complexity `O(1)`). If our queue doesn't have such reference, we will need to scan the whole list to find the final node (time complexity `O(n)`).

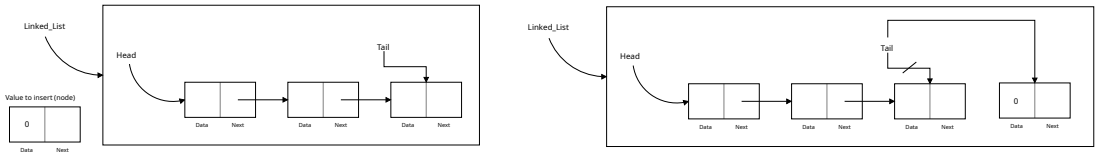


Figure 97: Inserting a new node at the end of a (double-ended) linked list

Inserting at an arbitrary position requires us to scan the list until we find the position that we want, after that we just need to split and rebuild the references correctly, which is a fast operation.

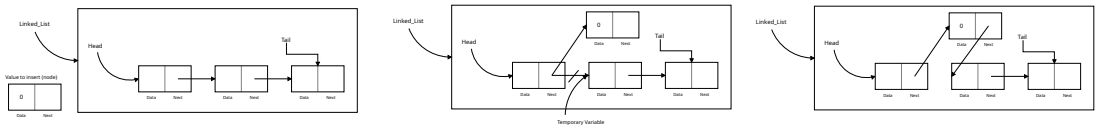


Figure 98: Inserting a new node at an arbitrary position in a (double-ended) linked list

Table 31: Performance table for Linked Lists

Operation	Average Cost
Indexing	O(n)
Insert/Delete At Beginning	O(1)
Insert/Delete At End	O(1) for double-ended, o(n) otherwise
Insert/Delete at position	time to search + O(1)

Table 32: Summary Table for Linked Lists

Container Name	Linked List
When To Use it	All situations that require quick insertions/removals, either on the head or the tail (used as stacks or queues).
Advantages	Very fast insertions/removals, quite fast iteration through all the elements.
Disadvantages	Slow indexing at an arbitrary position. Sorting can be complex.

4.16.5 Doubly-Linked Lists

A doubly-linked list is a variation of a linked list where each node not only has a reference to its successor, but also a reference to its predecessor. This allows for easy processing of the list in reverse, without having to create algorithms that entail a huge overhead.

All the operations of insertion, indexing and deletion are performed in a similar fashion to the classic singly-linked list we saw earlier, just with an additional pointer to account for.

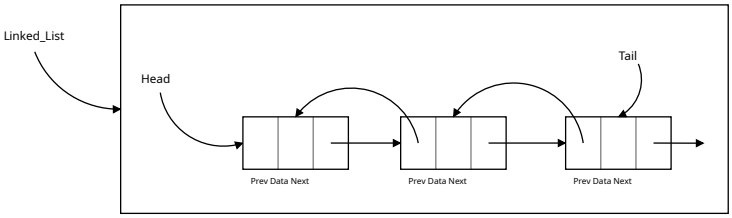


Figure 99: Doubly Linked List Reference Image

Table 33: Performance table for Doubly-Linked Lists

Operation	Average Cost
Indexing	O(n)
Insert/Delete At Beginning	O(1)
Insert/Delete At End	O(1)
Insert/Delete at position	time to search + O(1)

Table 34: Summary Table for Linked Lists

Container Name	Doubly-Linked List
When To Use it	All situations that require quick insertions/removals, either on the head or the tail (stacks or queues) or iterating through an entire list, forwards or backwards.
Advantages	Very fast insertions/removals, quite fast iteration through all the elements. Possibility of easily iterating the list in reverse order.
Disadvantages	Slow indexing at an arbitrary position. Sorting can be complex.

4.16.6 Hash Tables

Hash Tables are a good way to store **unordered data** that can be referred by a "key". These structures have different names, like "maps", "dictionaries" or "hash maps".

The idea behind a hash map is having a key subject to a *hash function*$_{[g]}$ that will decide where the item will be positioned in the internal structure.

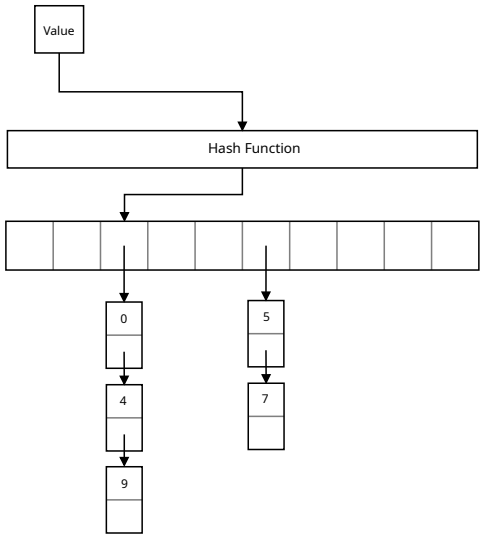


Figure 100: Hash Table Reference Image (Hash Table with Buckets)

The simplest way to implement a hash table is using an "array with buckets": an array where each cell has a reference to a linked list.

On average, finding an item requires passing the key through the hash function, such hash function will tell us where the item is in our internal structure immediately. Thus giving a time complexity of $O(1)$.

Inserting has more or less the same performance, the key gets worked through the hash function, deciding which linked list will be used to store the item.

Deletion works in the same fashion, passing the key through the hash function and then deleting the value; giving a time complexity of $O(1)$

Table 35: Performance table for Hash Tables

Operation	Average Cost
Searching	O(1)
Insert	O(1)
Delete	O(1)

Table 36: Summary Table for Hash Tables

Container Name	Hash Table
When To Use it	All situations that require accessing an element by a well-defined key quickly. Building unordered data sets.
Advantages	Fast insertions/removals, direct indexing (in absence of hash collisions) by key.
Disadvantages	In case of a bad hashing function, it reverts to the performance of a linked list, cannot be ordered.

4.16.7 Binary Search Trees (BST)

Binary search trees, sometimes called "ordered trees" are a container that have an "order relation" between their own elements.

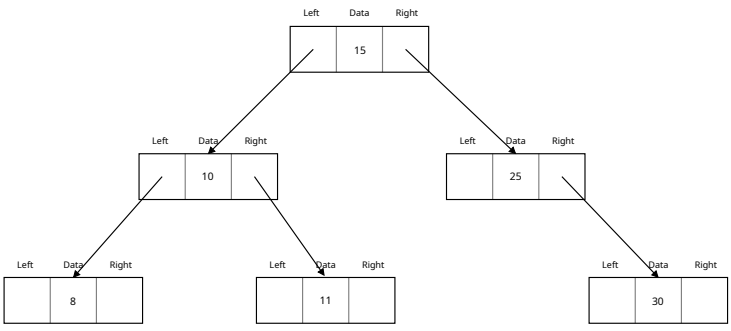


Figure 101: Binary Search Tree Reference

The order relation allows us to have a tree that is able to distinguish between “bigger” and “smaller” values, thus making search really fast at the price of a tiny slowdown in insertion and deletion.

Searching in a BST is easy, starting from the root, we check if the current node is the searched value; if it isn’t we compare the current node’s value with the searched value.

If the searched value is greater, we search on the right child. If it is smaller, we continue our search on the left child.

Recursively executing this algorithm will lead us to find the node, if present. Such algorithm has a $O(log(n))$ time complexity.

In a similar fashion, insertion will recursively check subtrees until the right spot of the value is found. The insertion operation has the same time complexity as searching: $O(log(n))$.

Deletion is a bit more conceptually complex, since it’s necessary to maintain the ordering of the nodes. Such operation has a time complexity of $O(log(n))$.

Table 37: Performance table for Binary Search Trees

Operation	Average Cost
Searching	O(log(n))
Insert	O(log(n))
Delete	O(log(n))

Table 38: Summary Table for Binary Search Trees

Container Name	Binary Search Tree
When To Use it	Situations that require good overall performance and requires fast search times.
Advantages	Good insertion and removal times, searching on this structure is fast.
Disadvantages	Given the nature of the data structure, there is no direct indexing, nor ordering.

4.16.8 Heaps

Heaps are a tree-based data structure where we struggle to keep a so-called “heap property”. The heap property defines the type of heap that we are using:

- **Max-Heap:** For each node `N` and its parent node `P`, we’ll always have that the value of `P` is always greater or equal than the value of `N`;
- **Min-Heap:** For each node `N` and its parent node `P`, we’ll always have that the value of `P` is always less or equal than the value of `N`;

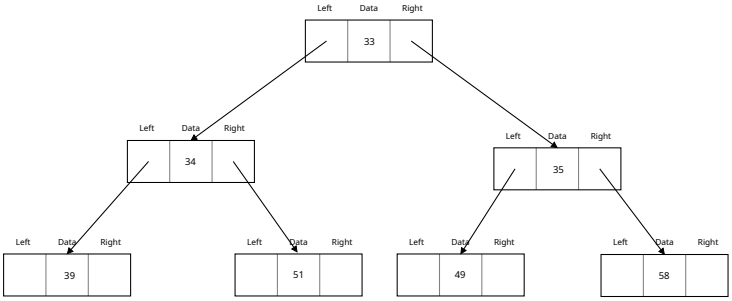


Figure 102: Heap Reference Image (Min-Heap)

Heaps are one of the maximally efficient implementation of priority queues, since the highest (or lowest) priority item is stored in the root and can be found in constant time.

Table 39: Performance table for Heaps

Operation	Average Cost
Find Minimum	$O(1)$ to $O(log(n))$, depending on the implementation
Remove Minimum	$O(log(n))$
Insert	$\Theta(1)$ to $O(log(n))$ depending on the implementation

Table 40: Summary Table for Heaps

Container Name	Heap
When To Use it	All situations where you require to find and/or extract the minimum or maximum value in a container quickly; like priority queues.
Advantages	Good general time complexity, maximum performance when used as priority queues.
Disadvantages	No inherent ordering, there are better solutions for general use.

4.16.9 Stacks

Stacks are a particular data structure, they have a limited way of working: you can only put or remove items on top of the stack, plus being able to “peek” on top of the stack.

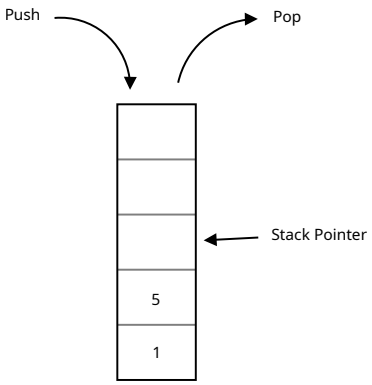


Figure 103: How a stack works

Stacks are LIFO (Last in - First Out) data structures, and can be implemented with both a linked list or a cleverly-indexed array.

Depending on the single implementation, the operation used to “pop” an item from the stack will also return the element, ready to be used in an upcoming computation.

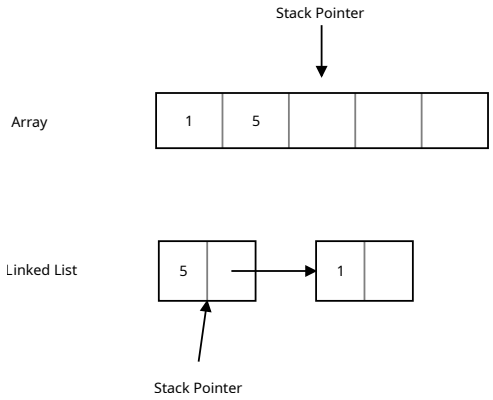


Figure 104: Array and linked list implementations of a stack

4.16.10 Queues

Queues are the exact opposite of stacks, they are FIFO (First in - First Out) data structures: you can put items on the back of the queue, while you can remove from the head of the queue.

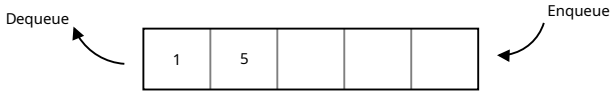


Figure 105: How a queue works

Depending on the single implementation, the operation used to “dequeue” an item from the queue will also return the element just removed, ready to be used in an upcoming computation.

As with stacks, queues leverage limitations in their way of working for greater control over the structure itself.

Usually queues are implemented via linked lists, but can also be implemented via arrays, using multiple indexes and index-wrapping when iterating.

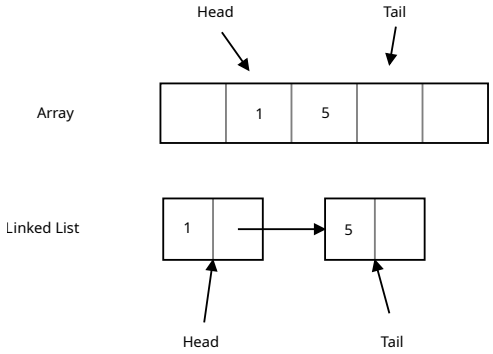


Figure 106: Array and linked list implementation of a queue

4.16.11 Circular Queues

Circular Queues are a particular kind of queues that are infinitely iterable, every time an iterator goes after the last element in the queue, it will wrap around to the beginning.

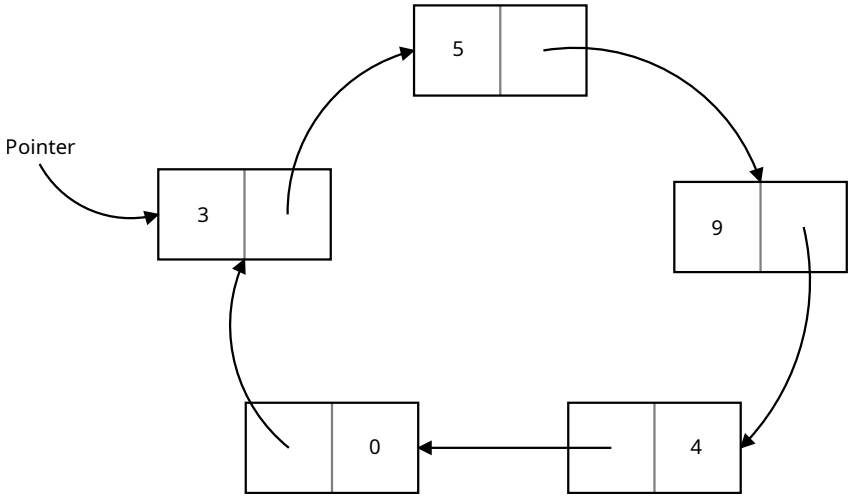


Figure 107: How a circular queue works

Circular Queues can be implemented via linked lists or cleverly indexed arrays, with all the advantages and disadvantages that such structures entail.

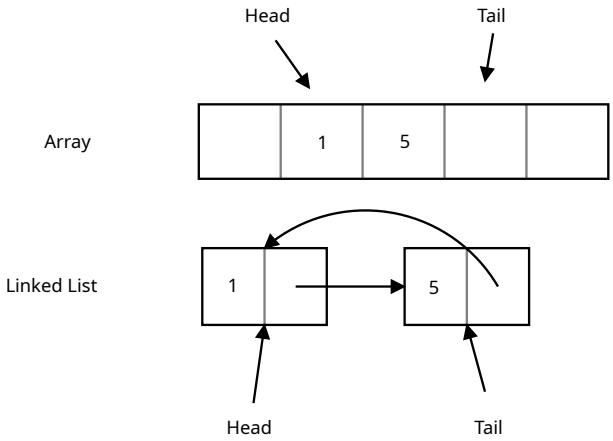


Figure 108: Array and linked list implementation of a circular queue

4.17 Equality vs. Identity

In many programming languages there is a difference between things that are “equal” and things that are “identical”.

This can translate in different operators in each programming language, for instance:

- `==` (equality) versus `is` (identity) in Python;
- `==` (equality) versus `===` (identity) in JavaScript;

So it is important to remember the difference.

Equality checks if two objects are “equal”, given a certain set of rules. For instance, two circles may be considered “equal” if they have the same values for center and radius.

Identity checks is two objects are actually the same object (usually by checking if the pointers refer to the same region in memory).

Let’s take the following code:

Listing 31: Checking identity vs. checking equality

```python
from typing import Self

class Circle:
    center: tuple[int, int] = (0, 0)
    radius: int = 0

    def __init__(self, center: int, radius: int):
        self.center = center
        self.radius = radius

    def __eq__(self, other: Self):
        """
        Here we apply the rules to decide if two circles are equal
        """
```

```
15          return (
16              self.center == other.center
17              and self.radius == other.radius
18          )
19
20
21  if __name__ == "__main__":
22      # Let's create two circles
23      A: Circle = Circle((10, 20), 30)
24      B: Circle = Circle((10, 20), 30)
25      # Let's check if they're equal
26      if A == B:
27          print("A and B are equal")
28      else:
29          print("A and B are different")
30      # Let's check for identity
31      if A is B:
32          print("A and B are identical")
33      else:
34          print("A and B are not identical")
35      # Let's print the memory ids
36      print("id(A) = ", id(A))
37      print("id(B) = ", id(B))
```

It creates a set of rules to define if two circles are equal (same center and radius), then it creates two circles with the same parameters and checks for equality and identity.

The result of running the code is the following:

```
penaz@PenazMW2 ~ python circles.py
A and B are equal
A and B are not identical
id(A) =  139582710072144
id(B) =  139582710031296
```

Figure 109: The result of running the identity vs equality code

As we can see, circles `A` and `B` are considered “equal”, according to the rules we have set, but they are not the same object (we have called the constructor twice), and thus the identity check fails.

4.18 Truthiness and “Falsiness”

Some programming languages try to make themselves more readable by shortcutting some boolean condition, using so-called “truthy” and “falsy” values.

These values are automatically converted into a boolean according to some rules that have been set by the language itself.

For instance some “falsy” values can be:

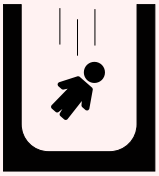
- The `false` keyword (obviously);
- The number `0` (zero);
- The empty string;
- The empty list (in Python, for instance);
- The language's "NULL" value (`None` for Python, `null` for JS, ...);

while some "truthy" values can be:

- The `true` keyword (again, obviously);
- Any number that is not zero;
- Any string with at least one character;
- A list with at least one element (in Python, for instance);
- Any object that is "non-NULL".

This allows us to write code that is a little bit more terse, improving readability.

Pitfall Warning!



Keep in mind the difference between equality and identity when dealing with "truthy" and "falsy" values. In some cases, terser code (that leverages the so-called "type coercion") might hide some corner cases that result in bugs.

4.19 Operators are functions too

In most programming languages operators are just another way to write a function. Something like `a = 2+2` could be written as `a=operator_add(2, 2)`.

This also means that comparison operators are functions, there's essentially no semantic difference between `a == 2` and `equals(a, 2)`. This has some interesting consequences.

If you start thinking of comparison operators as functions, you may ask yourself:

> If comparison operators are functions, can I assign their result to a variable?

And the answer is (most of the time) yes!

You will see this all throughout the book: assigning a complicated chain of comparisons into a variable and then using such variable in an if statement, or using a comparison statement in a return statement.

Here's an example:

Listing 32: Using operators as functions

```
// Operators can be treated as functions, that means you can
// Assign them to a variable.

```

```
// This...

if (a == b && c == d){
    // Do something...
}

// Is equivalent to this

let complex_condition = (a == b && c == d);

if (complex_condition){
    // ...
}

// ---------------8<---------------

// Also this...

function thing(a, b){
    if (a == b){
        return true;
    }else{
        return false;
    }
}

// Is equivalent to...

function thing(a, b){
    return a == b;
}
```

4.20 The principle of locality

This is one of the most-talked about principles in computer science: it usually refers to the principle of "memory locality", but it may also refer to other kinds, like "temporal locality" or "branch locality".

Let's analyze the most common ones.

- **Spatial locality:** (sometimes called "memory locality") if a certain region of storage (or memory) is referenced, there is a good probability that nearby regions of storage (or memory) will be referenced in the near future. This is true, for instance, in Arrays.
- **Temporal locality:** if a certain region of storage (or memory) is referenced, there is a good probability that the same region will be referenced in the near future. CPU caches leverage this principle by copying recently-used data into faster storage.

Temporal locality can be seen as a special case of spatial locality.

4.21 Treating multidimensional structures like one-dimensional ones

This is usually done when dealing with pointers, but we may need to use some math to deal with sprites and animations too.

As we'll see in the Sprite sheets section, it is more efficient to store sprites and animation frames in sprite sheets.

When dealing with frames of animation, we like our frames to be "one-dimensional", each "cell" represents a certain "time".

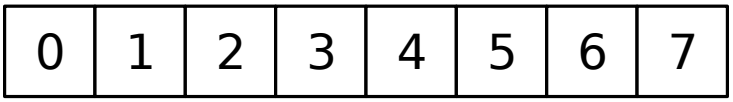


Figure 110: The "easy way" of dealing with frames

When dealing with sprite sheets, we may find that our animation has frames saved in a "matrix" of some sort, like so:

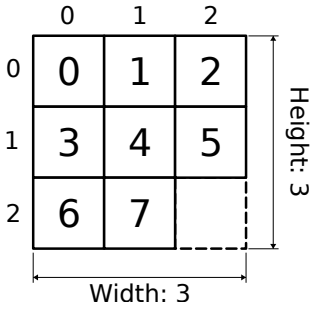


Figure 111: A sample sprite sheet with the same frames as before

The images we've just seen will help you understand how the following formulas work.

To convert from 2-dimensional $(row, column)$ coordinates to a single index, the formula is:

$$index = width \times row + column$$

Note!

Remember that in many programming languages arrays and similar structures are 0-indexed. This is the system that will be used here.
If you're using a language that indexes arrays starting from 1 (like Lua), these formulas need to be changed a bit.

So if I want to know the index of the 3rd element of the second row, with index (2,1), the formula becomes:

$$index = 3 \times 1 + 2 = 5$$

The inverse formula is the following:

$$\begin{cases} row = \lfloor \frac{index}{width} \rfloor \\ column = index \% width \end{cases}$$

So if we wanted to know the (row,column) position of the frame with index 7 we would have:

$$\begin{cases} row = \lfloor \frac{7}{3} \rfloor = \lfloor 2.33333 \rfloor = 2 \\ column = 7 \% 3 = 1 \end{cases}$$

Note!

This can be done with structures with n dimensions, but the formula becomes a lot more complex the more dimensions you add. We'll stop at 2 for now.

4.22 Data Redundancy

When dealing with certain structures, there are operations that are inherently complex to do: let's take for example counting the elements in a list:

Listing 33: Counting the elements in a list

```
class List{
    constructor(){
        this.nodeList = null;
    }
    // ...
    getLength(){
        let counter = 0;
        for (const item in this.nodeList){
            counter = counter + 1;
        }
        return counter;
    }
}
```

It's easy to see that an algorithm like this has a $\Theta(n)$ complexity, which may not be ideal for an operation as common as finding the length of a list.

This is where data redundancy comes into play: the length of a list is an intrinsic property of the list itself, so why not save it inside the "head" of our structure?

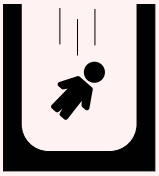
This will obviously require a bit more work in all the methods that will change the number of elements inside the list, since we need to keep the "length" property in sync with the actual length of the list, but in exchange we can count the elements in a list by doing a simple lookup.

Let's see an example implementation:

Listing 34: Counting the elements in a list with data redundancy

```
class List{
    constructor(){
        this.nodeList = null;
        this.length = 0;
    }
    // ...
    getLength(){
        return length;
    }

    addItem(node){
        // ... Normal operation ...
        // ...
        // We update our length counter
        this.length = this.length + 1;
    }

    removeItem(node){
        // ... Normal removal operation ...
        // ...
        // We update our length counter
        this.length = this.length - 1;
    }

    clear(){
        // ... Normal clear operation ...
        // ...
        // We clear the length too
        this.length = 0;
    }

    // ...
}
```

Pitfall Warning!



It is extremely important that we keep our "redundant properties" synchronized with the actual state of our objects, even when exceptions are raised. Not doing so will create bugs.

Let's consider another example: we have a standard linked list, like the one that follows:

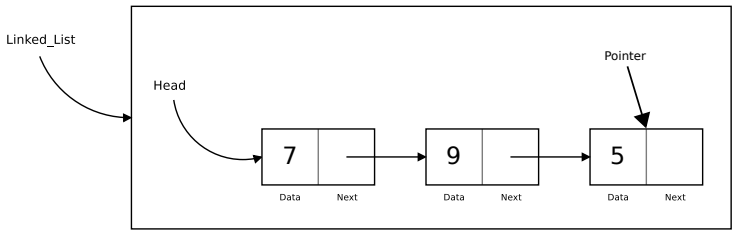


Figure 112: Singly-Linked List has no redundancy

Our "pointer" is pointing the node containing the number "5", and now we want to know the value of the node that precedes it. To do that we need to start from the head, saving in a temporary variable our nodes, until we find the node pointed by our "pointer".

Listing 35: Finding the previous element in a singly linked list

```
function get_previous_node(lst, current_node){
    let pointer = lst.head;
    let previous = null;
    while (pointer != current_node){
        previous = pointer;
        pointer = pointer.next;
    }
    return previous;
}
```

This operation has $O(n)$ complexity, which is not great. If we wanted to print a list in reverse with such technique, the situation would be even worse.

Doubly-linked lists are another example of data redundancy. We are saving the content of the "previous" node, so that we can do a simple lookup with complexity $O(1)$ and easily (and efficiently) do our "reverse printing".

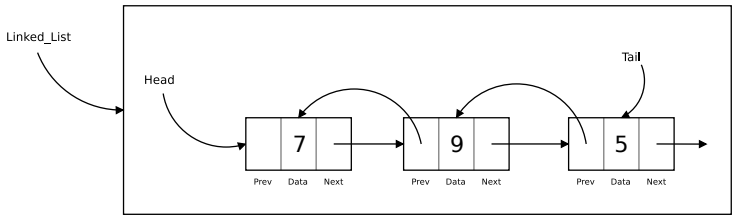


Figure 113: A doubly linked list is an example of redundancy

4.23 Introduction to Multi-Tasking

When it comes to humans, we are used to have everything at our disposal immediately, but when it comes to computers, each processing unit (CPU) is usually able to perform only one task at a time.

To allow for multi-tasking (doing many activities at once), the CPU switches between tasks at high speeds, giving us the illusion that many things are happening at once. There are many methods to ensure multi-tasking without *process starvation*[g], the most used is *pre-emption*[g] where there are forced context switches between processes, to avoid one hogging the CPU.

4.23.1 Multi-Threading vs Multi-Processing

Sometimes Multi-Threading and Multi-Processing are used interchangeably, but this is actually not correct. Let's see the differences between the two terms and how they contribute (in different ways) to allow multi-tasking.

Multi-Processing is a practice that makes use of multiple CPUs inside the same machine, this allows to process CPU-intensive calculations in a parallel manner, thus gaining performance in our software. This style of parallelization is usually done by spawning multiple processes, each of which will be run on a different CPU (or Core).

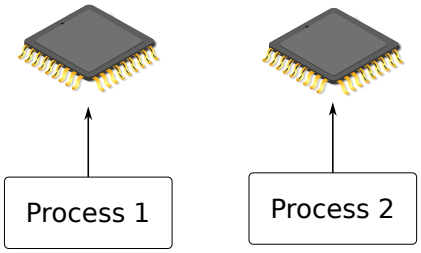


Figure 114: In a multi-processing environment, each CPU takes care of a task

Multi-Processing has some disadvantages: creating a process can be quite expensive and thus give us some tangible overhead if processes are created and destroyed often.

Multi-Threading is a programming practice that allows to run different "lines of execution" (called "threads"), inside of the same parent process, so to achieve the maximum possible CPU utilization.

Multi-Threading has the advantage of lower overhead, since threads are quite cheap to create, but also has some more limitations when the tasks to execute are "CPU-bound" (take a lot of CPU time).

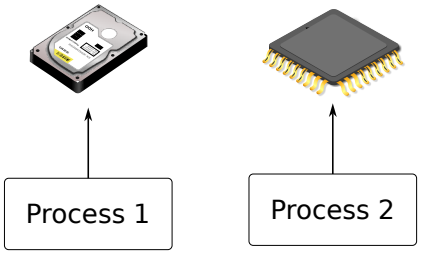


Figure 115: In multi-threading, the CPU uses I/O wait time to take care of another task

Multi-Threading works well when the threads are "I/O bound" (they use network or disk a lot, while the CPU usage is low), this means essentially that while one thread is waiting for I/O (like loading an asset), another thread can

perform other calculations on the CPU instead of just “wait for the I/O to finish”.

4.23.2 Coroutines

If you search for the word “coroutine” online, you will find a lot of extremely convoluted explanations involving the knowledge of the difference between *preemptive*[g] and *non-preemptive* multitasking, subroutines, threads and lots more. Let’s try to make sense of this.

First of all, coroutines are computer programs can run in multitasking (so it can run separated from our main game loop) which are used in non-preemptive multitasking. Differently from the preemptive style defined in the glossary, in non preemptive multitasking the operating system never forces a context switch, but it’s the coroutine’s job to **yield** the control over to another function.

Instead of “fighting for resources”, coroutines politely free the processor and give control of it to something else (could be the caller or another coroutine), this form of multitasking is often called **cooperative multitasking**.

A particularly interesting point of coroutines is the fact that their execution can be “suspended” and “resumed” without losing its internal state. Coroutines are used in more advanced engines (using the *Actor Model*) and in some particular situations. You may never need to use a single coroutine, or you may need to use them every day, so it’s worth knowing at least what they are.

4.24 Introduction to Multi-Threading

When it comes to games and software, we usually think of it as a single line of execution, branching to (not really) infinite possibilities; but when it comes to games, we may need to dip our toes into the world of multi-threaded applications.

4.24.1 What is Multi-Threading

Multi-Threading means that multiple threads exist in the context of a single process, each thread has an independent line of execution but all the threads share the process resources.

In a game, we have the “Game Process”, which can contain different threads, like:

- World Update Thread
- Rendering Thread
- Loading Thread
- …

Multi-Threading is also useful when we want to perform concurrent execution of activities.

4.24.2 Why Multi-Threading?

Many people think of Multi-Threading as “parallel execution” of tasks that leads to faster performance. That is not always the case. Sometimes Multi-Threading is used to simplify data sharing between flows of execution, other times threads guarantee lower latency, other times again we may *need* threads to get things working at all.

For instance let's take a loading screen: in a single-threaded application, we are endlessly looping in the input-update-draw cycle, but what if the "update" part of the cycle is used to load resources from a slow storage media like a Hard Disk or even worse, an optical disk drive?

The update function will keep running until all the resources are loaded, the game loop is stuck and no drawing will be executed until the loading has finished. The game is essentially hung, frozen and your operating system may even ask you to terminate it. In this case we need the main game loop to keep going, while something else takes care of loading the resources.

4.24.3 Thread Safety

Threads and concurrent execution are powerful tools in our "programmer's toolbox", but as with all powers, it has its own drawbacks.

4.24.3.1 Race conditions

Imagine a simple situation like the following: we have two threads and one shared variable.

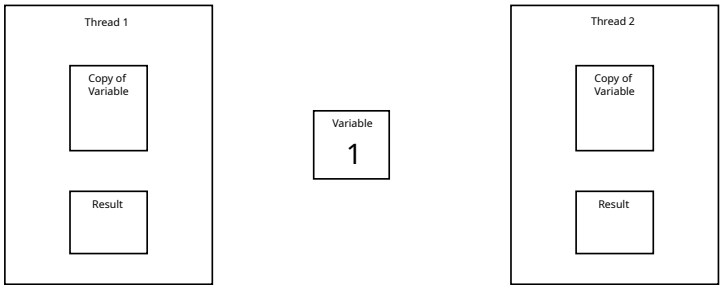


Figure 116: Two threads and a shared variable

Both threads are very simple in their execution: they read the value of our variable, add 1 and then write the result in the same variable.

This seems simple enough for us humans, but there is a situation that can be really harmful: let's see, in the following example each thread will be executed only once. So the final result, given the example, should be "3".

First of all, let's say Thread 1 starts its execution and reads the variable value.

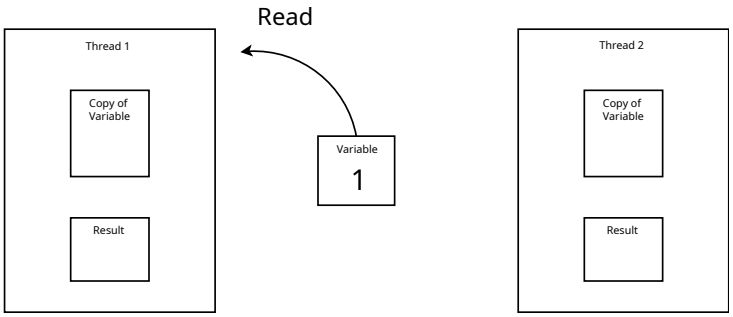


Figure 117: Thread 1 reads the variable

Now, while Thread 1 is calculating the result, Thread 2 (which is totally unrelated to Thread 1) starts its execution and reads the variable.

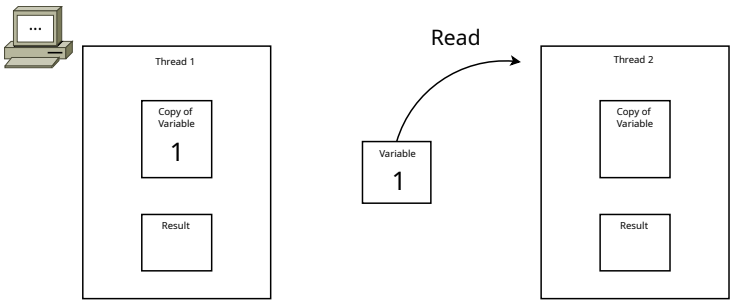


Figure 118: While Thread 1 is working, Thread 2 reads the variable

Now Thread 1 is finishing its calculation and writes the result into the variable.

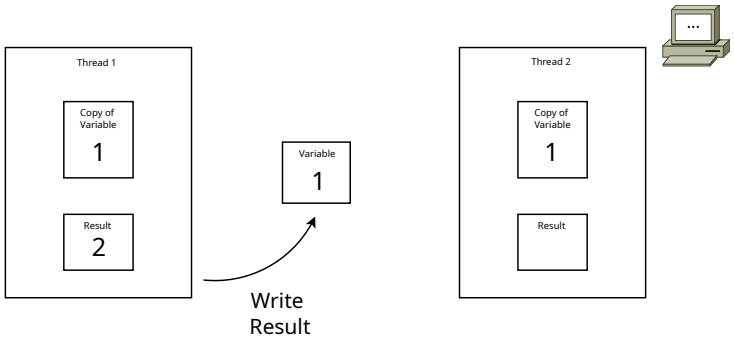


Figure 119: Thread 1 writes the variable

After That, Thread 2 finishes its calculation too, and writes the result into the variable too.

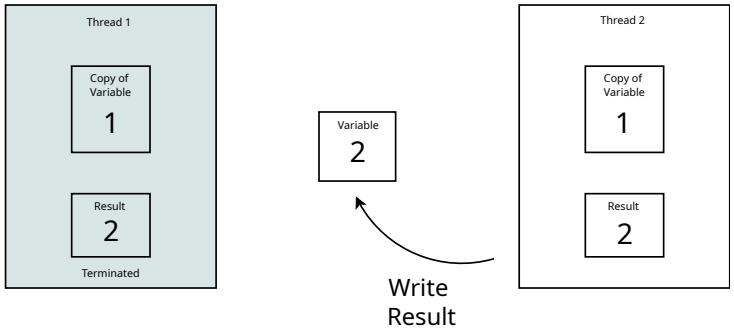


Figure 120: Thread 2 writes the variable

Something is not right, the result should be “3”, but it’s “2” instead.

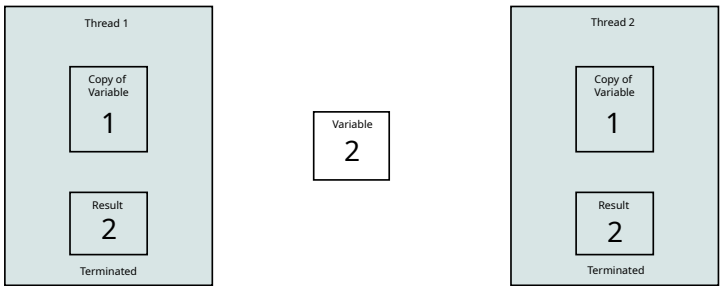


Figure 121: Both Threads Terminated

We just experienced what is called a **"race condition"**: there is no real order in accessing the shared variable, so things get messy and the result is not deterministic. We don't have any guarantee that the result will be right all the time (or wrong all the time either).

4.24.3.2 Critical Regions

Critical Regions (sometimes called "Critical Sections") are those pieces of code where a shared resource is used, and as such it can lead to erroneous or unexpected behaviors. Such sections must be protected from concurrent access, which means only one process or thread can access them at one given time.

4.24.4 Ensuring determinism

Let's take a look at how to implement multi-threading in a safe way, allowing our game to perform better without non-deterministic behaviors. There are other implementation approaches (like thread-local storage and re-entrancy) but we will take a look at the most common here.

4.24.4.1 Immutable Objects

The easiest way to implement thread-safety is to make the shared data immutable. This way the data can only be read (and not changed) and we completely remove the risk of having it changed by another thread. This is an approach used in many languages (like Python and Java) when it comes to strings. In those languages strings are immutable, and "mutable operations" only return a *new string* instead of modifying the existent one.

4.24.4.2 Mutex

Mutex (Short for **mut**ual **ex**clusion) means that the access to the shared data is serialized in a way that only one thread can read or write to such data at any given time. Mutual exclusion can be achieved via algorithms (be careful of *out of order execution*[g]), via hardware or using "software mutex devices" like:

- Locks (known also as *mutexes*)
- Semaphores
- Monitors
- Readers-Writer locks
- Recursive Locks

- ...

Usually these multi-threaded functionalities are part of the programming language used, or available via libraries.

Let's see how Mutex solve our concurrency problem.

As seen before, we have a shared variable and two threads that want to add one to it.

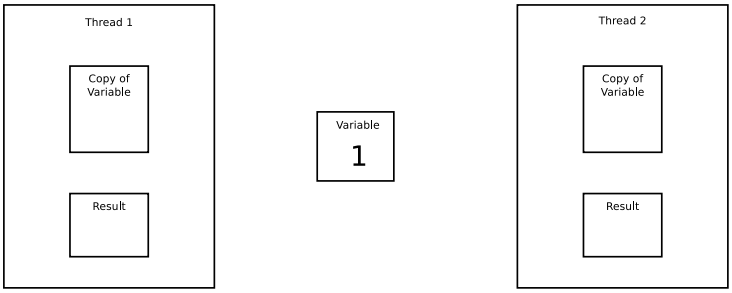


Figure 122: How mutex works (1/8)

Now the first thread reads the variable and "locks" the mutex (thus stopping other threads from accessing the variable).

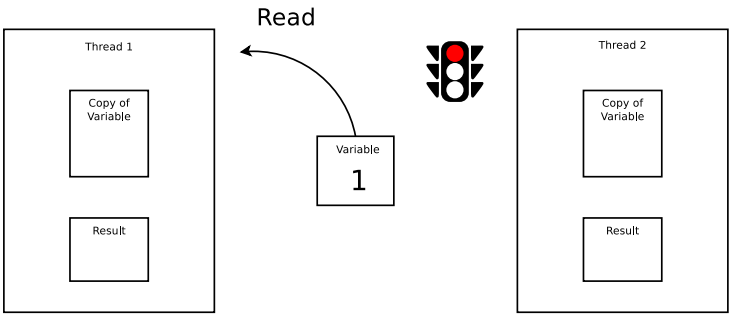


Figure 123: How mutex works (2/8)

When the second thread wants to access the "critical region", it will check on the Mutex, find it "locked" and be forced to wait: it cannot read the variable, because we would have a "race condition" otherwise.

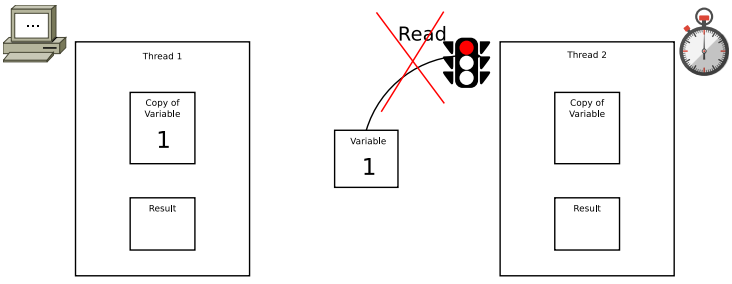


Figure 124: How mutex works (3/8)

As soon as the first thread finishes its job, it will write the result in the variable and "unlock" the mutex, allowing others to access the variable.

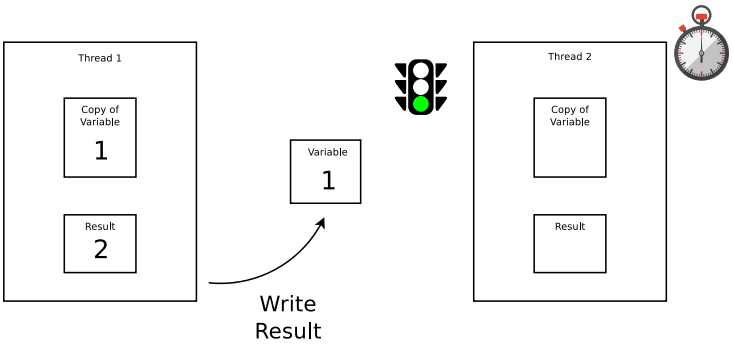


Figure 125: How mutex works (4/8)

Since the second thread was waiting, it will read the variable result (now 2) and “lock” the mutex for safety. The second thread entered the “critical region”.

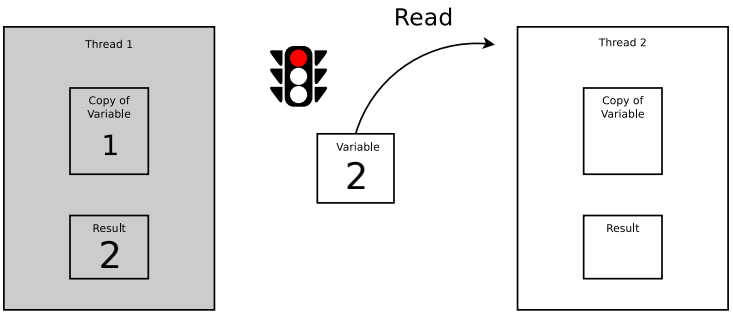


Figure 126: How mutex works (5/8)

The second thread will do its job as normal, if a third thread tried to access the variable, it would be stopped by the locked mutex.

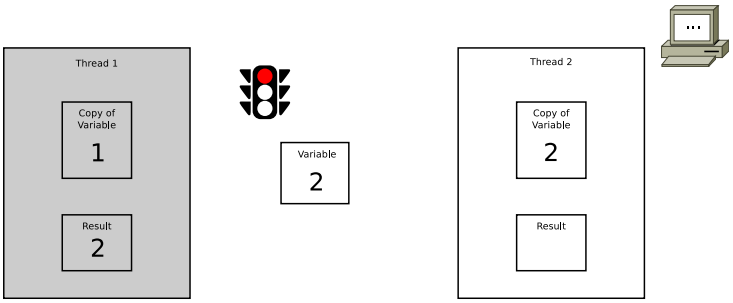


Figure 127: How mutex works (6/8)

When its job is done, the second thread will write to the variable and “unlock” the Mutex, thus allowing other threads or processes to access the variable.

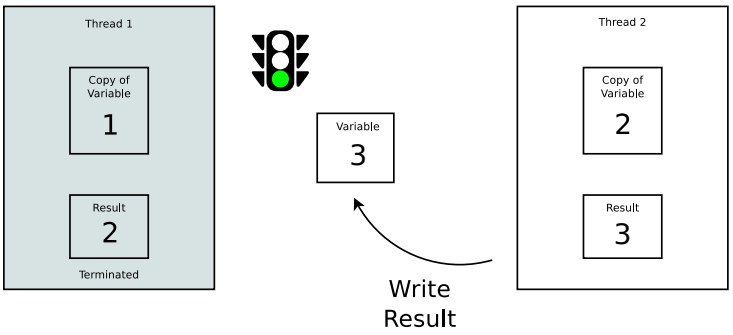


Figure 128: How mutex works (7/8)

Now both threads finished their jobs and the result inside the variable is correct.

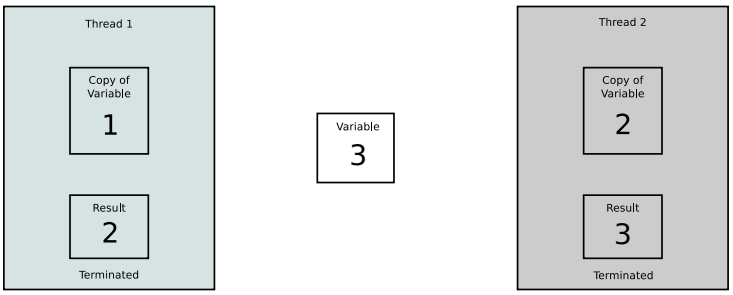


Figure 129: How mutex works (8/8)

4.24.4.3 Atomic Operations

[This section is a work in progress and it will be completed as soon as possible]

4.25 Developer Experience (DX/DevEx)

Much like “user experience”, developer experience (DevEx for short) is something that should be kept in high regard.

Developer experience encompasses everything that concerns the interaction between a developer and their work environment, which could be their tools, the work environment including culture, and anything that could affect their productivity and satisfaction.

This may also include the code itself: good code breeds other good code, while bad code will encourage bad practices (similar to the “Broken Window Principle”).

4.26 What to do when coding

Here are some small universal tips that you can use when coding that will help you, no matter the programming language you’re in. These will usually avoid a lot of headaches by covering cases that may get overlooked or by giving guidance on how to solve some problems.

4.26.1 Safe default initialization

Every time you make use of a variable, array or any data structure, you should always have it initialized to a known value.

Never leave variables in an uninitialized state: not all programming languages do what you might think, some may initialize an integer to a default value (zero), while others may leave it to whatever was in memory already (which would result in essentially a random value). If you make use of such variables as indexes inside arrays, you may find yourself in front of surprising behaviours.

The same goes for arrays, always make sure that all the array "cells" contain a known, valid and usable value from the very beginning, depending on your objective.

4.26.1.1 "Smart" default initialization

As a corollary of the "safe default initialization" theme, we can make use of a trick: if a certain array is known to contain mostly a certain value, provided some exceptions, it may be smarter to initialize it in a for loop and then change the exceptions manually.

Imagine an array containing "character expressions" that usually has a "neutral expression", but in a few occasions it may contain an angry or happy expression.

You may be tempted to initialize the array this way:

Listing 36: A naive array initialization

```
// ...
let tyler = [];
tyler.push(CharacterExpression.neutral);
tyler.push(CharacterExpression.neutral);
tyler.push(CharacterExpression.neutral);
// 31 more lines
tyler.push(CharacterExpression.angry);
tyler.push(CharacterExpression.sad);
tyler.push(CharacterExpression.neutral);
// 26 more lines
tyler.push(CharacterExpression.neutral);
// ... Next character ...
```

This way gives a bad developer experience: every time you encounter feature creep (and trust me, you will), you will have to update the size of the array, create a new line with the initialization value (provided you don't forget and end up with a crash due to out-of-bounds indexing) and find yourself with a ton of lines of code doing the same thing.

A smarter way to do this could be the following:

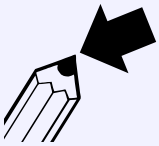
Listing 37: A smarter array initialization

```
// ...
const tyler_array_len = 64;
let tyler = [];
for (let i = 0; i < tyler_array_len; i++) {
    tyler.push(CharacterExpression.neutral);
}
// Here we take care of the exceptions
tyler[33] = CharacterExpression.angry;
tyler[34] = CharacterExpression.sad;
// ... Next character ...
```

Using a for loop to take care of the default values and then taking care of the exceptions is:

- **Shorter:** Doesn't matter how many items the array has, the default initialization code will stay the same;
- **Safer:** If you extend the array but forget to change a value, a character might have the wrong expression, but the game won't crash;
- **More flexible:** Since you won't have to worry about as much code as in the "naive version", it will be easier to expand upon your game.

Tip!



Use Enums for an even more readable code, since in most languages Enums map to integers, you can loop through an array of integers and then use the "named integers" offered by Enums to make the code even better.

4.27 What to avoid when coding

Here are some examples of what are usually considered "universally bad practices" when coding, no matter the language you're using. There are usually referred as "code smells" and should be avoided as much as possible (but nothing is imperative in programming).

4.27.1 Magic Numbers

Magic numbers (or more generally "magic constants") are numbers that can be found in your source code that have special meanings but such meaning cannot usually be quickly inferred without having to refer to somewhere else (either in the code or in the documentation).

Let's think of a simple piece of code:

Listing 38: A piece of code featuring some magic numbers

```
// ...
screen.initialize(640, 480, 32, false);
// ...
```

It is difficult to understand what each value passed to the function means, so we would need to refer to the documentation or where such function was coded to be sure of what each constant means.

And now here's the same piece of code, but the "magic numbers" have been externalized into constants. This usually doesn't affect performance (most compilers replace constants with their inline values) and it is much more readable.

Listing 39: A piece of code after magic numbers are removed

```
// ...
const SCREEN_WIDTH = 640;
const SCREEN_HEIGHT = 480;
const BITS_PER_PIXEL = 32;
const FULLSCREEN = false;
screen.initialize(SCREEN_WIDTH, SCREEN_HEIGHT, BITS_PER_PIXEL, FULLSCREEN);
// ...
```

Avoiding magic numbers helps a lot with Developer Experience, especially when the "meaning" of such numbers is declared in a place far away from where they are used.

This approach also has some additional advantages, besides readability:

- **Ease of maintenance:** If one day we want to up the resolution of our game to 800x600, we just need to change two numbers, instead of running around the whole program, replacing 640 with 800 and 480 with 600, risking to replace some value that is not the screen size;
- **Resilience:** Making a typo on a number is easy: you can type 460 instead of 640 and you wouldn't be the wiser until a bug pops out. Making typos on words that make sense is harder, and your IDE will warn you about uninitialized variables/constants;
- **Grouping:** You can group all "constants" at the top of the file (or in a separate file altogether) and facilitate change and review.

4.27.2 Code monoliths

Another thing that should usually be avoided is having long files. This usually is signal that whatever is in such file is taking care of too many things: you certainly can program an entire small game in a single file, but good luck with its maintenance.

Same goes for long classes, functions and pieces of code in general. These are usually called "code monoliths".

If classes are very long (as in "thousands of lines") maybe they're taking care of too many responsibilities, and need to be broken down into simpler pieces and then composed back together.

If functions are long maybe they're too complex, or they're doing more than one task (so they're like a small program by themselves), and may be useful to break them down in smaller functions that call each other, or are called by a function that works as "orchestrator".

Try to code your software with reuse in mind, if you find a pattern of very similar but slightly different pieces of code, maybe it’s time to abstract them into something more generic that can be fed some parameters to do the same thing.

5 A Game Design Dictionary

> Why should you make games? Do it to give players joy from your unique perspective and to have fun expressing yourself. You win and the players win.
>
> *Duane Alan Hahn*

In this section we will talk about platforms, input systems and game genres, in a quick fashion. This chapter will introduce you to the language and terms used in game design, this way the following chapters will be easier to comprehend.

We will talk about the differences and challenges deriving from each decision and the basic way game genres work. The objective of this chapter is giving you some terminology and knowledge about game design, before deep-diving into the topic.

5.1 Platforms

There are several different platforms a game can be developed for, and each one has its own advantages and drawbacks. Here we will discuss the most notable ones.

5.1.1 Arcade

Arcade cabinets have been around for decades, and have still a huge part in the heart of gaming aficionados with classic series going on like “Metal Slug”. The main objective of these machines is to make you have fun, while forcing you to put quarters in to continue your game.

These cabinets’ software is known to be very challenging (sometimes due to the fact that you’re popping quarters into the machine for the “right to play”), having some nice graphics and sound. Arcade games are usually presented in the form of an “arcade board”, which is the equivalent of a fully-fledged console, with its own processing chips and read-only memory.

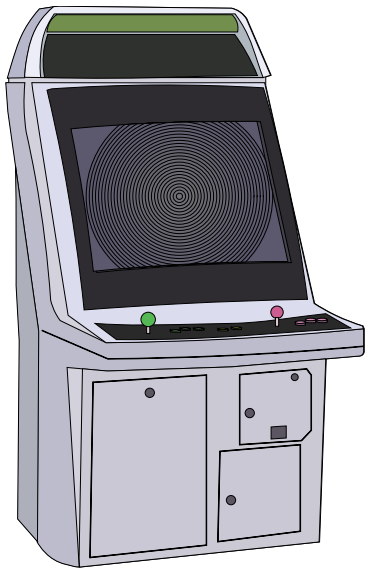


Figure 130: How an arcade machine usually looks like

In the case of arcades, the hardware is usually tailored to support the software; with some exceptions added later (like the Capcom Play System, also known as CPS), where the hardware is more stable between arcades, while the software changes.

5.1.2 Console

Consoles are a huge (if not the biggest) part in the video game industry. Their Hardware is dedicated solely to gaming (and some very marginal "multimedia functionalities") and it evolves in "generations": this means that each "generation" has a stable hardware programmers can study and exploit.

Figure 131: A portable console

This hardware stability is a double-edged sword: the hardware can be really hard to master at the beginning, resulting in some poor-performing games at the beginning of the generation, but when mastered the results are incredible. This feeds into a cycle that looks like the following:

1. New Generation is introduced
2. Initial confusion, with poor performance and graphics
3. Hardware is mastered and games have great performance/graphics
4. The games become "too big" for the current generation and a new generation must be introduced.

5.1.3 Personal Computer

Personal Computers are another huge part of the video game industry. They are extremely flexible (being general-purpose machines) but have a huge drawback: their hardware is not the same from one unit to the other. This means that the programmer needs to use "abstraction layers" to be able to communicate with all the different hardware.

Figure 132: A personal computer

This compounds with the fact that “abstraction layers” used by the developer (like SDL, SFML or GLFW) are running on top of other “abstraction layers”, like sound servers, device drivers, etc... which can be littered with bugs themselves. Just look at how many indirections we have on a modern Linux system (which is usually bundled with PulseAudio):

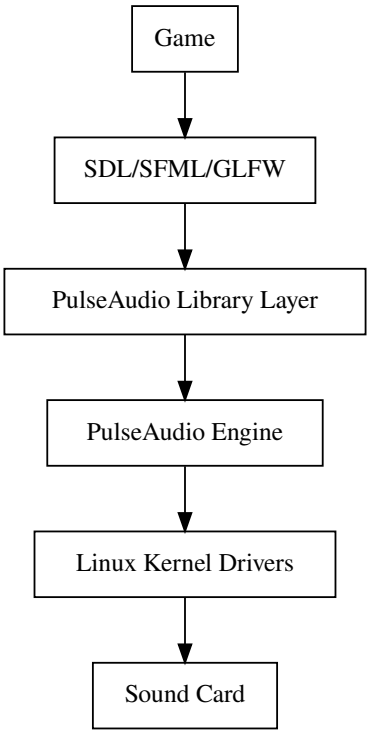


Figure 133: How many abstraction layers are used just for a game to be able to play sounds

This can have performance costs, as well as forcing the programmer to add options to lower graphic settings, resolution and more.

All of this just to be able to run on as many computers as possible. The upside is that when the computer is really powerful, you can get great performance and amazing quality, but that's a rare occasion.

5.1.4 Mobile

One of the most recent platforms game developers work on is right in your pocket: your smartphone.

Figure 134: A smartphone

Today's smartphones have enough power to run fully-fledged video games, on the go. Sadly the touch screen can prove to be really uncomfortable to use, unless the game is specially tailored for it.

5.1.5 Web

Another platform that has seen a massive rise in recent times is the Web: with WebGL and WebAssembly, fully-fledged games (including 3D games) can run on our browser, allowing for massively-multiplayer experiences (like Agar.io) without the hassle of manual installation or making sure the game is compatible with your platform.

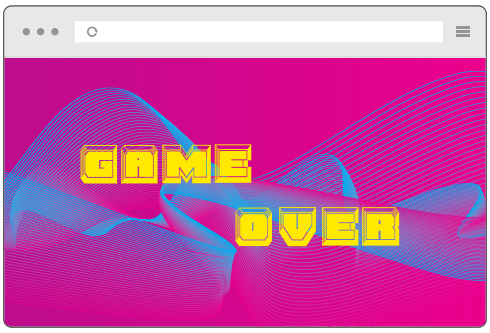


Figure 135: Fully fledged games can run in your browser nowadays

A drawback of the "web approach" is the limited performance that web browsers, WebGL and WebAssembly can give, as well as the need to download the game before being able to play (and sometimes you may need to re-download the game if you cleared your browser's cache).

5.1.6 Fantasy consoles

Since limitations breed creativity, the game development community has come up with a bunch of nonexisting consoles that have their own specifications, trying to match (in part) older systems, while letting developers still work on their games with modern tools.

Figure 136: When there is no console people just invent theirs, in software form

These systems have arbitrary limitations (like color palettes, resolution or number of sound channels) to imitate a certain "generation" of systems (be it 8 or 16 bit, usually).

Games for these consoles are distributed in "virtual carts" or "virtual floppies", which are essentially just files containing the game code and assets. These software consoles are developed with learning in mind, so each game can be played and then, at the press of a button, you can enter the editor mode, check the game's code and assets and learn from them.

5.2 Input Devices

A game needs a way to be interacted with: this "way" is given by input devices. In this section we will take a brief look at the input devices available in a game.

5.2.1 Mouse and Keyboard

One of the most common input devices, most of the currently available frameworks and engine have support for input via mouse and keyboard. These input methods are great for visual novels, point and click adventures, FPS/TPS games and anything that is considered to be "made for PCs".

5.2.2 Gamepad

One of the classics of input devices, works well with the majority of games: FPS/TPS games may need some aim assist mechanic in your game. Point and click adventures feel clunky with this input method.

As with Mouse and Keyboard, most of the currently available engines and frameworks support gamepads.

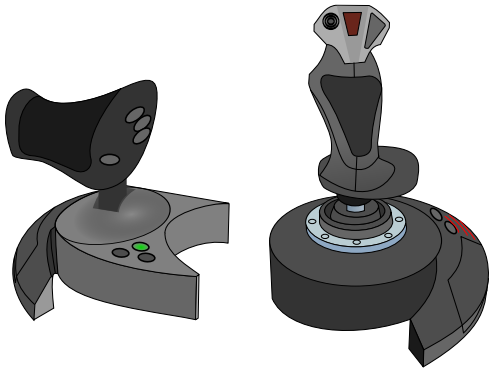
5.2.3 Touch Screen

With the coming of smartphones, touch screen is a new input device that we have to account for. Touch screens emulate computer mice well enough, although they lack precision.

The nature of being a mix between an input device and a screen brings a lot of new ways to experience a game if well done. Many times touch screens are used to simulate game pads: the lack of the tactile feedback given by buttons makes this simulation clunky and uncomfortable.

Some of the most recent framework and engines support touch screens, although there's an additional layer of complexity given by the specific operating system of the smartphone you're building for.

5.2.4 Dedicated Hardware



Some games require dedicated hardware to work at their best, if at all. Guitars (guitar hero), wheels for racing games, joysticks for flying simulators, arcade sticks for arcade ports...

Dedicated hardware requires precise programming, and is usually an advanced topic. On PCs many "dedicated input devices" are recognized as "game pads" and use an "axis" and "buttons" abstraction that makes coding easier.

5.2.5 Other Input Devices

A special mention is deserved for all the input devices that are "general purpose" (as in not "dedicated") but are still in a group outside what we saw so far.

In this group we see gyroscopes, accelerometers (like the Nintendo Wii/Switch JoyCons), sensors, IR, as well as other exotic hardware that can still be exploited in a video game.

5.3 Game Genres

Let's analyze some game genres to understand them better and introduce some technical language that may be useful in writing a Game Design Document.

These genres are quite broad, so a video game is usually a mix of these "classes" (like a strategy+simulation game).

5.3.1 Shooters

Shooters are games that involve... shooting. They can include any kind of projectile (bullets, magic from a fairy, arrows from a hunter) and can be crossed with any other genre (creating sub-genres in a way), like 2D platformers.

Some of the most known shooter genres are:

- **FPS** (first person shooters), 3D games where the game is shown from the point of view of the protagonist. This involves only seeing a HUD and the weapon, instead of the whole character;
- **TPS** (third person shooters), 3D games where the game is shown from a behind-the-character perspective. Some show the whole protagonist, while others adopt an over-the-shoulder perspective;
- **Top Down Shooters**, usually 2D games where you may be piloting a vehicle (space ship, plane, etc...) and shoot down waves of enemies, in this category we fit arena shooters (like Crimsonland) and space shooters (like Galaga);

- **Side scroller shooters**, usually 2D games and platformers, where you control the protagonist and shoot enemies on a 2D plane, in this category we find games like Metal Slug.

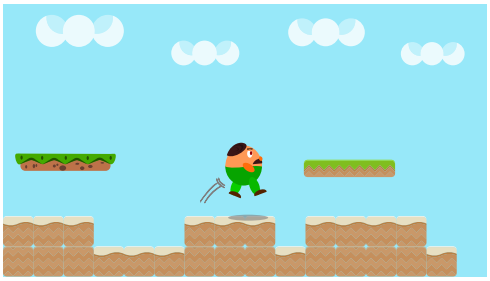
5.3.2 Strategy

Strategy games involve long-term planning and resource control, they are slower games, but can be really intense when played in competition with other players.

Some of the most popular strategy genres are:

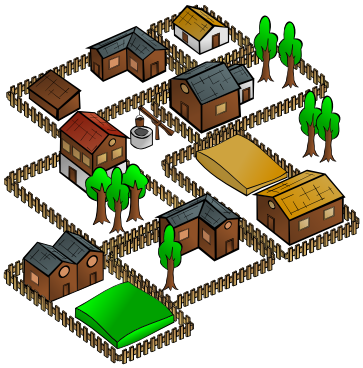
- **RTS** (real time strategy), where units are controlled in real time;
- **Turn-based strategy**, where units and resources are managed in turns;

5.3.3 Platformer



Platformer games involve difficult jumps and precise movement, they can both be 2D and 3D games. A prime example of platformer games is the Mario series: Mario 1,2,3 for 2D games and Mario 64 for 3D.

5.3.4 RPG



RPGs or “Role Playing Games” are games where you assume the role of a character in a fictional setting. In RPGs the world is well-defined and usually have some level or class system and quite advanced item management.

RPGs can be either action/adventure, with real-time actions, turn-based or hybrid, where the movement is done in real time but battles happen in turns. Some prime examples of RPG games are the Legend of Zelda series, as well as the Final Fantasy series.

5.3.5 MMO

MMO (Massively Multiplayer Online) is a term used for games that have a heavy multiplayer component via the internet. The most known MMO genre is MMORPGs (Massively Multiplayer Online Role-Playing Games).

5.3.6 Simulation

Simulation games cover a huge variety of games that are created to "simulate reality", in more or less precise ways. Among simulation games we can find:

- **Racing Games:** sometimes more simulative others more arcade-like, racing games simulate the experience of driving a vehicle, more or less realistic (from modern cars to futuristic nitro-fueled bikes);
- **Social Simulation:** simulating the interaction between characters, a pioneer on the genre is surely "The Sims";
- **Farming simulation:** simulating the quietude and work in the fields;
- **Business simulation:** like "game dev tycoon" or "rollercoaster tycoon";

But there are also other kinds of simulations, like Sim City, where you manage an entire city.

5.3.7 Rhythm Games

Rhythm games are based on the concept of following a music beat as precisely as possible, this can be also used as a "mechanic" in other types of games.

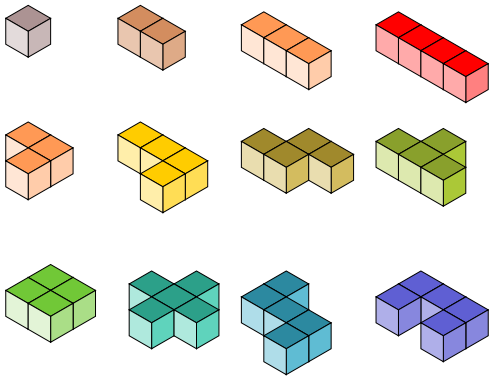
Some examples of Rhythm games are "Dance-Dance Revolution" (also known as DDR), as well as more innovative games like "Crypt of the Necrodancer" (a mix between rhythm game and dungeon crawler).

5.3.8 Visual novels

Visual novels are graphical adventures whose primary objective is "telling a story", they can be linear or have a "choose your own path" component. They usually feature multiple endings and hand-crafted still images as artwork.

The more modern versions feature more interactive components and fully-fledged 3D graphics, but what ties the genre together is usually a "point and click" style of gameplay.

5.3.9 Puzzle games



Puzzle games are centered about making the player think: they can test a lot of problem-solving skills from pattern recognition, to word completion, to logic.

Some example of puzzle games include Lemmings, Boulder Dash, any match-3 game (started with “Shariki”, followed by “Bejeweled” until the more modern titles for mobile phones), and Tetris.

Puzzle games can involve math (like Sudoku), Physics (like the game “Peggle”), Hidden objects or even programming (for instance “Shenzen I/O” for “realistic programming”, or “Opus Magnum” for a different approach).

Nothing stops other genres from including puzzle elements, but this small section is dedicated to the games that feature puzzle elements as their core mechanic.

5.4 Miscellaneous

Here we will talk about some other terms that you may hear in the game development and design world, but that don’t fit into a specific category.

5.4.1 Emergent Gameplay

Sometimes, when interacting with simple game mechanics, players can give life to complex situations. When that happens usually we talk about “emergent gameplay”.

Emergent gameplay can take place in open-ended games, where there are many solutions to a situation and none of them is “preferred by the game”. For instance, we can think of someone guarding a door, there are many ways to get through the guard, such as:

- Attacking the guard (and winning);
- Find an alternative path;
- Sneak around the guard to knock them unconscious;
- Find a way to make the guard leave their post;
- ...

Random Trivia!

A prime example of a game that leverages emergent gameplay is Minecraft. Players can either survive, build palaces, build redstone circuits and much more.

Part 2: Project Management

6 Project Management Basics and tips

> Those who plan do better than those who do not plan even though they rarely stick to their plan.
>
> *Winston Churchill*

Project management is a very broad topic but I feel that some basics and tips should be covered in this book. Knowing some project management can save you a lot of headaches and can make the difference between success and a colossal failure.

6.1 The figures of game design and development

Before delving into the topic at hand, we need to familiarize ourselves with the main figures that are involved in the process of game design and development, since you'll probably (if you are the only developer of your game) have to take upon all their tasks.

6.1.1 Producer/Project Manager

The producer is a figure that has experience in many fields and has an overall view of the project. They essentially keep the project together.

Their duties are:

- Team Building (and its maintenance too);
- Distributing duties and responsibilities;
- Relations with the media.

Under the term "project manager" you can find different roles, among them:

- Product Manager;
- Assistant Producer;
- Executive producer.

A good project manager will need tools to manage tasks (Like a Kanban Board), as well as tools that promote communication in the team (Chats, VoIP) and information repositories (having all information in the same place is

important!).

6.1.2 Game Designer

The game designer takes care of the game concept, usually (but not only!) working with really specific software, usually provided by the programmers in the team (like specific level editors).

They design balanced game mechanics, manage the learning curve and take care of level design too.

Under the “Game Designer” term you can find different roles, among them:

- Level Designer;
- World Builder;
- Narrative Designer;
- Quest/Mission Designer.

A good game designer must know mathematics, some scripting and be able to use planning tools (again, our friendly Kanban Board comes into play) as well as diagram drawing tools.

6.1.3 Writer

Writers are the ones who can help you give your game its own story, but also help with things that are outside the mere game itself.

Some of their jobs include:

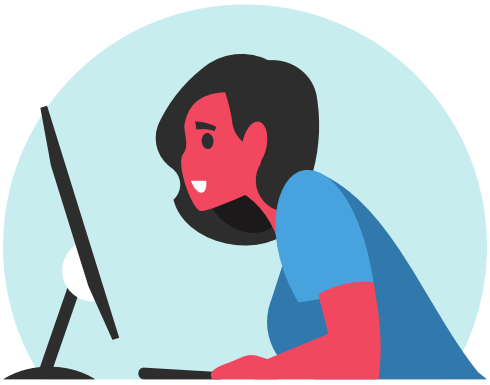
- Writing tutorial prompts;
- Writing narration;
- Writing dialogue;
- Writing pieces for the marketing of your game (sometimes known as “Copywriting”).

Under the term of “Writer” you can find more roles, like:

- Editor;
- Narrative Designer;
- Creative Writer.

A good writer must have good language skills, as well as creativity. They must be able to use planning programs (like everyone, communication is important) as well as writing programs, like LibreOffice/OpenOffice Writer.

6.1.4 Developer



Logic and mathematics are the strong suit of programmers, the people who take care of making the game tick, they can also have many specializations like:

- Problem Solver
- Game mechanics programmer;
- Controls programmer;
- AI developer;
- Visuals Programmer;
- Networking programmer;
- Physics programmer;
- ...

They must be familiar with IDEs and programming environments, as well as Source Control Tools (Like Git), knowledge of game engines like Unity is preferred, but also tied to the kind of game that is made.

6.1.5 Visual Artist

In 2D games visual art is as important as in 3D games and good graphics can really boost the game's quality greatly, as bad graphics can break a game easily.

Among visual artists we can find:

Both in 2D and 3D games:

- 2D Artists;
- Animators;
- Environment Artists;
- UI Artists/Designers;
- Conceptual Artists.

In 3D games:

- 3D Modelers;
- Texture Artists.

Visual Artists must be knowledgeable in the use of drawing programs, like Krita, GIMP or their commercial counterparts.

6.1.6 Sound Artist

As with graphics, sound and music can make or break a game. Sound artists may also be musicians, and their task is to create audio that can be used in a video game, like sound effects, atmospheres or background music.

Under the umbrella of a sound artist, you can find:

- Audio Engineers;
- Game Composers;
- Music Mixers;
- Voice actors;
- Audio Programmers.

The knowledge of DAW (Digital Audio Workstation) software is fundamental, as well as knowing some so-called "middlewares", like FMOD. Another important bit of knowledge is being able to use Audio editors effectively.

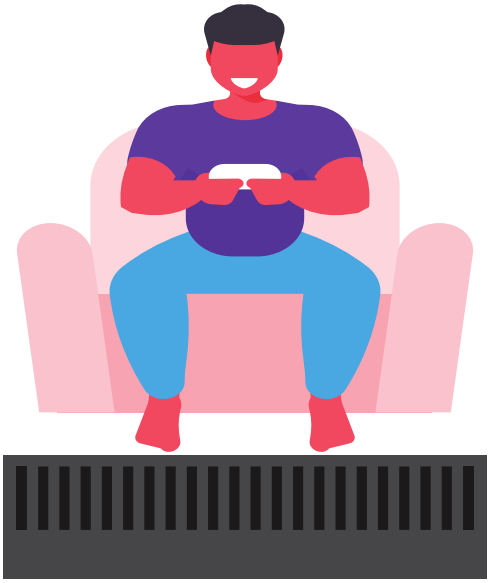
6.1.7 Marketing/Public Relations Manager

Even the best game in the world will get no attention without someone who takes care of letting people know about it. The marketing and public relations manager is responsible for promoting a game through various channels, from specialized outlets to social media.

Their job include:

- developing and executing marketing campaigns;
- creating press releases and other promotional materials;
- managing social media accounts;
- working with journalists and influencers to publicize the game;
- publicizing the game on dedicated trade shows.

6.1.8 Tester



Probably the most important job in a game development team, testing needs people with high attention to detail, as well as the ability to handle stress well.

Testers are able to find, describe and help you reproduce bugs and misbehaviors of your game.

6.2 Some general tips

6.2.1 Be careful of feature creep

The "it would be cool to..." trap, formally called "feature creep", is a huge problem in all projects that involve any amount of passion in them.

Saying "it would be cool to do `<insert something here>`: let's implement it!" can spiral out of control and make us implement new features forever, keeping us from taking care of the basics that make a good game (or make a game at all).

Try to stick to the basics first, and then eventually expand when your game is already released, if it's worth it: first make it work, then make it work well and only in the end make it elegant.

6.2.2 On project duration

When it comes to project management, it's always tough to gauge the project duration, so it can prove useful to remember the following phrase:

> "If you think a project would last a month, you should add a month of time for unforeseen events. After that, you should add another month for events that you really cannot foresee."

This means that projects will last at least 3 times the time you foresee.

That may seem a lot like an exaggeration, but unforeseen events happen and they can have a huge impact on the release of your game. It's better to err on the side of caution and even delay the release if something goes wrong. Shigeru Miyamoto said the following:

> A delayed game is eventually good, a bad game is bad forever.

so maybe being "abundant" with your time estimates is not that wrong.

6.2.3 Brainstorming: the good, the bad and the ugly

Brainstorming is an activity that involves the design team writing down all the ideas they possibly can (without caring about their quality yet).

This is a productive activity to perform at the beginning of the game development and design process, but it can be a huge source of feature creep if done further down the line.

After the initial phase of brainstorming, the team analyzes the ideas and discards the impossible ones, followed by the ones that are not "as good as they sounded at first". The remaining ideas can come together to either form a concept of a video game or some secondary component of it.

In short: brainstorming is a great activity for innovation, but since it's essentially "throwing stuff at a wall and see what sticks".

This activity can sometimes be either unproductive or "excessively productive": in both cases we end up with nothing of use in our hands.

6.2.4 On Sequels

In case your game becomes a hit, you will probably think about making a sequel: this is not inherently a bad thing, but you need to remember some things.

When developing a sequel, you will have to live up to your previous game, as well as the expectations of the players, and this becomes more and more difficult as the "successful sequels" go on.

Not only a sequel must be "as good or better" than its predecessor, but also it should add something to the original game, as well as the established lore (if there is any).

Your time and resource management must be top-notch to be able to “bring more with less”, since your need for resources cannot skyrocket without a very good reason.

Also don’t get caught in the some kind of “sequel disease” where you end up making a sequel just to “milk the intellectual property”: you will end up ruining the whole series: it may end up being hated by the ones who played the first games, and new players will be discouraged by a series that either overstays its welcome, or has one or more low-quality sequels.

6.3 Common Errors and Pitfalls

When you are making a new game, it’s easy to feel lost and “out of your comfort zone”, and that’s okay! It’s also easy to fall into traps and pitfalls that can ruin your experience, here we take a look at the most common ones.

6.3.1 Losing motivation

Sometimes it can happen to lose motivation, usually due to having “too much ambition”: make sure you can develop the kind of game you want to make, for instance leave multiplayer out of the question (multiplayer games are really hard and network code can be a real pain to work on). It will just suck up development time, and it isn’t that much of an important feature anyway (and it can still be implemented later, like it happened in *Stardew Valley*).

Like in music, many people prefer “mediocrity” to “something great”, so don’t force yourself to innovate: do things well enough and if the innovative idea comes, welcome it.

If you get tired, take a break, you’re your own boss, and no one is behind you zapping you with a cattle prod: just focus on making a good overall product and things will go well.

6.3.2 The “Side Project” pitfall

It happens: you have a ton of ideas for games of all kinds, and probably you’ll start thinking:

> What’s bad about a small “side project”? I want to change things up a bit...

You will end up having lots of “started projects” and nothing finished, your energy will deplete, things will become confusing and you won’t know what game you’re working on anymore.

Instead, make a small concept for the new mechanic and try to implement it in your current game, you may find a new mix that hasn’t been tried before, making your game that much more unique.

6.3.3 Making a game “in isolation”

While making a game you will need to gather some public for it, as well as create some hype around it: making a game on your own without involving the public is a mistake that deprives you of a huge source of suggestions and (constructive) criticism (as well as satisfaction, when you manage to get some people interested in your game).

Make your game public, on platforms like `itch.io` or `IndieDB`, get feedback and encouragement. Create trailers towards the end of development, put them on `YouTube` or `Vimeo` and if you want to go all out, get in touch with the press (locally

first) and create hype around your product.

6.3.4 (Mis)Handling Criticism

Among all the other things that are happening, we also need to handle feedback from our “potential players”, and this requires quite the mental effort, since we can’t make it “automatic”.

Not all criticism can be classified as “trolling”, and forging our game without listening to any feedback will only mean that such game won’t be liked by as many people as we would like, maybe for a very simple problem that could have been solved if only we listened to them.

At the same time, not all criticism is “useful” either, not classifying criticism as “trolling” does not mean that trolling doesn’t exist, some people will take pride in ruining other people’s mood, either by being annoying and uselessly critic, or by finding issues that don’t actually exist.

The question you should ask yourself is simple:

> Is this criticism I’m receiving constructive? Can it make my game better?

If the answer is no, then you may want to ignore such criticism, but if it is constructive, maybe you want to keep it in consideration.

6.3.4.1 Misusing of the Digital Millennium Copyright Act

This is what could be considered the apex of mishandling criticism: the usage of DMCA takedowns to quash criticism towards your game.

Note!

What follows **is not legal advice**. I am not a lawyer.

If you want to know more (as in quantity and quality of information), contact your favorite lawyer.

Sadly, mostly in the YouTube ecosystem, DMCA takedowns are often used as a means to suppress criticism and make video-reviews disappear from the Internet. Useless to say that this is **potentially illegal** as well as **definitely despicable**.

Takedowns according to the DMCA are a tool at your disposal to deal with copyright infringements by people who steal part (or the entirety of) your work, allowing (in the case of YouTube at the very least) to make the allegedly infringing material. This should be used carefully and just after at the very least contacting the alleged infringer privately, also because there is an exception to the copyright rule.

6.3.4.1.1 The Fair Use Doctrine

The so-called “Fair Use” is a limited exception to the copyright law that targets purposes of review, criticism, parody, commentary, and news reporting, for instance.

The test for “Fair use” has four factors (according to 17 U.S.C. §107):

1. **The Purpose and character of the use:** if someone can demonstrate that their use advances knowledge or the progress of arts through the addition of something new, it’s probably fair use. This usually is defined by the question “is the work **transformative** enough?”
2. **The nature of the copyrighted work:** For instance, facts and ideas are not protected by copyright, but only their particular expression or fixation is protected. Essentially you can’t really sue someone for making a game very similar to yours (For instance making a 2D sidescrolling, run’n’gun platformer).
3. **The amount and substantiality of the portion used in relation to the work as a whole:** If someone uses a small part (compared to the whole) of the work, and if that part is not really substantial, then it’s probably fair use.
4. **The effect on the potential market for the copyrighted work:** this defines if the widespread presence of the “allegedly infringing use” can hinder on the copyright owner’s ability to exploit (earn from) their original work.

There can also be some additional factors that may be considered, but these four factors above are usually enough to decide over the presence (or absence) of fair use.

6.3.4.1.2 The “Review Case”

Let’s take a simple example: a video-review on our brand new video game, that takes some small pieces of gameplay (totaling about 5 minutes), on video and comments on the gameplay, sound and graphics. A very common scenario with (I hope) an unsurprising turnout.

Let’s take a look at the first point: the purpose is criticism, the review brings something new to the table (essentially **it is transformative**): someone’s impression and comments about the commercial work.

Second point: the game is an interactive medium, while the review is non-interactive by nature, the mean of transmission is different.

Third point: considering the average duration of 8 to 10 hours of a video game, 5 minutes of footage amounts for around 0.8% to 1% of the total experience, that’s a laughable amount compared to the total experience.

Fourth Point: this is the one many people may get wrong. A review can have a huge effect on the market of a copyrighted work (a bad score from a big reviewer can result in huge losses), but that’s not really how the test works. The fourth test can usually be answered by the following questions:

> What’s the probability that someone would buy (or enjoy for free) the work from the alleged infringer, instead than from me (the copyright owner)?

This is called “being a direct market substitute” for the original work. The other question is:

> Is there a potential harm (other than market substitution) that can exist?

This usually is related to licensing markets. And here lies the final nail on the coffin: there is no direct market substitution and courts recognize that certain kinds of market don't negate fair use, and reviews are among those kinds of market. In essence **Copyright is not a shield against adverse criticism**.

6.3.5 Not letting others test your game

This is a common mistake when you are focused on making the game: using your own skill as a "universal measure" for the world's skill level. You may be an unknown master at 2D platformers, and as such what is "mildly difficult" for you may be "utterly impossible" for the average player. Or the opposite.

Try to keep the challenge constant through the levels, applying the usual slight upwards curve in difficulty that most games have (or check the section about difficulty curves for some ideas), and let others test your game.

A beta version with feedback capabilities (or just a beta version and a form or email address can do the trick too) is pure gold when it comes to understanding what your players think about the game's challenge level.

Remember: when a level is (perceived as unfairly) too hard, players will stop playing the game.

6.3.6 Being perfectionist

If you are called "perfectionist" by your friends, that should be a red flag in your game development process from the very beginning.

Finding yourself honing the game over and over, allocating countless hours (that always feel as "not enough") into making the game "better", will end up just sabotaging the development process itself.

Instead try to prefer a more "scientific approach", where you study your game's shortcomings (with the help of some testers, or "friend-made-tester"), order them by their "effort vs improvement" ratio and start with those who require the lowest effort compared to the improvements they bring.

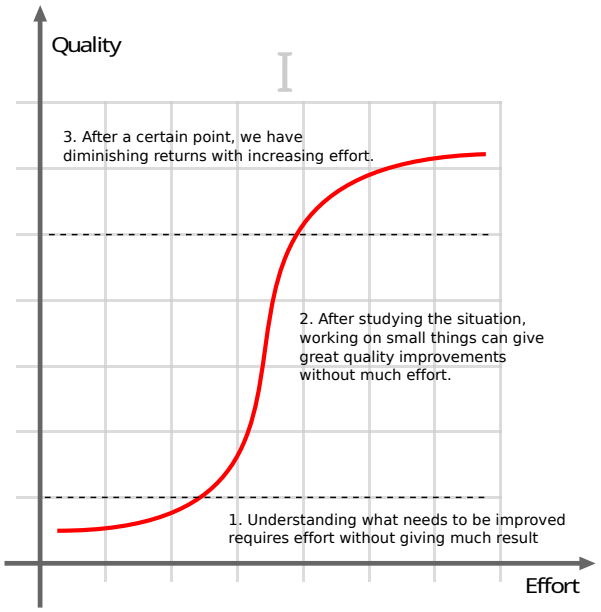


Figure 137: How to approach improvements on your game

You can see that you can get really good returns for relatively little effort, but if you're a perfectionist, you may want to push forward and put more and more hours, with diminishing returns.

This means that when you have:

- Good Visuals and Good Audio
- Working Gameplay
- A challenge that lasts the test of time
- The testing phase completed

You have a complete product. **Release it**. Updating it is very easy these days, and maybe that will give you the mental energy to undertake a new game. Maybe a sequel even?

6.3.7 Using the wrong engine

The game engine is one of the most important decisions you can take at the beginning of your game development journey. Realizing that you used the wrong engine after months of development can be a huge setback, as well as a "black hole" for your motivation.

Don't trust market hype over an engine, and don't trust the vendor's promises either.

Does the game engine have the features you will need **already**? No? Then your money should stay where it is, and you should look somewhere else.

If such engine's producer is promising the feature you want in future, don't trust it, that version may come, or it may never come at all. If you bought the engine and such feature won't ever be there, your money won't come back.

6.4 Software Life Cycle Models

When talking about project management (in itself or in the broader field of Software Engineering) it is really useful to talk about some guideline models that can be used to manage your project.

6.4.1 Iteration versus Increment

Before getting to the models, we need to discuss the difference between two terms that are often used interchangeably: “iteration” and “increment”.

Iteration is a non-deterministic process, during an iteration you are revisiting what you have already done, and such revisiting can include an advancement or a regression. While iterating, you have no idea when you will finish your job.

Increment is deterministic instead, with increments you are proceeding by additions over a base. Every increment creates a “new base” for the next increments, and increments are numbered and limited, giving you an idea of when you have to finish your job.

6.4.2 Waterfall Model

The Waterfall model, also known as “sequential model” is the simplest one to understand, easily repeatable (in different projects) and is composed by phases that are **strictly sequential**, which means:

- There is no parallelism;
- There is no overlap between phases;
- When a phase is completed, you cannot go back to it.

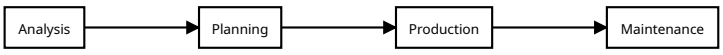


Figure 138: Diagram of the waterfall life cycle model

This makes the Waterfall life cycle model *extremely rigid*, everything needs to be carefully analyzed and documented (sometimes people define this model “document-driven”) and the coding is done only in its final phases.

In order to have a good result, this model requires quantifying some metrics (time spent, costs, …) and such quantification heavily relies on the experience of the project manager and the administrators.

6.4.3 Incremental Model

When a project of a certain size is involved, it’s a bad idea to perform the so-called “big-bang integration” (integrating all the components together). Such approach would make troubleshooting a nightmare, so it’s advisable to *incrementally integrate* the components.

The Incremental Model allows to have a “high-level analysis and planning”, after that the team decides which features should be implemented first. This way the most important features are ready as soon as possible and have more time to become stable and integrate with the rest of the software.

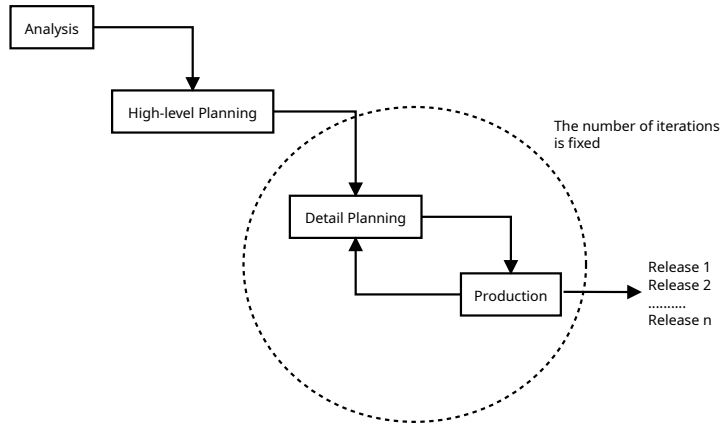


Figure 139: Diagram of the incremental life cycle model

This model can make use of strictly sequential phases (detail planning -> release -> detail planning -> release ...) or introduce some parallelism (for instance planning and developing frontend and backend at the same time).

As seen from the diagram, the high-level analysis and planning are not repeated, instead the detail planning and release cycle for a well-defined number of iterations, and on each iteration we will have a working release or prototype.

6.4.4 Evolutionary Model

It's not always possible to perfectly know the outline of a problem in advance, that's why the evolutionary model was invented. Since needs tend to change with time, it's a good idea to maintain life cycles on different versions of your software at the same time.

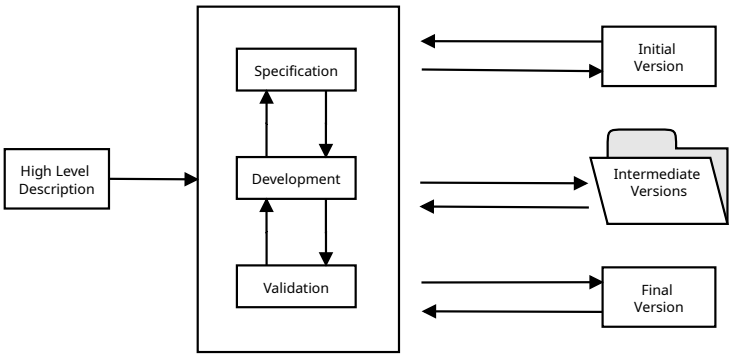


Figure 140: High-level diagram of the evolutionary life cycle model

Adding a way to implement the feedback you get from your customers and stakeholders completes the micro-managed part of the life cycle model, each time feedback and updates are implemented, a new version is released.

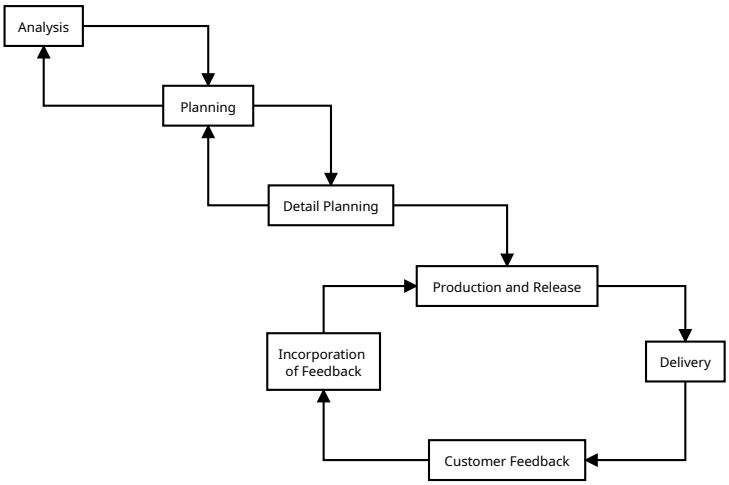


Figure 141: Diagram of the evolutionary life cycle model

6.4.5 Agile Software Development

Agile Software Development was born as a reaction to the excessive rigidity of the models we've seen so far. The basic principles of Agile Software Development are presented at the http://agilemanifesto.org website, but we will shortly discuss them below.

- Rigid rules are not good;
- A working software is more important than a comprehensive documentation;
- Seek collaboration with the stakeholder instead of trying to negotiate with them;
- Responding to change is better than following a plan
- Interactions and individuals are more important than processes and tools.

Obviously not everything that shines is actually gold, there are many detractors of the Agile model, bringing on the table some criticism that should be noted:

- The agile way of working entails a really high degree of discipline from the team: the line between "flexibility" and "complete lack of rules" is a thin one;
- Software without documentation is a liability more than an asset: commenting code is not enough - you need to know (and let others know) the reason behind a certain choice;
- Without a plan, you can't estimate risks and measure how the project is coming along;
- Responding to change can be good, but you need to be aware of costs and benefits such change and your response entail.

6.4.5.1 User Stories

Agile models are based on "User Stories", which are documents that describe the problem at hand.

Such documents are written by talking with the stakeholder/customer, listening to them, actively participating in the discussion with them, proposing solutions and improvements actively.

A User Story also defines how we want to check that the software we are producing actually satisfies our customer.

6.4.5.2 Scrum

The term “scrum” is taken from the sport of American Football, where you have an action that is seemingly product of chaos but that instead hides a strategy, rules and organization.

Let’s see some Scrum terminology:

- **Product Backlog:** This is essentially a “todo list” that keeps requirements and features our product must have;
- **Sprint:** Iteration, where we choose what to do to create a so-called “useful increment” to our product. Each Sprint lasts around 2 to 4 weeks and at the end of each sprint you obtain a version of your software that can be potentially sold to the consumer;
- **Sprint Backlog:** Essentially another “todo list” that keeps the set of user stories that will be used for the next sprint.

As seen from the terminology, the Scrum method is based on well-defined iterations (Sprints) and each sprint is composed by the following phases:

- **Sprint Planning:** You gather the product backlog and eventually the previous sprint backlogs and decide what to implement in the upcoming sprint;
- **Daily Scrum:** A daily stand-up meeting that lasts around 15 minutes where a check on the daily progress is done;
- **Sprint Review:** After the sprint is completed, we have the verification and validation of the products of the sprint (both software and documents);
- **Sprint Retrospective:** A quality control on the sprint itself is done, allowing for continuous improvement over the way of working.

6.4.5.2.1 Criticisms to the Scrum approach

The Scrum approach can quickly become chaotic if User Stories and Backlogs are not well kept and clear. Also, no matter how short it can be, the Daily Scrum is still an invasive practice that interrupts the workflow and requires everyone to be present and ready.

6.4.5.3 Kanban

Kanban is an Agile Development approach taken by the scheduling system used for lean and just-in-time manufacturing implemented at Toyota.

The base of Kanban is the “Kanban Board” (sometimes shortened as “Kanboard”), where plates (also called “cards” or “tickets”) are moved through swimlanes that can represent:

- The status of the card (To Do, Doing, Testing, Done)
- The Kind of Work (Frontend, Backend, Database, …)

- The team that is taking care of the work

The board helps with organization and gives a high-level view of the work status.

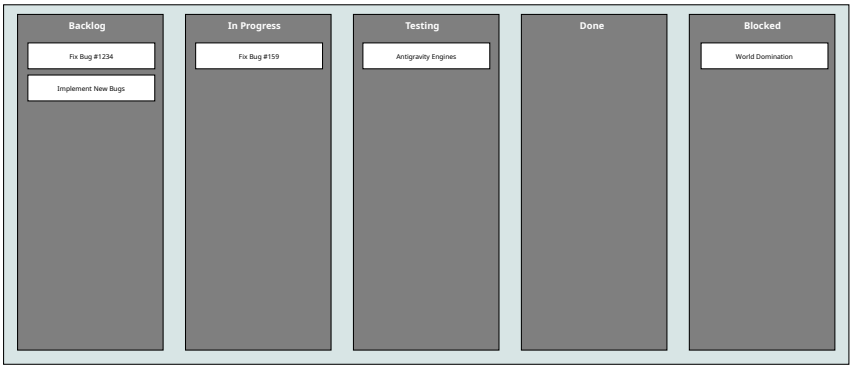


Figure 142: Example of a Kanban Board

6.4.5.4 ScrumBan

ScrumBan is a hybrid approach between Scrum and Kanban, mixing the Daily Scrum and Sprint Approach with the Kanban Board.

This approach is usually used during migration from a Scrum-Based approach to a purely Kanban-based approach.

6.4.6 Lean Development

Lean development tries to bring the principles of lean manufacturing into software development. The basis of lean development is divided in 7 principles:

- **Remove Waste:** “waste” can be partial work, useless features, waiting, defects, work changing hands...
- **Amplify Learning:** coding is seen as a learning process and different ideas should be tested on the field, giving great importance to the learning process;
- **Decide late:** the later you take decisions, the more assumptions and predictions are replaced with facts, Also strong commitments should happen as late as possible, as they will make the system less flexible;
- **Deliver early:** technology evolves rapidly, and the one that survives is the fastest. If you can deliver your product free from defects as soon as possible you will get feedback quickly, and get to the next iteration sooner;
- **Empower the team:** managers are taught to listen to the developers, as well as provide suggestions;
- **Build integrity in:** the components of the system should work well together and give a cohesive experience, giving the customer and impression of integrity;
- **Optimize the whole:** optimization is done by splitting big tasks into smaller ones which helps finding and eliminating the cause of defects.

6.4.7 Where to go from here

Obviously the models presented are not set in stone, but are “best practices” that have been proven to help with project management, and not even all of them.

Nothing stops you from taking elements of a model and implement them into another model. For example you could use an Evolutionary Model with a Kanban board used to manage the single increment.

6.5 Version Control

When it comes to managing any resource that is important to the development process of a software, it is vitally important that a version control system is put in place to manage such resources.

Code is not the only thing that we may want to keep under versioning, but also documentation can be subject to it.

Version Control Systems (VCS) allow you to keep track of edits in your code and documents, know (and blame) users for certain changes and eventually revert such changes when necessary. They also help saving on bandwidth by uploading only the differences between commits and make your development environment more robust (for instance, by decentralizing the code repositories).

The most used Version Control system used in coding is Git, it’s decentralized and works extremely well for tracking text-based files, like code or documentation, but thanks to the LFS extension it is possible for it to handle large files efficiently.

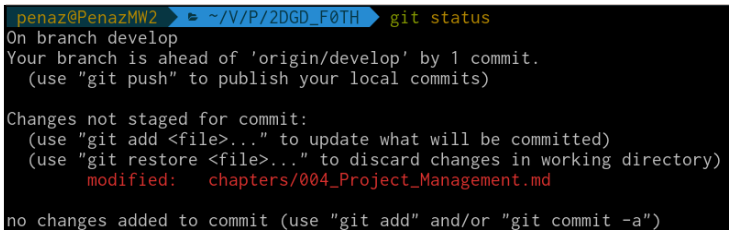


Figure 143: An example screen from Git, a version control system

Other used version control systems are Mercurial and SVN (subversion).

Another useful feature of many version control systems are remote sources, which allow you to upload and synchronize your repositories with a remote location (like GitHub, GitLab or BitBucket for instance) and have it safe on the cloud, where safety by redundancy is most surely ensured.

6.6 Metrics and dashboards

During development you need to keep an eye on the quality of your project, that’s when you need a **project dashboard**: but before that, you need to decide what your **quality metrics** are, that means the measurements that define if your project is “up to par” with what you expect or not.

6.6.1 SLOC

This is probably the simplest metric out there: The "Source Line of Code" (SLOC). It is used to measure the size of a program by counting its lines of code. Once Bill Gates said the following:

> Measuring programming progress by lines of code is like measuring aircraft building progress by weight.

An aircraft must be lightweight and robust, and being heavier than necessary will stop it from flying. The same reasoning should be applied here: a longer source code doesn't mean a better product.

It is important to strike a balance between "readability" and "brevity": your code should be short, but being source code, it is still meant for humans to read, so readability matters more than brevity.

Usually the SLOC metric is used to give a "order of magnitude" impression of the program: considering 2 programs that do exactly the same thing, one is 10.000 lines of code, the other one is 100.000, you may start to suspect that the bigger program is more (probably uselessly) complex and less maintainable.

6.6.2 Cyclomatic Complexity

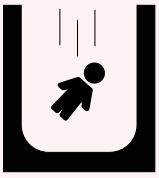
More precisely called "McCabe's Cyclomatic Complexity", this metric defines the number of linearly independent paths through a program's source code: the higher the metric, the higher is the number of paths a piece of code can take in its elaboration.

This means that a higher number of paths takes into account a higher number of conditions and decisions and when such number becomes too high, the code becomes hard to maintain.

The maximum complexity suggested is 10, although sometimes it's good to relax such metric to a maximum of 15. When the cyclomatic complexity becomes higher than the maximum value, it is suggested to split the module into smaller, more maintainable modules.

Your IDE, if advanced enough, should already be able to warn you of a high cyclomatic complexity.

Pitfall Warning!



Be mindful that cyclomatic complexity may have issues of "over-estimation" or "under-estimation", depending on a case-by-case basis. McCabe's cyclomatic complexity is far from a "silver bullet" that will suit all your needs, but as all other metrics, it can give a pointer over where refactoring may be necessary.

6.6.2.1 How cyclomatic complexity is calculated

Advanced Wizardry!

This section contains the technical explanation on how to calculate cyclomatic complexity. If you're not interested in this, feel free to gloss over this section.

As people say, an example is worth a thousand words, so let's take the following UML activity diagram, that represents a simple program (I made it a bit more complex for the sake of demonstration).

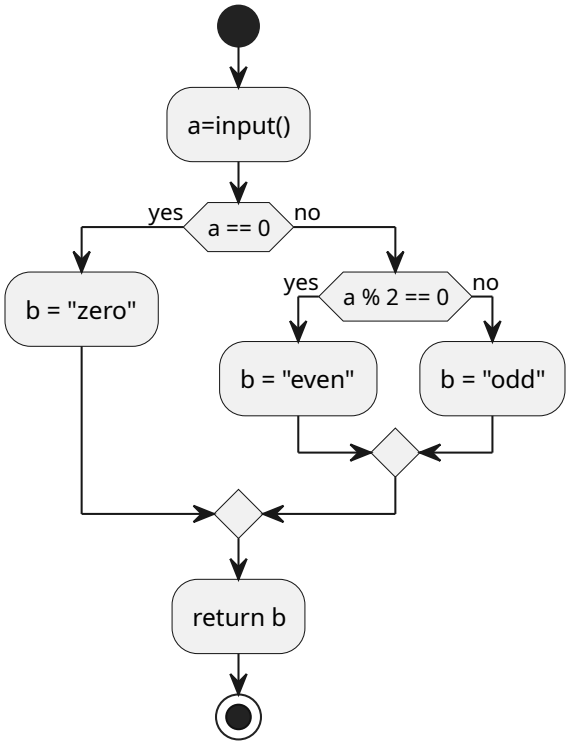


Figure 144: UML of the program which we'll calculate the cyclomatic complexity of

First of all, we need to convert it into the corresponding flow diagram, which usually means eliminating the start nodes and merge nodes used by UML. The result should look something like the following:

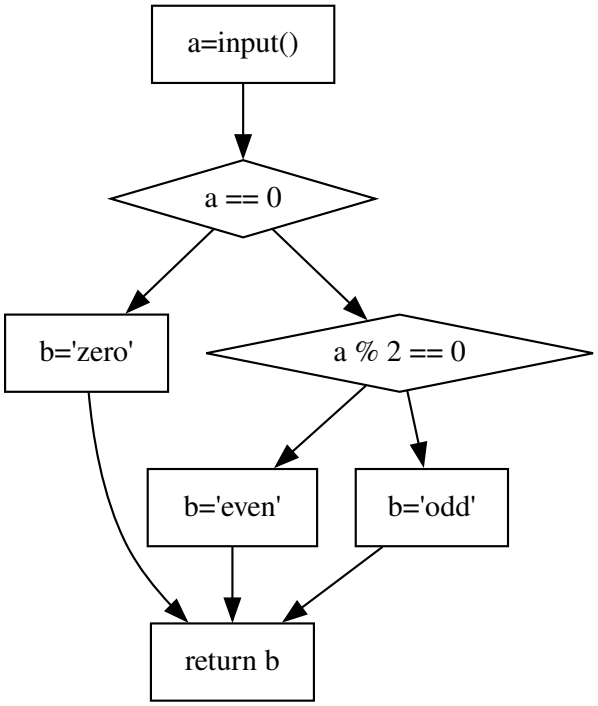


Figure 145: Flow diagram of the program we'll calculate the cyclomatic complexity of

Now we need to count 3 things:

- The number of "nodes": that is the number of boxes and diamonds in our flow diagram. In our case it is 7.
- The number of "edges": that is the number of arrows that connect the nodes in our flow diagram. In our case it is 8.
- The number of "exit points": usually that is the number of stop nodes in our UML diagram, in our flow diagram it's the number of return statements. In our case it is 1.

Now we need to apply the following formula: $C = E - N + 2 \cdot P$.

This formula can be explained as follows:

$$Cyclomatic\,Complexity\ =\ Edges\ -\ Nodes\ +\ 2\ \cdot\ Exit\ Points$$

In our case we have: $C = 8 - 7 + 2 \cdot 1 = 3$

Usually a complexity lower than 15 is considered OK, but also the lower the better.

6.6.3 Code Coverage

When you have a test suite, you may already be thinking about a metric that tells you how much of your code is tested. Well, here it is: the *code coverage* metric tells you what percentage of your code base has been run when executing a test suite.

That is both the useful and damaging part of this metric: *code coverage* doesn't tell you **how well** your code is tested, just **how much code was executed**, so it's easy to incur into what I like to call "incidental coverage": the code coverage presents a higher value, when the code is merely "executed" and not thoroughly "tested".

Code coverage is split in many "sub-sets", like:

- **Statement Coverage:** how many statements of the program are executed;
- **Branch Coverage:** defines which branches (as in portions of the if/else and "switch" statements) are executed;
- **Function Coverage:** how many functions or subroutines are called.

This is also why it's better to prepare unit tests first, and delay the integration tests for a while.

To know more about those terms, head to the testing section.

6.6.4 Code Smells

Code Smells is a blanket term representing all the common (and thus known) mistakes done in a certain programming language, as well as bad practices that can be fixed more or less easily.

Some of these smells can be automatically detected by static analysis programs (sometimes called Linters), others may require dynamic execution, but all code smells should be solved at their root, since they usually entail a deeper problem.

Among code smells we find:

- Duplicated Code;
- Uncontrolled Side Effects;
- Mutating Variables;
- God Objects;
- Long Methods;
- Excessively long (and thus complex) lines of code.

6.6.5 Coding Style infractions

When you are collaborating with someone, it is absolutely vital to enforce a coding style, so that everyone in the team is able to look at everyone else's code without having to put too much effort into it.

Coding style can be enforced via static analysis tools, when properly configured.

Counting (automatically) the number of coding style infractions can help you estimate how much effort working on the code is necessary, thus you would be able to foresee slowdowns in the development process.

6.6.6 Depth of Inheritance

Some people say that inheritance is evil and should be avoided, some other say it's good. As with all things, *in medio stat virtus* (virtue stands in the middle), sometimes inheritance is better left where it is, other times its usage

is necessary for things to make sense.

The *depth of inheritance* metric tells us how deep the inheritance hierarchy is, thus this metric will tell the us the strength of one of the possible dependency types. The deeper the inheritance, the more dependencies we have, which means that we have more classes that, if edited, will change the behavior of the "children classes".

It's better having a short inheritance depth, (although it's not necessarily wrong) having a longer chain of dependencies might mean we have a structural problem, where some classes are "too generic" and at the top of the hierarchy we have some kind of "universal object".

6.6.7 Number of methods / fields / variables

Let's talk numbers: having too many methods or fields in a class can be an indicator of a so-called "god object": an object that has too many responsibilities under its wing (does too many things), this is a breach of the *single responsibility principle* and should be avoided.

We can fix this by splitting the class into smaller classes, each with its own single responsibility.

A high number of local variables instead may point to a complexity issue: your algorithm may be more complex than needed, or needs to be split into different functions.

6.6.8 Number of parameters

This metric is specific for functions, when a function has a lot of parameters, it's harder to call and harder to understand. Functions should have no more than 5 parameters in most cases, more and it will be complex.

Some automated tools in your IDE may be able to warn you in case methods and functions have too many parameters.

To solve this issue, you may need to review the function (maybe it has too many responsibilities?) or pass a so-called "complex structure" to it (thus merging all the parameters into one).

6.6.9 Other metrics

The metrics listed above are not the only ones available to you, some IDEs have aggregated metrics (like the "maintainability index" in Visual Studio), while there may be other metrics you want to measure, some follow:

- **Lead Time:** Time elapsed between the start and end of a process (may be a ticket, or a task);
- **MTBF:** (Mean Time Before Failure) represents the mean time before the software crashes;
- **Crash Rate:** The number of times a software crashes, over the number of times it's used.

7 Writing a Game Design Document

> If you don't know where you are going. How can you expect to get there?
>
> *Basil S. Walsh*

One of the most discussed things in the world of Game Development is the so-called "GDD" or "Game Design Document". Some say it's a thing of the past, others swear by it, others are not really swayed by its existence.

Being an important piece of any software development process, in this book we will talk about the GDD in a more flexible way.

7.1 What is a Game Design Document

The Game Design Document is a Body Of Knowledge that contains everything that is your game, and it can take many forms, such as:

- A formal design document;
- A *Wiki*[g];
- A *Kanboard*[g];
- A collection of various files, including spreadsheets.

The most important thing about the GDD is that it contains all the details about your game in a centralized and possibly easy-to-access place.

It is not a technical document, but mostly a design document, technical matters should be moved to a dedicated "Technical Design Document".

7.2 Possible sections of a Game Design Document

Each game can have its own attributes, so each Game Design Document can be different, here we will present some of the most common sections you can include in your own Game Design Document.

7.2.1 Project Description

This section is used to give the reader a quick description of the game, its genre (RPG, FPS, Puzzle,...), the type of demographic it covers (casual, hardcore, ...). Additional information that is believed to be important to have a basic understanding of the game can be put here.

This section should not be longer than a couple paragraphs.

A possible excerpt of a description could be the following:

> This game design document describes the details for a 2D side scrolling platformer game where the player makes use of mechanics based on using arrows as platforms to get to the end of the level.
> The game will feature a story based on the central America ancient culture (Mayan, Aztec, ...).
> The name is not defined yet but the candidate names are:

7.2.2 Characters

If your game involves a story, you need to introduce your characters first, so that everything that follows will be clear.

A possible excerpt of a characters list can be the following:

> **Ohm** is the main character, part of the group called “The Resistance” and fights for restoring the electrical order in the circuit world.
> **Fad** is the main side character, last survivor and heir of the whole knowledge of “The Capacitance” group. Its main job is giving technical assistance to Ohm.
> **Gen. E. Rator** is the main antagonist, general of “The Reactance” movement, which wants to conquer the circuit world.

This can be a nice place where to put some character artwork.

If your game does not include a story, you can just avoid inserting this section altogether.

7.2.3 Storyline

After introducing the characters, it’s time to talk about the events that will happen in the game.

An example of story excerpt can be the one below:

> It has been 500 mega-ticks that the evil **Rator** and the reactance has come to power, bringing a new era of darkness into the circuit world.
> After countless antics by the evil reactance members, part of the circuit world’s population united into what is called “The Resistance”.
> Strong of thousands of members and the collaboration of *the Capacitance*, the resistance launched an attack against the evil reactance empire, but the empire stroke back with a carpet surcharge attack, decimating the resistance and leaving only few survivors that will be tasked to rebuild the resistance and free the world from the reactance’s evil influence.
> This is when a small child, and their parents were found. The child’s name, **Ohm**, sounded prophetic of a better future of the resistance.
> And this is where our story begins.

As with the Characters section, if your game does not include a story, you can just skip this section.

7.2.3.1 The theme

When people read the design document, it is fundamental that the game’s theme is quickly understood: it can be a comedy-based story, or a game about hardships and fighting for a better future, or maybe it is a purely fantastic game based on ancient history…

Here is a quick example:

> This is a game about fighting for a better future, dealing with hardships and the deep sadness you face when you are living in a world on the brink of ruin.
> This game should still underline the happiness of small victories, and give a sense of “coziness” in such small things, even though the world can feel cold.

If you feel that this section is not relevant for your game, you can skip it.

7.2.3.2 Progression

After defining the story, you should take care of describing how the story progresses as the player furthers their experience in a high-level fashion.

An example:

> The game starts with an intro where the ruined city is shown to the player and the protagonist receives their magic staff that will accompany them through the game.
> The first levels are a basic tutorial on movement, where the shaman teaches the player the basic movement patterns as well as the first mechanic: *staff boosting*. Combat mechanics are taught as well.
> After the tutorial has been completed, the player advances to the first real game area: **The stone jungle**.
> ...

7.2.4 Levels and Environments

In this section we will define how levels are constructed and what mechanics they will entail, in detail.

We can see a possible example here:

> The First Level (Tutorial) is based in a medieval-like (but adapted to the center-America theme) training camp, outside, where the player needs to learn jumping, movement and fight straw puppets. At the end of the basic fighting and movement training, the player is introduced to *staff boosting* which is used to first jump to a ledge that is too high for a normal jump, and then the mechanic is used to boost towards an area too far forward to reach without boosting.
> ...

Some level artwork can be included in this section, to further define how the levels will look and feel.

7.2.5 Gameplay

This section will be used to describe your gameplay. This section can become really long, but do not fear, as you can split it in meaningful sections to help with organization and searching.

7.2.5.1 Goals

> Why is the player playing your game?

This question should be answered in this section. Here you insert the goals of your game, both long and short term.

An example could be the following:

> Long Term Goal: Stop the great circuit world war
>
> Optional Long Term Goal: Restore the circuit world to its former glory.
>
> Short Term Goals:
>
> - Find the key to the exit
> - Neutralize Enemies
> - Get to the next level

7.2.5.2 Game Mechanics

In this section, you describe the core game mechanics that characterize the game, extensively. There are countless resource on how to describe game mechanics, but we'll try to add an example here below.

> The game will play in the style of the well-known match-3 games. Each match of 3 items will add some points to the score, and new items will "fall" from a randomly chosen direction every time.
> Every time an "L" or a "T" match is performed, a special item of a random color will be generated, when a match including this item is made, all the items in the same row and column will be deleted and bonuses will be awarded.
> Every time a match with 4 items in a row is performed, a special item of a random color will be generated, when a match including such item is made, all items in a 3x3 grid centered on the item will be deleted and bonuses will be awarded.
> Every time a match with 5 items in a row is performed, a special uncolored item will be generated, this can be used as a "wildcard" for any kind of match.
> In case the 5-item special is matched with any other special item, the whole game board will be wiped and a bonus will be awarded.
> ...

7.2.5.3 Skills

Here you will describe the skills that are needed by the users in order to be able to play (and master) your game.

This will be useful to assess your game design and eventually find if there are some requirements that are too high for your target audience; for instance asking a small child to do advanced resource management could be a problem.

This will also help deciding what the best hardware to use your game on could be, for instance if your game requires precise inputs for platforming then touch screens may not be the best option.

Here's an example of such section:

> The user will need the following skills to be able to play the game effectively:
>
> - Pressing Keyboard Buttons or Joypad Buttons
> - Puzzle Solving (for the “good ending” overarching puzzle)
> - Timing inputs well (for the sections with many obstacles)
>
> ...

7.2.5.4 Items/Powerups

After describing the basic game mechanics and the skills the user needs to master to be able to play the game effectively, you can use this section to describe the items and powerups that can be used to alter the core gameplay.

For example:

> The player can touch a globular light powerup to gain invincibility, every enemy that will touch the player will get automatically killed. The powerup duration is 15 seconds.
> Red (incendiary) arrows can be collected through the levels, they can get shot and as soon as they touch the ground or an enemy, the burst into flames, similarly to a match.
> ...

In this section you describe all items that can be either found or bought from an in-game store or also items derived from micro-transactions. In-game currency acquisition should be mentioned here too, but further detailed in the monetization section.

7.2.5.5 Difficulty Management and Progression

This section can be used to manage how the game gets harder and how the player can react to it. This will expand on game mechanics like leveling and gear.

This section is by its own nature quite subjective, but describing how the game progresses helps a lot during the tighter parts of development.

Below a possible example of this section:

> The game will become harder by presenting tougher enemies, with more armor, Health Points and attack. To overcome this difficulty shift, the player will have to create defense strategy and improve their dodging, as well as leveling up their statistics and buy better gear from the towns’ shops.
> In the later levels, enemies will start dodging too, and will also be faster. The player will need to improve their own speed statistic to avoid being left behind or “kited” by fast enemies.
> As the game progresses, the player will need to acquire heavy weapons to deal with bigger bosses, as well as some more efficient ranged weapons to counteract ranged enemies.
> ...

This section is good if you want to talk about unlocking new missions/maps/levels too.

7.2.5.6 Losing Conditions

Many times we focus so much on how the player will get to the end of the game that we absolutely forget how the player can *not* get to the end of the game.

Losing conditions must be listed and have the same importance of the winning conditions, since they add to the challenge of the game itself.

A possible example of how a "losing conditions" section could be written is the following:

> The game can be lost in the following ways:
>
> - Losing all the lives and not "continuing" (Game Over)
> - Not finding all the Crystal Oscillators (Bad Ending)

An interesting idea could be having an "endings" section inside your game, where all endings (both good, bad and neutral) are listed, encouraging the player to pull themselves out from the "losing condition" that is a bad ending.

7.2.6 Graphic Style and Art

Here we describe the ideas on how the game will look like. Describing the graphic style and medium.

Here is a possible example of the game:

> This is a 2D side scroller with a dark theme, the graphics should look gloomy and very reminiscing of a circuit board.
> The graphical medium should be medium-resolution pixel art, allowing the player's imagination to "fill in" the graphics and allowing to maintain a "classic" and "arcade" feeling.
> ...

7.2.7 Sound and Music

Sadly, in way too many games, music and sound is an afterthought. A good soundtrack and sound effect can really improve the immersion, even in the simplest of games.

In this section we can describe in detail everything about Music and Sound Effects, and if the section becomes hard to manage, splitting it in different sub-sections could help organization.

> Music should be based on the glitch-hop style, to complement the electronic theme. 8 or 16-bit style sounds inside the score are preferable to modern high-quality samples.
> Sound effects should appeal to the 8 or 16-bit era.
> Lots of sound effects should be used to give the user positive feedback when using a lever to open a new part of the level, and Extra Lives/1UP should have a jingle that overrides the main music.

7.2.8 User Interface

In this section we will describe everything that concerns the User Interface: menus, HUD, inventories and everything that will contribute to build the user experience that is not strictly tied to the gameplay.

This is especially important in games that make heavy use of menus, like turn-based strategy games or survival games where inventory management can be fundamental.

Let's see an example of how this section can be written:

> The game will feature a cyberpunk-style main menu, looking a lot like an old green-phosphor terminal but with a touch of futurism involved. The game logo should be visible on the left side, after a careful conversion into pixel-art. On the right, we see a list of buttons that remind old terminal-based GUIs. On the bottom of the screen, there should be an animated terminal input, for added effect.
> Every time a menu item is highlighted or hovered by the mouse, the terminal input will animate and write a command that will tie to the selected menu voice, such as:
>
> - Continue Game: `./initiate_mission.bin -r`
> - Start Game: `./initiate_mission.bin --new`
> - Options: `rlkernel_comm.bin --show_settings`
> - Exit: `systemcontrol.bin --shutdown`
>
> The HUD display should remind a terminal, but in a more portable fashion, to better go with the "portability" of a wrist-based device.

It's a good idea to add some mock designs of the menu in this section too.

7.2.9 Game Controls

In this section you insert everything that concerns the way the game controls, eventually including special peripherals that may be used.

This will help you focusing on better implementing the input system and limit your choices to what is feasible and useful for your project, instead of just going by instinct.

Below, a possible way to write such section

> The game will control mainly via mouse and keyboard, using the mouse to aim the weapon and shoot and keyboard for moving the character.
> Alternatively, it's possible to connect a twin-stick gamepad, where the right stick moves the weapon crosshair, while the left stick is used to move the character, one of the back triggers of the gamepad can be configured to shoot.
> If the gamepad is used, there will be a form of aim assistance can be enabled to make the game more accessible to gamepad users.

7.2.10 Accessibility Options

Here you can add all the options that are used to allow more people to access your game, in more ways than you think.

Below, we can see an example of many accessibility options in a possible game.

> The game will include a “colorblind mode”, allowing the colors to be colorblind-friendly: such mode will include 3 options: Deuteranopia, Tritanopia and Monochromacy.
> Additionally, the game will include an option to disable flashing lights, making the game a bit more friendly for people with photosensitivity.
> The game will support “aim assistance”, making the crosshair snap onto the enemy found within a certain distance from the crosshair.
> In order to assist people who have issues with the tough platforming and reaction times involved, we will include the possibility to play the game at 75%, 50% and 25% speed.

7.2.11 Tools

This section is very useful for team coordination, as having the same toolkit prevents most of the “works for me” situations, where the game works well for a tester/developer while it either crashes or doesn’t work correctly for others.

This section is very useful in case we want to include new people in our team and quickly integrate them into the project.

In this section we should describe our toolkit, possibly with version numbers included (which help reducing incompatibilities), as well as libraries and frameworks. The section should follow the trace below:

> The tools and frameworks used to develop the game are the following:
> **Pixel Art Drawing:** Aseprite 1.2.13
> **IDE:** Eclipse 2019-09
> **Music Composition:** Linux Multimedia Studio (LMMS) 1.2.1
> **Map and level design:** Tiled 1.3.1
> **Framework:** SFML 2.5.1
> **Version Control:** Git 2.24.0 and GitLab

7.2.12 Marketing

This section allows you to decide how to market the game and have a better long-term plan on how to market your game to your players.

Carefully selecting and writing down your target platforms and audience allows you to avoid going off topic when it comes to your game.

7.2.12.1 Target Audience

Knowing who is your target audience helps you better suit the game towards the audience that you are actually targeting.

Here is an example of this section:

> The target audience is the following:
> Age: 15 years and older
> Gender: Everyone
> Target players: Hardcore 2D platformer fans

7.2.12.2 Available Platforms

Here you describe the launch platforms, as well as the platforms that will come into the picture after the game launched. This will help long term organization.

Here is an example of how this section could look:

> Initially the game will be released on the following platforms:
>
> - PC
> - Playstation 4
>
> After launch, we will work on the following ports:
>
> - Nintendo Switch
> - XBox 360
>
> After working on all the ports, we may consider porting the game to mobile platforms like:
>
> - Android 9.0 +
> - iOS 11.0 +
>
> ...

7.2.12.3 Monetization

In this optional section you can define your plans for the ways you will approach releasing the game as well as additional monetization strategies for your game.

For example:

> The game will not feature in-game purchases.
> Monetization efforts will be focused on selling the game itself at a full “indie price” and further monetization will take place via substantial Downloadable Content Expansions (DLC)
> The eventual mobile versions will be given away for free, with advertisements integrated between levels. It is possible for the user to buy a low-price paid version to avoid seeing the advertisements.

7.2.12.4 Internationalization and Localization

Internationalization and Localization are a matter that can make or break your game, when it comes to marketing your game in foreign countries.

Due to political and cultural reasons, for instance you shouldn't use flags to identify languages. People from territories inside a certain country may not be well accepting of seeing their language represented by the flag of their political adversaries.

Another example could be the following: if your main character is represented by a cup of coffee, your game could be banned somewhere as a "drug advertisement".

This brings home the difference between "Internationalization" and "Localization":

Internationalization Making something accessible across different countries without major changes to its content
Localization Making something accessible across different countries, considering the target country's culture.

We can see a possible example of this section below:

> The game will initially be distributed in the following languages:
>
> - English
> - Italian
>
> After the first release, there will be an update to include:
>
> - Spanish
> - German
> - French

7.2.13 Other/Random Ideas

This is another optional section where you can use as a "idea bin", where you can put everything that you're not sure will ever make its way in the game. This will help keeping your ideas on paper, so you won't ever forget them.

We can see a small example here:

> Some random ideas:
>
> - User-made levels
> - Achievements
> - Multiplayer Cooperative Mode
> - Multiplayer Competitive Mode

7.3 Where to go from here

This chapter represents only a guideline on what a Game Design Document can be, feel free to remove any sections that don't apply to your current project as well as adding new ones that are pertinent to it.

A Game Design Document is a Body of Knowledge that will accompany you throughout the whole game development process and it will be the most helpful if you are comfortable with it and it is shaped to serve you.

Part 3: Game Development Basics

8 The Game Loop

> All loops are infinite ones for faulty RAM modules.
>
> *Anonymous*

8.1 The Input-Update-Draw Abstraction

Animations and movies are an illusion, and so are games. Games and movies show still images tens of times per second, giving us the illusion of movement.

Any game and its menus can be abstracted into 3 main operations that are performed one after the other, in a loop:

1) Process the user input
2) Update the world (or menu) status
3) Display (Draw) the updated world (or again, menu) to the screen

We call such abstraction the “game loop”.

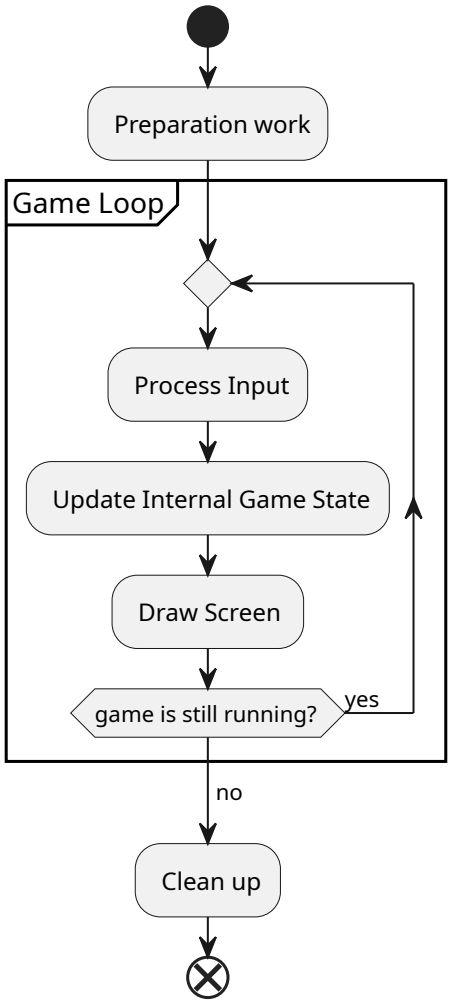


Figure 146: UML Diagram of the input-update-draw abstraction

So a pseudocode implementation of such loop would be something like the following:

Listing 40: Game Loop example

```
function game(){
    let game_is_running = true;
    while (game_is_running){
        process_user_input();
        update_world();
        draw();
    }
}
```

This abstraction will become really useful when dealing with many rows of code and keeping it neatly organized.

8.2 Input

8.2.1 Events vs Real Time Input

Some frameworks may be able to further abstract how they process input by giving an *API*$_{[g]}$ that allows to make use of **events**.

Most of the time, events will be put in a queue that will be processed separately. This way it's easier to program how to react to each event and keep our code neatly organized. The downside is that the performance of an event-driven input processing is directly tied to how many events are triggered: the more events are triggered, the longer the wait may be before we get to our processed input.

This usually depends on the implementation of the event queue: an event queue is less wasteful in terms of resources and allows for less coupled code, but the queue could be cluttered with events we're not interested in (for instance mouse movement events in a game that uses only keyboard for controls) so we need to take the time to configure our event handler to ignore certain events when not necessary.

Note!

A well-configured event-based input system **is the most efficient way of doing things**, allowing code to be executed only when necessary.

On the opposite side, we have so-called "real-time input", where at a certain point of our update routine, we check for the instantaneous status of the input peripherals and process it immediately. This allows for a faster, more reactive code and to apply some different logic (for instance pressing left and right on the keyboard can be coded to make the character stop). Besides being more immediate, this system shares a lot of traits with "polling" which can be performance-heavy, as well as inducing some undesired code coupling.

Again, a well-implemented and well-configured event-based system should feel no different from real-time input, with the advantage of having better performance and having less code coupling.

8.3 Timing your loop

When it comes to anything that remotely relates to physics (that includes video games), we need to set the relation to time in our loop. There are many ways to set our delta time (or time steps), we'll see some of the most common.

8.3.1 What is a time step

A time step (or delta time) is a number that will define "how much time passed" between two "snapshots" of our world (remember, the world is updating and showing in discrete intervals, giving the illusion of movement). This number will allow us to make our loop more flexible and react better to the changes of load and machines.

8.3.2 Fixed Time Steps

The first and simplest way is to use a fixed time step, our delta time is fixed to a certain number, which makes the simulation easier to calculate but also makes some heavy assumptions:

- Vertical Synchronization is active in the game
- The PC is powerful enough to make our game work well, 100% of the time

An example of fixed time step loop can be the following (assuming 60 frames per second or $dt = \frac{1}{60}$):

Listing 41: Game loop with fixed timesteps

```
let dt = 1.0/60.0;
let game_is_running = True;

while (game_is_running){
    process_user_input();
    update_world(dt);
    draw();
}
```

Everything is great, until our computer starts slowing down (high load or just not enough horsepower), in that case the game will slow down.

This means that every time the computer slows down, even for a microsecond, the game will slow down too, which can be annoying.

Note!

A similar problem can apply between different computers: if `computer A` can run the game at 30fps maximum, while `computer B` will run at 120fps (and we don't account for that), using fixed timesteps the game will run 4 times as fast on `computer B`.

8.3.3 Variable Time Steps

A way to limit the issues given by a fixed time step approach is to make use of variable time steps, which are simple in theory, but can prove hard to manage.

The secret is measuring how much time passed between the last frame and the current frame, and use that value to update our world.

An example in pseudocode could be the following:

Listing 42: Game loop with variable time steps

```
let game_is_running = True;

// We initialize our dt at 1/60th of a second for the first loop
let dt = 1.0/60.0;

while (game_is_running){
    // We get the system time in milliseconds
    let begin = Date.now();
    process_user_input();
    update_world(dt);
    draw();
    let end = Date.now();
    // We update our dt
    dt = end - begin;
}
```

This allows to smooth the possible lag spikes, even allowing us to disable Vertical Sync and have a bit less input lag, but this approach has some drawbacks too.

Since the delta time now depends on the speed of the game, the game can "catch up" in case of slowdowns; that can result in a slightly different feeling, depending on the framerate, but if there is a really bad slowdown `dt` can become really big and break our simulation, and collision detection will probably be the first victim.

Also this method can be a bit harder to manage, since every movement will have to be scaled with `dt`.

8.3.4 Semi-fixed Time Steps

This is a special case, where we set an upper limit for our time steps and let the update loop execute as fast as possible. This way we can still simulate the world in a somewhat reliable way, avoiding the dangers of higher spikes.

A semi-fixed time step approach is the following (assuming 60 fps or $dt = \frac{1}{60}$):

Listing 43: Game loop with Semi-Fixed time steps

```
let dt = 1.0/60.0;
let game_is_running = true;

// We bootstrap frametime for 1/60th of a second for the first frame
```

```
let frametime = 1.0/60.0;

while (game_is_running){
    // We get the system time in milliseconds
    let begin = Date.now();

    while (frametime > 0.0){
        let deltaTime = min(dt, frametime);
        process_user_input();
        update_world(deltaTime);
        frametime = frametime - deltaTime;
    }
    draw();
    let end = Date.now();
    // We memorize how long this frame lasted
    frametime = end - begin;
}
```

This way, if the loop is running too slow, the game will slow down and the simulation won't blow up. The main disadvantage of this approach is that we're taking more update steps for each draw step, which is fine if drawing takes more than updating the world. If instead the update phase of the loop takes more than drawing it, we will spiral into a terrible situation.

We can call it a "spiral of death", where the simulation will take Y seconds (real time) to simulate X seconds (of game time), with $Y > X$, being behind in your simulation makes the simulation take more steps, which will make the simulation fall behind even more, thus making the simulation lag behind more and more.

8.3.5 Frame Limiting

Frame limiting is a technique where we aim for a certain duration of our game loop. If an iteration of the game loop is faster than intended, such iteration will wait until we get to our target loop duration.

Let's again consider a loop running at 60fps (or $dt = \frac{1}{60}$):

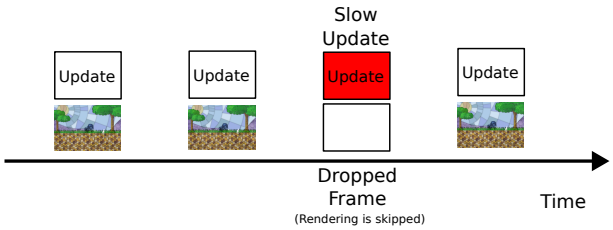
Listing 44: Game loop with Frame Limiting

```
let targetTime = 1.0/60.0;
let game_is_running = True;

// We bootstrap dt to 1/60th of a second for the first frame
let dt = 1.0/60.0;

while (game_is_running){
    // We get the system time in milliseconds
    let begin = Date.now();
    process_user_input();
    update_world(dt);
    draw();
    let end = Date.now();
```

```
14      // We update our dt
15      dt = end - begin;
16      // We assume we have a single-thread sleep() function
17      sleep(Math.max(targetTime - dt, 0));
18  }
```

Even if the frame is limited, it's necessary that all updates are tied to our delta time to work correctly. With this loop the game will run **at most** at 60 frames per second, if there is a slowdown the game will slow down under 60 fps, if the game runs faster it won't go over 60fps.

8.3.6 Frame Skipping/Dropping



A common solution used when a frame takes longer to update and render than the target time is using the so-called "frame dropping". The game won't render the next frame, in an effort to "catch up" to the desired frame rate.

This means that the old frame will stay on screen, which will cause a perceptible visual stutter.

8.3.7 Multi-threaded Loops

Higher budget (AAA) games don't usually use a variation of the "classic" game loop, but instead make use of the capabilities of newer hardware. Using multiple threads (lines of execution) executing at the same time, making everything quicker and the framerate higher.

Multi-threaded loops are created in a way that separates the input-update part of the game loop from the drawing part of it. This way the update thread can take care of updating our simulation, while the drawing/rendering loop can take care of drawing the result to screen.

The catch is that we can't just wait for the input-update thread to finish before rendering, that wouldn't make it quicker than just using a one-threaded game loop: instead we make the rendering thread "lag behind" the input-update thread by *1 frame* - this way while the input-update thread takes care of the frame number n, the drawing thread will be rendering the prepared frame number $n - 1$.

Thread						
Updating	1	2	3	4	5	6
Rendering		1	2	3	4	5

This 1-frame difference between updating and rendering introduces lag that can be quantified between *16.67ms* (at 60fps) and *33.3ms* (at 30fps), which needs to be added with the 2-5 ms of the LCD refresh rate, and other factors that

can contribute to lag. In some games where extreme precision is needed, this could be considered unacceptable, so a single-threaded loop could be considered more fitting.

8.4 Issues and possible solutions

In this section we have a little talk about some common issues related to the game loop and its timing, and some possible solutions

8.4.1 Frame/Screen Tearing

Screen tearing is a phenomenon that happens when the "generate output" stage of the game loop happens in the middle of the screen drawing a frame.

This makes it so that a part of the drawn frame shows the result of an output stage, while another part shows a more updated version of the frame, given by a more recent game loop iteration.

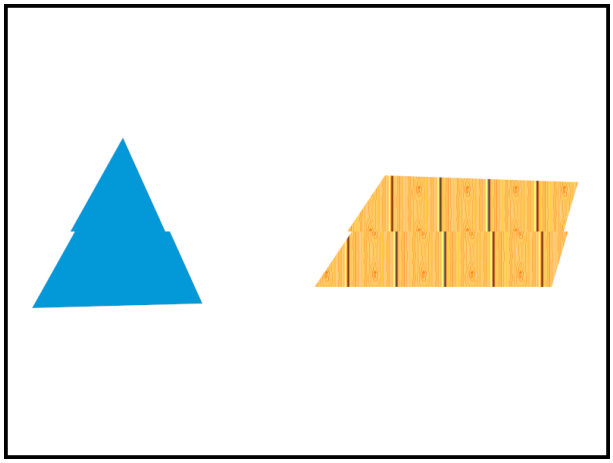


Figure 147: An example of screen tearing

A very common fix for this phenomenon is **double buffering**, where two color buffers are used. While the first is shown on screen, the game loop updates and draws on the second color buffer.

When comes the time to draw the color buffer on screen, an operation called "flipping" is performed, where the second color buffer is shown on screen, so that the game loop can draw on the first color buffer.

To make the game even smoother, a technique called "triple buffering" can be used, which adds a third color buffer is used to make the animation smoother at the cost of a higher input lag.

8.5 Drawing to screen

When drawing to screen, the greatest majority of games make use of what is called the "painter's algorithm", which looks something like the following:

1. Clear the screen
2. Draw The Farthest Background
3. Draw The Second Farthest Background
4. Draw The Tile Map
5. Draw The enemies and obstacles
6. Draw The Player
7. Display everything on screen

If we divide each “layer” we can see how the painter’s algorithm works:

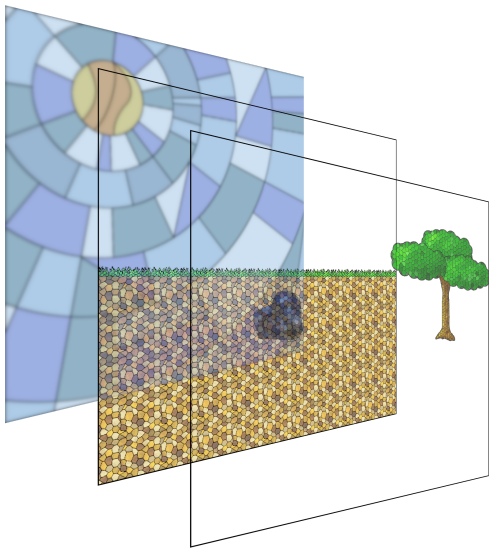


Figure 148: A small example of the “painter’s algorithm”

Just like a real painter, we draw the background items before the foreground ones, layering each one on top of the other. Sometimes games make use of priority queues to decide which items to draw first, other times game developers (usually under the time constraints of a game jam) just hard-code the draw order.

8.5.1 Clearing the screen

Special note about clearing the screen: this is an operation that sometimes may look useless but, like changing the canvas for a painter, clearing the screen (or actually the “buffer” we’re drawing on) avoids a good deal of graphical glitches.

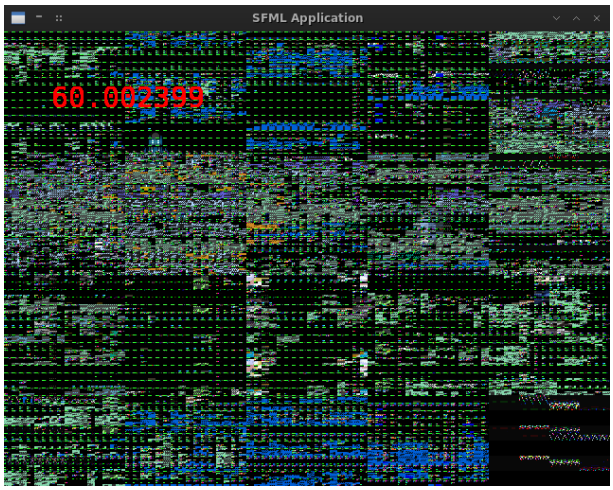


Figure 149: How not clearing the screen can create glitches

In the previous image, we can see how a black screen with only a FPS counter can end up drawing all kinds of glitches when the screen buffer is not cleared: we can clearly see the FPS counter, but the rest of the screen should be empty, instead the GPU is trying to represent residual data from its memory, causing the glitches.

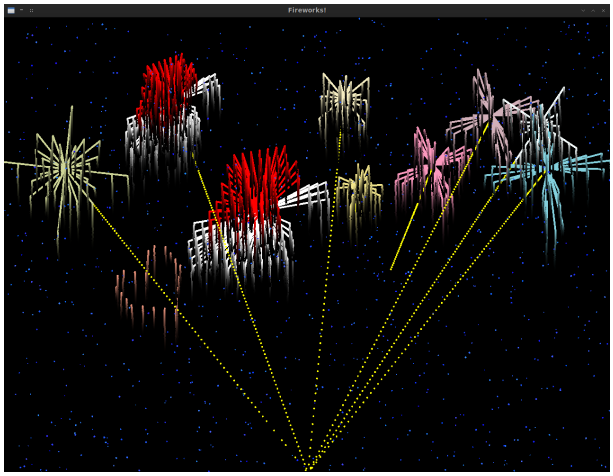


Figure 150: Another type of glitch created by not clearing the screen

If you forget to clear your screen or set a background every frame, the old buffer data will remain on screen, creating a “trail-like” effect on your game, which is probably undesirable.

9 Collision Detection and Reaction

> Every detection of what is false directs us towards what is true: every trial exhausts some tempting form of error.
>
> *William Whewell*

When it comes to collision management, there are two main phases:

- **Collision Detection:** you find out which game objects collided with each other;
- **Collision Reaction:** you handle the physics behind the collision detected, making the game objects react to such collision.

Collisions don't only happen between game objects (two fighters hitting each other), but also between a character and the world (or they would end up just going through the ground).

In this section we'll talk about some ways you can detect and react to collisions.

9.1 Why Collision Detection is done in multiple passes

Collision detection algorithms can be quite costly, even more when you are using a brute force approach, but it's possible to have a more precise collision detection at a lower cost by combining different collision detection algorithms.

The most common way to apply a multi-pass collision detection is by dividing the process in a "broad" and a "fine" pass.

The broad pass can use a very simple algorithm to check for the possibility of a collision, the algorithms used are usually computationally cheap, such as building quad trees.

When the simpler algorithm detects the possibility of a collision, a more precise algorithm is used to check if a collision really happened, usually such finer algorithms are computationally expensive and will benefit from the first "broad pass" filter, thus avoiding useless heavy calculations.

Note!

In this chapter we'll see the easier narrow-pass detection first, followed by the more complex broad-pass algorithms, but remember that a good collision detection system does a "broad-pass" first, before delving into the "narrow-pass".

9.2 Narrow-Phase Collision Detection: did it really collide?

First of all, we need to see how we can make sure that two objects really collide with each other.

Sometimes this presents a (quite common) problem when it comes to precision: computers have no knowledge of infinity (due to their finiteness, see computers are (not) precise). This means that we may need to give some leeway

and define an “acceptable error” in our calculations, thus we will create a “small enough value” (which in math is represented by the Greek letter “epsilon”: ϵ) and change our algorithms accordingly.

9.2.1 Collision Between Two Points

This is the simplest case: points are mono-dimensional objects, and the only way two points can collide is when they have the same coordinates.

An example algorithm would be the following:

Listing 45: Point to point collision detection

```
function point_collision(A, B){
    return A.x == B.x && A.y == B.y;
}
```

This algorithm consists in a constant number of operations, so it runs in O(1).

Since numbers in computers can be **really** precise, a collision between two points may be a bit too precise, so it could prove useful to have a “buffer” around the point, so that we can say that the two points collided when they’re **around the same place**.

In this case, it may prove to be a lot more useful to do a point vs circle detection, or even a circle vs circle collision detection, in that case the “radius” would be the “approximation” of a point.

If instead you want to use a different method that doesn’t involve square roots, you can use epsilon values to have an approximation of the collision. In this case the collision area won’t be round, but square.

Listing 46: Point to point collision detection with epsilon values

```
def point_collision(A, B):
    epsilon = 0.0001  # Let's take a sufficiently low value
    # If both coordinates are "close enough", we trigger a collision.
    # We take the absolute value, just in case some subtractions end up being negative.
    return abs(A.x - B.x) <= epsilon and abs(A.y - B.y) <= epsilon
```

9.2.2 Collision Between A Point and a Circle

Now a circle comes into the mix, a circle has two major characteristics: a **center** and a **radius**.

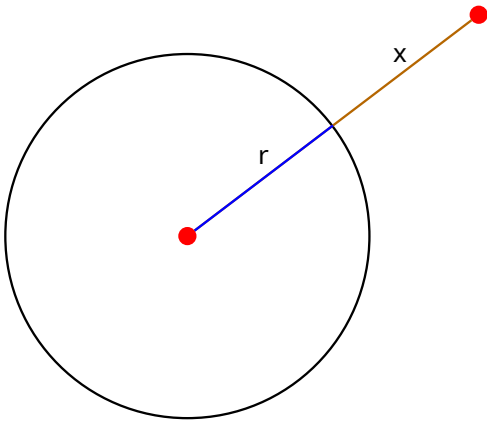


Figure 151: Reference image for Point-Circle Collision detection

We can see that the distance between the center of a circle and our point can be expressed with a formula:

$$d = r + x$$

Where `r` is the circle radius and `x` is the difference of the distance between the center of the circle and the point (which can be negative):

$$x = d - r$$

The point is inside the circle when $x \leq 0$, which means:

$$x \leq 0 \Leftrightarrow d - r \leq 0 \Leftrightarrow d \leq r$$

We can express this in a few words:

> A point is considered inside of a circle when the distance between the point and the center of the circle is *less than or equal* to the radius.

So we need a function that calculates the distance between two points, and then use it to define if a point is inside a circle.

An example could be the following:

Listing 47: Point to circle collision detection

```
class Circle{
    // Let's define a circle class/structure
    constructor(){
        this.center = new Point();
```

```
        this.radius = 0;
    }
}

function distance(A, B){
    // Calculates the distance between two points
    return Math.sqrt((A.x - B.x) ** 2 + (A.y - B.y) ** 2);
}

function circle_point_collision(A, B){
    return distance(A.center, B) <= A.radius;
}
```

Although slightly more heavy, computation-wise, this algorithm still runs in O(1).

9.2.3 Collision Between Two Circles

Let's add another circle into the mix now, and think in more or less the same way as before:

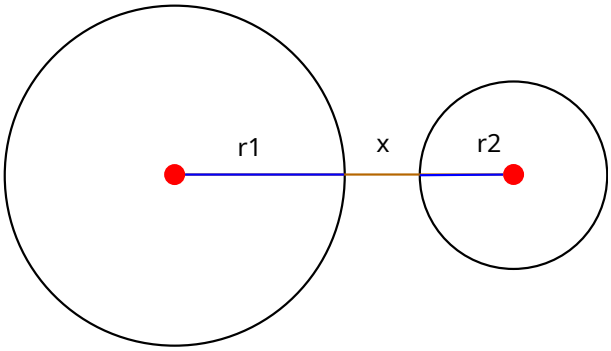


Figure 152: Reference image for Circle-Circle collision detection

We can see the distance between the center of the circles as expressed with the following formula:

$$d = r_1 + x + r_2$$

Where r_1 and r_2 are the radii, and x is defined as follows:

$$x = d - (r_1 + r_2)$$

As before, our x can be negative, which means that the circles are colliding if $x \leq 0$, which means:

$$x \leq 0 \Leftrightarrow d - (r_1 + r_2) \leq 0 \Leftrightarrow d \leq r_1 + r_2$$

We can express the concept in words again:

> Two circles are colliding when the distance between their centers is less or equal the sum of their radii

In pseudo code this would be:

Listing 48: Circle to Circle Collision Detection

```
class Circle{
    // Let's define a circle class/structure
    constructor(){
        this.center = new Point();
        this.radius = 0;
    }
}

function distance(A, B){
    // Calculates the distance between two points
    return Math.sqrt((A.x - B.x) ** 2 + (A.y - B.y) ** 2);
}

function circle_circle_collision(A, B){
    return distance(A.center, B.center) <= A.radius + B.radius;
}
```

Again, this algorithm performs a number of operations that is constant, so it runs in O(1).

9.2.4 Collision Between Two Axis-Aligned Rectangles (AABB)

This is one of the most used types of collision detection used in games: it's a bit more involved than other types of collision detection, but it's still computationally easy to perform. This is usually called the "Axis Aligned Bounding Box" collision detection, or AABB.

Let's start with a bit of theory. We have two squares:

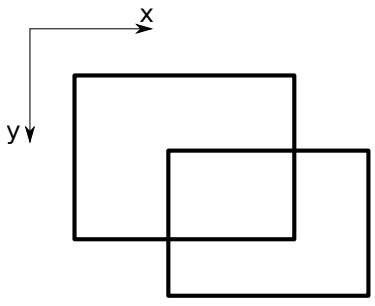


Figure 153: Example used in the AABB collision detection

To know if we may have a collision, we need to check if one of the sides is "inside" (that means between the top and bottom sides) of another rectangle:

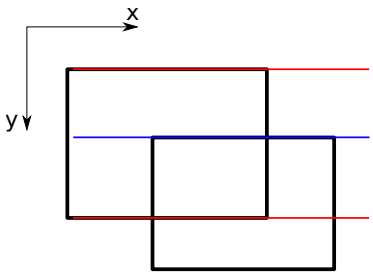


Figure 154: Top-Bottom Check

In this case we know that the “top side” of the second rectangle (highlighted in blue) has a `y` coordinate between the first rectangle’s top and bottom sides’ `y` coordinates (highlighted in red).

Though this is a necessary condition, this is not sufficient, since we may have a situation where this condition is satisfied, but the rectangles don’t collide:

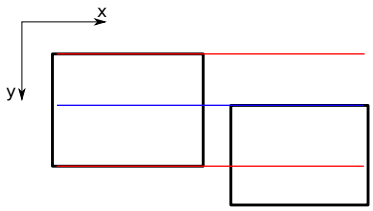


Figure 155: Top-Bottom Check is not enough

So we need to check the other sides also, in a similar fashion:

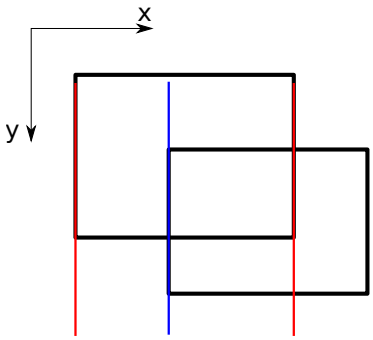


Figure 156: An example of a left-right check

This has to happen for all four sides of one of the rectangle.

Now we can try putting down a bit of code, we’ll assume that rectangles are defined by their top-left corner (as usually happens) and their width and height:

Listing 49: Axis-Aligned Bounding Box Collision Detection

```
1 class Point{
2     // Rewritten as a memo
```

```
    constructor(){
        this.x = null;
        this.y = null;
    }
}

class Rectangle{
    constructor(){
        this.corner = new Point();
        this.width = 0;
        this.height = 0;
    }
}

function rect_rect_collision(A, B){
    if ((A.corner.x < B.corner.x + B.width) &&
       (A.corner.x + A.width > B.corner.x) &&
       (A.corner.y < B.corner.y + B.height) &&
       (A.corner.y + A.height > A.corner.y)){
        return True;
    }else{
        return False;
    }
}
```

This complex conditional checks 4 things:

- The left side of rectangle A is **at the left** of the right side of rectangle B;
- The right side of rectangle A is **at the right** of the left side of rectangle B;
- The top side of rectangle A is **over** the bottom side of rectangle B;
- The bottom side of rectangle A is **underneath** the top side of rectangle B.

If all four checks are true, then a collision happened.

The best way to understand this algorithm properly is to test it by hand and convince yourself that it works.

This is a very light algorithm but can quickly become heavy on the CPU when there are many objects to check for collision. We'll see later how to limit the number of checks and make collision detection an operation that is not as heavy on our precious CPU cycles.

9.2.5 Line/Point Collision

We can represent a segment by using its two extreme points, which proves to be a quite inexpensive way to represent a line (it's just two points). Now how do we know if a point is colliding with a line?

To know if a point is colliding with a line we need... Triangles!

Every triangle can be represented with 3 points, and there is a really useful theorem that we can make use of:

> The sum of the lengths of any two sides must be greater than, or equal, to the length of the remaining side.

So, given a triangle ABC:

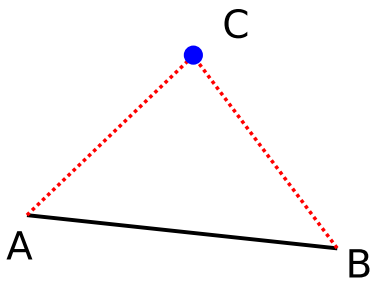


Figure 157: Example of the triangle inequality theorem

We get the following 3 inequalities:

$$\overline{AB} + \overline{BC} \geq \overline{AC}$$

$$\overline{AC} + \overline{BC} \geq \overline{AB}$$

$$\overline{AB} + \overline{AC} \geq \overline{BC}$$

What is more interesting to us is that when the one of the vertices of the triangle is **on** its opposite side, the triangle degenerates:

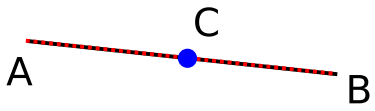


Figure 158: Example of a degenerate triangle

And the theorem degenerates too, to the following:

$$\overline{AC} + \overline{BC} = \overline{AB}$$

So we can calculate the distance between the point and each of the two extremes of the line and we know that when the sum of such distances is equal to the length of the line, the point will be colliding with the line.

In code, it would look something like the following:

Listing 50: Line to Point Collision detection

```
1 class Point{
2     // Rewritten as a memo
3     constructor(){
```

```
        this.x = null;
        this.y = null;
    }
}

class Line{
    constructor(){
        this.A = new Point();
        this.B = new Point();
    }
}

function distance(A, B){
    // Calculates the distance between two points
    return Math.sqrt((A.x - B.x) ** 2 + (A.y - B.y) ** 2);
}

function line_point_collision(pt, ln){
    // First, let's calculate the length of the line
    let length = distance(ln.A, ln.B);
    // Now let's calculate the distance between the point pt
    // and the point "A" of the line
    let pt_a = distance(ln.A, pt);
    // Same Goes for the distance between pt and "B"
    let pt_b = distance(ln.B, pt);
    // Now for the detection
    if (pt_a + pt_b == length){
        return true;
    }else{
        return false;
    }
}
```

It could prove useful to put a “buffer zone” in here too, so that the collision detection doesn’t result too jerky and precise. In that case you may want to take a look at the line vs circle algorithm, in that case the radius would be the “approximation” of the point.

9.2.6 Line/Circle Collision

As in the previous paragraph, we memorize a line as a pair of Points, so checking if the circle collides with either end of the line is easy, using the Point/Circle collision algorithm.

Listing 51: Partial Implementation of a Line to Circle Collision Detection

```
class Point{
    // Rewritten as a memo
    constructor(){
        this.x = null;
        this.y = null;
```

```
    }
}

class Line{
    constructor(){
        this.A = new Point();
        this.B = new Point();
    }
}

class Circle{
    // Let's define a circle class/structure
    constructor(){
        this.center = new Point();
        this.radius = 0;
    }
}

// ...

function line_circle_collision(circle, line){
    let collides_A = circle_point_collision(circle, line.A);
    let collides_B = circle_point_collision(circle, line.B);
    if (collides_A || collides_B){
        return true;
    }
    // ...
}
```

Now our next objective is finding the closest point **on the line** to the center of our circle. The details and demonstrations on the math behind this will be spared, just know the following:

Given a line $\overline{AB}$ between points $A = (x_1, y_1)$ and $B = (x_2, y_2)$ and a point $P = (x_k, y_k)$, the point on the line closest to P has coordinates:

$$x = x_1 + u \cdot (x_2 - x_1)$$

$$y = y_1 + u \cdot (y_2 - y_1)$$

With:

$$u = \frac{(x_k - x_1) \cdot (x_2 - x_1) + (y_k - y_1) \cdot (y_2 - y_1)}{||B - A||^2}$$

That's a lot of math!

We need to be careful though, cause this formula gives us the point for an *infinite* line, so the point we find could be

outside of our line. We will use the line/point algorithm to check for that.

After we made sure the point is on the line, we can measure the distance between such point and the center of our circle, if such distance is less than the radius, we have a hit! (Or just apply the circle/point collision algorithm again).

The final algorithm should look something like this:

Listing 52: Line to circle collision detection

```
class Point{
    // Rewritten as a memo
    constructor(){
        this.x = null;
        this.y = null;
    }

    constructor(x, y){
        this.x = x;
        this.y = y;
    }
}

class Line{
    constructor(){
        this.A = new Point();
        this.B = new Point();
    }
}

class Circle{
    // Let's define a circle class/structure
    constructor(){
        this.center = new Point();
        this.radius = 0;
    }
}

function distance(A, B){
    // Calculates the distance between two points
    return Math.sqrt((A.x - B.x) ** 2 + (A.y - B.y) ** 2);
}

function line_point_collision(line, point){
    // ...
}

function circle_point_collision(circ, point){
    // ...
}

function line_circle_collision(circle, line){
```

```
    // We check the ends first
    let collides_A = circle_point_collision(circle, line.A);
    let collides_B = circle_point_collision(circle, line.B);
    if (collides_A || collides_B){
        return true;
    }
    // We pre-calculate "u", we'll use some variables for readability
    let x1 = line.A.x;
    let x2 = line.B.x;
    let xk = circle.center.x;
    let y1 = line.A.y;
    let y2 = line.B.y;
    let yk = circle.center.y;
    let u = ((xk - x1) * (x2 - x1) + (yk - y1) * (y2 - y1))/(distance(line.A, line.B)) ** 2;
    // Now let's calculate the x and y coordinates
    let x = x1 + u * (x2 - x1);
    let y = y1 + u * (y2 - y1);
    // "Reuse": we'll use some older functions, let's create a point, with the coordinates we
     found
    let P = new Point(x,y);
    // Let's check if the "closest point" we found is on the line
    if ((line_point_collision(line, P)) == false){
        // If the point is outside the line, we return false, because the ends have already been
      checked against collisions
        return false
    }else{
        // Let's Reuse the Point/Circle Algorithm
        return circle_point_collision(circle, P);
    }
}
```

9.2.7 Point/Rectangle Collision

If we want to see if a point collides with a rectangle is really easy, we just need to check if the point's coordinates are inside the rectangle.

Listing 53: Point/Rectangle collision detection

```
function pointRectCollision(x1, y1, rectx, recty, rectwidth, rectheight){
    // We check if the point is inside the rectangle
    return x1 >= rectx && x1 <= rectx + rectwidth && y1 >= recty && y1 <= recty + rectheight;
}
```

9.2.8 Point/Triangle Collision

A possible way to define if a point is inside a triangle, we can use a bit of geometry.

We can use *Heron's formula* to calculate the area of the original triangle, and compare it with the sum of the areas created by the 3 triangles made from 2 points of the original triangle and the point we are testing.

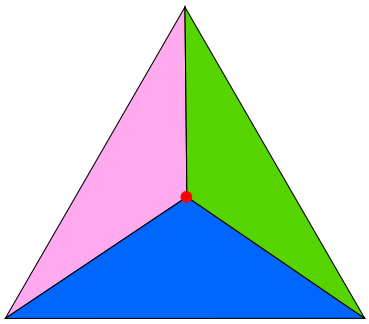


Figure 159: Point/Triangle Collision Detection: division into sub-triangles

If the sum of the 3 areas (represented in different colors in the figure) equals to the original calculated area, then we know that the point is inside the triangle.

Let's see the code:

Listing 54: Point/Triangle Collision Detection

```
function point_triangle_collision(px, py, x1, y1, x2, y2, x3, y3){
    let original_area = Math.abs((x2 - x1) * (y3 - y1) - (x3 - x1) * (y2 - y1));
    let area1 = Math.abs((x1-px)*(y2-py) - (x2-px)*(y1-py));
    let area2 = Math.abs((x2-px)*(y3-py) - (x3-px)*(y2-py));
    let area3 = Math.abs((x3-px)*(y1-py) - (x1-px)*(y3-py));
    if (area1 + area2 + area3 == original_area){
        return true;
    }else{
        return false;
    }
}
```

Let's see how we can change the algorithm to accommodate for some leeway, since the we may be requiring too much precision from our algorithms. We can do that by using epsilon values.

Our main test is that the sum of the area of the 3 triangles we create (A_1, A_2, A_3) is equal to the area of the original triangle (A_0), in math terms:

$$A_1 + A_2 + A_3 = A_0$$

We can also rewrite such equation this way:

$$A_1 + A_2 + A_3 - A_0 = 0$$

Due to possible precision issues we know that there are some values where the equation above is not true, so we choose a "low enough error" that we are willing to accept, for example $\epsilon = 0.0001$, and use this test instead:

$$|A_1 + A_2 + A_3 - A_0| < \epsilon$$

Which can be expanded (if you want) to

$$-\epsilon < A_1 + A_2 + A_3 - A_0 < \epsilon$$

The code wouldn't change much, but for sake of clarity, here it is:

Listing 55: Point/Triangle Collision Detection with epsilon

```
function point_triangle_collision(px, py, x1, y1, x2, y2, x3, y3){
    // We accept anything that is closer than 1/1000th of unit
    const epsilon = 0.0001;
    let original_area = Math.abs((x2 - x1) * (y3 - y1) - (x3 - x1) * (y2 - y1));
    let area1 = Math.abs((x1-px)*(y2-py) - (x2-px)*(y1-py));
    let area2 = Math.abs((x2-px)*(y3-py) - (x3-px)*(y2-py));
    let area3 = Math.abs((x3-px)*(y1-py) - (x1-px)*(y3-py));
    if (Math.abs(area1 + area2 + area3 - original_area) < epsilon){
        return true;
    }else{
        return false;
    }
}
```

9.2.9 Circle/Rectangle Collision

First of all we need to identify which side of the rectangle we should test against, so if the centre of the circle is to the right of the rectangle, we will test against the right edge of the rectangle, if it's above we'll test against the top edge and so on...

After that, we just perform some math on the distances and calculated values to detect if the circle collides with the rectangle.

Listing 56: Rectangle to Circle Collision Detection

```
class Point{
    // Rewritten as a memo
    constructor(){
        this.x = null;
        this.y = null;
    }
}

class Rectangle{
    constructor(){
        this.corner = new Point();
```

```
        this.width = 0;
        this.height = 0;
    }
    // ...
}

class Circle{
    // Let's define a circle class/structure
    constructor(){
        this.center = new Point();
        this.radius = 0;
    }
}

function circle_rectangle_collision(circ, rect){
    // Detects a collision between a circle and a rectangle

    // These variables are used as the coordinates we should test against
    // They are temporarily set to the circle center's coordinates for a reason we'll see soon
    let tx = circ.center.x;
    let ty = circ.center.y;

    // Let's detect which edge to test against on the x axis
    if (circ.center.x < rect.corner.x){
        // We're at the left of the rectangle, test against the left side
        tx = rect.corner.x;
    }else if (circ.center.x > rect.corner.x + rect.width){
        // We're at the right of the rectangle, test against the right side
        tx = rect.corner.y + rect.width;
    }

    // Same thing on the vertical axis
    if (circ.center.y < rect.corner.y){
        // We're above the rectangle, test against the top side
        ty = rect.corner.y;
    }else if (circ.center.y > rect.corner.y + rect.height){
        // We're below the rectangle, test against the bottom side
        ty = rect.corner.y + rect.height;
    }

    // Let's get the distance between the testing coordinates and the circle center
    let distanceX = circ.center.x - tx;
    let distanceY = circ.center.y - ty;
    let distance = Math.sqrt(distanceX ** 2 + distanceY ** 2);

    // Note that if the center of the circle is inside the rectangle, the testing coordinates
     will be the circle's center itself, thus the next conditional will always return true

    if (distance <= circ.radius){
        return true;
```

```
61      }
62
63      // Default to false in case no collision occurs
64      return false;
65  }
```

9.2.10 Line/Line Collision

Line/Line collision is quite simple to implement once you know the inner workings of geometry, but first we need to explain the thought behind this algorithm, so... **math warning!!**

Let's look at the following image:

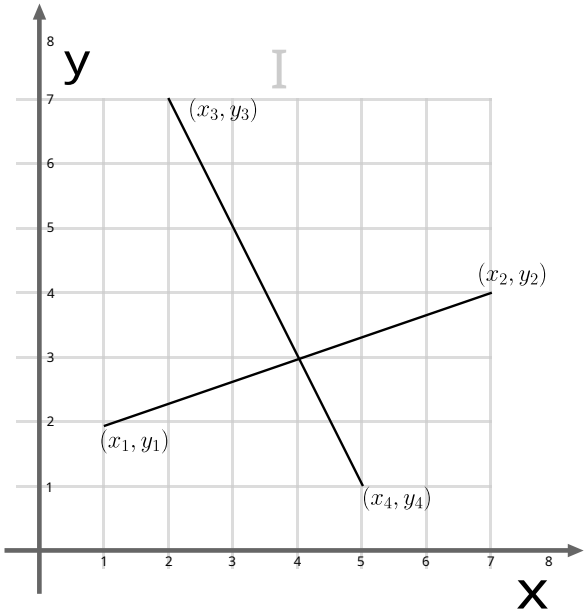


Figure 160: Example image for line/line collision

A generic point P_a of line A can be represented with the following formula:

$$P_a = P_1 + u_a \cdot (P_2 - P_1)$$

which translates into the coordinate-based equations:

$$\begin{cases} x_a = x_1 + u_a \cdot (x_2 - x_1) \\ y_a = y_1 + u_a \cdot (x_2 - x_1) \end{cases}$$

This makes us understand that any point of line A can be represented by its starting point P_1, plus a certain fraction (represented by u_a) of the vector represented by $P_2 - P_1$.

This also means that $0 \leq u_a \leq 1$, else the point won't be on the segment.

In the same way, a generic point P_b of line B can be represented with:

$$P_b = P_3 + u_b \cdot (P_4 - P_3)$$

which becomes:

$$\begin{cases} x_b = x_3 + u_b \cdot (x_4 - x_3) \\ y_b = y_3 + u_b \cdot (x_4 - x_3) \end{cases}$$

The two lines will collide when $P_a = P_b$, so we get the following equations:

$$\begin{cases} x_1 + u_a \cdot (x_2 - x_1) = x_3 + u_b \cdot (x_4 - x_3) \\ y_1 + u_a \cdot (y_2 - y_1) = y_3 + u_b \cdot (y_4 - y_3) \end{cases}$$

That need to be solved in the u_a and u_b variables.

The result is:

$$\begin{cases} u_a = \frac{(x_4 - x_3) \cdot (y_1 - y_3) - (y_4 - y_3) \cdot (x_1 - x_3)}{(y_4 - y_3) \cdot (x_2 - x_1) - (x_4 - x_3) \cdot (y_2 - y_1)} \\ u_b = \frac{(x_2 - x_1) \cdot (y_1 - y_3) - (y_2 - y_1) \cdot (x_1 - x_3)}{(y_4 - y_3) \cdot (x_2 - x_1) - (x_4 - x_3) \cdot (y_2 - y_1)} \end{cases}$$

Substituting either of the results in the corresponding equation for the line will give us the intersection point (which may be useful for some particle effects).

Now some notes on our solution:

- If the denominator for the equations for u_a and u_b equals to zero, the two lines are parallel
- If both the numerator and denominator for u_a and u_b are equal to zero, the two lines are coincident
- If both $0 \leq u_a \leq 1$ and $0 \leq u_b \leq 1$ then the two segments collide.

Now we can translate all this math into code:

Listing 57: Implementation of the line/line collision detection

```
function lineLineCollision(x1, y1, x2, y2, x3, y3, x4, y4){
    // Let's calculate the denominator, this will allow us to avoid a
    // "divide by zero" error
    let den = ((y4 - y3) * (x2 - x1) - (x4 - x3) * (y2 - y1));

    if (den == 0){
        // The lines are parallel
```

```
        return false;
    }

    let uA = ((x4 - x3) * (y1 - y3) - (y4 - y3) * (x1 - x3)) / den;
    let uB = ((x2 - x1) * (y1 - y3) - (y2 - y1) * (x1 - x3)) / den;

    // Let's see if uA and uB tell us the lines are colliding
    if ((uA >= 0 && uA <= 1) && (uB >= 0 && uB <= 1)){
        return true;
    }

    // If not, they don't collide
    return false;
}
```

This collision detection algorithm can be useful for line-based puzzle games, line the untangle puzzle.

9.2.11 Line/Rectangle Collision

Given the previous explanation about the Line/Line collision detection, it's quite easy to build a Line/Rectangle algorithm; distinguishing the cases where we want to account for a segment being completely inside of a rectangle or not.

Listing 58: Implementation of the line/rectangle collision detection

```
function lineLineCollision(x1, y1, x2, y2, x3, y3, x4, y4){
    // our previous implementation of the line/line collision detection
    // ...
}

function pointRectCollision(x1, y1, rectx, recty, rectwidth, rectheight){
    // our previous implementation of a point/rectangle collision detection
    // ...
}

function lineRectangleCollision(x1, y1, x2, y2, rectx, recty, rectwidth, rectheight){
    // If we want to test if a line is completely inside of a rect, we just need
    // to see if any of its endpoints is inside the rectangle
    if (pointRectCollision(x1, y1, rectx, recty, rectwidth, rectheight) || pointRectCollision(x2
    , y2, rectx, recty, rectwidth, rectheight)){
        // At least one of the ends of the segment is inside the rectangle
        return true;
    }
    // Now to test the rectangle against the line, if it's not completely inside
    let left = lineLineCollision(x1, y1, x2, y2, rectx, recty, rectx, recty + rectheight);
    let right = lineLineCollision(x1, y1, x2, y2, rectx + rectwidth, recty, rectx + rectwidth,
    recty + rectheight);
    let top = lineLineCollision(x1, y1, x2, y2, rectx, recty, rectx + rectwidth, recty);
    let bottom = lineLineCollision(x1, y1, x2, y2, rectx, recty + rectheight, rectx + rectwidth,
     recty + rectheight);
```

```
23
24      if (left || right || top || bottom){
25          // We hit one of the sides, we are colliding
26          return true;
27      }
28
29      // In any other case, return false
30      return false;
31  }
```

This can prove useful to test for “line of sight” inside an AI algorithm.

9.2.12 Point/Polygon Collision

Here we are, the most complex matter when it comes to narrow-phase collision detection: detecting collisions between arbitrary convex polygons.

Note!

In this book we will focus on convex polygons “without holes”, which is the most common situation you’ll find yourself in.

First of all, we will start by talking about some theorems and requirements that will help us on the way to build a “polygon vs polygon” collision detection algorithm.

9.2.12.1 Jordan Curve Theorem

Let’s imagine a plane, like our 2D screen: if we draw a non-self-intersecting, continuous loop in the plane we obtain a *Jordan Curve*. This curve separates the plane in two distinct regions: the “inside” and the “outside”.

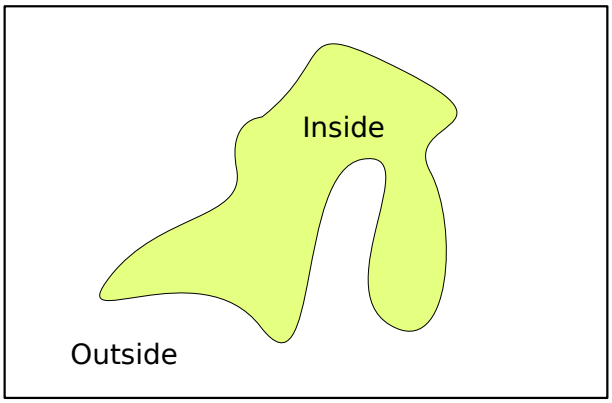


Figure 161: Example of a Jordan Curve

Any non-self-intersecting polygon (be it convex or non-convex) can be seen as a Jordan curve, this means that we can easily identify (programmatically) if a point is inside or outside the polygon. At least in the “convex” case.

Let's take a convex polygon, and a point inside such polygon: we can see that if we choose a point outside the polygon (non-colliding) we can strike a line between the "inside point" and the chosen point, and such line will intersect one of the polygon's edges. This gives us an idea on how to check for "point vs. polygon".

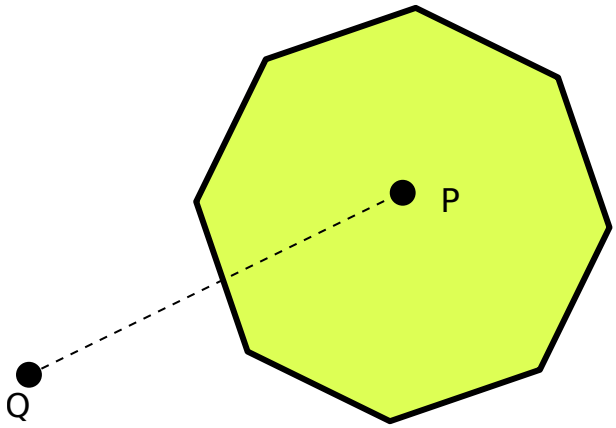


Figure 162: A simple case where a point is outside the polygon

This doesn't happen if the point is inside the polygon, obviously:

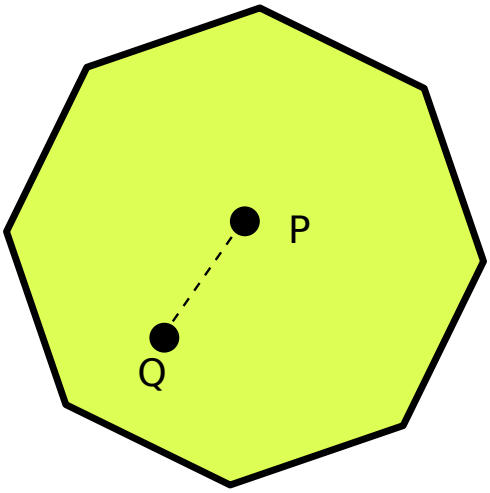


Figure 163: A simple case where a point is inside the polygon

This is all well and good, but we have two problems on hand:

- Finding a point inside the polygon;
- We have a non-convex polygon;

Let's leave the first problem aside, since talking about it may end up being confusing and just empty talk (or writing, being this a book), and let's focus on the second problem.

If we have a non-convex polygon, we may end up with a line that intersects the polygon's perimeter even if the point is colliding:

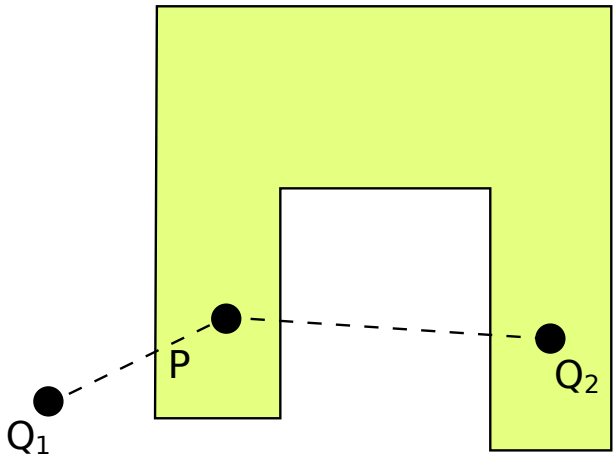


Figure 164: How a non-convex polygon makes everything harder

Here we call P the "point inside the polygon" while Q_1 and Q_2 are the points we are testing: as we can see Q_2 triggers our "non-colliding" test even though it is inside the polygon.

Can you see what can help us solving this issue? I'm sure you have a *number* of ideas in mind, we'll talk about it in the non-convex polygon collision detection section.

9.2.12.2 Thinking outside the box: polygon triangulation

As you can see, as simple as it can be, the Jordan curve theorem poses some problems that may be a bit out of our reach as of now, so let's try to find a less ideal but easier to understand solution.

Let's now limit ourselves to convex polygons, which (again) is the most common situation.

We can take inspiration from 3D graphics, where any solid shape (and thus the polygons that make those up) are decomposed to a bunch of triangles. Nothing stops us from doing the same and taking any polygon and decomposing it to a group of triangles, like follows:

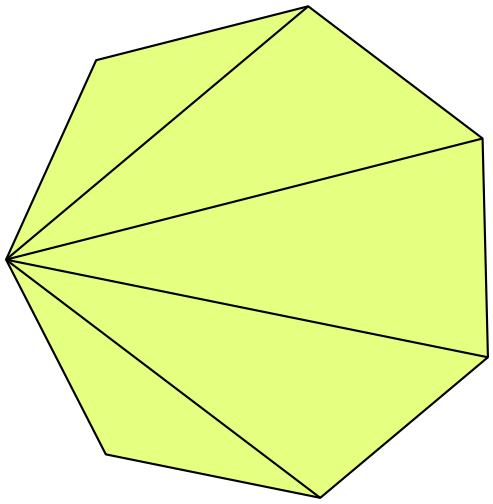


Figure 165: Decomposing a polygon into triangles

This specific triangulation is called “fan triangulation” and it is chosen for its $\Theta(n)$ (where n is the number of vertices) execution time.

9.2.12.3 Bounding Boxes

Before making our poor CPU undertake big calculations, we may want to check if there is even a possibility of a collision, maybe with a simpler algorithm.

The great majority of the lifetime of our game objects is spent not colliding with anything, so if we can easily exclude a collision before starting complex algorithms, our game will just benefit from it.

We can take our complex polygon and give it a “bounding box”, any point that is inside such box *has a possibility* of colliding with our polygon, but any point *outside* the bounding box *surely will not collide*.

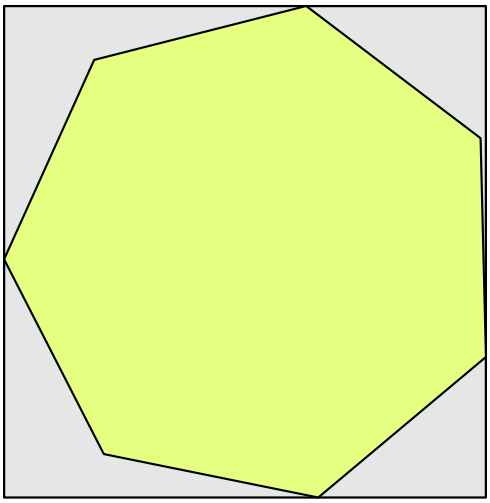


Figure 166: Example of a polygon with its bounding box

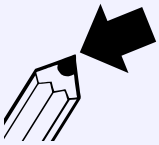
How do we calculate a bounding box? Simple, we just need 4 coordinates:

- The smallest x (which we’ll call x_{min})
- The smallest y (y_{min})
- The biggest x (x_{max})
- The biggest y (y_{max})

The vertices of our bounding box will always be:

$$A(x_{min}, y_{min})\ B(x_{max}, y_{min})\ C(x_{max}, y_{max})\ D(x_{min}, y_{max})$$

Tip!



Thanks to how rectangles work, we can just use the points A and C to build a rectangle: since they contain all 4 coordinates, we can infer B and D from them.

This is simple to achieve: we just need to loop over all the vertices and find our coordinates. The algorithm here below:

Listing 59: How to find the bounding box of a polygon

```
class Point{
    // Rewritten as a memo
    constructor(){
        this.x = null;
        this.y = null;
    }
}

class Rectangle{
    constructor(){
        this.corner = new Point();
        this.width = 0;
        this.height = 0;
    }

    static from_points(topleft, bottomright){
        // ...
    }
}

function bounding_box(vertices){
    // First we create and bootstrap the variables
    let xmin = vertices[0].x;
    let xmax = vertices[0].x;
    let ymin = vertices[0].y;
    let ymax = vertices[0].y;
    // Now we iterate through all the other vertices
    for (const vertex in vertices){
        if (vertex.x < xmin){
            xmin = vertex.x;
        }
        if (vertex.x > xmax){
            xmax = vertex.x;
        }
        if (vertex.y < ymin){
            ymin = vertex.y;
        }
        if (vertex.y > ymax){
```

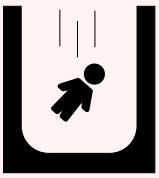
```
            ymax = vertex.y;
        }
    }
    // Now we can build the needed points for the bounding box
    A = new Point();
    C = new Point();
    A.x = xmin;
    A.y = ymin;
    C.x = xmax;
    C.y = ymax;
    // We build our bounding box
    let boundingBox = Rectangle.from_points(A, C);
    // and return it
    return boundingBox;
}
```

To check if the collision "may happen", we can just use a simple Point vs Rectangle collision check.

9.2.12.4 Point/Polygon collision detection using triangulation

Finally, after all the math and preparations, we can start working towards our collision detection algorithm.

Pitfall Warning!



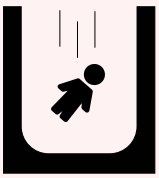
This algorithm works only with convex polygons that have no holes, also it probably is not the most efficient way to check for collisions between a point and a polygon.
This is more akin to an exercise in creativity and less about "notions": we found a simple solution to a complex problem. Even if it is not the most efficient, it may be "efficient enough".

9.2.12.4.1 The "Polygon" class

Differently from previous classes and structures, the "polygon" class will need a little more work. This is because we are going to do more than just merely memorize vertices.

First of all we need an **ordered** list (or array) of vertices, which will be represented by points. Secondly, we need facilities to calculate list of triangles, as well as their areas.

Pitfall Warning!



You may be tempted to memorize the “triangles” that are an output of the “fan triangulation”, as well as their areas. This may be a good idea if well managed, but we will need to take care of “moving” those triangles and manage when the polygon gets deformed: in that case all the triangle areas will have to be recalculated.
Same goes for the bounding box, which will change in size when the polygon rotates or deforms. In this book we will try to keep the class as generic as possible (as well as simple), thus we will just recalculate everything every frame as needed.

Thirdly, we need the constructor to do some math before we can use the polygon. Finally we need to integrate a “fanning” function.

Whew... That’s a lot of work, but here’s the code for the polygon class:

Listing 60: A (not so) simple polygon class

```
class Polygon{
    constructor(){
        this.vertices = [];  // An array of Point() classes
    }

    calculate_bounding_box(){
        // This function calculates the bounding box
        // -------------------------
        // First we create and bootstrap the variables
        let xmin = vertices[0].x;
        let xmax = vertices[0].x;
        /*
         * ...
         * see the bounding box algorithm for the full version
         * ...
         */
        // We build our bounding box
        let boundingBox = Rectangle.from_points(A, C);
        // and return it
        return boundingBox;
    }

    do_fanning(){
        /*
         * This function iterates over the vertices and returns
         * an array of triangles corresponding to the "fan triangulation"
         */
        // We fix the "base" of the fan on the first vertex
        let root_vertex = vertices[0];
        let temp_triangles = [];
        // Now we iterate through all the other vertices
        for (let j = 2; i < vertices.length; j++){
```

```
            // j goes from the third vertex, to the last
            // j - 1 goes from the second to the second to last
            temp_triangles.push(Triangle.from_points(root_vertex, j - 1, j));
        }
        // In the end, we will have the triangles array, we can just return it
        return temp_triangles;
    }
}
```

9.2.12.4.2 The algorithm

After all this preparation, we are finally ready for the algorithm, which will happen in two passes:

1. A “broad”-ish pass, where we compare the point to the polygon’s bounding box
2. A “proper-narrow” pass, where we do a series of triangle vs point collision detections

Here’s the code:

Listing 61: Polygon vs Point collision detection

```
// ...
function polygon_point(poly, point){
    // First of all, we get the polygon's bounding box
    let bounding_box = poly.calculate_bounding_box();
    // Then we do a simple point vs. rectangle check
    if (!point_rectangle(bounding_box, point)){
        // We are not even in the bounding box, we can't collide
        return false;
    }
    // If instead we are in the bounding box, we need to get the "fan triangulation"
    let triangles = poly.do_fanning();
    // Now we check, for each triangle, if the point collides
    for (const triangle in triangles){
        if (point_triangle(triangle, point)){
            // We found the "slice" of the polygon that the point collides with
            return true;
        }
    }
    // If we pass all triangles without a hit, we are in the bounding box
    // but outside the polygon, that's the worst case, and we are not colliding
    return false;
}
```

9.2.12.4.3 Performance analysis

The algorithm seems fairly simple, but we may want to check its performance to see how efficient it is. In this analysis `n` will be the number of vertices, while `m` is the number of triangles.

The best case is that the point we're testing is outside the polygon's bounding box: this means that we calculate the bounding box (which is $\Theta(n)$) and we check the point against it (which is $\Theta(1)$), thus our best case (lower bound) is $\Omega(n)$.

The worst case is when the whole algorithm is performed to the end, which means the point is inside the bounding box, but outside the polygon: this means we calculate the bounding box ($\Theta(n)$), check against it ($\Theta(1)$), do the "fan triangulation" ($\Theta(n)$), check each triangle without finding any collision ($O(m)$) and get to the end. Out worst case (upper bound) is $O(n+m)$.

Considering the fact that the number of triangles `m` is tied to the number of vertices `n` by the formula (valid for simple convex polygons)

$$m = n - 2$$

We have an upper bound of $O(n+m) = O(n+n-2) =\sim O(n)$, this is because the constant gets "squashed by the linear behaviour" of `n`, and $2 \cdot n$ behaves asymptotically in the same way as n when the dataset grows.

Even though we have a tight bound of $\Theta(n)$ in our entire algorithm (which means the amount of calculations goes up slowly with the addition of new vertices), we need to be mindful of the amount of calculation that is done, including some heavy operations like square roots.

9.2.13 Circle/Polygon Collision

Now that we got one of the hardest topics out of the way, we can focus on other types of collision detection between arbitrary convex polygons: one of those is the "circle vs polygon" collision detection.

Let's see an example image first:

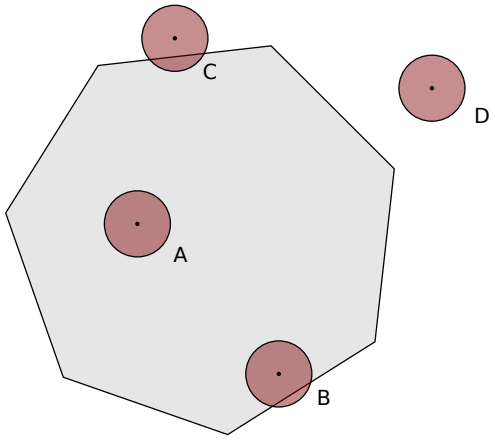


Figure 167: Example image used for circle/polygon collision detection

Here we can see four different cases of collision (or lack thereof) between a circle and a polygon (if you're particularly acute, you may have noticed we're missing a 5th case, but we'll talk about it shortly):

- **Case A:** The circle is completely inside the polygon;
- **Case B:** The circle is partially inside the polygon, with the center being **inside** the polygon;
- **Case C:** The circle is partially inside the polygon, with the center being **outside** the polygon;
- **Case D:** The circle is completely outside the polygon.

Case A and B can be solved together with a point/polygon check, where the point is the center of the circle, while case C can be solved by a line/circle check between the circle and all the edges of the polygon.

What about the "missing 5th case"? Here it is:

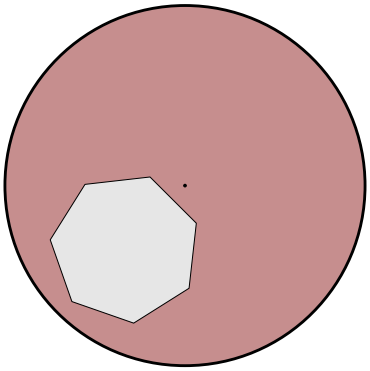


Figure 168: An edge case of the circle/polygon check

In this case the circle contains the polygon completely, with its center outside of the polygon area, so the check used in cases A and B wouldn't work and neither would the one used in case C.

This is a really rare edge-case, since usually the game does its checks so fast that you'd end up in case C long before this edge-case sees the light of day. In the event this happens, we just need to check if any of the vertices of the polygon is inside the circle.

Here's the full algorithm:

Listing 62: Polygon vs Circle collision detection

```
// ...
function circle_polygon(poly, circ){
    // Case C (and partly B) are less resource-intensive than
    // a point/polygon check, so let's do them first
    for (let i = 0; i < poly.vertices.length; i++){
        // We iterate through all the vertices
        let j = i + 1;
        // If we get to the end, we wrap around j
        if (j == poly.vertices.length){
            j = 0;
        }
        let temp_line = Line.fromPoints(poly.vertices[i], poly.vertices[j]);
        // In case we find a hit, we already know there is a collision
        if (line_circle_collision(circ, temp_line)){
            return true;
```

```
16          }
17      }
18      // Now Let's check for cases "A" and "B"
19      if (polygon_point(poly, circ.center)){
20          // If the center is inside the polygon, we have a collision
21          return true;
22      }
23      // Now let's check for the rare edge-case: if this case happens, all the vertices
24      // are inside the circle, so we can only check one of them
25      if (circle_point_collision(circ, poly.vertices[0])){
26          // If any vertex is inside the circle, we have a collision, so we check the first
27          return true;
28      }
29      // If none of the checks above returned, we don't have a collision (case D)
30      return false;
31  }
```

9.2.14 Line/Polygon Collision

The line vs polygon collision detection algorithm is not really different from the ones we have seen previously. Let's take a look at an image with all the cases we can think about:

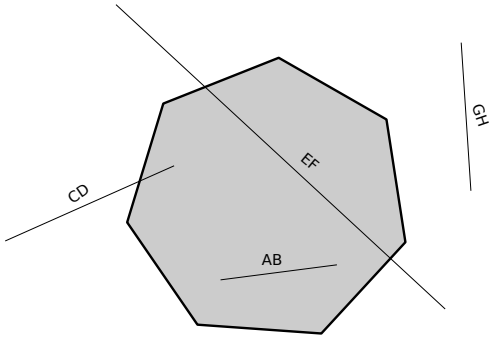


Figure 169: Example image used for line/polygon collision detection

Here we can see 4 cases (this time for real):

- **Line $\overline{AB}$:** The segment is completely inside the polygon (including its ends);
- **Line $\overline{CD}$:** The segment is partially inside the polygon (one of its ends is inside the polygon);
- **Line $\overline{EF}$:** The segment crosses the polygon, but both its ends are outside the polygon;
- **Line $\overline{GH}$:** The segment is completely outside the polygon;

We can solve the cases involving the lines $\overline{AB}$ and $\overline{CD}$ by checking if either of the ends is inside the polygon, using a point/polygon collision check.

The case involving the line $\overline{EF}$ can be solved by a line/line collision check between the $\overline{EF}$ and all the edges of the polygon.

Let's take a look at the full algorithm:

Listing 63: Polygon vs Line collision detection

```
// ...
function line_polygon(line, poly){
    // First of all, let's check if either of the line ends are inside the polygon
    // This covers cases AB and CD
    if (polygon_point(poly, line.A)){
        // One of the ends is inside the polygon, we have a hit
        return true;
    }
    if (polygon_point(poly, line.B)){
        // One of the ends is inside the polygon, we have a hit
        return true;
    }
    // Now we check for case EF
    for (let i = 0; i < poly.vertices.length ; i++){
        // We iterate through all the vertices
        let j = i + 1;
        // If we get to the end, we wrap around j
        if (j == poly.vertices.length){
            j = 0;
        }
        let temp_line = Line.fromPoints(poly.vertices[i], poly.vertices[j]);
        if (line_line_collsion(temp_line, line)){
            return true;
        }
    }
    if (line_line_collision(temp_line, line)){
        return true;
    }
    // If none of the previous checks was triggered, we don't have a collision
    return false;
}
```

9.2.15 Polygon/Polygon Collision

Here we are, the final frontier, polygon vs polygon collision detection. We went through a lot of pages of notions and reasoning to get here, now we have the tools to undertake one of the more complex collision detection methods.

Remember: we are checking if two **convex** polygons are colliding, let's see an example image first.

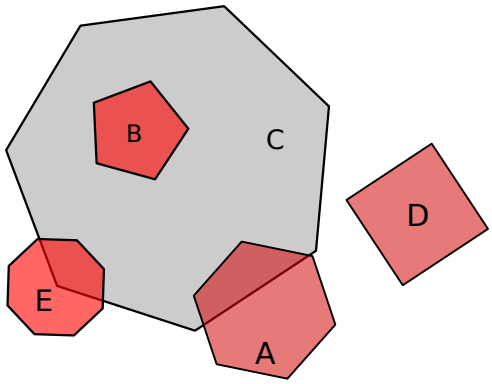


Figure 170: Example image used for polygon/polygon collision detection

We can see 4 cases here, from the simplest to the hardest:

- The Square **D** is outside the polygon;
- The Pentagon **B** is completely inside the polygon
- The Octagon **E** is colliding with the heptagon **C** and a vertex of **C** is inside of **E**;
- The heptagon **C** is colliding with the hexagon **A**, but none of the vertices of **C** are inside of **A**;

We can easily solve the cases involving **A** and **E** with a "polygon vs line" collision detection, while the case involving **B** can be checked by doing a "polygon vs point" check.

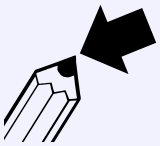
Let's take a look at the algorithm:

Listing 64: Polygon vs Polygon collision detection

```
// ...
function polygon_polygon(p1, p2){
    // First we do a polygon vs line check for all the edges
    for (let i = 0; i < p2.vertices.length; i++){
        let j = i + 1;
        if (j == p2.vertices.length){
            // Wrap around in case we get to the end
            j = 0;
        }
        let temp_line = Line.fromPoints(p2.vertices[i], p2.vertices[j]);
        if (polygon_line(p1, temp_line)){
            // We have a hit
            return true;
        }
    }
    // Now we check in case one polygon contains the other, we can just check a single vertex
    if (polygon_point(p1, p2.vertices[0]) || polygon_point(p2, p1.vertices[0])){
        return true;
    }
    // None of the checks was triggered, there is no collision
    return false;
}
```

As you can see, the algorithm is quite short, but it builds on a lot of previous algorithms that we already studied, so there is a lot of "hidden complexity" behind these few rows of code.

Tip!



We can make the algorithm perform a bit better by adding a check between the (axis aligned) bounding boxes first: this will drastically reduce the amount of "line vs polygon" and "point vs polygon" checks, at the expense of a slightly heavier algorithm when a collision happens.

9.2.16 Non-convex polygons collision

Let's go back to our previous example, using a non-convex polygon: we have an "inside point" and two points to test, one inside and one outside.

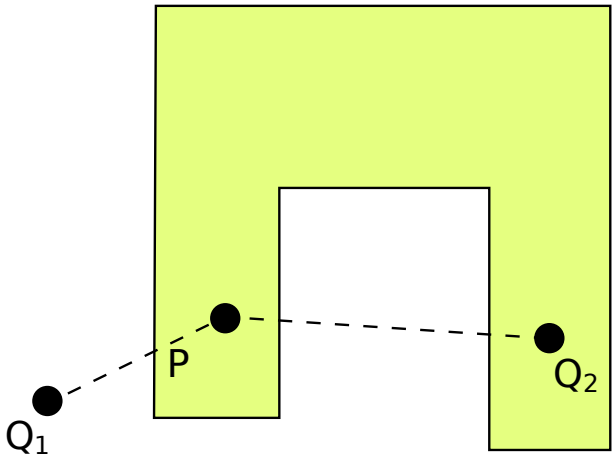


Figure 171: How a non-convex polygon still makes everything harder

The trick is counting the number of times our "segment between the points" hits the perimeter of the polygon:

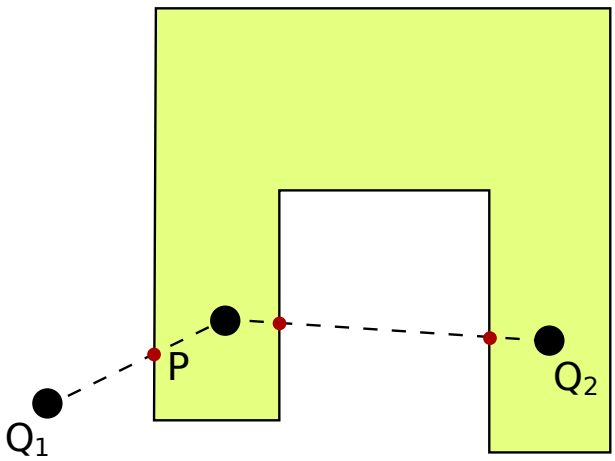


Figure 172: Counting how many times we hit the perimeter gives us the result

If the number of “hits” is odd, we know the point tested is outside, if the number of “hits” is even, the point is inside the polygon.

The previous statement fails when we hit a vertex in our way: we can’t really count it as a “double hit”, because there’s the possibility that we are hitting it while “entering” the polygon.

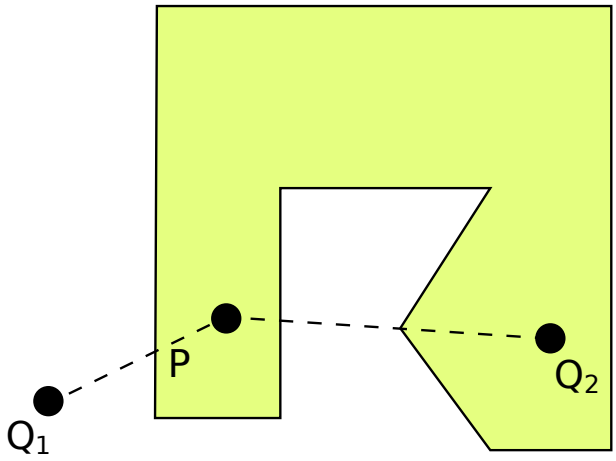


Figure 173: Issues with vertices make everything even harder

If we counted the vertex hit as a “double hit”, we would end up having a point “inside the polygon” figuring as a “point outside the polygon”.

The complications and edge cases are many and beyond the scope of this book, so we’ll stop here and instead continue with the ways we discussed earlier.

9.2.16.1 Polygon triangulation: the return

We can extend the reasoning we made with simple convex polygons earlier to all simple polygons (so we can include non-convex ones too): any non-self-intersecting polygon without holes can be decomposed into triangles.

The only limitation we have is the method: the “fan triangulation” method works only with convex polygons and a very limited set of non-convex ones; so we need to find a different way of triangulating those polygons.

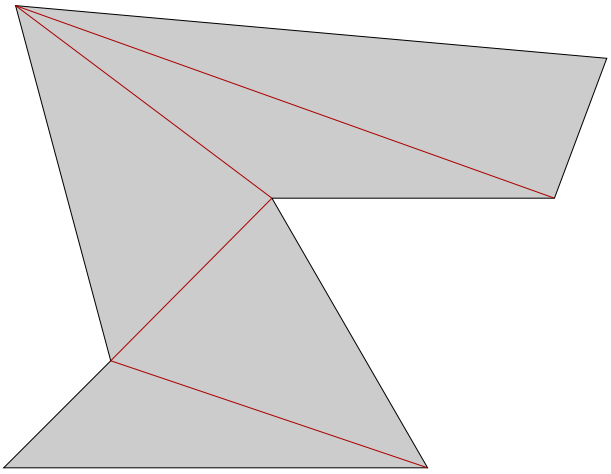
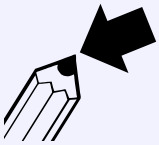


Figure 174: Triangulating a non-convex polygon

Triangulation methods include “ear clipping” and “monotone polygon triangulation”, but their implementation is beyond the scope of this book.

Tip!



You can always take any type of polygon (even with holes) and decompose it into a bunch of convex polygons that can be fan-triangulated. Many graphical libraries represent polygons based on the fan-triangulation method.

9.2.17 Pixel-Perfect collision

Pixel perfect collision is the most precise type of collision detection, but it’s also by far the slowest.

The usual way to perform collision detection is using **bitmasks** which are 1-bit per pixel representation of the sprites (white is usually considered a “1” while black is considered a “0”).

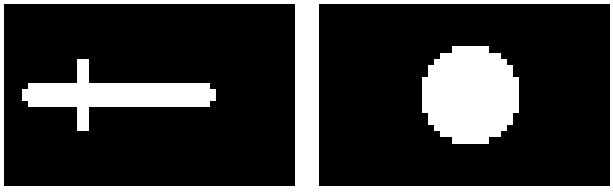


Figure 175: Two Bitmasks that will be used to explain pixel-perfect collision

A logic “AND” operation is performed, pixel-by-pixel, on the bitmasks; with the sprite position taken in consideration, as soon as the first AND operation returns a “True” a collision occurred.

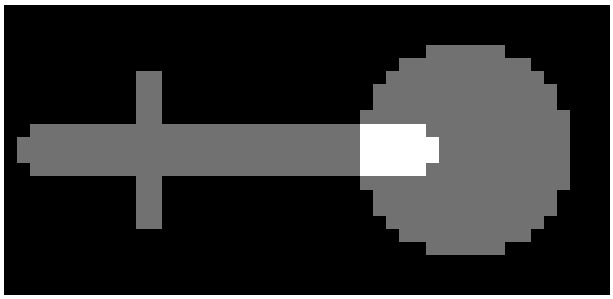


Figure 176: Two Bitmasks colliding, the 'AND' operations returning true are highlighted in white

Listing 65: Example of a possibile implementation of pixel perfect collision detection

```
class Color{
    constructor(){
        this.colorData = [];
    }

    isWhite(){
        // ...
    };
}

class Bitmask{
    constructor(){
        this.data = [];  // An array of Color() classes
    }
    getColor(x, y){
        // ...
    };
    // ...
}

class Sprite{
    constructor(){
        this.bitmask = new Bitmask();
        this.x = null;
        this.y = null;
        this.width = 0;
        this.height = 0;
    }
}

function pixel_perfect_collision(A, B){
    // Calculate the intersecting rectangle to limit checks
    let x1 = max(A.x, B.x);
    let x2 = min((A.x + A.width), (B.x + B.width));

    let y1 = max(A.y, B.y);
    let y2 = min((A.y + A.height), (B.y + B.height));
```

```
38
39      // For each pixel in the intersecting rectangle, let's check
40      for (let y = y1; y <= y2; y++){
41          for (let x = x1; x <= x2; x++){
42              // We're working in the intersecting triangle, so we'll need to
43              // rework our coordinates
44              let a = A.bitmask.getColor(x - A.x, y - A.y);
45              let b = B.bitmask.getColor(x - B.x, y - B.y);
46
47              if (a.isWhite() && b.isWhite()){
48                  return true;
49              }
50          }
51      }
52
53      // If no collision is occurred by the end of the checking, we're safe
54      return false;
55  }
```

This algorithm has a time complexity of $O(n \cdot m)$ where n is the total number of pixels of the first bitmask, while m is the total number of pixels in the second bitmask.

9.3 Broad-phase collision detection: is a collision even possible?

Now we need to find which game objects collided, and this can be easily one of the most expensive parts of our game, if not handled correctly.

This section will show how knowing which items will surely **not** collide can help us optimize our algorithms.

We need to remember that each object (as good practices suggest) know only about themselves, they don't have "eyes" like us, that can see when another object is approaching them and thinking "I'm gonna collide". The only thing we can do it having "someone else" take care of checking for collisions.

As an example, we'll take the following situation:

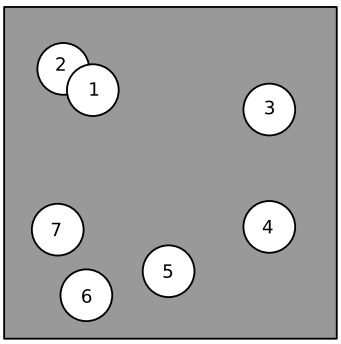


Figure 177: Example for collision detection

We can evidently see how circles 1 and 2 are colliding, but obviously our game won't just "know" without giving it a way to think about how two objects collide.

9.3.1 The Brute Force Method

The simplest method is the so-called "brute force" method: you don't know which items may collide? Just try them all.

So if we consider a list of 7 game objects, we'll need to see if 1 collides with 2, 1 collides with 3, ..., 2 collides with 1, ...

An algorithm of this type could be the following:

Listing 66: Brute Force Method of collision search

```
function is_collision(A, B){
    // Defines how two items collide (being circles, this could be a difference of radii)
    // ...
}

let items = [item1, item2, ...];

function get_colliding_items(items_to_check){
    colliding_items = [];

    for (const A in items_to_check){
        for (const B in items_to_check){
            if (A != B){
                // We avoid checking if an item collides with itself, for obvious reasons
                if (is_collision(A, B)){
                    // We keep together the pair of items that collided
                    colliding_items.push([A, B]);
                }
            }
        }
    }
    return colliding_items;
}
```

This algorithms runs in $O(n^2)$, because it checks every item with every other, even with itself.

In this example, the algorithm completes in 49 steps, but you can imagine how a game could slow down when there is an entire world to update (remember the collision detection, among with other updates and rendering/drawing, must happen in less than 16.67 and 33.33ms, so if you can save time, you totally should).

9.3.2 Building Quad Trees

A nice idea would be being able to limit the number of tests we perform, since the brute force method can get really expensive really quickly.

When building quad-trees, we are essentially dividing the screen in “quadrants” (and if necessary, such quadrants will be divided into sub-quadrants), detect which objects are in such quadrants and test collisions between objects that are inside of the same quadrant.

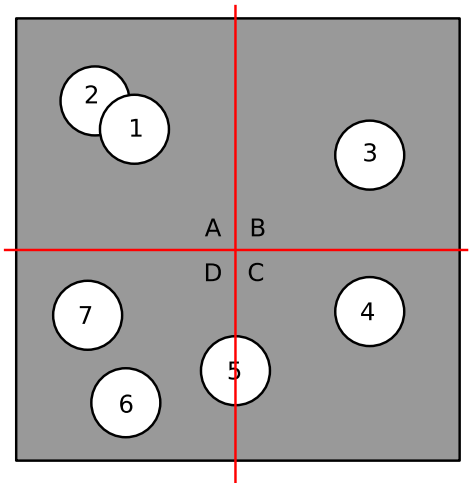


Figure 178: Graphical example of a quad tree, overlaid on the reference image

And here below we can see how a quad tree would look, in its structure:

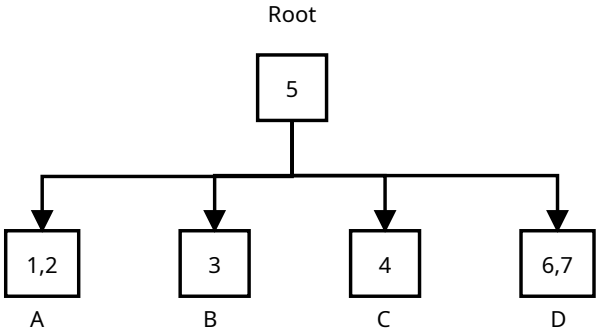


Figure 179: A quad tree

The rules to follow in a quad tree are simple, both in filling and retrieval. When we are filling a quad tree:

- Each node starts by being inserted in the root;
- If the root is “full” (exceeds a set quantity of nodes), it “splits” into 4 sub-trees;
- If a node would fit in two quadrants (like #5), it gets put inside the parent of both quadrants.

When we are retrieving the nodes we will know that an object inside a certain node can collide only with the objects in the same nodes or in the subtree rooted at such node.

With the original brute force method, we will make at most 49 tests for 7 items (although it can be optimized), while with quad trees we will perform:

- 6 Tests against node 5 (5-1, 5-2, 5-3, 5-4, 5-6, 5-7);

- 1 Test against node 1 (1-2);
- 1 Test against node 2 (2-1);
- No tests against node 3, because it's on its own and there are no subtrees;
- No tests against node 4, for the same reason;
- 1 Test against node 6 (6-7);
- 1 Test against node 7 (7-6).

For a total of 10 tests, which can be further optimized by avoiding testing pairs of objects that have already been tested. But this is if we want to test all objects for collision against all other objects (thus it is a somewhat more optimized "brute force").

9.3.2.1 A more precise definition

To be more precise, quad-trees are part of the group of "spatial acceleration structures". They are structures that are usually used on top of other containers (like arrays) to accelerate or reduce the number of accesses.

For example, you may have an existing array and using pointers you can use a quad-tree to quickly refer to the place in memory a certain object is.

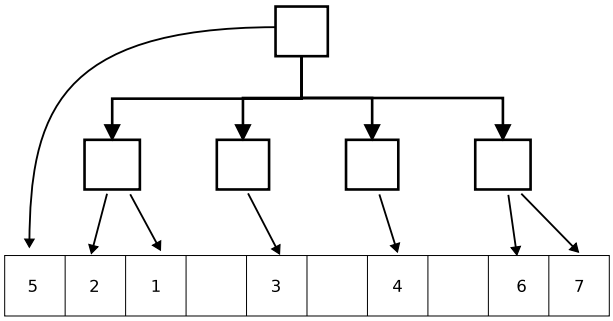


Figure 180: Quad trees as spacial acceleration structures

Redundancy will help us making things quicker and easier, adding a pointer from the underlying data structure back to the quad-tree will help us understanding where an object is positioned.

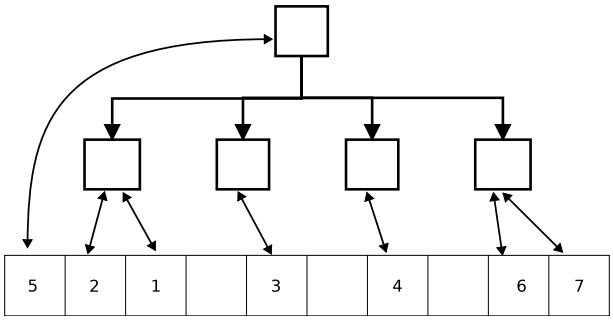


Figure 181: Redundancy in quad-tree pointers

9.3.2.2 Querying quad trees

Where quad trees shine is when we have an object and we want to check for collisions with any other object.

Using our "back pointer" we can refer back to the quad tree and severely limit the number of collision tests: any object will be able to collide only with its ancestors or descendants.

[This section is a work in progress and it will be completed as soon as possible]

9.3.3 Building AABB-Trees

Another way to efficiently execute a broad-phase collision detection is by building trees containing Axis-Aligned Bounding Boxes.

The main idea is similar to what we've seen with binary search trees, mixed with the quad-trees we've just talked about: we are trying to keep track of objects that are close together (like Quad-Trees do) and when searching, we try to eliminate a good portion of data each time we descend the tree (similarly to binary search trees).

This is done by calculating a "cost function" every time we insert an object into the tree: our objective is making the cost as little as possible. An idea for the cost function could be the size of the rectangle (expressed by its perimeter, or just $width + height$).

Our example image, would be represented this way:

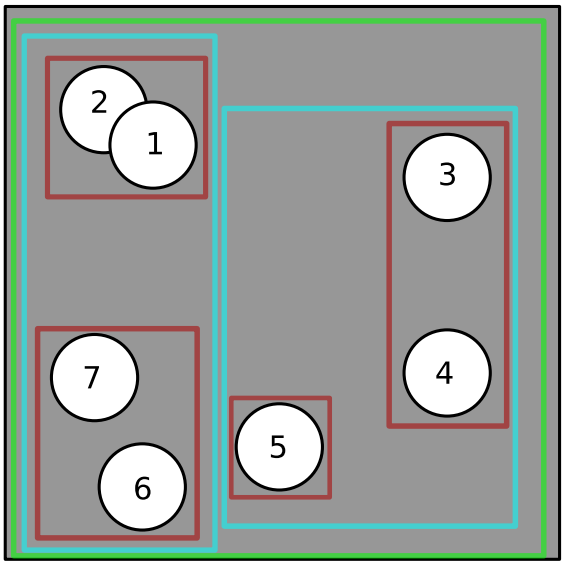


Figure 182: How an AABB-tree would process our example image

This can look a bit confusing, let's see how the tree would look like:

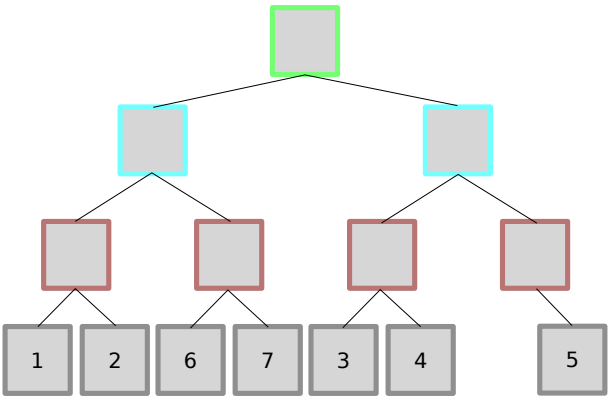


Figure 183: How a possible AABB-tree structure would look like

The performance of this tree is tightly related to its "balancing": differently from other types of "balanced trees", AABB-trees rely on how evenly each parent node is split by its children (instead of the usual "depth" metric). If an AABB-tree doesn't split evenly, the algorithm won't be able to "exclude" as many nodes on each iteration, thus degrading to a brute-force method (trying the given AABB against all other bounding boxes).

9.3.3.1 Querying AABB-trees

The idea behind this type of tree is making queries as fast as we can, and that can be done by checking on smaller rectangles on every iteration of our search algorithm. For instance we can find a list of possible colliding entities with a given bounding box in only a few tests (in our example).

Let's take for instance a circle "P" that is exactly between the points 3 and 4:

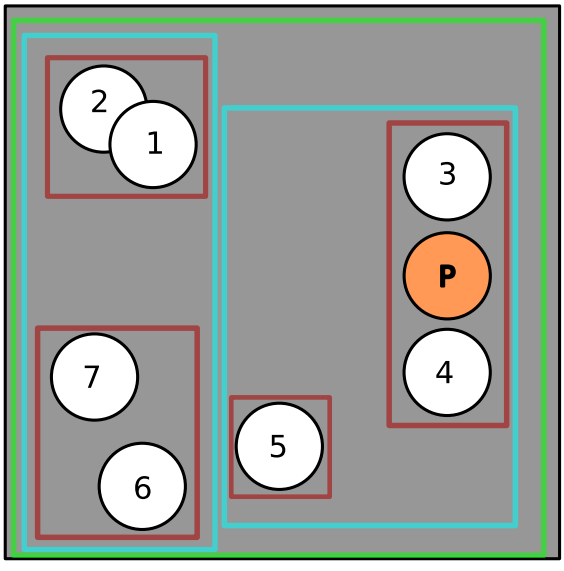


Figure 184: Example of a search in an AABB-Tree

First we do the root test, to see if it may collide with any of the 7 circles we have (if it was outside of the green

rectangle, we would have finished already). Then we do the “left (cyan) child” test, in this case we’re not colliding with the relative bounding box, so we keep going.:

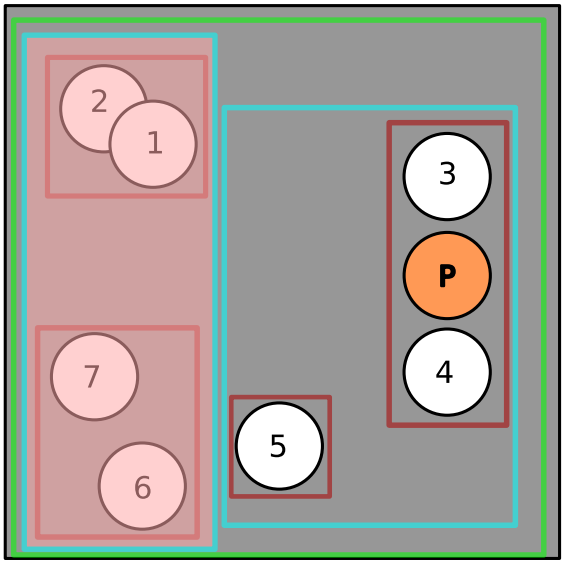


Figure 185: Querying an AABB-tree (1/3)

This way we excluded 1,2,6, and 7. We now do the “right (cyan) child” test, we’re colliding with the relative bounding box, we continue on this branch.

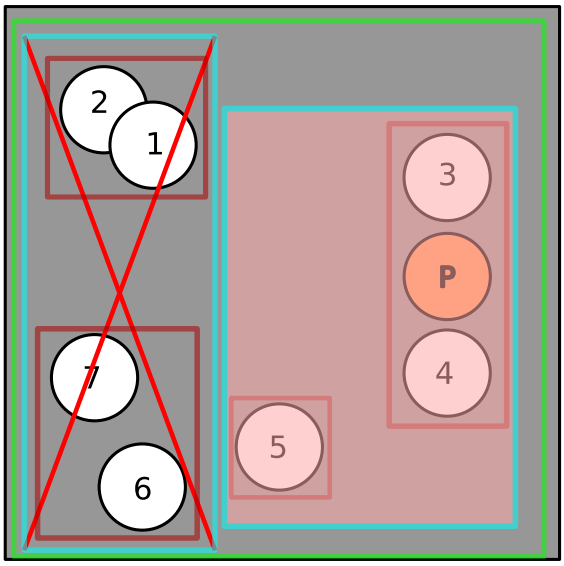


Figure 186: Querying an AABB-tree (2/3)

We do the “left (red) child” test, we’re colliding with the relative bounding box, now we can do a narrow-phase collision detection with the leaves of this node (and in the meantime we also excluded 5).

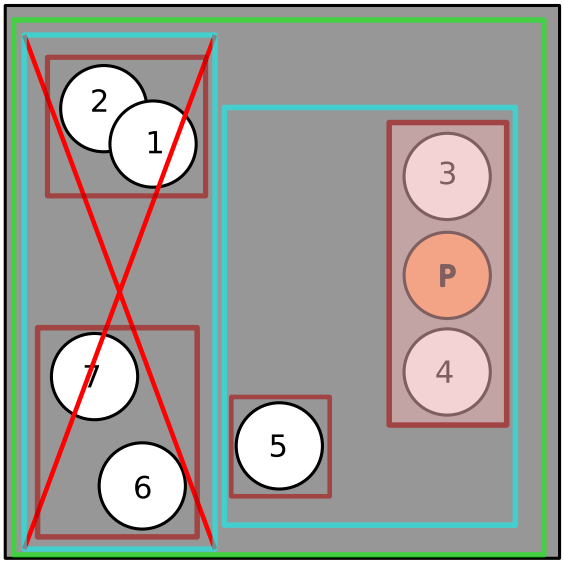


Figure 187: Querying an AABB-tree (3/3)

[This section is a work in progress and it will be completed as soon as possible]

9.3.4 Collision groups

[This section is a work in progress and it will be completed as soon as possible]

9.4 Other Collision Detection Methods

9.4.1 Calculating the position of tiles

When you are using tiles to build a level, being able to use quad trees or brute force methods to limit the number of collision checks inside your game may be harder than other methods.

Using a bit of math is probably the easiest and most efficient method to find out which collisions happened.

Let's take an example level:

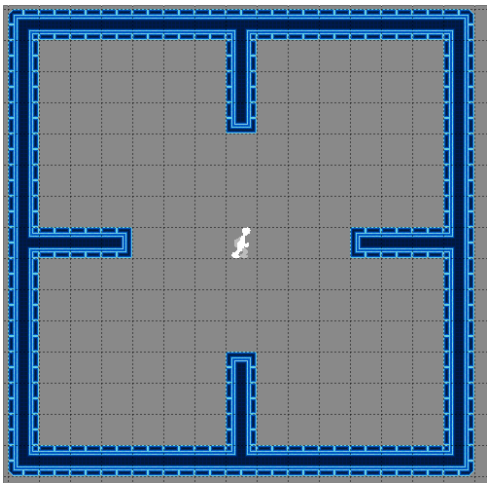


Figure 188: Example tile-based level

If a game entity is falling, like in the following example:

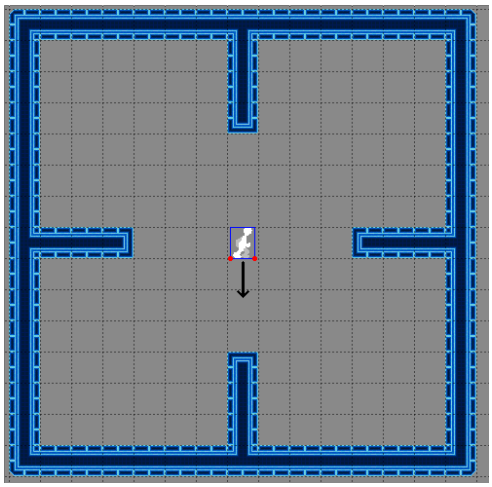


Figure 189: Tile-based example: falling

Using the simple AABB collision detection, we will need to check only if the two lowest points of the sprite have collided with any tile in the level.

First of all let's consider a level as a 2-dimensional array of tiles and all the tiles have the same size, it is evident that we have two game entities that work with different measures: the character moves pixel-by-pixel, the ground instead uses tiles. We need something to make a conversion.

Assuming `TILE_WIDTH` and `TILE_HEIGHT` as the sizes of the single tiles, we'll have the following function:

Listing 67: Converting player coordinates into tile coordinates

```
1 const TILE_WIDTH = 32;
2 const TILE_HEIGHT = 32;
3
```

```
function convert_pixels_to_tile(x, y){
    // Converts a point into tile coordinates
    let tile_x = Math.floor(x / TILE_WIDTH);
    let tile_y = Math.floor(y / TILE_HEIGHT);
    let to_return = [tile_x, tile_y];
    return to_return;
}
```

To know which tiles we need to check for collision, we just have to check the two red points (see the previous image), use the conversion function and then do a simple AABB check on them.

Listing 68: Tile-based collision detection

```
const TILE_WIDTH = 32;
const TILE_HEIGHT = 32;


class Rectangle{
    // A rectangle will represent the player
    constructor(){
        this.corner = new Point();
        this.width = 0;
        this.height = 0;
    }
}

function convert_pixels_to_tile(x, y){
    // Converts a point into tile coordinates
    let tile_x = Math.floor(x / TILE_WIDTH);
    let tile_y = Math.floor(y / TILE_HEIGHT);
    let to_return = [tile_x, tile_y];
    return to_return;
}

// We assume the player is falling, so no check will be shown here
let points_to_check = [
    Point(player.corner.x, player.corner.y + player.height),
    Point(player.corner.x + player.width, player.corner.y + player.height),
]
for (const point in points_to_check){
    let detected_tile_coordinates = convert_pixels_to_tile(point.x, point.y);
    let detected_tile = get_tile(detected_tile_coordinates[0], detected_tile_coordinates[1]);
    if (AABB(player, detected_tile.rectangle)){
        // React to the collision
        // ...
    }
}
```

Considering that this algorithm calculates its own colliding tiles, we can state that its complexity is `O(n)` with `n` equal

to the number of possibly colliding tiles calculated.

If an object is bigger than a single tile, like the following example:

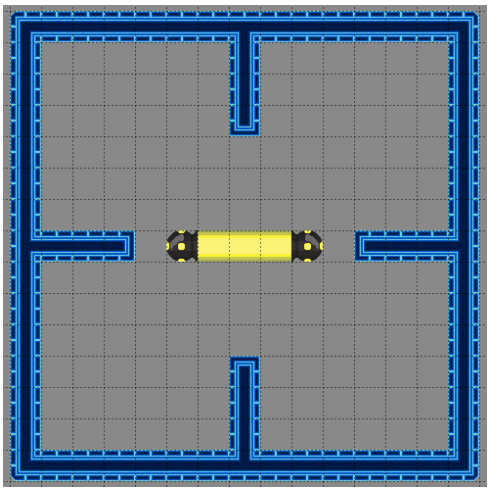


Figure 190: Example tile-based level with a bigger object

We will need to calculate a series of intermediate points (using the `TILE_WIDTH` and `TILE_HEIGHT` measures) that will be used for the test

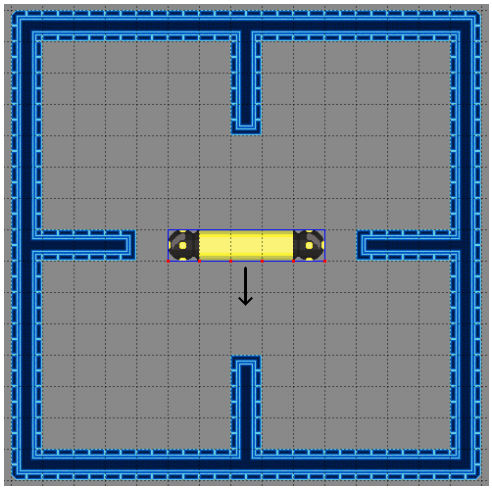


Figure 191: Tile-based example with a bigger object: falling

And using the same method the colliding tiles can be found without much more calculations than the previous algorithm, actually we can use exactly the same algorithm with a different list of points to test.

9.4.2 The “Tile + Offset” Method

This is a really good trick that works well for games that are heavily based on grids: the player can move only in the four cardinal directions and movement is tile-based.

By “tile-based movement” I mean that if you press any direction for even the smallest amount of time (even a single frame), the player will move in that direction by a tile (however big it may be).

The idea behind this kind of collision detection is very simple: some tiles are marked as walls. When the player wants to move in a certain direction, the game will check the tile in the chosen direction, if it’s a wall the movement will be blocked, if it’s passable the game will tween (usually using an offset parameter) the player travelling between tiles.

Listing 69: Tile + Offset collision detection

```
class TiledPlayer{
    constructor(){
        this.offset = new Vector2D(0, 0);
        this.current_position = new Vector2D(10, 10);
        this.next_position = new Vector2D(10, 10);
    }

    update(dt){
        // ...
        // Check which direction is the player going
        if (KEYBOARD.Up_Arrow_Pressed){
            this.offset.y = -1;
        }
        if (KEYBOARD.Down_Arrow_Pressed){
            this.offset.y = 1;
        }
        if (KEYBOARD.Right_Arrow_Pressed){
            this.offset.x = 1;
        }
        if (KEYBOARD.Left_Arrow_Pressed){
            this.offset.x = -1;
        }
        // Get the destination tile
        this.next_position = this.current_position + this.offset;
        // Is the tile a wall?
        if (!MAP.get_tile(this.next_position).isWall()){
            // No, move the player to the new tile
            this.current_position = this.next_position;
        }
        // ...
    }
}
```

This code shows only how to update the internal status of the player, which is what we care about. As you can see, the code is extremely simple, which makes for a great collision detection algorithm that doesn’t use a lot of resources. This algorithm can be extended and improved by handling collision with anything else that isn’t a wall (maybe enemy units?).

9.5 Collision Reaction/Correction

When you are sure, via any algorithm, that a collision has occurred, you now have to decide how to react to such collision. You may want to destroy the player or the target, or you may want to correct the behaviour, thus avoiding items getting inside walls.

9.5.1 HitBoxes vs HurtBoxes

First of all, we need to explain the difference between a "HurtBox" and a "HitBox".

Such difference can be more or less important, depending on the game that is coded, and sometimes the two concepts can be confused.

A **HitBox** is a shape (usually a rectangle, see Collision Between Two Axis-Aligned Rectangles (AABB)) that is used to identify where a certain entity can *hit* another entity. For the player a "hitbox" could encase their sword while attacking.

A **HurtBox** is instead a shape that is used to identify where a certain entity can *get hurt* by another entity. For the player a "hurtbox" could be their body.

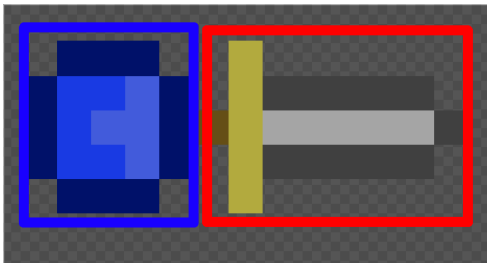


Figure 192: Example of a hitbox (red) and a hurtbox (blue)

9.5.2 Collision Reaction Methods

It has happened: a collision occurred and now the two objects are overlapping.

How do we react to this event in a convincing (not necessarily "realistic") and efficient manner? There are a lot of methods to react to collisions and below we will show some of the most used, along with some interesting ones.

We will use the following image as reference for each collision reaction:

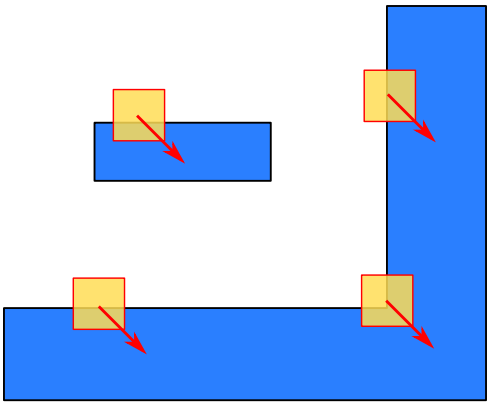


Figure 193: Images used as a reference for collision reaction

We will study each case separately, at the time the collision is detected (so the two objects are already interpenetrating), and each case will be a piece of this reference image.

9.5.2.1 A naive approach

This is the simplest method we can think of: as soon as the object gets inside of a wall, you push it back to one of the edges of the block, while keeping an eye on the direction it's moving.

9.5.2.1.1 How it works

This works when you treat the `x` and `y` axis separately, updating one, checking the collisions that come up from it, update the other axis and check for new collisions.

Listing 70: Code for the naive collision reaction

```
// Naive collision reaction with rectangles
function update(dt){
    // ...
    player.position = player.position + player.speed * dt;
    // Refer to your favourite collision detection and broad/fine passes
    if (collision(player, object)){
        if (player.x_speed > 0){  // going right
            player.position.x = object.rectangle.left;  // reset position
            player.x_speed = 0;  // stop the player
        }
        if (player.x_speed < 0){  // going left
            player.position.x = object.rectangle.right; // reset position
            player.x_speed = 0;  // stop the player
        }
    }
    // Again, refer to your favourite collision detection and broad/fine passes
    if (collision(player, object)){
        if (player.y_speed > 0){  // going down
            player.position.y = object.rectangle.top;  // reset position
            player.y_speed = 0;  // stop the player
```

```
21          }
22          if (player.y_speed > 0){  // going up
23              player.position.y = object.rectangle.bottom;  // reset position
24              player.y_speed = 0;  // stop the player
25          }
26      }
27      // ...
28  }
```

9.5.2.1.2 Analysis

Let's see how this method reacts in each situation.

When we are trying to slam against the wall, this method works as follows:

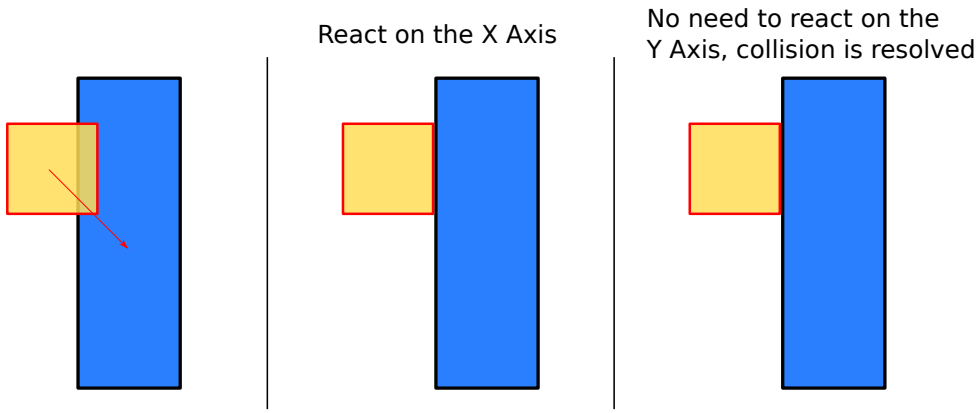


Figure 194: How the naive method reacts to collisions against a wall

1. We separate our position vector in its x and y components.
2. We check for collisions, and if so, we react on the x axis in a direction opposite to the x component of the velocity.
3. We check for collisions again, if there are any, we react on the y axis, in a direction opposite to the y component of the velocity.

9.5.2.1.3 Problems

Problems arise when we try to use the same method to react to a collision on a horizontal plane. In that case reacting on the x axis first will bring some unexpected surprises.

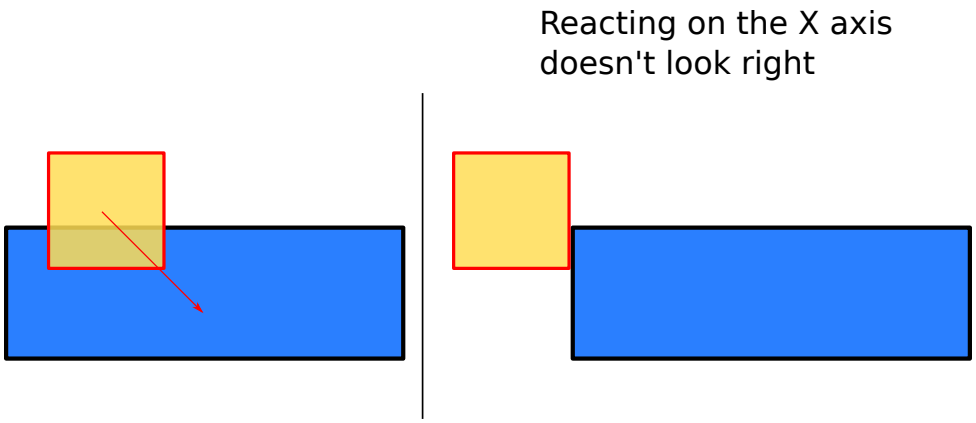


Figure 195: How the naive method reacts to collisions against the ground

We need to find a way to decide which axis we should correct first.

9.5.2.2 Shallow-axis based reaction method

This method works in a similar fashion to the naive method, but prioritizes reactions on the axis that shows the shallowest overlap.

This requires measuring how much the objects overlap on each axis, which can be a little more involved, but not really expensive.

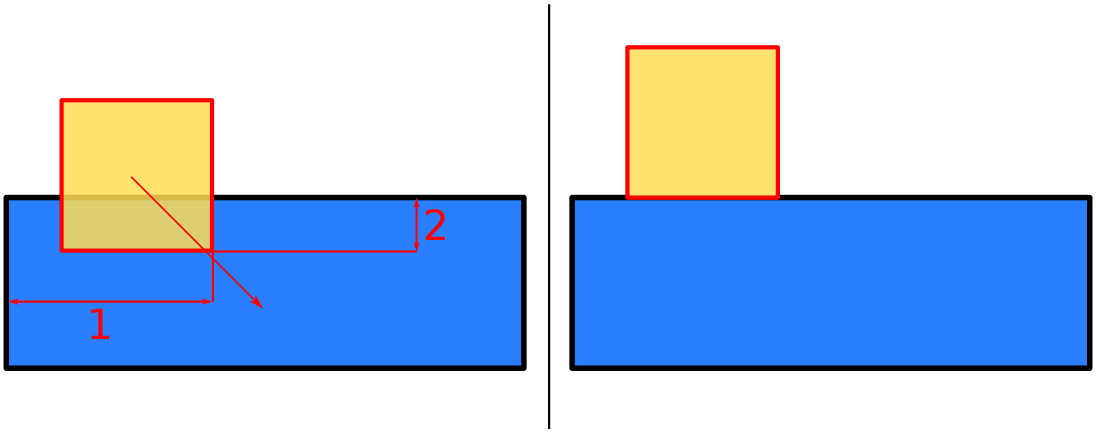


Figure 196: Example of shallow-axis based reaction on a horizontal plane

In the previous picture, we can see how the algorithm chooses to solve the collision on the y axis first and only on the x axis after; but since solving the y axis solves the collision, no reaction is performed on the x axis.

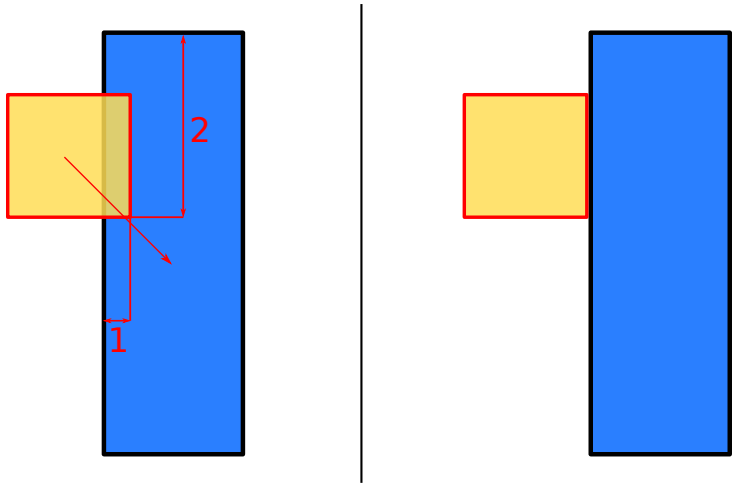


Figure 197: Example of shallow-axis based reaction on a vertical plane

In this new situation, the algorithm chooses to solve the collision on the x axis first; but since solving the x axis solves the collision, no reaction is performed on the y axis.

Listing 71: Possible implementation for a shallow axis collision reaction

```
function solve_collision(player, object){
    /*
     * This algorithm solves a collision between the player
     * and an unmovable object
     * We are assuming the player is moving
     */
    // The overlap will help us decide how to react
    let overlap = get_overlap(player, object);
    if (overlap.x > overlap.y){
        // Y is the "shallow axis"
        if (player.speed.y > 0){
            // Player is going towards the bottom of screen
            player.rect.bottom = object.rect.top;
        }else{
            // Player is going towards the top of the screen
            player.rect.top = object.rect.bottom;
        }
    }else{
        // X is the "shallow axis"
        if (player.speed.x > 0){
            // Player is going right
            player.rect.right = object.rect.left;
        }else{
            // Player is going left
            player.rect.left = object.rect.left;
        }
    }
}
```

9.5.2.3 Interleaving single-axis movement and collision detection

This is a method quite simple to understand: you split the movement in its x and y components, move on the first component, check and react, move on the other component, check and react again.

9.5.2.3.1 How it works

This works by treating the x and y axes separately, updating one, checking the collisions that come up from it, update the other axis and check for new collisions.

Listing 72: Code for interleaving movement and collision reaction

```
// Interleaving movement and collision reaction with rectangles
function update(dt){
    // ...
    player.position.x = player.position.x + player.x_speed * dt;
    // Refer to your favourite collision detection and broad/fine passes
    if (collision(player, object)){
        if (player.x_speed > 0){  // going right
            player.position.x = object.rectangle.left;  // reset position
            player.x_speed = 0;  // stop the player
        }
        if (player.x_speed < 0){  // going left
            player.position.x = object.rectangle.right; // reset position
            player.x_speed = 0;  // stop the player
        }
    }
    player.position.y = player.position.y + player.y_speed * dt;
    // Again, refer to your favourite collision detection and broad/fine passes
    if (collision(player, object)){
        if (player.y_speed > 0){  // going down
            player.position.y = object.rectangle.top;  // reset position
            player.y_speed = 0;  // stop the player
        }
        if (player.y_speed > 0){  // going up
            player.position.y = object.rectangle.bottom;  // reset position
            player.y_speed = 0;  // stop the player
        }
    }
    // ...
}
```

9.5.2.3.2 Analysis

Let's see how this method reacts in each situation.

When we are trying to fall on the ground, this method works as follows:

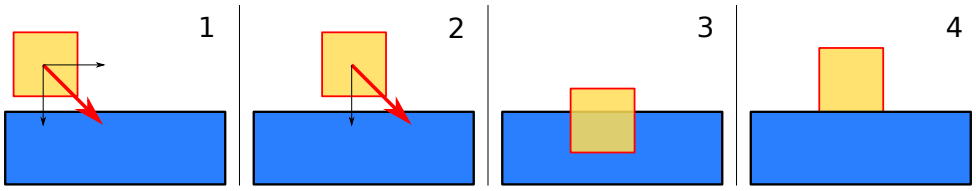


Figure 198: How the the interleaving method reacts to collisions on a horizontal plane

1. We divide the movement vector in its x and y components.
2. We move along the x axis and check for collisions, in this case there are none.
3. We move along the y axis, after checking for collisions we find that we are colliding on the ground.
4. We react to the collision by moving the sprite on top of the ground.

9.5.2.4 The “Snapshot” Method

This method is a bit more involved, but allows for a finer control over how you go through or collide with certain obstacles.

The secret to this method is taking a snapshot of the object’s position before its update phase and do a series of comparisons with the position after the update.

Listing 73: Example of the ”snapshot” collision reaction method

```
// Snapshot collision reaction
// All the sprite origins are on the top-left corner of the entity
let snapshot = player_instance.copy();  // The "snapshot"

// Update the player_instance here
player_instance.position = player_instance.position + (velocity * dt);

// Now check for collisions
// ...
for (const block of player_instance.get_collision_list()){
    if ((snapshot.y >= block.y + block.height) && (player_instance.y < block.y + block.height)){
        // We are coming on the block from below, react accordingly
        // Ignoring this reaction will allow players to phase through blocks when coming from below
        player_instance.position.y = block.y + block.height;
    }

    if ((snapshot.y + snapshot.height <= block.y) && (player_instance.y + snapshot.height > block.y)){
        // We are coming on the block from above
        player_instance.position.y = block.y;
        player_instance.on_ground = true;
    }

    if ((snapshot.y + snapshot.width <= block.x) && (player_instance.x > block.x)){
        // We are coming on the block from left
```

```
25          player_instance.position.x = block.x - player_instance.width;
26      }
27
28      if ((snapshot.y >= block.x + block.width) && (player_instance.x < block.x + block.width)){
29          // We are coming on the block from right
30          player_instance.position.x = block.x + block.width;
31      }
32  }
```

This method solves the problem given by platforms that can be crossed one-way, since (differently from methods based on the direction of velocity) you have an additional information: if you were colliding with the object in the previous frame.

[This section is a work in progress and it will be completed as soon as possible]

9.5.3 When two moving items collide

So far we've seen methods that involve a moving object colliding with a stationary one, but what if we wanted to react to a collision between two moving objects?

Some more math will be needed but it's not extremely difficult to pull off.

First of all, we need to find the "collision vector" (we'll call that u_{coll}), which is simply a vector that is calculated using the difference of the objects' positions. We'll need just the direction, so we will normalize it too (so it will become $\hat{u}_{coll}$).

Let's imagine two objects, with the following positions: $A(x_1, y_1)$ and $B(x_2, y_2)$

$$u_{coll} = (x_2 - x_1, y_2 - y_1)$$

$$\hat{u}_{coll} = \frac{u}{||u||}$$

Now we need to know how the objects are moving in relation to each other, this will allow us to see if and how we need to react. Let's calculate the "relative velocity" of the objects.

$$v_{rel} = (v_{x2} - v_{x1}, v_{y2} - v_{y1})$$

Now we need to see how the relative velocity affects the collision, which means we need to project such velocity onto the collision vector. Sounds like a job for the dot product.

$$s = \hat{u}_{coll} \cdot v_{rel}$$

s can be called "the speed of collision" (it's a scalar number, not a vector) and tells us what we need to know: if

$s > 0$ then the objects are moving away from each other already and we don't need to do anything. If $s < 0$ then the objects are moving towards each other

To react to objects that are moving towards each other, we just need to change their velocity by a factor of $s \cdot \hat{u}_{coll}$.

Here's how the code for reacting to the collision of two moving objects looks like:

Listing 74: Code for the collision reaction between moving objects

```
// First, we prepare some useful functions
function dot_product(u, v){
    return (u.x * v.x) + (u.y * v.y);
}

function scale_vector(factor, v){
    return Vector2D(
        x = factor * v.x,
        y = factor * v.y
    );
}

function magnitude(v){
    return Math.sqrt(dot_product(v, v));
}

// ...
if collides(obj1, obj2){
    // Here we know that obj1 and obj2 are colliding, and we assume
    // they are moving

    // Since the "position" field is a vector, we can easily calculate "ucoll"
    let ucoll = obj2.position - obj1.position;
    // Now we calculate its relative unit vector
    let unit_ucoll = ucoll / magnitude(ucoll);
    // Let's calculate the relative velocity of the objects, since
    // the "velocity" field is a vector, that's easy
    let vrel = obj2.velocity - obj1.velocity;
    // Now we calculate s
    let s = dot_product(unit_ucoll, vrel);
    // If s < 0, we need to change the velocity of the objects
    if (s < 0){
        obj2.velocity += scale_vector(s, unit_ucoll);
        obj1.velocity += scale_vector(s, unit_ucoll);
    }
    // ...
}
// ...
```

There are a couple of considerations to make with this simplified example:

- The moving objects are assumed to have the same mass. If they don't the calculation will change a bit.
- The collision is assumed to be "perfectly elastic", that means (in math terms) that the "restitution factor" will always be 1 (none of the impact is "absorbed").

Making a proper physics engine is beyond the scope of this book, so we'll leave it at the simpler version.

9.6 Common Issues with time-stepping Collision Detection

The methods we saw so far when checking for collisions are called "time-stepping techniques" due to the fact that each loop we "take a snapshot" of the situation and analyze it, this opens the door to a series of issues that may be annoying and we may find in our game development endeavors.

9.6.1 The "Bullet Through Paper" problem

The "bullet through paper" (sometimes called "tunneling") is a common problem with collision detection, when an obstacle is really thin (our "paper"), and the object is really fast and small (the "bullet") it can happen that collision is not detected.

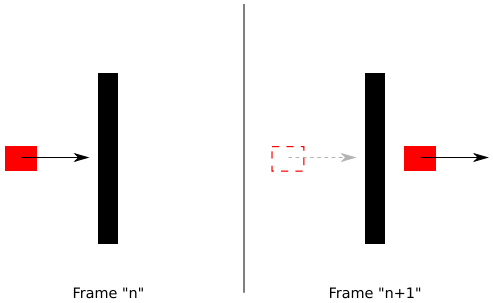


Figure 199: Example of the "Bullet through paper" problem

The object is going so fast that it manages to go through the entirety of the obstacle in a single frame.

Possible solutions to this problems are various, some even going out of the realm of the so-called "time-stepping techniques" (like speculative contacts or ray casting) that can be very expensive from a computational standpoint.

Such solutions should therefore be enabled (or implemented) only for fast-moving objects and only if necessary, since resources and time are at a premium in most cases.

9.6.2 Precision Issues

Sometimes it can happen that the position is reset incorrectly due to machine precision or wrong rounding, this can lead to the character that looks spazzy or just going through the floor at random times. The solution to these issues is making sure that the position and state are set correctly so that there are no useless state changes between frames.

Sometimes the "spazziness" of the character derives from the fact that collision reaction sets the character one pixel over the floor, triggering the "falling" state, the next frame the state would be changed to "idle" and then in

the frame “n+2” the cycle would restart with collision reaction putting the character one pixel over the floor.

9.6.3 One-way obstacles

Some of the methods exposed can be used only with completely solid obstacles. If you want to make use of platforms that you can cross one-way you should pay attention, since you may get teleported around when your velocity changes direction.

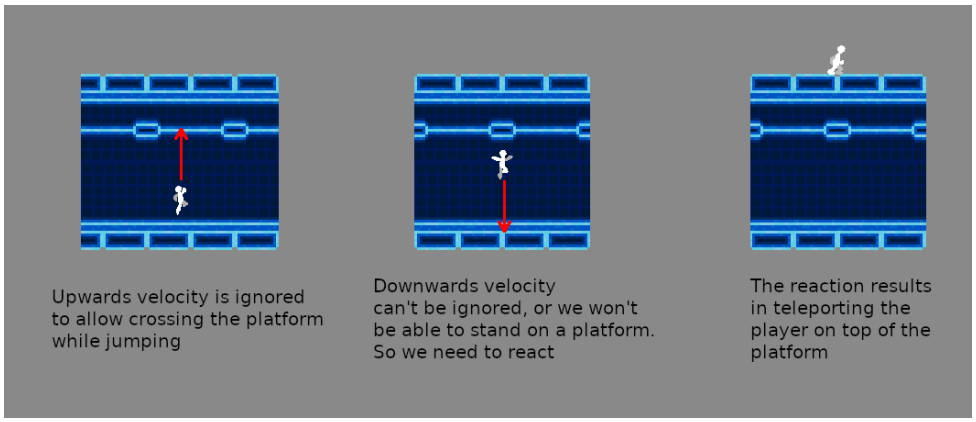


Figure 200: How velocity changing direction can teleport you

In the previous example we try to jump on a platform by going through it, but our jump quite doesn’t make it. Since velocity has changed direction, we end up being teleported over the platform, which is considered a glitch.

9.7 Separating Axis Theorem

We have taken an in-depth look at a series of specialized algorithms, but there is a more generic theorem that allows us to determine if two convex polygons are colliding: The *Separating Axis Theorem* or SAT. This theorem states:

> If two convex objects are not penetrating, there exists an axis for which the projection of the objects will not overlap.

This is connected to a simpler “human” explanation, which is:

> If two convex polygons are not colliding, then you can draw a straight line between them.

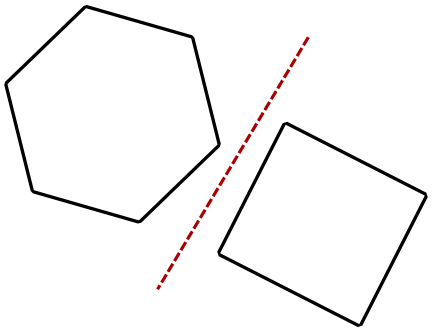


Figure 201: Example of how you can draw a line between two convex non-colliding polygons

Before delving further into the matter, let's see what we need to know:

- The difference between a Convex and a Concave Polygon
- What a Projection is
- Some Vector Maths

9.7.1 Why only convex polygons?

To explain this, we'll use the "human explanation": if one of the shapes is concave, there is a possibility that the polygons are not colliding, but we cannot draw a straight line between them.

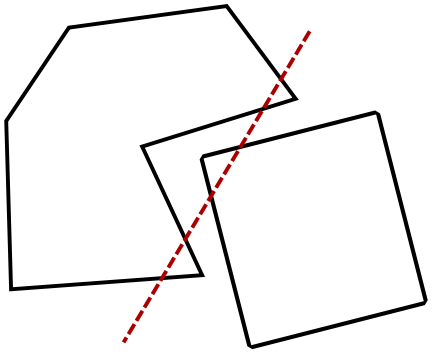
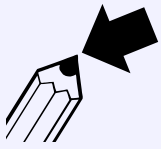


Figure 202: Why the SAT doesn't work with concave polygons

Thus our algorithm would return a collision where there is none.

Tip!



This problem can be solved by "decomposing" the concave polygons in two or more convex polygons, but for the sake of simplicity we'll assume all polygons we are checking for collisions are convex.

Now let's check the more "technical explanation".

9.7.2 How it works

Let's read the definition of the separating axis theorem again and break it down:

> If two convex objects are not penetrating, there exists an axis for which the projection of the objects will not overlap.

The first part defines the condition: in case two objects are not colliding, then what follows is true.

For what we were concerned so far, axes were "aligned to the screen boundaries", but axes can actually have different orientations and we can project shapes onto them.

The condition in our definition is represented as follows:

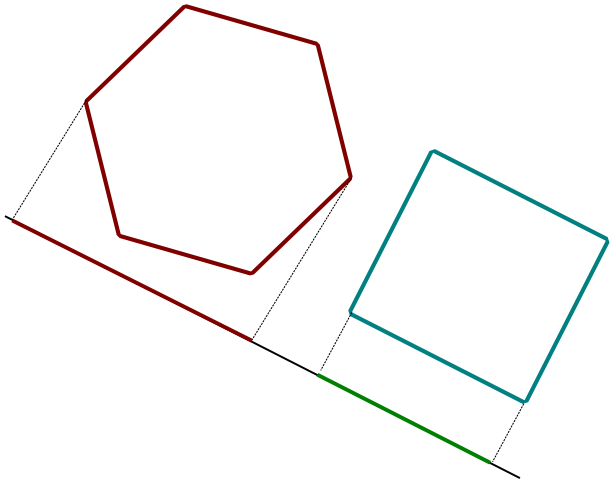


Figure 203: How the SAT algorithm works

As we can see, we have found an axis (which in this case is slanted) where the projection of the two shapes don't overlap. The presence of this axis where the projections don't overlap is guaranteed by the fact that the two polygons don't collide.

Random Trivia!

We can now easily see why the "human explanation" is (for our own purposes) equivalent to the "technical" one: we just need to take a single point inside the "gap" between the two projections and strike a line perpendicular to our axis.
That's our "separating axis".

9.7.2.1 Finding the axes to analyze

Now we only have a problem: we definitely can't spend an infinite amount of time trying all possible combinations in the hope of finding an axis where the projections don't overlap.

The fact is: we don't need to try them all. Actually we need to try just a few, as many as the sides of the polygons involved.

The axes we need to check are actually the axes parallel to the “normal of the polygon’s edges”. In layman’s terms: the axes we need to check are parallel to lines which are perpendicular to the edges of our polygons.

Let’s take it step by step, first we find the “normals”, which are just unit vectors perpendicular to the edges of our polygons.

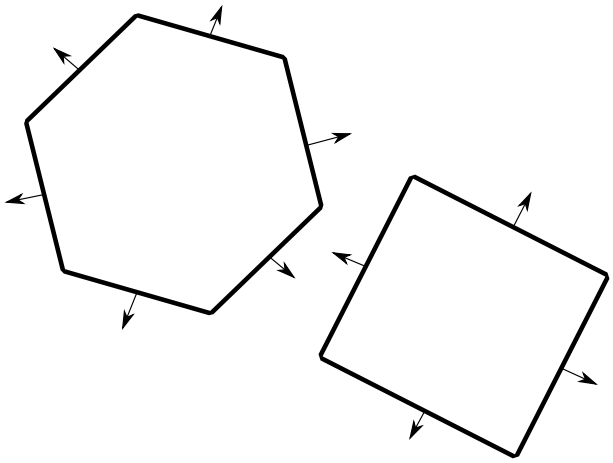


Figure 204: Finding the axes for the SAT (1/2)

Now we just have to strike axes parallel to those normals, and those are the axes we will need to check against.

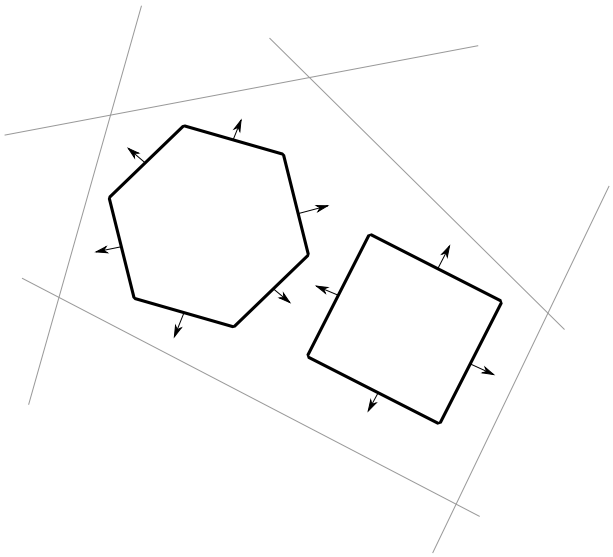
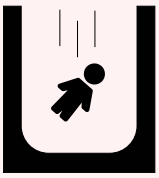


Figure 205: Finding the axes for the SAT (2/2)

In the previous pictures, I chose axes around the two polygons, for the sake of clarity.

Pitfall Warning!



Do not think that the axes we found are 5: there actually are 10. This is due to the fact that the figures I chose (for the sake of cleanliness) are a rectangle and an hexagon, which have edges that are parallel in groups of two.

9.7.2.2 Projecting the shapes into the axes and exiting the algorithm

Now, for each axis we found, we need to perform a projection of the two polygons onto such axis.

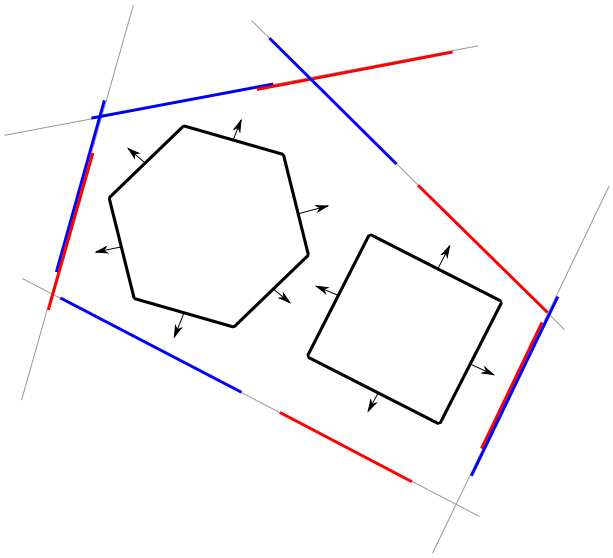


Figure 206: Projecting the polygons onto the axes

Now we consider each axis on its own and see if the projections overlap.

As soon as we find an axis where the two projections don't touch (overlap), we know that the two polygons are not colliding. Thus we exit the algorithm.

If all the axes we scan have overlapping projections, we can say that the polygons we're analyzing are colliding.

In the example, we can find two axes that have non-overlapping projections, thus the worst case is that the algorithm misses both of them 3 times in a row and exits at the fourth iteration.

Random Trivia!

If you use Axis-Aligned rectangles as your "polygons", you will notice how the Separating Axis Theorem will degenerate into something very similar to a simple AABB collision detection.
The only difference is that we're checking a condition where the rectangles **don't** collide.

Due to its nature, this algorithm has higher efficiency when there are few collisions, since it exits as soon as we find a separating axis (a gap in the projections).

9.7.2.3 From arbitrary axes to “x and y”

The only thing that remains is how to switch from an “arbitrary axis” to our usual “x and y” axes. Here projections will help us again: we can simply project our projections.

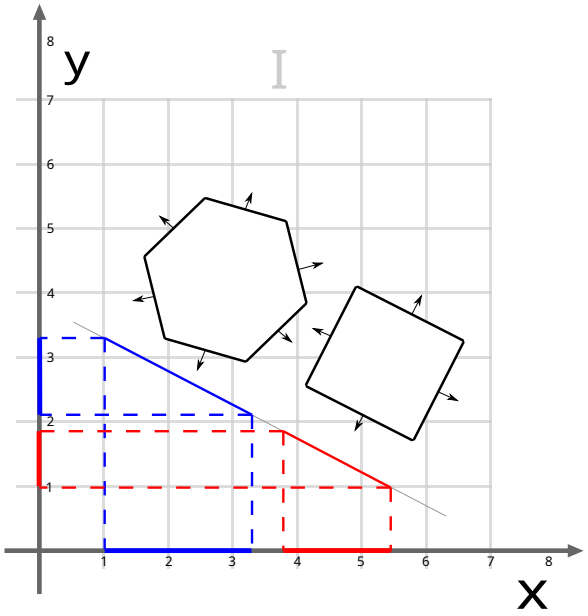


Figure 207: Projecting our projections onto the x and y axes

If we look closely, we’re just projecting polygons onto a bunch of axes so that they get “flattened to lines”, then we’re projecting such lines onto the x and y axes to see if there those lines are touching or not.

9.8 Ray Casting

Sometimes it can necessary to use unusual techniques to detect collisions: ray casting is one of them. If well used (and with some “illusion magic”), ray casting can be a nice way to solve the “bullet through paper” problem.

9.8.1 What is Ray Casting?

Mostly used in 3D, ray casting is a technique where you cast an imaginary ray (usually of light) until it hits something, but its uses can go beyond that.

Let’s take for example shooting a simple bullet: this can give some issues when the “bullet” is small and fast, as explained earlier in The bullet through paper problem.

First of all, let’s put up some (arbitrary) constraints that will help us making the computation easier and better performing without giving away our tricks too easily.

Our bullet will shoot from the barrel of our gun (duh!), but we also define a point where the bullet will despawn: this will limit our ray length and make our algorithm perform better. We can still give an excuse such as “bullets are affected by gravity” (which actually is true), and maybe use it as a difficulty management technique (stopping people from sniping the enemy can make the game harder and force the player to play the game the way we, the developers, want).

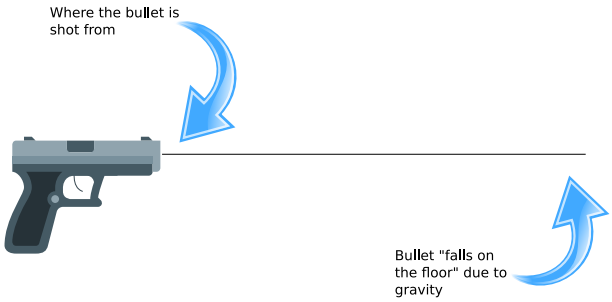


Figure 208: How Ray Casting Works: Gun (1/2)

Attached to our gun, is an invisible line (our ray), that will follow every movement of the gun itself

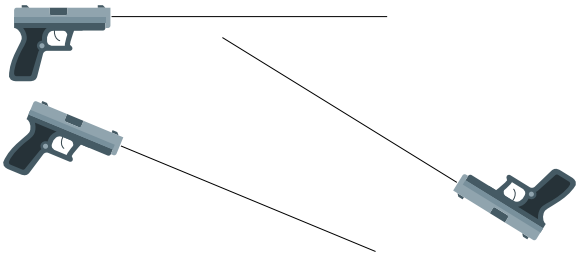
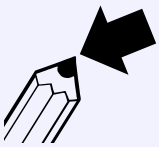


Figure 209: How Ray Casting Works: Gun (2/2)

When we want to shoot the gun, instead of using the previously stated “time-stepping techniques”, we perform a line-to-rectangle (or line-to-circle, or whatever we find best) collision detection, at the same time we play a really fast animation of the bullet shooting along the cast ray. If the cast ray hits an enemy, they’ll die (or get destroyed).

Tip!



If you find that the bullet animation won’t align well with the enemy dying, the animation may not be fast enough. Some games even give up showing the bullet at all, and instead show a white line for a split second, that fades away. The effect works really well!

[This section is a work in progress and it will be completed as soon as possible]

9.8.2 Other uses for ray casting: Pseudo-3D environments

[This section is a work in progress and it will be completed as soon as possible]

10 Scene Trees

> Trees sprout up just about everywhere in computer science...
>
> *Donald Knuth*

10.1 What is a scene

A scene usually represents a single screen in our game: it can be a menu or a single level. Many engines (like Godot) make use of this kind of abstraction to "break down" a game into more manageable pieces.

The problem is that single scene can contain tens if not even hundreds of elements, thus efficient management is necessary to avoid losing track of pieces of your game, as well as simplifying the drawing routines.

10.2 Scene trees and their functionalities

A scene tree is "yet another abstraction layer": pieces of your level are arranged in a parent-child relationship, which encourages composition-based approaches heavily, making the code more flexible and easier to maintain.

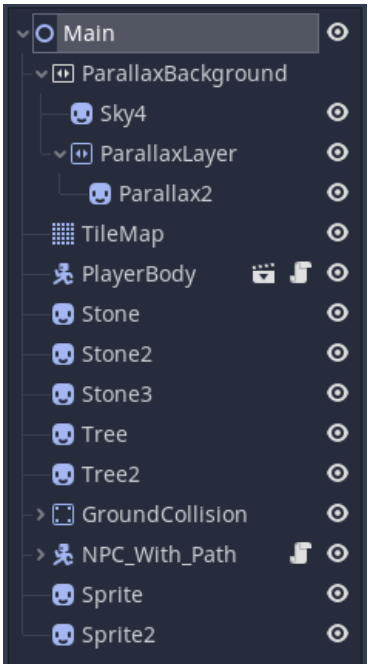


Figure 210: How a scene tree looks (specifically in Godot)

Each scene tree contains one or more "nodes" that represent a component of our level, like a sprite. These nodes can be grouped "logically" but scene trees can bring a lot more to the table.

10.2.1 How scene trees can make drawing entities easier

Let's imagine a game like the famous Galaxian: we have a ship that shoots aliens, and sometimes aliens can react by "breaking formation" and attacking the player. Sometimes a single alien can break formation, sometimes it's a group of three.

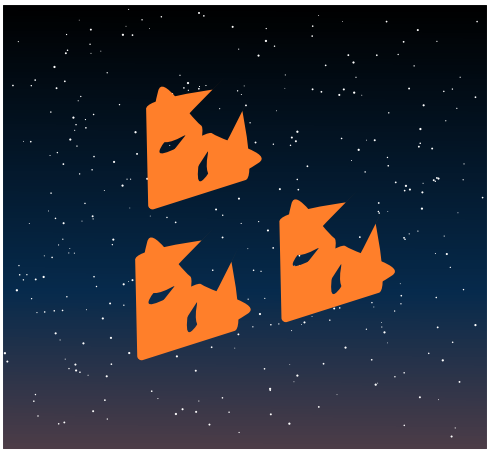


Figure 211: Example of a ship attack formation

The situation here is more complex than it seems: this “troop” has a “captain” leading two other ships, who are following at a fixed distance and angle: so if the captain moves, the “soldier ships” move, if the leader rotates, the “soldier ships” will rotate accordingly.

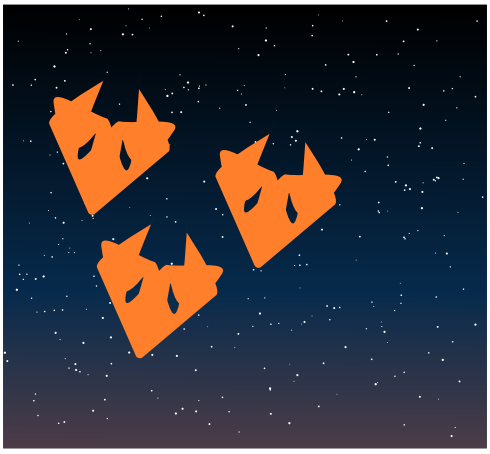


Figure 212: What happens when the ship attack formation rotates

This can quickly get messy, since we’ll have to rotate the leader according to the screen, then rotate the “soldier ships” according to the leader first, and then to the screen.

Scene trees can be used to make things easier, each node will rotate in relation to its parent.

10.3 Implementing a scene tree

[This section is a work in progress and it will be completed as soon as possible]

11 Cameras

> Nothing's beautiful from every point of view
>
> *Horace*

The great majority of games don't limit the size of their own maps to the screen size only, but instead they have maps way bigger than the screen.

To be able to manage and display such maps to the screen, we need to talk about cameras (sometimes called "viewports"): they will allow us to show only a portion of our world to the user, making our game much more expansive and involving.

11.1 Screen Space vs. Game Space

Before starting with the most used type of cameras, we need to distinguish between what could be called "screen space" and what is instead "game space" (sometimes called "map space").

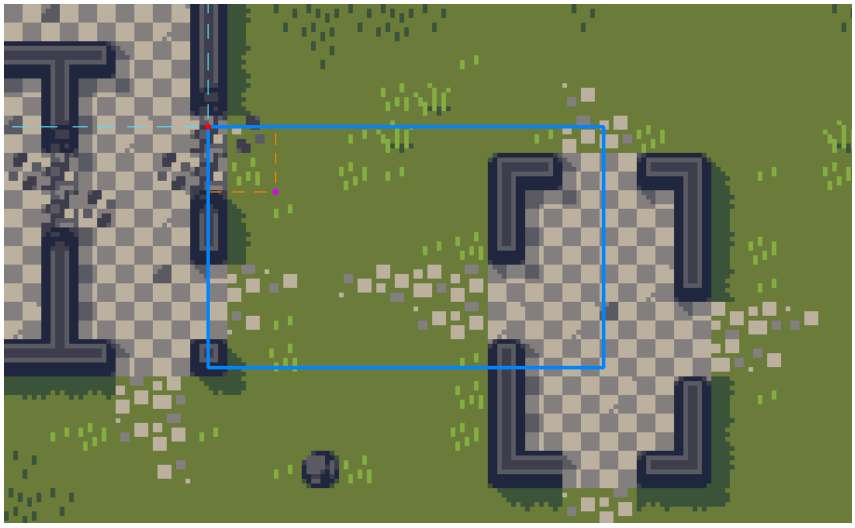


Figure 213: Reference Image for Screen Space and Game Space[1]

We talk about "game space" when the coordinates of a point we are talking about are referred to the top-left corner *of the entire game (or level) map*.

Instead we talk about "screen space" when the coordinate of such point are referred to the top-left corner *of the screen*.

Looking at our reference image, we can see how different the coordinates of the magenta dot are in screen space and in map space.

It is possible to convert screen space to map space and vice-versa by accounting for the viewport offset (represented by the red dot in the reference image), like follows:

[1] Jawbreaker tileset, listed as public domain at https://adamatomic.itch.io/jawbreaker

$$screen\ coordinates = map\ coordinates - viewport\ coordinates$$

$$map\ coordinates = screen\ coordinates + viewport\ coordinates$$

In a more friendly way, we can see our viewport as a "window" that moves around the map. Alternatively, we can see it as a viewport that is still all the time but has the map scrolling under it.

11.2 Cameras and projections

Advanced Wizardry!

This subsection gives a general idea on how cameras work in a 3D engine, but it is definitely useful to better understand how cameras work in general.

Cameras are just an approximation of of how we see things as humans. This approximation is due to a number of tradeoffs made to make things seem realistic, but avoid the issues that reality brings with itself.

Let's look at how a person sees, in a somewhat schematic way:

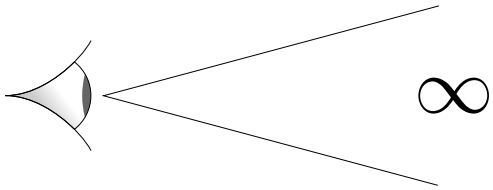


Figure 214: How a person sees things

A person can see anything directly in front of their eyes, to infinity (or at least until something blocks their vision, like a mountain, a building or fog).

We're definitely having a couple of problems: the first one is that we cannot represent infinity on a computer. If we try to represent everything from the camera's point of view to infinity, we won't be able to play the game at all.

The second issue is very close objects: in real life an object that is right up to your face will cover your entire vision. This may not be something that you want.

This is why computers render only things between two given planes, like the following:

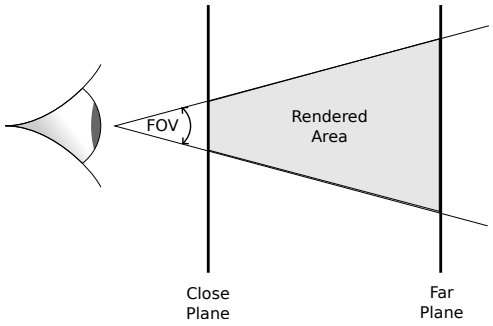


Figure 215: How videogame cameras see things

A videogame camera renders only what is situated between a “close plane” and a “far plane” (this means that objects too close or too far from the camera will not be rendered). Moreover objects are projected onto the screen, which may deform them if odd “Field of View” (FOV)[g] values are used.

11.3 Most used camera transitions and types

11.3.1 Static Camera

This is the simplest camera we can implement: each level has the size of the screen (or of the virtual resolution we decided, see Virtual Resolution), and every time we go out of the map, the screen fades to black and back to the new “room”.

[This section is a work in progress and it will be completed as soon as possible]

11.3.2 Grid Camera

This is an improvement on the static camera formula, each level (or room) has the size of the screen (or virtual resolution we chose), every time we go out of the map, the screen scrolls into the new section. This camera is used by the first Legend Of Zelda game for the Nintendo Entertainment System.

[This section is a work in progress and it will be completed as soon as possible]

11.3.3 Position-Tracking Camera

This camera is a bit more involved: the viewport tracks the position of the player and moves accordingly, so to keep the character centered on the screen. There are two types of position tracking cameras that are used in video games: horizontal-tracking and full-tracking cameras.

This type of camera can has some serious drawbacks when sudden and very quick changes of direction are involved: since the camera tracks the player all the time, the camera can feel twitchy and over-reactive; this could cause uneasiness or even nausea.

11.3.3.1 Horizontal-Tracking Camera

Horizontal-tracking cameras keep the player in the center of the screen horizontally, while jumps don't influence the camera position. This is ideal for games that span horizontally, since we won't have the camera moving when jumping and temporarily hiding enemies we may fall on.

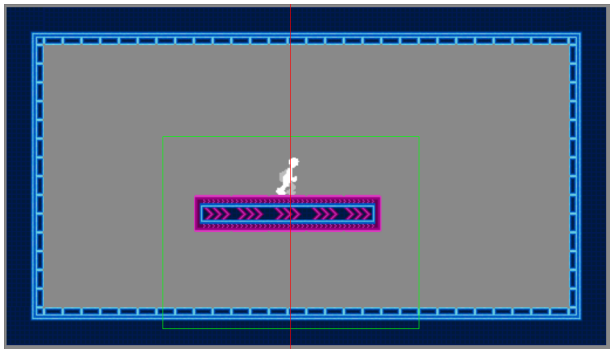


Figure 216: Example of an horizontally-tracking camera

This is the camera used in the classic Super Mario Bros. for the Nintendo Entertainment System.

[This section is a work in progress and it will be completed as soon as possible]

11.3.3.2 Full-Tracking Camera

Sometimes our levels don't span only horizontally, so we need to track the player in both axes, keeping it in the center of the screen at all times. This is good for platformers that don't require extremely precise maneuvering, since precise maneuvering could result in way too much movement from the camera.

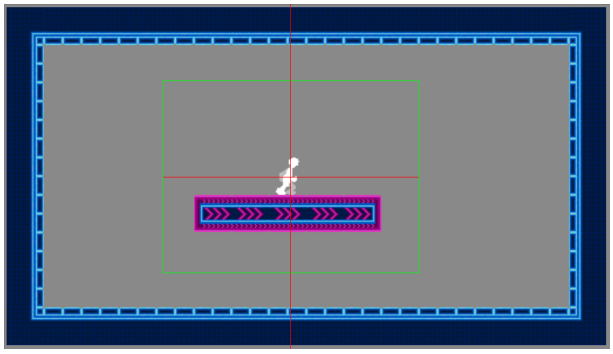


Figure 217: Example of a full-tracking camera

[This section is a work in progress and it will be completed as soon as possible]

11.3.4 Camera Trap

The "Camera Trap" system was invented to eliminate, or at least mitigate, the issues given by the position tracking camera. The playable character is encased in a "trap" that, when "escaped" makes the camera catch up in an effort to put the player back in such "trap".

The trap is represented by an invisible rectangle which can be visualized on screen in case you need to debug your camera.

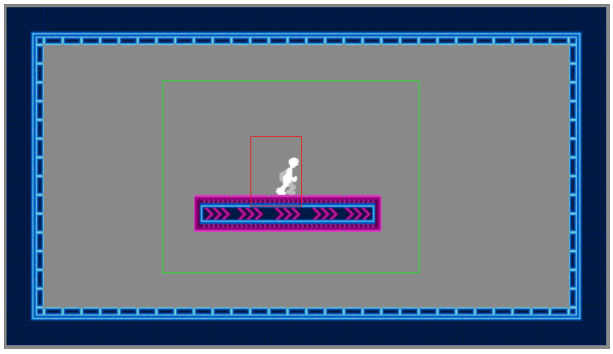


Figure 218: Example of camera trap-based system

This allows the camera to be less twitchy, giving a more natural sensation. Furthermore you can size the camera trap according to the type of game you are coding: slow-paced games can have a larger camera trap, allowing for the camera to rest more on the same screen, while faster paced games can have a smaller camera trap for faster reaction times.

[This section is a work in progress and it will be completed as soon as possible]

11.3.5 Look-Ahead Camera

This is a more complex camera that is implemented when the playable character moves towards a certain direction very quickly. The Look-Ahead camera is used to show more space in front of the player, giving more time to react to upcoming obstacles or enemies.

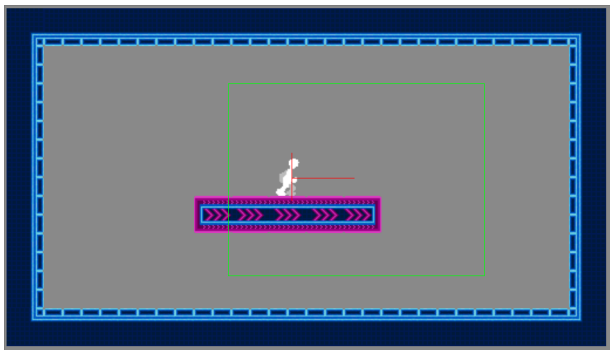


Figure 219: Example of look-ahead camera

This camera needs a good implementation when it comes to changing direction: having a sudden change of direction in the player character should have a slow panning response from the camera towards the new direction, or the game will feel nauseating.

So this camera is not ideal for games that require precision platforming, since the continuous “corrections” required to hit a tight platform would move the camera around too much, giving the player nausea.

[This section is a work in progress and it will be completed as soon as possible]

11.3.6 Hybrid Approaches

There are hybrid approaches to cameras too, mixing and matching different types of camera can give your game an additional touch of uniqueness. For instance in “Legend of Zelda: A link to the past”, the camera is a mix between a “camera trap” and a “grid camera”, where each zone is part of a grid, and inside each “grid cell” we have a tracking system based on the “camera trap”.

This allows the game to have a more dynamic feel, but also saves memory, since the SNES had to load only one “zone” at a time, instead of the whole map.

Another idea would be using an “out-of-center” camera trap that changes position according to how the player “escapes the camera trap”, thus solving some of the biggest issues of the look-ahead camera.

Feel free to experiment and invent!

11.4 Clamping your camera position

Whichever type of camera you decide to make use of (besides the static and grid cameras), there may be a side effect that could not be desirable: the camera tracking could follow the player so obediently that it ends up showing off-map areas.

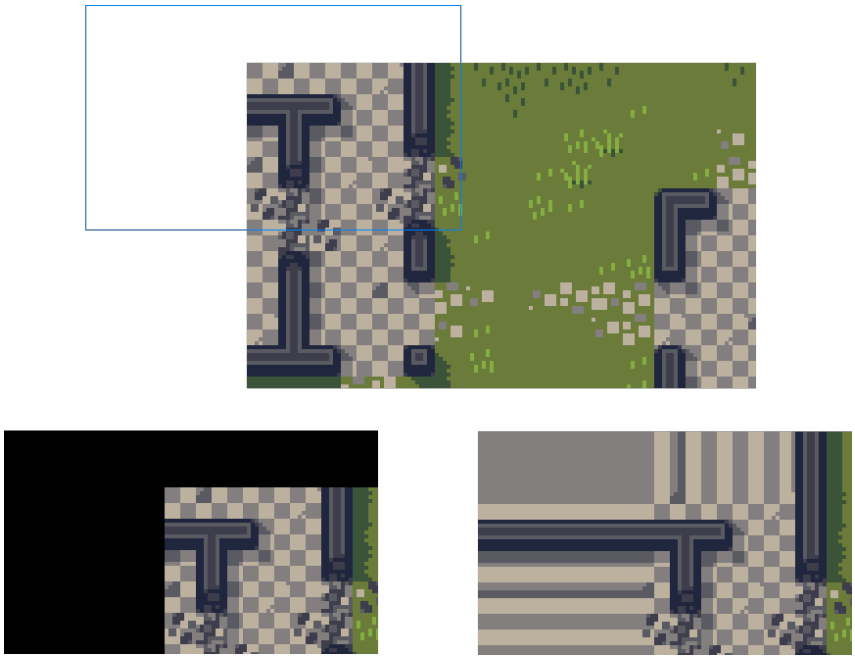


Figure 220: How the camera may end up showing off-map areas

Off-map areas may be shown as black in many cases, but in other cases (when the screen is not properly “cleared”) the off-map area can show glitchy versions of the current map.

In this case, it will be necessary to “clamp” the camera position, this way it will still follow the player, but won’t show off-map areas.

This usually just involves a check on the viewport boundaries against the map boundaries, followed by a reset of the coordinates to the map boundaries.

12 Game Design Tips

> There are three responses to a piece of design - yes, no, and WOW! Wow is the one to aim for.
>
> *Milton Glaser*

Game design is a huge topic, in this section we will just dip our toes into the argument, talking about some genres and features in games, including some tips and tricks that can make the difference between a “good” and a “bad” experience.

In this section we will also talk about level design tips, tricks and common pitfalls to avoid. We will talk about tutorials, entertaining the player and ways to reward them better.

12.1 Tutorials

12.1.1 Do not pad tutorials

Tutorials are meant to introduce the player to the game’s mechanics, but a bad tutorial can ruin the experience. Tutorials should be comprehensive but also compact, padding tutorials should absolutely not be a thing.

Gloss over the simpler things (usually the ones that are common to the genre) and focus more on the unique mechanics of your game.

Avoid things like:

> Use the “right arrow” button to move right, the “left arrow” button to move left, use “up arrow” to jump, use “down arrow” to crouch

Instead use:

> Use the arrows to move.

And eventually present the more complex mechanics in an “obstacle course” fashion.

12.1.2 Integrate tutorials in the lore

Tutorials are better when well-integrated in the lore, for instance if your game features a high-tech suit maybe you should make a “training course” inside the structure where such suit was invented.

By integrating the tutorial into the game world, it will feel less of a tutorial for the player, but more like training for the game’s protagonist.

12.1.3 Let the player explore the controls

Sometimes it’s better to allow the player to explore the controls, by giving them a safe area to try: this area is usually a tutorial or a specific training area.

It can prove more effective to avoid spoon-feeding your player with all the moves, and just let them explore the core mechanics of the game by themselves, eventually assisted by an in-game manual of some sort.

So instead of doing something like (thinking about a 2D tournament fighter):

> Do →↘↓ + A to do a chop attack
> Do →↗↑ + A to do an uppercut
> ...

Try something like:

> Do →↗↑ + A to do an uppercut
> Try more combination with your arrows and the attack buttons for more moves
> Check the move list in the pause menu

12.2 Consolidating and refreshing the game mechanics

12.2.1 Remind the player about the mechanics they learned

There's a latin saying that goes "repetita juvant", which means "repeating does good".

A good idea is to sprinkle around different levels concepts that have been learned previously, so to remind and consolidate them. This is more effective when done shortly after learning a new mechanic.

12.2.2 Introduce new ways to use old mechanics

After a while, old mechanics tend to become stale, to rejuvinate them we can apply such mechanics to new problems. Changing their use slightly can make an old experience new again.

For instance, knowing that shooting our magic beam against something on the ceiling will make it drop (usually killing an enemy), we can make the player use such envinronmental interactivity to drop a suspended weight to open a door, or shoot a bell to "force" a change of guard so to sneak stealthily.

12.3 Rewarding the player

12.3.1 Reward the player for their "lateral thinking"

A good idea could be rewarding the player for not throwing themselves "head first" into the fight, but instead thinking out of the box and avoid the fight altogether, or just win it differently.

Putting a very powerful enemy in front of some treasure (for instance currency used in-game) can seem unfair, unless you place an unstable stalactite that can be shot with your magic beam.

Your magic beam won't deal enough damage to the enemy to kill it before such enemy takes your life, but a stalactite on their head will do the trick, and the reward for such lateral thinking will be a heap of coins (or gems, or whatever currency you invented).

Figure 221: Example of how to induce lateral thinking with environmental damage [2] [3] [4]

Giving tips to the player by breaking the fourth wall can be another idea, a rock or a patch of dead grass conveniently shaped like an arrow could point towards a secret room that has a fake wall.

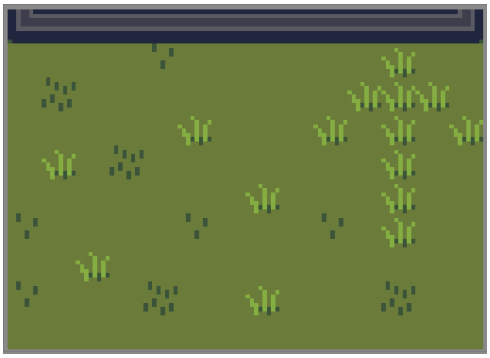


Figure 222: Example of how to induce lateral thinking by "breaking the fourth wall" [5]

This last tip should be done very subtly, so not to ruin the immersion. Unless your game takes advantage from these kind of things (for instance games based on comedy).

12.3.2 Reward the player for their tenacity

After suggesting to reward players for not butting head-first into fights, now I'm going to suggest the exact opposite (in a way): reward your players for their tenacity.

Beating a tough boss with a certain (weak) weapon, or just the plain tenacity and skill that is needed to undertake a hard task, such feats should be rewarded: for instance with a powerful weapon that can be used after some level-ups.

[2] 32x32 Chests attribution: Bonsaiheldin (http://nora.la), hosted at opengameart
[3] Simple SVG ice platformer tiles, listed as "Public Domain (CC0)" at OpenGameArt.org
[4] Fossil (Undead) RPG Enemy Sprites attribution: Stephen Challener (Redshrike), hosted by OpenGameArt.org
[5] Jawbreaker tileset, listed as public domain at https://adamatomic.itch.io/jawbreaker

12.3.3 Reward the player for exploring

Exploration can lead the player to discover secrets, which can range from simple gear, to pieces of unexplored environment, or even pieces of the game's lore.

World exploration should not be limited to simple secrets, a nice idea could be finding a path towards something that is usually considered "environmental damage" (like a catapult in the background) so that the player can deactivate it.

Thinking out of the box can lead to some really interesting results when it comes to this tip.

12.3.4 Reward the player for not immediately following the given direction

This is an extension of the previous point, the player should be rewarded for their exploratory efforts, even more when those efforts mean not immediately following the direction given by the designer.

"Thinking differently" should be rewarded with challenge and rewards up to said challenge. If the mission tells a player to climb up a tower, the more curious players could be led to hit the tower's underground dungeon before going on with the mission. A nice challenge in such dungeon with a fittin reward could expand on the game experience.

12.3.5 Reward the player for not trusting you entirely

Sometimes it can be fun, for both the game designer and the player, to play a bit of a trick to the player themselves.

Some famous games, like DOOM and Dark Souls, use secrets-in-secrets to trick players into thinking they found something valuable, while hiding something way more important. Let's see the example below.

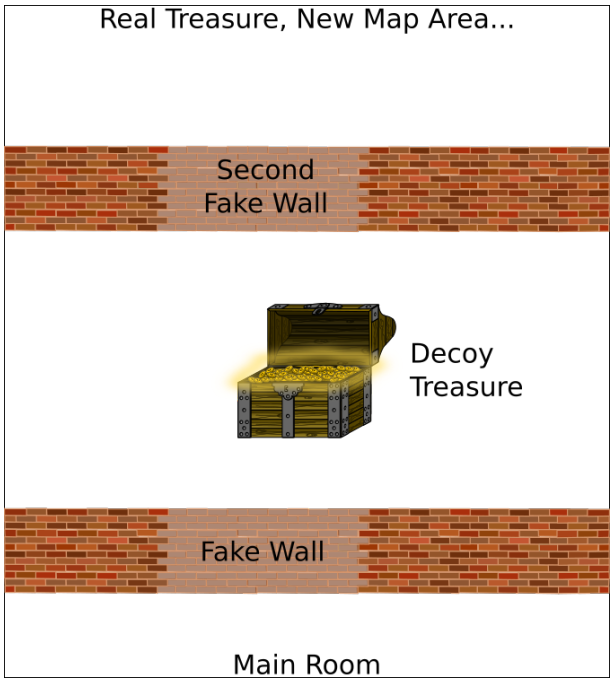


Figure 223: Example of secret-in-secret

We can see how we hid a secret inside of another secret and used a piece of valuable (but not *too valuable*) treasure to make the player think they found the secret, while the real secret is hiding behind another fake wall.

12.3.6 Reward Backtracking (but don't make it mandatory!)

To make the game's experience broader and richer, you may want to reward the player's exploration efforts by hiding treasure behind some backtracking.

For instance you can show the player a locked door somewhere in the level, such door will unlock and open after beating a boss monster or a wave-based challenge in the next room and hide some weapons that would otherwise be unlocked further into the game.

A nice idea would be "suggesting" to the player that something interesting happened, by playing the sound of the door opening as soon as the event is triggered. Another idea would be showing the player that the door opened (for instance if you're in an open area, the player would be able to see clearly an open gate that was definitely closed before).

The most important thing to remember is that all of this needs to be optional, a reward for the player's willingness to explore your levels further: avoid making backtracking mandatory, this will only feel like you're "padding the game" with nothing worth of note.

The player is paying you with their time and effort, it's only right that you pay them back with a pleasurable experience.

Random Trivia!

A nice example of backtracking bonuses (although mandatory to 100% the game) is used in the level "Sphynxinator" in Crash Bandicoot 3: Warped.
When you start the level, you can run backwards and you'll find 4 crates (which are necessary to get the "Gem" and 100% the game, but not mandatory for the normal ending), one of which is an extra life.
The backtracking is really short and gives you a nice bonus.

12.3.7 The "lives" system

Extra ships, 1-ups, extends, continues: these are all instances of what we can call the "lives system". This system gives a more "arcade feel" to your game and adds an important challenge factor to it.

Without something that threatens a game over, beating the game is no longer a challenge, but it's a matter of time. When overcoming a challenge is inevitable, it is not a challenge anymore, and the player will end up losing interest.

This is what the "lives system" is for: it's a "sword of Damocles", hanging over the player's head, continuously threatening a "game over" and pushing the player to do their best in order to get as far into the game as possible.

"Continues" are just "a lives system for your lives", they're in a very limited number (or have a price, like putting another quarter into the arcade cabinet) and allow you to "continue the game" with a new set of "lives" without

losing your progress.

As with all things in video games, it doesn't need to bring real challenge, but just the "illusion" of it.

Furthermore, lives and continues are a great tool to reward your player for their efforts: giving them an extra life every 20.000 points, granting a continue for a no-hit boss battle, putting a bunch of 1-ups in a hard-to-reach place are all great ways to challenge and reward your player and give your game more depth.

12.3.7.1 1-UPs

When a life system is in place, getting an extra life (a so-called 1-UP) is cause for celebration, since it allows the player to get further into the game or play with new and bolder strategies or just feel more at ease.

There are many ways you can reward the player with an extra life, such as:

- Finding a secret;
- Reaching a certain score threshold (for example every 100.000 points);
- Finding a certain item (a "physical 1-UP");
- Complete a certain combo-chain (for example kill over 8 enemies without touching the ground);
- ...

No matter how the 1-UP is achieved, this should be celebrated with a jingle that is very recognizable: this will allow the player to "know" that they got a 1-UP without thinking too hard about it. Not "celebrating" this event would make it "ordinary" and uninteresting, while it's extremely important in the grand scheme of things.

Some games even go as far as temporarily pause the game while the (short) jingle plays, that how important an extra-life is: "Stop everything! We got a 1-UP here!".

12.3.7.2 Other approaches

There are different approaches to a "lives system" that don't necessarily involve lives. The main objective is creating a mechanic that rewards the player for doing the right thing and punish them for doing the wrong thing (although not too harshly, or the player will stop playing).

One such approach was used in the first System Shock (1994) game: the space station the game takes place in is divided into floors. Each floor (with few exceptions) have a "cyborg conversion chamber": if you die your body will be brought to a conversion chamber in the floor you're in and will become a cyborg, serving the enemy. Being converted into a cyborg is the losing condition, and the punishment is being sent to the main menu and being forced to load a save file.

Here's the catch: the "cyborg conversion chambers" are fashioned out of "restoration bays", that means that disabling the cyborg conversion process will allow the player to be immediately resurrected at the nearest restoration bay, although they won't start at full health. This rewards the player for finding such restoration bays by giving them essentially infinite lives.

[Do you know more about this? You can contribute, this book is open source!]

12.4 Loading Screens

Loading screens can be subject to design too and deciding what to put in them can really enhance the player experience with your product.

12.4.1 What to put in a loading screen

"What can we put in a loading screen?" The answer may sound obvious to some, but a simple loading screen has a lot going on. Let's think about it, the most barren loading screen imaginable has at least two elements:

- An animation, to make the loading screen less boring and to ensure the player that our game didn't lock up;
- Some kind of progress indicator, to let the player know how far the loading routine has gone.

But we can put lots more into it, let's take a

- **Story:** In story-heavy games, it may be a good idea to put a reminder, a briefing or just a couple sentences telling what's happening story-wise. This was done in DOOM (2016).
- **World Building:** If the story is not-so-linear, a good idea could be just telling some facts about the world of the game, what some primary NPCs like, their habits, etc...
- **Tips:** Putting some tips to help the player is one of the most common things done with loading screens, these tips should be short and useful (no "press F to kill your enemy silently", that's basic controls).
- **Minigames:** If your game may take a potentially long time to load, making the player play a simple minigame (maybe with rewards) while they wait. This was done on the Playstation 2 (and can be enabled on the PC version) of Okami, where you can earn demon fangs with two minigames.
- **Status-related sentences:** This is something that can serve both as a "loading screen filler" as well as "hidden debug feature"
  - **Actual loading information:** Some players may like knowing what their PC is doing, so showing "loading backgrounds" or "loading sound data" is a nice screen filler and can give your players a pointer in case the game locks up while loading.
  - **Funny phrases:** Instead of boring, actual loading information, you can put funny phrases like "inserting buckazoids" (Space Quest, anyone?), if you have a list of funny phrases connected to actual loading checkpoints, it will function as a small debug helper.

12.4.2 Letting the player "exit" the loading screen

In some cases it can be a good thing to have a loading screen fade into the next stage (or area) directly, while in other situations it may be wiser to prompt the player to "exit" the loading screen themselves (maybe by a "Press any key to continue" prompt).

The most important factor is whether the areas we will load into are safe: if the player is not ready, they may get killed by enemies, and that feels unfair.

You should use a "press to continue prompt" when at least one of the following conditions applies:

- **Any area we load to may be unsafe:** We don't want our players to take a small break, walk around the

room and come back to a dead character;

- **There is text on the loading screen:** Be it a tip, world building or story, the player may be reading it, and taking away the text will annoy them.

12.4.3 Avoiding a loading screen altogether

Dynamic loading is a technique that is usually implemented in 3D games, but nothing stops you from using it in your 2D game!

The main issue is avoiding the player noticing that you unloaded a piece of the map that you previously visited, while loading the next piece. This means:

- **Avoiding "popping":** the player won't appreciate seeing pieces of the game appear or disappear in front of their eyes;
- **Avoiding slowdowns:** the player will immediately realize something is going on if the game slows down or drops frames at a certain point. Plus it will feel like the game is not well-optimized.

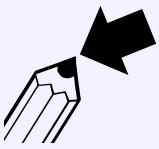
Popping can be avoided in many ways, the most common one can be summarized with "what's outside the screen doesn't exist": if more than a single room fits in your screen, what's outside the screen space is a good candidate for garbage collection.

An interesting idea could be making use of the player's point of view to try and foresee which room the player will head to next. With a clever use of doors and "cone of vision" you can unload rooms that are inside the screen space.

This can be done by "cutting" the player's field of vision using doors and unloading a kind-of far-away room that may be seen if the door was open.

Another way could be using so-called "points of no return": rooms where you can't go back, forcing the player to continue on a certain almost-linear path. Be careful to not use too many of them, though! Players don't really appreciate seeing possibilities cut off from them.

Tip!



Elevators make good dynamic loading rooms: you're changing floors, so it makes sense to not being able to see anything outside and having an elevator animating while changing floors can "hide" that a loading operation is happening.

12.5 Designing the story and gameplay flow

When we are preparing the terrain for our game, it is vital to have an idea of how the story and the gameplay will unfold. There are lots of different types of gameplay, here we present some of them.

12.5.1 Linear Gameplay

This is the simplest type of gameplay design: all story events come one after the other, without any possibility of deviating from the flow.

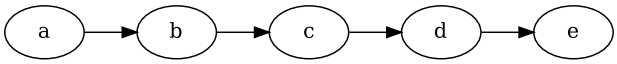


Figure 224: Example Scheme of linear gameplay

Very much like a presentation, there is no branching, but such linearity can present some advantages, like ease of testing and possibility of applying traditional storytelling tools which have been developed for thousands of years.

Table 42: Summary of linear gameplay

Gameplay Flow Type	Linear Gameplay
Advantages	Simple and cheap to test, traditional storytelling tools can be used easily.
Disadvantages	No replayability, the gameplay may not feel very “interactive”

12.5.2 Branching gameplay

Going towards more complex flow types, we can use branching to allow for more interactivity.

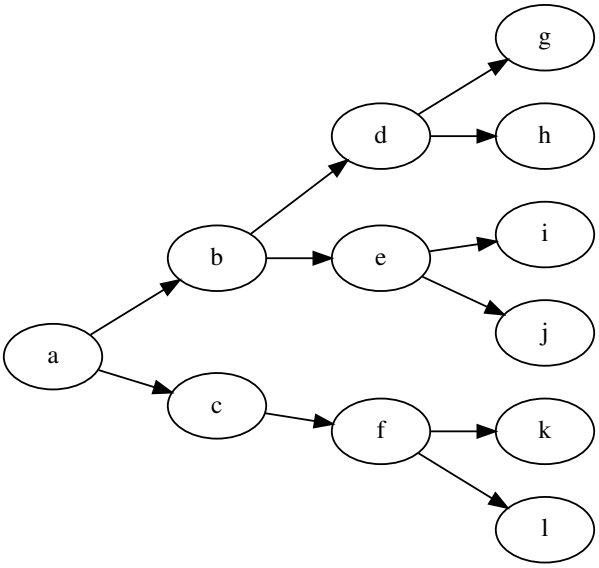


Figure 225: Example Scheme of branching gameplay

This type of gameplay flow allows for a lot of interactivity by crafting the game in a way that player decisions have a direct influence on the story flow.

This gameplay flow is harder (and thus more costly) to test, but allows for multiple endings.

Gameplay Flow Type	Branching Gameplay

Advantages	Simple to implement, allows for a strong feel of interactivity, allows for a lot of replayability by giving gameplay paths.
Disadvantages	Hard and costly to test, can get out of hand if not managed correctly.

12.5.3 Parallel gameplay

The Branching gameplay flow has a huge disadvantage: it can be really hard to manage and doesn't really suit well "more linear" games.

Here's where parallel gameplay comes into play.

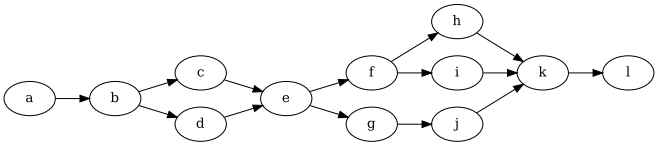


Figure 226: Example Scheme of parallel gameplay

In this flow style, there are branches running "parallel" to one another, but merge into "mandatory events" (which are usually story related). This way we have varied gameplay while keeping the story essentially linear.

Gameplay Flow Type	Parallel Gameplay
Advantages	Moderately expensive to test, some traditional storytelling tools can be used, story is easier to manage.
Disadvantages	Replayability suffers from a story standpoint. If not well-made the player will feel like the story is "on rails" from the get go.

12.5.4 Threaded gameplay

A different kind of gameplay is the "threaded" version, where there are many "beginnings", "middles" and "endings", usually done by playing different characters.

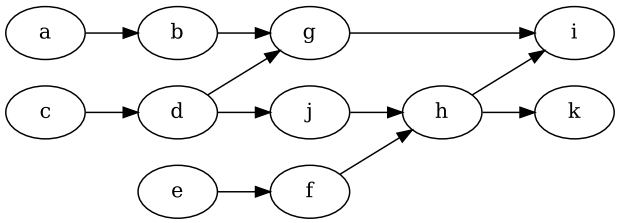


Figure 227: Example Scheme of threaded gameplay

This gives more replayability by giving many different and intertwining stories that allow to better understand a "bigger picture" of some sort. This gameplay flow can be costly, since it requires testing all the possible paths and crossings.

Gameplay Flow Type	Threaded Gameplay
Advantages	Good replayability, great for giving many "sides" to a story.
Disadvantages	Testing all the paths can be costly, more difficult to manage.

Random Trivia!

This was done in Resident Evil 6, where different characters (and teams) have different stories that overlap.

12.5.5 Episodic gameplay

A more "object-oriented" approach to storytelling can be done by making small "episodes" (like mini-stories) with many entry and exit points.

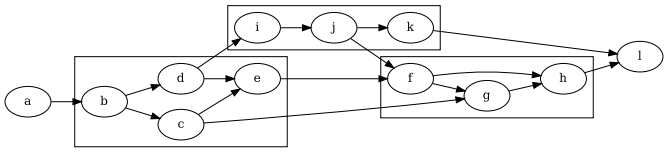


Figure 228: Example Scheme of episodic gameplay

We need to be mindful of loops (we don't want to replay an episode that was already completed) when laying out our episodes. This gameplay flow allows for great interactivity, but kind of "forces" replaying the game to see all the possible episodes.

Gameplay Flow Type	Episodic Gameplay
Advantages	Great interactivity.
Disadvantages	Tends to "force" replaying the game to see all episodes and paths, hard (and thus costly) to test and manage.

12.5.6 Adding parallel paths

Nothing forbids us to mix and match methods to create something that suits our game better.

A very much appreciated and used gameplay flow is having a linear story with lots of "side quests" to give some diversion from normal gameplay, as well as replay value, since people are bound to miss some side quests.

12.5.7 Looping Gameplay

This is a typical gameplay flow of roguelike games, where the player has to play the same game many times, beginning to end, eventually advancing a "bigger story".

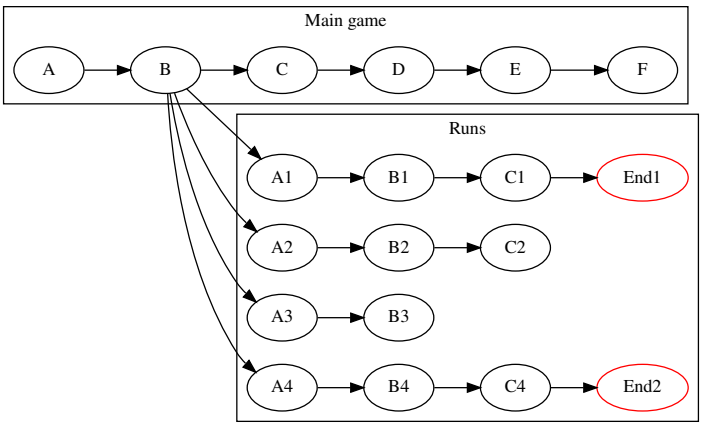


Figure 229: Example Scheme of looping gameplay with a overarching story

The most important thing when laying out a looping kind of gameplay is that the world needs to change between each "run": either by adding new weapons/items/collectibles or by unlocking a new part of the story (new levels, for instance) or adding new characters. Each run should feel like unique by itself.

Gameplay Flow Type	Looping Gameplay
Advantages	Great replayability.
Disadvantages	Needs a lot of care in laying out how the runs evolve between one and the next: if all the runs "feel the same" the player will abandon the game.

12.5.7.1 Soft-reset mechanics

Pretty often games adopting looping gameplay flow have one or more "soft-reset" mechanics. This mechanic consist in starting the whole game from the beginning, while having some advantages that come from the previous runs.

A soft-reset run may see:

- Main character having a higher experience gain;
- Starting with a "box of tools". For instance you start the new run with a weapon that had to be unlocked in the previous run;
- Main character starting with a higher attack;
- ...

This adds a new dimension to the gameplay: funnily enough, a player may progress through the entire game faster by resetting it.

This adds more decision making (thus more power) to the player, who can take the risk and "invest time" by soft-resetting the game (where the investment is the time spent getting back to the point they were before) to have higher gains in the long run.

Bonus points if you can add a reason for soft-resetting in the game's story.

There is no real limit to the quantity of soft-resets you can stack on top of each other, although it's advisable that each soft-reset has a suitable "reward" and different "reset harshness".

For instance you can have a soft-reset where you lose your weapons and levels, but have a 2% attack bonus. Then you can have a second level of soft reset where you lose money, weapons, levels and something else in exchange for 10% higher experience gain. Balance is the key.

12.6 Some game genres and their characteristics

12.6.1 Roguelikes and Rogue-lites

Roguelike games are usually games that involve dungeon-crawling and procedurally generated levels, usually with a fantasy background. In this small section we will take a look at the features that characterize roguelike games.

The most accepted interpretation of a roguelike game is the "Berlin Interpretation", which is based on the features that follow. When games diverge from these features, but are still loosely based on the classic roguelike design, they are usually called "rogue-lites" or "roguelike-likes".

12.6.1.1 Use of pseudo-randomness and procedural generation

This is done to increase replayability: the dungeons (or levels alike) are generated procedurally, with a tinge of randomness added to them. Joining procedural generation and pseudo-randomness is better than simple pseudo-randomness, since the rules applied will make the level beatable without special equipment, as well as lead to more aesthetically pleasing levels overall.

12.6.1.2 Permadeath

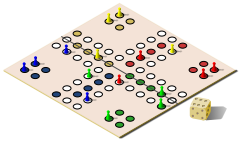
In the great majority of roguelike games, the death of a character is permanent. When a character dies, the player will have to begin a new "run": the levels will be generated anew and the available loot will change too.

Usually permadeath is joined with an erasure of the savefile connected to the "failed run", this avoids so-called "save-scumming": a practice where players would load back their savefile repeatedly to achieve better results (which is usually considered akin to cheating, in the roguelike field).

Another way to stop "save-scumming" is deleting the savefile when loading it, so when you save the only thing you can do to keep your savefile is exiting the game.

Permadeath makes the "save game" functionality more of a "suspension of the gameplay" instead of giving the player a recoverable state they can limitlessly return to.

12.6.1.3 Turn-based Gameplay



Like tabletop games, the gameplay of roguelikes is usually turn-based: this allows the player to take as much time as needed to take a decision.

12.6.1.4 Lack of mode-based gameplay

Roguelikes don't have a real concept of "progression": they allow you to do anything from the get-go, without blocking any action just because you're at a certain point in the game.

12.6.1.5 Multiple ways to accomplish (or fail!) a task

Roguelikes usually allow you to complete a task in many different ways, so many in fact that it seems the developers thought of everything. Let's take for example a locked door, a roguelike game would give you many options:

- Find the key or trigger to open such door;
- Lockpick it;

- Burn it down;
- Find a way around it;
- Kick it down;
- ...

This also means that you have to be careful with your actions: if a weapon freezes entities when it touches their flesh, you better have a pair of gloves handy (or you may end up frozen yourself!).

12.6.1.6 Resource Management is key

Resource Management in roguelike games is vital: usually they feature a hunger mechanic, as well as healing items, weapons and various loot that the player must sort through to be able to survive. The player will be forced to leave some loot on the floor of the dungeon, or choose between a known weapon and something unknown that may be weaker or “cursed”.

12.6.1.7 Peace was never an option

Most roguelike games are based on hack and slash mechanics, where your main goal is killing monsters. In this kind of games, “peaceful options” don’t exist (although they may exist, in a somewhat temporary fashion, to put leverage on some stealth mechanics - like getting a better weapon to kill a powerful enemy by first sneaking around them).

12.6.1.8 Dealing with the unknown

Roguelike games are heavily based on the concept of “unknown”: you need to explore an unknown place, finding loot which powers are unknown and should be identified. Magical items change with every run, and give just vague descriptions (like “a red potion”) which may heal in one run and kill you in another.

Furthermore items can be subject to change, acquiring or losing traits due to environmental alterations or player modification.

12.6.2 Deck builders

[This section is a work in progress and it will be completed as soon as possible]

12.7 Tips and Tricks

12.7.1 General Purpose

These tips and tricks are good for any kind of game: from the simplest platformer to twin stick shooters, to strategy games. These are good starting points to make your game feel more complete and fun to the player.

12.7.1.1 Make that last Health Point count

Players love that rush of adrenaline they get when they escape a difficult situation with just one health point. That “just barely survived” situation can be “helped” by the game itself: some programmers decide to program the last HP in a special way.

Some prefer giving the last health point a value that is higher than the other health points (kind of like a “hidden health reserve”), others instead prefer giving a brief period of invincibility when that last “1HP” threshold is hit.

These small devices allow you to give players more of those “near death” experiences that can give players that confidence boost to keep them playing through a hard stage, while at the same time, reducing the chance that they will rage-quit.

Random Trivia!

This was implemented in both DOOM and Assassin’s creed, where the last portion of health had more “hit points”.
In Bioshock when you take your last point of damage, you get about 1 or 2 seconds of invulnerability.

12.7.1.2 Avoiding a decision can be a decision itself

An interesting way to make the characters from a game seem more real, is registering the “lack of response” or “lack of action” in the game’s AI or dialogue tree.

This means that “ignoring” has consequences, and inaction is in and itself an action of “doing nothing” which should be accounted for, just like ignoring someone in real life can have serious consequence or where someone may prefer to do nothing instead of taking one of many bad decisions.

Random Trivia!

This trick is used in the game “Firewatch”, where not responding to a dialogue prompt is a noted decision.

12.7.1.3 Telegraphing

Players hate the feeling of injustice that pops out when a boss pulls out a surprise attack, that’s why in many games where precise defense movement is required bosses give out signals on the nature of their attack.

This “telegraphing” technique, allows for that “impending danger” feel, while still giving the player the opportunity to take action to counteract such attack.

Telegraphing is a nice way to suggest the player how to avoid screen-filling attacks (which would give the highest amount of “impending danger”).

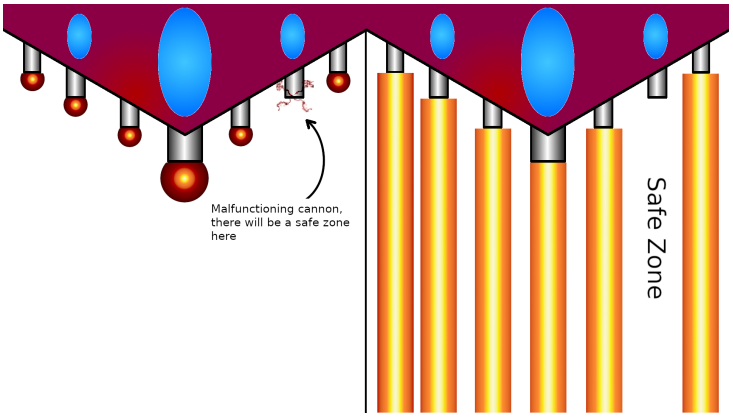


Figure 230: Example of a telegraphed screen-filling attack in a shooter

Another form of telegraphing is showing where the attacks will come from, using a “charging up animation”: this will attract the player’s attention towards those spots and help them gauge the next attack.

Random Trivia!

A form of telegraphing was used in the Bioshock series: the first shots of an enemy against you always miss, that is used to avoid “out of the blue” situation, which somehow communicates both the presence and position of enemies.

12.7.1.4 Minigames

Many times underrated, minigames are a really vital part of a great game experience.

Minigames can be a fun diversion from the main game, extending the engagement time, as well as a priceless resource for bigger open-ended games: you can use common “low-level” materials to feed into the minigame to get

better materials, weapons or prizes.

This is a win/win situation, you throw away unused materials to get useful tools, materials or cosmetics, also playing into the mechanism that maybe some people will get things wrong and need the “low-level” materials again, further lengthening the engagement of your game.

12.7.1.5 When unlockables are involved, be balanced

When you’re creating a game that involves “unlockables” (for instance a roguelike where you unlock more items for the upcoming runs), you should absolutely balance your unlockables in a way that compels the player to unlock them.

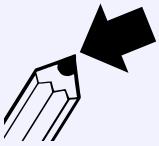
If you hide a “negative item” behind an unlockable, the player will actively avoid doing the actions that lead to unlocking such item. This is especially true now, in the age of widespread Wikis.

If you have to implement unlockables you should either:

- **Make the unlocked item a “good item”:** this will naturally compel the player to unlock such item to make the subsequent runs easier and more fun and varied;
- **Make the unlocked item a “neutral item” with situational good outcomes:** the player will be less attracted by these items, but the situational good outcomes (we can call them “interactions” or “synergies”) can make the player willing to put in the effort to unlock such item;
- **Unlock 2 items at once: a “good” one and a “bad” one:** the player may be less attracted by this “good+bad” combination, but may still be willing to go through with the unlock effort for the sake of the “good item” and also for added variety in their next runs;
- **Make the unlocked item a “bad item that can become good”:** this way the player may be attracted by trying to make a synergy using a “bad item” to cancel its bad effects while keeping the good ones. This allows the player to get engaged in “making builds” for their character.

If you really want put a “good item” behind “a gate”, a good idea would make the “bad item” a pre-requisite for unlocking a “very good item”. Alternatively, you can unlock the bad item “on the way” to unlocking a “good item”: for instance you can make “beat the first boss 5 times” a requirement for the bad item to be unlocked, while “beat the second boss 10 times” could be a requirement for the “good item” to be spawned.

Tip!



Remember: you should always account for wikis, some people think that wikis “ruin the surprise” of the game, while others use wikis just out of curiosity, some again instead use wikis as a “guide” to make the game easier or organize their strategy better.

12.7.2 Shooters

12.7.2.1 Make the bullets stand out

One of the most annoying things that can happen when you're running-and-gunning your way through a level is being hit by a not-so-visible bullet.

Your own bullets, as well as (and most importantly!) the enemies' should stand out from the background and the other sprites, so that the player can see and avoid them.

Some people may want to ask why your own bullets should stand out too, the answer is: so you can easily aim for your targets and distinguish your own bullets from the enemies'.

12.7.3 RPGs

12.7.3.1 Grinding and farming

When it comes to games part of the RPG genre, two words must be in your dictionary: **grinding** and **farming**, both as a player and as a game developer.

Sometimes used as synonyms and similar in execution, these terms are actually different and have different objectives. Let's see how.

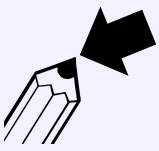
12.7.3.1.1 Grinding

Much like "grinding an axe", grinding in RPGs entails cleaning areas from enemies repeatedly (either by re-playing missions or just doing random encounters) with the objective of earning "experience points", thus making yourself stronger.

Grinding is somewhat a "self-leveling game design hinge", allowing you to have some leeway when designing the difficulty of your levels: if a player likes having an easier time, they will "grind themselves" to a higher level; if instead they prefer a challenge, they will power through the "easier parts" until they find the challenge they seek (due to being probably "underleveled").

We can also use "designed grinding", (as well as "level gates", where you need to have a certain amount of experience to continue) to pace our game and eventually even lengthen the experience a bit.

Tip!



When designing your levels and "designing your grind", you need to be mindful of your target audience.
Some cultures are used to (and enjoy) a higher amount of grinding than others, so too low of an amount may feel unsatisfactory to them, while an amount too high may be frustrating.

You should also be very careful on "forcing grinding" on your players: players like having choice and really dislike having anything forced on them, and this can change with your target audience.

Random Trivia!

Super Hydlide for the Sega Genesis/MegaDrive is one of the games that had its experience requirements tailored to the tastes of the market it was targeted to.
Considering the Japan release as the baseline, the US release sees its experience requirements halved as well as the given experience doubled.

12.7.3.1.2 Farming

Farming entails the same actions as grinding, but here we are using enemies as "farming animals", the objective is obtaining a certain amount of materials to obtain a weapon or item.

The most important aspect of "designing farming" is definitely reward the player for their farming: if a "special item" requires a lot of materials (and thus a lot of farming), such item should be worth the effort, or the player will feel cheated out of their time, effort and materials.

Note!

You should be mindful that some players will exploit some of your more complex mechanics to be able to farm for items and currency faster. This happened in the game "The Witcher III", where players used to kill cows and then meditate to make such cows respawn.

Random Trivia!

In the game "The Witcher III", precisely in patch 1.05, a mechanic to prevent such exploitation was introduced. The "Bovine Defence Force Initiative" consisted of a respawning moderately-leveled monster (called "Chort") that attacks the player, thus making the farming very dangerous for low level players.
On the flip side, higher level players could exploit this endlessly respawning monster to gain Chort Hides, which are worth more currency than Cow Hides. This was patched by making only one Chort spawn.

[Do you know more about this? You can contribute, this book is open source!]

12.7.3.2 Leveling Curves

When it comes to RPGs there are different ways to "design the leveling curve" in a game, depending on how you prefer designing your game.

12.7.3.2.1 "Exponential" curve

One of the most known ways to shape your leveling curve is making each further "skill level" require more experience points than the previous one, for example:

Table 48: An example of exponential level curve

Level	Experience Points
1	5.000
2	10.000
3	20.000
4	40.000
...	...

This type of leveling curve entails that each newer (harder) enemy gives out a higher amount of experience (not necessarily the quantity to make the experience feel “linear”).

The exponential nature of this level progression allows the “push” players towards harder enemies, since grinding lower-tier enemies becomes less and less efficient the more your level increases.

12.7.3.2.2 “Level-based” experience rewards

As an addition or an alternative to the exponential curve, some games try to “push” the players more towards harder enemies by scaling (down) their experience points with your level.

This method allows you to further shape the curve not only to try and prevent mindless grind of low-tier enemies, but also by rewarding challenging harder enemies.

This method can be implemented in many ways, one of them could be assigning a level to the enemy itself and then scale the experience rewarded by the difference between the player level and the enemy level.

Table 49: An example of “level-based” experience rewards

Player Level - Enemy Level	Experience Variation	Experience Reward
< -3 (very underleveled)	+ 600%	35.000
-2 (moderately underlevel)	+ 300%	20.000
-1 (slightly underlevel)	+ 100%	10.000
0 (ideal)	0%	5.000
+1 (slightly overlevel)	- 50%	2.500
+2 (moderately overlevel)	- 84%	800
> +3	Fixed Value	1

This can be done in many ways, from changing the formula itself to having “hidden status effects” that change the experience output according to the level difference.

12.7.3.3 “Mastering”

An interesting mechanic to add in RPGs is “mastering”, as in “mastering the usage of a tool”, this mechanic entails having set bonuses after a certain “mission” is completed with a certain tool.

Some examples could be:

- Kill 1000 enemies with a spear → 20% more attack power using spears;
- Resist 2000hp of damage while wearing a heavy armor → you can wear heavy armor without running speed penalties;
- Kill 50 high-rank demons with a shotgun → shotgun reloads twice as fast.

This can add a new level of depth in your game, as well as giving the player more choice over how they want to shape their character: do they want to spend the effort to get a “mastering bonus” or is their time spent better elsewhere?

Random Trivia!

Tales of Berseria includes a very interesting “mastering” mechanic: once you master a piece of equipment, its bonus “sticks” to the character and is kept even after the piece of equipment is removed. This stimulates the player to try new equipment to “collect bonuses”.
On the flip side, if you keep the same piece of equipment, such bonus is doubled. This way the player can decide between a “double bonus” or “enlarging their bonus pool”.

12.8 Perceived Fairness

12.8.1 You don’t need precise collision detection

Here we go, spouting bold claims again. Here’s a bolder one: you don’t **want** precise collision detection. It’s slower, harder to implement and most of all, the player may get annoyed at it.

In the heat of a gaming session, with all the action going on, the player may become a bit “blind” to small things: this means that they will perceive as arbitrary anything that is not “evident”.

Let’s take this arrow vs “generic jumping man” example:

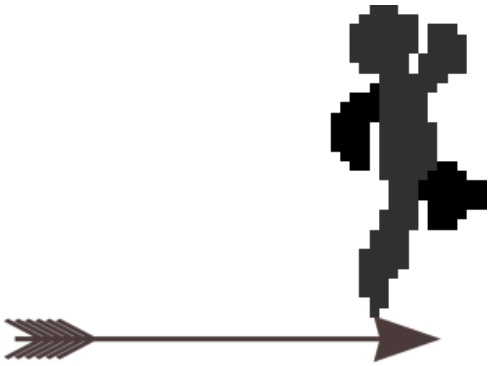


Figure 231: A pixel perfect detection would trigger in this case

With still images, things are obvious: that's a collision, it's on the tip of the character's toe, but it's a hit. It may be 1 or 2 pixels (it's actually 3), but it is still a hit.

Now let's imagine the image moving: the arrow darting from left to right while the character is ascending, it looks like a near-hit, the player will appreciate the adrenaline rush of a near-death. If we go and declare that as a "hit", the player won't understand why, they will say "that's unfair, it just missed me". Our collision detection was **too precise**.

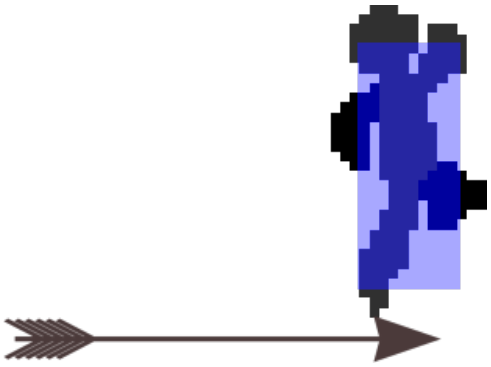


Figure 232: A smaller hitbox may save the player some frustration

Sometimes we just need a hitbox that is small enough to avoid these "looks-like-a-miss" incidents: the instances where the collision detection triggers are evident and the player will appreciate the sensation given by the near-deaths.

Obviously you need to strike a balance, if the hitbox is so small that evident hits are counted as misses (with few exceptions, like bullet hells) it will break the immersion.

12.8.2 Immediate dangers should be well visible

One of the biggest frustrations a player can encounter is definitely being damaged (or killed) by a hidden object.

To avoid these "unfair shots", we should draw immediate dangers "late" in the drawing phase of the game loop, but there is an issue: we don't want to break the immersion.

Let's take the following example: a bomb gets spawned just behind a chest by an enemy, and our character is dangerously close to it.

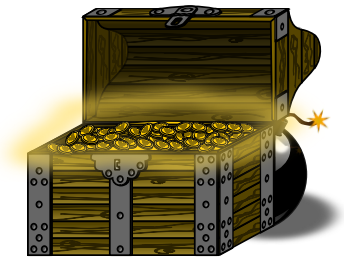


Figure 233: A bomb spawned behind a chest, drawn in a realistic order

If the bomb is "behind" the chest, we can't suddenly make it pop "in front of" the chest, that would definitely ruin the immersion, also it may end up messing with the forced perspective that some 2D games use.

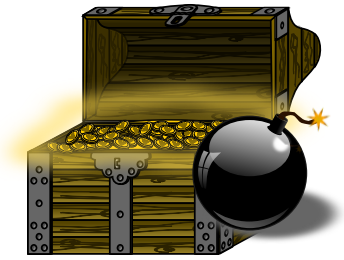


Figure 234: Moving the bomb in front of the chest may ruin immersion

As you can see, even though the bomb has a shadow, it looks like the bomb and the shadow are "floating mid-air", thus ruining immersion.

Different games implement different solutions to the problem, some prefer highlighting the danger with an outline drawn over everything, something like the following:

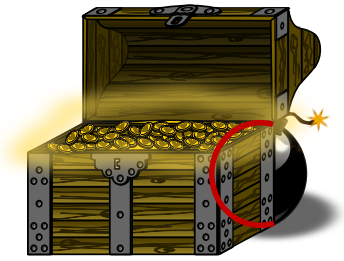


Figure 235: Highlighting the hidden part of a danger can be useful

Other instead prefer making the "foreground objects" semi-transparent, so that the player can see what lies behind.

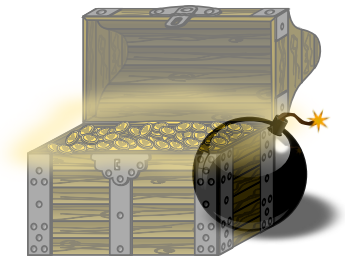


Figure 236: Making objects transparent is a solution too

This solution is usually applied when the player themselves are behind the obstacle, giving a more “interactive” and “less confusing” feel to the entire game.

12.9 Miscellaneous

This section denotes some various things that don’t really fit in “tips and tricks” but are still related to game design.

12.9.1 You cannot use the “Red Cross” in games

Note!

What follows **is not legal advice**. I am not a lawyer.

If you want to know more (as in quantity and quality of information), contact your favourite lawyer.

When you are developing a game, you will be tempted to use the famous “red cross” symbol on your health packs or health-related items.

Don’t do that

The red cross symbol (a red cross over white background) is not in the public domain, but it’s actually a symbol governed by the ICRC (International Committee of the Red Cross) and you may get in legal trouble for misusing it.

Halo and Doom changed their health packs symbol, from the red cross to a “red H” and a “pill” respectively.

The ICRC enforcement of this rule is inconsistent, but it would technically be a violation of the First Geneva Convention, chapter VII, articles 44 and 53.

Also states themselves (like Canada) tend to have rules of law regulating the use of the symbol in more detail.

12.9.2 Auto-saving

Some people may consider auto-saving a simple “quality of life improvement”, but it can also save the players a lot of frustration in case your game crashes: trust me, no matter how good your programming is, your game will crash (it may be a buggy graphics driver, an edge case that hits 0.0001% of the time or just bad luck).

If possible, you should provide the player with both an auto-save feature and a “manual save” one, this way the player can save where they want but also have a back-up just in case.

To implement an auto-saving feature, we need a slot to auto-save into, so we can choose one of two ways:

- **Choosing the save slot when starting a new game:** this means that the auto-save feature will auto-save and overwrite the selected save slot at every major event, which may be not desired. This is where the manual saving feature comes handy: allowing the player to save manually will also allow them to create a backup savefile.
- **Dedicated “auto-save” slot:** this leaves the manual saving feature intact, but also adds a “special saving slot” the player can’t save onto. This slot is dedicated to the most recent auto-save (regardless of the save slot we load from).

12.9.3 Feedback is important

It is extremely important to add feedback to actions, such as hits: a good visual feedback and the right sound can make all the difference in the world for your game.

The most common visual reaction to a hit is lighting up (by adding a white overlay) the sprite that got hit: this way it is really evident that a hit happened.

The visual feedback should also mirror the effectiveness of the hit too. An explosive weapon should do tons more damage than a single bullet: if this doesn’t happen the weapons will feel unbalanced and just badly designed.

13 World Building

> Nobody believes me when I say that my long book is an attempt to create a world in which a form of language agreeable to my personal aesthetic might seem real. But it is true.
>
> *J.R.R. Tolkien*

Good part of game design consists in creating a world that is coherent and makes sense in the players' minds.

If a world doesn't abide by its own rules well enough, the so-called "suspension of disbelief" is interrupted and the player will begin to think with the logic of the real world and will be pulled out of the world you have created.

This may create different kinds of issues and confusion or boredom or even have the player begin to engage in so-called "meta-gaming[g]", where the player begins using real-world knowledge to step around the game's rules.

13.1 Character Design

Characters, both playable and not, are one of the pillars onto which games build on. If the characters are shallow, uninteresting or don't fit, the player will have one less reason to stick with playing your game until the end.

Here are some tips to make your characters more interesting and add more depth to the world you are building.

13.1.1 Make your characters suck at something

Perfect is boring.

One of the best ways you can make your characters interesting is by making them really bad at something. It can be a valiant knight, renowned for his strength and valor, but for some reason he's rough and rude, for instance.

Another idea could be having a warrior that is the strongest the world has ever seen, but they're brash, arrogant and absolutely unreasonable in their actions, to the point where the player may even question their sanity.

Having "defects" makes the character more relatable and realistic, or at least gives more depth to them. Plus when a character has shortcomings, you can leverage that for comedic effect.

[Do you know more about this? You can contribute, this book is open source!]

13.2 Level Design

After writing your characters, they need a place to live in. Their world is composed of smaller sections (which are called "maps" or "levels"), that's where level design comes in.

Anything that touches the level itself, from the simple visual aspect to the history of the world that a level is part of, is part of the bigger class of "level design" and in this section, we'll check out some small tips to make your world better.

13.2.1 Anything can tell a story

Any object can tell its own small piece of story about the world you're building.

What's the reason behind a small fetch quest that you've been tasked to complete? Maybe that apple that you were tasked to retrieve is a very rare item that can cure a rare disease, or a rusty sword you found in a dungeon might have been owned by the most renowned king in the past.

Even something as simple as a somewhat detailed description of the item may give it more depth. Here's a small example:

> Seeds of the Lifeblood Tree: A small seedling coming from a shrub that used to be cultivated centuries ago. Its thick, blood-colored sap is said to be the blood of the Earth God, able to cure wounds and poisoning alike. Due to the 3rd Pontian-Saarland war, the scorches caused by the battles destroyed all known places where this shrub could be found. This seed pod seems to have survived the fires and may still sprout.

Random Trivia!

The "Dark Souls" series brings this concept to the extreme: there is no narration and almost the entirety of the story is told through the description of the objects you find along the way. It is your task to piece everything together and discover what the lore is.

13.2.2 Details matter

Details can, in some cases, matter as much as the bigger picture: a small animation of grass rustling in the wind or moving when characters cross it can make the world feel a lot more cohesive and cared for.

Some people may gloss over such details, but others may really appreciate the added care and thought that went onto creating your world.

Some details that you can include may be:

- Trees rustling in breezy or windy areas;
- Leaves falling in autumn (you can use Particle systems for that);
- Grass rustling when an entity crosses it (or when a smaller entity hides in it);
- "Dangling Physics": anything that can reproduce the behaviour of a dangling object when touched by someone;
- "Breakable Objects": an object that can fall and shatter if bumped, or something that explodes if shot (even just a bottle);
- Soundscapes, like having reverb in very big empty spaces, making metal bunkers sound tinny and dry ...;
- Much, much more.

[Do you know more about this? You can contribute, this book is open source!]

13.3 Quest design

After you create your world, you need to give the player something to do in such world. That's where your quest design capabilities will shine.

In this section you will find some tips and tricks to make quests more fun, meaningful and enjoyable.

13.3.1 You can make "busy work" interesting too

Nobody likes fetch quests. Or at least, "fetch quests for the sake of fetching something" are boring.

You can put a twist on the so-called "fetch quests" by adding a story layer on them: why does the quest giver want the item you're looking for? What if, after finding the item, you started being attacked by a certain group of people, seeking to steal the item from you?

One way of making fetch quests more interesting is using them to introduce a bigger story: fetching an item, being attacked by a certain group (like a secret organization) and having the item stolen from you, tracking down the group and uncover a secret plot to topple the king.

The only limit is your fantasy.

[Do you know more about this? You can contribute, this book is open source!]

13.4 Story Design

Some (not all!) games have a story. Guiding your player through it might seem a bit daunting, but there are some tips you can follow to make your life a bit easier.

13.4.1 Use the "illusion of choice" to gently push the story forward

The simple sequence of "this event" then "that event" and then "that other event" can get really old, really quickly.

This is where the "illusion of choice" comes in and may help you making your story a bit more interesting while keeping it somewhat linear.

Let's imagine a story where your character must be brought to the police station to further the story: it's a good idea to have the police question the character in front of their home first to set the scene.

You can have the character do various things that lead them to the same ending:

- Tell the truth, but such truth is not believable (or the police is misguided). The police doesn't believe the player and brings them in for questioning.
- Admit involvement, but deny any responsibility. The police will take the player's involvement at face value and brings them in for further questioning.
- Make up a lie. The police notices inconsistencies in the player's responses and brings them in for questioning.

As you can see, we have given the player three different options that lead to the same outcome (eventually through different dialogue).

[Do you know more about this? You can contribute, this book is open source!]

Part 4: Creating your assets

14 Creating your assets

> Art is not what you see, but what you make others see.
>
> *Edgar Degas*

14.1 Assumptions

This book (for now) assumes you already have some minimal knowledge of some matters, including but not limited to:

- Bars (in music)
- Beats (in music)
- Basic music notation (either *letter notation*[g] or the *modern music notation*[g])

14.2 Graphics

14.2.1 Some computer graphics basics

Before we start doing anything, we need to know some basics about graphics. In this section we will introduce some terms like "bit depth" (or "color depth"), "filtering", "sprite sheets", "virtual resolution" and others.

We will see a lot of stuff, from a bird-eye view of how graphics are stored in video memory to normal maps and more.

14.2.1.1 The "color wheel"

We can't start talking about colors without mentioning primary and secondary colors, and have a "color wheel" accompany the explanation.

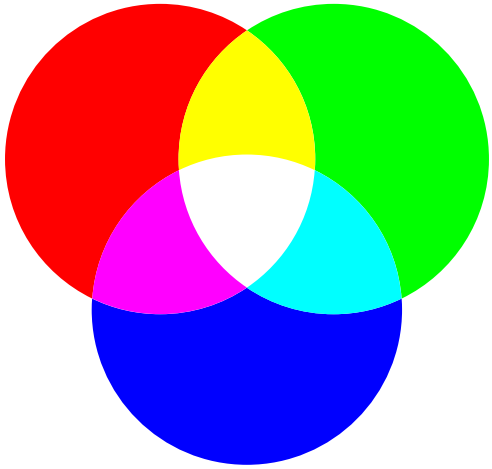


Figure 237: The "color wheel" for screens

In art classes, you've probably learned that primary colors were Magenta, Yellow and Cyan. On screens it's a bit different (this is due to how light works, compared to paint): on screens we have Red, Green and Blue as primary colors (thus making up the "RGB system").

In fact if you looked really close on old CRT screens, you would have probably seen a lot of red-green-blue pixels that together, seen from afar, would have made up the millions of colors that you've seen for a long time.

By mixing up the 3 primary colors, you would have obtained the secondary colors for light, which are... Magenta, Yellow and Cyan!

Random Trivia!

This is due to how light and paint work differently. It is said that "lights are additive" and "paints are subtractive": if you mix Magenta, Yellow and Cyan paints using minuscule dots (like in printing), you will see that Red, Green and Blue are formed. This is obviously an idealization, the colors you'd actually obtain by mixing paints by hand would be closer to Orange, Purple and Dark Green. In short, this is how ink-jet printing works (also a Black component is added, thus forming the CMYK model).

14.2.1.2 Color representations

In computers, there are two main representations for colors: the RGB and the HSV representations.

14.2.1.2.1 RGB Representation

We can memorize our colors by addressing their primary color components: Red, Green and Blue. This is usually done with 3 pairs of bytes, one for each "color channel" (component).

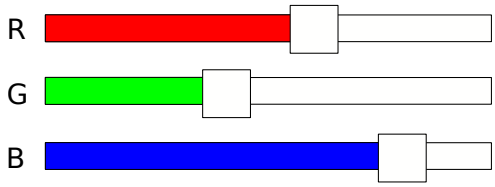


Figure 238: An example of an RGB picker

This means that you can represent color with a 3-tuple: (RRR, GGG, BBB), where each channel can take a value from 0 to 255 (or 00 to FF in hexadecimal, in that case the color is usually prepended by a `#` symbol) if we consider the more commonly used "16 million colors" scheme.

- Pure red is represented with the 3-tuple $(255, 0, 0)$ (or `#FF0000` in hex)
- Pure green with $(0, 255, 0)$ (or `#00FF00`)
- Pure blue with $(0, 0, 255)$ (or `#0000FF`).

Black is the absence of any color component, which means it's represented with the $(0, 0, 0)$ 3-tuple (of `#000000`), while White is represented with $(255, 255, 255)$ (`#FFFFFF`).

14.2.1.2.2 RGBA Representation

Sometimes we need to represent transparency, in that case we need an extra pair of bytes to do so. In this case we talk about the "RGBA system" (Red-Green-Blue-Alpha).

Colors are represented by a 4-tuple: (RRR, GGG, BBB, AAA), where each channel can take a value from 0 to 255 (or 00 to FF in hexadecimal).

14.2.1.2.3 HSV Representation

Another way to represent colors is using the HSV system (Hue, Saturation, Value). Sometimes this system is also called HSB (Hue, Saturation, Brightness).

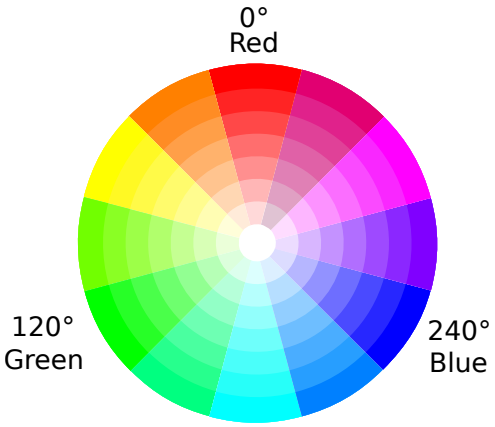


Figure 239: A simplified "slice" of an HSV representation

Using the reference image above, the Hue is selected by choosing an angle on the circle: we find pure red at $0°$, then proceeding counterclockwise we have pure green at $120°$, blue at $240°$ and then we go back to red at $360°$ (which is back at $0°$). This means that it's represented as a value between 0 and 359.

Saturation can be chosen by getting farther or closer to the center, with the minimum saturation being in the center and maximum at the outside. Saturation can be described as the "colorfulness of something compared to its brightness", which would mean that the color feels "less white-y" the higher the saturation. It is represented with a number between 0 and 100.

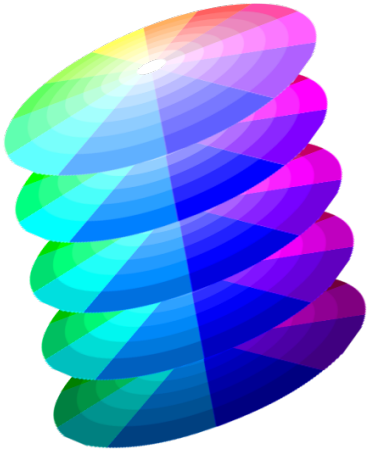


Figure 240: More slices of the HSV representation show how value changes

Value (or brightness) is a bit different: the HSV representation is actually a cylinder (thus the use of "slice" in the

figure description): you can imagine a stack of slices that get darker and darker the closer we get to the bottom of the cylinder. This too is represented with a number between 0 and 100.

Note!

Following this definition we can define pink as a "less saturated" red, as long as the brightness stays high (something like $(0, 50, 100)$). If the brightness is low (towards the lower half) we would obtain a dull brick red (for instance $(0, 50, 50)$).

14.2.1.3 Primary, Secondary and Tertiary colors

Primary colors are the basis of our color system and their definition is complex and outside the scope of this book. Let's just assume that they are the colors that we can mix together to "make other colors".

On screens, the primary colors are Red, Green and Blue; while in classic painting green is usually substituted by Yellow. If we wanted to be more precise, painting uses Magenta, Cyan and Yellow.

Secondary colors are made by mixing, in equal parts, two primary colors: this will give us more colors to work with.

Tertiary colors are obtained by mixing a secondary color with a primary color, in equal parts, thus obtaining even more hues to play with.

14.2.1.4 Analogous Colors

Analogous colors are tertiary colors that are next to each other on the color wheel. For instance Red, Orange and Vermilion (sometimes called Red Orange) are analogous colors.

They are good to create harmonious, almost-monochromatic compositions, since analogous colors are very common in nature.

14.2.1.5 Complementary Colors

Contrasting (complementary) color pairs were used in impressionism for their "eye-catching" character, they are created starting from the 3 primary colors (in screens: Red, Green, Blue), choosing one and combining the other two in a "secondary color".

Complementary colors are positioned on opposite sides of the color wheel. The color star (an alternative to the color wheel to represent colors) makes this "opposition" very easy to see:

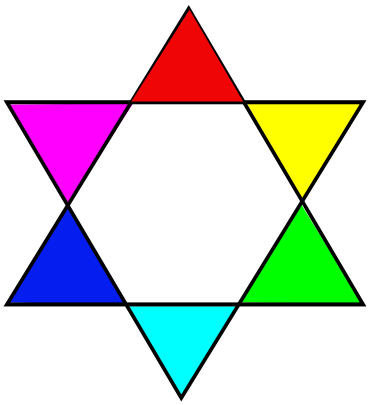


Figure 241: The color star shows how complementary colors are on opposite sides

Some complementary color for screens (color addition) pairs are:

- **Red and Cyan**: Choose red, then green+blue gives cyan;
- **Green and Magenta**: Choose green, then red+blue gives magenta;
- **Blue and Yellow**: Choose blue, then red+green gives yellow.

While, talking about colors made by paint (color subtraction) we have three following color pairs:

- **Magenta and Green:** Choose magenta, then yellow+cyan gives green;
- **Yellow and Purple:** Choose yellow, then magenta+cyan gives purple;
- **Cyan and Orange:** Choose cyan, then magenta+yellow gives you orange.

Complementary colors tend to attract the viewer's eye in the points of intersection of their hues.

14.2.1.6 Color Depth

Raster graphics use bits to represent the color of each single pixel, the amount of bits used for each pixel is known as *color depth* (or "bit depth").

The color depth used for our images can influence the performance and look of our game: more bits means more color choices, but also more memory occupied by those colors. Coupled with high resolutions, an image can easily (if uncompressed) weigh MB worth of memory.

Let's see the most common color depths used in history, first we start with a full-color reference image:

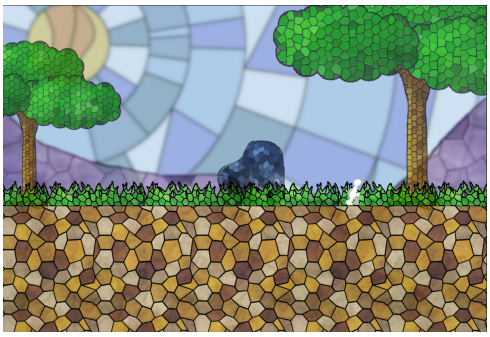


Figure 242: Reference image that we will for bit depth comparison

1-Bit color: Each pixel gets only 1 single bit, which means it's either black or white. This was used mostly in text-based systems, but some indie games manage to get great results out of just two colors.

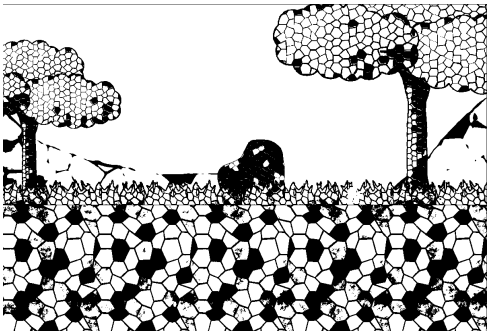


Figure 243: Reference image, quickly converted to 1-bit color depth

2-Bit color: Each pixel can select from a palette of 4 colors (usually fixed). This was used by CGA cards on the IBM, and usually the colors could be chosen from one of four available palettes: "white, black, light cyan, light magenta" or "yellow, black, light red, light green". The other two palettes are just a "low intensity variant" of the ones we've just seen.

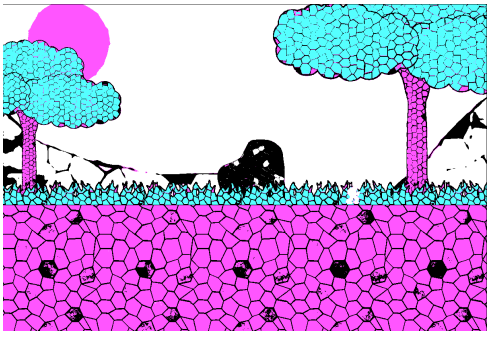


Figure 244: Reference image, converted to 2-bit color depth in CGA style

Random Trivia!

CGA was limited in the number of colors used, but due to a "defect" in the composite NTSC output (called "composite artifact colors" or "cross-color artifacting") these 4 colors could be used carefully to create completely new colors.

4-Bit color: Here we start seeing 16 colors, usually chosen from a selection of fixed palettes. This was used by EGA cards on the IBM, as well as the Commodore 64 (along with "color cells").

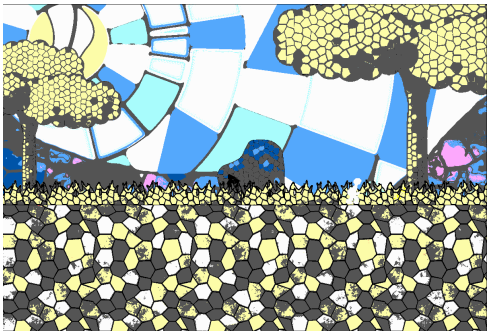


Figure 245: Reference image, converted to a 4-bit color depth in EGA style

8-Bit color: 256 colors chosen from a fully programmable palette (that's some luxury right there!). Used on VGA cards at low resolution.

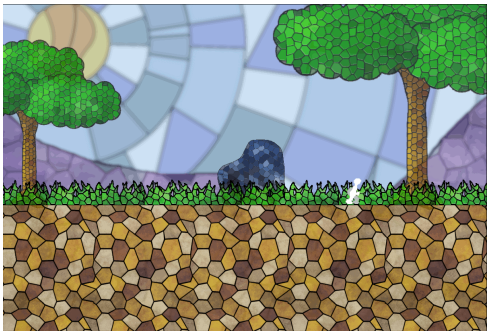


Figure 246: Reference image, converted to an 8-bit color depth

The previous image looks a lot like the reference image, save for some artifacting. Since at higher color depths we won't see much difference, let's just list the remaining ones:

- **16-Bit color:** Sometimes called "High Color", allows for up to 32768 colors with transparency, or up to 65536 without transparency.
- **24-Bit color:** Sometimes called "True Color", this is the most used color depth, allowing for over 16 million colors. Each red-green-blue channel has 256 possible values.
- **30-Bit color:** Sometimes called "Deep Color", this format allows for 10 bits per channel and over 1 billion colors on screen. Supported by many graphic cards and some high-end mobile phones.

- **48-Bit color:** This format allows for hundreds of thousands of billions of colors (if you want to read the number, it's around 281474976710656). This is used by image editing software to avoid loss of data while working with colors.

14.2.1.7 Direct Color vs. Indexed Color

There are two main ways to represent color in digital images.

The first is "Direct Color", which usually allows 256 values (from 0 to 255 included) for each color channel (red, green and blue), for a total of over 16 Million colors.

Each single color is identified by its value, which can be a waste of space and memory when the image has few well-defined colors.

The second way to store images is with "indexed color": in this case a "palette" of colors is created and each pixel color refers to an index to such palette. This allows for smaller images, at the expense of the number of available colors. If you want to add a new color to the picture, first you need to add it to your palette if there is space.

14.2.1.8 Lossless Formats

There are a few ways to store information on a computer, either you store them raw, or you use some tricks to make such information occupy less space in your drive.

When it comes to storing information, lossless formats are usually uncompressed or make use of clever tricks to compress your images without losing information in the process.

In computer graphics, lossless formats include:

- JPEG 2000 (in its lossless form);
- Portable Network Graphics (PNG);
- Free Lossless Image Format (FLIF);
- PDF (in its lossless form);
- TARGA Files (Truevision TGA, with TGA file format);
- TIFF (in its lossless form).

14.2.1.9 Lossy Formats

When it comes to compressing information, the best way to store the least amount of information possible is to actually *not store them*. Lossy file formats get rid of some information present in the picture, in a more or less evident way (depending on the compression ratio) to lower the file size.

In computer graphics, lossy file formats include:

- JPEG (in its most common "lossy" form).

14.2.1.10 Transparency

Usually you need to have transparency in your artwork, for instance for your sprites. There are different ways to get transparency in your artwork, depending on the image format you're using and the support offered by the engine/framework you're using.

14.2.1.10.1 Alpha Transparency

This is the most common type of transparency available today: along with the usual Red-Green-Blue (RGB) channels, the image has an additional "Alpha" channel. Sometimes images with "Alpha Transparency" are also referred as "RGBA" images.

This allows to set the transparency precisely and allows for "partial transparency" too, which means that we are able to create shadows and semi-transparent surfaces.

The PNG format is one of the many image formats that supports alpha transparency.

14.2.1.10.2 Indexed Transparency

Normally used in GIF images, "Indexed Transparency" is the product of some limitation imposed in the format itself: you can only choose from a limited palette of colours to paint your picture.

If you want to have transparency in your picture, you will need to sacrifice a color and tell the format that such color is the "transparency color". In many images a very bright, evident color (like magenta) is used. Such color will not be painted, thus giving the transparency effect.

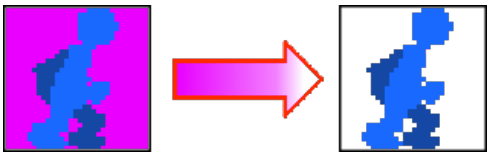


Figure 247: Indexed transparency takes a color and "marks it" as transparent

This also mean that we cannot make semi-transparent surfaces, since only that specific color will be fully transparent, and that's it.

14.2.1.11 Texture Filtering

Sometimes your images will need to be scaled or filtered to avoid annoying artifacts, in this small chapter we will see some filters and how they look.

14.2.1.11.1 Nearest Neighbor Filtering

[This section is a work in progress and it will be completed as soon as possible]

14.2.1.11.2 Bilinear Filtering

[This section is a work in progress and it will be completed as soon as possible]

14.2.1.11.3 Trilinear Filtering

[This section is a work in progress and it will be completed as soon as possible]

14.2.1.12 What is a sprite?

While reading this book, you might already have seen the term "sprite" being used a lot, but what is a sprite?

A sprite is a 2D image (or animation), that is composited with other sprites or textures into a larger scene. In a game, usually they are composed by at least 2 elements:

- A texture (which represents the image itself);
- A position (usually represented with an (x, y) pair).

In some media, you may find other terms that are used as synonyms of "sprite", like:

- Player Graphics/Missile Graphics: These terms come from the Atari line of computers and consoles;
- Objects (or OBJs): These terms usually come from the Nintendo line of consoles, for instance when referring to the memory region where sprite attributes are stored; in that case it was called OAM (Object Attribute Memory);
- MOBs (or Movable Object Block): Coming from the MOS Technology data sheets. Strangely enough Commodore, the main user of MOS products, used the term "sprite" instead.

14.2.2 General Tips

In this section we will take a look at some basic art tips that will give you some indication on how to create your own art for your very own game. This will be pointers to keep you going.

14.2.2.1 Practice, Practice, Practice…

As with all forms of art, the secret to "getting good" is practice. There's no way to avoid it, your first piece may been nice or flat out terrible, your next one will be better, and the one after that will be even better… Hard work always beats "talent" in the long run.

14.2.2.2 References are your friends

References are a tricky topic, some artists swear by it, others oppose them fiercely.

In my opinion, looking at a real-life version of what you want to draw can be one of the most useful things you can do when drawing. Even when drawing something that involved a huge amount of fantasy, having a reference can give you indications on shapes and sizes.

It doesn't have to be a one-to-one reference either, you can get ideas for your dragon from crocodiles and lizards, or even snakes!

14.2.2.3 Don't compare your style to others'

One of the most frustrating things that can happen when learning something new, is comparing yourself to another artist and saying "I should be able to draw like them".

Everyone has their own unique style, and you should work on what makes it unique, instead of comparing your style to others.

14.2.2.4 Study other styles

This ties a bit to the previous point, you should not compare to others, but you should also take some time to look at other people's work, find what you like about it and implement it into your own art style.

Looking around you can help you grow as an artist and aid you in the difficult seeking of your own art style.

14.2.2.5 Learn to deconstruct objects into shapes

Every complex object can be deconstructed into simpler shapes: circles, ellipses, triangles, rectangles, squares...

You can use such simple shapes, overlapping them to create a skeleton of your subject (living or not), so that you can draw it in an easier way: with the layer system introduced by huge part of the current drawing applications, you don't have to do precision work when it comes to removing such skeleton.

14.2.2.6 Graphics Smearing

No matter how good our graphics are and how high our frame rate, there are very few things that give the same effect of "smearing" when it comes to animations: deforming an image to make it look more dynamic when animated by simulating "motion blur".

Sometimes graphics smearing are referred to "smear frames" or "elongated inbetweens".

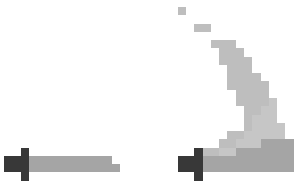


Figure 248: Example of using smearing for animating a sword slash

Smearing has been used in animation for decades (since the 1930s!): just take any famous animated movie from the late 20th century and pause at any moment, you'll probably find one. Some are actually quite funny to look at, others may be even disturbing!

Random Trivia!

It has been used in 3D graphics too, for example in the famous game “Crash Bandicoot” features smearing for the spin move: changing the model with a “smeared version” was less resource-intensive than more realistic “motion blur” techniques back in the PS1 era, plus it matched the “cartoony mood” of the game.

There are programs dedicated to help you with animating and smearing pixel art, but don’t fear to experiment by manually smearing frames using your favourite paint program: some interesting results (even if not the ones you were looking for at that moment) might arise!

14.2.3 Sprite sheets

Every time we create a sprite, we need some amount of memory to store its information, and to match the hardware constraints most of the time a sprite’s image must be padded with unused pixels.

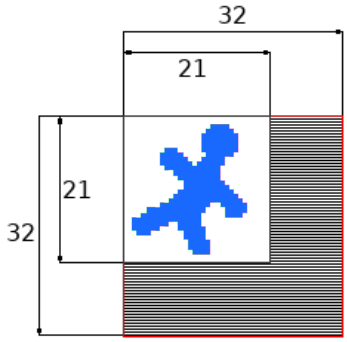


Figure 249: Example sprite that gets padded to match hardware constraints

Each sprite image that gets stored (and there could be potentially hundreds) wastes more and more memory, so we need a way to store sprites more efficiently. In the previous example we can see how a 21x21 sprite gets padded towards a 32x32 size, the sprite uses 52% more memory than it should!

Enter the humble Sprite Sheets.

We save our sprites (as well as animation frames) into a single drawing, called a “sprite sheet”. By composing a sprite sheet with several smaller images of the same size, we just need to adapt our rendering to draw a portion of such sprite sheet on our screen. The sprite sheet is the only thing that will need to be adapted to match our hardware constraints, saving memory.

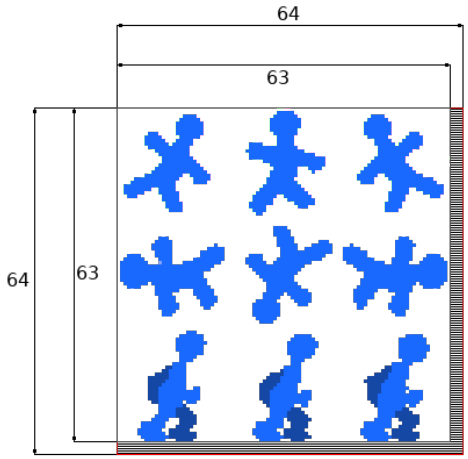


Figure 250: Example spritesheet that gets padded to match hardware constraints

In the previous example, the sprite sheet occupies only 1.5% more memory than it should. That's a great improvement.

This way, instead of having a lot of references to sprites to draw, each one wasting its own memory, we just need the reference to the sprite sheet and a list of coordinates (rectangles, most probably) to draw.

Libraries like OpenGL support "sprite atlases" (or sprite batches), allowing for the graphics card to take care of drawing (after preparing the batch) while the CPU can use more of its cycles to take care of input, movement and collisions.

Sprite sheets are also used in 3D games, usually under the name of "texture atlases": the objective of a texture atlas is reducing I/O operations and context switching by leveraging the principle of locality.

14.2.3.1 Sprite Sheet Gotchas

There are some issues when working with sprite sheets that we need to look for. They are definitely not deal-breakers but without care they may be the origin of hard-to-debug problems.

The first is texture compression: with some compression algorithms or certain compression factors, you may find out that each "sprite" contained in a spritesheet may end up influencing all the other sprites.

Let's take the following example:

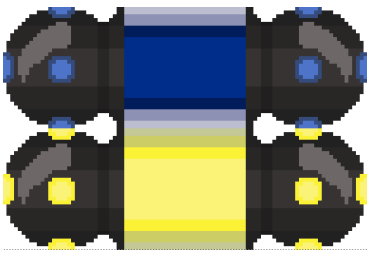


Figure 251: Two platforms in a spritesheet

Now let's compress the image heavily using JPEG and see what happens:

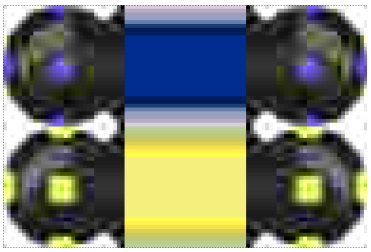


Figure 252: Two platforms in a spritesheet after heavy compression

As you can see, the compression artifacts bring some yellow onto the blue platform, which is not what we wanted.

The second issue can be thought as something similar to the first: when using mipmapping (pre-calculates sequences of images with progressively lower resolutions), usually in 3D games, the sprites may again influence each other.

14.2.4 Virtual Resolution

There are times where having the crispest graphics around is not a good idea, for instance in Pixel-Art games.

Times where it's a better idea to keep your game low-resolution, be it for a matter of performance (your first games won't be extremely optimized and will be slow even at low map sizes and resolutions) or to keep your pixel-art crisp and "pixelly".

This clashes with the continuous push towards higher resolutions, which can mess up your game in a variety of ways, like the following ones.

If the game is forced in windowed mode, you'll have a problem like the following:

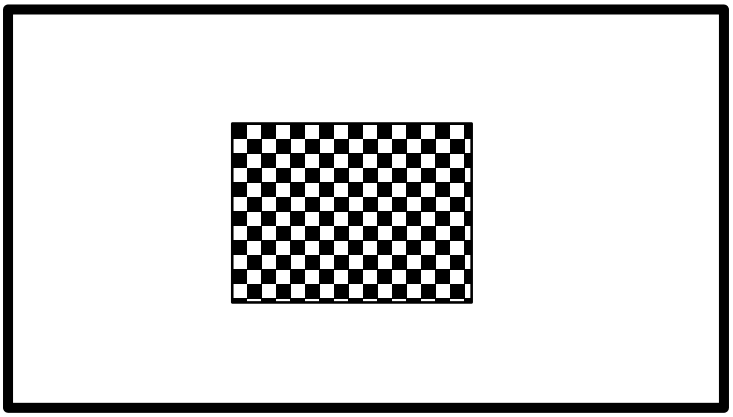


Figure 253: Windowed Game Example - A 640x480 game in a 1920x1080 Window

The game window is way too small and hard to see, and can get even smaller if the $HUD_{[g]}$ takes even more space out of the window. This can be mitigated by calculating the position of each element in comparison to the window size, this although can result in items too small (or too big if downscaling), like the following HUD example.

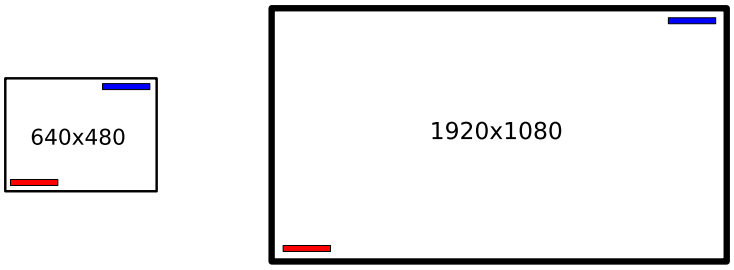


Figure 254: Fullscreen Game Example - Recalculating items positions according to the window size

This is where **virtual resolution** comes into play, the frame is rendered in a virtual surface which operates at a "virtual resolution" and the whole frame is drawn and upscaled (or downscaled) to the screen, without having to recalculate anything in real time (thus making the game faster and lighter).

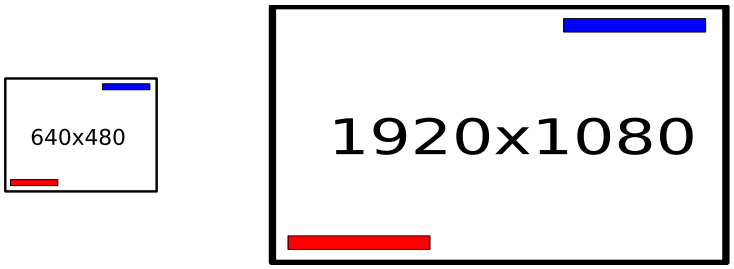


Figure 255: Fullscreen Game Example - Virtual Resolution

The items get scaled accordingly and there is no real need to do heavy calculations. Virtual Resolution allows for different kinds of upscaling, with or without filtering or interpolation, for instance:

- **No filtering** - Useful for keeping the "pixeliness" of your graphics, for instance in pixel-art-based games. It's fast.;
- **Linear, Bilinear, Trilinear Filtering** - Gives a more soft look, slower;
- **Anisotropic Filtering** - Used in modern 3D games, highest quality but also among the slowest, it is usually used when rendering sloped surfaces (from the player's point of view).

For more details, check the filtering section.

14.2.5 Using limited color palettes

[This section is a work in progress and it will be completed as soon as possible]

14.2.5.1 Dithering

Dithering (usually called "Color Quantization") is a technique used to give the illusion of a higher "color depth" when you're using a limited palette of colors.

The usage of dithering introduces patterns into the image when the pixels are visible, if instead the pixels are small enough the pattern will look like a new color, without actually introducing a new color into the palette.

Using two different levels of blue, we can use a dithering pattern to obtain a new tone of blue, like the following:

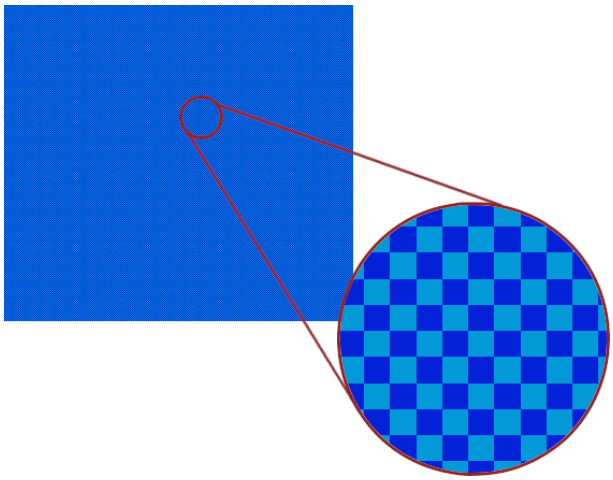


Figure 256: Dithering example

It's possible to study how your palette reacts to dithering using a dithering table, this will give you an idea of the colors available via dithering.

You can see a simple example of a dithering table here:

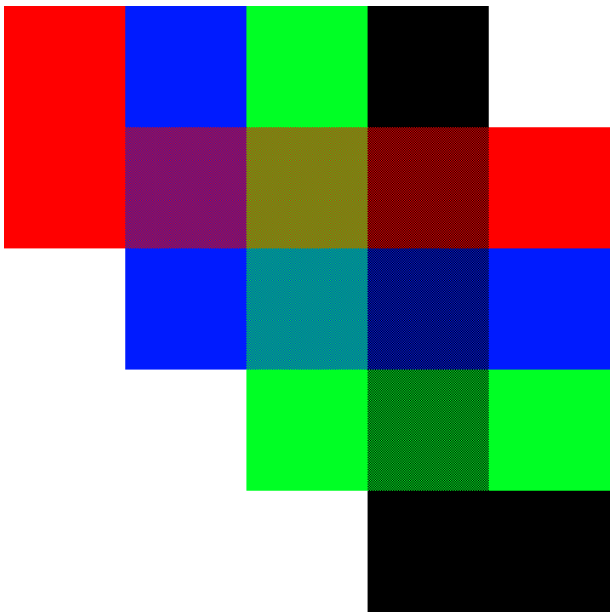


Figure 257: Dithering Table Example

You can see the palette colors on the top row and the left column, then you can see how dithering (in this case a simple checkerboard pattern) allows for different colors to pop out.

There are different dithering patterns, that allow for different type of colors, intensity and patterns:

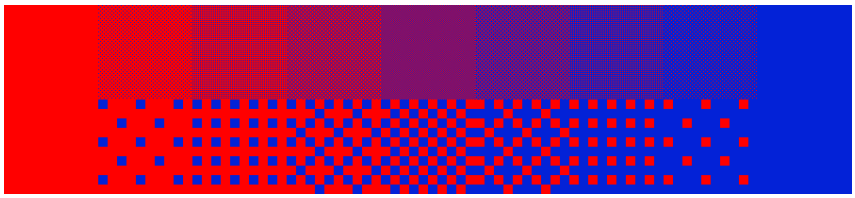


Figure 258: Some more dithering examples

14.2.5.2 Palette Swapping

Palette swapping is a technique mostly used in older video games, where an already-existing graphic (like a player character's sprite) is reused with a different palette (combination of colors).

This palette swap makes the new graphic quite recognizably distinct from the original graphic. This technique was normally used to tell apart first and second player.

Random Trivia!

A prime example of this is the video game that (re)started it all: Super Mario Bros. Mario and Luigi are exactly the same sprite, but Luigi uses a different palette.

Some other video games use palette swapping to indicate their status (like using a green or purple-based palette to indicate the "poisoned" status), or indicate a difference in their statistics (like a red-based palette to indicate an enhanced attack statistic), in other occasions different palettes are used to distinguish stronger versions of the same enemy.

Other franchises, like Pokémon, use palette swaps to introduce "special versions" of some entity (in the case of Pokémon, a shiny Pokemon).

Palette Swapping can be used in more creative ways, though. Going back to Super Mario Bros. you can see that the clouds and the bushes in the levels are exactly the same graphic, just with a different palette. Same goes for the underground bricks and the overworld bricks: they just have a different color.

14.2.6 Layering and graphics

There are some things that should be kept in mind when drawing layers for your game, here we talk about some key points about layering and graphics.

14.2.6.1 Detail attracts attention

When it comes to games, it's easy to get too excited and craft your work with the highest amount of detail possible, but there is a problem: detail tends to attract players' attention.

If you put too much detail in the background, you're going to distract them from the main gameplay that happens in the foreground, which can prove dangerous: the graphics can get messy and you can even get to the point of not

being able to distinguish platforms from the background.

So a golden rule could be:

> Use high detail in the foreground, gameplay-heavy elements - use less detail in the backgrounds

A good idea to make the background a bit less detailed is using blurring, which allows to keep the overall aesthetic but makes it look “less interesting” than what’s in the foreground.

This doesn’t mean the background should be devoid of detail, just don’t overdo it.

14.2.6.2 Use saturation to separate layers further

Bright colors attract attention as much as detail does, so a good idea is making each background layer more “muted” (color-wise) than the ones in foreground.

The main technique to make backgrounds more muted is lowering saturation, blending the colors with grey: this will make the background less distracting to the player.

So another rule can be written as:

> Layers farther away should have lower color saturation than the layers closer to the camera

14.2.6.3 Movement is yet another distraction

As detail and saturation are due to attract attention from the player, movement is another one of those “eye-catchers” that can make the gameplay more chaotic and difficult for the player.

Small amounts of movement are OK, but fully-fledged animations in the background will prove distracting.

Let’s take note of rule number 3 then:

> Try to avoid movement in the background

14.2.6.4 Use contrast to your advantage

Complementary colors tend to attract a lot of attention in the points of intersection of their hues.

If backgrounds feature complementary colors, it may distract the player from the main gameplay.

Figure 259: Which spaceship is easier to spot at a glance?

Our rule number four should then be:

> Keep backgrounds low-contrast to avoid distracting players

Also the opposite rule may apply:

> Keep the main gameplay elements contrasting, so to attract the attention towards them

An orange-robed character will be easier to follow on a blue-ish background, for instance.

14.2.6.5 Find exceptions

Nothing is ever set in stone, and no rules should keep you from playing a bit outside the box and fiddling with some exceptions to a couple rules.

14.2.6.6 Summarizing Layering

Let’s take the following image:

Figure 260: How contrast and detail can help distinguishing foreground and background [6]

You can notice that the grass in the foreground has a lot more hues of gold and brown, the trees and the grass are darker and more saturated compared to the background, which is more "muted".

If we were to break down the image into its main layers, from furthest to nearest, we would obtain something like this:

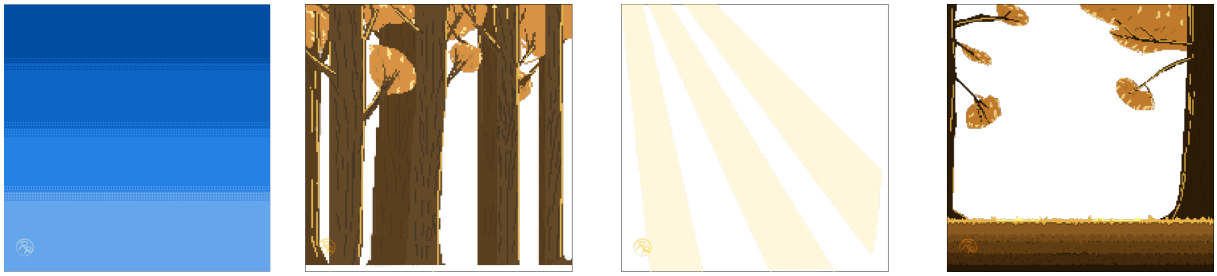


Figure 261: Breaking down the image allows us to see the differences between the layers [7]

From left to right we have:

- **the background:** our sky box, there is very little detail here, just enough to blend the color bands together to make a cohesive piece;
- **the farthest layer:** the trees have very few hues in their bark and leaves, little detail is added, just enough to make out lights and shadows;
- **light rays:** this semi-transparent layer is extremely simple, but being so light and monochromatic it adds variety to make the image interesting without distracting from the foreground too much;

[6] Image by DashingRaven (https://linktr.ee/DashingRaven) used with explicit permission
[7] Image by DashingRaven (https://linktr.ee/DashingRaven) used with explicit permission

- **the foreground:** this layer features the highest amount of detail, the most evident is the grass (but if you look closely, you may notice some texture in the tree bark too), as well as the most saturated and darkest colors in the picture. The contrast with the other layers makes the foreground "pop up".

To summarize, we can make a handy diagram that will give us a "rule of thumb" when drawing layers for our game:

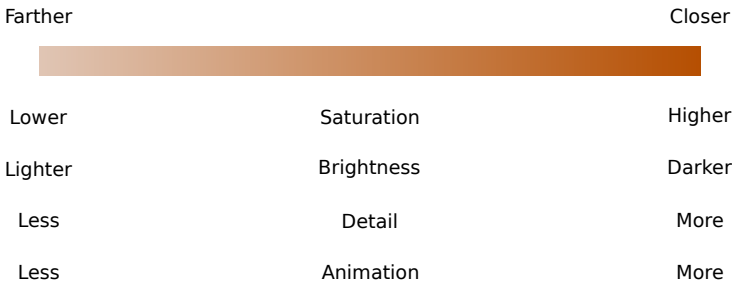


Figure 262: A diagram to show how each section affects our perception of a layer

In short, the closer an object is:

- the more saturated its colors;
- the darker it looks;
- the more detail it has;
- the more it is animated;

14.2.7 Pixel Art

[This section is a work in progress and it will be completed as soon as possible]

14.2.7.1 What pixel art is and what it is not

[This section is a work in progress and it will be completed as soon as possible]

14.2.7.2 Tools

When it comes to pixel art, your most used tools will be:

- **Pencil:** Its hard borders are ideal for base colors and outlines, allowing per-pixel editing without worrying about blur radius;
- **Brush:** The brush tool's soft borders are great for shading, in case you want to go for a more "hi-res" look;
- **Paint Bucket:** This is great for filling outlined ares with a base color. This allows also to create "zones" that help you distinguish between different parts of a sprite;
- **Eraser:** Great tool for removing stray pixels! Just make sure the eraser is set to delete the whole pixel and not just "make it more transparent looking".

Opacity options for tools are great for shading: you can take the same color and blend it with other existing colors, or make it stronger by going through the same area more than once. This makes the process more akin to real painting.

14.2.7.3 Layers

As with all other drawing styles, layers are your best friends in pixel art too. Just make sure to start with a transparent layer, since some drawing programs (for memory and CPU efficiency) won't let you easily add a "transparent background layer" afterwards.

Layers allow you to work on something new without affecting what's already "ready", as well as separate parts of a character (arms, legs, torso and head, for instance).

You can always "flatten the image" (merge all the layers into one) later.

14.2.7.4 Sub-pixel animation with palette cycling

When drawing pixel-art, artists may face what looks to be an insurmountable issue: pixels are a discrete measure which cannot be divided further.

This means that subtle movements may end up really difficult to represent: a character may look like it's laughing or shrugging, while it's only meant to breathe calmly (for instance in an idle animation).

This is where sub-pixel animation comes into play: the basic idea is to play with the colors inside what you're drawing in a way that simulates movement, but the overall shape of the subject doesn't change (or is delayed in such a way that the change of shape looks smoother).

This technique is called "palette cycling": by alternating two shades of the same colors in rapid succession, you are able to simulate animations that give the illusion of being "smaller than a single pixel".

[This section is a work in progress and it will be completed as soon as possible]

14.2.8 Normal Mapping

If you want your 2D game to look amazing, you can't escape shaders. One of the ways to use shaders is calculating light, but since we're in a 2D environment, we don't have "normal vectors" to use to calculate how the light interacts with our 2D objects.

Enter normal maps: these are sprites that "map" their color channels to the direction of the "normal vector": this means that communicate direction with color. Awesome, isn't it?

Random Trivia!

If you're curious, normal maps usually have this color to direction mapping:

- x (which can go from -1 to +1) is mapped to the red channel (which can go from 0 to 255)
- y (which can go from -1 to +1) is mapped to the green channel (which can go from 0 to 255)
- z (which can go from 0 to -1) is mapped to the blue channel (which can go from 128 to 255)

This is an example of a texture, along with a possible normal map:

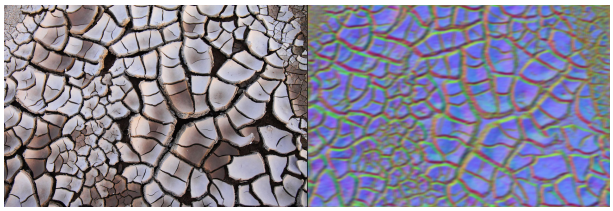


Figure 263: A texture (on the left), with a possible normal map (on the right)

There are many ways to get a normal map: you can try to get a program to generate one for you, make the object in a 3D modeler and extract the normal map from there, or just draw it by hand.

If you choose the last option some tools, like Aseprite, have a “normal mapping” mode that shows you this special color picker:

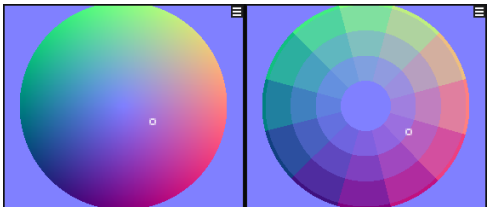
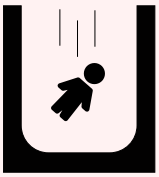


Figure 264: Aseprite’s normal mapping color picker (both in its normal and discrete versions)

Just imagine this color picker like a “3D sphere” and pick the color of the face of the “3d surface” you’re trying to draw.

Pitfall Warning!



Be careful with your shaders, some may expect one or more of your channels to be “flipped”.

14.2.8.1 A simple example

Let’s take a simple box, with no shading, like the one below:

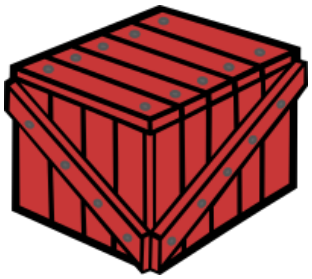


Figure 265: A box that will be used to show how normal maps influence light

Now we'll shine a light on the box, without any normal map: this will happen twice:

In the first example (left) the light will be a round gradient that will come from the top right corner of the image, while in the second example the light will be a bit stronger and coming from the top left corner. This is the result.

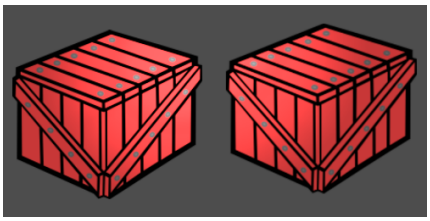


Figure 266: How the lack of normal mapping makes lighting look artificial

Now we'll draw the simplest normal map possible: just filling the 3 faces of the box that we can see with the (kind of) corresponding colors from our "3d sphere" (the normal mapping color picker), we can see the result is very different

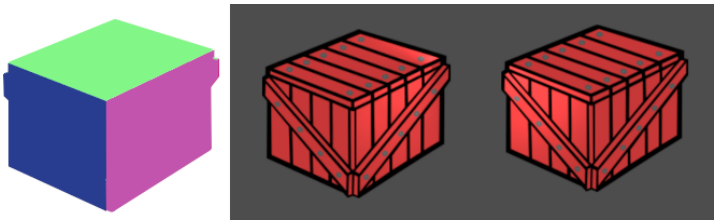


Figure 267: How normal mapping changes lighting

Now let's make something a bit more detailed, by highlighting the faces of the cross-braces on the sides of the box, the way they're lit it's again different:

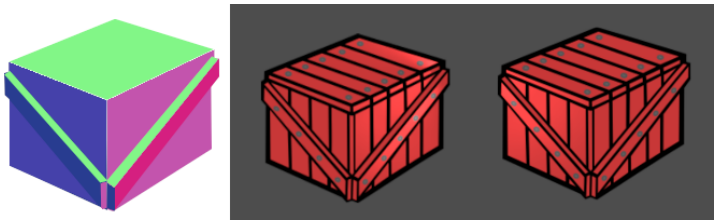


Figure 268: A more detailed normal map results in better lighting

You can get as detailed as you want, but remember that it may have some performance impact if you go overboard with many sprites.

14.2.9 Blending surfaces

Sometimes we need sprites (or textures) to influence each other in a different way than simple "draw `x` instead of `y`". This may be useful for applying ambient lighting and shadow effects to sprites.

Doing so on a pixel-by-pixel basis is inefficient (in a similar way to what ray casting does with collision detection), thus it's better to leave such task to the hardware, usually the GPU.

Thus the concept of "surface blending" was born: a way to take two surfaces and have a result that is essentially a mathematical operation of the single pixel colors.

This is usually supported by the engine or library that you use, and doesn't need to be manually programmed.

14.2.9.1 Addition

Surface addition is very simple: each pixel color (usually as an RGB 3-tuple) from the first surface is taken and each channel value is summed to the values of the second surface (with a maximum value of 255 for each channel).

So if we were to take two overlapping rectangles, one red (with RGB tuple $(255, 0, 0)$) and one green (with RGB tuple $(0, 255, 0)$):

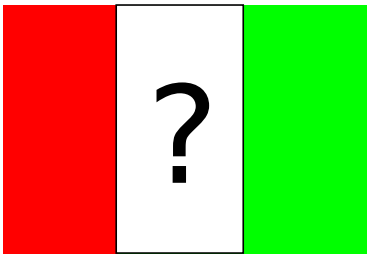


Figure 269: Surface blending - addition (1/2)

We can take the tuples and calculate the resulting color:

$$\begin{array}{rrrr} & 255 & 0 & 0 \\ + & 0 & 255 & 0 \\ \hline & 255 & 255 & 0 \end{array}$$

The resulting color is yellow.

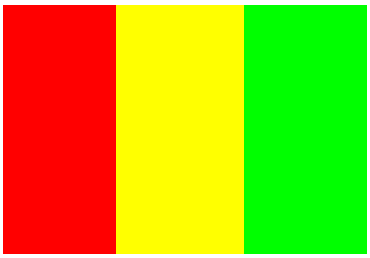


Figure 270: Surface blending - addition (2/2)

So if you were to use a gray sprite as your second surface (which usually has all 3 channels set at the same value), the resulting surface will have lighter pixels where the two sprites overlap, creating a "lighting effect".

Figure 271: Simple lighting effect using 2 surfaces and blending

This approach is a little bit limited: you need to align the surfaces correctly, or artifacts may be visible, but it's not that hard.

14.2.9.2 Subtraction

Similarly to what happens with addition, you can "subtract" colors too.

So if we were to take two overlapping rectangles, one yellow (with RGB tuple $(255, 255, 0)$) and one red (with RGB tuple $(255, 0, 0)$):

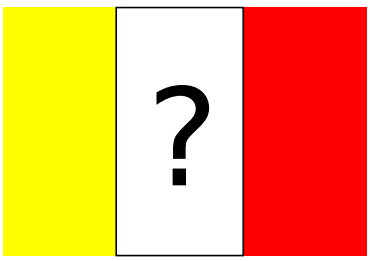


Figure 272: Surface blending - subtraction (1/2)

We can take the tuples and calculate the resulting color:

$$\begin{array}{rrrr} & 255 & 255 & 0 \\ - & 255 & 0 & 0 \\ \hline & 0 & 255 & 0 \end{array}$$

The resulting color is green.

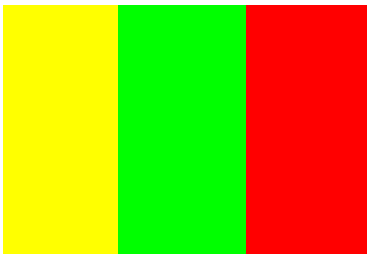
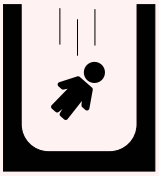


Figure 273: Surface blending - difference (2/2)

This can be useful for shadows, for instance:

Figure 274: Simple shadow effect using 2 surfaces and blending

Pitfall Warning!



"Subtraction" and "difference" may refer to different blending modes: in the "difference" case, it may happen that the first surface is subtracted from the second or vice-versa, depending on which one returns a non-negative result.

14.2.9.3 Alpha Blending

Another way to blend surfaces is via "Alpha Blending": the way it is implemented changes depending on implementation (engine, libraries, etc...) but they all share some common points.

Instead of having 3 channels for red, green and blue, there is an additional channel dedicated to the "alpha value": this value ranges from 0 (fully transparent) to 255 (fully opaque).

By overlaying surfaces with alpha channels, you can achieve transparency effects, which may be useful for water effects, for instance.

14.2.10 Tips and Tricks

This section contains various tips and tricks used by artists to create certain effects inside video games, demonstrating how sometimes something really simple can have a great effect on the game experience.

14.2.10.1 Tiles

14.2.10.1.1 Getting started

When you are starting to make a new tileset, it's a good idea to begin with a base sized 5x5 tiles or more (so if your single tile is 32x32 pixels, the image will be at least 160x160 pixels): this can give you variations on the tileset that won't make the "tiling" (repetitions) so easy to catch when the map is made.

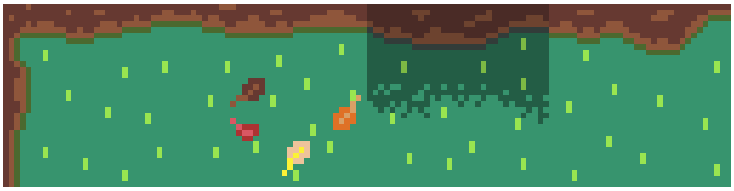


Figure 275: Example of tile "alternatives"

If you think something does not fit in a tileset, try to think what the surrounding area would look like: darker grass could be overgrown, a darker spot in the water could signify deeper water, shadows and light help too.

When it comes to "corner tiles", using the same tile, rotated in 90 degree steps is a great basis to build upon, after that you can edit the tiles accordingly.

This "rotation trick" can be used for most of the tiles you create, let's take for instance the following diagram, representing some tiles:

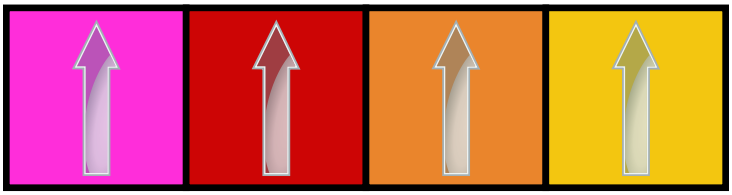


Figure 276: Tile "Rotation Trick" (1/3)

We can take the tiles and rotate them, to get something like the following:

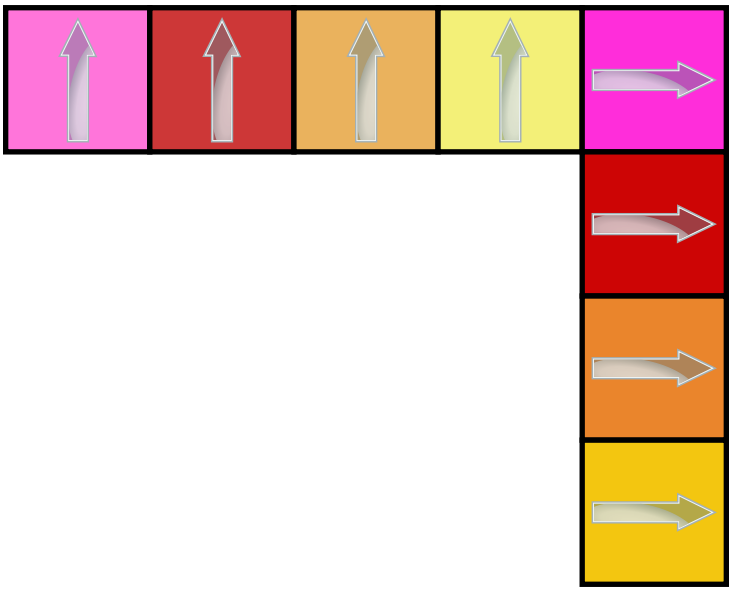


Figure 277: Tile “Rotation Trick” (2/3)

If we continue with copying, rotating and pasting, we can obtain a great basis for our tileset:

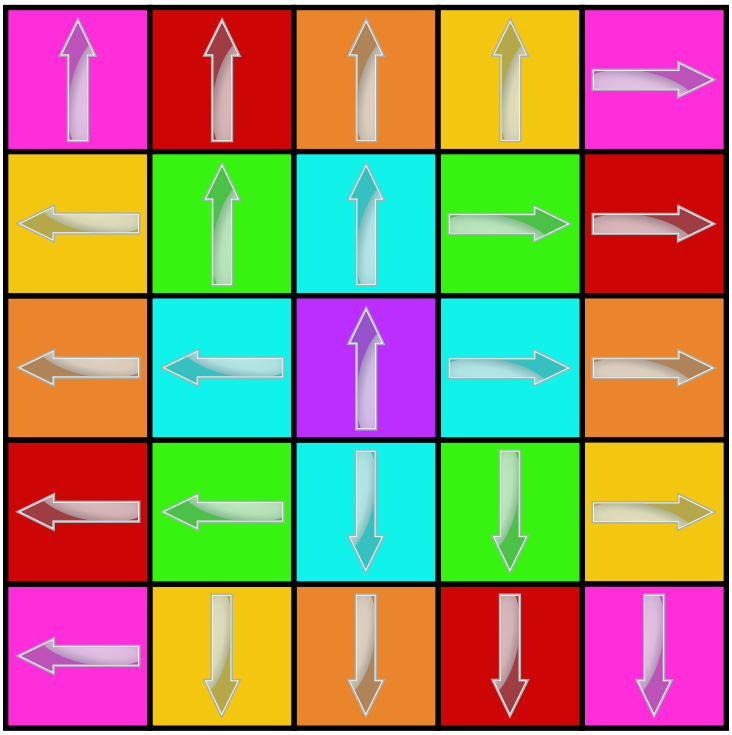


Figure 278: Tile “Rotation Trick” (3/3)

After that we can edit and make it so tiles are seamless, while putting the minimum amount of necessary effort to create something convincing.

14.2.10.1.2 9-slice Scaling

[This section is a work in progress and it will be completed as soon as possible]

14.2.10.1.3 Creating “Inside rooms” tilesets

In many cases, when dealing with tile-based games, we need to create a tileset that is good to represent “inside” environments, like a basement, a cave or the inside of a building. A simple way to reach that goal is creating a set of black and transparent tiles that can be overlaid on another tileset, like the following:

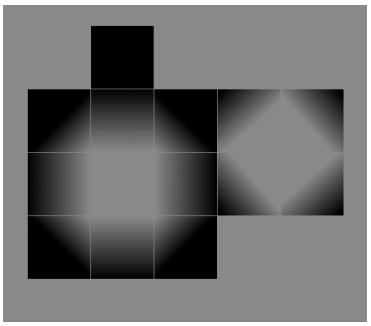


Figure 279: Example of black and transparent tileset used in “inside rooms”

Such tiles can then be overlaid onto something like the following:

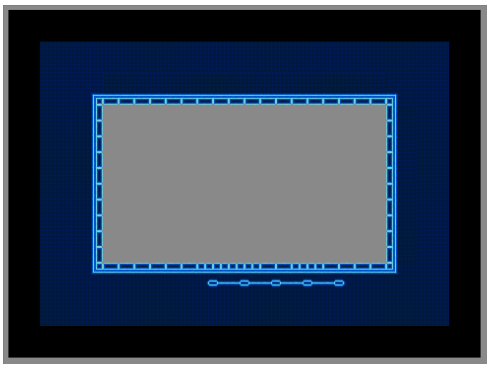


Figure 280: Example of incomplete “inside room”

And we obtain the following result:

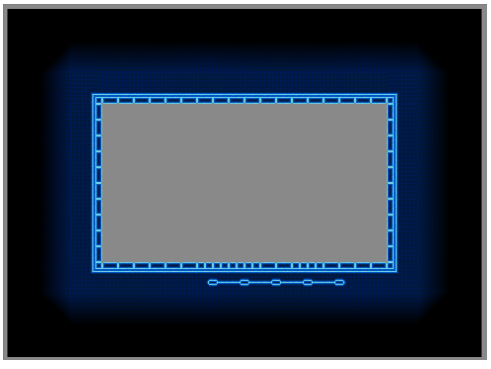


Figure 281: Example of “inside room” with the black/transparent overlay

As you can see, something as simple as some shadow can really sell the effect of being “inside a dark place”.

14.2.10.2 Sprites and icons

14.2.10.2.1 Shape first

When you’re making some kind of sprite or icon, you should always get the basic shape of the object down first, then you can give the object more depth and detail with colors. This will help you understanding the space occupied by your object.

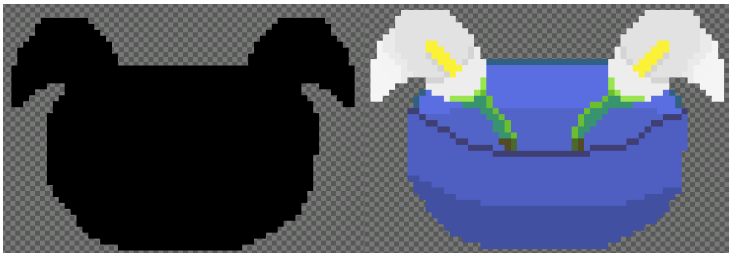


Figure 282: How color can completely change an object

It’s a puppy! It’s a jester! It’s a... hastily drawn flower vase?

This is the power of color, you can change the entire nature of an object by changing how it’s colored, but having the basic shape of the object down first will help you a long way.

14.3 Sounds And Music

The best gameplay and graphics mean nothing if they are not accompanied by good audio. Audio can make or break a game, as does music: some games have really memorable soundtracks even if their graphics may not be “top-notch”.

In this section you will find the basics, tips and tricks to make your own music and sound effects.

14.3.1 Some audio basics

Before creating sounds and music, we need to clarify some terminology, as well as learn some basics before diving into FM synthesis like wave forms. After that we can learn about trackers and Software DAWs.

In this section we will learn about sample rate, bit depth, lossy/lossless formats and clipping, among other things.

14.3.1.1 Sample Rate

Differently from Analog Audio, which is continuous (as in has an infinite amount of detail), Digital Audio is a stream of numbers (ones and zeros) that is “discrete” in nature. That means that we blast these numbers thousands of times a second to be able to build a decent sounding sound.

The number of times we record such numbers from our digital microphone (as well as the number of times we blast such numbers back from our speakers) is called **sample rate** and it is measured in *Hz*.

The more the samples per second, the more detail we can squeeze into our audio files, but at the same time the bigger the file will become too.

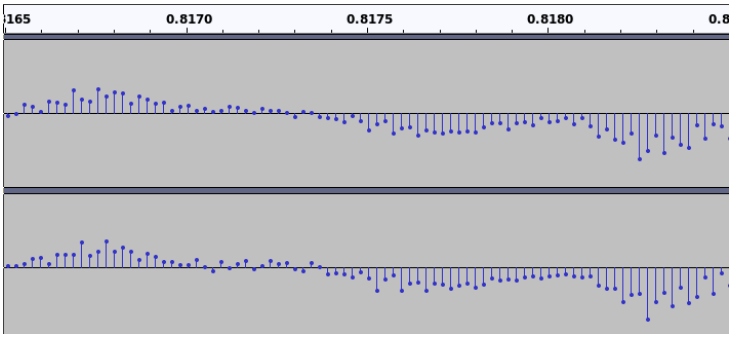


Figure 283: Graphical Representation of Sample Rate (44.1KHz)

In normal CD-Audio, we have a sample rate of 44100 Hz, which means that we recorded a sample 44100 times in a single second.

When making our game's audio, we should always stay around such value, since going lower would make the audio sound worse, since we lower the amount of information the audio itself has.

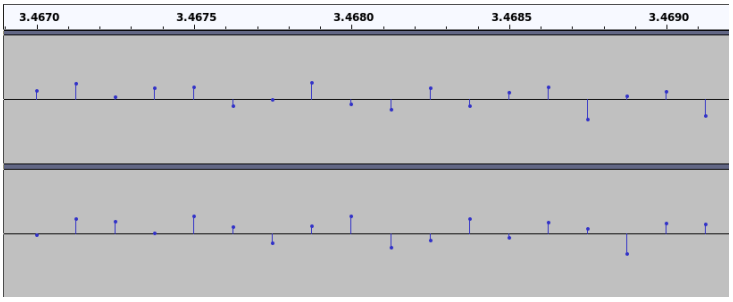


Figure 284: Graphical Representation of Sample Rate (8KHz)

Also we should avoid using weird sample rates, here are some the most commonly used:

- 44.1kHz (or 44100 Hz if you prefer), used in the CD Audio format;
- 48kHz, used in Pro Audio contexts;

While working with audio (mixing and editing), we can go higher:

- 88.2kHz - Double the CD Audio standard, used to record and recreate more frequencies;
- 96kHz used in some serious Professional audio contexts.

14.3.1.2 Bit Depth

Along with sample rate, there is another value in audio that expresses its quality: the bit depth.

The bit depth defines how many “volume values” we can have in a single sample, which shapes the quality of the sound in a different way than sample rate.

If our audio has a 1-bit bit depth, each sample will have only 2 values:

- **0:** Mute
- **1:** Blast at full volume

Which strongly reduces the quality of the audio.

Random Trivia!

In Pokèmon Yellow for GameBoy, Pikachu’s voice was encoded with in 1-bit depth.

If we had a 2-bit depth, we could make each sample have more volume values:

- 00: Mute
- 01: 33% volume
- 10: 66% volume
- 11: 100% volume

Usually audio has a 16 Bit depth, but more modern systems make use of 24 Bits or even 32 Bits.

14.3.1.3 Lossless Formats

As with graphics, there are audio formats that allow you to store uncompressed information, as well as compressed (but without losses) sounds. The most common lossless audio formats include:

- WAV (uncompressed);
- AIFF (usually uncompressed, but AIFF-C supports both lossless and lossy compression);
- FLAC (lossless compression).

14.3.1.4 Lossy Formats

As with graphics, there are also “lossy formats” that allow us to store information in even less space by getting rid of information that is considered outside our hearing spectrum, for instance. Some of the most known are:

- Mpeg Layer 3 (MP3);
- OGG Vorbis;
- Windows Media Audio (WMA);
- Advanced Audio Codec (AAC).

14.3.1.5 Clipping

Clipping is a phenomenon when you're trying to output (or record) a wave that exceeds the capacity of your equipment. As a result, the wave is "clipped" (cut) and there is a very audible distortion.

You surely heard clipping in audio before, usually when people scream on a low-quality microphone and the audio gets distorted.

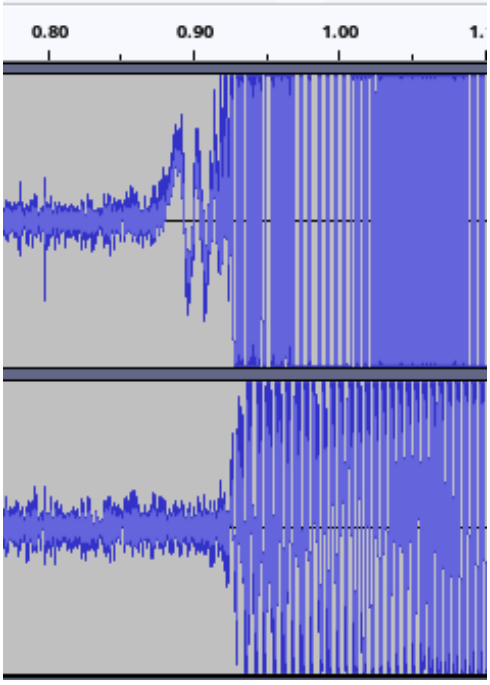


Figure 285: Example of audio clipping

The best way to repair clipping is to re-record the audio completely, although some tools can help in case you absolutely cannot re-record the audio.

Also you should be wary of clipping, because there may be cases where it damages your audio equipment.

14.3.2 Sound Synthesis

Now we're entering technical territory. We're going to talk about sound synthesis: the art of creating sounds, also called "sound synthesis".

14.3.2.1 AM Synthesis

The first, and technically simplest way to generate sound is via AM (amplitude modulation) synthesis.

With this technique you take the wave form created by an *oscillator*$_{[g]}$ and modulate its amplitude (volume) according to a second wave form.

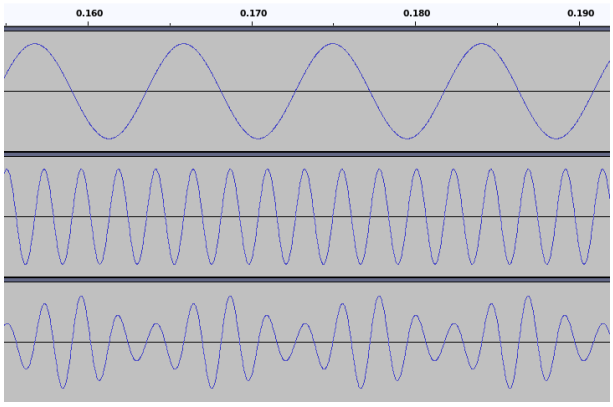


Figure 286: Example of AM Synthesis

In this example we see a 440Hz sine wave (in the middle) having its amplitude (quite heavily) modulated by a 110Hz sine wave (on top): the resulting wave form on the bottom has a "tremolo effect" to it.

14.3.2.2 FM Synthesis

With this technique you take the wave form created by an *oscillator*$_{[g]}$ (called "carrier frequency") but instead of modulating its amplitude, you modulate its frequency (pitch) according to another wave (called "modulator frequency").

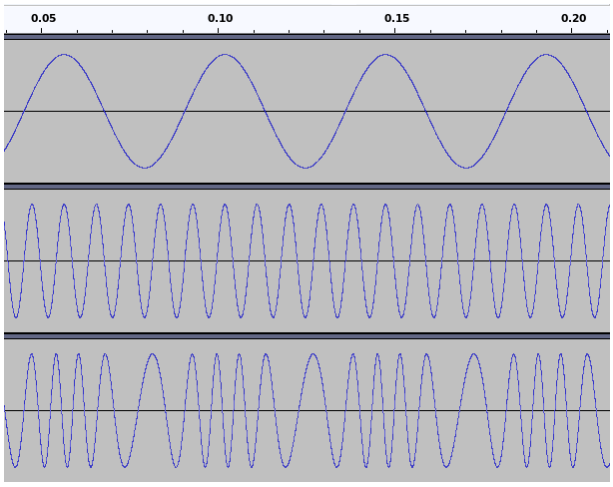


Figure 287: Example of FM Synthesis

In this example (for sake of visibility) we have a 110Hz sine wave (in the middle) having its frequency (again, heavily) modulated by a 22Hz sine wave (on top): we can see the result in the bottom of the figure.

This frequency modulation happens so fast that we end up with something that sounds completely different from the original wave form.

14.3.2.3 FM Synthesis vs Sample-based music

FM synthesis was invented to circumvent one of the biggest issues that plagued the 8 and 16-bit era: lack of space.

Games were saved on small cartridges (which ranged, on average, from 4Kb on the Vic20 to 2MB on the Genesis/MegaDrive), such cartridges had to contain graphics, music and the entire code of the game.

Back then there were no compressed music files, not that it would have helped much, so instead of memorizing the song itself, the instructions to play the song would be saved.

Let's make a simple example: we need to make a very simplistic soundtrack, composed of a pure sine wave that represents the "A above middle C" (or A_4) for an indefinite amount of time. We could either save all the samples (and thus waste precious cartridge space), or we could just save the following "code":

Listing 75: How FM music may be saved on an old console

```
stop all channels;
select the sinewave waveform on channel 1;
set the frequency of channel 1 at 440Hz;
start channel 1;
```

All music made via FM synthesis is nothing more than a bunch of instructions for the FM chip on how to work.

The next step forward was on the Commodore Amiga, where the first sample-based music started: we save small pieces of PCM audio (called samples), rework them a bit using ADSR envelope and pitch-shifting and call them "instruments". Such instruments are then used to compose the track.

The music heard from these systems is fruit of a "hybrid approach": small pieces of (sometimes recorded) audio actually exist in the track, and they're re-used, pitch shifted and reworked all around the track. This makes for very small files (around 10 to 100Kb) with a lot of flexibility.

Modern music is essentially made up of a huge, monolithic sample, usually in the form of an MP3 file or something equivalent, recorded from real-life instruments or synthesized, but instead of "saving the instructions" like in the old days, we just save the entire track as a PCM sample.

14.3.3 Basic Wave Forms

It is important to know how the main wave forms look and sound in order to understand how to create your own instruments, as well as having a further insight on how older 8-bit games sounded.

It is suggested to look for each sine wave on the internet and hear how it sounds, here we will briefly talk about the main waveforms, their shape and uses.

14.3.3.1 Sine Wave

A sine wave has an amplitude that follows a trigonometric sine wave, it sounds really "pure" and is usually used at 440Hz to tune instruments in A.

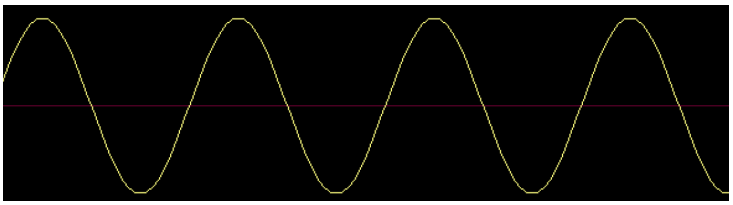


Figure 288: How a sine wave looks

In games it is usually used to give out the impression of a flute-like instrument.

14.3.3.2 Square Wave

A square wave looks... square-like. It is one of the most used waves in 8-bit music, sounds a bit rougher than a sine wave and is used for beeps and blips.

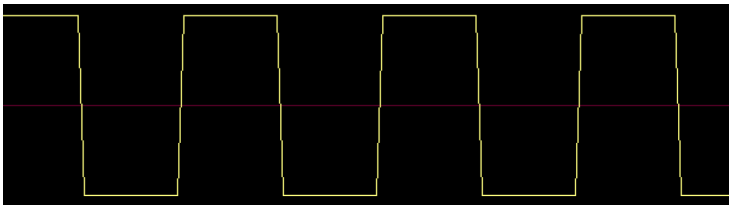


Figure 289: How a square wave looks

In game music it is normally used as lead instrument, and with various modulations, it can sound like a xylophone or a piano, or at least a very artificial rendition of those.

The NES had 2 voices (or channels) dedicated to square waves.

14.3.3.3 Triangle Wave

A triangle wave is another very used wave in 8-bit music, given it's very "muted" characteristics it can be used to give songs a "bass track" of some sort.

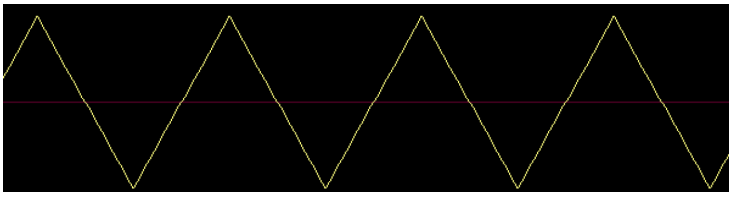


Figure 290: How a triangle wave looks

The NES had one channel completely dedicated to triangle waveforms.

14.3.3.4 Sawtooth Wave

Sawtooth waves were a staple of the Commodore64 era, with its “gritty”, “bzzt” sound which can sound a lot like a trombone on long notes.

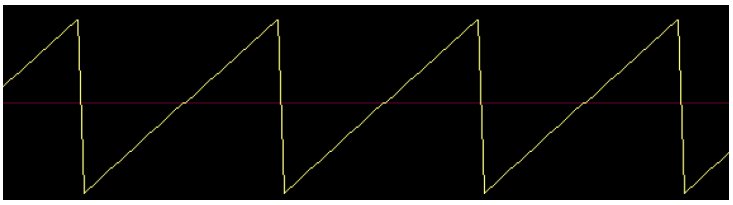


Figure 291: How a sawtooth wave looks

14.3.3.5 Noise

Noise is not a real, static waveform, but more like what comes out when the amplitude has a random value on each sample.

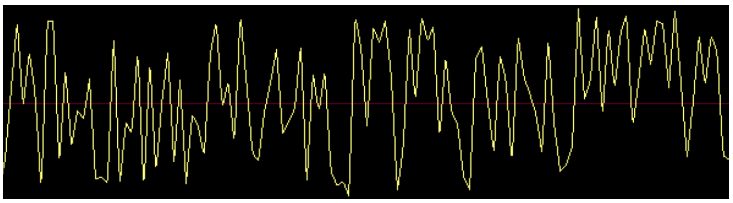


Figure 292: How a noise wave looks

Noise can be used (with the right modulation and processing) to simulate percussions.

The NES had one channel completely dedicated to noise waveforms.

14.3.4 ADSR Envelope

Now that we’ve seen how waveforms look like, we need to understand how such sounds change over time. The sound envelope is used to describe exactly that.

The most common way to control the signal envelope is through four parameters: Attack, Decay, Sustain and Release. Thus the name “ADSR envelope”.

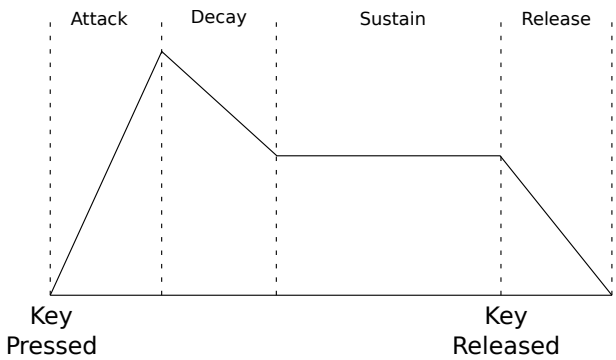


Figure 293: Representation of an ADSR Envelope

14.3.4.1 Attack

The attack is the measure of time the sound takes from zero to its initial peak, when the key is pressed.

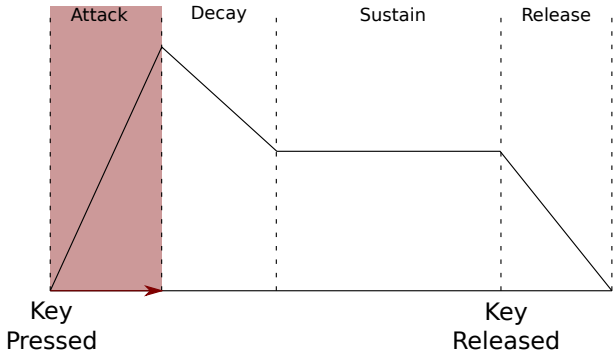


Figure 294: Attack on ADSR Envelope

The longer the attack, the slower the sound will "rise" when a key is pressed.

14.3.4.2 Decay

After the attack, comes the decay, which is the measure of time it takes the sound to drop to "sustain level" after the initial peak.

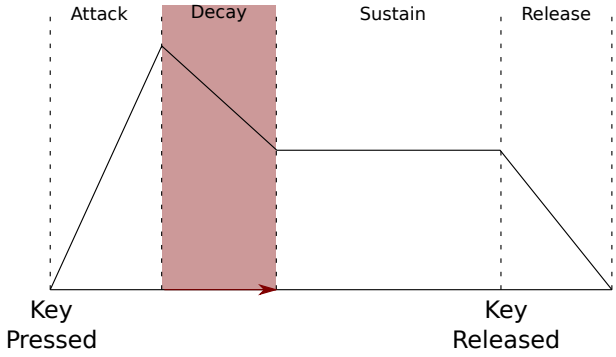


Figure 295: Decay on ADSR Envelope

The longer the decay, the slower the sound will drop to sustain level.

14.3.4.3 Sustain

After the decay is completed, we are now sustaining the signal. Sustain is *not* a measure of time, but **it is a measure of volume**.

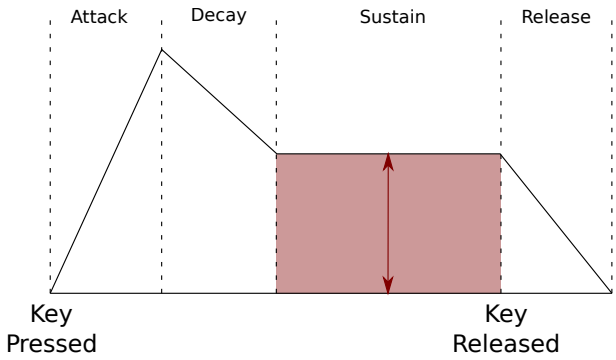


Figure 296: Sustain on ADSR Envelope

The higher the sustain level, the louder the signal will be when at "sustain level". This signal will last until we release the key.

14.3.4.4 Release

After we release the key, the sound will have to fade out somehow. Release is the measure of time it takes the sound to go from sustain level back to zero.

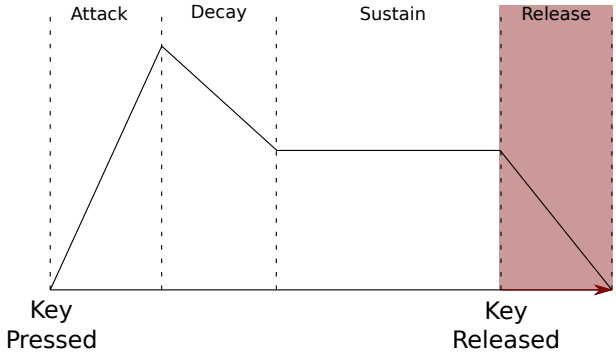


Figure 297: Release on ADSR Envelope

The higher the release time, the longer the sound will take to "fade out".

14.3.5 Digital Sound Processing (DSP)

Let's think about a simple situation: we want to play a "walk" sound effect: every time our character's foot hits the ground, we play a "step" sound effect.

If we play the same sound over and over, it will become boring really quickly, breaking the immersion. An idea could be saving different sounds for a footstep and then every time the player takes a step, a random footstep sound will be played.

This solves the problem, at a cost: we need to use more memory to keep track of such sounds, multiply that for tens of sound effects and the game may run out of memory on low-end systems (or we can "run out" of patience in creating tens of variants of sound effects).

An alternative solution could be using DSP: editing the sound sample in real time to add more variety and depth while saving memory, the trade-off would be CPU time, but it's an acceptable deal.

14.3.5.1 Reverb

When you take a stroll on a sidewalk, you have a certain "openness" on the footstep sounds you hear, but that surely changes if you're walking with hard shoes on a hard floor inside of a small cave. You can hear a lot of reverb and echo at every single step.

Reverb is the first of the sound effects that we encounter in our journey: it allows to give more depth to our sound effects, making it sound like we're inside of a small cave or a very large room.

14.3.5.2 Pitch Shift

A way to give more variety to a sound effect without much work is using pitch shift to make our sound a bit higher or lower, randomly, so that each step is slightly different from the other: this way our ears will get less tired of hearing said sound effect.

Pitch shifting must be used with caution, since abusing it will distort the sound effect and break the immersion in our game.

Another example of pitch shift is used in racing games, where the car roar is pitch-shifted up or down according to the acceleration given to the car.

14.3.5.3 Filtering

Another sound effect we can use is filtering, which are divided in 3 main sections, according to the frequency they "allow to pass through":

- **Low Pass Filter:** This filter allows low frequencies to pass through unfiltered, while the frequencies higher than a defined threshold will be cut. This allows for effects where the bass is unaltered but higher frequencies are cut away;
- **High Pass Filter:** Opposite of the previous filter, this filter allows high frequencies to pass through unaltered while the frequencies lower than a defined threshold will be cut;
- **Band Pass Filter:** A combination of the two previous filters, this filter let's through all the frequencies between two defined threshold values. This allows for more interesting effects like a music sounding through an old radio.

An interesting example is when an explosion happens near the player, in that case the "stun" effect is given by using a low pass filter on an explosion sound (which makes it sound really low and muffled), eventually a slowdown is applied and a secondary sound effect of a very high pitch sound is added (something similar to what you hear when your ears are ringing).

14.3.5.4 Doppler Effect

To give more depth to your sound effects, you can use pitch shift to create the “Doppler Effect” that we hear in real life when a police car passes by: when the car approaches us the pitch is higher, when the car is in front of us we hear the siren as it should be, and when the car passes us we hear a lower pitched version of the siren sound effect.

The Doppler effect can be really useful when applied to car racing games again, when we overtake one of our opponents (or one of our opponents overtakes us) using the Doppler effect can help the player feel more “immersed” in the experience.

The Doppler effect would actually apply to light too, but we would need to have something travelling at a really high speed or said object to be really far away (like a planet).

14.3.6 Simulating older consoles’ audio

Sometimes we want to give our players a sensation of nostalgia, or we just want to limit ourselves to get the most out of our creativity (creativity comes from limitations), so we may decide to emulate the audio of a famous console (or at least remember of it in some way).

It’s not sufficient to “make it sound” the same way (same pitch, same general sound) but we also need to adhere at the limitations of the consoles. Usually such limitations are in the number of channels (which means the number of notes that can played at once by all instruments), but sometimes it’s more structural.

Note!

These is just a simplified version of the limitations of each console. Also if you want you can freely break any of the “rules” and make something original that has the “taste” of something “classic”.

Let’s take a look at some of the most famous.

14.3.6.1 Commodore Vic20

The Commodore Vic20 is one of the first famous home computers; its audio comes from the VIC chip, the same chip that takes care of the video output.

The VIC chip has 3 channels dedicated only to square/pulse waves, with a range of 3 octaves where each octave is a single octave apart from the others (So its octave structure would be something like 1st - 3rd - 5th).

The VIC chip also features a noise channel, for a total of 4 voices. Remember that these voices are shared between music and sound effects.

14.3.6.2 Commodore 64

Probably the most famous home computer of the 8-bit era and had an amazing sound chip for the time: the timeless SID chip, which was used for audio output and controlling paddles/joysticks.

The SID chip (in its two main iterations: the MOS 6581 and the 8580) has 3 channels, each one can be programmed to use one of four wave forms:

- Square/pulse wave
- Triangle
- Sawtooth
- Noise

also the SID chip features ADSR controls for each channel, giving even more control and possibilities.

Random Trivia!

The MOS 6581 (the first SID Chip model) had a flaw in its volume register that was very special: if you could change the value of the $D418 register fast enough, you could play audio samples with a 4-bit resolution, effectively giving the C64 a 4th channel for playing PCM samples!

This issue was fixed in the later MOS 8580 revision, but it can be "added back" by adding a resistor on the board.

But one thing that makes the SID chip very special is the ability to reprogram (and thus change) the instruments on the fly. This real-time programming capability makes it possible to give the "illusion of more instruments"

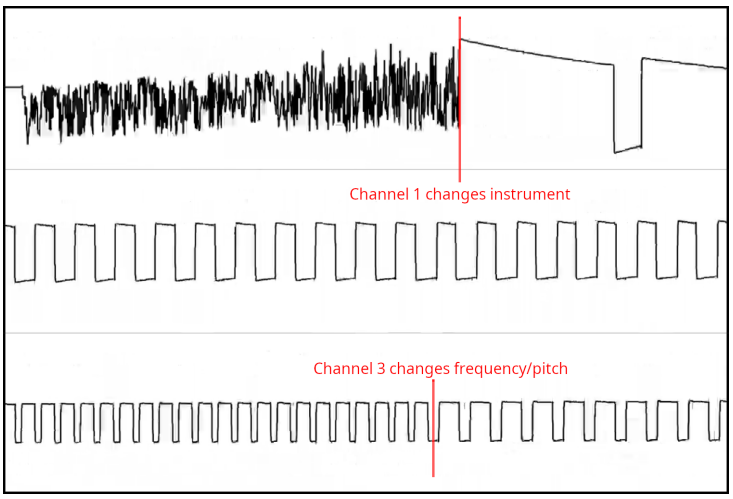


Figure 298: A freeze frame of a C64 song, you can see the instruments changing

Note!

You can hear the difference between a somewhat "simple song" like "Monty on the run" and something more complex, like the "R-Type" title theme (check the oscilloscopes). To see how "volume samples" worked, check the oscilloscopes of "Hot Rod" and "Netherworld" (both title themes).

14.3.6.3 Commodore Amiga

This is another famous home computer, although some could argue that it was being produced during the fall of Commodore, it is still the cradle of sample-based music and music trackers.

The Commodore Amiga's sound chip, name Paula, had a 4-channel PCM sample-based sound system, where each sample has an 8-bit resolution. Nothing stops people from just mixing more samples together and give the illusion of more channels.

Another limitation of the Paula chip is that 2 channels are strictly dedicated to the "left" stereo channel, while the other 2 are for the "right" stereo channel.

14.3.6.4 Sega Master System / GameGear

The Sega Master System is a quite famous 8-bit console, which had moderate success, and has a lot in common (hardware-wise) with the portable Sega GameGear and those similarities extend to the sound chip too.

The sound chip used is an equivalent of the Texas Instruments SN76489 (also known as PSG) which features 3 channels dedicated to square/pulse waves + a noise channel.

Random Trivia!

To be precise, the ancestor of the Master System, the SEGA SG-1000, used a real TI-SN76489, while the Master System uses a "clone" integrated into its VDP (Video Display Processor). Also the Japanese version of the Master system had an FM synthesizer chip too! To be precise it was the Yamaha YM2413, which gave a sound similar to the newer Sega Genesis/MegaDrive. Sadly such chip was removed for the international release.

14.3.6.5 Sega Genesis/MegaDrive

Probably Sega's most famous console: the Genesis/MegaDrive is a bit of a weird beast when it comes to sound. You'll see why.

Mainly the console uses a Yamaha YM2612 chip for sound, which offers 6 programmable FM channels + 1 DAC (digital to analog converter) that can play small samples. The sound chip is technically stereo, but the feature is underused due to the fact that, in the original console, stereo sound could be heard only through the headphone jack.

In addition, mostly for Master System compatibility, the console features a TI-SN76489 chip equivalent (integrated into its VDP), adding 3 square wave channels and a noise channel.

Random Trivia!

To underline how important the sound was in this console, just think that sound had its own dedicated fully-fledged CPU! It was a Zylog Z80, the same used as main CPU in the Sinclair ZX Spectrum.

14.3.6.6 NES

Probably the most famous console in the world, the NES had a limited but interesting toolkit for its sound.

The base console had 5 channels, distributed as follows:

- 2 channels dedicated to square waves;
- 1 channel only for triangle waves;
- 1 noise channel;
- 1 channel dedicated to playing small digital sound samples (used normally for drums).

But what's most interesting is that such capabilities could be extended by cartridge hardware, the most famous is probably the Konami VRC6, which added 2 more square wave channels, as well as a sawtooth one, used in Castlevania III.

Random Trivia!

Just to underline how extensible this system was, the Konami VRC7 contained a sound chip that provided a 6-channel FM synthesizer. Sadly its extended audio capabilities were used in a single, japan-exclusive game.

14.3.6.7 SNES

The SNES is a huge step forward in time for audio on Nintendo systems, featuring 8 channels that make use of 8-bit samples.

The S-SMP Chip also features a variety of filters and effects, so you have pretty much full freedom except for the number of channels.

14.3.6.8 Game Boy

The Game Boy is definitely not a step forward in audio technology, due to the lower amount of channels and less available waveforms. Yet, this console has a distinctive sound and some amazing tracks (just listen to any song from Bionic Commando, for instance).

In this console you have:

- 2 square wave generators;
- 1 noise generator;
- 1 PCM 4-bit wave sample, this allows the programmer to create their own wave forms, for instance;
- 1 audio input from the cartridge although this has never been used commercially

All the square wave and the noise generator feature envelopes (ADSR), while the first square wave generator also features a frequency sweeping functionality (allowing for slides between notes).

The Game Boy has a lot of obscure audio behaviour, but following the limitations above should give you enough of an idea to make something that sounds like it.

14.3.6.9 AdLib / SoundBlaster

The AdLib and SoundBlaster cards are based on the Yamaha YM3812 chip, which features 9 channels that use a digital oscillator. Given the high number of channels and the freedom given by them, it's pretty easy to get a result that sounds like old DOS game as soon as you get the tone down.

14.3.7 "Swappable" sound effects

Back to our walking example, an idea to increase the variety of sound effects at our disposal would be keeping a list of "swappable sounds": sounds that are still part of the class we're considering, but are radically different.

For instance we could have different walking sounds for different floors, so that walking on grass and walking on a stone pavement will be different. In this case it would be useful to make the sounds configurable and give the sound manager the chance to inspect what type of floor we're walking on.

An example of "swappable sound effects" configuration is given in the following file, which is written in YAML:

Listing 76: Example of swappable sound effects

```
footsteps:
  grass:
    - grasswalk1.wav
    - grasswalk2.wav
  stone:
    - stonewalk1.wav
    - stonewalk2.wav
  metal:
    - metalstep1.wav
    - metalstep2.wav
```

Making a configuration file instead of hard-coding the elements allows for easy extensibility and modding, which everyone loves (See Designing entities as data).

14.3.8 Some audio processing tips

Sometimes you may find yourself being a bit confused about what to do with your audio samples and music, so here's a small list of tips to make your life that tiny bit easier when it comes to processing audio.

14.3.8.1 Prefer cutting over boosting

Sometimes we may find our audio samples lacking that "punch" they would need, the first idea we may have would be to use a "bass boost" filter to make the low frequencies more prominent. Most of the time, this is not a good idea, since boost filters can create artifacts.

It's better to cut the higher frequencies instead, and eventually boost the entire volume of the sample during mixing. This way the nature of the sample doesn't get tainted by boosting, and we obtain the result we wanted.

14.3.8.2 Spatial sound is not only for 3D Games

If you have a 2D game, that doesn't mean you can't immerse the player further using spatial sound. This works well with top-down games (where you see the character from the top of their head) or games that make use of ray casting to create pseudo-3d environments.

Imagine that you have a source of sound in front of you.

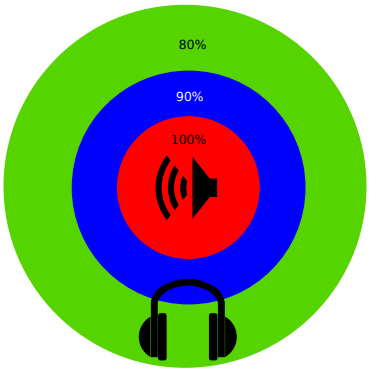


Figure 299: A source of sound right in front of you sounds the same in both ears

Both your ears are more or less the same distance from the sound source, hence they will hear the sound at the same volume.

But if you are at a different location, like the following example

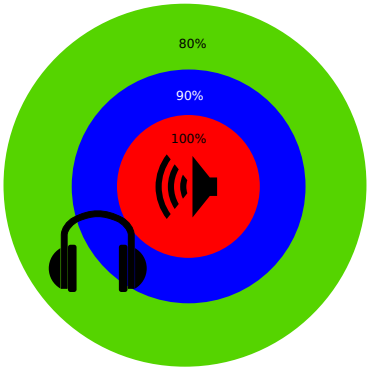


Figure 300: A source of sound out of center sounds different in each ear

Each ear would hear the sound with a different volume, with the ear "farthest from the source" hearing the sound at a lower volume.

This is the beginnings of spatial sound in a nutshell.

This can be expanded and made as complex as you want:

- Technically speaking, each ear would hear the sound at a slightly different delay (although considering an average distance between ears of 20cm - less than 10 inches, the delay between each ear would be about

0.0006 seconds);

- The sound gets lowered in volume and muffled if it goes through walls;
- If you're peeking from a corner with only one eye (like you're spying), one ear would be behind a wall, thus hearing less than the other;
- Sounds reflect on surfaces, thus a sound may feel like it comes from another direction than the one where the source is, thus a second "echoed source" should be mixed in;
- Different sounds react differently to walls: lower frequencies travel through walls more easily, while high frequencies get filtered more. This means that booming sounds can be perceived farther away than for instance, whistles or high-pitch squeals.

It can be as simple or as complex as you want, but the more complex something becomes, the more diminishing your returns will probably be.

[Do you know more about this? You can contribute, this book is open source!]

14.3.9 DAW Basics

14.3.9.1 What is a DAW Software?

Digital Audio Workstation Software (DAW Software) are pieces of software that have extensive recording, playback and editing features, allowing you to create your own songs, given some instrument samples (or pre-recorded tracks).

They also feature mixing facilities, waveform display and track controls, some even feature (the software equivalent of) effect racks, such as equalizers, to further the possibilities of creating your work in the best way possible.

14.3.9.2 The Piano Roll

Have you ever seen one of those pianos that seem to have an integrated music box? That is a so-called "piano roll" and is used to store music and play it back on the piano.

Most Software DAW have a similar abstraction, called "piano roll" too.

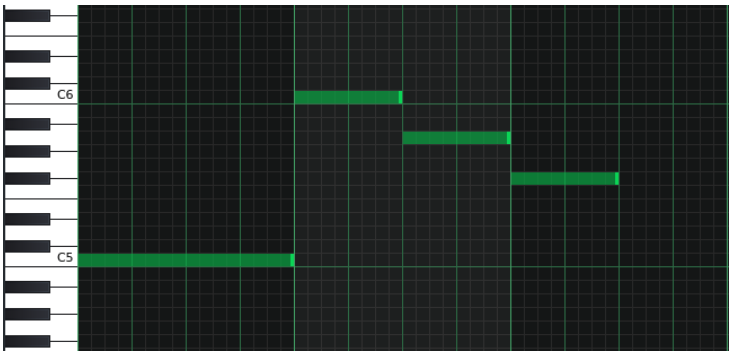


Figure 301: Example of a piano roll in LMMS

In the previous image, we can see four notes, each defined by its vertical position (respectively a C5, C6, A5, F#5) and length (respectively a single whole/semibreve and three half/minim notes). In the previous example each vertical

green bar represents $\frac{1}{4}$ and each light green bar represents a beat. This should help you imagining how to compose something.

Different systems have different piano roll variations. For instance:

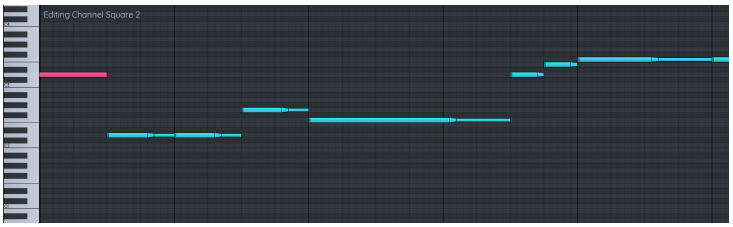


Figure 302: Example of a piano roll in FamiStudio

This Software DAW has a different shape for the notes, meaning a certain kind of effect is applied to them. All the basic principles still apply, though.

The piano roll abstraction allows you to edit your music easily, by grabbing notes and moving them, or making them longer by dragging their edges.

14.3.10 Music Tracker Basics

14.3.10.1 What is a Music Tracker Software?

Born with the 4-voice sampling system in the Commodore Amiga, Music Trackers are essentially a type of music sequencer. The great majority of music trackers have the majority of their screen occupied by the tracker version of a music sheet.

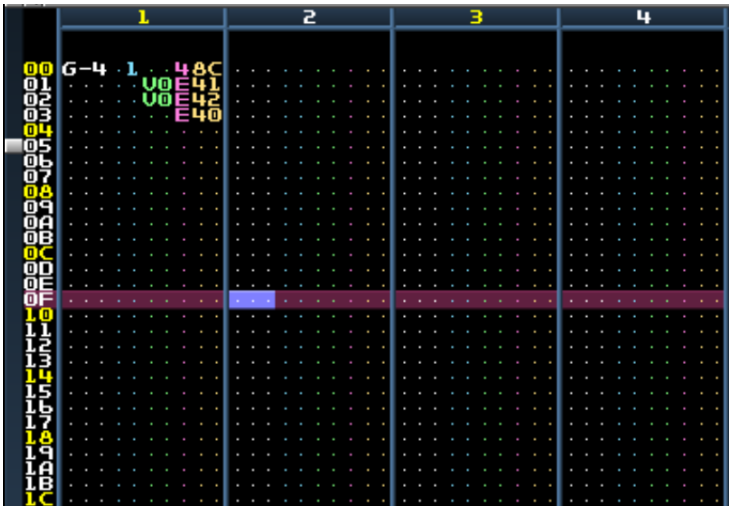


Figure 303: A screen from MilkyTracker

Differently from how the great majority Software DAW work, notes are positioned in channels at certain points of a timeline that spans vertically.

Usually each channel contains 64 rows (16 beats), but it can be changed to the composer's preference.

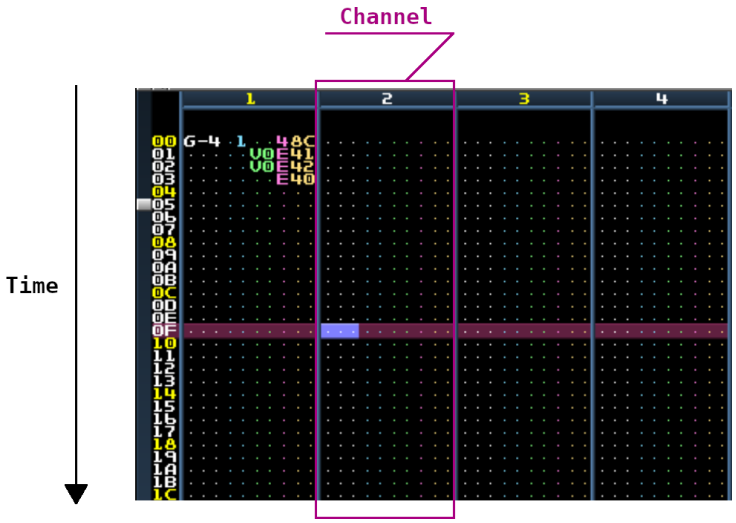


Figure 304: Simple overview of a tracker

Each row of a channel contains instructions for the tracker to execute, in the form of notes, instruments and commands (or effects).

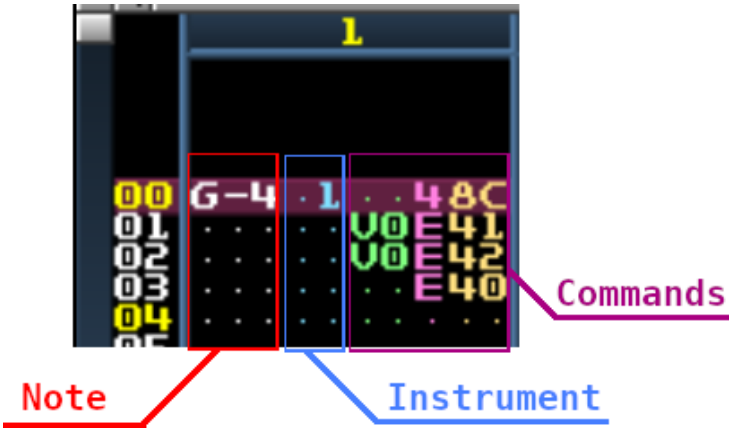


Figure 305: How each tracker channel is divided

Notes are written in the usual "Letter Notation" that you see in many music environments, while instruments are enumerated, then there are commands: commands can instruct the tracker to apply a temporary effect on the note, like portamento or vibrato.

In the previous example, there is a "vibrato" command going on, starting with the `48C`: `4` is the "vibrato command", `8` is the vibrato speed and `C` (hex for "12") is the vibrato depth.

The vibrato continues with a `V0 E41` command pair, where `V0` changes the vibrato depth to `0` (the speed is the same defined in the previous command), while `E41` is a "vibrato control command" (`E4`) which changes the waveform of the vibrato to `1` which is "ramp down".

The next command does more or less the same thing, besides changing the vibrato waveform to “square”.

The `E40` command resets the vibrato waveform to the default “sine wave”.

[This section is a work in progress and it will be completed as soon as possible]

14.3.10.2 Samples

Samples are the basis of a music tracker: they are essentially wave forms which can be sped up or slowed down to create different notes. Without any sample, you wouldn’t have any instrument, which in turn would mean you’d have no sound at all.

Usually samples come in the form of small digital sound files, most trackers allow the sample to be looped (wholly or in part) to simulate a “sustain” effect.

[This section is a work in progress and it will be completed as soon as possible]

14.3.10.3 Instruments

An instrument is a set of a sound sample, with some effects applied by default (if you want). Essentially an instrument is a “container” for a sample and some parameters to allow the change of pitch and effects.

[This section is a work in progress and it will be completed as soon as possible]

14.3.10.4 Channels

A “channel” (also called a “voice”), is a space where one sample is played back at a time. One channel is not “fixed” to a certain instrument and modern music trackers can mix an unlimited number of channels. Many times music makers limit themselves to a certain number of channels to achieve a “retro feeling” or to challenge themselves.

[This section is a work in progress and it will be completed as soon as possible]

14.3.10.5 Patterns

A pattern is essentially a piece of a song: a group of tracks with their own instruments, settings and notes written in them. The “pattern” abstraction allows you to easily repeat pieces of a song by just referring to the pattern.

[This section is a work in progress and it will be completed as soon as possible]

14.3.11 Basic Rhythms

When composing music, we may not know where to start: this is the objective of this section: give you some easy rhythms to start with. Here I will use LMMS’s beat+bassline editor to represent the notes, as it’s the easiest to understand.

Remember to check your tempo too, since it may make the difference between something akin to the “house” genre and something more “techno”.

14.3.11.1 Four on the floor

This is the most basic rhythm there is: let's consider a situation where we have 16 beats per bar (so each note is 1/16th):

Figure 306: A single bar in our basic rhythm

The four on the floor rhythm uses a kick drum in the 1st, 5th, 9th and 13th beat, giving a constant rhythm.

Figure 307: A basic four on the floor rhythm

This is practically the basic of all dance-based music.

14.3.11.2 Four on the floor with off-beat hi-hats

A basic rhythm like the "four on the floor", by itself, can prove to be quite boring. To spice it up, we can add some closed hi-hats in the off-beats, like the following:

Figure 308: Four on the floor with off-beat hi-hats

If you listen closely, a lot of music has this basic pattern mixed into it, in one way or another.

14.3.11.3 A simple rock beat

By using hi-hats, a kick drum and a snare drum, you can create a very simple rock beat. Keep 4 beats in a bar, each beat put a hi-hat, on 1 and 2 put a kick drum and on 3 a snare drum.

Figure 309: A simple rock beat

[Do you know more about this? You can contribute, this book is open source!]

14.3.12 Adaptive Music

Adaptive Music (sometimes called “dynamic music” or “interactive music”) is a background music track that reacts to the events of the game. Such reactions can involve volume, rhythm, tune, adding new instruments (adding drums, for instance).

Random Trivia!

A simple example of Adaptive Music can be found in Super Mario World, where hopping on Yoshi will add or change the drum beats on the level’s music track.

[This section is a work in progress and it will be completed as soon as possible]

14.4 Fonts

14.4.1 Font Categories

Before starting with fonts and the ways they can be integrated in your game, we should start with some definitions and categorizing fonts by their own characteristics.

14.4.1.1 Serif and Sans-Serif fonts

In typography, *serifs* are small strokes that get attached to larger strokes in a letter (or symbol) of certain fonts. The font families that make use of serifs are called **serif fonts** or **serif typefaces**.

Serif fonts look more elegant and give a “roman” feeling (in fact, serif fonts are also called *roman* typefaces) and are good for games that take place in historical settings or need a semblance of pretend “historical importance” (in their own world).

DejaVu Serif

Figure 310: Example of a serif font (DejaVu Serif)

Serif fonts look better on paper and could come out as a bit harder to read on screens. A famous serif font is Times New Roman.

On the opposite side, we have **sans-serif fonts**, where such small strokes are absent. Sans-Serif fonts seem easier to read on screens and look simpler, but they don’t look as good on paper, when long text bodies are involved.

DejaVu Sans

Figure 311: Example of a sans-serif font (DejaVu Sans)

A famous sans-serif font is Arial.

14.4.1.2 Proportional and Monospaced fonts

The majority of fonts used today are **proportional**, where each letter occupies its own space proportional to its own width. Examples of proportional fonts are Times New Roman and Arial.

Figure 312: Example of a proportional font (DejaVu Serif)

Notice the difference in width between certain pairs of letters, like “i” and “o” or “a” and “l”.

Proportional fonts are good for general text that don’t have any particular constraint.

On the opposite side, there are **monospaced** fonts, also called **fixed-width** fonts. In these font families, each letter occupies the same amount pre-defined width.

Figure 313: Example of a monospaced font (Inconsolata)

Again, notice how all letters occupy the same horizontal space.

Monospaced fonts are used for computer texts, coding and ascii-art. Examples of monospaced fonts are Courier New and Inconsolata.

14.4.2 Using textures to make text

If you want to make text to show on the screen, an idea could be creating a sprite sheet that contains all the characters that you want to use, then you can split it at runtime and (knowing that each letter is in a grid) you can index each letter by its position and write text.

Let’s imagine the simplest case: we need to use only uppercase letters, and our sprite sheet is 1 row by 26 columns:

Figure 314: A simple spritesheet for rendering text using textures

That looks awfully similar to an array, doesn’t it? We just need to know how big each tile is (and the easiest way to do so it making them all the same size) and their index in the array, connect them to a letter and we can make text!

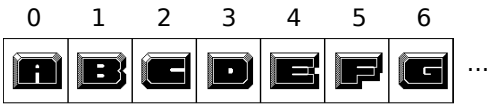


Figure 315: Indexing our spritesheet for rendering

So, if we were to write the word “HELLO”, we would need the letters at index 7, 4, 11 (twice) and 14. If the tiles were 32 x 32 pixels, we would need a surface that is 32 pixels tall and 160 pixels wide.

The code to make text from a texture could look something like this:

Listing 77: A simple algorithm to create a text using a texture

```
let TILE_WIDTH = 32;
let TILE_HEIGHT = 32;
SPRITESHEET = engine.load_spritesheet("resources/font.png");

function get_cell_from_letter(letter){
    /*
     * Here we will use a bit of ASCII magic, each letter
     * has a "number attached to it", related to its position
     * in the ASCII table. A starts at position 65 and each letter
     * follows, until Z at position 90.
     * This means we can "convert" each letter to its ASCII index,
     * subtract 65 and have our "array index"
     */
    let ascii_index = letter.charCodeAt(0);
    let array_index = ascii_index - 65;
    return array_index;
}

function make_text(string to_write){
    // First, we create the surface to write the text onto
    let surface_width = TILE_WIDTH * to_write.length;
    let final_text = new Surface(surface_width, TILE_HEIGHT);
    // This will keep track of the left side of the first free "cell"
    let current_pixel = 0;
    // Now we iterate each letter of the word we want to render
    for (let i=0; i < to_write.length; i++){
        letter = to_write.charAt(i);
        // First, we need to find which cell corresponds to the letter
        let letter_cell = get_cell_from_letter(letter);
        // After that, we need to get the subsurface, which contains only the letter
        // It will start at letter_cell * TILE_WIDTH
        graphical_letter = SPRITESHEET.get_subsurface(
            left = letter_cell * TILE_WIDTH,
            top = 0,
            width = TILE_WIDTH,
            height = TILE_HEIGHT
        );
        // Now that we have the subsurface, we can draw it on our final surface, to render the
    text
        final_text.draw(graphical_letter, {current_pixel, 0});
        // Now we prepare for the next loop, by increasing the current pixel by 32 (thus moving
        // our "cursor" right)
        current_pixel += TILE_WIDTH;
    }
```

```
44      // After the loop exits, our text is ready to be used
45      return final_text;
46  }
```

And that's how you get a text from a texture. Pun (maybe) not intended.

Rendering style	Spritesheet-based
When to use it	When you need very complex shapes and colors (even multiple colors in the same letter).
Advantages	Customizability. Simplicity (once you have created the rendering functions). May save some space compared to other methods (you only have the characters that you need). Editing the font doesn't need extra programs (you just need your usual spritesheet editor).
Disadvantages	Scaling and resizing can be problematic: scaling up may look fuzzy (if filtering is applied) or pixellated (if no filtering is applied), scaling down can result in loss of shape and detail.

14.4.3 Using Fonts to make text

Another way to make text is using fonts: there are a couple of formats you can use, like TTF and OTF.

The advantage of this system is that usually well supported by the engine or library that you use. This means that you don't need to write code exclusive to font rendering, which means you don't have to write code that may contain bugs itself.

There are also some limitations, due to how fonts are built: you can only use a single color, for instance.

Rendering style	Font-based
When to use it	When you want fonts that won't lose detail or crispness when they get resized.
Advantages	Rendering functions are usually part of the engine or library. Fonts resist resizing very well, being usually based on vector graphics.
Disadvantages	Editing fonts requires specific programs. Can only use one color at a time. Limited customizability.

14.5 Shaders

14.5.1 What are shaders

Shaders are technically small programs that (usually) run inside your Graphics Card (GPU). In gaming they are usually used for post-processing and special effect, allowing to free the CPU from a lot of workload, using the specialized GPU capabilities.

Shaders can be classified in different groups:

- **2D Shaders:** These shaders act on textures and modify the attributes of pixels.
  - **Pixel (Fragment) Shaders:** Used to compute color and other attributes relating to a single output pixel. They can be used for blur, edge detection or enhancement and cel/cartoon shading.
- **3D Shaders:** These shaders act on 3D models or other geometry.
  - **Vertex Shaders:** These shaders are run once per vertex given to the GPU, converting the 3D position in virtual space to the 2D coordinates of your screen. These shaders cannot create any new geometry.
  - **Geometry Shaders:** These shaders can create new primitives, like points or triangles.
  - **Tessellation Shaders:** These shaders allow to divide simple meshes into finer ones at runtime, this allows the meshes closest to the camera to have finer details, while the further ones will be less detailed.
  - **Primitive Shaders:** Akin to the computing shaders, but have access to data to process geometry.
- **Computing Shaders:** These shaders are not limited to graphics, but are related to the world of GPGPU (General Purpose computing on GPU), these can be used to further stages in animation or lighting.

14.5.2 Shader Programming Languages

There are numerous programming languages, depending on the platform and libraries you are using to program your game.

If you are using OpenGL, you should use the official OpenGL Shading Language, called **GLSL**.

Listing 78: Simple GLSL Fragment shader

```
#ifdef GL_ES
precision lowp float;
#endif

uniform float u_time;

void main() {
    gl_FragColor = vec4(1.0,0.0,0.0,1.0);
}
```

If you are using Direct3D, you should instead use the “High Level Shader Language”, also called **HLSL**.

If instead you want to use Vulkan, you will need to use the **SPIR-V** (Standard Portable Intermediate Representation) format, but the good news is that (at the time of writing) you can convert your GLSL code into SPIR-V and use it with Vulkan.

Modern engines, like Unity and Unreal Engine also include GUI node-based editors that help you create new shaders by using directed graphs, without using any code.

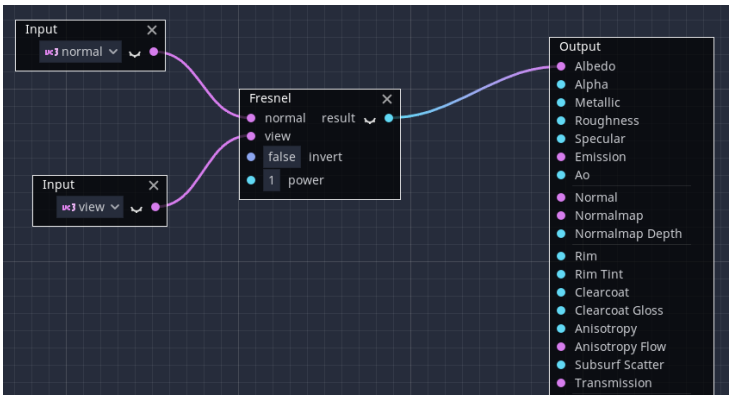


Figure 316: Godot's "Visual Shader" Editor

14.5.3 The GLSL Programming Language

GLSL is a programming language that is syntactically close to C, thus if you know a bit of the C programming language, GLSL will at least "look" familiar.

[This section is a work in progress and it will be completed as soon as possible]

14.5.3.1 The data types

There are many data types in GLSL, here is a quick list of data types supported in GLSL.

14.5.3.1.1 Non-vector types

Among these we find "the usual suspects", that means:

- **bool** a standard conditional type, it can be true or false;
- **int** a standard 32-bit integer, with sign;
- **uint** a 32-bit integer without sign;
- **float** single precision floating point number;
- **double** double precision floating point number.

14.5.3.1.2 Vector Types

Non-vector types can be aggregated into vectors, which can have 2, 3 or 4 elements:

- **bvec2 bvec3 bvec4** vector of booleans, with 2,3 or 4 elements;
- **ivec2 ivec3 ivec4** vector of signed integers, with 2,3 or 4 elements;
- **uvec2 uvec3 uvec4** vector of unsigned integers, with 2,3 or 4 elements;
- **vec2 vec3 vec4** vector of single-prevision floating point numbers;
- **dvec2 dvec3 dvec4** vector of double-prevision floating point numbers.

14.5.3.1.3 Matrices

Matrices are groups of floating points (either single or double prevision) that have $n \times m$ size.

The type is `matnxm` (for instance `mat2x3` represents a 2×3 matrix). A shortcut for an $n \times n$ matrix is `matn` (so `mat3` would represent a 3×3 matrix).

[This section is a work in progress and it will be completed as soon as possible]

14.5.4 Some GLSL Shaders examples

[This section is a work in progress and it will be completed as soon as possible]

Part 5: Advanced Topics

15 Design Patterns

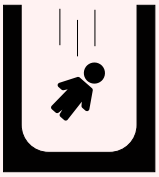
> Patterns mean "I have run out of language."
>
> *Rich Hickey*

Design Patterns are essentially "pre-made solutions for known problems" and can help decoupling elements of your game, increasing maintainability.

Note!

This book will introduce design patterns as they were explained in the famous "Gang of Four" book: "Design Patterns: Elements of Reusable Object-Oriented Software". In some languages (for instance ones that have functions as first-class objects) such patterns can take different forms. This book will tell you which pattern does what, along with a rough implementation, but it's up to you to check the most efficient one in your favourite language.

Pitfall Warning!



When people find out about design patterns, they usually come up with the "everything is a nail syndrome". Don't overuse design patterns only because it looks "cool" or "because it may solve a future problem". What matters is your game, right now.
Design patterns are not a cure-all, they can introduce overhead and could lead to over-engineering: balance is key when it comes to creating a game (or any software in general).
Leave future problems to your future self.

15.1 Creational Design Patterns

Creational patterns is a category of design patterns that deal with object creation mechanisms, that means creating objects in a way that best suits the single situation.

15.1.1 Singleton Pattern

Sometimes it can be necessary to ensure that there is one and only instance of a certain object across the whole program, this is where the *singleton* design pattern comes into play.

To make a singleton, it is necessary to hide (make private) the class constructor, so that the class itself cannot be instantiated via its constructor.

After that, we need a static method that allows to get the singleton's instance, the method needs to be static to make it callable without an instance of the singleton class.

The UML diagram for a singleton is really simple.

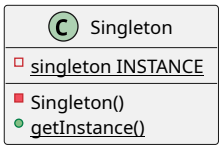


Figure 317: The UML diagram for a singleton pattern

Some singleton implementations may instantiate themselves immediately, which is not always necessary, in that case a good idea could be implementing what is called “lazy loading”, where the instantiation happens the first time you ask the object for its own instance.

Listing 79: Example of a singleton pattern with lazy loading

```
class LazySingleton {

    // This way the constructor always returns the singleton instance
    constructor(){
        // Multi-threading: manage race conditions
        // ----- Critical region start -----
        if (!Singleton.INSTANCE){
            Singleton.INSTANCE = this;
        }
        // ----- Critical region end -----
        return Singleton.INSTANCE;
    }

    static getInstance(){
        return Singleton.INSTANCE;
    }
}
```

If multiple threads are involved in using a lazy-loading singleton, you may need to take care of preventing *race conditions*[g] that could result in multiple instances of the singleton being created.

Many critics consider the singleton to be an “anti-pattern”, mostly because it is really overused and adds a possibly unwanted “global state” (see it as a global variable, but in an object-oriented sense) into the application.

Before applying the singleton pattern, ask yourself the following questions:

- Do I really need to **ensure** that only one instance of this object is present in the whole program?
- Would the presence of more than one instance of this object be detrimental to the functionality of this program? How?
- Does the instance of this object **need** to be accessible from everywhere in the program?

Table 52: Summary table for the Singleton Pattern

Pattern Name	Singleton
When to Use it	In all situations that strictly require one instance of an object, accessible globally.
Advantages	Allows the class to control its own instantiation, allows for easier access to the sole instance of a class.
Disadvantages	Introduces some restrictions that may be unnecessary, introduces a global state into the application.

15.1.2 Dependency Injection

Dependency Injection is a very simple concept that is really hard to explain. It is essentially used to avoid having classes build instances of services (which may be other classes or functions) inside of themselves and instead receiving such services from outside.

Let's make a concrete example. You have a class that takes care of everything concerning a file upload: from getting it from the internet, to logging to saving it to the hard disk.

A first implementation would look something like the following:

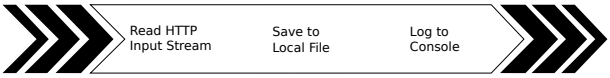


Figure 318: A naive implementation of a local file upload system

What would happen if, instead of a hard disk you need to transfer the files to an external service like S3, or maybe it just needs to be saved into memory for further processing? You would probably need to duplicate the class to allow for these new "services".

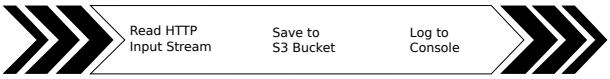


Figure 319: A naive implementation of a file upload system on S3

A better solution for reuse would be having the "file saving service" separated from the entire "file upload" class, and instead having this service "injected" onto the "file upload" class. This can happen via setter functions, via constructors, builders, factories or even interface injection (where it's the dependency interface that provides a method to inject the dependency).

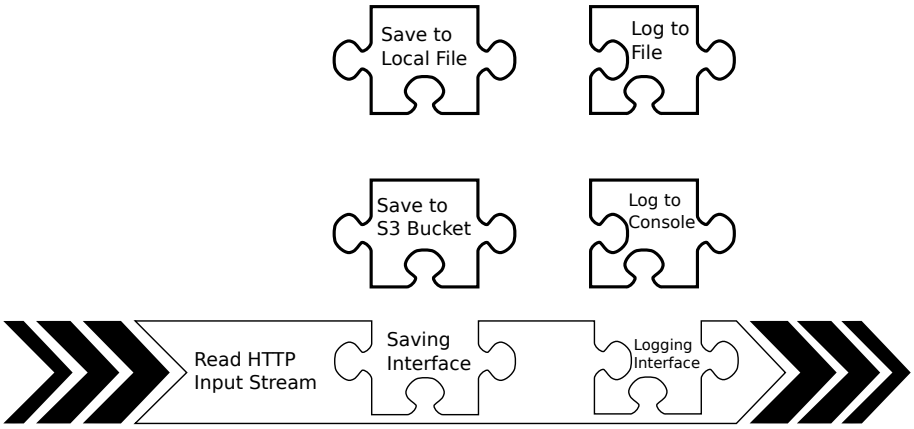


Figure 320: Using Interfaces and DI to build a flexible file upload

Since now the “file upload” class doesn’t depend on how or where the file is saved, you can also substitute such “file saving service” with a mock during tests.

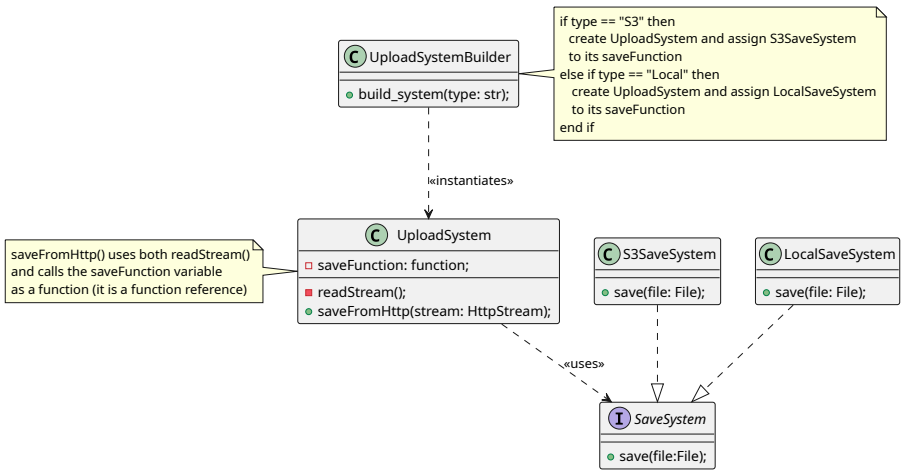


Figure 321: Possible class structure for a DI file upload

Dependency injection can be divided in two main categories:

- **Setter injection:** In this case, an object has specific `set()` methods that allow you to set the dependency after the object has been constructed. This allows also for changing the dependency on the fly (useful to change the effect of a weapon, for instance);
- **Constructor injection:** In this case, the injection happens in the class’s constructor. The injected functionality is decided before the object that needs the dependency is constructed.

[This section is a work in progress and it will be completed as soon as possible]

Table 53: Summary table for the Dependency Injection Pattern

Pattern Name	Dependency Injection

When to Use it	In all situations that require a degree of configurability or where the behaviour of the code needs to be changed without direct editing.
Advantages	Decreased coupling, reusability, maintainability, flexibility, less boilerplate code, allows for concurrent development of services, allows for easier unit testing.
Disadvantages	Behaviour and construction are separated, which may make tracing code harder. May hinder IDE automation if implemented using reflection or dynamic programming.

15.1.3 Prototype

Sometimes, in our game, we need to decide which objects to create at runtime, as well as instantiate dynamically loaded classes. In these cases the prototype pattern comes to the rescue: we define a "prototype" that allows to create classes by cloning itself.

There is the UML diagram for the pattern:

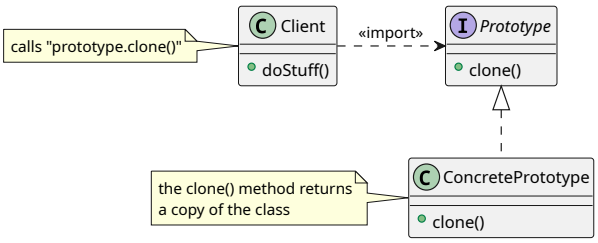


Figure 322: Diagram of the Prototype Pattern

Table 54: Summary table for the Prototype design pattern

Pattern Name	Prototype
When to Use it	When you need to either decide the objects to create at runtime or instantiate dynamically loaded classes.
Advantages	Decoupling, added flexibility.
Disadvantages	May become overused, depending on the situation can be difficult to implement.

[This section is a work in progress and it will be completed as soon as possible]

15.2 Structural Design Patterns

Structural patterns is a category of design patterns that deals with the relationships between entities, with the objective of simplifying the realization of such relationships.

15.2.1 Flyweight

Sometimes it may be necessary to keep track of a large number of very similar objects.

Imagine a lot of sprites of trees that have the same texture and size, but have different positions: it could prove to be really resource-heavy to keep all the sprite objects in memory, each one with its own copy of the texture and size. This could prove to be performance-heavy too, since all those textures will have to be moved to the GPU.

Here comes the Flyweight pattern: we try to share as much of the so-called "intrinsic state" of the objects between the object that contain the so-called "extrinsic state".

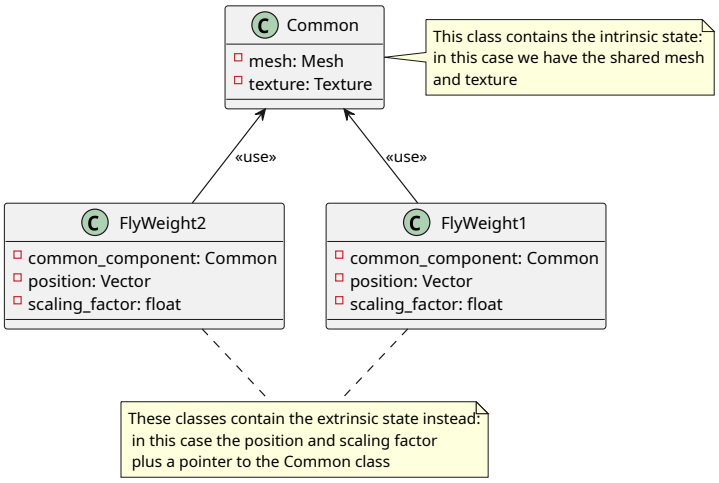


Figure 323: UML Diagram of the Flyweight pattern

Below is an example code for the flyweight pattern.

Listing 80: Code for a flyweight pattern

```
class Common{
    // Contains the common data for a 3D Object to be replicated
    constructor(){
        this.mesh = new Mesh();
        this.texture = new Texture();
    }
}

class FlyWeight{
    // Contains only the necessary data to create an instance of the item
    constructor(common_ptr){
        this.common_pointer = common_ptr;
        this.position = new Vector2();
        this.scale_factor = 1;
    }
}
```

Random Trivia!

This is just speculation, but SFML's graphics system may make use of the Flyweight pattern: you need to load the image into a "Texture" first (which does all the low-level lifting) and then you can instance an "Image" class which is more high-level. Many images can refer to the same texture (which may be a Sprite Sheet).

Table 55: Summary table for the Flyweight Pattern

Pattern Name	Flyweight
When to Use it	When you need to support a large number of similar objects efficiently, when you need to avoid creating a large number of objects.
Advantages	Saves memory when a large number of similar objects is involved, avoids some of the overhead given by the creation of many objects.
Disadvantages	The intrinsic state must be "context independent", so it cannot change (or all the flyweights that refer to that state will change too). Flyweight instantiation requires particular attention in multi-threaded environments, due to the shared memory.

15.2.2 Component/Composite Pattern

When building any game entity, we find that the complexity of the game entity itself literally explodes: a monolithic class can include loads of different operations that should stay separate, such as:

- Input Handling
- Graphics and Animation
- Sound
- Physics
- ...

At this point our software engineering senses are tingling, something is dangerous here.

A better alternative in bigger projects is splitting the monolithic class and create different components and allow for their reuse later. Enter the Component pattern.

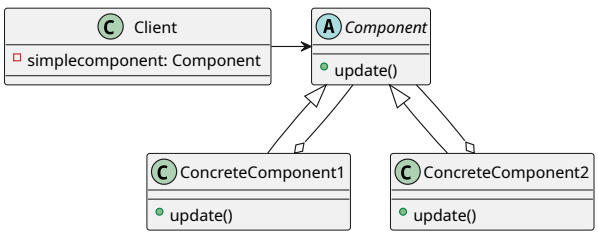


Figure 324: Diagram of the Component Design Pattern

The client is connected to a list of Components that have the same interface (in the previous case, the `update()`

method), so each Game Entity can become a “container of components” that define its behaviour.

For instance, instead of having all the functionalities listed above, our game entity could have the following components:

- Input Component
- Graphics Component
- Sound Component
- Physics Component

Which can be reused, extended and allow for further flexibility and follows more closely the DRY principle.

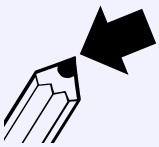
Here we can take a look at a sample implementation of the Component Design Pattern:

Listing 81: Example Implementation Of the Component Pattern

```
class Component{
    // Defines the abstract class/interface for the component pattern
    constructor(){
        // In javascript there are no real facilities for abstract classes
        // So we need to play around
        if (new.target == Handler){
            // The new.target changes in derivative classes
            throw new TypeError("Cannot construct abstract objects")
        }
    }

    update(){
        // Do nothing, this is an abstract class
    }
}

class ConcreteComponent1 extends Component{
    // Defines the concrete component number 1
    update(){
        // Do Stuff
    }
}

class ConcreteComponent2 extends Component{
    // Defines the concrete component number 2

    // The component can contain a list of other components that get updated
    constructor(){
        super();
        this.list = [component1, component2, ..., component9];
    }

    update(){
        for (const component in this.list){
```

```
35              component.update();
36          }
37
38          // Do Other Stuff
39      }
40  }
41
42  class Client{
43      constructor(){
44          this.first_component = new ConcreteComponent1();
45          this.second_component = new ConcreteComponent2();
46      }
47
48      update(){
49          // This is the Client's update function
50          this.first_component.update();
51          this.second_component.update();
52      }
53  }
```

Tip!



You can also think about components as “capabilities”: objects can be “movable”, so they have an input or physics component, they can be “drawable” so they have a graphics component, etc...

Table 56: Summary table for the Component/Composite design pattern

Pattern Name	Component/Composite
When to Use it	When you need to deal with a part-whole hierarchy where each component needs to be treated equally.
Advantages	Decoupling, added flexibility.
Disadvantages	On bigger systems, the management may become really complex.

15.2.3 Decorator

There are some cases where we need to add or remove behaviours from a class at runtime, dynamically. The decorator pattern gives a flexible alternative to subclassing and addresses this need.

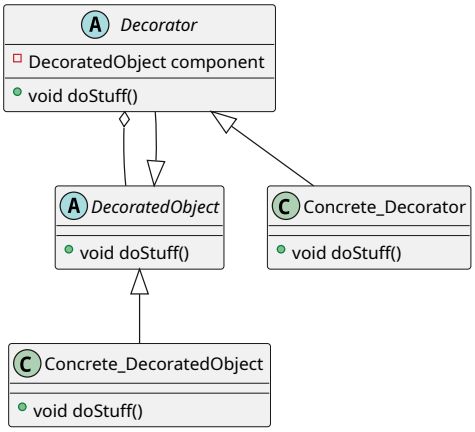


Figure 325: Diagram of the Decorator Pattern

As you can see the decorator makes heavy use of abstract classes and interfaces, which most programming languages implement without any issue.

Table 57: Summary table for the Decorator design pattern

Pattern Name	Decorator
When to Use it	When you need to add or remove functionalities from a class dynamically. Very useful for applying memoization techniques.
Advantages	Decoupling, added flexibility.
Disadvantages	Usage of interfaces or abstract classes can seem a bit daunting at the beginning, it may cause an explosion in number of classes.

[This section is a work in progress and it will be completed as soon as possible]

15.2.4 Adapter

Let's face it, not everything is straight out compatible with everything else. It happens with power plugs, why wouldn't it happen in a world as varied as software development?

Sometimes we need an adapter, and that's exactly what this design pattern is: provide a layer of compatibility between two incompatible interfaces.

The adapter design pattern can be implemented in two ways, but first let's check the summary table.

Table 58: Summary table for the Adapter design pattern

Pattern Name	Adapter
When to Use it	When you need to provide a layer of compatibility between two incompatible interfaces or you need to provide an "alternative interface" to a class.

Advantages	Decoupling, added flexibility, better compatibility, code reuse.
Disadvantages	Needing many adapters may mean there is a deeper structural problem with your program.

15.2.4.1 Object Adapter

The “object adapter” is the version where the adaptor delegates the task to the adaptee at runtime. To do so, it has an instance of the adaptee class as its class field.

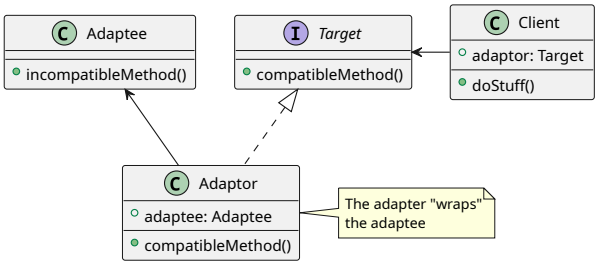


Figure 326: Diagram of the Object Adapter Pattern

[This section is a work in progress and it will be completed as soon as possible]

15.2.4.2 Class Adapter

The “class adapter” version instead inherits the adaptee class at compile-time. Since the adaptor inherits from the adaptee class, the adaptee’s methods can be called directly, without needing to refer to a class field.

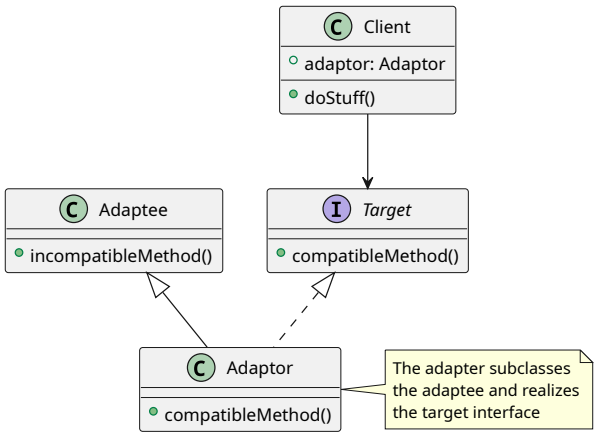


Figure 327: Diagram of the Class Adapter Pattern

[This section is a work in progress and it will be completed as soon as possible]

15.2.5 Facade

There are times where you have a very complex library, with a very complex interface, that is extremely complex to interact with. The Facade pattern hides such complexity behind a simple-to-use interface that works by delegation.

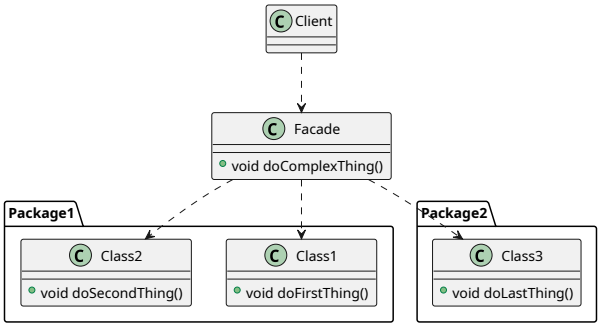


Figure 328: Diagram of the Facade Pattern

This pattern should be used with extreme care and only when necessary, since adding "levels of indirection" will make the code more complex and harder to maintain.

Table 59: Summary table for the Facade design pattern

Pattern Name	Facade
When to Use it	When you need to present a simple interface for a complex system or you want to reduce the dependencies on a subsystem.
Advantages	Decoupling, added readability.
Disadvantages	May become overused, delegating adds a bit of overhead, sometimes it may be wrongly used where either an adapter or a decorator is needed. May become a single point of failure[g].

Here's an example of a possible facade pattern:

Listing 82: Example Implementation Of the Facade Pattern

```
class FirstService{
    // Implementation here...
}

class SecondService{
    // Implementation here...
}

class Facade{
    /*
     * This class hides the complexities of using
     * FirstService and SecondService from the user
```

```
13      * by "wrapping" them in a comfortable startAll
14      * function
15      */
16
17     constructor(){
18         this.service1 = new FirstService();
19         this.service2 = new FirstService();
20     }
21
22     startAll(){
23         /*
24          * The facade starts all the services and does
25          * some status checking, this is hidden from the
26          * user.
27          * Returns true if all services started successfully
28          * false otherwise
29          */
30         let firstServiceStarted = this.service1.start();
31         if (!firstServiceStarted){
32             return false;
33         }
34         let secondServiceStarted = this.service2.start();
35         if (!secondServiceStarted){
36             return false;
37         }
38         // Here everything started successfully
39         return true;
40     }
41 }
```

15.2.6 Proxy

Sometimes, the access to a certain object may be problematic or not directly possible, for instance when:

- The object is available only remotely;
- The object needs a form of access control;
- The object is too heavy to complex to be available in memory at all times.

In these cases, the proxy pattern is the pattern that may solve your issues: in its most general form, a proxy is an interface to another object. Such interface may have additional operations attached to it (for instance for access control), or it could represent a “virtual object”, where the real object is located somewhere else.

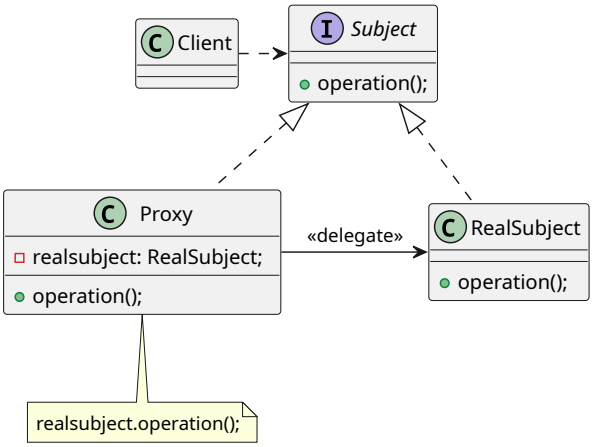


Figure 329: Diagram of the Proxy Pattern

This can be very useful in implementing multiplayer systems: a well-designed system won't care if the input controlling a character come from the keyboard, a gamepad, another computer (while playing multiplayer) or a file (for replays, for instance). The proxy pattern will make it so that everything has the same interface and the "user object" (as in the object that uses the proxy) won't have to worry about what's on "the other side of the proxy".

Listing 83: Example Implementation Of the Proxy Pattern

```
class User{
    // This class represents a user, with their permissions
    // ...
    can_do_thing(){
        // Changes according to user permissions
        // ...
    }
}

class Request{
    // This class represents a web request
    constructor(user){
        this.user = user;
    }
}

class WebPage{
    /*
     * Represents a call to a web page
     */
    // This function gets implemented in the concrete classes;
    get(request){
        throw "Cannot use get() from abstract class"
    }
}

class WebPageProxy extends WebPage{
```

```
    // Represents an authentication proxy for a web page
    get(request){
        // Get the requesting user
        requesting_user = request.user;
        if (requesting_user.can_do_thing()){
            return super.get(request);
        }else{
            return HttpResponse("This user cannot access this page");
        }
    }
}

let user = User();
// The following line is valid due to polimorphism
let page = WebPageProxy();
// If the user has permission the page will come through.
page.get(Request(user));
```

The proxy pattern is also very useful for lazy-loading: the Proxy object will pretend to be the original object, until the object is really needed (thus moving the instantiation weight to the "first use" of an object), then it will just "pass through" every command.

Pattern Name	Proxy
When to Use it	When you need to represent a remote object, a complex/heavy object that can't fit in memory or if you need to control access to an object. Useful for lazy loading, replays or remote play.
Advantages	Decoupling.
Disadvantages	When used for lazy-loading, "passing through" function calls will add function pointers to the stack, which may slow down performance.

15.3 Behavioural Design Patterns

Behavioural patterns is a category of design patterns that deals with communication between objects. This is done by identifying common communication patterns between such objects and abstracting it.

15.3.1 Command Pattern

It may be necessary, during our software development, to abstract our functions into something that can be assigned and treated as an object.

Many programming languages now feature functions as "first class citizens", allowing to treat functions as objects: assigning functions to variables, calling functions, lambdas, inline functions, functors, function pointers...

The command pattern allows us to abstract a function (or any executable line of code) into its own object that can be handled as such, allowing us to package a request into its own object for later use.

This pattern can be useful to code GUIs, making actions in our games that can be undone, macros, replays and much more.

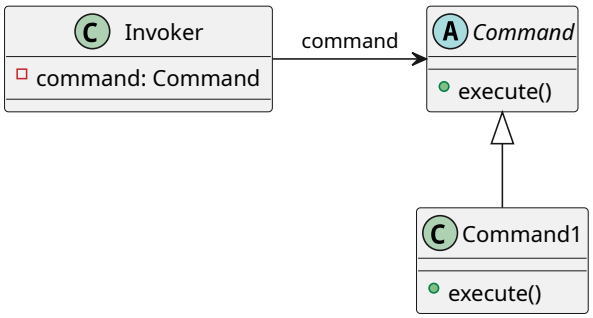


Figure 330: UML diagram for the Command Pattern

Listing 84: Example code for the Command Pattern

```
class Command{
    // This is the abstract class that will be used as interface
    constructor(){
        // In javascript there are no real facilities for abstract classes
        // So we need to play around
        if (new.target == Handler){
            // The new.target changes in derivative classes
            throw new TypeError("Cannot construct abstract objects")
        }
    }

    execute(){};
}

class JumpCommand extends Command{
    // This will implement the execute method
    execute(){
        jump();
    }

    jump(){
        // DO STUFF
    }
}
```

Table 61: Summary table for the Command Pattern

Pattern Name	Command
When to Use it	In all situations where you want to avoid coupling an invoker with a single request or when you want to configure an invoker to perform a request at runtime.

Advantages	Allows for encapsulation, less coupling, more flexibility and customization at runtime.
Disadvantages	Late binding and objects may introduce some overhead.

15.3.2 Observer Pattern

The observer pattern is used to implement custom event handling systems, where an object automatically reacts to events generated by another object.

There are 2 main objects used in an observer pattern:

- **The Subject**: sometimes called “Observed Object”
- **The observer**: sometimes called “Dependent Object”

The subject is the creator of a “stream of events” that is consumed by the observer objects.

The subject implements in its structure a list of observers that will be notified when a change occurs, as well as methods to register (add) a new observer as well as to unregister (remove) an existing observer, while the observers will implement a method that will be called by the subject, so that the observers can be notified of such change.

Here we can see an UML diagram of the observer pattern:

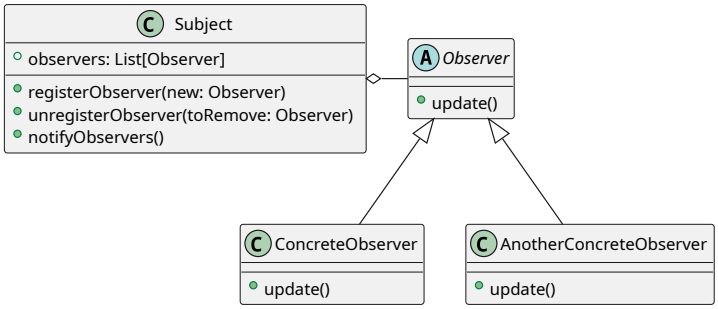


Figure 331: The UML diagram of the observer pattern

Here we can see the Observer abstract class (it can be an interface), a concrete subject and two Concrete Observers that implement what required by the Observer.

Here we can see an implementation of the observer pattern:

Listing 85: Code for an observer pattern

```
class Subject{
    /* This is the observed class that contains the list of observers and
     * the notifyObservers method */

    constructor(){
        this.observers = [];
    }
```

```
8
9       register_observer(observer){
10          observers.push(observer);
11      }
12
13      notifyObservers(){
14          for (const observer in this.observers){
15              observer.update();
16          }
17      }
18  }
19
20
21  class Observer{
22      /* This is the class that contains the update method, used to force
23       * an update in the observer */
24
25      update(){
26          console.log("I have been updated!");
27      }
28  }
29
30
31  let subject = new Subject();
32  let observer = new Observer();
33  subject.register_observer(observer);
34  subject.notifyObservers();
```

If needed, you can pass information between the subject and the observers just by calling each `update()` method with the necessary arguments.

Table 62: Summary table for the Observer Pattern

Pattern Name	Observer
When to Use it	Event Handling systems, making objects react to other objects' actions
Advantages	Decoupling, added flexibility, more performing than if statements for conditions that happen rarely.
Disadvantages	Can be a bit hard to set up, makes the architecture more complex, if un-registration is not done well there could be serious memory leaks (even in garbage-collected languages).

15.3.3 Strategy

In some situations it may be necessary to select a single algorithm to use, from a family of algorithms, and that decision must happen at runtime.

In this case, the *strategy pattern* (also known as the "policy pattern"), allows the code to receive runtime instructions over what algorithm to execute. This allows for the algorithm to vary independently from the client that makes use of such algorithm.

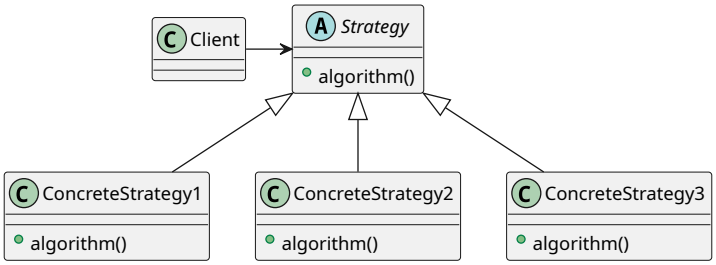


Figure 332: The UML diagram of the strategy pattern

Listing 86: Code for a strategy pattern

```
class Strategy{
    // This class defines the strategy interface the client will refer to

    constructor(){
        // In javascript there are no real facilities for abstract classes
        // So we need to play around
        if (new.target == Handler){
            // The new.target changes in derivative classes
            throw new TypeError("Cannot construct abstract objects")
        }
    }

    algorithm(){
        // This algorithm will be implemented by the subclasses
    }
}

class ConcreteStrategy1 extends Strategy{
    algorithm(){
        // Real implementation of the algorithm
        // DO STUFF
    }
}

class ConcreteStrategy2 extends Strategy{
    algorithm(){
        // Real implementation of the algorithm
        // DO STUFF SLIGHTLY DIFFERENTLY
    }
}

// Example Usage
function main(){
```

```
    let to_execute = null;
    if (condition){
        to_execute = new ConcreteStrategy1();
    }else{
        to_execute = new ConcreteStrategy2();
    }
    to_execute.algorithm();  // This will execute 1 or 2 depending on "condition"
}
```

Table 63: Summary table for the Strategy Pattern

Pattern Name	Strategy
When to Use it	Every time you need to decide which algorithm to execute at runtime.
Advantages	Decoupling, added flexibility.
Disadvantages	Can cause proliferation of similarly-looking concrete strategies, late binding on functions and the object oriented nature of the pattern could create some overhead.

15.3.4 Chain of Responsibility

Sometimes we have the necessity of handling conditionals that are themselves connected to runtime conditions. This is where the *chain of responsibility pattern* comes into play, being essentially an object-oriented version of an `if ... else if ... else` statement.

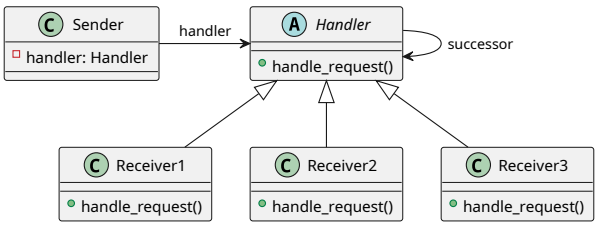


Figure 333: UML Diagram of the Chain of Responsibility Pattern

As can be seen from the diagram, the sender is not directly connected to the receiver, but instead it's connected to a "Handler" interface, making them independent.

As with a chain of responsibility in a company relays a task to "higher ups" if the task cannot be handled, the chain of responsibility pattern involves each received reviewing the request and if possible, process it, if not possible, relay it to the next receiver in the chain.

Listing 87: Code for a chain of responsibility pattern

```
class Handler{
    // This is the handler abstract/class interface that the sender connects to
    constructor(){
        // In javascript there are no real facilities for abstract classes
```

```
        // So we need to play around
        if (new.target == Handler){
            // The new.target changes in derivative classes
            throw new TypeError("Cannot construct abstract objects")
        }
        this.next = null;  // The next handler in the chain
    }

    handle_request(){
        if (condition){
            // In case I can handle this request
            return this.real_handler();
        }

        if (this.next != null){
            return next.handle_request();
        }
    }

    real_handler(){
        // This function gets implemented in the concrete classes
    }

    add_handler(new_handler){
        this.next = new_handler;
        return this.next;  // Allows for chaining .add_handler().add_handler()...
    }
}
```

Table 64: Summary table for the Chain of Responsibility Pattern

Pattern Name	Chain of Responsibility
When to Use it	When you need to implement flexible if...else if...else statements that change on runtime. When you want to decouple a sender from a receiver.
Advantages	Decoupling, added flexibility.
Disadvantages	Some overhead is added by the objects and late binding, could lead to proliferation of similar-looking handlers/receivers.

15.3.5 Visitor

[This section is a work in progress and it will be completed as soon as possible]

15.4 Architectural Design Patterns

Architectural patterns are a category of design patterns that provide reusable solutions to recurring problems in creating software systems and structures. They can address problems like minimizing a defined risk or dealing with

performance limitations, for instance.

15.4.1 Service Locator

The service locator pattern is very useful when you need to manage connections between objects, usually at runtime: instead of hard-coding each reference of each service, you have a “central hub” that you can ask such connections to.

This allows you to swap out (or modify) individual parts of your program without affecting the rest of your software, which can prove useful for testing.

[This section is a work in progress and it will be completed as soon as possible]

16 Useful Containers and Classes

> Eliminate effects between unrelated things. Design components that are self-contained, independent, and have a single, well-defined purpose.
>
> *Anonymous*

In this chapter we will introduce some useful data containers and classes that could help you solve some issues or increase your game's maintainability and flexibility.

16.1 Resource Manager

A useful container is the "resource manager", which can be used to store and manage textures, fonts, sounds and music easily and efficiently.

A resource manager is usually implemented via generic programming, which helps writing DRY code, and uses search-efficient containers like hash tables, since we can take care of loading and deletion during loading screens.

First of all, we need to know how we want to identify our resource; there are many possibilities:

- **An Enum:** this is usually implemented at a language-level as an "integer with a name", it's light but every time we add a new resource to our game, we will need to update the enum too;
- **The file path:** this is an approach used to make things "more transparent", but every time a resource changes place, we will need to update the code that refers to such resource too;
- **A mnemonic name:** this allows us to use a special string to get a certain resource (for instance `skeleton_spritesheet`), and every time our resource folder changes, we will just need to update our loading routines (similarly to the Enum solution).

Secondarily, we need to make sure that the container is **thread-safe** (see more about multi-threading in the multi-threading section), since we will probably need to implement a threaded loading screen (see how to do it here) to avoid our game locking up during resource loading.

[This section is a work in progress and it will be completed as soon as possible]

16.2 Animator

This can be a really useful component to encapsulate everything that concerns animation into a simple and reusable package.

The animation component will just be updated (like the other components) and it will automatically update the frame of animation according to an internal timer, usually by updating the coordinates of the rectangle that defines which piece of a sprite sheet is drawn.

[This section is a work in progress and it will be completed as soon as possible]

16.3 Finite State Machine

A finite state machine is a model of computation that represents an abstract machine with a finite number of possible states but where one (or a finite number) of states can be "in execution" at a given time.

We can use a finite state machine to represent the status of a player character, like in the following diagram:

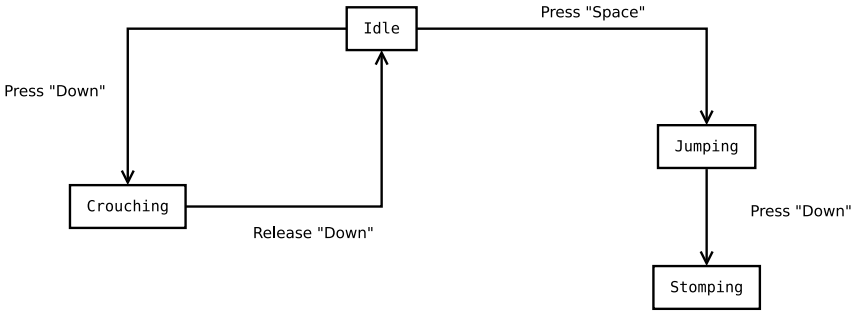


Figure 334: Diagram of a character's state machine

Each state machine is made out of two main elements:

- **states** which define a certain state of the system (for the diagram above, the states are: Idle, Crouching, Jumping and Stomping);
- **transitions** which define a condition and the change of state of the machine (for the diagram above there are two "Press Down" transitions, one "Release Down" and one "Press Space")

State machines are really flexible and can be used to represent a menu system, for instance:

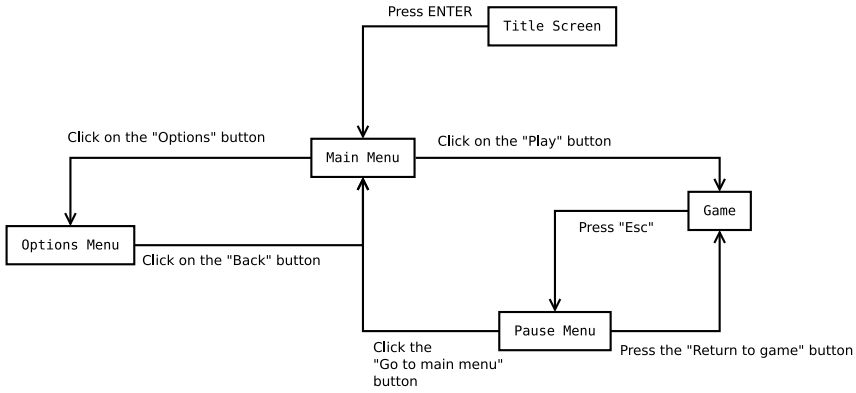


Figure 335: Diagram of a menu system's state machine

In this more convoluted diagram we can see how pressing a certain button or clicking a certain option can trigger a state change.

Each state can be created so it has its own member variables and methods: in a menu system it can prove useful to have each state have its own `update(dt)` and `draw()` functions to be called from the main game loop, to improve on the code readability and better usage of the nutshell programming principle.

Listing 88: A simple finite state machine

```
class FSM{
    /*
     * This class defines a Finite State Machine
     * The currently active state is represented by a function
     * pointer
     */
    constructor(){
        this.current_state = null;
    }

    setState(f){
        /*
         * Sets the state, from this point on, update will
         * change its strategy
         */
        this.current_state = f;
    }

    update(dt){
        // If there is a current state, execute it
        if (this.current_state != null){
            this.current_state(dt);
        }
    }
}
```

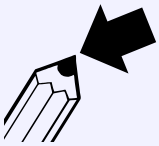
The finite state machine can be used as a brain inside of anything, and use functions to commutate how things react to external events:

Listing 89: An example of usage of a FSM

```
class Enemy{
    /*
     * Represents a simple enemy
     */
    // ...

    constructor(x, y){
        this.position_x = x;
        this.position_y = y;
        PURSUETIME = 10.0;
        pursue_timer = new Timer();
        brain = new FSM();
        this.brain.setState(this.patrol);
    }

    sees(other){
        /*
```

```
18          * Checks if the enemy sees an entity
19          */
20         // ...
21     }
22
23     patrol(dt){
24         // Normal patrolling of the enemy
25         // Move, turn, path find...
26         if (this.sees(player)){
27             // ...
28             // Pursue for xx seconds
29             this.pursue_timer.set(this.PURSUETIME);
30             this.pursue_timer.start();
31             // Change FSM State
32             this.brain.setState(this.pursue);
33         }
34     }
35
36     pursue(dt){
37         // Tries to pursue the enemy
38         if (this.sees(player)){
39             // Continue Pursuing, by resetting the timer
40             this.pursue_timer.set(this.PURSUETIME);
41             // ...
42         }
43         // ...
44         // If the enemy is not in sight for xx seconds
45         if (this.pursue_timer.is_finished()){
46             // go back to patrolling
47             this.brain.setState(this.patrol);
48         }
49     }
50
51     update(dt){
52         // The enemy update function
53         // ...
54         this.pursue_timer.update(dt);
55         this.brain.update(dt);
56         // ...
57     }
58 }
```

Tip!



If, for some reason, you want even more abstraction and separation, you can use the basics of a strategy pattern to implement your Finite State Machine.

16.4 Menu Stack

Although menus can be represented via a finite state machine, the structure of an User Interface (UI) is better suited for another data model: the stack (or rather, something similar to a stack).

A stack allows us to code some other functions in an easier way, for instance we can code the "previous menu" function by just popping the current menu out of the stack; when we access a new menu, we just push it into the menu stack and the menu stack will take care of the rest.

Unlike the stacks we are used to, the menu stack can also be accessed like a queue (first in - first out) so you can draw menus and dialog windows on top of each other, while the last UI element (on top of the stack) keeps the control of the input-update-draw cycle.

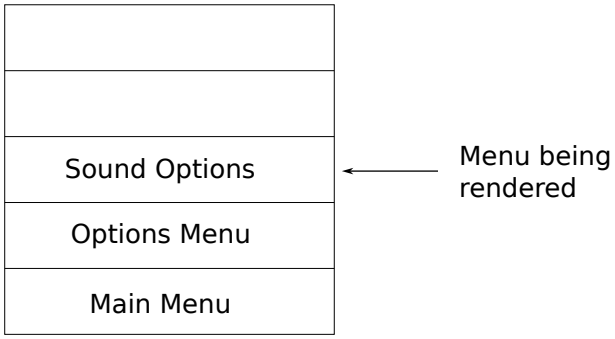


Figure 336: Example of a simple menu stack

In the menu stack we also have some functionalities that may not be included in a standard stack, like a "clear" function, which allows us to completely clean the stack: this can prove useful when we are accessing the main game, since we may not want to render the menu "below" the main game, wasting precious CPU cycles.

[This section is a work in progress and it will be completed as soon as possible]

16.5 Particle Systems

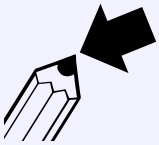
Many special effects in games (like fire, dust and even falling leaves) are done using particle systems. Particle systems consist mainly of 2 parts: the particles themselves and the emitters. This is where we will be focusing.

16.5.1 Particles

The basic building block of our particle system is the particle: which can represent a speck of dust, a puff of smoke or a single leaf.

Figure 337: Some examples of particles

Tip!



Particles are a great candidate for a flyweight pattern. They usually share the same texture, which is the heaviest component they have.

Let's see an example of a particle class:

Listing 90: A simple particle class

```
class Particle{
    /*
     * This is a simple particle class, it contains a reference to
     * its texture, as well as some state
     */
    constructor(texture, position, velocity, acceleration, lifespan = 2000){
        // We prepare the particle for usage
        this.texture = texture;
        this.position = position;
        this.velocity = velocity;
        this.acceleration = acceleration;
        this.lifespan = lifespan;  // About 2 seconds by default
    }

    update(dt){
        // We update the velocity (assuming dt is in milliseconds)
        this.velocity = this.velocity + this.acceleration;
        // Then the position
        this.position = this.position + this.velocity * dt;
        // Now we update the lifespan of the particle;
        this.lifespan = this.lifespan - dt;
    }

    is_dead(){
        // Returns a boolean representing if the particle is dead
        return this.lifespan <= 0;
    }

    setPosition(position){
        // Sets the particle position
        this.position = position;
    }
}
```

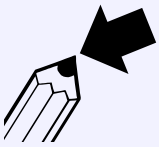
Particle can be even more complex: for instance you could add a variable to track the particle's rotation (useful for falling leaves). Also if you want to take care of resetting the particle status (for recycling), you may need to memorize the initial status in more variables, as well as include some more utility methods.

Listing 91: A more complex particle class

```
class Particle{
    /*
     * This is a simple particle class, it contains a reference to
     * its texture, as well as some state
     */

    constructor(texture, position, velocity, acceleration, lifespan = 2000, rotation = 0, a_vel = 0, a_accel = 0){
        // We prepare the particle for usage
        // The particle texture
        this.texture = texture;
        // Position, velocity and acceleration on the 2D plane
        this.position = position;
        this.velocity = velocity;
        this.acceleration = acceleration;
        // Lifespan of the particle
        this.lifespan = lifespan;  // About 2 seconds by default
        // Current angle of rotation, and relative velocity and acceleration
        this.rotation = rotation;
        this.angular_velocity = a_vel;
        this.angular_acceleration = a_accel;
        // We also prepare the reset variables: the position will be set by the emitter
        this.initial_lifespan = lifespan;
        this.initial_velocity = velocity;
        this.initial_a_vel = a_vel;
        this.initial_rotation = rotation;
    }

    update(dt){
        // We update the velocity (assuming dt is in milliseconds)
        this.velocity = this.velocity + this.acceleration;
        // Then the position
        this.position = this.position + this.velocity * dt;
        // Then the rotation
        this.angular_velocity = this.angular_velocity + this.angular_acceleration ;
        this.rotation = (this.rotation  + this.angular_velocity * dt) % 360;  // Wrap to zero when at 360 degrees
        // Now we update the lifespan of the particle;
        this.lifespan = this.lifespan - dt;
    }

    is_dead(){
        // Returns a boolean representing if the particle is dead
        return this.lifespan <= 0;
    }

    reset(){
        // This function resets the initial status of the particle
        this.velocity = this.initial_velocity;
```

```
48          this.rotation = this.initial_rotation;
49          this.a_vel = this.initial_a_vel;
50          this.lifespan = this.initial_lifespan;
51      }
52  }
```

Tip!



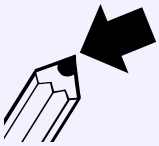
If you want your particles to behave in a more “randomized way”, you may want to save less “initial status variables” and delegate more of the “reset logic” to the particle system.

16.5.2 Emitters

A particle emitter represents a spot that “emits particles”, it instantiates a list of particles and defines their initial settings. An emitter usually takes care of ignoring (not rendering) “dead particles” (the ones whose lifespan is over) as well as the ancillary cleanup.

Emitters can emit a stream of particles, as well as just a burst of them, or even both: it all depends on what you want your system to do.

Tip!



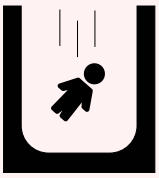
Don’t think that you need a lot of particles to make a convincing effect: a well-coded particle emitter can make a convincing (or at least enjoyable) effect with 5 to 10 particles per emitter.

Listing 92: A simple particle emitter class

```
class Emitter{
    /*
     * This is a simple particle emitter, it contains a list
     * of particles and it updates and manages them
     */

    constructor(location){
        this.origin = location;
        this.particles = new Array(8);  // We prepare 8 spaces for particles
    }

    update(dt){
        // Update the entire system, by updating each particle
        for (const particle of this.particles){
            if (!particle.is_dead()){
                particle.update(dt);
            }
        }
```

```
    }
}
```

Pitfall Warning!



Try to avoid destroying “dead particles” and instantiate new ones every time, the continuous creation and destruction of classes will slow down your game considerably. It’s much better to instantiate all the needed particles and “recycle” the dead ones as necessary, by resetting their properties.

If you want to recycle particles, your emitter will be a little more complex.

Listing 93: A more complex (and complete) particle emitter class

```
class Emitter{
    /*
     * This is a simple particle emitter, it contains a list
     * of particles and it updates and manages them
     */

    constructor(location, one_shot = false){
        this.origin = location;
        this.particles = new Array(8);  // We prepare 8 spaces for particles
        // Defines if this emitter streams continuously or only a burst of particles
        this.one_shot = one_shot;
    }

    update(dt){
        // Update the entire system, by updating each particle
        for (const particle of this.particles){
            if (this.one_shot){
                if (particle.is_dead()){
                    continue;
                }else{
                    particle.update(dt);
                }
            }else{
                if (particle.is_dead()){
                    particle.reset();  // Resets the state of the particle
                    particle.setPosition(this.origin);
                }
                particle.update(dt);
            }
        }
    }
}
```

16.5.3 Force Application

Until now we've had constant acceleration applied to our particles, without keeping track of one of the most basic principles of physics: Newton's second law of motion.

As a reminder, here's the formula:

$$force = mass \cdot acceleration$$

To make it useful for our purposes, we will have "force" as the input of our function, while acceleration is its output, so:

$$acceleration = \frac{force}{mass}$$

This way we can influence the acceleration of a particle by using a force (represented as a vector), with a function similar to the following:

Listing 94: A particle with mass and force application

```
class Particle{
    /*
     * This is a simple particle class, now it has some mass
     * and a force application function
     */
    // ...

    constructor(texture, position, velocity, acceleration, lifespan = 2000, rotation = 0, a_vel = 0, a_accel = 0, mass = 1){
        // We prepare the particle for usage the same way as earlier
        // ...
        this.mass = mass;
    }

    // ...

    applyForce(force){
        // This function influences the acceleration by applying force
        let da = force / this.mass;
        this.acceleration = this.acceleration + da;
    }
}
```

This way, you can change how a particle behaves: for instance, you can apply a lateral force to a falling leaf to simulate wind. This way the falling leaves will not seem boring.

16.6 Timers

Timers are an essential component in many game mechanics: Coyote Time and Jump Buffering are two prime examples of timer-based mechanics.

If we want to execute a function every few seconds we need timers.

If we need to cap how many bullets we can shoot from a weapon, guess what? Timers.

Making a timer is not as complicated as it may seem, we need:

- A variable that keeps track of the time passed;
- A variable that keeps track of how much time needs to pass before the function gets executed;
- A pointer to said function;
- A boolean to track whether the timer is active or not;
- A boolean to decide whether the timer should be “one shot” or “continuous”.

Listing 95: A simple timer class

```
class Timer{
    /*
     * This is a simple timer class that executes a function after
     * a certain amount of time
     */
    constructor(){
        this.time = 0;
        this.set_time;
        function_to_execute = function(){};
        this.one_shot = false;
        this.active = true;
    }

    constructor(time, func, one_shot, active){
        // We prepare the timer and memorize the setting
        this.time = time;
        this.set_time = time;
        // The function pointer should already be prepared with the arguments
        this.function_to_execute = func;
        // Is this timer one-shot then disable?
        this.one_shot = one_shot;
        if (one_shot == undefined){
            this.one_shot = false;
        }
        // Does this timer need to be active when constructed?
        this.active = active;
        if (active == undefined){
            this.active = false;
        }
    }

```

```
32      update(dt){
33          if (!this.active){
34              // We return directly if the timer is disabled
35              return;
36          }
37          // Like any other entity, we update it
38          this.time = this.time - dt;
39          // When the timer "ticks", we execute the function
40          if (this.time <= 0){
41              this.function_to_execute();
42              if (this.one_shot){
43                  // If this timer is a one-shot, we disable it
44                  this.active = False;
45              }
46              // We reset the timer (we may need to re-activate it manually later)
47              this.time = this.set_time;
48          }
49      }
50  }
```

16.6.1 Accounting for “leftover time”

This timer is a nice and simple solution, but it has a small flaw: when the timer is set to execute continuously and the function is executed, it doesn’t account for “leftover time”. This may be easier to understand with an example.

Let’s imagine that we have a timer that shoots a bullet every quarter of a second (250ms), our game is running at a steady 30fps (which means each frame takes 33.33ms), our timer’s internal counter will behave like this:

- **Frame 1:** 250 - 33.33 = 216.67
- **Frame 2:** 216.67 - 33.33 = 183.34
- ...
- **Frame 7:** 50.20 - 33.33 = 16.87
- **Frame 8:** 16.87 - 33.33 = -16.46 (Trigger the function and reset the timer)
- **Frame 9:** 250 - 33.33 = 216.67

Our timer doesn’t account for the over 16ms leftover that we had between frame 8 and frame 9, thus our timer will be imprecise. This may seem an easy fix at first glance, but it is not.

16.6.1.1 A naive solution

The first solution that would come to mind would be substituting the timer variable assignment with a sum, thus frames 7,8 and 9 would look like this:

- **Frame 7:** 50.20 - 33.33 = 16.87
- **Frame 8:** 16.87 - 33.33 = -16.46 (Trigger the function and reset the timer, by adding 250)
- **Frame 9:** 233.54 - 33.33 = 200.21

Here is the code:

Listing 96: A naive approach to account for leftover time

```
class Timer{
    /*
     * ...
     * This is the same as the older version
     * ...
     */

    update(dt){
        if (!this.active){
            // We return directly if the timer is disabled
            return;
        }
        // Like any other entity, we update it
        this.time = this.time - dt;
        // When the timer "ticks", we execute the function
        if (this.time <= 0){
            this.function_to_execute();
            if (this.one_shot){
                // If this timer is a one-shot, we disable it
                this.active = False;
            }
            // We reset the timer differently, by adding the "set time"
            this.time = this.time + this.set_time;
        }
    }
}
```

But what happens if we have a sudden lag spike, longer than the timer itself?

On **Frame 7** we have 50.20 - 500 = -449.8, due to a Lag spike: last frame took a lot longer to process, we have to execute our function and reset the timer, adding 250.

On **Frame 8** we have -199.8 - 33.33 = -233.13 : The timer is trying to catch up to the lag spike, since the trigger condition is still happening, we execute the function again and add 250 to the timer.

On **Frame 9** we have 16.87 - 33.33 = -16.46 We've almost caught up with the lag spike, but we need to execute the function a third time and add 250 to our timer.

Frame 10 behaves normally with 233.54 - 33.33 = 200.21.

Due to the lag spike the function gets executed three times, which is the technically correct way to "catch up" with the number of times the timer *should have* triggered. But this may be an undesirable side effect.

If our game is already slowing down, executing even more functions won't help, so a better approach would definitely be avoiding calling functions more than we strictly need.

16.6.1.2 A different approach

To avoid this "catching up", there are many ways, I'm going to write two of them in this book. The first is quite simple: we add the set timer in a loop until we reach a value higher than zero.

Listing 97: A possible solution to account for leftover time

```
class Timer{
    /*
     * ...
     * This is the same as the older version
     * ...
     */

    update(dt){
        if (!this.active){
            // We return directly if the timer is disabled
            return;
        }
        // Like any other entity, we update it
        this.time = this.time - dt;
        // When the timer "ticks", we execute the function
        if (this.time <= 0){
            this.function_to_execute();
            if (this.one_shot){
                // If this timer is a one-shot, we disable it
                this.active = False;
            }
            // We reset the timer differently, by adding the "set time" until we have a positive
 value
            while (this.time <= 0){
                this.time = this.time + this.set_time;
            }
        }
    }
}
```

This approach has a very minor issue: we are using a loop, so the further we stray away from zero, the more times we will have to add. A second approach would be calculating a "multiplier" and directly apply that to the added value, thus avoiding a loop.

Listing 98: Another possible solution to account for leftover time

```
class Timer{
    /*
     * ...
     * This is the same as the older version
     * ...
     */

```

```
    update(dt){
        if (!this.active){
            // We return directly if the timer is disabled
            return;
        }
        // Like any other entity, we update it
        this.time = this.time - dt;
        // When the timer "ticks", we execute the function
        if (this.time <= 0){
            this.function_to_execute();
            if (this.one_shot){
                // If this timer is a one-shot, we disable it
                this.active = False;
            }
            // We reset the timer differently, by adding the "set time" with a multiplier
            // this.time is guaranteed to be negative or zero, by dividing by a negative number
            // we have a positive multiplier
            let multiplier = ceil(this.time / -this.set_time);
            this.time = this.time + (multiplier * this.set_time);
        }
    }
}
```

This second approach has an issue too: we will need to calculate the ceiling of a value, which may require a bit more CPU time (although most modern CPUs don't require more than a single cycle to do so).

Both approaches are valid and for longer timers even the "naive" approach is valid and fast. The choice is up to your personal taste and sensibility.

16.7 Inbetweening

Inbetweening, also known as "tweening", is a method that allows to "smear" a value over time, this is usually done with animations, where you set the beginning and end position of a certain object, as well as the time the movement should take, and let the program take care of the animation.

This is particularly useful in animating $UI_{[g]}$ objects, to give a more refined feel to the game.

Here we will present some simple tweenings that can be programmed, and explain them.

Let's start with a *linear* tweening, usually the following function is used:

Listing 99: Linear Tweening

```
function linearTween(time, begin, change, duration){
    return change * (time / duration) + begin;
}
```

Let's explain the variables used:

- **time**: The current time of the tween. This can be any unit (frames, seconds, steps, ...), as long as it is the same unit as the "duration" variable;
- **begin**: represents the beginning value of the property being inbetweened;
- **change**: represents the change between the beginning and destination value of the property;
- **duration**: represents the duration of the tween.

Note!

The measure (time / duration) represents the "percentage of completion" of the tweening.

In some cases a Linear tweening is not enough, that's where *easing* comes into play.

Before introducing easing let's analyze the function again, if you try plugging in some data into the function, you will find that there is always going to be:

$$(change\ in\ property) \cdot (factor) + (beginning\ value)$$

So we can use our function substituting `begin` with 0 and `change` with 1 to calculate `factor` and have a code similar to this one:

Listing 100: Example of a simple easing function

```
// We calculate the tweening factor
let factor = linearTween(time, 0, 1, duration);
// Now we change the property we're tweening
object.property = property_original_value + (destination_value - property_original_value) *
    factor;
```

With linear tweening, the function degenerates to $\frac{time}{duration}$, but now we can replace our linear tween with the following function:

Listing 101: Ease-in

```
function easeIn(time, duration, power){
    return (time/duration) ** power;
}
```

By changing the `power` parameter, we change the behaviour of the easing, making the movement slower at the beginning and pick up the pace more and more, until the destination is reached. This is called a "ease-in".

For an "ease-out", where the animation starts fast and slows down towards the end, we use the following function instead:

Listing 102: Ease-out

```
function easeOut(time, duration, power){
    return 1 - (1 - (time / duration)) ** power;
}
```

With some calculations, and if statements on the time passed, you can combine the two and get an “ease-in-out” function.

Listing 103: Ease-in-out

```
function easeInOut(time, duration, power){
    let threshold = duration / 2;
    if (time <= threshold){
        return easeIn(time, duration, power);
    }else{
        return easeOut(time,duration, power);
    }
}
```

Obviously these functions have an issue: they don’t clamp the value between 0 and 1, that will have to be done in the movement function or by adding a check, or using some math, for instance using `min(calculated_value, 1)`.

Listing 104: Clamping values

```
function clamp(value, min, max){
    // Clamps "value" so it is always between "min" and "max"
    if (value < min){
        return min;
    }
    if (value > max){
        return max;
    }
    return value;
}
```

In that case, calling the clamping function with values `0` and `1` would solve the issue.

A think that many people tend to forget, but that is really important is that you can tween any property of any entity or object in your game, for instance:

- The position of a UI item;
- The width of a health bar;
- The rotation of an on-screen map or compass;
- The colour of the sky while doing a day-to-night transition.

In short: any numeric value that can transition “smoothly” between two values in a certain amount of time can be tweened.

16.7.1 Bouncing

Easing is a great way to make more "natural looking" transitions, but in nature nothing is absolutely precise: we may want to overshoot a bit and correct it so it looks a bit like a "bounce".

The idea behind this is the following: when around 80% of the time has passed, our property should have overshot its maximum value by something between 10% and 20% (more would look unnatural). The remaining 20% of the time will be used to "ease back" to the final value. This will give us the bouncy feeling we want.

Here is a quick example of a function that will help us create a bounce effect: the `t` variable represents the time of the animation that has passed, with 0 being the beginning and 1 being the end of the animation itself.

Listing 105: A bouncy tween function

```
function bounce_tween(t){
    // This constant will allow us to overshoot the max value by around 10%
    const c = 1.70158;

    return 1 + (c + 1) * (t - 1) ** 3 + c * (t - 1) ** 2;
}
```

16.8 Chaining

[This section is a work in progress and it will be completed as soon as possible]

17 Artificial Intelligence in Videogames

> The question of whether a computer can think is no more interesting than the question of whether a submarine can swim.
>
> *Edsger W. Dijkstra*

In this part of the book we will take a look at some data structures and algorithms that will help you building your game's Artificial Intelligence.

17.1 Path Finding

Path Finding is that part of AI algorithms that takes care of getting from point A to point B, using the shortest way possible and avoiding obstacles.

17.1.1 Representing our world

Before undertaking the concept of path finding algorithms, we need to decide in which way we should represent our world to the AI we want to code. There are different ways to do it and here we will discuss some of the most used ones.

17.1.1.1 2D Grids

The simplest way to represent a world to an AI is probably using 2D grids: we can represent the world using a 2-dimensional matrix, where every cell is a free space, an obstacle, a start or goal cell.

This representation works well with top-down tile-based 2D games (with tile-based or free movement).

Listing 106: Possible representation of a 2D grid

```
class Grid_2D{
    // Represents a 2D grid of "Tile" classes

    constructor(rows, cols){
        // Prepares the memory for the grid
        this.grid = Array.from(Array(rows), () => new Array(cols));
        this.height = rows;
        this.width = cols;
    }

    getCell(row, col){
        // Gets a cell from the 2D Grid
        if ((row >= 0 && row < height) && (col >=0 && col < width)){
            // We better check if we are inside the grid
            return grid[row][col];
        }else{
            return null;
        }
    }

```

```
    getAdjacentCells(row, col){
        /* Returns a list of cells adjacent the ones we give
        REMEMBER: We index at 0 so the first row is 0, the last one is at
        "height - 1", same goes for columns */
        let toReturn = [];
        if ((row >= 0 && row < height) && (col >=0 && col < width)){
            // We better check if we are inside the grid
            if (row > 0){
                // We are not on the first row, we can add the cell above
                toReturn.push(this.getCell(row - 1, col));
            }
            if (row < height - 1){
                // We are not on the last row, we can add the cell below
                toReturn.push(this.getCell(row + 1, col));
            }
            if (col > 0){
                // We are not on the first column, we can add the cell on the left
                toReturn.push(this.getCell(row, col - 1));
            }
            if (col < width - 1){
                // We are not on the last column, we can add the cell on the right
                toReturn.push(this.getCell(row, col + 1));
            }
        }
        /* If the checks went well, toReturn will have
        a list of the adjacent cells, if not it will be empty */
        return toReturn;
    }
}
```

Even though this is probably the most straightforward way to represent a world in many cases, most of the algorithms used work on a graph structure instead of a 2D grid. It shouldn't be too hard to adapt the algorithms presented here to work with this structure.

17.1.1.2 Path nodes

A more flexible way to represent our world is using "Path nodes", where each "path node" is represented by a node in a graph.

This type of graph-based abstraction is the most used when teaching path finding algorithms like `A*` or `Dijkstra`.

You can see more about graphs in the dedicated section in the Data Structures section.

17.1.1.3 Navigation meshes

Navigation meshes are used to solve a problem that can arise when we try to represent our world using path nodes: we can't represent "safe areas" (where the AI-driven entity can cross) without using possibly thousands of path nodes.

Navigation meshes are constituted by a collection of convex polygons (the meshes) that define the “safe areas”, each mesh has no obstructions inside of itself, so the AI-driven entity can safely cross the entire mesh freely.

This abstraction allows to use `A*` and `Dijkstra` algorithms, but instead of trying to navigate a graph, you look for a path between meshes (which are saved in a graph structure).

To make the abstraction easier to understand, let’s take a look at the following map.

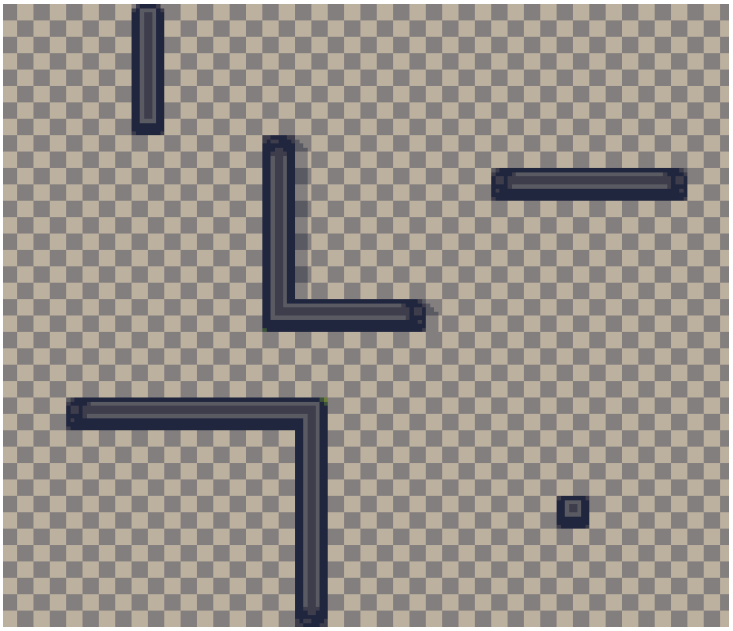


Figure 338: Map we will create a navigation mesh on [8]

The first step is to divide the map into convex polygons, in our case we will use rectangles.

[8] Jawbreaker tileset, listed as public domain at https://adamatomic.itch.io/jawbreaker

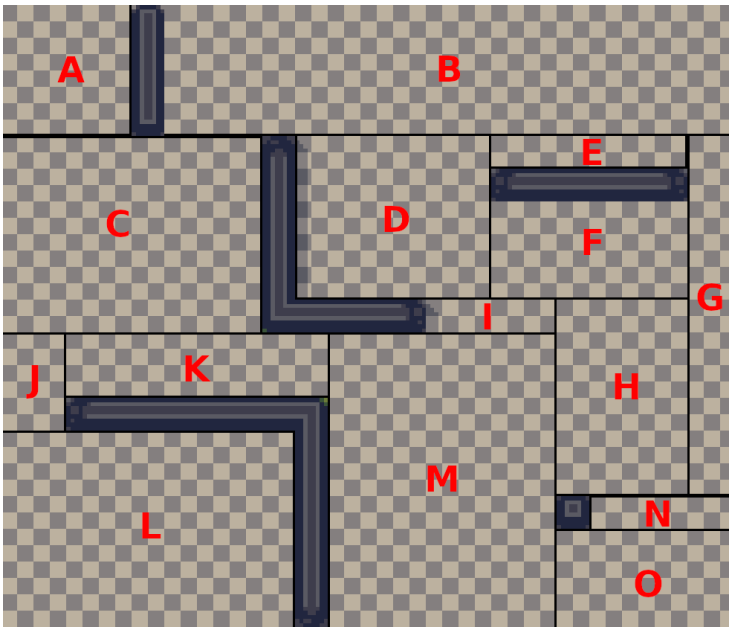


Figure 339: Dividing the map into many convex polygons and labelling them [9]

Now each rectangle is a node of our graph (and will have to be treated accordingly by the AI, knowing it can navigate freely inside each rectangle), now we need to connect each node, following the limitations given by the walls.

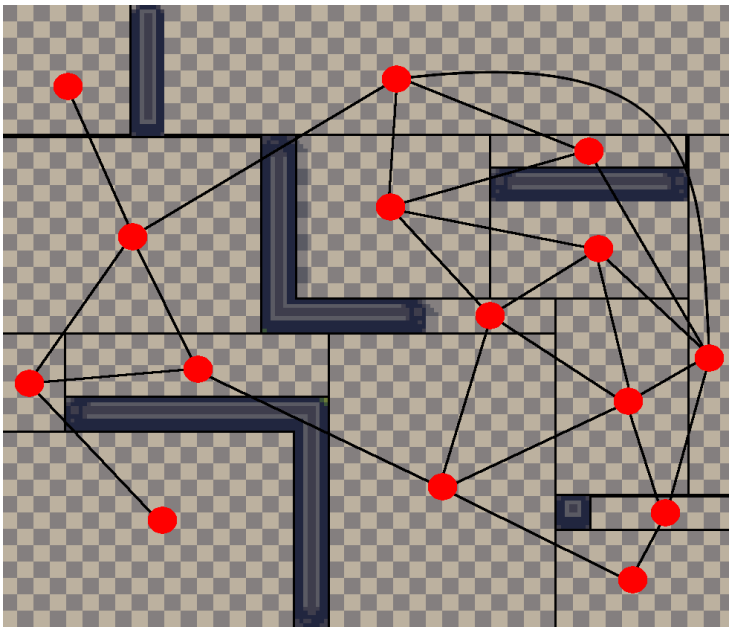


Figure 340: Creating the graph [10]

The graph will be look something like this, as a data structure:

[9] Jawbreaker tileset, listed as public domain at https://adamatomic.itch.io/jawbreaker
[10] Jawbreaker tileset, listed as public domain at https://adamatomic.itch.io/jawbreaker

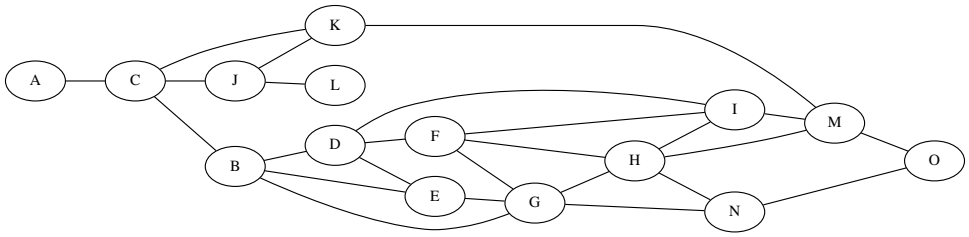


Figure 341: The final data structure

[This section is a work in progress and it will be completed as soon as possible]

17.1.2 Heuristics

In path finding there can be "heuristics" that are accounted for when you have to take a decision: in path finding an heuristic $h(x)$ is an estimated cost to travel from the current node to the goal node.

An heuristic is admissible if it *never overestimates* such cost: if it did, it wouldn't guarantee that the algorithm would find the best path to the goal node.

In this book we will present the most common heuristics used in game development.

17.1.2.1 Manhattan Distance heuristic

The Manhattan Distance heuristic doesn't allow diagonal movement (allowing it would allow the heuristic to overestimate the cost), and for a 2D grid is formulated as follows:

$$h(x) = |start.x - goal.x| + |start.y - goal.y|$$

Graphically:

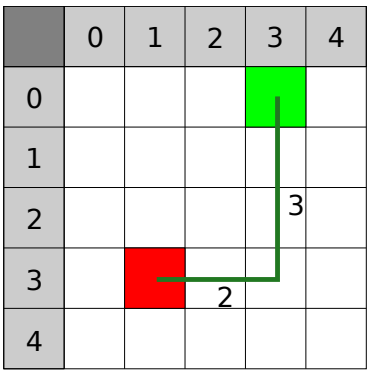


Figure 342: Example of Manhattan distance

This means that if our character moves only left-right and up-down, the Manhattan distance won't overestimate the movement cost, making it admissible.

Listing 107: Example code calculating the Manhattan distance on a 2D grid

```
class Tile{
    constructor(){
        this.x = null;
        this.y = null;
    }
}

function manhattan_distance(start, goal){
    return Math.abs(start.x - goal.x) + Math.abs(start.y - goal.y);
}
```

17.1.2.2 Euclidean Distance heuristic

Euclidean Distance works well when diagonal movement in a 2D grid is allowed, Euclidean distance is calculated with the standard distance formula:

$$h(x) = \sqrt{(start.x - end.x)^2 + (start.y - end.y)^2}$$

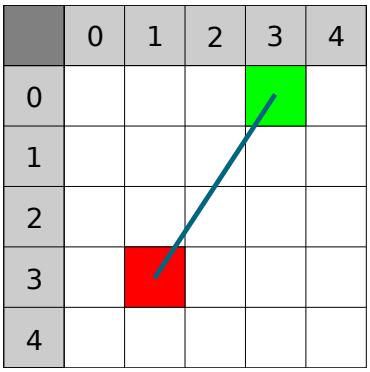


Figure 343: Example of Euclidean Distance

Listing 108: Example code calculating the Euclidean distance on a 2D grid

```
class Tile{
    constructor(){
        this.x = null;
        this.y = null;
    }
}

function euclidean_distance(start, goal){
    return Math.sqrt((start.x - goal.x) ** 2 + (start.y - goal.y) ** 2);
}
```

Differently from the Manhattan distance, the Euclidean distance won't overestimate the cost in any case, and it is usable even if diagonal movement is not happening, but it will be more "computationally intensive" to calculate.

17.1.3 Algorithms

Before getting to the algorithms, we need to consider two supporting data structures that we will use:

- **Open Set:** a sorted data structure that contains the nodes that currently need to be considered. It should be an heap or any kind of structure that can be quickly be sorted as we will have to often refer to the node/cell/mesh with the lowest heuristic.
- **Closed Set:** a data structure that will contain all the nodes/meshes/cells that have already been considered by the algorithm. This structure should be easily searchable (like a binary search tree), since we will often need to see if a certain node/cell/mesh is part of this set.

We will reference this image when we check what path the algorithm will take:

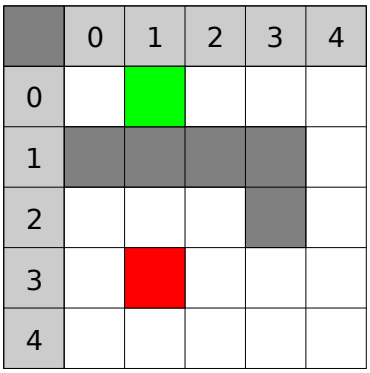


Figure 344: Pathfinding Algorithms Reference Image

The shaded rows represent the indexes we'll refer to when operating the algorithm.

In each algorithm there will be the path taken by its implementation, so we invite you to execute the algorithm's instructions by hand, taking account of the heuristics pre-calculated and shown in the images below.

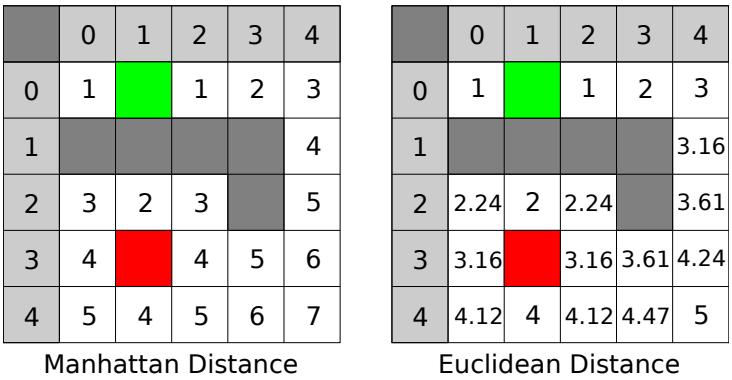


Figure 345: Pathfinding Algorithms Heuristics Reference Image

17.1.3.1 A simple "Wandering" Algorithm

This is not a real "pathfinding" algorithm, as much as something that should give the impression of people "wandering around". This algorithm is really simple and can be summarized in 2 rules:

1. Choose one direction at random, if you can't go that way continue your search clockwise;
2. You cannot go back, unless it's the only direction available.

This is good when used on mazes, let's try an example:

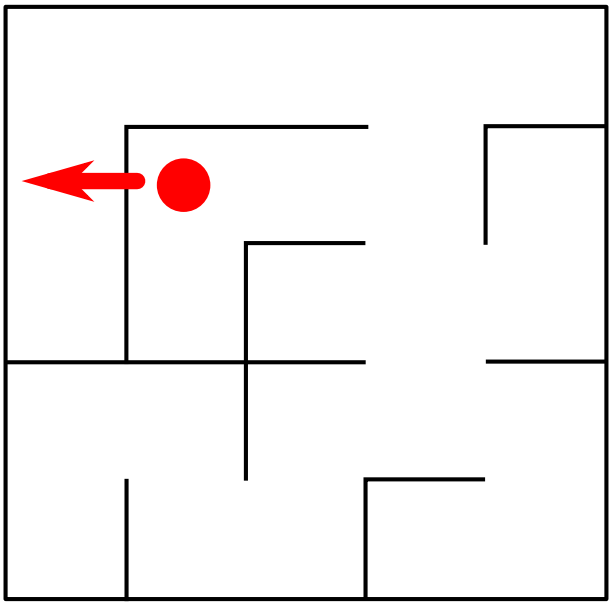


Figure 346: Simple wandering algorithm 1/2

In this case, our random choice is forwards, but that leads us to a wall, so we continue our search clockwise:

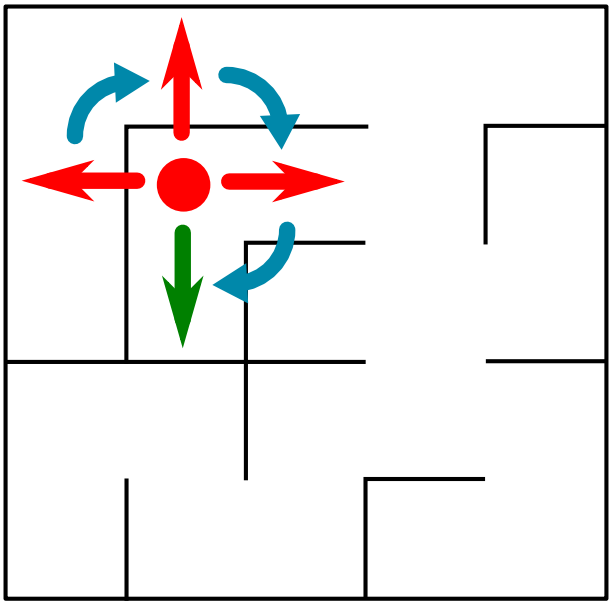


Figure 347: Simple wandering algorithm 2/2

After going clockwise once, we hit another wall, so we continue, this direction is good, but it would lead us backwards, so we ignore it. The only direction possible is left.

This algorithm is far from perfect: if we build a very simple maze, we can break the algorithm:

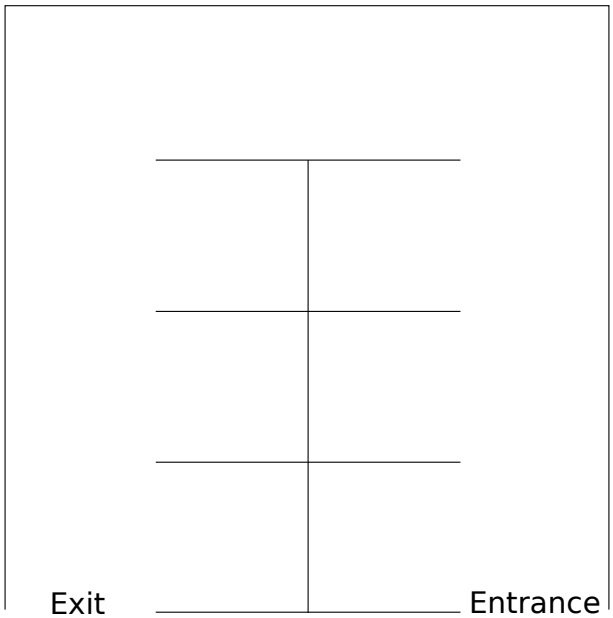


Figure 348: This maze breaks our wandering algorithm

If you calculate the probabilities, you can see that starting from the entrance, the AI has a 75% chance of going into one of those alcoves and only a 25% chance of going forwards.

This means that an entity has only a $25\%^6 = 0.0244\%$ chance of finishing the maze without ever turning left. Not only that, but every time the entity exits an alcove, it has a 50% chance of turning either direction (turning left means going forwards, going right means going backwards), which means that there is a 37.5% chance (75% chance turning left and 50% turning right) that the entity will go back towards the entrance of our maze.

Let's see a possible implementation of this algorithm.

Listing 109: Implementation of a simple wandering algorithm

```
const DIRECTIONS = ["NORTH", "EAST", "SOUTH", "WEST"];

class AIEntity{
    constructor(){
        // 0=North, 1=East, ...
        this.forward_direction_index = 0;
        this.current_cell = [1,0];
    }
    // ...
    get_adjacent_cell(cell, direction){
        // Returns the adjacent cell in said direction
        let cell_copy = cell.copy();
```

```
        if (direction == "NORTH"){
            cell_copy.y -= 1;
        }
        if (direction == "SOUTH"){
            cell_copy.y += 1;
        }
        if (direction == "WEST"){
            cell_copy.x -= 1;
        }
        if (direction == "EAST"){
            cell_copy.x += 1;
        }
        return cell_copy;
    }

    is_valid(cell){
        /* Returns true if the cell is valid, aka
         * does not have a wall and does not go backwards */
        if (cell.is_wall()){
            // The cell is a wall
            return False;
        }
        if (cell == this.get_adjacent_cell(this.current_cell, DIRECTIONS[(
    forward_direction_index + 2) % 4])){
            // We're going backwards, we don't want that
            return False;
        }
        // In all other cases, it's valid
        return True;
    }

    update(dt){
        // ...
        // Choose a random direction
        let chosen_direction = DIRECTIONS[Math.random() * DIRECTIONS.length];
        let i = 0;
        let next_cell = this.get_adjacent_cell(current_cell, chosen_direction);
        while (!this.is_valid(next_cell) && i != 4){
            chosen_direction = DIRECTIONS[(DIRECTIONS.indexOf(chosen_direction) + 1) % 4];
            next_cell = this.get_adjacent_cell(current_cell, chosen_direction);
            i = i + 1;
        }
        if (i == 4){
            // We exhausted the possibilities, go backwards
            chosen_direction = DIRECTIONS[(forward_direction_index + 2) % 4];
            next_cell = this.get_adjacent_cell(current_cell, chosen_direction);
        }
        // Move
        this.move_to(next_cell);
    }
```

```
62 }
```

17.1.3.2 A slightly better “Wandering” algorithm

As we have seen, the previous wandering algorithm has a very heavy bias, let’s plan for another algorithm that works a bit better and has a lower bias.

1. Check for all valid directions around you.
2. If no valid direction is found, go back.
3. If at least one valid direction is found, choose a random one between the valid directions found.

This algorithm relies on the fact that randomly selecting an item from a list containing a single item will return that single item in 100% of the cases.

Let’s see a code implementation of such algorithm.

Listing 110: Implementation of a better wandering algorithm

```
1  const DIRECTIONS = ["NORTH", "EAST", "SOUTH", "WEST"];
2
3  class AIEntity{
4      constructor(){
5          // 0=North, 1=East, ...
6          this.forward_direction_index = 0;
7          this.current_cell = [1,0];
8      }
9      // ...
10     get_adjacent_cell(cell, direction){
11         // Returns the adjacent cell in said direction
12         let cell_copy = cell.copy();
13         if (direction == "NORTH"){
14             cell_copy.y -= 1;
15         }
16         if (direction == "SOUTH"){
17             cell_copy.y += 1;
18         }
19         if (direction == "WEST"){
20             cell_copy.x -= 1;
21         }
22         if (direction == "EAST"){
23             cell_copy.x += 1;
24         }
25         return cell_copy;
26     }
27
28     is_valid(cell){
29         /* Returns true if the cell is valid, aka
30          * does not have a wall and does not go backwards */
31         if (cell.is_wall()){
```

```
32              // The cell is a wall
33              return False;
34          }
35          if (cell == this.get_adjacent_cell(this.current_cell, DIRECTIONS[(
        forward_direction_index + 2) % 4])){
36              // We're going backwards, we don't want that
37              return False;
38          }
39          // In all other cases, it's valid
40          return True;
41      }
42
43      get_available_directions(cell){
44          /* Returns a list of available directions */
45          let result = [];
46          for (const direction of DIRECTIONS){
47              if (is_valid(this.get_adjacent_cell(cell, direction))){
48                  result.push(direction);
49              }
50          }
51          return result;
52      }
53
54      update(dt){
55          // ...
56          // Get a list of the available directions
57          let available_directions = this.get_available_directions(current_cell);
58          let chosen_direction = "NORTH";  // Just a default
59          if (available_directions.length === 0){
60              // No directions are available, let's go back
61              chosen_direction = DIRECTIONS[(forward_direction_index + 2) % 4];
62          }else{
63              // Choose a random direction among the available ones
64              chosen_direction = available_directions[Math.random() * available_directions.length
        ];
65          }
66          // Move
67          next_cell = this.get_adjacent_cell(current_cell, chosen_direction);
68          this.move_to(next_cell);
69      }
70  }
```

This algorithm chooses between all available directions with the same probability, and has a minor bias towards going to the maze entrance.

17.1.3.3 The Greedy “Best First” Algorithm

This is a *greedy algorithm*[g] that searches the “local best” (what is best in a certain moment, without planning future decisions) that makes use of heuristics.

For each of the neighbouring cells/meshes/nodes that have not been explored yet, the algorithm will take the one that has the lowest heuristic cost. Since this algorithm doesn't make any planning, this can lead to results that are not optimal, usually translating in entities hugging walls to reach their goal, as well as taking longer paths.

In this algorithm we will use a "Node" structure composed as follows:

Listing 111: The node structure used in the greedy "Best First" algorithm

```
class Node{
    constructor(){
        this.parent = null;  // This will be used to build the path
        this.h = null;  // The h(x) value for the node
    }
}
```

Let's look at the algorithm itself:

Listing 112: The greedy "Best First" algorithm

```
// We bootstrap the variables
let openSet = [];
let closedSet = [];
let currentNode = start;
closedSet.push(currentNode);  // This will avoid re-analizing this node
do{
    for (const n of currentNode.getAdjacentList()){
        if (closedSet.includes(n)){
            // We already analyzed this node, continue to next n
            continue;
        }else{
            n.parent = currentNode;
            if (!openSet.includes(n)){
                n.h = getHeuristics(n, end);  // Computes the value of n's h(x)
                openSet.push(n);
            }
        }
    }
    if (openSet.length === 0){
        // We exhausted all possibilities
        break;
    }
    // Select a new "currentNode"
    // We order the openset by h, the first element will be the one with the lowest total cost
    openSet.sort((first, second) => {
        if (first.h < second.h){
            return -1;
        }
        if (first.h > second.h){
            return 1;
        }
```

```
32          return 0;
33      });
34      // Shift pops and returns the first element of the array
35      currentNode = openSet.shift();
36      closedSet.push(currentNode);
37  }while (currentNode != end);
38
39  if (currentNode == end){
40      /* We reached the end and solved the path, we need to do a stack
41         reversal to find the path */
42      let finalPath = [];
43      let n = end;
44      while (n != null){
45          finalPath.push(n);
46          n = n.parent;  // We use "parent" to run the found path backwards
47      }
48  }else{
49      // We cannot find a path between "start" and "end"
50      // ... Behave accordingly
51  }
```

An interesting idea to optimize this algorithm and avoid the final “stack reversal” would be to find the path starting from the end node, towards the starting node.

In the image below we can see the path taken by the algorithm, and how it is not the most optimal path.

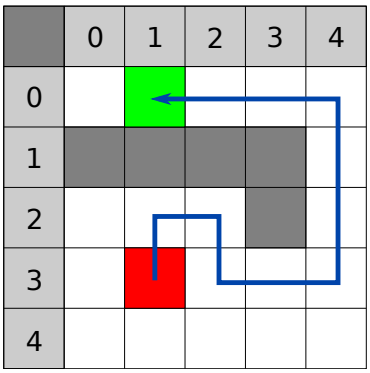


Figure 349: The path taken by the greedy “Best First” algorithm

17.1.3.4 The Dijkstra Algorithm

The idea behind the Dijkstra algorithm is having a “cost” component that expresses the cost that has to be incurred when traveling from the start node to the current node. This will allow our previous algorithm to evolve and take the shortest path possible.

To be able to keep track of such “path-cost” component, we will use a different “Node” structure from the one used in the greedy “best-first” algorithm:

Listing 113: The node structure used in the Dijkstra Algorithm

```
class Node{
    constructor(){
        this.parent = null;  // This will be used to build the path
        this.g = null;  // The path cost value for the node
    }
}
```

The idea behind the whole algorithm is that "if we find a quicker way to get from the start node to the current node, we should take it".

Let's take a look at the algorithm:

Listing 114: The Dijkstra Algorithm

```
let openSet = [];
let closedSet = [];
let currentNode = start;
closedSet.push(currentNode);
do{
    for (const n of currentNode.getAdjacentList()){
        if (closedSet.includes(n)){
            continue;
        }else if (openSet.includes(n)){  // Check if this path is better
            let new_g = getPathCost(n, start);
            if (new_g < n.g){
                // We found a better path from start to currentNode
                n.parent = currentNode;
                n.g = new_g;
            }
        }else{
            n.parent = currentNode;
            n.g = getPathCost(n, start);
            openSet.push(n);
        }
    }

    if (openSet.length === 0){
        // We exhausted all possibilities
        break;
    }
    // We order the openset by g, the first element will be the one with the lowest total cost
    openSet.sort((first, second) => {
        if (first.g < second.g){
            return -1;
        }
        if (first.g > second.g){
            return 1;
        }
```

```
35          return 0;
36      });
37      // Shift pops and returns the first element of the array
38      currentNode = openSet.shift();
39      closedSet.push(currentNode);
40  }while(currentNode != end);
41
42  // Reconstruct the path like in the greedy "best-first" algorithm
```

As with the greedy “best-first” algorithm we can optimize the “stack reversal” stage by starting from the end node.

Below we can see the path taken by the algorithm:

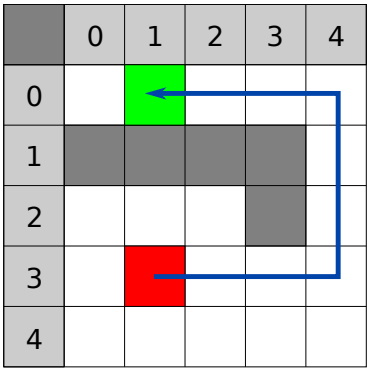


Figure 350: The path taken by the Dijkstra Algorithm

17.1.3.5 The A* Algorithm

The A* Algorithm joins the “path-cost” idea with the heuristic to have a more efficient path-finding algorithm.

The algorithm is really similar to Dijkstra, but it orders the open set by a new formula f, that is calculated as follows:

$$f(x) = g(x) + h(x)$$

Where $g(x)$ is our path cost and $h(x)$ is the heuristic we selected.

Let’s see the code:

Listing 115: The A* Algorithm

```
1  let openSet = [];
2  let closedSet = [];
3  let currentNode = start;
4  closedSet.push(currentNode);
5  do{
6      for (const n of currentNode.getAdjacentList()){
7          if (closedSet.includes(n)){
8              continue;
```

```
        }else if (openSet.includes(n)){  // Check if this path is better
            let new_g = getPathCost(n, start);
            if (new_g < n.g){
                // We found a better path from start to currentNode
                n.parent = currentNode;
                n.g = new_g;
                n.f = n.g + n.h;
            }
        }else{
            n.parent = currentNode;
            n.g = getPathCost(n, start);
            n.h = getHeuristicCost(n, end);
            n.f = n.g + n.h;
            openSet.push(n);
        }
    }

    if (openSet.length === 0){
        // We exhausted all possibilities
        break;
    }
    // We order the openset by f, the first element will be the one with the lowest total cost
    openSet.sort((first, second) => {
        if (first.f < second.f){
            return -1;
        }
        if (first.f > second.f){
            return 1;
        }
        return 0;
    });
    // Shift pops and returns the first element of the array
    currentNode = openSet.shift();
    closedSet.push(currentNode);
}while(currentNode != end);

// Reconstruct the path like in the greedy "best-first" algorithm
```

The path taken by the A* Algorithm is exactly the same as the one taken by the Dijkstra Algorithm, but the heuristic helps the A* Algorithm in visiting less nodes than its counterpart.

17.1.3.5.1 Dijkstra Algorithm as a special case of the A* Algorithm

The Dijkstra Algorithm can be implemented with the same code as the A* Algorithm, just by keeping the heuristic cost $h(x) = 0$.

The absence of the heuristics (which depends on the goal node) leads the Dijkstra Algorithm to visit more nodes, but it can be useful in case there are many valid goal nodes and we don't know which one is the closest.

17.2 Finite state machines

We can use finite state machines, introduced previously, to define some quite complex Artificial Intelligence schemes.

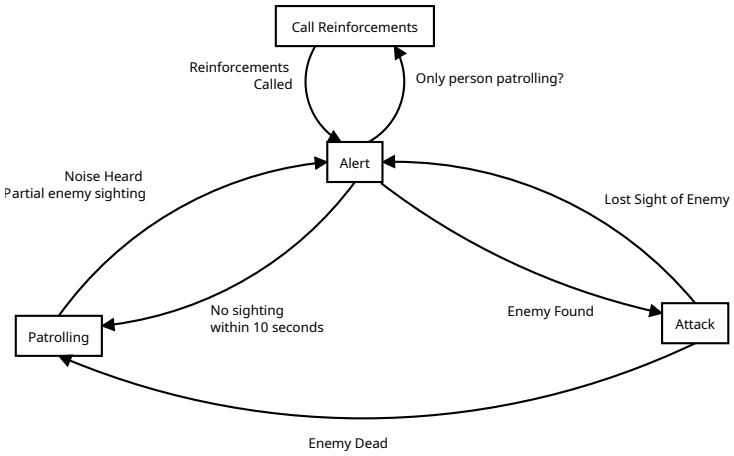


Figure 351: Finite state machine representing an enemy AI

We can see in the previous image how we can use conditions as transition between different "states of mind" of our enemy AI, making it act differently.

The enemy will be patrolling by default, but if the player is heard or seen the enemy will enter its "alert state", where it will either call for backup or actively search for the player. As soon as the player is found, the enemy will attack and try to kill the player.

If you want a refresher on finite state machines, check the finite state machine section some chapters earlier.

17.3 Decision Trees

Decision trees are a structure used to define the decision process of an AI-controlled entity.

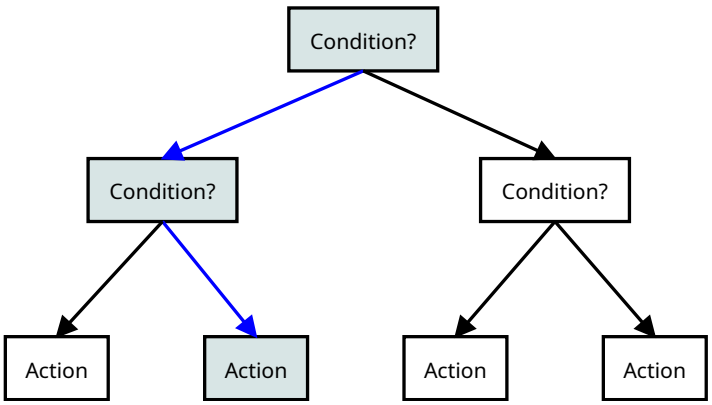


Figure 352: Example of a decision tree

Decision trees are always evaluated from root to leaf, and each node represents a condition that can be more or less complex. In the image above we used a simple "binary tree" to represent conditions that can be answered with

“yes” or “no”.

[This section is a work in progress and it will be completed as soon as possible]

17.4 Behaviour Trees

Similar in structure to Decision Trees, *Behaviour Trees* are different in their evaluation.

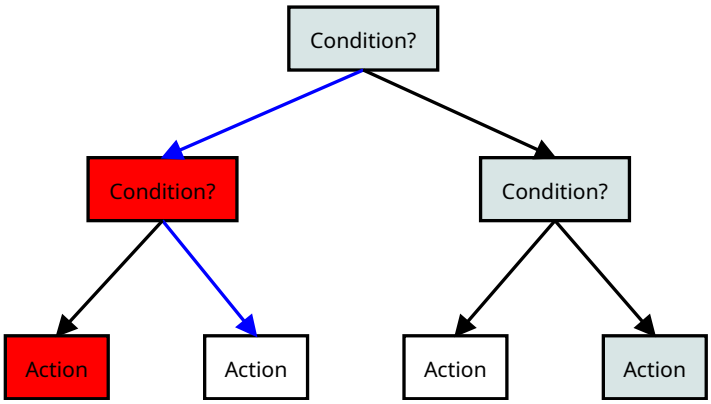


Figure 353: Example of a behaviour tree

First of all, the child nodes of a behaviour tree are ordered by priority, if a child node has all of its conditions met its internal state is changed to “running” and the chosen “behaviour” is returned to the caller.

The next evaluation, the tree is again evaluated, in order, if a “running” node is met, such behaviour will continue to persist for the current frame. In case a “higher priority” behaviour is met, that one behaviour will start running (instead of the behaviour chosen earlier).

In case a node doesn’t have all of its conditions met, the algorithm will return to the parent node, which will chose the next node, in priority order.

[This section is a work in progress and it will be completed as soon as possible]

17.5 Tips and tricks

As with every activity in video game development, the development of AI has some tricks that can make a character seem “smarter” than they actually are.

Again, this involves a fair bit of deception.

17.5.1 “Jump when the player shoots”

A really simple pattern that can be applied to boss fights in 2D games is making the boss jump when the player shoots a projectile: this, in conjunction with the normal boss patterns, makes the boss appear “aware” of the projectiles and is actually trying to dodge them.

This pattern has an obvious drawback: the player can ensure that they hit the boss by jumping themselves before shooting. This can be mitigated by adding a degree of randomness to the boss jumping.

To develop this kind of AI pattern, the boss entity needs to be aware of what the player is doing. This seems the ideal situation for the Observer Design Pattern.

Listing 116: Example code for ”jump when player shoots” AI pattern

```
class Player{
    // ...
    registerShootingObserver(callback){
        // Function used to register an observer that will be called when the
        // player shoots a projectile.
    }
}

class JumpingBoss{
    // A boss that jumps when the player shoots. Sometimes.


    constructor(x, y, player){
        // ...
        this.player_shot = false;
        this.y_velocity = 0.0;
        this.on_ground = false;
        player.registerShootingObserver(this.setPlayerShot);
        // ...
    }

    setPlayerShot(){
        // Sets a state that tells the AI that the player shot a bullet
        this.player_shot = true;
    }

    jump(){
        // Sets the boss velocity to -10, making it jump
        if (this.on_ground){
            this.y_velocity = -10;
        }
    }

    update(dt){
        // ...
        if (this.player_shot == true){
            if (Math.floor(Math.random() * 5) + 1 == 1){
                // Jump only 20% of the times the player shoots
                this.jump();
            }
        }
        // We reset player_shot to false, if we didn't the boss would jump
        // a lot more often than 20% of the time
```

```
44          this.player_shot = false;
45          // ...
46      }
47  }
```

17.5.2 Distance-based patterns

You can influence the AI of an enemy by their distance from the player's position. In the following sections we will use "Boss" to indicate the enemy, for simplicity.

17.5.2.1 "Ranged pattern"

The first pattern in the class of "distance-based" ones will be called the "ranged pattern". That is because this pattern fits well with bosses that use ranged attacks. Since in the real world ranged units have issues with dealing with melee attacks, they will try to "keep their distance".

The main idea is that every frame, the boss will calculate how far the player is (in a AABB fashion, to save on resources): if the player is too close, the boss will try to increase their distance to the player. This will require calculating the direction the boss will have to move and some state variables to tie the thing together.

Listing 117: Example code for a "ranged" AI pattern

```
1  class Boss{
2      constructor(player){
3          this.player = player;
4          this.player_too_close = false;
5          this.base_movement_velocity = 10;
6          this.too_close_space = 20;
7          this.velocity = Vector2D();
8          this.position = Vector2D();
9      }
10
11     update(dt){
12         // ...
13         if (Math.abs(player.position.x - this.position.x) < this.too_close_space){
14             if (Math.abs(player.position.y - this.position.y) < this.too_close_space){
15                 // The player is too close
16                 if (Math.random(1,6) == 1){
17                     // Add a bit of randomization
18                     this.player_too_close = true;
19                 }
20             }
21         }
22         // We're using a variable to preserve the "too close" state between frames
23         if (this.player_too_close){
24             // The player is too close, make some distance
25             let distance = this.position - player.position;
26             // Make it a direction
```

```
            let direction = distance.normalize();
            // This is the direction the boss should go now, transfer it to velocity
            this.velocity = direction * this.base_movement_velocity;
        }
        // ...
        // The boss and player now have moved, let's see if they're far enough
        // ...
        if (Math.abs(player.position.x - this.position.x) > this.too_close_space){
            if (Math.abs(player.position.y - this.position.y) > this.too_close_space){
                // The player is far enough now
                this.player_too_close = false;
            }
        }
    }
}
```

17.5.2.2 “Melee pattern”

The second pattern will be called “melee pattern” and will work exactly the opposite of the “ranged pattern”: if the player is too far, the boss will try to close in. This is ideal when the player uses melee weapons (or the player uses ranged weapons).

The way this will be implemented is the same as the previous pattern, just in reverse.

Listing 118: Example code for a ”melee” AI pattern

```
class Boss{
    constructor(player){
        this.player = player;
        this.player_too_far = false;
        this.base_movement_velocity = 10;
        this.too_far_space = 30;
        this.velocity = Vector2D();
        this.position = Vector2D();
    }

    update(dt){
        // ...
        if (Math.abs(player.position.x - this.position.x) > this.too_far_space){
            if (Math.abs(player.position.y - this.position.y) > this.too_far_space){
                // The player is too close
                if (Math.random(1,6) == 1){
                    // Add a bit of randomization
                    this.player_too_far = true;
                }
            }
        }
        // We're using a variable to preserve the "too far" state between frames
        if (this.player_too_far){
```

```
24              // The player is too far, close in
25              let distance = player.position - this.position;
26              // Make it a direction
27              let direction = distance.normalize();
28              // This is the direction the boss should go now, transfer it to velocity
29              this.velocity = direction * this.base_movement_velocity;
30          }
31          // ...
32          // The boss and player now have moved, let's see if they're close enough
33          // ...
34          if (Math.abs(player.position.x - this.position.x) < this.too_far_space){
35              if (Math.abs(player.position.y - this.position.y) < this.too_far_space){
36                  // The player is close enough now
37                  this.player_too_far = false;
38              }
39          }
40      }
41  }
```

18 Other Useful Algorithms

> Fancy algorithms are slow when n is small, and n is usually small.
>
> *Rob Pike*

18.1 World Generation

18.1.1 Midpoint Displacement Algorithm

As the name implies, this algorithm entails recursively taking the midpoint between two extremes and "displace" it.

Let's see how it works first. We have a completely flat terrain:

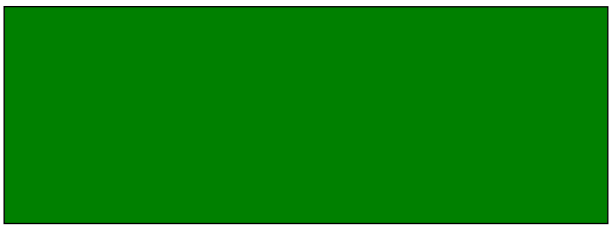


Figure 354: How the Midpoint Displacement Algorithm Works (1/4)

Then we take the midpoint between the two extremes and move it (up or down) by a random amount (between two sensible extremes):

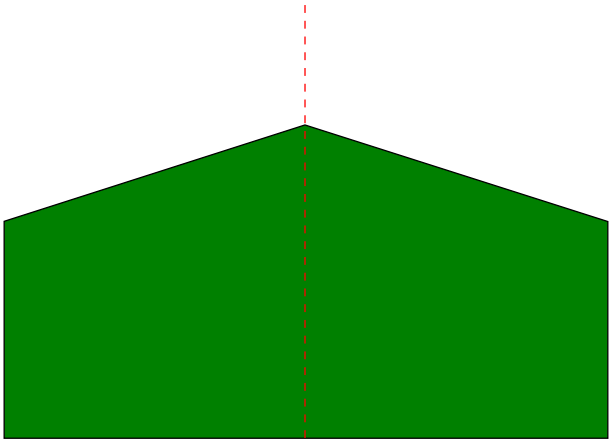


Figure 355: How the Midpoint Displacement Algorithm Works (2/4)

In this case we moved it up by a certain measure, now we take the two midpoints on the left and right sides of the previous midpoint, and displace them too:

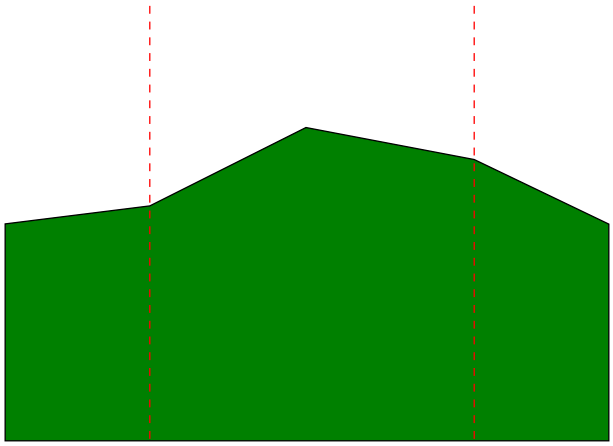


Figure 356: How the Midpoint Displacement Algorithm Works (3/4)

Then we take the midpoints between the segments we created, and displace them again:

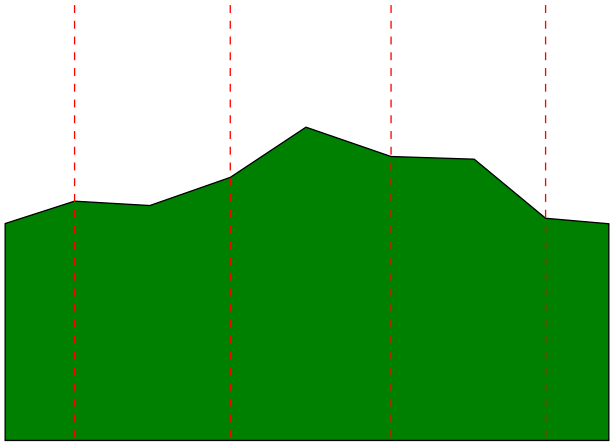


Figure 357: How the Midpoint Displacement Algorithm Works (4/4)

Each displacement is usually done by a lower amount of the previous ones, so that the first displacements give a "general shape" of the terrain, while the ones further down the line are going to give "detail" to our terrain.

The algorithm terminates when we reached a pre-defined number of subdivisions, sometimes called "octaves". In the previous example, we have 4 octaves.

Let's take a look at a possible implementation of the midpoint displacement algorithm:

Listing 119: Example implementation of the midpoint displacement algorithm

```
let MIN = 0;
let MAX = 100;
let OCTAVES = 5;

// This will contain the "heights" of our terrain
let terrain = [];

```

```
// We start by deciding the start and end "heights" of our terrain
terrain[0] = Math.random() * (MAX - MIN) + MIN;
terrain[31] = Math.random() * (MAX - MIN) + MIN;
// We interpolate all the missing values
interpolate(terrain, 0, 31);


function midpoint_displacement(begin, end, octave){
    // Get the midpoint
    let midpoint = Math.floor((end - begin) / 2);
    // Get the midpoint value
    let value = (Math.abs(terrain[end] - terrain[begin])) / 2;
    // Get the possible displacement
    let displacement = MAX / octave;
    // Displace by a random amount
    value = value + Math.random() * 2 * displacement - displacement;
    // Apply the value
    terrain[midpoint] = value;
    // Interpolate the values between begin and midpoint
    for (let i = begin + 1; i < midpoint;  i++){
        terrain[i] = interpolate(terrain, begin, midpoint);
    }
    // Interpolate the values between midpoint and the end
    for (let i = midpoint + 1; i < end;  i++){
        terrain[i] = interpolate(terrain, midpoint, end);
    }
    // Recursion on the subtree
    if (octave < OCTAVES){
        // Recur left
        midpoint_displacement(begin, midpoint, octave + 1);
        // Recur right
        midpoint_displacement(midpoint, end, octave + 1);
    }
}
```

Note!

This algorithm can be extended to create 2D height-maps (which can be used in turn to create 3D ground) and noise textures quite easily, but it also presents some artifacts that can be noticeable. The diamond-square algorithm solves this issue.

18.1.2 Diamond-Square Algorithm

The diamond-square algorithm is an evolution in 2D of the midpoint displacement algorithm (so far, we just changed one value, in one dimension).

The algorithm iteratively repeats two steps:

- a **diamond step** where you find all the squares, take the midpoint and set it as the average of the four corners, plus a random displacement;
- a **square step** where you find all the diamonds, take the midpoint and set it as the average of the four corners, plus a random displacement.

Let's see how it works.

First of all, we bootstrap our square with arbitrary values at the four corners, this will be our starting point.

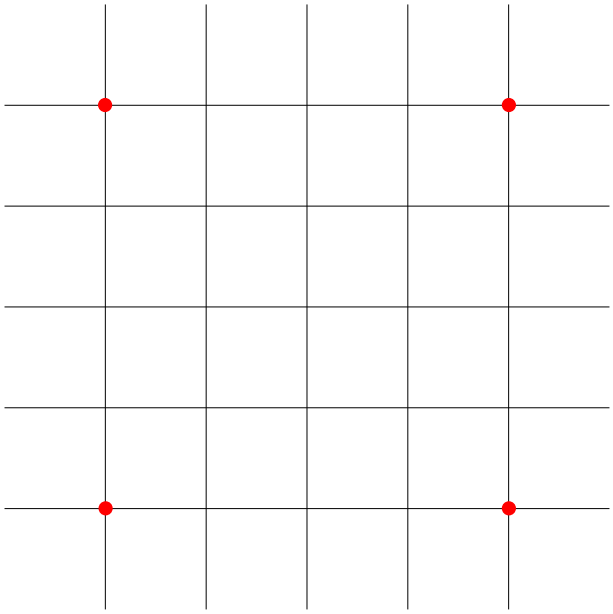


Figure 358: How the diamond-square algorithm works (1/5)

Now we perform the first "diamond step": we have only one big square, with 4 corners, we identify its center and set its value to the average of the corners and apply a random displacement.

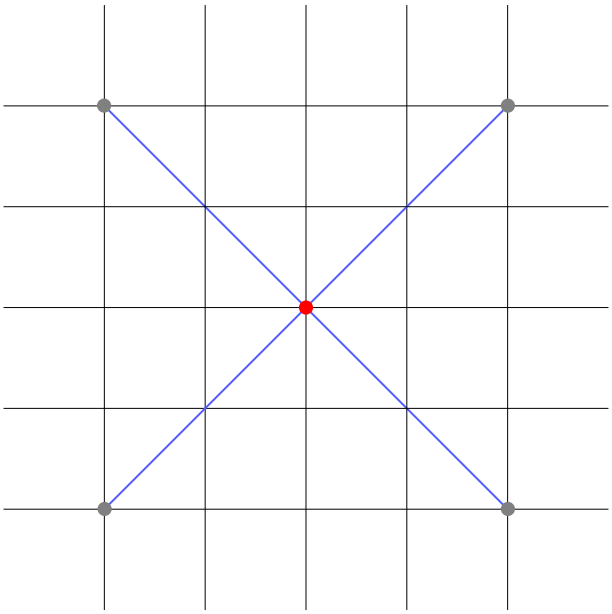


Figure 359: How the diamond-square algorithm works (2/5)

Now we can perform our first “square step”: we have 4 diamonds, we identify their centres (in red) and set them to the average of their neighbours, plus a random displacement.

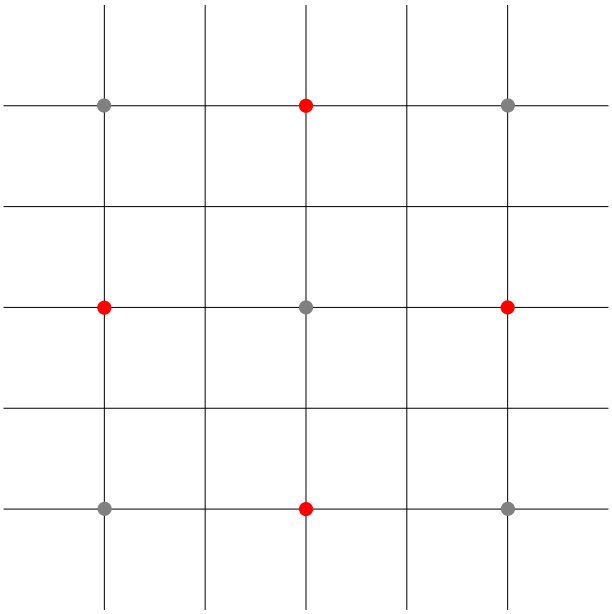


Figure 360: How the diamond-square algorithm works (3/5)

Now we can iterate with another “diamond step” on the four new smaller squares we have, get their centers and again average and displace.

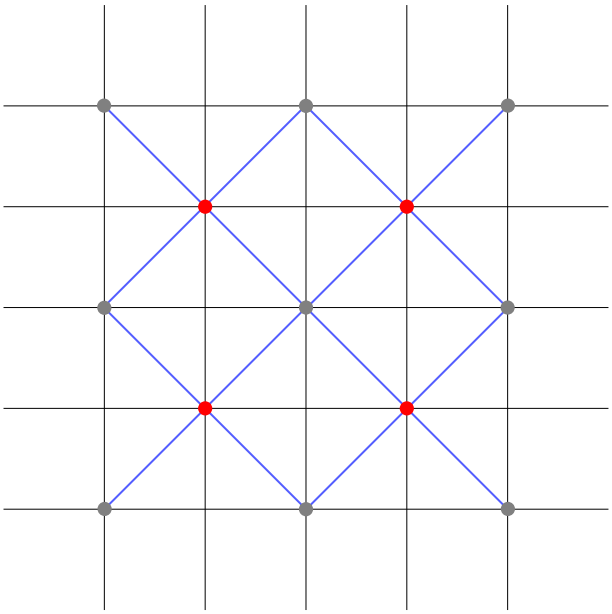


Figure 361: How the diamond-square algorithm works (4/5)

As the last step, for this 5x5 grid, we perform another “square step”, finding 12 smaller diamonds, and again getting the average of the corners and apply a displacement.

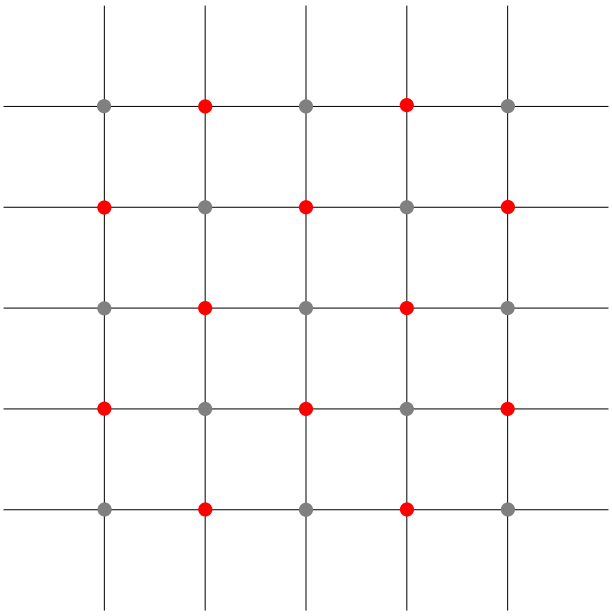


Figure 362: How the diamond-square algorithm works (5/5)

[This section is a work in progress and it will be completed as soon as possible]

18.1.3 Maze Generation

Maze generation is the base of a great majority of dungeon generation systems, you can create a maze, carve out a few rooms, put an entrance and an exit and you have a nice quick dungeon!

There are many ideas that can be used to generate a maze, some are based on a prepared map that gets refined into a maze, some other are based on walls instead of tiles, here we will see some of the many algorithms that exist.

18.1.3.1 Randomized Depth-First Search (Recursive Backtracker)

The Depth-First Search (DFS) algorithm is known in the world of tree and graph structure as a traversal algorithm. We can use a randomized DFS algorithm as a simple maze-generation algorithm.

The Randomized DFS Algorithm is usually implemented using the backtracking technique and a recursive routine: such algorithm is called “Recursive Backtracker”.

The idea behind the algorithm is, starting from a defined “cell”, to explore the grid randomly by choosing an available direction, digging a path.

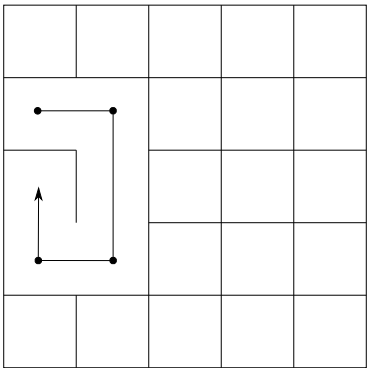


Figure 363: How the recursive backtracker algorithm works (1)

When the algorithm detects that there is no available direction that means that the “head” of our digger is hitting against already explored cells or the map borders.

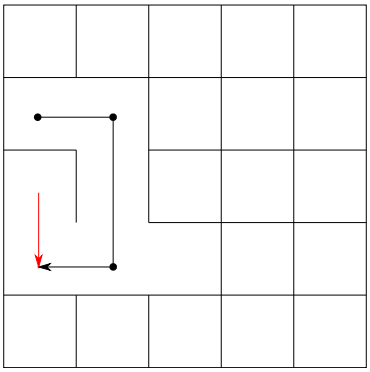


Figure 364: How the recursive backtracker algorithm works (2)

In such case, we “backtrack” until we find a cell with at least one available direction and continue our exploration.

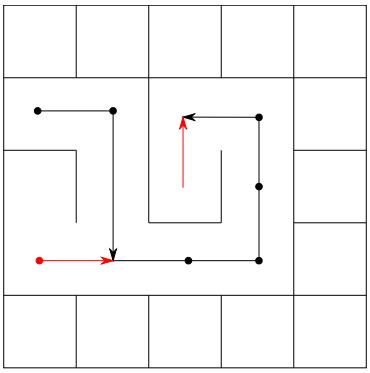


Figure 365: How the recursive backtracker algorithm works (3)

This “digging and backtracking” keeps going until there are no other cells that have not been visited.

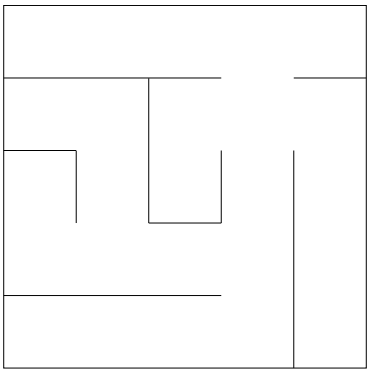


Figure 366: How the recursive backtracker algorithm works (4)

In some versions of the algorithm we need to also keep track of cells that will be used as “walls”, so the actual implementation varies.

Listing 120: Example implementation of recursive backtracker maze generation

```
class Maze{
    constructor(w, h){
        /*
         * A simple constructor
         */
        this.width = w;  // Needs to be an odd number
        this.height = h;  // Needs to be an odd number

        // Create a 2D matrix containing the maze data
        // False = Path, True = Wall
        this.cells = [];
        // Create a 2D array filled with "True"
        for (let i = 0; i < h; i++){
            this.cells[i] = new Array(w).fill(true);
        }
    }
```

```
17
18      make_path(x, y){
19          /*
20           * Given a cell coordinates, sets the cell as a path
21           */
22          cells[y][x] = false;
23      }
24
25      make_wall(x, y){
26          /*
27           * Given a cell coordinates, sets the cell as a wall
28           */
29          cells[y][x] = true;
30      }
31
32      is_wall(x, y){
33          /*
34           * Returns true if the given cell coordinates are inside the maze
35           * boundaries and if the selected cell is a wall
36           */
37          // Let's check if the coordinates are within the maze grid
38          if ((x >= 0) && (x < width) && (y >= 0) && (y < height)){
39              // if they are, then we can check if the cell is a wall
40              return cells[y][x];
41          }else{
42               // If we're outside the maze boundaries, we return false
43               return false;
44          }
45      }
46
47      draw_maze(x, y){
48          /*
49           * Draws a maze using the recursive backtracker algorithm and
50           * system stack recursion.
51           *
52           * This version will dig walls 2 cells at a time, thus leaving some cells
53           * acting as "walls" instead of having a more complex structure.
54          */
55          // We dig a path in the current cell
56          make_path(x, y);
57          // We create a list of available directions (x and y)
58          directions = [
59              {"x": 1, "y": 0},
60              {"x": -1, "y": 0},
61              {"x": 0, "y": 1},
62              {"x": 0, "y": -1}
63          ];
64          // And we shuffle them
65          directions = shuffle_array(directions);
66
```

```
67          // While there is at least one direction available
68          while (directions.length != 0){
69              // We take the last item in our directions list (which is random)
70              direction_to_try = directions.pop();
71
72              // Calculate the new node's coordinates using the chosen direction.
73              // We are doubling the movement in each direction, so some cells
74              // can act as walls
75              let next_x = x + (direction_to_try.x * 2);
76              let next_y = y + (direction_to_try.y * 2);
77
78              // If the node we found is a wall, it means we didn't visit it
79              // (and it's inside our maze boundaries)
80              if (is_wall(next_x, next_y)){
81                  // We have found a new node to dig towards
82
83                  // Since we are moving 2 cells at a time, we need to carve
84                  // the cell that "links" our "current node" and the "next node"
85                  let link_cell_x = x + direction_to_try.x;
86                  let link_cell_y = y + direction_to_try.y;
87                  make_path(link_cell_x, link_cell_y);
88
89                  // Repeat the carving process with the new coordinates
90                  draw_maze(next_x, next_y);
91              }
92          }
93          // If all available directions are exhausted, we return (thus popping
94          // one layer from the system stack: we go back one step)
95          // If we are back at the starting cell, the algorithm terminates
96          return;
97      }
98  }
```

This algorithm can involve a big deal of recursion, which can lead to a *stack overflow*[g] in your program, stopping the algorithm from working and your game in its entirety. It is possible to work around this issue by using an explicit stack, instead of using the call stack.

Listing 121: Example implementation of recursive backtracker with an explicit stack

```
1  class Maze{
2      constructor(w, h){
3          /*
4           * A simple constructor
5           */
6          this.width = w;  // Needs to be an odd number
7          this.height = h;  // Needs to be an odd number
8
9          // Create a 2D matrix containing the maze data
10         // False = Path, True = Wall
11         this.cells = [];
```

```
12          // Create a 2D array filled with "True"
13          for (let i = 0; i < h; i++){
14              this.cells[i] = new Array(w).fill(true);
15          }
16          this.carved_passages = [];
17      }
18
19      // ...
20      // Same as the previous example
21      // ...
22
23      draw_maze(x, y){
24          /*
25           * Draws a maze using the recursive backtracker algorithm and
26           * system stack recursion.
27           *
28           * This version will dig walls 2 cells at a time, thus leaving some cells
29           * acting as "walls" instead of having a more complex structure.
30          */
31          // First thing, we push our cell into our "carved passages",
32          // this will simulate the system stack
33          this.carved_passages.push({"x": x, "y": y});
34          // When the "carved_passages" array is empty, we are back at the start of the maze
35          let stored_x = x;
36          let stored_y = y;
37          let next_x = null;
38          let next_y = null;
39          // While the array is not empty
40          while (!carved_passages.length != 0){
41              // We dig a path in the current cell
42              make_path(stored_x, stored_y);
43              // We create a list of available directions (x and y)
44              directions = [
45                  {"x": 1, "y": 0},
46                  {"x": -1, "y": 0},
47                  {"x": 0, "y": 1},
48                  {"x": 0, "y": -1}
49              ];
50              // And we shuffle them (it needs a specialized function, cause javascript doesn't
    have one)
51              directions = shuffle_array(directions);
52
53              // While there is at least one direction available
54              while (directions.length != 0){
55                  // We take the last item in our directions list (which is random)
56                  direction_to_try = directions.pop();
57
58                  // Calculate the new node's coordinates using the chosen direction.
59                  // We are doubling the movement in each direction, so some cells
60                  // can act as walls
```

```
                next_x = stored_x + (direction_to_try.x * 2);
                next_y = stored_y + (direction_to_try.y * 2);

                // If the node we found is a wall, it means we didn't visit it
                // (and it's inside our maze boundaries)
                if (is_wall(next_x, next_y)){
                    // We have found a new node to dig towards

                    // Since we are moving 2 cells at a time, we need to carve
                    // the cell that "links" our "current node" and the "next node"
                    let link_cell_x = stored_x + direction_to_try.x;
                    let link_cell_y = stored_y + direction_to_try.y;
                    make_path(link_cell_x, link_cell_y);

                    // Repeat the carving process with the new coordinates
                    // we do this by pushing into the stack
                    carved_passages.push({"x": next_x, "y": next_y});
                }
            }
            // If all available directions are exhausted, we pop
            // one layer from our stack: we go back one step
            if (directions.length == 0){
                let popped_vector = carved_passages.pop();
                stored_x = popped_vector.x;
                stored_y = popped_vector.y;
            }else{
                // If not, we continue
                stored_x = next_x;
                stored_y = next_y;
            }
        }
    }
}
```

This algorithm, being taken from a Depth-First search algorithm, is biased towards creating very long corridors.

18.1.3.2 Randomized Kruskal's Algorithm

This algorithm is based on a randomized version of the minimum-spanning tree algorithm known as Kruskal's algorithm.

The algorithm needs the following data structures to work:

- One structure that contains all the "walls" of our maze, this can be a list
- One structure that allows for easy joining of disjoint sets, this will contain the cells

Initially all the cells are separated by walls, and each cell is its own set.

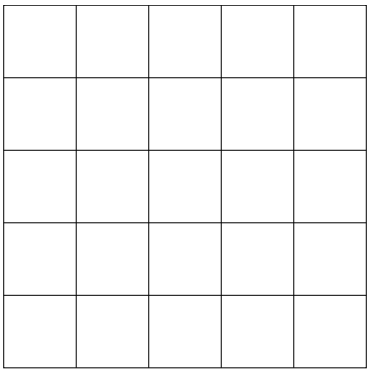


Figure 367: How the Randomized Kruskal's Algorithm Works (1/6)

Now we select a random wall from our list, if the cells separated by such wall are part of different sets, we delete the wall and join the cells into a single set.

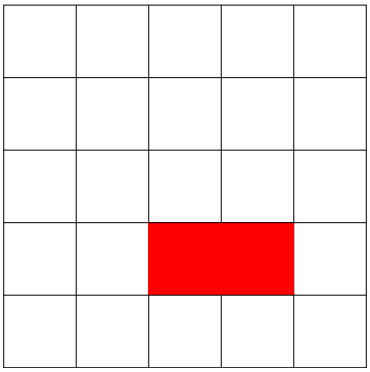


Figure 368: How the Randomized Kruskal's Algorithm Works (2/6)

The "different sets" check allows us to avoid having loops in our maze (and also deleting all the walls, in some cases). Next we select another wall, check if the cells divided by the wall are from different sets and join them.

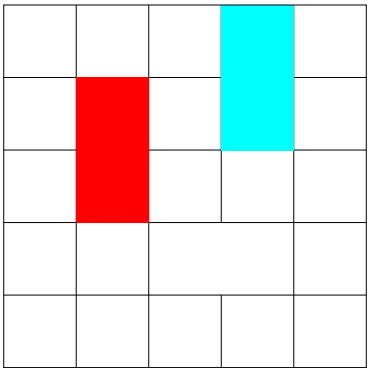


Figure 369: How the Randomized Kruskal's Algorithm Works (3/6)

This doesn't look much like a maze yet, but by uniting the cells we can start seeing some short paths forming in our maze.

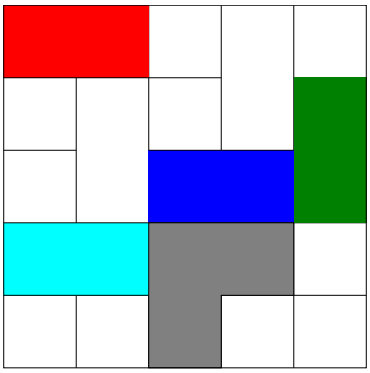


Figure 370: How the Randomized Kruskal's Algorithm Works (4/6)

The black cells are starting to develop a path, as stated earlier. As the sets get bigger, there will be less walls we can "break down" to join our sets.

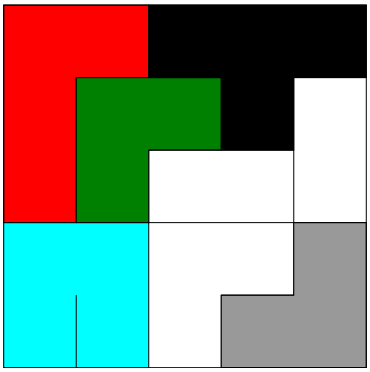


Figure 371: How the Randomized Kruskal's Algorithm Works (5/6)

When there is only one set left, our maze is complete.

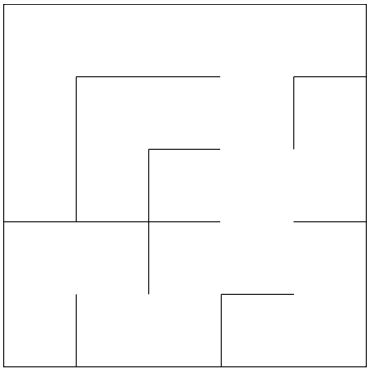


Figure 372: How the Randomized Kruskal's Algorithm Works (6/6)

Being based on a minimum-spanning tree algorithm, this algorithm is biased towards creating a large number of short dead ends.

Now let's see an example implementation of the Randomized Kruskal's Algorithm:

[This section is a work in progress and it will be completed as soon as possible]

18.1.3.3 Recursive Division Algorithm

This algorithm is a bit similar to the recursive backtracker, but instead of focusing on passages, this algorithm focuses on walls: the idea is recursively dividing the space available with a horizontal or vertical wall which has a "hole" placed randomly.

This algorithm can give better results when the choice between "vertical" and "horizontal" walls is biased by the sizes of the sub-areas given by the last division.

Let's see how the algorithm works.

Starting from an empty maze, with no walls, we decide the direction (horizontal or vertical) of our first wall and add it, in a random position, making sure that there's an opening in such wall.

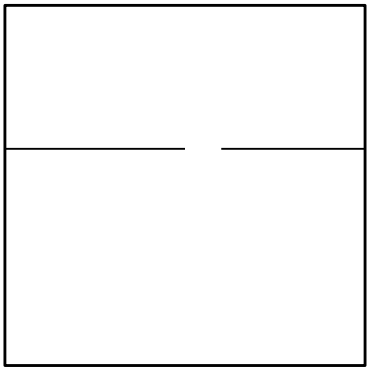


Figure 373: How the Recursive Division Algorithm Works (1/6)

We select one of the two sub-areas we find, recursively and we add another wall in a random position and with a random direction.

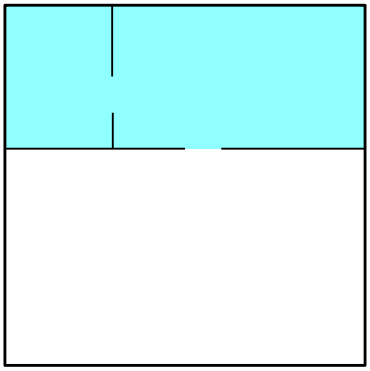


Figure 374: How the Recursive Division Algorithm Works (2/6)

We select one of the two sub-sub-area, and add another wall, with a random position and direction.

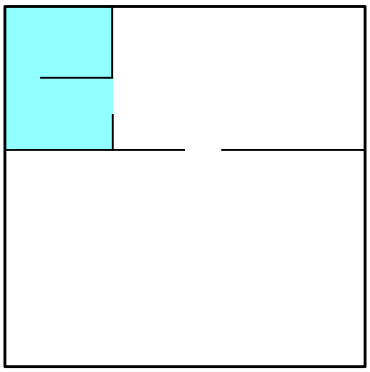


Figure 375: How the Recursive Division Algorithm Works (3/6)

We keep on diving each sub-area recursively, adding walls, until the sub-area had one of its 2 dimensions (horizontal or vertical) equal to 1 cell.

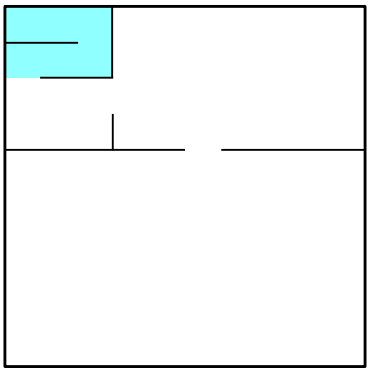


Figure 376: How the Recursive Division Algorithm Works (4/6)

When that happens, we backtrack to one of the previous sub-sections and continue.

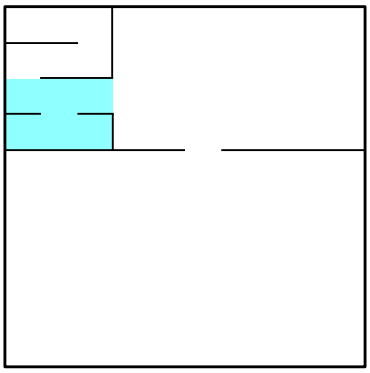


Figure 377: How the Recursive Division Algorithm Works (5/6)

This keeps going until the maze is complete.

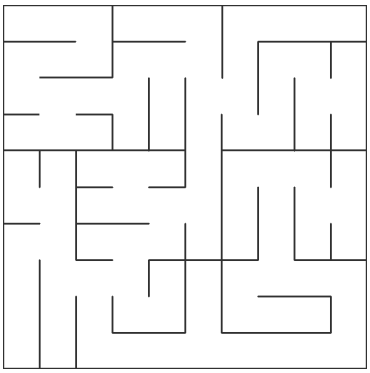


Figure 378: How the Recursive Division Algorithm Works (6/6)

Although it's one of the most efficient algorithms out there (considering that it can easily be converted to a multi-threaded version), given its nature, this algorithm is naturally biased towards building long walls, which give the maze a very "rectangle-like" feeling.

This bias is more noticeable with bigger mazes, like the following one.

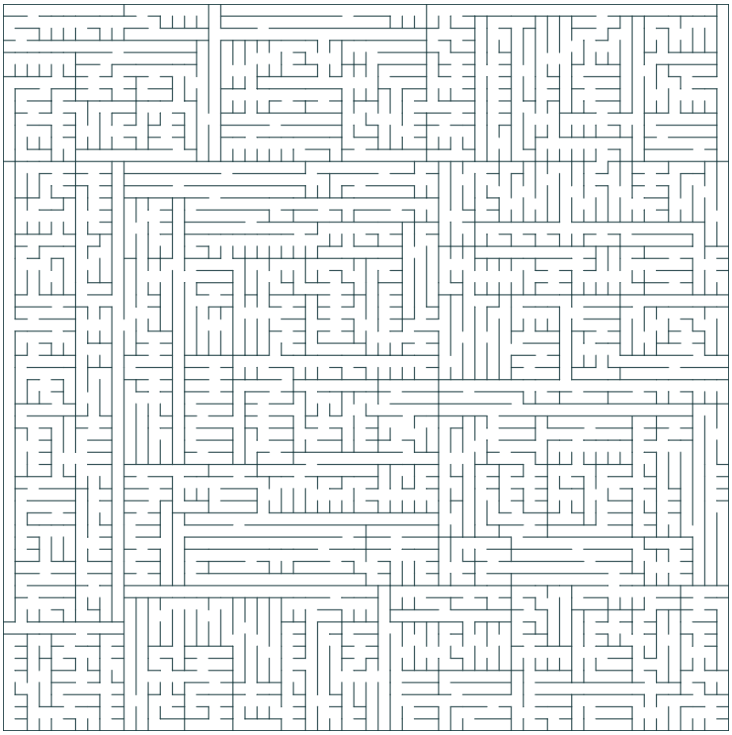


Figure 379: The bias of Recursive Division Algorithm

Let's see an example implementation of this algorithm:

[This section is a work in progress and it will be completed as soon as possible]

18.1.3.4 Binary Tree Algorithm

This is another very efficient “passage carver” algorithm: for each cell we carve a passage that either leads upwards or leftwards (but never both).

Let’s see how the algorithm works: we start from the bottom-right cell of the maze (the last one).

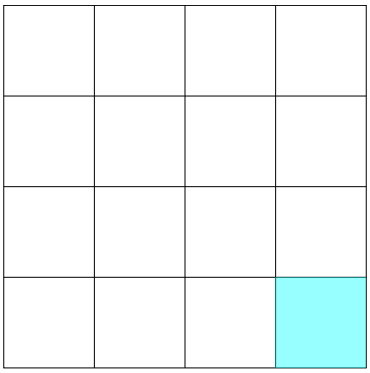


Figure 380: How the Binary Tree Maze generation works (1/6)

Now we decide, randomly, to carve a passage either upwards or leftwards (we will not carve a passage that “creates a hole in a wall”). In this case we go leftwards.

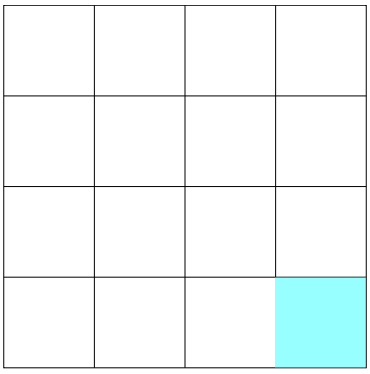


Figure 381: How the Binary Tree Maze generation works (2/6)

Now let’s go to the second-to-last cell...

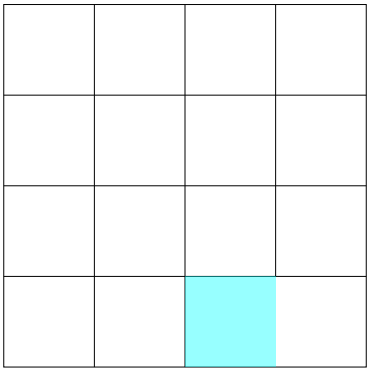


Figure 382: How the Binary Tree Maze generation works (3/6)

And again, decide randomly to carve a passage either upwards or leftwards, this time we chose upwards.

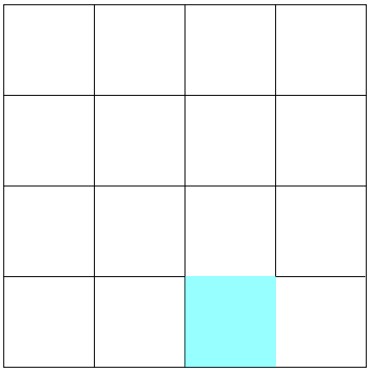


Figure 383: How the Binary Tree Maze generation works (4/6)

Again we go to the “previous” cell and continue with our process, until we hit the left wall (which will force us to carve a passage upwards) or the top wall (which will force us to go left); when we hit both the top and the left walls, we stop.

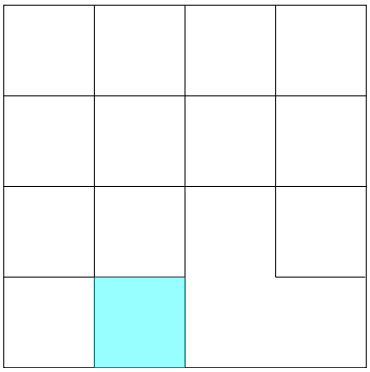


Figure 384: How the Binary Tree Maze generation works (5/6)

Here is the result of the algorithm:

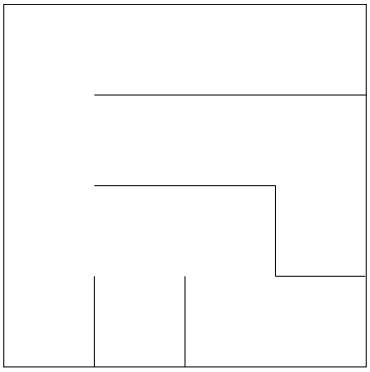


Figure 385: How the Binary Tree Maze generation works (6/6)

Given its deep roots into the computer science “Binary Tree” structure (where the root is the upper-left corner), this

algorithm shows only half of the cell types available in mazes: there are no crossroads and all dead ends will either have a passage upwards or leftwards (but again, never both at the same time).

Let's see an example implementation of the "binary tree algorithm":

[This section is a work in progress and it will be completed as soon as possible]

18.1.3.5 Eller's Algorithm

Eller's algorithm is the most memory-efficient maze-generation algorithm known so far: you generate the maze row-by-row, without needing to memorize the whole maze in memory while creating it.

To start, we decide the width and height of the maze, and create a single row, large as the width we want. Then we set each cell to be its own "set".

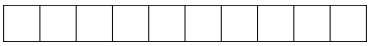


Figure 386: How Eller's Maze Generation Algorithm Works (1/7)

Now we scroll through the cells and randomly join adjacent cells that are part of two different "sets".

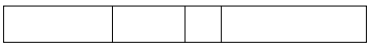


Figure 387: How Eller's Maze Generation Algorithm Works (2/7)

After joining we create some "holes" in the bottom wall, making sure that each "set" has at least one hole to get to the next row.

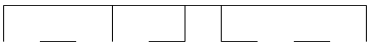


Figure 388: How Eller's Maze Generation Algorithm Works (3/7)

After that we start creating the next row, connecting the cells that have a "hole" with the previous row and assigning them the same set. In the picture the gray cells didn't get a set assigned yet.

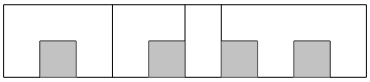


Figure 389: How Eller's Maze Generation Algorithm Works (4/7)

After that we assign a new set to the remaining cells.

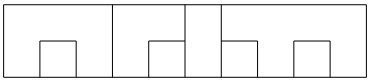


Figure 390: How Eller's Maze Generation Algorithm Works (5/7)

At this point we just need to iterate, ignoring the previous row: we join adjacent cells that are not part of the same "set" (we grayed out the previous row).

Figure 391: How Eller's Maze Generation Algorithm Works (6/7)

Then we create "holes" for each set and prepare the next row. In case we want the maze to be wholly interconnected then if the row is the last row, we can just join all the cells.

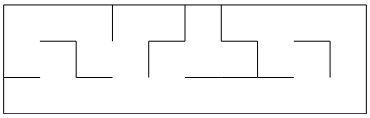


Figure 392: How Eller's Maze Generation Algorithm Works (7/7)

Obviously we can repeat the iteration as many times as we want, and we get a maze as big as we want. This algorithm has no obvious biases and is good for very efficient dungeon generation, if you add rooms, for instance.

Let's see a possible implementation of this strange, but interesting algorithm:

[This section is a work in progress and it will be completed as soon as possible]

18.2 Dungeon Generation

[This section is a work in progress and it will be completed as soon as possible]

18.3 Noise Generation

"Noise" can be a very important part of game development: we can create textures with it, or even use it to generate worlds. In this section we will take a look at how to create "noise" efficiently and with the desired result: from completely randomized to more "natural looking" noise we can use to create maps.

18.3.1 Randomized Noise (Static)

The simplest kind of noise we can generate is also known as "static", for each unit of our elaboration (it can be a pixel, for instance), we generate a random number between two bounds.

Here is an example of random noise:

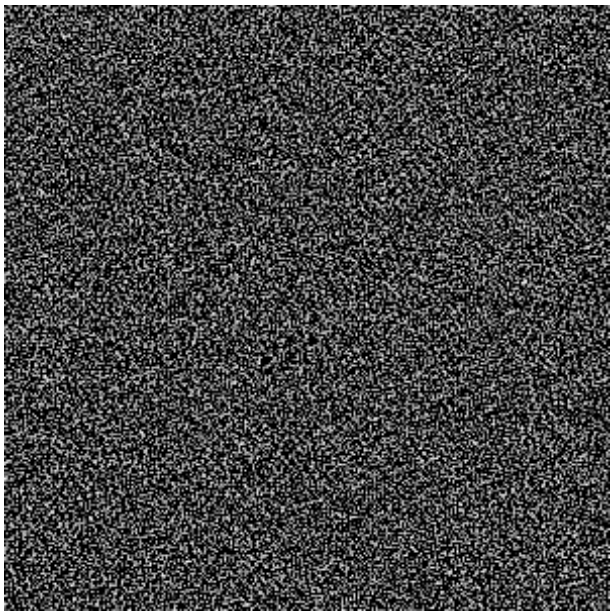


Figure 393: Example of Random Noise

We can create some “TV-like” static with a few lines of code, like the following:

Listing 122: Example implementation of randomized noise

```
let WIDTH = 800;
let HEIGHT = 600;
// We create an empty texture (assuming it can be seen as a 2D array)
let texture = new Texture(WIDTH, HEIGHT);

// Now we iterate through each pixel of the texture
for (const row in texture){
    for (const pixel in row){
        // We create a random gray color (0 is black, 255 is white)
        let rand_gray_tone = Math.random() * 255;
        // Most colors are made of Red Green and Blue, by placing them at the
        // same value, we get a tone of gray
        let rand_color = new Color(rand_gray_tone, rand_gray_tone, rand_gray_tone);
        pixel.setColor(rand_color);
    }
}
```

18.3.2 Perlin Noise

[This section is a work in progress and it will be completed as soon as possible]

18.4 Animation

18.4.1 Skeleton animation

[This section is a work in progress and it will be completed as soon as possible]

19 Procedural Content Generation

> Science is what we understand well enough to explain to a computer. Art is everything else we do.
>
> *Donald Knuth*

19.1 What is procedural generation (and what it isn't)

Sometimes you hear "procedural generation" being thrown around as a term describing that some part of a video game is generated with an element of randomness to it.

This isn't entirely true, since "procedural generation" suggests the presence of a "procedure to generate the item", in short: an algorithm.

A procedurally generated weapon is not statically created by an artist, but instead by an algorithm that puts together its characteristics. If the algorithm has the same data in its input, then the same item will be generated as an output.

When you introduce an element of randomness (or more precisely **pseudo-randomness**) you have what is called "random generation".

Let's make a simple example: we want our Super-duper-shooter to make use of procedural/random generation to create your weapons. The following example will clarify the difference in algorithms between procedural and random generation, all weapons have a body, a scope, a barrel and an ammo magazine.

This is a possible algorithm for a procedural weapon:

Listing 123: Example procedural weapon creation

```
function createProceduralWeapon(){
    // The "class" and "new" keywords are supported in ES6/ECMAScript 2015
    let wp = new Weapon();
    // Load some pre-defined components
    wp.loadBody("body0001.png");
    wp.loadScope("scope0051.png");
    wp.loadBarrel("barrel0045.png");
    wp.loadAmmoMagazine("mag0009.png");
    // set weapon damage to 45
    wp.setDamage(45);
    // set weapon range to 15
    wp.setRange(15);
    // set weapon spread to 23
    wp.setSpread(23);
    return wp;
}
```

This instead is a possible algorithm for a random weapon, for simplicity we assume that the pieces are all compatible:

Listing 124: Example Randomized weapon creation

```
function createRandomizedWeapon(){
    // The "class" and "new" keywords are supported in ES6/ECMAScript 2015
    let wp = new Weapon();
    // Load some randomized components
    wp.loadRandomBodyFrom("weaponBodies/shotguns");
    wp.loadRandomScopeFrom("weaponScopes/shotguns");
    wp.loadRandomBarrelFrom("weaponBarrels/shotguns");
    wp.loadRandomAmmoMagazineFrom("weaponMagazines/shotguns");
    // set weapon damage to a value between 35 and 50
    wp.damage = Math.random() * 16 + 35;
    // set weapon range to a value between 13 and 18
    wp.range = Math.random() * 5 + 13;
    // set weapon spread to a value between 20 and 30
    wp.spread = Math.random() * 10 + 20;
    return wp;
}
```

As you can see, the algorithms are very similar to each other, but the second one has an element of randomness added to it.

So, as a memorandum:

Procedural generation is **consistent**, even though something is generated in real time, given the same input the same output will be returned.

Random generation is usually **not consistent**, although it is possible to control the random generator (via its seed) to obtain deterministic results, given the same input.

Seeding a random number generator correctly can allow you to generate a huge universe without storing it into memory, for instance; although the edits to such universe will have to be saved in some other way.

19.2 Advantages and disadvantages

As with everything, procedural and random generation has its advantages and disadvantages, which will be explained below.

19.2.1 Advantages

19.2.1.1 Less disk space needed

Using algorithms to build worlds and items means generating them mostly in real-time, which means we don't have to save them to hard-disk, since if the algorithm is not randomized, you can always re-create the same worlds and items when requested. This was more pressing at the times of the NES, where game sizes were usually around a couple hundreds of KB.

19.2.1.2 Larger games can be created with less effort

When a world is handcrafted, everything has to be placed and textured manually, which takes time and money. This obviously puts a superior limit to how big these worlds can be.

When procedural (and randomized) generation comes into play, there is no theoretical limit to how big these worlds can be (considering an infinitely powerful hardware).

Same goes for items, each handcrafted item takes time and money, while using procedural generation you can re-use components of said items to generate a potentially infinite number of new items that have certain characteristics.

19.2.1.3 Lower budgets needed

Creating a video game is expensive, in fact the so-called "AAA" games costs are in the order of millions of dollars. Using procedural and random generation you can create variations of your resources (textures, for instance), lowering costs.

19.2.1.4 More variety and replayability

When a world and its objects are handmade, the game experience is bound to be fixed: same items to collect, same world, same overall experience. Procedural and random generation can bring some sense of "unknown" to the game every time you play. This also enhances the replayability value of the game.

19.2.2 Disadvantages

19.2.2.1 Requires more powerful hardware

Procedural generation makes use of algorithms, and such algorithms can be really taxing on the computer hardware, so loading times might increase or users with less powerful computers might experience stutters as their computer cannot "keep up" with the game demands.

19.2.2.2 Less Quality Control

Computers are able to crunch numbers at an incredible rate, but they lack creativity. In a procedurally generated world you lose the "human touch" that can introduce subtleties and changes that can be brought by a good designer with experience.

At the same time, there is a variation in user experience, so you cannot guarantee the same gameplay quality to all players. Some players may find a really easy map to play in, while others might find a really hard map that prohibits such gameplay.

19.2.2.3 Worlds can feel repetitive or "lacking artistic direction"

Consequence of having less quality control, worlds and items might feel like they "lack artistry", as well as being repetitive.

If you use procedural and randomized generation, you have the chance of generating incredibly large worlds with a huge variety of items with less resources and algorithms; that's where our human nature of "recognizing patterns"

crashes the party: repeating patters are really easy to spot and can remove us from the game's atmosphere and introduce us to one of our worst enemies: boredom.

This can become even worse if we try to "find the middle ground" and build our levels using hand-made "chunks", joined together. If we want to avoid our player getting bored by repeating "pieces of level" we will need to build a lot of chunks that fit together to make something interesting and new almost every time.

19.2.2.4 You may generate something unusable

In extreme cases, there is a possibility that we end up generating an unplayable world, or useless items: terrain too high to climb, walls blocking a critically-necessary area, dungeon rooms with no exits, etc...

19.2.2.5 Story and set game events are harder to script

Being uncertain, procedural generation makes set events harder to script, if not impossible. In this case it's more common to use a mix of procedural generation and pre-made game elements, where the fixed elements are used to drive the narrative and the procedurally generated elements are used to create an open world for the player to explore and vary its gameplay experience.

19.3 Where it can be used

Procedural (and random) generation can be used practically anywhere inside of a video game, some examples could be the following:

- **World Generation:** Using an algorithm called "Perlin noise", you can generate a so-called "noise map" that can be used to generate 3D terrain, using giving areas with higher concentration a higher height. For dungeon-crawling games you might want to use a variation of maze generation algorithms, and so on so forth;
- **Environment Population:** You can use an algorithm to position certain items in the world, and if an element of randomness is required, positioning items in a world is certainly a very easy task and can add a lot to your game, but be careful not to spawn items into walls!;
- **Item Creation:** As stated previously, you can use procedural generation to create unique and randomized items, with different "parts" or different "stats", the possibilities are endless!;
- **Enemies and NPCs:** Even enemies and NPCs can be affected by procedural (and randomized) generation, giving every NPC a slightly different appearance, or scaling an enemy size to create a "behemoth" version of a slime, maybe by pumping its health points too, randomizing texture colors, again the possibilities are endless;
- **Textures:** It's possible to colorize textures, giving environments different flavours, as well adding a layer of randomness to a procedurally generated texture can greatly enhance a game's possibilities;
- **Animations:** An example of procedurally generated animations are the so-called "ragdoll physics", where you calculate the forces impacting a certain body (and it's "virtual skeleton"). A simpler way could be making the program choose randomly between a set of pre-defined "jumping animations" to spice up the game;
- **Sounds:** You can use sound manipulation libraries to change the pitch of a sound, to give a bit of randomness to it, as well as using "sound spatialization" by changing the volume of a sound to make it come from a certain

place, compared to the player's position;

- **Story:** In case you want to put some missions behind a level-gate, you can use procedurally generated missions to allow the players to grind for experience and resources, so they are ready for the upcoming story checkpoints;
- **Difficulty Management:** Procedural generation can be involved into difficulty management by handing the enemy parameters and spawning to suit our needs.

Random Trivia!

Want to see procedural generation brought to the extreme? Check out `.kkrieger` (by `.theprodukkt`, a subdivision of the demo group Farbrausch): it's a first person shooter that weighs 96Kb (no, it's not a typo) that makes **massive** use of procedural generation, from textures, to meshes, to sound.

19.4 Procedural Generation and Difficulty Management

As stated above, we can use elements of procedural generation to aid us in managing the difficulty of our game, tweaking the challenge we are offering to our players.

19.4.1 Static difficulty

Sometimes called "Algorithmic Difficulty", is the kind of difficulty management seen in many games: elements and enemies are placed in a way that gives the player an ascending difficulty curve, with different parameters.

Such parameters build the abstract concept of "difficulty level" (beginner, normal, advanced, master, ...): each difficulty level contains a set of parameters that change how the game feels.

In practice usually you assign a "difficulty level" to an area and make the game handle the enemy spawn accordingly; sometimes such "area level" is used to make the game spawn randomized items with a certain power level, given a certain amount of displacement of its statistics.

19.4.2 Adaptive Difficulty

Not really considered "procedural generation", adaptive difficulty makes use of different algorithms (taken on a lease from AI programming) to tweak the game difficulty in reaction to how the player plays.

If the player progresses quickly, the game will become harder, instead if the player tends to lose lives, the game will become easier and less random.

The objective of adaptive difficulty is to create an "optimal experience" for everyone, by determining if the current game is "too easy" or "too hard" for the player.

19.4.2.1 Rubberbanding

Rubberbanding (not to be confused with "network lag", sometimes called "rubberbanding") is an instance of adaptive difficulty, mostly used in racing games, where the opponents' abilities (like speed in racing games) are "tweaked"

to keep the game challenging (but not unfair).

This usually ends up with opponents getting faster the further in the lead the player is (sometimes even going over the maximum speed) or going slower when the player falls behind too much.

You can see as a virtual "rubber band" that ties your opponents to you, the more the rubber band is stretched, the more the opponents are "attracted" to you (or you to them, it can work both ways).

Random Trivia!

"Rubberbanding" is not limited to racing games. For instance NBA Jam tweaks player skills to keep the game competitive and enjoyable by using rubberbanding: if you're too far ahead, the opponents will get a skill boost.

19.4.3 Static vs. Adaptive Difficulty

Each approach has its own advantages and shortcomings, which can make one or the other better suited for your game.

Static difficulty is easy to create and leaves choice to the players between varying levels of difficulty; maybe someone wants a more "relaxing experience" instead of being continuously challenged.

The biggest shortcoming is that each level of difficulty is an estimate of its difficulty, so an "easy mode" may be way too easy, while the "normal" mode may be too hard for someone. It's hard to find the balance.

Static difficulty can be planned by using difficulty curves, which can be part of either a technical or a proper game design document.

Also there's all the work dedicated to program the parameters for each level of difficulty.

Adaptive difficulty is harder to code and sometimes can lead to great results, but it also completely invalidates the concept of "grinding" in an RPG game, for instance. If you try to become stronger by undertaking easier quests, you will find that the quests keep getting harder the stronger you get.

Adaptive difficulty doesn't allow the player to "grind their way out" of a difficult part of the game.

20 Developing Game Mechanics

> Fools say that they learn by experience. I prefer to profit by others' experience.
>
> *Otto von Bismarck*

In this section we will take a look at how to develop some known game mechanics, with pieces of code to aid you.

20.1 General Purpose

20.1.1 I-Frames

I-Frames (also known as "invincibility frames") is a term used to identify that period of time after being hit, where the character either flashes or becomes transparent and is immune to damage.

This mechanic can be seen as "giving the player an advantage" but instead it has deeper roots into "fairness" than "difficulty management", let's see why.

Let's assume that our main character has 100 health points, and touching an enemy deals 5 points of damage. In absence of I-Frames, this would translate into 5 points of damage every frame, which would in turn come out between $5 \cdot 30 = 150$ and $5 \cdot 60 = 300$ points of damage per second (at respectively 30 and 60fps).

The average human reaction time is around 1 second, this would mean that touching an enemy would kill us before we even realize we are touching such enemy.

Checking if we're still colliding with an enemy after receiving damage is not a good strategy, since that would allow the player get only one point of damage from a boss, and then carefully stay inside the boss's hitbox while dealing damage to the enemy. Thus allowing the player to exploit the safeguard.

Giving a brief period (usually between 0.5 and 2 seconds) of invincibility after being hit, allows the player to understand the situation, reorganize their strategy and take on the challenge at hand. After the invincibility period, the player will take damage again, patching the exploit we identified earlier.

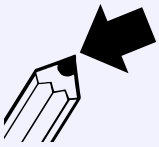
I-Frames can be easily implemented via timers, in a way similar to the following:

Listing 125: Example of I-Frames Implementation

```
const INVINCIBILITY_TIME = 0.75;  // Seconds of invincibility
// ...
function update(dt){
    let inv_time = 0;
    // ...
    if (inv_time <= 0){
        // Check for collision
        // ...
        // Collision has been detected here, we have a hit
        inv_time = INVINCIBILITY_TIME;  // Start of the invincibility period
    }else{
        // We are currently invincible
```

```
13          inv_time = inv_time - dt;  // We decrease the invincibility time
14      }
15  }
```

Tip!



Remember: feedback is important! You need to let the player know when they are invincible due to i-frames. This can be done by making the player semi-transparent, flashing or anything else that can indicate a "different status".

20.1.2 Tilemaps

Tilemaps are a really interesting abstraction that allows us to draw maps by using pre-made "tiles" instead of having to draw them "pixel-by-pixel".

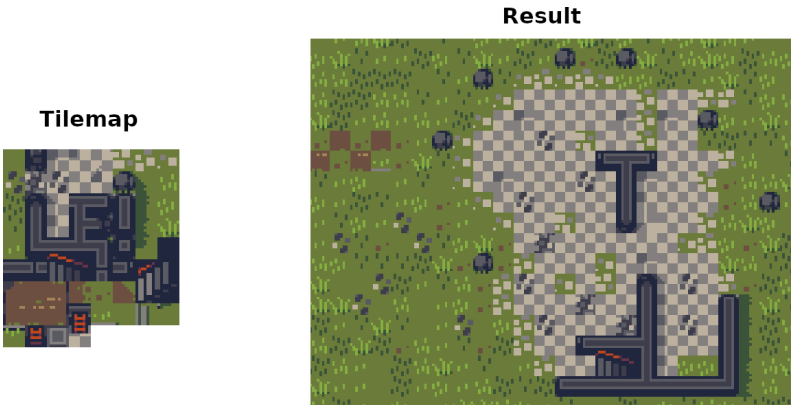


Figure 394: Example of a tileset and a tilemap drawn with it [11]

This also allows us to have a new coordinate system that works "using tiles", which could be preferable than single pixels (since we may put properties on our tiles, like a "solid" property for collision detection).

Another advantage of tilemaps is the ability to use a small texture to draw gigantic maps without adding much data in memory (tilemaps are a fantastic example of the "Flyweight Pattern").

20.1.2.1 Rectangular Tilemaps

Rectangular tilemaps are the most commonly used tile maps in game development: it's easy to translate back and forth between "screen pixels" and "tiles", and if the tilesets are well-made everything looks seamless.

This has the advantage of using less memory (we need to save only the tileset, plus a few coordinates and pointers), thus making our game perform better.

[11] Jawbreaker tileset, listed as public domain at https://adamatomic.itch.io/jawbreaker

Random Trivia!

Super Mario Bros. uses maps that are based on square tiles. Even the pipes are tiles: this allows the game to have variable lengths of pipes without increasing the number of tiles.

[This section is a work in progress and it will be completed as soon as possible]

20.1.2.2 Hexagonal Tilemaps

Sometimes you may want to underline a "tabletop" game feel, in that case a hexagonal tilemap (sometimes called "hexmap") may be a great idea (at the cost of more complicated algorithms).

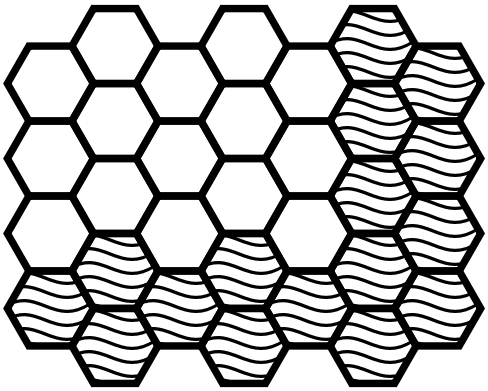


Figure 395: Simple structure of a hexmap

Hexmaps allow for a different kind of movement (the player can move to up to 6 directions, instead of 4), which makes for an interesting remix of the classic tile-based mechanics.

To be able to work with tilemaps, we need to get acquainted with the concept of "outer circle" of a polygon, which is the circle that intersects the edges of a polygon. In the case of our hexagon

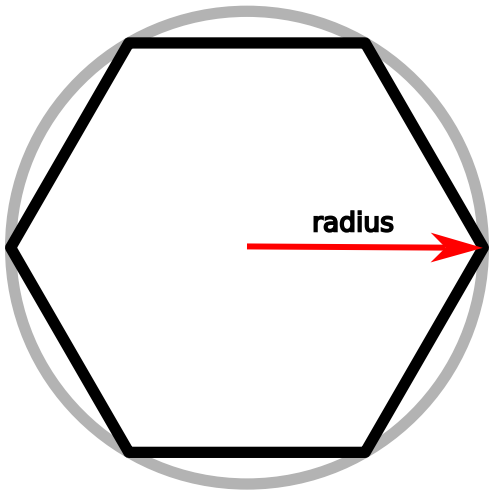


Figure 396: The outer circle or an hexagon

Now we can work out how to measure the space occupied in the Cartesian system by using the radius of the outer circle. We will obtain the following results:

$$size_1 = 2 \cdot radius$$

$$size_2 = \sqrt{3} \cdot radius$$

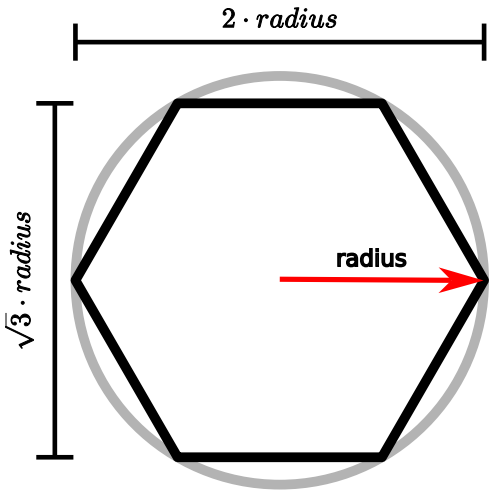


Figure 397: The size of an hexagon, calculated

With this, we can calculate the distances between the centres of the outer circles, in a way that allows us to create our hexmap.

$$dist_1 = \frac{3}{4} \cdot size_1 = \frac{3}{4} \cdot 2 \cdot radius = \frac{3}{2} \cdot radius$$

$$dist_2 = size_2 = \sqrt{3} \cdot radius$$

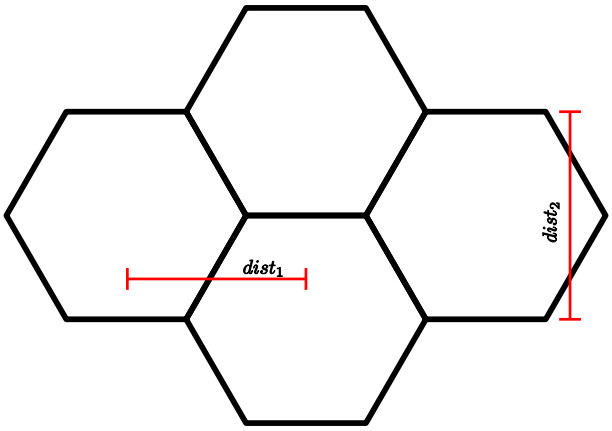


Figure 398: Making a hexmap

Note!

If you want to turn the hexagons “pointy side up”, you just need to switch over the formulas.

[This section is a work in progress and it will be completed as soon as possible]

20.1.2.3 Isometric Tilemaps

In some cases you may want to try and give your game a more “premium” feel: isometric maps can help you in that. The game is technically 2D, but the way the tiles are designed makes it look like it’s a 3D game!

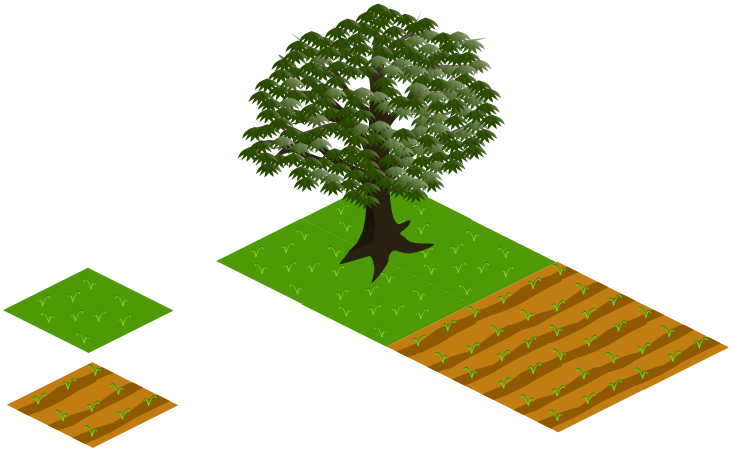


Figure 399: A simple isometric tiles and a tilemap

Isometric tilemaps make use of more difficult algorithms, considering the odd shape the tiles are (usually they’re diamond-shaped).

Random Trivia!

Diablo 2 is actually a 2D game that uses isometric tiles. Every item and character is a pre-rendered sprite: this means that every item drop is also pre-rendered and stored with some defined degrees of rotation to give a more variegated feeling.

[This section is a work in progress and it will be completed as soon as possible]

20.1.3 Scrolling Backgrounds and Parallax Scrolling

20.1.3.1 Infinitely Scrolling Backgrounds

When doing any kind of game that features a scrolling background, you should construct your art accordingly, allowing for enough variety to make the game interesting while avoiding creating huge artwork that weighs on the game's performance.

In a game that uses a scrolling background, the background used should be at least two times the screen size, in the scrolling direction (Virtual Resolution can prove really useful in this case) and the image should have what are called "loop points".

Loop points are points where the image repeats itself, thus allowing us to create an image that is virtually infinite, scrolling through the screen. To have a so-called "loop point" the image should be at least twice the size of the screen, in the scrolling direction.

The image below shows a background and its loop points.

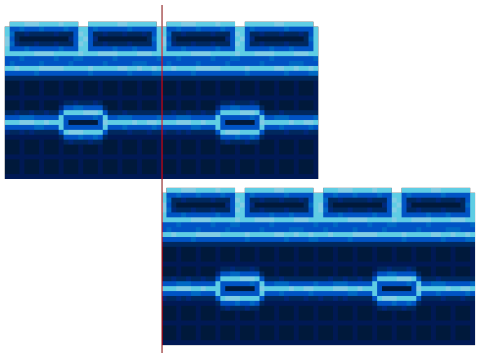


Figure 400: Demonstration of an image with loop points

To make an image appear like it's scrolling infinitely we need to move it back and forth between loop points when the screen passes over them.

For ease of explanation let's consider a screen scrolls towards the right, when we have reached a loop point, we reset the image position back to the position it was at the beginning and, since the image has been crafted to loop, the player won't notice that the background has been reset.

Listing 126: Example of an infinitely scrolling background implementation

```
let background_x_offset = 0.0;  // The x offset of the background
const BACKGROUND_X_SIZE = 512;  // The horizontal size of the background
const LOOP_POINT = 256;  // The horizontal loop point of the image
const DISTANCE_FACTOR = 0.5;  // The background moves at half the player speed

function update(dt){
    // ...
    // In case we're moving right, the background scrolls left slightly
    if (player.speed_x > 0){
        // Update player's position and state
        // ...
        // Update the background position
        background_x_offset = background_x_offset - player.speed_x * DISTANCE_FACTOR * dt;
        // If we passed the image's loop point
        if (background_x_offset <= - LOOP_POINT){
            // We reset the coordinates, keeping note of the remainder
            background_x_offset = background_x_offset % LOOP_POINT;
        }
    }
    // In case we're moving left, the background scrolls right slightly
    if (player.speed_x < 0){
        // Update player's position and state
        // ...
        // Update the background position
        background_x_offset = background_x_offset - player.speed_x * DISTANCE_FACTOR * dt
        if (background_x_offset >= 0){
            // We reset the coordinates, keeping note of the remainder, just backwards
            background_x_offset = background_x_offset - BACKGROUND_X_SIZE;
        }
    }
}

function draw(){
    // ...
    // Draw the background
    screen.draw(background, (background_x_offset, 0));
    // ...
}
```

20.1.3.2 Parallax Scrolling

Parallax in games is an effect that can give more depth to our environment: it looks like objects farther away are moving much slower than the objects closer to us.

This can be used to our advantage, along with some other tricks to enhance the perception of depth explained in the chapter dedicated to graphics.

Creating a parallax effect is quite easy: first we need at least two background layers (although three seems to be the best compromise between performance and depth of the effect):

- The **sprite layer** that will represent the closest layer to us, that will move at a certain speed that will be decided while developing the game;
- A **moving background** that will move slower compared to the sprite layer, thus giving the parallax effect;
- A **fixed background** that will represent our horizon and the farthest objects.

For the sake of clarity, we will re-use an image presented earlier to explain the "painter's algorithm":

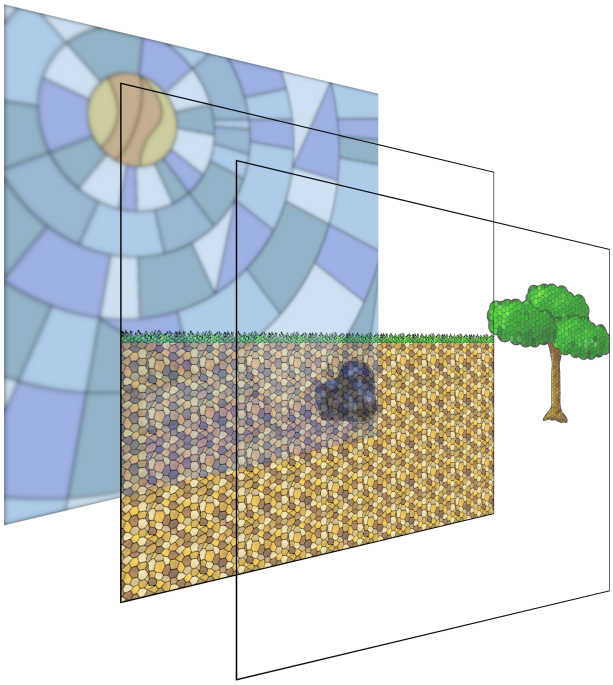


Figure 401: How we can split our game into layers

As stated earlier, a third optional background can be added to deepen the parallax scrolling effect, such background can take any of these positions:

- **Above** the sprite layer: in this case this "foreground layer" will need to move **faster** than the sprite layer and it should include very unobtrusive graphics, to avoid hiding important gameplay elements (like enemies);
- **Between the sprite layer and the first moving background**: in this case, the optional background should move **slower** than the sprite layer, but **faster** than the first moving background;
- **Between the first moving background and the fixed background**: in this case, the optional background will have to move **slower** than the first moving background.

The backgrounds should move all in the same direction, depending on the direction our character is moving: if our character is moving right, our moving backgrounds should move left.

Here's an example of how a simple parallax scrolling can be implemented between two rectangles:

Listing 127: Example of parallax scrolling implementation

```
let running = true;

```

```
class Rectangle{
    // ...
}

// We create a display surface of 640x480 pixels
let screen_surface = engine.set_display((640, 480));

// We keep the second rectangle a bit lower to be able to see both
let rectangle_1 = new Rectangle(x=245,
                                y=100,
                                width=150,
                                height=100,
                                fill_color=(0, 0, 255));
let rectangle_2 = new Rectangle(x=245,
                                y=120,
                                width=150,
                                height=100,
                                fill_color=(0, 255, 0));
let rectangle_1_speed = 7;
let rectangle_2_speed = 14;

// This will be 1 for right and -1 for left
let movement_direction = 0;


// For ease, we assume we have an event queue we can process and we won't
// take care of framerate limiting
while (running){
    // --------------- INPUT ---------------
    for (event of event_queue){
        if (event.type == QUIT){
            // We are quitting the game
            running = false;
        }
        if (event.type == KEYPRESS){
            // We are pressing a key
            if (event.key == RIGHT){
                // We are pressing the right key (moving the camera rightwards)
                movement_direction = -1;
            }
            if (event.key == LEFT){
                // We are pressing the left key (moving the camera leftwards)
                movement_direction = 1;
            }
        }
        if (event.type == KEYRELEASE){
            if (event.key == RIGHT or event.key == LEFT){
                movement_direction = 0;
            }
        }
```

```
    }
    // --------------- UPDATE ---------------
    if (movement_direction != 0){
        rectangle_1.x += rectangle_1_speed * movement_direction;
        rectangle_2.x += rectangle_2_speed * movement_direction;
    }
    // --------------- DRAW ---------------
    // Fill the display with black
    screen_surface.fill((0, 0, 0))
    // Draw the rectangles
    rectangle_1.draw_on(screen_surface);
    rectangle_2.draw_on(screen_surface);
    // Show the result on screen
    screen_surface.display();
}
```

20.1.4 Avoid interactions between different input systems

This is a small improvements that can be done on menu systems: if the player is using a keyboard to navigate the menu, the mouse should not interact with the menu.

In many frameworks when a fullscreen game window "captures" the mouse cursor, this is put on the center of the screen, which could be where a menu item is positioned.

Now imagine you are "flowing through" the menu, trying to load a saved file and the cursor is detected pointing at the "delete save file" option; you are briskly walking through the menu and what you think is the "do you want to load this file?" dialog is actually asking "do you want to **delete** this save file?". You select "yes" by pressing enter on your keyboard and your save file is gone!

This is an extreme edge case, but it could happen. Even if it is a minor annoyance like starting a new save file when instead you want to load an existing one, it diminishes the quality of your experience.

20.1.5 Sprite Stenciling/Masking

[This section is a work in progress and it will be completed as soon as possible]

20.1.6 Loading screens

If you load your resources in the same thread that executes the main game loop, your game will lock up while loading, which may trigger windows to ask you if you want to terminate the task. In this case it is better to dip our toes into multi-threading and create a proper loading screen.

The loading screen will be composed of two main components:

- **The graphical loading screen:** that will show the progress of the resource loading to the user, as well as tips and animations;

- **The actual resource loading thread:** that will take care to load the resources to the right containers, as well as communicating the global loading status to the loading screen in the main game loop.

We can represent the two “loops” in the following UML diagram:

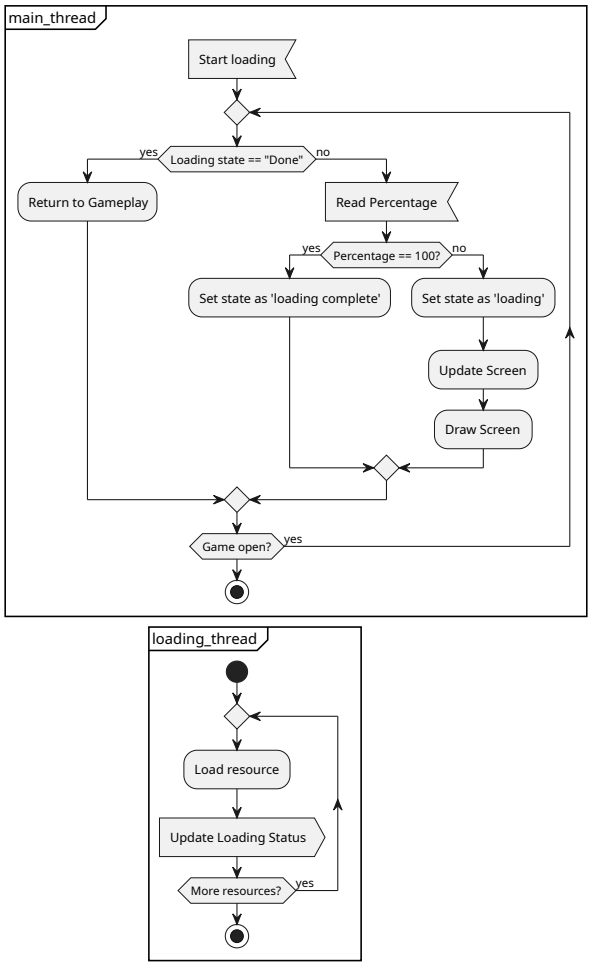


Figure 402: Rough UML diagram of a multi-threaded loading screen

[This section is a work in progress and it will be completed as soon as possible]

20.1.7 Simulating Inertia

After learning how to move something on a screen, the next step is making the movement less “jarring” by introducing inertia. Before throwing solutions around, let’s see what the problem is.

When we press a button, without inertia, our character starts moving at full speed towards the direction we defined, and it will stop as soon as we let go of the button. We can represent such behaviour with the following chart:

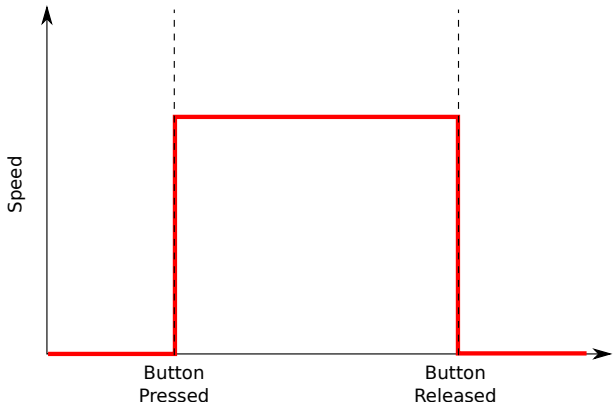


Figure 403: Example chart of how movement without inertia looks

This could be jarring on its own, but the situation gets more serious the higher the speed difference: one thing is having a character being able to run from a standstill and stop immediately, but it feels all the more jarring if a character can turn 180 degrees on its path without the slightest hint of inertia (we assume that positive speed means “going right”, while negative speed means “going left”):

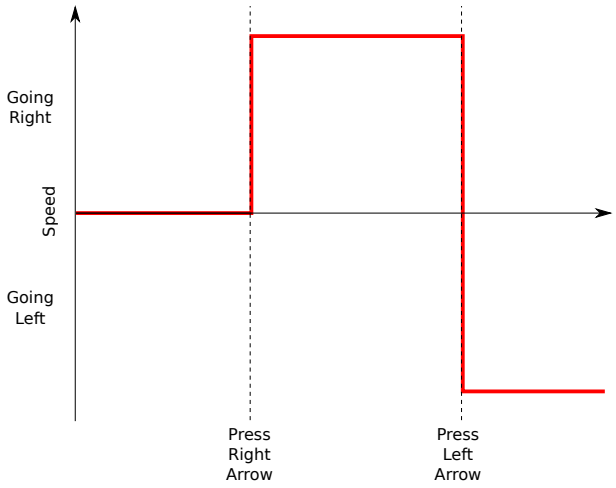


Figure 404: Example chart of how movement without inertia looks: reversing directions

If this is connected to a fully-tracking camera system your player is in for a ride rivaling the deadliest rollercoasters in the world. What we want is the speed curve to behave more like the following (here too we assume that positive speed means “going right”, while negative speed means “going left”):

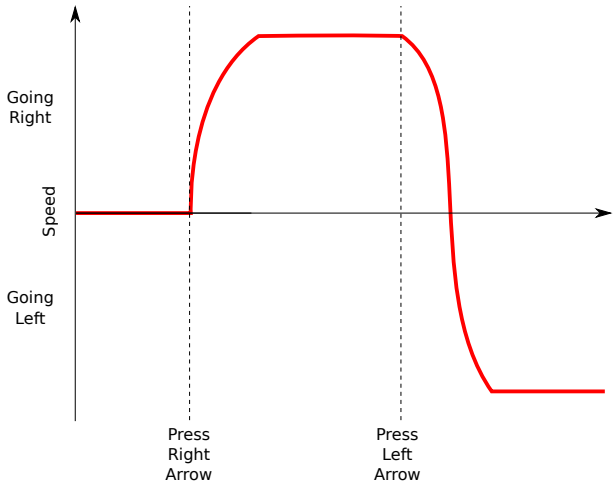


Figure 405: Example chart of how movement with inertia looks

A “softer” transition between directions can be a good way to avoid nausea as well as making the game behave more realistically, the change of direction can be also coupled with a skidding animation to make it even more convincing.

Random Trivia!

Inertia is so important (and common) that even the famous “Super Mario Bros.” (1983) for the NES features it, as well as a “skidding animation”.

In this section we will look at how to simulate inertia in a 1-dimensional space, where we can only move left or right.

When simulating inertia, the first things we need to know are:

- The top speed
- The acceleration rate
- The deceleration rate
- The direction we’re going
- The direction we want to go

Let’s think of the following situation, our character is running rightwards at a velocity v, measured in pixels per frame:

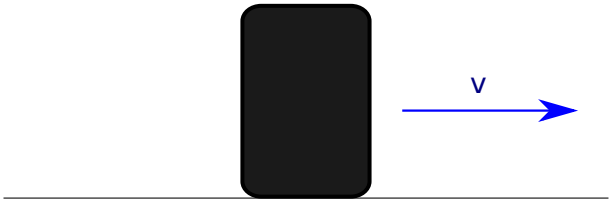


Figure 406: Example of character running

This means that the character’s x coordinate is moving every frame using the formula:

$$x_{n+1} = x_n + v$$

Now we suddenly want the player to start walking leftwards: we need to apply an acceleration a in that direction:

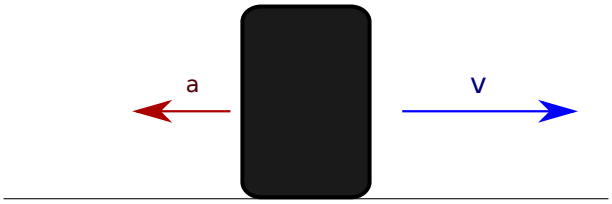


Figure 407: Applying an acceleration to a character running

The new acceleration will influence the velocity, frame by frame, with the formula

$$v_{n+1} = v_n + a$$

Since velocity and acceleration have opposite directions, the acceleration we're applying will start "eating away the velocity" frame by frame, until our character starts moving leftwards. This "eating away" phase is what gives the feeling of inertia.

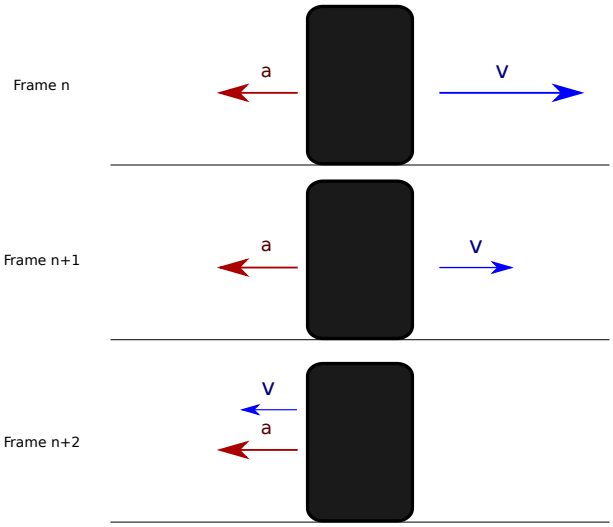


Figure 408: Applying an acceleration frame by frame leads to the feeling of inertia

Being acceleration and velocity both vectors, we can apply an acceleration both in a one-dimensional way (like a 2D platformer) or a 2-dimensional way (like a space shooter) and the formulas will still be valid.

Deceleration is a special case of what we've seen so far, with the exception that the acceleration will always have direction opposite to velocity and as soon as velocity reaches zero, we stop applying it.

Now we can start writing some code:

Listing 128: Code for simulating inertia

```
// ...
class Player{
    constructor(){
        this.MAX_SPEED = 50.0;  // Maximum speed
        this.ACCEL = 15.0;  // The acceleration rate
        this.DECEL = 30.0;  // The deceleration rate
        this.input_accel = new Vector2();  // Defines the direction we are accelerating
        this.velocity = new Vector2();  // Defines the direction and magnitude of our speed
        this.position = new Vector2();  // Defines our current position, in (x,y) coordinates
        this.is_moving = False;  // Tells us if we're moving
    }

    handle_input(){
        // First of all, we need to zero the input_accel, or we'll be working on "residual data"
        input_accel = Vector2.ZERO;
        // Now we can handle movement
        if (KEYBOARD.Left_Arrow_Pressed){
            this.input_accel.x = this.input_accel.x - 1;
        }
        if (KEYBOARD.Right_Arrow_Pressed){
            this.input_accel.x = this.input_accel.x + 1;
        }
        if (KEYBOARD.Down_Arrow_Pressed){
            this.input_accel.y = this.input_accel.y + 1;
        }
        if (KEYBOARD.Right_Arrow_Pressed){
            this.input_accel.y = this.input_accel.y - 1;
        }
        // If any component of the acceleration vector is not zero, we are moving
        if (this.input_accel != Vector2.ZERO){
            this.is_moving = true;
        }
    }

    handle_movement(dt){
        if (this.is_moving){
            // Vectors will take care of summing forces
            this.velocity = this.velocity + ACCEL * dt * this.input_accel;
            // We need to clamp the speed, to avoid going too fast
            this.velocity.clamp(MAX_SPEED);
        }else{
            // We are stopping, let's subtract the deceleration
            let velocity_value = this.velocity.length() - DECEL * dt;
            if (velocity_value < 0){
                // If, After decelerating, we have a negative value, we need to make it zero or
    the object will start moving backwards
                velocity_value = 0;
            }
            // We are just changing the length of the vector, so we can just clamp its length
```

```
49              this.velocity.clamp(velocity_value);
50          }
51
52          // Now it's time to move the object
53          this.position = this.position + this.velocity;
54      }
55  }
```

20.1.8 Corner correction

Let's consider the following situation: our character is jumping, but due to the player being a bit too eager on their jump, the collision boxes are still slightly overlapping:

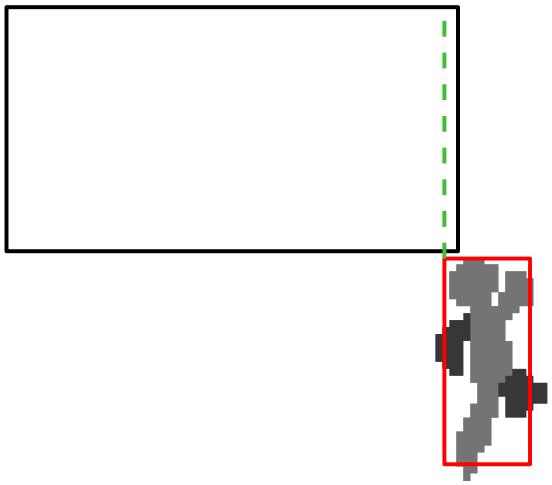


Figure 409: What would be a good collision response for this situation?

The character's head is just slightly hitting the corner of the geometry, but since collision detection doesn't really "care" about the kind of movement you're doing the jump will be stopped.

Wouldn't it be better if instead the character was just "slightly pushed" to the right so to complete the jump?

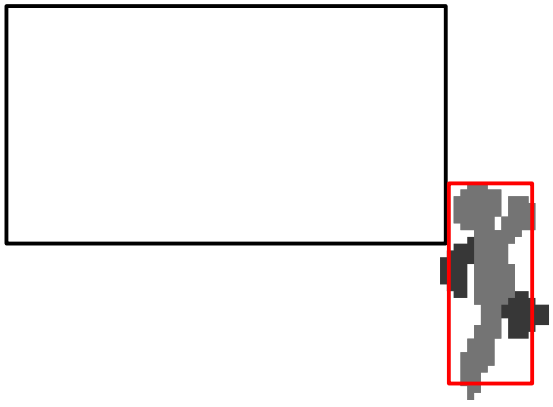
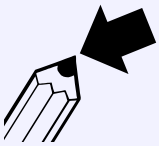


Figure 410: Corner correction makes for a more fluid experience

That would make it so the player doesn't get frustrated at a way-too-precise collision detection (remember me saying "you don't **want** precise collision detection?) and the game flow would be a lot smoother.

Tip!



This doesn't help only with side-scroller style run'n'jump games: if you're making a top-down game (like a 2D RPG) using tiles where movement is not tile-based. This will give you a smoother gameplay.

A possible implementation of a corner-correction algorithm, specifically for avoiding the jumping problem, could be the following:

Listing 129: Possible implementation of a simple corner correction

```
function update(dt){
    // Using the brute force checking for simplicity
    for (const block in player.findCollisions(blocks)){
        // ...
        if (player.is_jumping){
            // We are jumping, we need to check if we are ascending
            // this way we will avoid "bonking our head" on a pixel
            if (player.velocity.y < 0){
                // We know we are ascending, let's check how far we are
                // from the borders and react accordingly
                if (player.position.x > block.rect.right - 5){
                    // The player's left side is penetrating the block by
                    // less than 5 pixels, let's correct it
                    player.position.x = block.rect.right;
                }else if (player.position.x + player.width < block.rect.left + 5){
                    // The player's right side is penetrating the block by
                    // less than 5 pixels, let's correct it
                    player.position.x = block.rect.left - player.width;
                }else{
                    // The player is totally colliding with (bonking its head on) the block,
                    // without need for corner correction, let's just act normally
                    player.velocity.y = 0;
                    player.position.y = block.rect.bottom;
                }
            }
        }
    }
    // ...
}
```

20.2 2D Platformers

20.2.1 Simulating Gravity

Gravity in a 2D platformer is quite easy to simulate: you just need to apply a constant acceleration towards the direction your gravity is pulling (it doesn't have to be towards the bottom of the screen!) and move your objects accordingly to such acceleration.

Your acceleration should not be precise (like the physics constant $9.81m/s^2$), you don't want to make a physics engine: you want to make a somewhat convincing (or even better: entertaining) approximation of reality.

This is usually done before the player movement is used to update the character's status (but after the player input has been captured). Remember to add this acceleration before the collision detection is processed.

A useful precaution to avoid the bullet through paper problem when you are working with long falls: put a limit at the fall velocity (kind of like air friction limits an object's fall velocity) of your objects. By applying a hard limit to the velocity, your gravity will be realistic but won't break your simulation.

Listing 130: Code for applying gravity to an object

```
const GRAVITY_ACCELERATION = 10;
const MAX_FALL_VELOCITY = 500;
// ...
// Apply Gravity
speed_y = speed_y + GRAVITY_ACCELERATION;
// Cap the fall speed
if (speed_y > MAX_FALL_VELOCITY){
    speed_y = MAX_FALL_VELOCITY;
}
// ...
```

20.2.2 Avoiding "Floaty Jumps"

The previous trick shows a physics-accurate jumping: if we plot the height against time, we would get something that represents the curve of jump like the following:

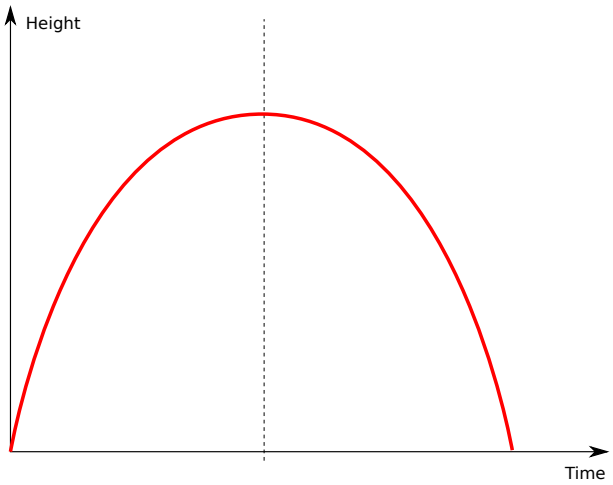


Figure 411: Plotting a physics-accurate jump

Although this can give the sensation that the character we're controlling is "floaty", which is not fun. In this case it's a better idea to enhance gravity when falling, to give the character some more "weight", which would be represented, more or less, by the following curve:

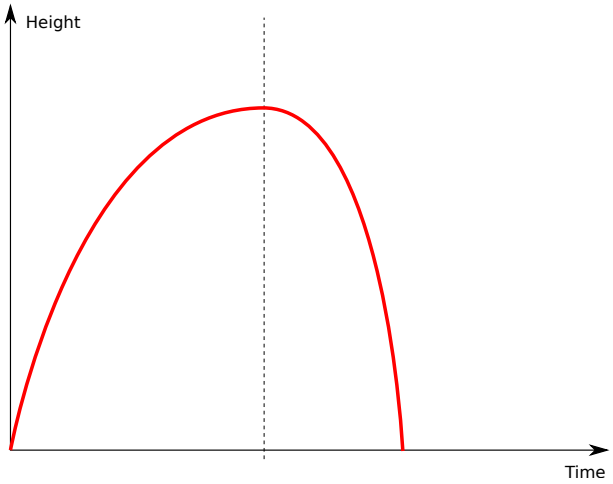


Figure 412: Plotting a jump with enhanced gravity

This can be obtained with few simple lines of code, not very different from the gravity example of earlier:

Listing 131: Code for jump with enhanced gravity while falling

```
const GRAVITY_ACCELERATION = 10;
const MAX_FALL_VELOCITY = 500;
const GRAVITY_FALL_MULTIPLIER = 1.5;
// ...
// Are we jumping?
if (speed_y < 0){
    // Apply Gravity Normally
    speed_y = speed_y + GRAVITY_ACCELERATION;
}else{
```

```
    // We're falling, enhance gravity
    speed_y = speed_y + GRAVITY_ACCELERATION * GRAVITY_FALL_MULTIPLIER;
}
// Cap the fall speed
if (speed_y > MAX_FALL_VELOCITY){
    speed_y = MAX_FALL_VELOCITY;
}
// ...
```

In this example we are assuming that the framework used uses the screen coordinate system, and jumping brings the player from bottom towards the top of the screen. If you want different behaviour (like gravity inversion in puzzle games), something a tiny bit more involved may be in order.

20.2.3 Making jumps “float differently”

As an addendum to the previous section, you can change how gravity is applied at the peak of the jump to give the player more time to correct their movement.

This can be done by reducing gravity when the jump is peaking, thus obtaining a plot similar to the following:

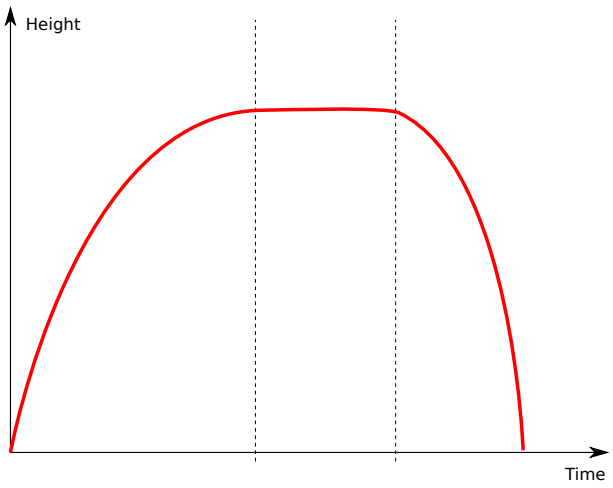


Figure 413: Plotting a jump with multiple gravity changes

This is an example of a jump with multiple “gravity changes”:

Listing 132: Code for jump with more gravity while falling and less when peaking

```
const GRAVITY_ACCELERATION = 10;
// We consider the jump "peaking" when the speed is between -50 and 50
const PEAKING_SPEED = 50
const MAX_FALL_VELOCITY = 500;
const GRAVITY_FALL_MULTIPLIER = 1.5;
const PEAKING_MULTIPLIER = 0.5;
// ...
// Are we jumping?
```

```
if (speed_y < - PEAKING_SPEED){
    // We're rising: apply Gravity Normally
    speed_y = speed_y + GRAVITY_ACCELERATION;
}else if (speed_y > PEAKING_SPEED){
    // We're falling, enhance gravity
    speed_y = speed_y + GRAVITY_ACCELERATION * GRAVITY_FALL_MULTIPLIER;
}else{
    // Our jump is peaking, lower gravity to give more time
    speed_y = speed_y + GRAVITY_ACCELERATION * PEAKING_MULTIPLIER;
}
// Cap the fall speed
if (speed_y > MAX_FALL_VELOCITY){
    speed_y = MAX_FALL_VELOCITY;
}
// ...
```

20.2.4 Ladders

[This section is a work in progress and it will be completed as soon as possible]

20.2.5 Walking on slanted ground

[This section is a work in progress and it will be completed as soon as possible]

20.2.6 Stairs

[This section is a work in progress and it will be completed as soon as possible]

20.2.7 Ledge Grabbing

[This section is a work in progress and it will be completed as soon as possible]

20.2.8 Jump Buffering

A nice trick used mostly in 2D platformers to allow for smoother gameplay is “jump buffering”, also known as “input buffering”.

Normally when a character is mid-air, the jump button does nothing, in code:

Listing 133: Code for jumping without buffering

```
function update(dt){
    // ...
    if (controls.jump.isPressed()){
        if (player.on_ground){
            // Jump
        }
    }
    // ...
```

```
}
```

Jump Buffering consists in allowing the player to “buffer” a jump slightly before the character lands, making the controls a bit less stiff and the gameplay more fluid.

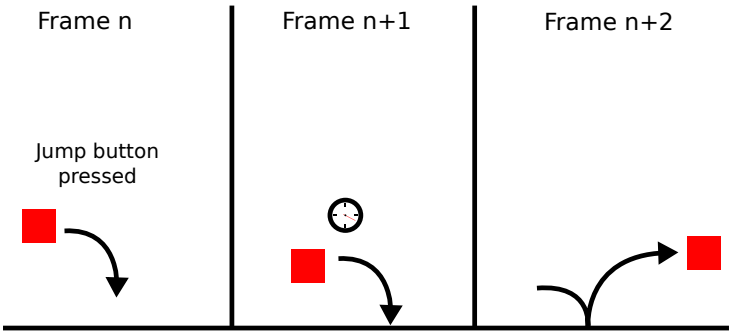


Figure 414: Example of how jump buffering would work

Jump buffering usually is put into practice using a timer, in a fashion similar to the following:

Listing 134: Jump buffering example

```
// ...
const jumpBufferTime = 5.0;
// ...
function update(dt){
    // ...
    if (controls.jump.isPressed()){
        player.hasBufferedJump = true;
        player.jumpBufferCountdown = jumpBufferTime;
    }
    // Take note on how this piece is outside the "jump is pressed" section
    if (player.hasBufferedJump){
        player.jumpBufferCountdown = player.jumpBufferCountdown - dt;
    }
    if (player.on_ground){
        if (player.jumpBufferCountdown > 0.0){
            // Jump
            player.jumpBufferCountdown = 0.0;
            player.hasBufferedJump = false;
        }
    }
    // ...
}
```

20.2.9 Coyote Time

Coyote time (also known as “edge tolerance”) is a technique used to allow a player to jump a few frames after they fall off a platform, allowing for a more fluid gameplay.

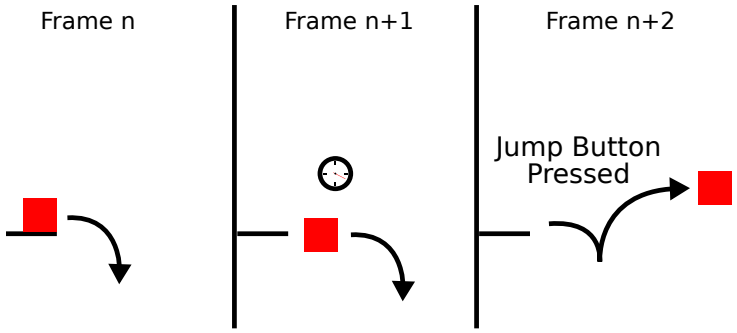


Figure 415: Example of how coyote time would work

The trick is starting a countdown as soon as the player leaves a platform without jumping, then if the player presses the jump button while that time is still going, they will perform the jump action, like they still were on a platform.

Listing 135: Coyote time code example

```
class Player{
    constructor(){
        this.coyote_time_started = False;
        this.coyote_time = 0;
        this.onground = False
        this.has_jumped = False
    }
    // ...
    update(dt){
        // ...
        if (this.onground){
            // Do stuff when player is on ground
            // ...
        }else{
            if (!this.has_jumped){
                // Player is not on ground and has not jumped, the player's falling
                if (!this.coyote_time_started){
                    this.coyote_time_started = true;
                    this.coyote_time = 5;
                }else{
                    this.coyote_time = this.coyote_time - dt;
                }
            }
        }
    }

    jump(){
        // This function takes care of jumping
        // ...
        if (this.coyote_time > 0){
            // Do Jump
        }
        // ...
```

```
34      }
35  }
```

20.2.10 Timed Jumps

A way to extend the mobility and challenge of a 2D platformer game is allowing players to jump higher the more the jump button is pressed: this allows the character to perform low and high jumps without much effort, making timing the jump button press a variable that adds to the challenge of a game.

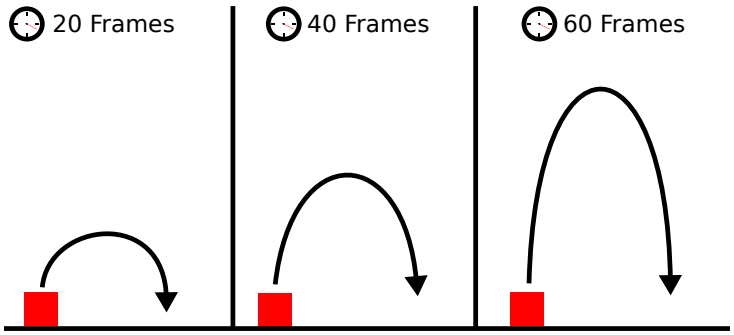


Figure 416: Example of how timed jumps would work

To work well, timed jumps need to be implemented by tracking the jump button's `onPress` and `onRelease` events. When the jump button has just been pressed, the character's `Y` velocity will be set, as soon as the button is released, such velocity will be capped, shortening the jump height.

Listing 136: Example code of how timed jumps work

```
class Player{
    constructor(){
        this.JUMP_VELOCITY = -12.0;
        this.y_speed = 0;
    }
    // ...
    onJumpKeyPressed(){
        /* The jump key has just been pressed (doesn't account the jump key being
        pressed from previous frames) */
        this.y_speed = JUMP_VELOCITY;
    }

    onJumpKeyReleased(){
        // The jump key was just released, cut the y_speed so the jump is lower
        if (this.y_speed < JUMP_VELOCITY / 2){
            // The speed is higher than the cutoff speed (in absolute value)
            this.y_speed = JUMP_VELOCITY / 2;
        }
    }
}
```

20.2.11 Wall Jumps

[This section is a work in progress and it will be completed as soon as possible]

20.2.12 Screen Wrap

[This section is a work in progress and it will be completed as soon as possible]

20.3 Top-view RPG-Like Games

20.3.1 Managing height

When it comes to Top-view RPG games, height is a way to give your game a lot more visual appeal. Let's see how we can manage height in our games.

20.3.1.1 Faking it

The simplest way to manage height in a Top-down RPG game is to not do so at all. If you have a good tileset the player may not even realize it.

Let's take the following simple example:

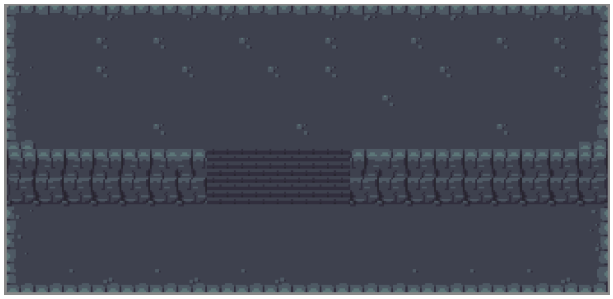


Figure 417: A simple example of fake height in RPG [12]

This can be seen as a "flattened" screen, where there are few collisions, while the tileset will "sell the effect".

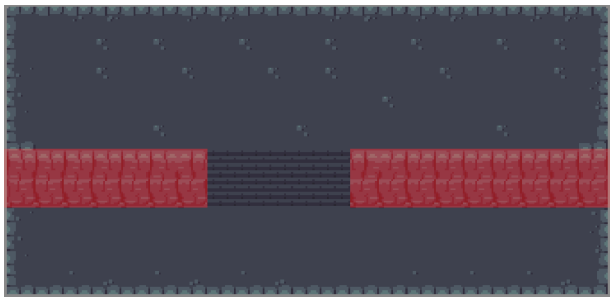


Figure 418: How few collisions may "sell" the effect of height [13]

[12] Dungeon Tileset, listed as Public Domain at https://opengameart.org/content/dungeon-tileset
[13] Dungeon Tileset, listed as Public Domain at https://opengameart.org/content/dungeon-tileset

In the previous image, the red sections are the tiles where collision is present: the stairs have nothing special, they are treated as any other “flat ground”, but the texture sells the effect of stairs.

We can go as complex as we want:

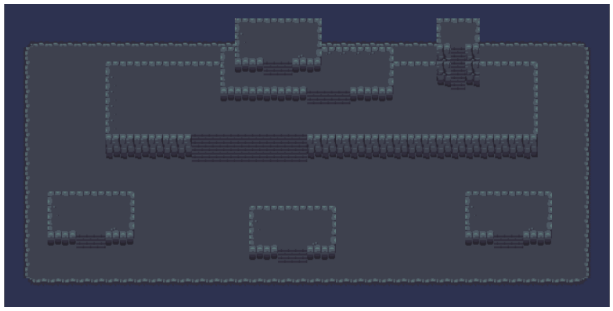


Figure 419: A more complex example of fake height [14]

And still have a somewhat simple and flat path, from bottom to top.

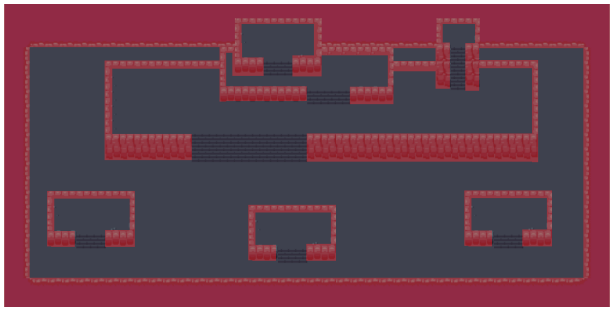


Figure 420: Even with complex tilemaps, the texture sells the height effect [15]

If you’re good at map building, this simple way of doing things can get you far enough to make a convincing effect and give variety to your environments without having to delve into complex algorithms that involve swapping layers or something similar.

20.3.1.2 Managing height for real

[This section is a work in progress and it will be completed as soon as possible]

20.4 Rhythm Games

20.4.1 The world of lag

Welcome to the world of rhythm games, as with all new experiences we shall start with… the final boss: Lag.

Lag will be one of the most problematic things you will have to account for: things are not as easy as you may imagine when it comes to implementing a rhythm game. Let’s see how to account for it, and eventually how to limit its effect on the player experience.

[14] Dungeon Tileset, listed as Public Domain at https://opengameart.org/content/dungeon-tileset
[15] Dungeon Tileset, listed as Public Domain at https://opengameart.org/content/dungeon-tileset

20.4.1.1 Input Lag

First of all: the ever-present "input lag": there is a certain time window between the moment the use presses a button and the moment the game receives such input. In the middle we find electrons running at breakneck speed through our keyboard circuitry, through the cable, to the motherboard, then the CPU, input abstraction layers in our OS, and finally the input system in our game.

And we didn't reach the game update stage yet.

Also we are not even accounting for the reaction time (about one second) from when the player sees something on screen and when they react.

Input lag is something that we cannot avoid, but there are countermeasures, as we will see below.

20.4.1.2 Video Lag

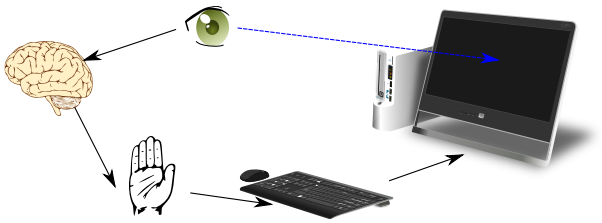


Figure 421: Reference image for video lag

As with the input lag, there is also a not-negligible video lag. The game has to prepare the image, send it to the video card, the card has to render it, apply effects and then send it to the screen, where the liquid crystals (or whatever technology we will have in the future) will have the re-align to create the colored pixels on screen.

20.4.1.3 Audio Lag

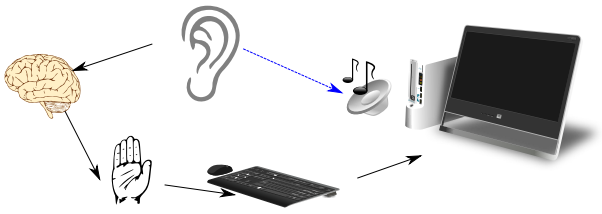


Figure 422: Reference image for audio lag

When the audio doesn't exactly match with the video, we talk about "audio lag", this has to be accounted for if you want to have a good rhythm game. In that case, there is a need to compensate for the audio lag, by starting each sound effect (or music) earlier or later by a well-defined amount of milliseconds.

20.4.1.4 Lag Tests

When it comes to lag, it is really difficult to estimate how the computer will react to our game, so we need a metric that will tell us what corrections we need to apply.

Such corrections are estimated comparing video and audio to the input: this way we will keep everything synchronized to the player input, making the game feel tighter.

First kind of test is done "video vs. input", the player has to push a button when something on the screen happens (like pushing rhythmically with a dot changing color), this way we can account for the video lag, compared to the input. This means we will obtain a $(video + input)$ lag measurement.

The second test done is the "audio vs. input" one, the player has to push a button when a sound cue happens on their speakers/headphones (like pushing rhythmically with a beep), this way we can account for the audio lag, compared to the input. This way we will obtain a $(audio + input)$ lag measurement.

By simple math we can account for the "video vs. audio" lag, like follows:

$$(video + input) - (audio + input)$$

$$video + input - audio - input$$

$$video + \cancel{input} - audio - \cancel{input}$$

$$video - audio$$

[This section is a work in progress and it will be completed as soon as possible]

20.4.2 Synchronizing with the Music

[This section is a work in progress and it will be completed as soon as possible]

20.4.2.1 Time domain vs. Frequency Domain

When we listen to music, we are essentially streaming a bunch of numbers as the time goes forward, so we can plot the amplitude of our waveform against time, as follows:

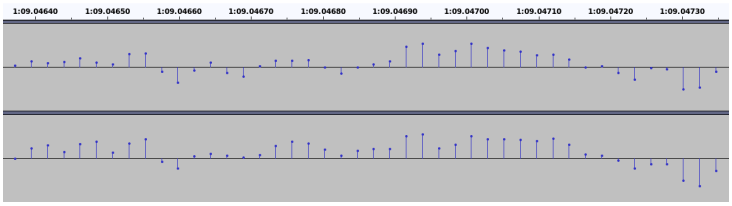


Figure 423: Plotting amplitude against time

In this case, when the time is the “independent variable” that we use to base our work, it’s said we’re working in *time domain*.

When we are working with games, we don’t really care about what will happen (music-wise) 5 minutes from now, instead we care about other things that are happening now. In that case, it may be interesting to work in *frequency domain*, which can look something like this:

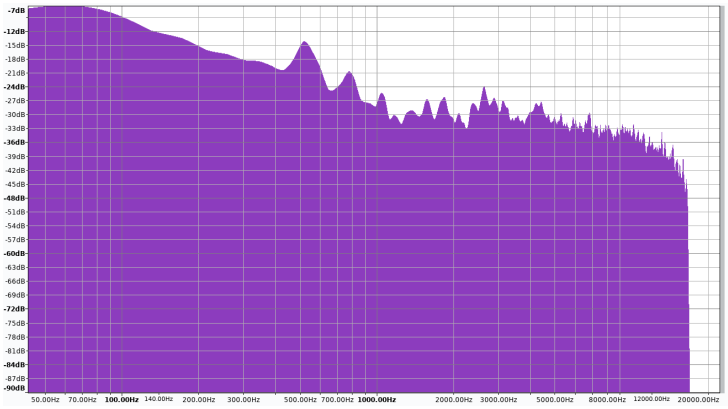


Figure 424: Plotting frequency domain

We can switch back and forth between the two domains with “transforms”, the most used is the Fourier Transform, and one of the most used algorithms to do it on computer is “FFT” (Fast Fourier Transform).

20.4.2.2 The Fast Fourier Transform

[This section is a work in progress and it will be completed as soon as possible]

20.4.2.3 Beat Detection

[This section is a work in progress and it will be completed as soon as possible]

20.5 “Bullet Hell” Style Games

The common definition of a “bullet hell” game is usually the one of a scrolling (usually space-themed) shooter with a very high level of difficulty and lots of enemy bullets on screen (hence the name).

20.5.1 Bullets

When it comes to this kind of game, it is vital that the enemy bullets are **well visible** (as stated in the shooters section in the "game design" chapter), this usually means that their color is brighter and has a lot of contrast with the background and the sprites on screen.

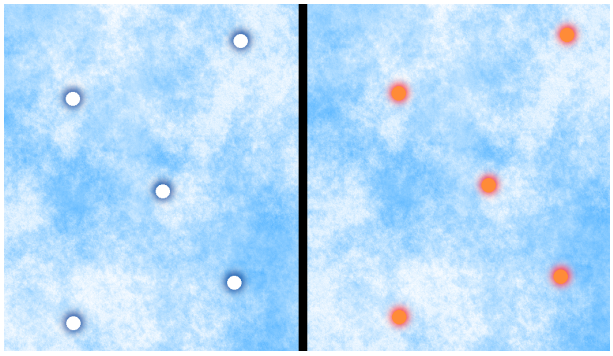
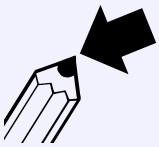


Figure 425: Example of how to better "highlight" bullets

Having "evident" enemy bullets makes the situation easier to assess, even when the situation becomes really chaotic. If you zoom out (or get your reading support farther from your eyes) you can see that the "non-highlighted" bullets (on the left side) tend to "blend in", while the "highlighted" (on the right side) version stay visible.

To highlight bullets, you can use "complementary colors", as shown in the use contrast to your advantage section.

Tip!



Bullet visibility is so important that in many games bullets are the last thing to be drawn before the player: this means they're drawn over explosions, other enemies and your own bullets too.

If you let players lose sight of bullets by drawing graphical effects over them, the game will feel unfair.

20.5.2 The Ship Hitbox

In the bullet hell genre usually the player ship's (or character of some kind) hitbox is usually much smaller than the visible sprite, this makes the game a little bit "easier than it seems", but at the same time it doesn't mean that the game is easy either.

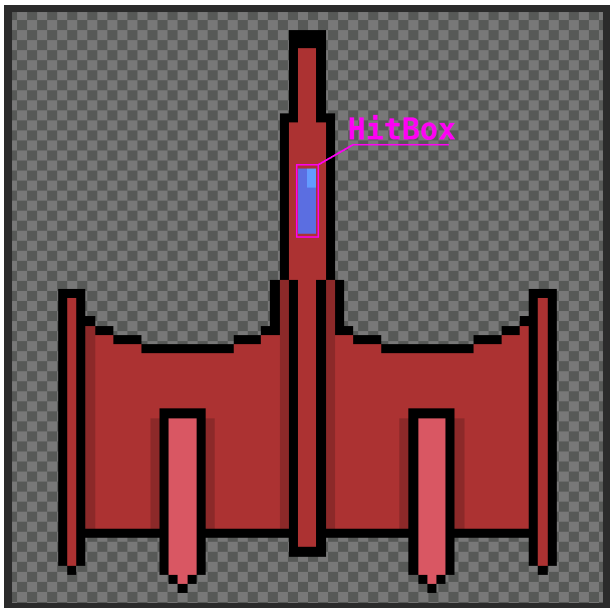


Figure 426: An example of a Bullet Hell ship hitbox

In this image, the ship's hitbox is limited to the cockpit, some games prefer some area that could be considered the "ship engine" while others just have a "core" of some sort.

Many games of the genre even make the hitbox a single pixel!

20.5.3 Screen-clearing bombs

Another mechanic used in bullet hell games are "screen-clearing bombs": these are used to rid the screen of the gigantic number of bullets on it, to give the player some breathing room.

In some games bombs may be also used to destroy small enemies and damage bigger ones. The screen clearing move can happen in many ways: the most common is just making the bullets disappear, but other games prefer turning the "destroyed bullets" into small collectibles that can give the player points.

20.5.4 Clearing bullets on pattern changes

Some bullet hell games feature multi-phase bosses, where the boss changes attack strategy, and thus their bullet pattern and speed, at certain points of the fight (usually when reaching a certain amount of health left). This may create some issues to the player, since the new bullets may cover all "escape routes" willingly left by the previous bullets, thus making it impossible to not die.

A simple and effective strategy is clearing the screen of the enemy bullets automatically when the boss changes phase (sometimes transforming the bullets into collectibles for score), this will allow for a quick breather to the player, as well as a somewhat smooth transition to the new phase.

20.5.5 Find other chances to clear some bullets

Some games find creative ways to clean up a screen cluttered with bullets: for instance some bullets can turn into collectibles when a pickup is touched by the ship.

Random Trivia!

In ZenoDyne R powerups are real, "physical" objects, and as such they block incoming bullets, so they can be strategically used as a "shield", and then pick them up at the last second.

Some games like to clear the screen (without giving out collectibles) at the beginning of a boss fight, to give a "clean slate" to start the boss with.

20.5.6 Turn enemy bullets into collectibles at the end of a boss fight

An interesting form of bonus that is often present in bullet hell games is turning all the boss's bullets on screen into collectibles at the end of a boss fight.

Since this genre of game gets progressively harder the more bullets are on screen, this small trick rewards players for being good at dodging, while players who used screen-clearing bombs will have a smaller bonus.

20.5.7 The "Chain Meter"

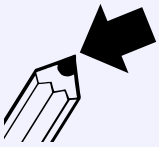
This is a mechanic used in many bullet hell games: the chain meter is a meter that gains value according to the number of enemies you kill in a certain amount of time and giving a score multiplier according to it.

This meter will automatically discharge with time, making it hard to keep up a high score multiplier, adding challenge to the player and rewarding them for being good at destroying enemies in large numbers fast.

Usually the meter has 5 levels, starting from level 1, when the meter is full, the meter "gains a level" and a score multiplier is applied accordingly. For instance we can have:

- Level 1: 1x multiplier (normal score)
- Level 2: 2x multiplier (double score for each killed enemy)
- ...
- Level 5: 5x multiplier

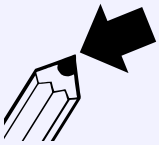
Tip!



You can code the "discharge" so it is faster at higher levels. This will bring even more challenge at keeping the level high.

When the player dies, the counter gets completely emptied and thus the multiplier gets reset to 1x.

Tip!



Alternatively, you can halve the level of the meter on player's death.

20.5.8 Managing the player's death

It is very common in the "bullet hell" genre to punish the player's death with a strong cut at the ship's power.

This has a problem: a player dying may spiral into a fully-fledged game over because the ship is now extremely underpowered compared to the stage the player died in.

A solution often used in this genre of games is having a dying player's ship have a random chance of releasing a random number of powerups and bomb pickups on death, thus allowing the now-weakened player to "regain some strength" and continue their game.

20.5.9 The Enemy AI

Probably the hardest part to develop in a "bullet hell" style game is the enemy AI and how to make the enemy bullets form a pattern that is hard but not impossible to dodge.

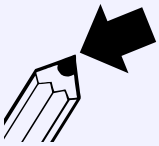
Since the gameplay is very hard to balance, this genre seldom sees a "procedural game" (an exception that comes to my mind is "Task force Kampas", which features procedural levels, but handmade bosses).

A way to make the game feel more fair is programming the AI so it doesn't shoot "on the way out" of the screen. Each enemy essentially has 3 phases in its AI:

1. Enter the screen
2. Fight (usually by shooting a single pattern or a continuous stream of the same pattern)
3. Exit the screen (or die)

When the enemy exits the screen, it should stop shooting and just orderly leave.

Tip!



If your game features enemy turrets, they should stop shooting when they are behind the player's ship: the player is already busy enough handling shots from the front. Shooting from behind makes the game unfair.

20.5.10 Be fair to the player, but also to the computer

The title may be a bit awkward: how can you "be fair to a computer"?

Computers don't have feelings, but players do. And letting the players kill the enemy before the AI activates takes away all the challenge from the game itself: the enemies become cannon fodder when the player's weapons have

enough “power” to instantly kill most of the enemy forces.

So to apply this, you should probably make the enemies invincible until they’re fully on screen: this way the player sees them and doesn’t kill them beyond the top of the game area.

20.5.11 Inertia

Control is everything in a game where a pixel can be the difference between life and death of your ship/character. This means that heavy inertia does not play well with the “bullet hell” genre.

This also doesn’t mean that you can’t apply any, just be careful and don’t go overboard. When a player dies because their ship went too far due to inertia, they will get mad at the game, and by transitive property, at the devs.

20.5.12 Some examples

There are games that make the most of the “bullet hell” mechanics to give player more challenge, or risk/reward choice.

One game is “Touhou”, which has a “grazing” mechanic: if a bullet slightly grazes (but does not hit) your hitbox, you will see some sparks and get a bonus in points.

Another title that makes the most of giving the player a “risk vs reward” choice is Ikaruga, with it’s “polarity” mechanic. Your ship has two sides: black and white, each side is able to absorb (and so is also immune) to the bullets of the same color, but also does more damage to the enemies of the opposite color.

20.6 Match-x Games

20.6.1 Managing and drawing the grid

When it comes to a match-x game, a good data structure for the play field is a matrix.

In most programming languages, a matrix is saved as an “array of arrays”, where you have each element of an array representing a row, and each element of a row is a tile.

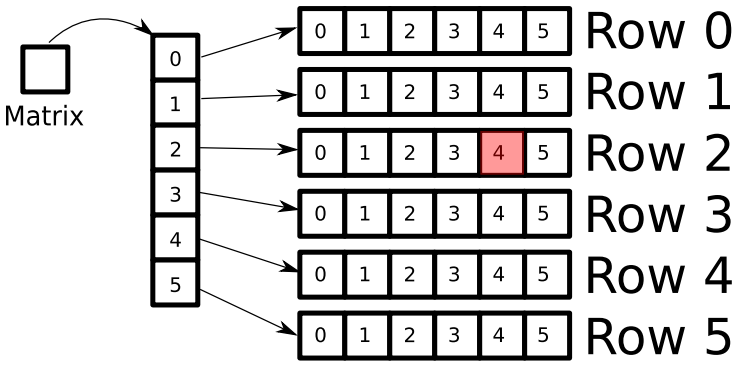


Figure 427: Example of a matrix, saved as “array of arrays”

This is a really nice way to interpret a grid, but at the same time it can open the door to some pitfalls if you’re not

careful.

Many programming languages allow for direct access to an element inside an “array of arrays” by using multiple access operators (usually the `[]` operator) in a row.

Usually each element you access with the first `[]` operator represents the rows, while the second time you use `[]` you will access the columns, this will make it so you need to access an element directly as follows: `matrix[y][x]` where “y” is the row number and “x” is the column number, which can prove counter-intuitive.

In the previous example, if we want to access the highlighted item, at the third row (indexed at 2), and in the fifth column (indexed at 4). We have to use `matrix[2][4]`, which is the opposite of what many people are used to when they think in `(x,y)` coordinates.

Try to keep visual representation and data structures separated in your mind, to avoid confusion.

20.6.2 Finding and removing Matches

If you’re doing a simple match-x game where you can only match tiles horizontally or vertically, the algorithm to find matches is quite simple, both conceptually and computationally.

The main idea is dividing the “horizontal matches” from the “vertical” ones. This will allow to simplify the algorithm and avoid some pitfalls (unless you want to give bonuses for “T-shaped” and “L shaped” matches).

For horizontal matches the idea is running through each row, keeping some variables representing the length of the match, as well as the color of the current “ongoing” match. As soon as we find a different color, if the length of the “ongoing” match is higher than “x” (usually 3), we save the references to the tiles involved for later removal.

Similarly we can do the same algorithm for vertical matches, by running through each column and saving the matches.

Here is a pseudo-code example:

Listing 137: Finding horizontal matches in a match-3 game

```
function findHorizontalMatches(){
    let matchLength = 0;
    const minMatchLength = 3;
    matches = [];
    for (const row in matrix){
        let lastMatchingTile = null;
        for (const column in row){
            let currentTile = matrix[row][column].tile;
            if (currentTile == lastMatchingTile){
                matchLength = matchLength + 1;
            }else{
                if (matchLength >= minMatchLength){
                    // We need to memorize all the tiles involved in the match
                    for (let i = column - matchLength; i <= column; i++){
                        matches.push(matrix[row][i]);
```

```
                    }
                }else{
                    // No matches, reset the counter and set the current tile as last matching
                    matchLength = 1;
                    lastMatchingTile = currentTile;
                }
            }
            // We need to account for the right-hand border corner case
            if (column == matrix[row].length){
                if (matchLength >= minMatchLength){
                    // We need to memorize all the tiles involved in the match
                    for (let i = column - matchLength; i <= column; i++){
                        matches.push(matrix[row][i]);
                    }
                }
            }
        }
    }
    return matches;
}
```

Let's talk a second about the last rows in the algorithm: they are specifically tailored to address a corner case that happens when there is a match that ends on the right border of the screen.

If such code was not there, the match number would grow by one, then the for loop would reset everything and we'd lose such match.

Similarly, we can make an algorithm that allows for vertical matches to be memorized for later removal:

Listing 138: Finding vertical matches in a match-3 game

```
function findVerticalMatches(){
    let matchLength = 0;
    const minMatchLength = 3;
    matches = [];
    for (const column in matrix){
        let lastMatchingTile = null;
        for (const row in column){
            let currentTile = matrix[row][column].tile;
            if (currentTile == lastMatchingTile){
                matchLength = matchLength + 1;
            }else{
                if (matchLength >= minMatchLength){
                    // We need to memorize all the tiles involved in the match
                    for (let i = row-matchLength; i <= row; i++){
                        matches.push(matrix[i][column]);
                    }
                }else{
                    // No matches, reset the counter and set the current tile as last matching
                    matchLength = 1;
```

```
                    lastMatchingTile = currentTile;
                }
            }
            // We need to account for the bottom border corner case
            if (row == matrix.length){
                if (matchLength >= minMatchLength){
                    // We need to memorize all the tiles involved in the match
                    for (let i = row-matchLength; i <= row; i++){
                        matches.push(matrix[i][column]);
                    }
                }
            }
        }
    }
    return matches;
}
```

Both algorithms run in $O(n)$, where "n" is the number of tiles on the screen.

Now we can proceed to remove every tile that has been memorized as "part of a match", the quickest way may be to set such tile to "null" (or an equivalent value for your programming language).

20.6.2.1 Why don't we delete the matches immediately?

We could, but that would open the door to a pitfall that could be tough to manage: in case of a "T" match, we would find that the "horizontal matches" algorithm deletes part of said match, and the "vertical matches" algorithm wouldn't be able to complete the "T match", because the necessary tiles are deleted.

Let's see the image to understand better:

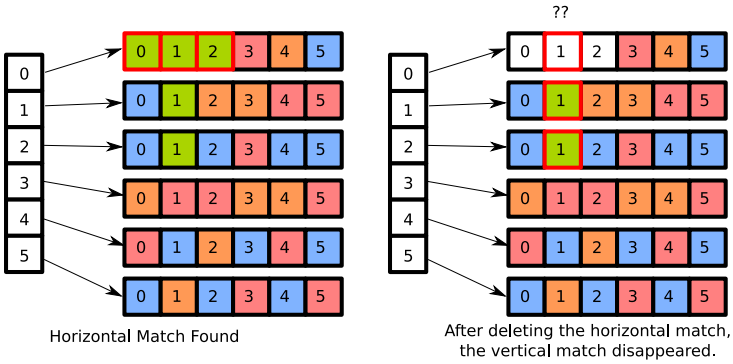


Figure 428: What happens when deleting a match immediately

As visible from the first image, there is a T-shaped match involving cells 0,1,2 of row 0, cell 1 of row 1 and cell 1 of row 2.

If we deleted the horizontal match immediately, we would lose the possibility of completing the vertical match (highlighted in the second image).

Instead we memorize everything first, and then delete all the matches at once, without the risk of losing anything.

20.6.3 Replacing the removed tiles and applying gravity

At this point, it is easy to make the "floating tiles" get into the right position: the hardest part is taking care of the graphics inbetweening that will give us that "falling effect" that we see in many match-x games.

Here is a possible implementation of the algorithm:

Listing 139: Eliminating matches and preparing the tween table

```
function removeMatches(){
    for (const tile in matches){
        matrix[tile.y][tile.x] = null;
    }
    matches = [];
}

function findFallingTiles(){
    // Our falling tiles list, will be used for tweening
    let falling_tiles = {};
    // We scroll each column of the board
    for (constcolumn in matrix){
        let found_space = false;
        let spaceY = 0;
        // We scroll each row, from bottom to top
        let y = matrix.length();
        while (y > 0){
            let tile = matrix[y][x];
            if (found_space){
                // If the current tile is not a space, bring it to the lowest space
                if (tile != null){
                    // Put it in the correct spot
                    matrix[spaceY][column] = tile;
                    tile.y = spaceY;

                    // Set the old position to empty
                    matrix[column][y] = null;

                    // Set the tween starting position for later
                    falling_tiles[tile] = y;

                    // We reset the found_space for next loop
                    found_space = false;
                    // We need to re-scan this tile (it will be empty, but there may be more
    tiles above)
                    y = spaceY;

                    // Reset spaceY for next loop
                    spaceY = 0;
                }
```

```
            }else if (tile == null){
                found_space = true
                // In case we didn't find a space yet, this is the one
                if (spaceY == 0){
                    spaceY = y;
                }
            }
            // We go up one tile
            y = y - 1;
        }
    }
    return falling_tiles;
}
```

We can make use of the table to tween the graphics with our favourite method: the map we used will contain the starting position of the tween, while the end position will be the position set on the tile itself.

After the graphics tweening, we need to create the new tiles that will go and fill up the holes that have been created by our matches and moved tiles.

Listing 140: Creating new tiles and preparing another tween table

```
function createNewTiles(){
    // Our falling tiles list, will be used for tweening
    let falling_tiles = {};
    // We scroll each column of the board
    for (const column in matrix){
        for (const tile in column){
            if (tile == null){
                let new_tile = Tile.random();
                new_tile.y = - 64; // A value that is out of the board
                matrix[column][tile] = new_tile;

                // Add this tile to the falling tiles mapping
                falling_tiles[new_tile] = tile.y;
            }
        }
    }
    return falling_tiles;
}
```

After creating the tiles and tweening them in place, it will be necessary to check for more matches that have been created from the falling tiles (and eventually notify some kind of “combo system” to apply a “score multiplier system” or even an achievement system using the Observer Pattern).

[This section is a work in progress and it will be completed as soon as possible]

20.7 Cutscenes

When you want to advance your storyline, a great tool is surely the undervalued cutscene. The game temporarily limits its interactivity and becomes more “movie-like”, making the storyline go forward. Cutscenes can be scripted or just true video files; in this chapter we will analyze the difference between the two, advantages and disadvantages of both and how to implement each one, from a high-level perspective.

20.7.1 Videos

The easiest way to implement cutscenes in most engines and frameworks, is to use videos. Many frameworks have native support for reproducing multimedia files with just a few lines of code, which makes this the preferred choice when it comes to the code.

The bad thing is that videos are a “static” format. They have their own resolution, their own compression and characteristics, this means that when a video is presented at a higher resolution that is own native one, we’re bound to have artifacts due to upscaling.

[This section is a work in progress and it will be completed as soon as possible]

20.7.2 Scripted Cutscenes

[This section is a work in progress and it will be completed as soon as possible]

20.8 Savefiles

[This section is a work in progress and it will be completed as soon as possible]

20.8.1 Serialization

[This section is a work in progress and it will be completed as soon as possible]

20.8.2 Deserialization

[This section is a work in progress and it will be completed as soon as possible]

20.8.3 Tips and tricks

[This section is a work in progress and it will be completed as soon as possible]

20.8.3.1 Add a version to your savefiles

[This section is a work in progress and it will be completed as soon as possible]

21 Dealing with piracy

> One thing that we have learned is that piracy is not a pricing issue. It's a service issue. The easiest way to stop piracy is not by putting anti piracy technology to work. It's by giving those people a service that's better than what they're receiving from the pirates.
>
> *Gabe Newell*

When you're selling a videogame (or anything that is covered by copyright), you will sooner or later have to deal with piracy.

Piracy usually refers to the act of distributing (either physically or digitally) copyrighted works (including music, software and movies) without the copyright holder's permission, usually regardless of whether such distribution is done for free or not.

Note!

There are some exceptions to forbidding the physical distribution of copyrighted goods: you usually can sell your regularly purchased physical books, movies and games, but you cannot make a copy of them and distribute that. For more information about the matter search for the terms "first-sale doctrine".

21.1 Objectives of anti-piracy measures

Many people think that the main objective of anti-piracy measures is to stop people from copying the game and distribute it for free, thus damaging the companies that develop and publish the game.

That is true in part, the real objective is to stop people from distributing the game in the first few weeks of a game's life cycle: that's where most of its earnings lie.

A game usually makes from 50 to 70% of its earnings in the first month, if we don't consider DLCs and expansions, so if an anti-piracy system is able to ward off pirates for that period, the majority of earnings can be saved.

After that period, the damage from cracked anti-piracy is diminished, since the interest towards a game will fade with time. After a while, you may even think about just removing anti-piracy to improve performance (its impact is always non-zero) or to boost preservation efforts.

21.2 Alternatives to anti-piracy measures

Even though piracy infringes on someone's copyright and right to earn money from their hard work, "wagging their finger" and implementing strict anti-piracy measures might not be the best solution.

Piracy may hide some underlying issues: it is a "service problem"; we are missing something that may be considered fundamental by others.

21.2.1 Regional Pricing

Not all regions have the same per-capita income. Let's make an example with data from the year 2025, both for income and price for triple-A games.

If we consider a triple-A game that costs about 80$ we can find out that:

- The US has a monthly per-capita income of about 5400$. A 80$ game would take up about 1.5% of such income. Some might consider it a fair price for entertainment, some may not (I think 80$ is too high, but I'm used to the good old times of 60$ games).
- Albania has a monthly per-capita income of about 850 to 1000$. An 80$ game would take up about 8% to 9.4% of such income. Starting to see a problem?
- In Brazil there is a monthly income that averages at about 612$. An 80$ game would take up a staggering 13% of such income. If a person earned minimum wage (barely scraping 275$) buying such a game would be crazy (taking up about 30% of their income).

So the lower-income countries may see a higher piracy rate for a simple reason: they cannot afford the game.

Having regional pricing that make your games possible to purchase in all regions can be a good way to cater to this audience: people who want to support you, but can't. Sometimes a simple currency-conversion is not enough, since it doesn't always keep track of the spending power of people.

21.2.2 Give out free demos

Some people may just want to try a game, and it appears that giving out free demos is a practice that has been lost to time.

As usual, when a market vacuum is created someone will fill it. Pirates might be someone who fills such market vacuum by giving out full versions of games.

By giving out a scope-limited demo (the first few levels or the first chapter) you take away such market vacuum and give prospective players a chance to try your game before committing to a full purchase.

21.2.3 No DRM entirely

Let's face it, if someone doesn't want to pay for a game, they will find a way to not pay for it, either by finding a cracked copy, or by cracking it themselves (provided they have tools and knowledge on how to do so). It's a sad reality.

Considering how invasive DRM systems have become (including but not limited to Kernel-mode drivers which sometimes almost behave like spyware) and their consequences on performance, people are becoming more and more conscious.

So the choice of using no DRM at all might be considered favourably by the general public, to the point it might be considered a positive feature.

21.2.4 Poisoning the well

I'm putting this as an "alternative" because you don't really need a DRM to enact this.

An alternative could be putting a "pirate version" out there yourself: many groups see pirating software as challenges to overcome, thus putting out a fake "pirate version" of the game may stop the pirates from targeting your game for a bit.

The secret is that such "special version" is actually different from the retail version: it is built in such a way that it either crashes randomly, has a game-breaking bug down the story or misbehaves in such a way that makes it impossible for the person who downloaded the pirated copy to enjoy the game as it should be. Your fantasy is your friend here, just make things believable enough that the bug you introduce can be chalked to a "bad release" or something else but a honeypot.

It's a dirty tactic, but it can prove to be effective.

Random Trivia!

Game dev Tycoon has implemented a similar tactic, but also had some "call home" code that notified the developers about how many copies of their game were pirated. As much as it might seem innocent from an outside view (it's just statistics), this wasn't well-received by all gamers (including the ones who purchased the game) because of its spyware-like behaviour.

21.3 Implementing anti-piracy measures

If you really want to implement anti-piracy in your game, you may want to take some precautions and make things harder for pirates, just to earn enough time to last for the majority of the sales period.

Note!

Note from the author: I don't like DRM systems in general. I find their modern DRM behaviour invasive and bordering on spyware-like. For fairness I will report on DRM and anti-piracy systems, but I don't really encourage their use, since it's really easy to border on "abuse".

21.3.1 Third-party DRM systems

The best way to do something, is to make someone else do it.

As much as trying to make your own anti-piracy and Digital Rights Management system might sound exciting, it may soon become extremely tedious and not as effective as some solutions offered by players who have been working on the matter for years.

Steam has its own DRM which is, for better or worse, well-understood by pirates and easily defeated, but it's always better than "no DRM" when it comes to protecting your own copyright.

There are third party solutions that I won't list here, but if you really want to implement one, you can look for more information online: giving out names might date this book since solutions change quickly.

21.3.2 Debugger detection

One way to stop the more "naive" trials to crack your game is debugger detection.

Implementations vary with your programming language, but here are various methods:

- On Linux you can check `/proc/self/status` and check for a value in the `TracePid` field, if it's not zero then a debugger is attached;
- On Windows, there is a handy `IsDebuggerPresent` function that returns a boolean;
- With JavaScript you can create a small script that takes the current date and time, calls the `debugger` keyword and then takes the date and time again. If the difference between the two dates is longer than an arbitrary length of time, a debugger might be attached. This works because the `debugger` keyword triggers the debugger only if a browser console is open.

Once you find that a debugger is attached, you can decide the best course of action, usually by closing the program or forcibly crashing it.

21.3.3 Checksums

A possible check you can do to prevent, or at least limit, piracy is checksums: your program will need the ability to inspect its own executable and its parts, take the bytes that make it up and get a result.

Such result is the "summary" of your own game, a function or a bunch of functions. If such "summary" (which is called "checksum") is wrong, you'll know that the executable has been tampered with.

Checksum algorithms are beyond the scope of this book, thus we won't present any, but there are lots of interesting resources online about the matter.

21.3.4 When an anti-piracy flag is triggered, be generic

When implementing anti-piracy checks, you should not show any piracy-related message: those messages can be traced in the executable file (the `strings` GNU/Linux command can show all the "printable strings" inside an executable) and thus give a lead on where your anti-piracy code is located, making its removal that much easier.

When an anti-piracy is triggered, you should try and hide it behind a generic error that is already used somewhere else (the more often such error is used, the better, like hiding a tree in a forest). Such error can crash the program and deter some people by making them think that the pirated copy is low quality.

Another way could be tinkering with the game variables and make the game misbehave worse and worse until it's unplayable or it crashes.

21.3.5 Obfuscation and redundancy of checks

As stated in the previous paragraph, it's quite easy to extract the printable strings from an executable file, so trying to make those strings not meaningful (or "obfuscated") is definitely a tool to slow down piracy of your game.

Another means of obfuscation is making parts of the code needlessly complex (usually the ones related to piracy checks), and never referring to the piracy checks directly, where possible. This is another layer of obfuscation.

The final objective is sending the pirates in a wild goose-chase, where they go through reams of code to get almost nowhere (like going through a stack of functions just to get to an index that is the current one, plus two) and have them try to wrap their head around code that is much too complex for what it should do.

Also there should be many (possibly different) versions of your "piracy check code" that should be replicated throughout your game, so that if a version is "patched out", there will be 5 more that can be triggered somewhere down the line. This is called "redundancy".

It is willfully inefficient and "not dry", but it is necessary when you want to slow down pirates.

21.3.5.1 Play with the pirates

Another way could be toying with the people who pirate the game: to make it more or less obvious that you know they're pirating the game.

Here are some examples:

- Game dev tycoon: after playing a while, your studio will go bankrupt due to piracy;
- The Talos Principle: you get stuck in an elevator;
- Crysis Warhead: guns shoot chickens which deal no damage;
- Alan Wake: the character wears a pirate-like eyepatch for the whole game;
- Serious Sam 3: an invincible, super-fast and enraged Arachnid spawns, following you for the whole time you are alive.

Your game does not need to become unplayable, that will only annoy the players and make them hate you, but a stern warning essentially saying "hey, we know you didn't buy this game" might have a bigger effect than you think.

Some situations may even lend them to occasions where the players are led to publicly oust themselves as people who are not playing a legitimate copy: if a problem pops out that seems a bug, but is actually something that happens only to pirated games, the player may open a support ticket asking for information which will just get a "if you stop pirating the game, it will fix itself" as a response.

21.3.6 Don't stop the pirates immediately

When a piracy flag is triggered, you shouldn't immediately crash the game. Crashing the game immediately will tell whoever is trying to pirate the game that their edits to the executable have been detected, giving them an immediate feedback on their job.

If you slowly begin introducing issues, or wait for a few minutes or hours before crashing your game, you will slow down the pirates.

Even better if, when a piracy check is triggered and a game crashed, that every save file is wiped, forcing the pirate to re-play a good portion of the game at least once (before they start backing up their save files) to see if their edits work.

Another interesting thing to implement would be wiping the saves at the game boot if a flag has been set (for instance in the settings, with a name completely unrelated to piracy, again “hiding a tree in the forest”), making restoring save files harder or even meaningless.

Part 6: Refining your game

22 Balancing Your Game

> The trick to balance is to not make sacrificing important things become the norm
>
> *Simon Sinek*

An imbalanced game is a frustrating game, and most of the time balancing a game is one of the toughest challenges a game developer/designer can find themselves to have to face.

Let's talk about some principles and guidelines that can help you balancing your game and keep your players challenged but not frustrated.

22.1 Do not annoy the player

The "master principle" everyone should follow (in my humble opinion) is "do not, under any circumstance, annoy the player".

You should not trade the "fun" of your game for any other mechanic (like showing an advertisement to allow them to continue playing): that is equivalent to betraying your player, makes the game feel unfair and un-fun.

Here are some examples of mechanics that will surely annoy the player:

- **Sudden spikes in difficulty:** when you have a sudden spike in difficulty, the player feels stumped and the game tends to lose its charm, you are "interrupting the flow" of the game by placing an arbitrary hurdle on your players' road;
- **Off-screen instant-death traps:** having something deadly that pops out from off-screen and kills the player is unfair and will make your players scream in agony and vexation, if you want to place some obstacles that pop from off-screen you should "telegraph" them. "Telegraphing" is a technique where you send a warning signal to the player that danger is coming. For instance a huge laser that instantly kills you should be preceded by a couple seconds by a yellow "!" signal on the right side of the screen, where the laser is due to strike. Another way to telegraph said laser would be to illuminate the part of the screen that is about to be hit, like the light of the laser is coming up;
- **Arbitrary invisible time limits:** If you suddenly interrupt the player's game with a "time up" and you have no countdown on the screen, the player will get frustrated, guaranteed;
- **Taking control away from the player:** Not allowing the player to move (getting blocked by an enemy and killed) or just not allowing the player to adjust their jump mid-air is a surefire way to make them not play your game anymore.

22.2 Favour the player when possible

In the process of balancing a game, as a game developer/designer you will surely find yourself in front of the following decision time and time again:

> Shall I favour the game's precision or should I give some leeway to the player?

The answer is the latter 99% of the time.

Giving some leeway to the player, for instance by having a more generous hit-box that allows you to stay alive even if a bullet grazes your character makes the game seem more "fair".

There are infinite ways to make a game challenging without having to force the player into accepting very precise hit-boxes or extremely tight gameplay.

22.3 Difficulty curves

When designing our game, it may be useful (sometimes mandatory) to have a high-level view of how our game's difficulty will evolve as the game itself is played. If we take a Cartesian plane and define time as the x axis, while the "perceived difficulty" is plotted on the y axis, we would obtain a **difficulty curve**, a high-level representation of how difficulty evolves as the game is played.

Knowing some basic difficulty curves, as well as their pros and cons, may give you an idea of how you want to build and balance your game. This section will be heavy on charts, so be prepared!

22.3.1 Simple Lines

Let's start with simple lines, they can be straight lines or simple curves that don't feature any waviness or wobbliness. These are usually the simplest to learn but that doesn't mean they are free from complicated drawbacks. Let's check some out.

22.3.2 Flat Line

The first curve is the "flat line", which is a simple horizontal line that spans the whole playtime. It can't get simpler than that.

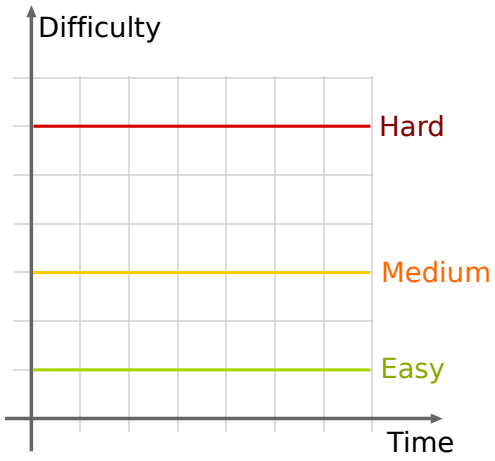


Figure 429: A Flat line difficulty curve

When the player selects a difficulty level, the difficulty stays around that value (with no real perceivable change) for the entire playthrough. That means that there is no "evolution" to take care of and no long-term balancing to perform.

This curve also represents a way of balancing your game that gets boring rather quickly, since the player gets better at the game as time passes, but the game doesn't "follow them" by giving them a higher challenge.

22.3.2.1 Linear Increase

To solve the issue of the flat line, you can add a linear increase to your difficulty in an effort to "keep up" with the player.

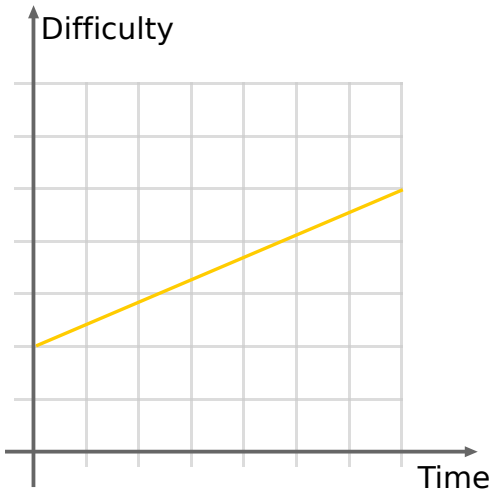


Figure 430: A linearly increasing difficulty curve

This curve is usually easy to manage, giving a lot of control over the initial difficulty and its evolution. The player is challenged for longer periods of time, since the game becomes more difficult the further the player plays it.

The biggest drawback of this kind of curve is its predictability: after a while, a somewhat "expert" player can predict "by feel" the upcoming challenges and prepare as a consequence, thus "squashing" the final part of the curve.

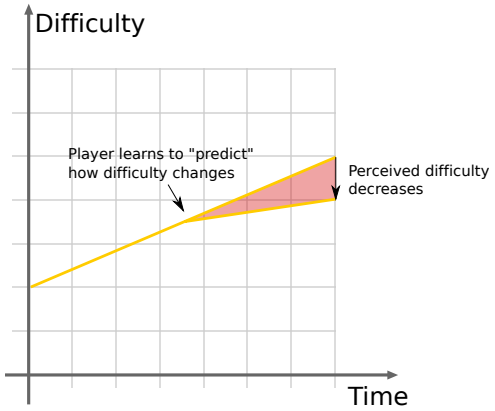


Figure 431: As the player learns to predict, the difficulty curve changes from our design

22.3.2.2 Logarithmic Line

The logarithmic line is usually presented as a "guide" for more advanced types of curves. This is due to the fact that it has some major issues.

Figure 432: A Logarithmic difficulty curve

The beginning of the game has a steep learning curve, which eases up as the game goes on. This means that the game is really hard at the beginning but the challenge dies down towards the end, which can make for a game very difficult to learn but not much more.

This can be a good curve if you want to "test the might and patience" of your players, but if not paired with a different approach in the late game, it may end up being boring in the long run.

22.3.2.3 Exponential Line

The complete opposite of the logarithmic line is the exponential one. This has a lot more use in game design that the previous example.

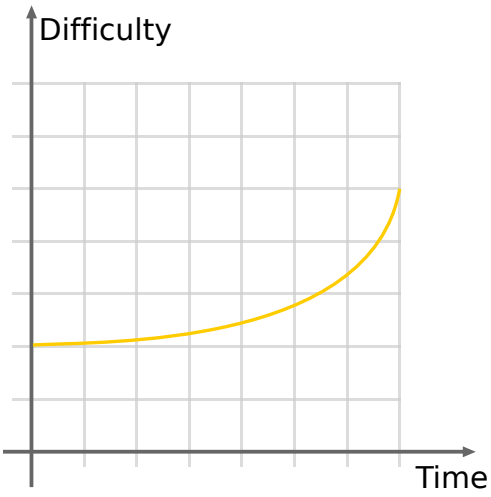


Figure 433: An exponential difficulty curve

The exponential difficulty curve gives the player a very relaxed beginning as well as a late game that can get really hard really fast (to the point that it can be too hard). This curve is the literal definition of a game that is "easy to learn but hard to master".

22.3.3 Wave patterns

The difficulty curves that we've seen so far have all one thing in common: they are simple and feature no real "lack of predictability", which can make a game a bit boring in the long run. Not because it's not challenging, but because it's predictable.

22.3.3.1 Linearly Increasing wave

Adding some waviness to the linearly increasing line can add some spice to the game very easily.

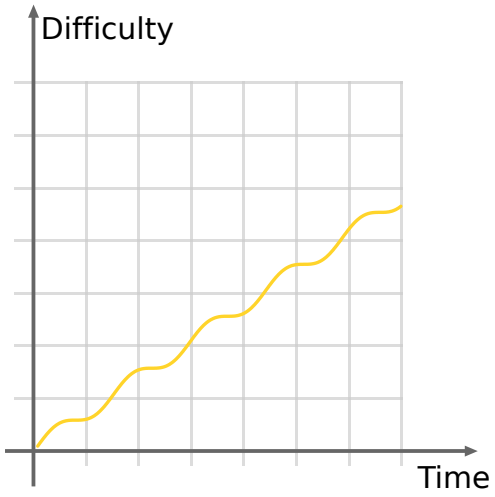


Figure 434: A linearly increasing wavy difficulty curve

This is a very efficient way of working, since it makes things more interesting, but if not implemented correctly it can lead to very high difficulty during the late game, since the "wave" may compound with an already high difficulty level.

22.3.3.2 Logarithmically Increasing wave

To try and fix the issues from the linearly increasing wave pattern, we may want to tie our difficulty to a logarithmic line.

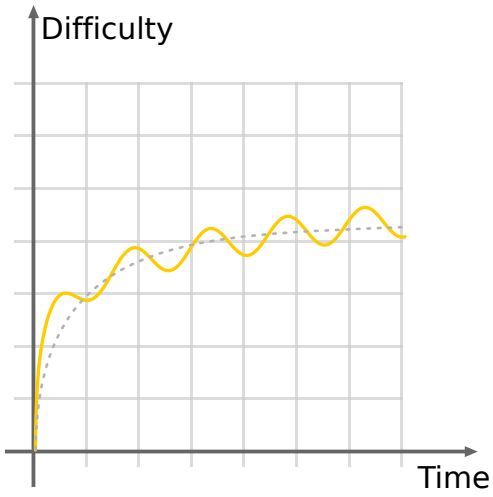


Figure 435: A Logarithmically increasing wavy difficulty curve

This kind of difficulty curve tends to “squash” the challenge towards the mid-to-late game, thus making the game a bit less difficult if the “wave” compounds with an already high difficulty level.

As a drawback, this curve may feel more “predictable” towards the late game, since the difficulty tends to get very “horizontal” towards the end; the wavy pattern helps keeping the predictability at bay, thus lengthening the enjoyability of the game.

22.3.4 Interval Patterns

For games that involve some random generation, like roguelites, we may want to “clamp” the difficulty between a “minimum” and a “maximum” but still allow for “runs” that feel different in difficulty from each other.

In this section we will show only wavy patterns, to exemplify the most “difficult to design” patterns, butt all patterns apply to simple lines too.

22.3.4.1 Simple Interval

The simplest way to implement an interval pattern is just defining a minimum and a maximum difficulty and setting the difficulty in such interval.

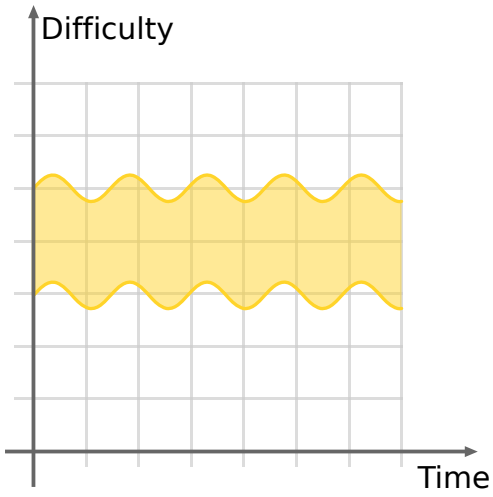


Figure 436: A simple wavy difficulty interval

This pattern is good for unpredictable challenges, but it is so unpredictable that you have no control over the initial difficulty either. This means that you may have a run of your game starting way too hard, while the next one may end up being very easy.

22.3.4.2 Widening Interval

To solve the lack of control over the initial difficulty, you may want to shape your interval like a letter “V” (just on its side).

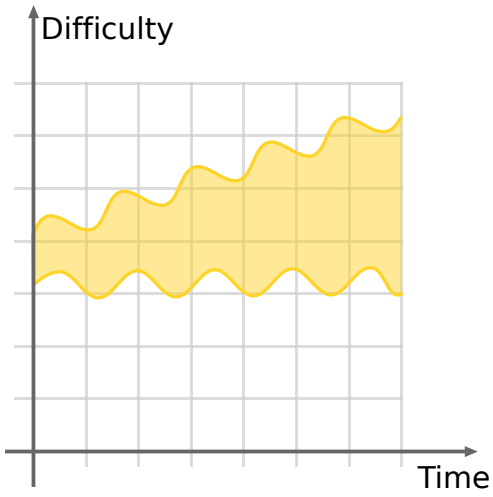


Figure 437: A widening and wavy difficulty interval

The widening interval allows you to have almost total control over the initial difficulty, while still keeping an unpredictable challenge in the mid and late game. The fact that the pattern widens towards the late game may end up being a drawback in some situations, since the game may have a really easy or really hard “ending”. This makes for a sometimes inconsistent experience.

22.3.4.3 Widening Interval with Logarithmic trend

When things tend to get out of control towards the late game, logarithmic curves come to our rescue and this is one of those times.

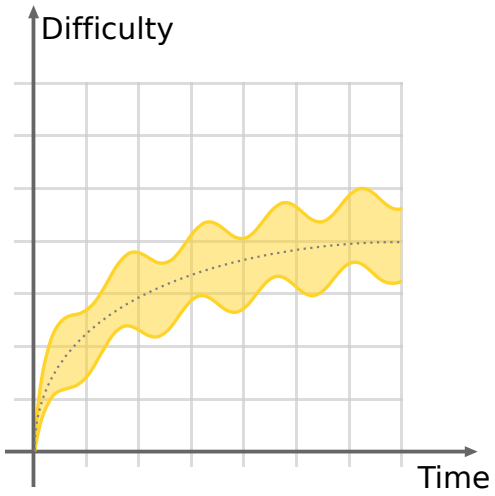


Figure 438: A widening wavy difficulty interval with a logarithmic trend

By tying our widening interval to a logarithmic line we have a way to better control how the game's difficulty evolves in the mid-to-late game. This gives the game's difficulty an "increasing trend" and coupled with a wavy difficulty line it can still be unpredictable enough to be enjoyable.

Such control doesn't come cheap though, since having so many things to control (the initial difficulty level, how much the curve widens, how fast things evolve) can be really difficult.

22.3.5 This is not everything

The difficulty curves that we've seen so far are definitely not the only ones that exist in video game development. You can mix and match until you reach a result that may look fun (in theory) and appeal to the player base that you have chosen.

Here we take a look at some more elements and curves that don't fit the previous description.

22.3.5.1 Sawtooth pattern

Every time we introduce a new mechanic, it may be useful and fun to let the player make large use of it, thus making the game a bit easier.

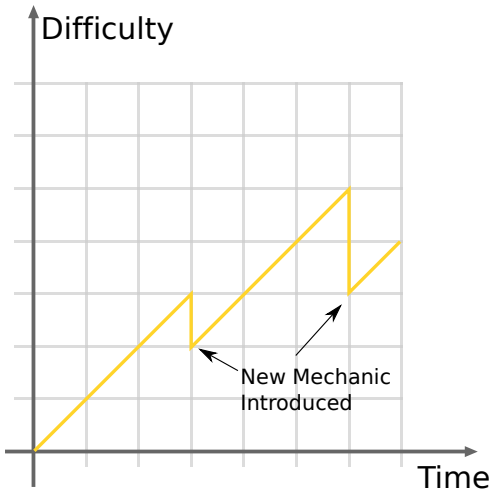


Figure 439: A sawtooth difficulty curve

This gives our curve a sawtooth-like shape, where the game gets slightly easier every time a new mechanic (like a powerup, or a tool) gets introduced, just to climb higher than the previous maximum. This can give the player an idea of "a reward for doing something difficult", and such reward is the new mechanic.

22.3.5.2 What not to do

When designing a video game, there are at least as many things you should as the ones you may want to do. One of the things you shouldn't do at all is adding "difficulty spikes" to your gameplay.

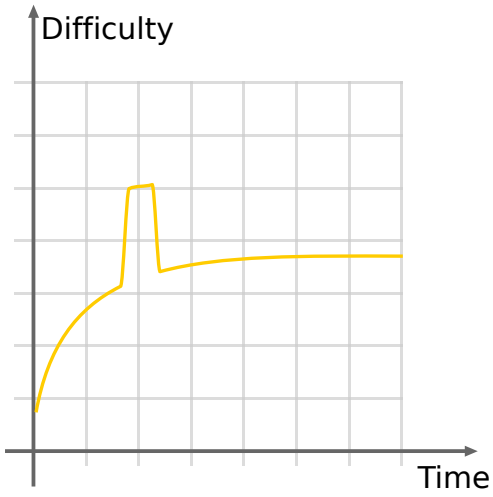


Figure 440: Difficulty spikes are not good

Difficulty spikes don't look good in graphs and don't make a game challenging or fun, they interrupt the natural flow of the game and end up frustrating the player. This may include an extremely precise jump in a 2D platformer just after a series of simple levels (even worse if such jump if far from the last checkpoint), or a very difficult boss that has no real place being there it is (difficulty-wise).

Another thing that you may want to avoid is making the game easier for experts: this may include adding a secret stash of collectibles (like powerups or skill points) in a place where only expert (or very very good) players can reach.

Furthermore, you should avoid punishing players who “don’t play that well” further than the minimum necessary: losing a life is already a strong “punishment”, if you make them lose all their gear without possibility of recovery (this goes for skills too), your game will be put on the shelf by the majority of your player base.

Avoid letting players “skip learning skills”, since they will find themselves in a world of trouble as soon as such skill is necessary to continue the game. The player will feel lost at first, then think that the game glitched out and only then (if they didn’t uninstall the game already) they make backtrack to look for something they missed.

Try to avoid overloading the players with information: when dealing with a tutorial that lasts longer than it should or that presents way too much information at once, players will lose focus and will tend to skip steps just to “get over it”.

22.3.5.3 Beyond difficulty

Difficulty is not everything in a game: a game may greatly enjoy from other elements, like comedy or just the feeling of relax that may come from a farming game. Some players really enjoy escaping the hectic city life to lose themselves in the rhythm of nature (although it is a simplified an virtual version of it).

Other games benefit from collectathon traits: deck-building games are a prime example. You start with a basic “deck of cards” which have certain powers, as you play the game more cards unlock and soon enough the player is enjoying the feel of strategy that comes from “building the perfect deck”.

22.4 Economy

Some games (not only MMOs) feature an “economy” side to their gameplay: this can prove to be something really difficult to balance without creating a virtual financial disaster.

This section will give you some basics to get things right.

22.4.1 Supply and Demand

Every economy is (at least in part) governed by the laws of supply and demand, which can be graphically represented in the following graph:

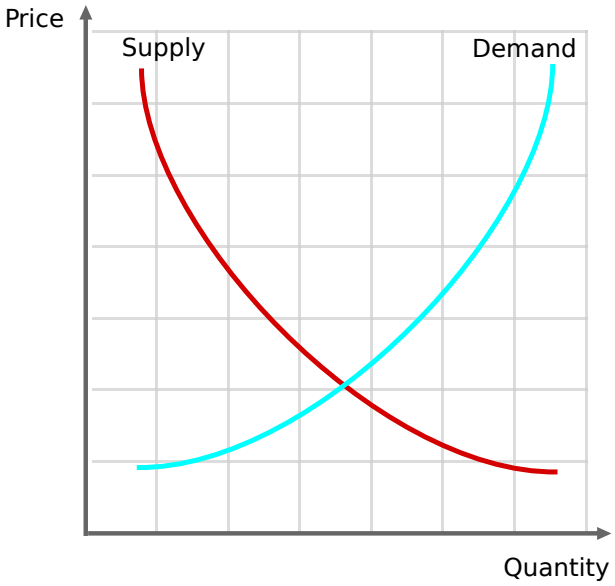


Figure 441: A simplified vision of supply and demand

Note!

This is an oversimplification of how the market (and the economy in general) works, just enough to keep you far away from the most common issues.

We can get some takeaways from such graph:

- If the demand is low (noone wants the product), suppliers will try to "boost it" by lowering prices;
- If the demand is high (many want the product), suppliers will try to earn more by boosting prices;
- If the supply is low (the product is rare), people will value it more (the price will be higher);
- If the supply is high (the product is really common), people will value it less (paying it less).

This also shows that artificially keeping the supply low will make the product feel more valuable, allowing to ask for higher prices.

Another thing to remember: money is a good too, and is subject to the same laws.

22.4.2 Money sources and sinks

Any artificial economy is usually composed by 2 "components":

- **Money Sources:** they create money from nothing, these can be quest givers, treasure chests and the like;
- **Money Sinks:** places that "destroy money", these are NPC salesmen at the market (that create items from nothing), fortune machines, anything that takes or exchanges money for something else.

Sources and sinks are extremely important and should be carefully balanced, since an imbalance in the quantity of money created and destroyed can have catastrophic effects. Among those, uncontrolled inflation and deflation are

the most prominent.

22.4.3 Inflation

Inflation is a phenomenon where prices usually rise uncontrollably: this means that money "lost its value".

This is usually due to the massive presence of money in the economy, so in a source/sink view, the money sources emit much more money than what the sinks can consume.

As a consequence fixed-price operations (like if you put "repair a weapon" at a fixed 50 golds) become incredibly cheap, while products in the market become prohibitly expensive.

In a supply/demand perspective, there is big supply of money which triggers little demand for it (since it's so common), while there is a big demand for products (thus raising the prices).

This may end with people having loads of money and noone accepting them for trades. Bartering may arise as an alternative to money.

22.4.4 Deflation

Deflation is a phenomenon where prices usually have a drop: this means that money has "too much value".

This has the exact opposite causes of inflation: there is too little money in the economy, so the money sources don't emit enough money and there are too many sinks that can consume it.

As a consequence fixed-price operations become extremely expensive (if you have 100 gold, paying 50 gold to repair a weapon may seem a lot), while products in the market become extremely cheap.

Again, in a supply/demand perspective, there is a low supply of money (making it more valuable), while demand is really high.

This can trigger "money hoarding" thus freezing the economy, sometimes bartering can arise as an alternative way to exchange goods without involving the "precious precious money". Some operations that require a minimum amount of money may even get locked because of the little amount of money circulating.

22.5 A primer on Cheating

Cheating is the act of fraudulently altering the game's mechanics in advantage of the player, performed by the players themselves.

It is something that many game developers and designers have to battle against, so here are some suggestions and tips to limit cheating in your game.

This section will just give a primer on the types of cheating we can find, since knowledge of something is the best weapon against it; so questions like "how to cheat" (or "how to hack") are outside the scope of this book.

22.5.1 Information-based cheating

Information-based cheats are all those cheats that rely additional information to the cheater, such information can give a sizeable advantage. A possible example is a cheat that removes the so-called “Fog of War” in a real-time strategy (RTS) game: having possibility of seeing all the enemy units allows the cheater to put up some countermeasures against the units that are being created.

These cheats include also x-ray hacks, all cheats that invalidate invisibility (as the server or peer would still need to transmit the coordinates of the hidden unit) and anything that can show information that is not meant to be shown to the user.

A possible solution is for the game to just “not transmit” the data, making the cheat useless, but sometimes that is just not possible.

22.5.2 Mechanics-based cheating

Another category of cheats is comprised of all those hacks that alter the game mechanics themselves, like killing all the players on the map. These kind of cheats are usually made possible by exploits or just because the cheater owns the server people are playing on.

These kinds of cheats can easily hinder the playability of a game, or even make it outright unplayable.

A possible solution to these cheats would be using a cheat-detection program (which would start a “cat and mouse” game, where hacks are updated to avoid detection, and detection programs are updated to detect new hacks) and also inserting some client-side verification of server commands (in case the server contains the “authoritative game state”); for instance if all players are killed at the same time, the clients could flag the server as possibly cheating.

22.5.3 Man-in-the-middle

This attack is well known in the networking environment: an external machine is used to route and intercept all the traffic directed to the game. This can be a real issue since the attacking program is “outside of the game environment”, making nearly all cheat-detection programs useless.

A man-in-the-middle attack can also be used to further exploit the game and find new vulnerabilities.

A possible solution could be completely encrypting all the game’s traffic, but that will be an issue since encryption takes away precious CPU cycles, and this could lead to an hindered gaming experience.

22.5.4 Low-level exploits

These kind of attacks don’t target the game itself, but tend to attack the technology that the game is using: they can range from brute force attacks (like using Denial of Service attacks) to more articulated actions that may break the equilibrium of the game.

This is not your game’s fault: it’s a problem with the technology stack used, the libraries or even the fact that the software is running on a computer. These exploits are really engrained into computers themselves, but they can still be mitigated with a little bit of care and attention.

22.6 How cheating influences gameplay and enjoyability

22.6.1 Single Player

Cheating in single player is an act that doesn't usually do a massive amount of damage, and such damage is usually confined inside the single "single-player" game.

Playing outside of the rules can be really fun (that's one of the principles the "glitch hunters" love: doing something outside of what another person imposed them), for instance some people cheat in games to bring some mayhem into their gameplay, or they use cheats implemented inside the game itself for a comedic factor (like the omnipresent "giant head" cheat).

Sometimes cheating happens because the game is unbalanced and players get annoyed at it, an instance of this happening could be when a game has a great story and gameplay but there is a boss that is so hard the game just stops there. You want to see how the story continues, but the game has gone so much out of balance you are willing to break its own mechanics to be able to continue it.

In this case the approach you should have is rebalancing the game, instead of limiting your players.

When it comes to cheat prevention, usually the first order of action is giving the game the ability to "check the validity" of an instruction.

For instance if a player character has its coordinates at (5,5) on frame n and coordinates at (1500, 5) at frame $n+1$, there is something fishy going on, since maybe the player can only move 500 pixels per second (while it moved 995 in one frame: $\frac{1}{60}$ of a second).

Such checks will slow down the processing, but will allow you to put a limit to cheating, possibly intervening in an active way, by resetting the space walked to the maximum amount possible in one frame, although this could give some issues with slower computers and variable time steps.

22.6.2 Multiplayer

When it comes to multiplayer and "leaderboards", cheating can be create some major damage to the game's enjoyability. It is honestly disheartening seeing a level that has been completed in 0 seconds on top of the leaderboard, totally unreachable with normal gameplay.

When competitive gameplay comes into the picture, playing against a cheater is frustrating and maddening, you feel powerless, the game is not fun and sometimes it even feels "broken", even though it is stable and playable.

Here we will distinguish between the two main forms of multiplayer: Peer-to-peer gameplay and dedicated servers.

22.6.2.1 P2P

Peer-to-peer multiplayer is the economically cheapest and easiest way to implement multiplayer, two or more computers (or consoles) are on "the same level" and communicate directly with each other, without a tertiary server in the middle.

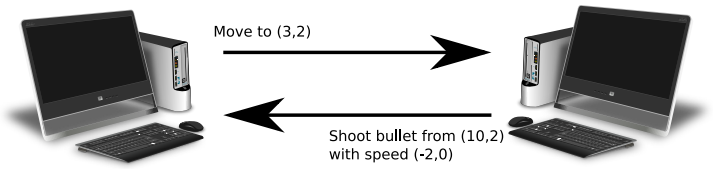


Figure 442: Example of a P2P connection

The main difficulty in preventing cheating is that there is no “authoritative game state”, the program cannot know if either player is cheating besides having an array of “possible actions”, like in single player, but with the added difficulty of network lag.

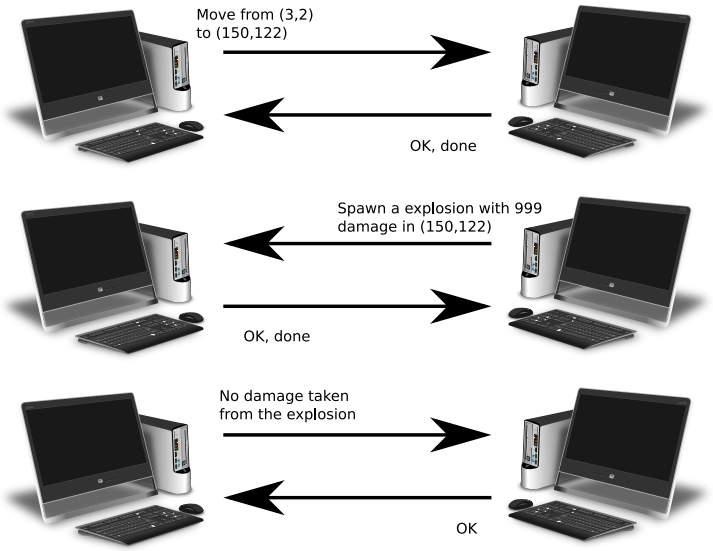


Figure 443: Two cheaters meet in P2P

Giving such “authoritative game state” to either of the players is not a good idea, because that way they would be able to cheat themselves and since they’re the “game master”, everything they do would be accepted.

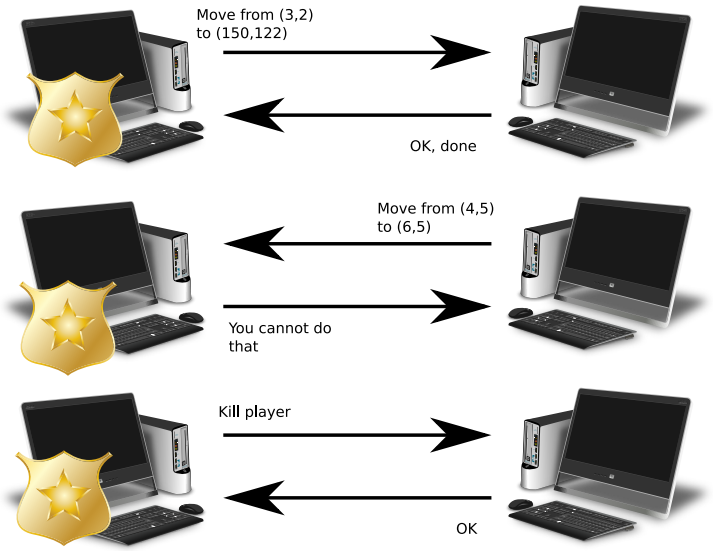


Figure 444: What would happen if one of the Peers had the authoritative game state

This is also the reason why many games that make use of P2P connections have implementations of anti-cheat systems that are shoddy at best.

22.6.2.2 Dedicated Servers

Dedicated servers is usually the best way to prevent cheating, a tertiary server is added to the mix, and said server is either controlled by the game creators or uses a software specifically tailored to work as a “multiplayer server”.

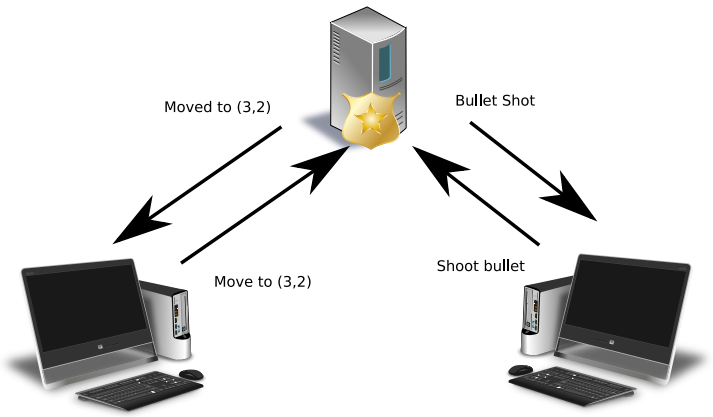


Figure 445: Example of a dedicated server

Such server contains the authoritative game state and decides what is right and what is wrong, or either what is possible and not possible.

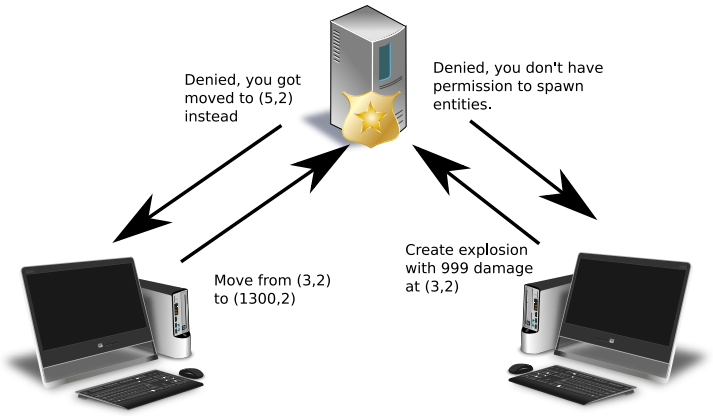


Figure 446: Two cheaters vs. Dedicated server

Usually a dedicated server software has been specifically made to limit cheating, as well as offering better performance than the P2P counterpart (it doesn't have to run graphics, for instance).

If a consistent leaderboard and lack of cheating is important (for instance in a heavily competitive multiplayer game), you should probably choose this option.

This section assumes that the game is using a third-party game server, and none of the players has direct access to said server, as this would enable cheating on the owner's part.

22.7 Cheating protection

Protecting your game, leaderboards and community from cheating is hard and there are many ways to prevent it. Dedicated servers are one of the many ways, cheat engine detection (there are many commercial solutions) but none of these solutions is cheater-proof.

If your game has 1 million players (I hope it does!) and there is a 0.0001% probability of people cheating at it, you have a statistical certainty that there will be a cheater among them. Cheating will happen and usually cannot be completely prevented, but that shouldn't discourage you.

It should be hard to cheat at a game but, most of all, it should be harder to do it undetected.

So there are some other tools in your toolbox that you can use, for instance you can save a "lightweight" replay of the game session (if your game has a leaderboard), that way anyone who wants to enter the leaderboard will also send a replay of their gaming session.

This has multiple advantages: from the community side you have players can "learn tricks from the best", but also can report who evidently cheats, because the replay would show it.

Usually these "lightweight" replays are done by recording the position of the player and its state, as well as the initial state of the game. Add the fact that the game is deterministic and you have the equivalent of a recording of the gameplay.

Even better, you can record the inputs of the player and see if "a simulation" done with those inputs validates against the positions and actions recorded on the replay: this way if someone modified their game to make themselves invincible or faster, the simulation would fail and the replay wouldn't validate.

Random Trivia!

It was recently discovered that the game "Trackmania" records inputs as well as the position of the vehicle, this allows the game to validate the replays against the most common forms of cheating

We can be sure that the simulation would be equal to a video because a game (as well as any program) is deterministic: given the same initial state and inputs, the game will always end up the same way. This is true even if random numbers are involved (that is why they're called "pseudorandom"), see "random numbers on computers are not really random".

22.7.1 Debug Mode vs Release Mode

Sometimes it may prove useful (and sometimes it is just plain necessary) to have some "cheats" to use as "shortcuts" while doing some development tasks.

Some of these give you invulnerability to make sure that you can test the balance of the weapons without worrying about dying in the harder stages, some other times you need a "level skip code" to quickly get to later levels when you have made incompatible edits to the save file format.

When using these codes it is vital to have a build flag to distinguish between a "release build" and a "debug build", this way it is possible to completely strip out the debug code from the build, thus "reducing the attack surface": you can't abuse code that is not there.

This is more difficult when dedicated servers are involved, since it would be necessary to have 2 copies of the server: a "release build" with all the debug code stripped out, and a "debug build" that allows for "debug cheats". Problem is that not having such "cheats" could make moderators' work a lot harder (let's consider cases like Minecraft servers, where administrators need to be able to fly around to be able to scout possible cheaters "by eye").

22.8 Some common exploits

In this section we will take a look at some really common exploits that can be used to break the balance of a game. By knowing these kinds of exploits you will be able to plan ahead and avoid annoying (or embarrassing) situations.

22.8.1 Integer Under/Overflow

This is probably one of the most common exploits that you can find in games. Computers have limited memory, thus they have a limit on the numbers that can be represented: what happens when such limit is exceeded?

It might be useful to remember the section talking about two's complement.

22.8.1.1 How the attack works

Let's imagine a simple management game: you can earn money from various activities and you spend money on staff wages. There is a button that allows you to increase staff wages by 1 unit. Staff wages are saved using (for some reason) only 3 binary digits.

The game calculates your monthly earnings as follows:

$$earnings = income - wages$$

Now that we have set up the environment, let's see how it can be broken.

Let's assume we have only one staff member how is paid 3 units per month. What would happen if we paid them another unit? In decimal it would be easy: $3 + 1 = 4$ but in binary it is a lot more complicated:

$$011 + 001 = 100$$

If we look at the previous table, that's the two's complement representation of -4! By raising the staff wages we've ended up with the staff paying us! This is called an "integer overflow".

This can work both ways: if we subtracted 1 from -4 in binary, we would go back to $+3$. This is what happens when an integer "underflows".

Random Trivia!

There is a story about the first "Civilization" game having an integer underflow bug (called "Nuclear Gandhi"): a civilization's "aggression value" was saved as an 8-bit unsigned integer. Gandhi had its initial aggression score set as 1 and when India achieved democracy, such score is lowered by 2 points, causing the "aggression value" to underflow to 255, making Gandhi the most aggressive leader in the game.
This is actually not true, but "Nuclear Gandhi" was included as an Easter egg in the following games, as a joke.

22.8.2 Repeat attacks

This affects for the most part multiplayer games that make use of the internet. Let's imagine the following situation: you have an online game that follows this procedure:

1. You accept a mission;
2. The mission is added to your journal;
3. You perform the mission's tasks;
4. You turn in the mission;
5. You receive experience and gold from the mission;

6. The mission is removed from your journal.

Between points 4 and 5 there is a synchronization effort between your client and the dedicated server: if the server doesn't confirm that the mission is really turned in, no experience is received and the mission stays in the journal, ready to turn in.

What if there was no control (on the server side) for turning in the mission more than once? This is what repeat attacks exploit.

In our example, you would get a program that throttles the connection to the point that the network code of the game is suffering heavily, but doesn't disconnect. At this point you just turn in the mission a lot of times (and such mission will stay in the journal because the client didn't receive an answer from the server yet). Since the server doesn't have a check for multiple mission turn-ins, it will return orders to the client for adding more experience and gold.

23 Accessibility in video games

> Accessible design is good design - it benefits people who don't have disabilities as well as people who do. Accessibility is all about removing barriers and providing the benefits of technology for everyone.
>
> *Steve Ballmer*

23.1 What accessibility is and what it is not

Let's start with what accessibility is not: accessibility is not making a game playable by people with disabilities. To be precise, it is not only that.

Accessibility is an inclusive practice where you make sure there are no (or at least the fewest number of) barriers that prevent someone with interacting with your game.

Many times "good design" and "customization options" bleed into "accessibility" too, as we can see from the sections below.

23.2 UI and HUD Scaling

This is one of those "customization options" that bleeds well into "accessibility". Some people may like smaller Heads-Up Displays (HUDs) and general UI, to give more space to gameplay, while others may want to have bigger UI because of some visual impairment.

UI and HUD scaling is a great customization for any player, not only the ones who suffer from some kind of impairment or disability.

23.3 Subtitles

This feature is really useful if you can't afford to keep the volume too high without disturbing someone (and don't have headphones). Another great use for subtitles is to expand the game's audience to non-native English speakers, who may have a bit of trouble with pronunciation.

Some games further expand their demographic by using speech in a certain language and enabling support for subtitles in another language, other times it can be used as a story ploy: in Rayman 2 the characters speak "Raymanian" (sometimes called Wandayē), a fantasy language which is subtitled in the language of choice.

Another use of subtitles is to describe world sounds for hearing-impaired users.

23.4 Mappable Buttons

This is another one of those "customization options" that falls into the "accessibility features" too. Giving the player the possibility of changing their button mapping is essential to guarantee that a larger number of players is able to enjoy the game you created.

Modern engines usually include input handlers that already support mappable buttons, you just need to "transpose" such functionality in your own game by creating the right menus.

23.5 Button Toggling

In some games, a player may have necessity to keep a button pressed to perform a certain action (like sprinting).

Some games allow buttons to work as "toggles", so one press enables sprinting, another press disables it. This can be both a functionality for "comfort" as well as a feature that allows the game to be more accessible to players with difficulty when it comes to muscle control.

23.6 Dyslexia

One often overlooked issue when it comes to text is dyslexia. In this small section we will take a quick look at some accessibility options that can help with this problem.

23.6.1 Text Spacing

A good start would be having the possibility of customizing the space between lines of text: this way the possibility of mixing up different lines of text is greatly reduced. This also helps other people who prefer a smaller UI, but at the same time find difficulty in reading very "condensed" text.

23.6.2 Fonts

Another great tool could be font customization: there are specialty fonts dedicated to greatly reduce letter-swapping and confusion, thanks to letters that have a thicker bottom (to give a clearer baseline) and different letter shapes.

One of such specialty fonts is "OpenDyslexic" (available at https://www.opendyslexic.org/), which is free to use, being licensed under the SIL-OFL license.

23.7 "Slow Mode"

Having an option that allows to slow down the game is an authentic boon for anyone who wants to enjoy a video game but has some issues when it comes to reaction time or muscle control, but this makes the game more enjoyable for audiences who may not appreciate a high level of challenge but still want to play your game.

23.8 Colorblind mode

When it comes to color, everyone sees colors differently but others may not be able to distinguish between certain colors at all: that's where a colorblind mode comes into play.

Having the game palette adapt so it's easier to distinguish colors for colorblind users is definitely one of those accessibility options that may require a bit more work but could make some players really happy too.

23.9 No Flashing Lights

When it comes to photosensitivity, (very dangerous) seizures are not the only outcome possible: some people may get really strong headaches from seeing intense visual effects and flashing lights.

Having an option to disable or tone down flashing lights and intense (sometimes even overbearing) visual effects can make a game a lot more enjoyable to someone who has some kind of sensitivity to light.

23.10 No motion blur

There are some people that like motion blur, some who don't and some who can't physically stand it: making motion blur a toggle can be both a quality of life improvement as well as something that can help people who have issues with sight.

Making it a slider so the player can "dial it in" is a great thing too: it adds a level of customization, as well as being a performance settings too (motion blur can slow down the game on older platforms).

23.11 Reduced Motion

Some people may be particularly sensitive to cameras or texture patterns that involve lots of motion, this could cause discomfort in the player or even full-blown motion sickness. Having an option to reduce motion (like avoiding certain camera rotation mechanics or making some patterns static) can really help users enjoy your game without having the game "hurt" them.

Random Trivia!

Sonic 4: Episode 2 has a level (precisely "Death Egg. Mk. II") which has a really cool mechanic where the camera follows the ground you're riding on, rotating with it. This is a really cool effect, but this is one of the effects that could cause motion sickness. A trigger to disable the camera rotation (and maybe take a slight hit in "realism") could help a lot, even though it could introduce some control scheme challenges.

23.12 Assisted Gameplay

Having an "aim assist" option that allows for the crosshair to "snap to an enemy" when it is "close enough" it's great for people who have problems with coordination, but it's also great for players who prefer to play with a controller instead of a mouse.

Any kind of "assisted gameplay" can help both people with and without disabilities.

23.13 Controller Support

Supporting only one certain type of "control device" is not great for accessibility; some people may prefer or need a controller of some sort, be it a gamepad or something more sophisticated, having support for "alternative inputs" is always a great idea.

23.14 Some special cases

A special mention goes to the VR game “Moss”, where a character (named Quill) communicates both emotionally and gives clues on the puzzles using the American Sign Language (ASL).

[Do you know more about this? You can contribute, this book is open source!]

24 Testing your game

> The bitterness of poor quality remains long after the sweetness of meeting the schedule has been forgotten.
>
> *Anonymous*

When you want to assure the quality of your product, you need to test it. Testing will be probably be the bane of your existence as a developer, but it will also help you finding bugs and ensure that they don't come back.

Remember that you should be the one catching bugs, or your players will do that for you (and it won't be pretty).

Let's take a deep dive into the world of testing!

24.1 When to test

24.1.1 Testing "as an afterthought"

Most of the time, testing is left "as an afterthought", something that comes last, or at least something that is done way too late in the development process.

Testing is made difficult because the code is all interconnected, making it really hard (if not impossible) to separate and test each component.

Leaving testing as an afterthought is not a good idea, let's see the alternatives.

24.1.2 Test-Driven Development

Some swear by the "Test-Driven Development" (from now on "TDD") approach. TDD consists in developing the test before writing a single line of code, then after the test is ready, create the code that solves that test.

That way the test will "drive us" to the solution of the problem. At least that's what it is supposed to do.

As a critique to the TDD approach, I personally think that people will end up trying to "solve the test", instead of "solving the problem". This means that sub-optimal solutions may be adopted, and "edge cases" will be missed.

24.1.3 The "Design to test" approach

To get the best of both worlds, we need to work a bit more on our software design, by designing by having its testing in mind: functions should be self-contained, the weakest amount of coupling should be present (if any) and it should be possible to mock elements easily.

Paying attention and going the extra mile to create an architecture that is easy to test will reward you in the long run.

24.1.4 You won't be able to test EVERYTHING

Before trying to test anything, remember that testing is **hard**. You won't be able to test everything, and the situation will be worse when you will be in a time crunch to deliver your game.

Remember that hacking is not always a bad thing, sometimes cutting corners will get your game shipped and having a solid (and tested) basis will help you with that too.

24.2 Mocking

Before talking about the nitty-gritty of testing, we need to talk about “Mocking”.

Mocking is a procedure that is usually performed during tests, that substitutes (in-place) an object or function with a so-called “mock”. A mock is something designed to have no real logic, and just return a pre-defined result, or just have a pre-defined, very simple, behaviour.

Mocking will help you “detach” objects that depend on each other, substituting such dependencies with “puppets” (the mock objects) that behave consistently and are not affected by bugs that may be present in the object that you are mocking.

24.3 Types of testing

Let’s take a look at how we can test our game, because (as with many things in life), testing can be more than meets the eye. In this section we will talk about both manual and automated testing, the difference between them and what method suits what situation.

24.3.1 Automated Testing

Automated testing is usually performed when new code is pushed to the repository, or on-demand. Automated testing makes use of a “test suite”: a bunch of tests that are run on the game’s elements to test their correctness.

Here we can see a small example of a simple function, along with a test, in pseudo-code:

```
function sum_two_numbers(a, b):
    // This function takes two numbers and sums them
    return a + b

function test_sum():
    // This function tests the sum function
    int result = sum_two_numbers(2, 2)
    assert result == 4
```

As we can see, the test makes use of the “assert” statement, which in many languages raises an error when the “assert” is false. This means that if, for some reason, the `sum_two_numbers` function was edited to return “2+2=5”, an exception would be raised when the test is run.

Care should be taken when making automated tests: they should be as simple as possible, to avoid the presence of bugs in the tests themselves, also, like all code in the world, it’s subject to its own maintenance: you should thoroughly comment your tests, and if the tested component changes, the connected test should change too, or it may fail.

It may seem a lot of effort coding automated tests, but such effort will be rewarded with lower maintenance effort in the long run: the (sometimes considerable) effort you've put into coding automated tests will avoid a huge deal of manual testing later on and ensure that loss of quality happens a lot less often.

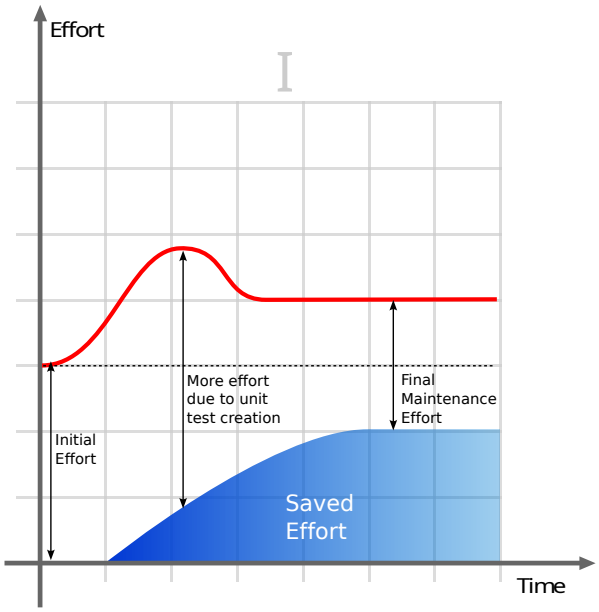


Figure 447: How the effort put in automated testing can give good returns

24.3.2 Manual Testing

Sometimes also called "play testing", this is usually divided in time periods, where more and more people participate:

- **In-house testing** with a small team of dedicated testers;
- **Closed Beta** where a small number of selected players is able to test the game, report bugs and issues. Usually at this stage, the game is already mostly finished and playable;
- **Open Beta** similar to the closed beta, but players can freely subscribe to play the game.

We will talk specifically about each one of these test types in detail in the following sections.

24.4 Unit Testing

Unit Testing takes care of testing the smallest component possible inside your system: that usually means a single function. Such "units" must be separated from all their dependencies (via mocking) to avoid bugs in their dependencies to interfere with our testing efforts.

Many programming languages have their own unit testing frameworks, among the most used:

- **Python:** Unittest (in the standard library);
- **Javascript:** unit.js;
- **C++:** Boost Testing Library;
- **Java:** JUnit;

- **Lua:** luaunit.

Remember: during unit testing, you need to make sure that the unit that you're testing has its dependencies mocked, or the results of your test will depend on the performance and correctness of said dependencies.

24.5 Integration Testing

Integration testing is a step further from "unit testing", here you take different "units", put them together and check how they behave as a group.

There are many approaches to integration testing, the most used are:

- **Big bang integration:** The majority of the units are put together to form a somewhat complete system (or at least a major part of it), after that the integration testing starts. This can lead up to an "explosion of complexity" if the results are not accurately recorded.
- **Bottom-up:** Test the "lowest level" components first, this should help testing the "higher level" ones. After that you test the components that are "one level above" the previous ones. Keep going until the component at the top of the hierarchy is tested.
- **Top-down:** Opposite of the previous approach, you test the integrated modules first, then you branch into the lower-level modules, until reaching the bottom of the hierarchy.
- **Sandwich Testing:** a combination of Bottom-Up and Top-Down.

24.6 Regression Testing

Regression testing is a bit of an outlier in our "specific to general" testing structure, but that's because their objective is different.

Regression testing (sometimes also called *non regression testing*, you'll see why) is used to avoid our software from regressing into previous bugs.

This means that every time you find a serious bug in your software, you should fix it and make a test that will check for you if said bug is resurfacing.

With time, bugs and regression tests will accumulate, which usually means that automation is involved (like continuous integration and delivery) to execute them at each push or at regular intervals.

24.7 Playtesting

Automated testing won't be able to help with how a game "feels" to the player, for that you need to a thorough play testing strategy. Here we will talk a bit of different strategies that can be mixed and matched to get the best out of it.

24.7.1 In-House Testing

The first sessions of play testing should be done in-house, with a dedicated play testing team that has great reporting capabilities, and the product should already include tools that allow for quick reporting of bugs and issues that arise,

as well as a good logging system set to its `DEBUG` level for maximum detail.

Close collaboration with the testing team is vital for a good game to be released, instead of seeing them as "the ones that give you more work", try looking at them as "the ones that will ensure your game gets a lot of praise".

24.7.2 Closed Beta Testing

Happening after the in-house testing, usually done with multiplayer games, the "Closed Beta" phase is done with a selected group of players that try the game and report each and every issue with it, as well as bugs.

The product should have an easy way to directly report bugs and issues form inside the game itself, with the possibility of attaching a detailed log with the "ticket".

24.7.3 Open Beta Testing

Differently from the "Closed Beta" phase, and usually coming after that, the "Open Beta" phase doesn't have a hard limit on the number of players that can take part to the beta testing.

The product should have the same characteristics that are used in the "closed beta" phase, plus possibly a higher degree stability.

Open Beta is usually done to test the robustness of the network system at higher loads, and having a possibly large (maybe larger than expected) player base can be a real test for the infrastructure.

24.7.4 A/B Testing

A/B testing is a particular kind of testing that doesn't involve the "solidity of the game", as much as the enjoyability of some of its features. In A/B testing users are randomly presented with one of two versions of a certain feature; with a slight difference (usually a single variable that affects the experience is changed) between each of the two "versions".

A/B testing is normally used in an user experience research setting, but it can be used in game development too, to see which version (usually called *variant*) is better suited for the game, or even prepare a "segmented strategy" where one of the two variants could find place in a certain situation (for example a "simplified control scheme" vs a "full control scheme").

25 Profiling and Optimization

> The real problem is that programmers have spent far too much time worrying about efficiency in the wrong places and at the wrong times; premature optimization is the root of all evil (or at least most of it) in programming.
>
> *Donald Knuth - Computer Programming as an Art*

25.1 Profiling your game

25.1.1 Does your application really need profiling?

In this section we will have a small check list that will let us know if our video game really needs to be profiled. Sometimes the FPS counter is trying to tell us a different story than the one we have in our heads.

25.1.1.1 Does your FPS counter roam around a certain "special" value?

There are cases where the FPS counter shows a low counter, but it stays around a certain value. This means that the FPS value is artificially limited somewhere, either by VSync or something else.

Some special values you may see are:

- 25 FPS: PAL refresh rate
- 29.970 FPS: NTSC Refresh Rate
- 30 FPS: Used in some games
- 50 FPS: Used in some games
- 60 FPS: Used in most games
- 75 FPS or 80 FPS: Used in some LCD Monitors
- 90 FPS: Used mostly in VR games
- 144 FPS: Used in more modern, high-refresh rate monitors
- 240 FPS: Used in the most recent high-end games and monitors

25.1.1.2 Is the animation of your game stuttering but the FPS counter is fine?

If your animation stutters or its speed varies according to the load of your platform but your FPS counter is still stuck at the maximum allowed framerate, you may have forgotten to tie the animation to the delta-time in your game loop. Check the timing your game loop section for more information.

[This section is a work in progress and it will be completed as soon as possible]

25.1.2 First investigations

First of all, we need to understand what is the bottleneck of your game: check your task manager and see how your game is performing.

25.1.2.1 Is your game using 100% of the CPU?

Is your game using 100% of the CPU (if you're on Linux, you may see percentages over 100%, that just means your game is using more than one CPU core)?

First of all, you should check if you're using the frame limiting approaches offered by your framework or game engine: if they're not active, your game will run "as fast as possible", which means it will occupy all the CPU time it can. This can result in high FPS count (in the thousands) but high energy consumption and slowdowns in other tasks.

If you have taken all the frame limiting approaches as stated above, that may mean that the game is doing a lot of CPU work and you may need to make the game perform less work for each frame. In this case profiling tools are precious to find the spots where the CPU spends most of its time: Valgrind or GProf are great profiling tools.

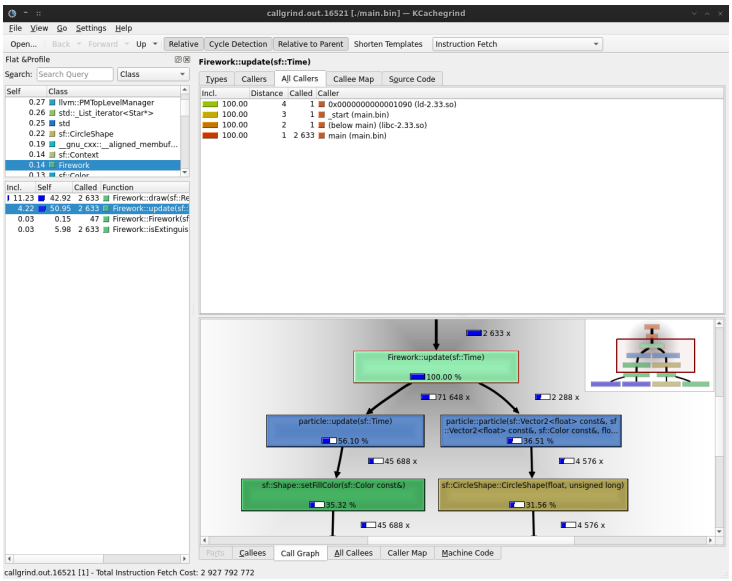


Figure 448: Using Valgrind's Callgrind tool and Kcachegrind we can see what is bogging down our game

25.1.2.2 Is your game overloading your GPU?

If instead your game is not using all of the CPU computing power, you may have a problem on the GPU: your game may be calling the drawing routines too often. The less a game has to communicate with the hardware, the higher the performance. In that case using Sprite Atlases and other "batching techniques" that allow to draw many objects with only one call will help your game perform better.

25.1.2.3 Is your game eating up more and more RAM as it's running?

Your game starts well enough, but after just a few minutes it starts slowing down and becomes choppy. Your may have a memory problem at hand.

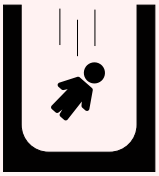
If your game supports windowed mode, keep your task manager (or "top"/"htop"/"bpytop" if you're on Linux) open and look at your game's process: does the memory used by your game increase as you're playing it?

If so, you may be having a so-called *memory leak*[g]: somewhere during its running cycle, your game forgets to clean up something, which stays resident in memory until your game closes. The result, after creating and deleting a lot

of entities and leaving a lot of *garbage* behind is that the total memory used increases.

This is especially common in languages like C++, where there is no automatic "garbage collecting" and having cases of so-called *unreachable memory*[g] can really mess up your memory usage.

Pitfall Warning!



Some people call unreachable memory cases "*dangling pointers*[g]", but technically they are two different (and opposite) things.
Check the glossary for more information.

If you suspect a memory leak, you may want to take a look at these sections:

- Entity Cleanup and Memory Leaks
- Using memory analyzers to detect leaks
- Resource Pools

[Do you know more about this? You can contribute, this book is open source!]

25.2 Optimizing your game

After accurate profiling, you need to intervene and try to get more out of your code. In this section we'll talk about some guidelines and tips on how to optimize your game.

25.2.1 Working with references vs. returning values

Depending on the programming language you're using, and the amount of internal optimization its compiler/interpreter has, you may have the possibility to choose between two main ways of working, when it comes to functions:

- Returning a value from a function;
- Passing a reference to variables into the function and use that reference in your function (for instance in C++).

"Value Copying" can be a real resource hog when your functions work with heavy data. Every time you return a value, instead of working on a reference, you are creating a new copy of the data you're working on, that will be later assigned.

This can happen also when passing parameters to a function (in this case you say the "parameter is passed by value"): a new copy of the parameter is created locally to the function, using up memory. "Value Copying" can help when you don't want to modify the data outside your function, but is a waste when instead you **want** to modify such values.

Using things like "references", "constant references" and "pointers" can be really precious in making your game leaner memory-wise, as well as saving you all the CPU cycles wasted in memory copying.

25.2.2 Optimizing Drawing

This heavily depends on the type of framework and engine you are using, but a good rule of thumb is using the lowest amount of calls to the draw routines as possible: drawing something entails a great amount of context switching and algorithms, so you should do it only when necessary.

If your engine/framework supports it, you should use sprite atlases/batches, as well as other interesting structures like Vertex Arrays (used in SFML), which can draw many elements on the screen with only one draw call.

Another way to optimize your drawing routine is avoiding to change textures often: changing textures can result in a lot of context changes (like copying the new texture from the RAM to the GPU memory), so you should use only one oversized texture (in the form of a Sprite Sheet) and draw only a part of it, changing the coordinates of the rectangle that gets drawn. This way you'll save the PC a lot of work.

25.2.2.1 Off-screen objects

Make sure that your engine doesn't try to draw objects on off-screen area (maybe on virtual surfaces): drawing is an expensive operation and we should do it on the smallest possible set of objects, which is the visible screen area (the viewport).

Drawing objects doesn't change their internal state, so you can keep updating the objects and then draw them only when they fall (even just partially) inside the display's viewport.

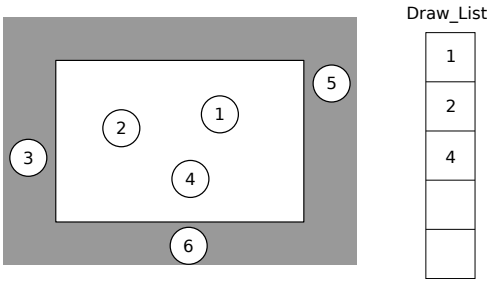


Figure 449: Not putting off-screen objects in the drawing queue can be a good optimization

Some engines may already take care of this optimization, but some lower-level libraries may also leave that optimization to you. A good way to test is drawing thousands (or even millions, with the help of a `for` cycle) of sprites off-screen without any update code (maybe in a specific project) and see if the engine slows down considerably the more entities are added.

25.2.3 Reduce the calls to the Engine Routines

Some engines have routines that introduce sanity checks, logic optimizations and more, and calling such routines more than necessary can burden your game's performance, even worse when you're calling them per-frame.

If you want to move a character diagonally both up and right, don't do this:

Listing 141: Double Engine Movement Call

```
function update(dt){
    let vector_up = new Vector(0, -1);
    let vector_right = new Vector(1, 0);
    // ...
    characterController.Move(vector_up * dt);
    characterController.Move(vector_right * dt);
    // ...
}
```

As all the sanity checks in the `Move` function will be executed twice per frame (since we're in the "Update" function). Instead you should get the resulting movement vector first, and then use the `Move` function only once:

Listing 142: Single Engine Movement Call

```
function update(dt){
    let vector_up = new Vector(0, -1);
    let vector_right = new Vector(1, 0);
    // ...
    let total_movement = vector_up + vector_right;
    characterController.Move(total_movement * dt);
    // ...
}
```

This way instead we're doing sanity checks and related operations only once, moving the character in its final position without wasting resources.

25.2.4 Entity Cleanup and Memory leaks

One of the biggest scourges in software development (and an even bigger one in game development) are memory leaks: the program allocates memory but doesn't release it properly.

Memory management (as well as any kind of "resource management") can be summarized in 4 phases:

- Acquisition;
- Initialization;
- Usage;
- Release.

This is especially annoying when languages that don't have automatic garbage collection (like C++) are involved, but it can affect any programming language. Memory management is hard, and we should always release any resource that we acquire as soon as we're done using it, but that's not always easy: for instance when loading and unloading levels is involved.

As mentioned before, this problem affects all languages, since some resources may be acquired by some "active code" that is actually never running, thus preventing the garbage collector from working as it should.

Besides "being careful" with your resource management, you can check for memory leaks by using specific tools.

25.2.5 Using analyzers to detect Memory Leaks

When developing a game, there are a lot of tools that allow you to inspect your game and find possible memory leaks. Some are "static scanners" while other (usually called "dynamic testing tools") require the game to be running.

25.2.5.1 Static Scanners

These tools analyze the code without running it, checking the style and common bugs that can be inserted by mistake. An example of these static tools are "linters" (or linting tools).

Most of these tools are included in IDEs but some (like LLVM's scan-build) are standalone.

Figure 450: An example screen from LLVM's scan-build

25.2.5.2 Dynamic testing tools

Some tools require the game to be running, some general-purpose ones are used to find memory leaks (like Valgrind), while others have more specific purposes and are usually integrated into the engine.

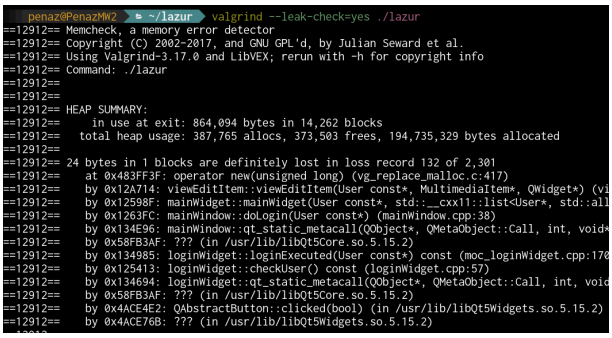


Figure 451: A screenshot from Valgrind, looks like we have a memory leak here

These more specific tools can track the FPS, memory as well as the calls done to each function, allowing you to track down what is bogging down your game.

Figure 452: A screenshot from Godot's profiler

25.2.6 Resource Pools

Among the most performance-hungry operations in computers we find instantiation and destruction of objects: they involve context switches in the CPU, memory allocation/freeing and a lot of other things.

If you find yourself needing to instantiate and destroy a lot of objects of the same type, you may want to consider a "resource pool" for such object.

A resource pool is a group of objects that is instantiated once, ready to use and kept in memory (eventually without updating the internal status of the "inactive objects") until needed.

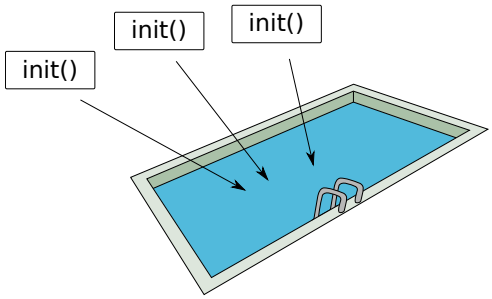


Figure 453: A resource pool instantiates objects and "keeps them" ready when needed

When you need one of the objects, instead of instantiating it (and thus allocating memory, changing CPU context, etc...) you just "pull" an item from the pool and change its internal state as needed (since the memory is already instantiated). This allows you to "move" the cost of instantiating the object to some place in your code where some delays are expected (for instance the loading screens, after you loaded your resources).

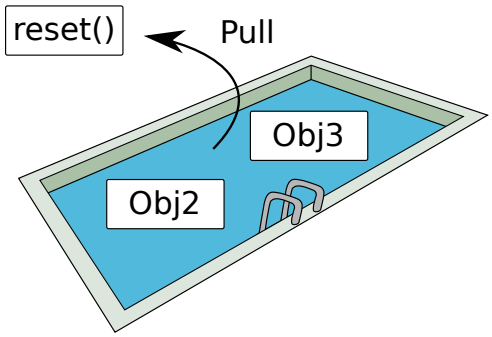


Figure 454: Pulling an object from a resource pool

When you're done, instead of destroying the class (thus calling memory free methods and changing the CPU context again), you "return" the item to the resource pool, ready for another round. This allows you to "move" the cost of destroying the object somewhere where slowdowns are acceptable, for instance (again) loading screens.

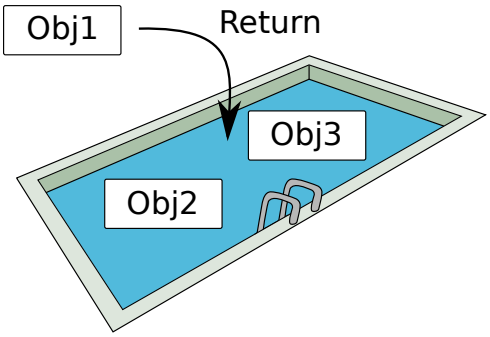


Figure 455: Returning an object from a resource pool

Particle systems are a prime example of resource pools: instead of continuously creating and destroying particles, you create all the particle object in advance to recycle and reuse during the game.

25.2.7 Lookup Tables

Inside older games, where CPU cycles were at a premium, a widely used trick to gain performance were "lookup tables".

These tables would store the result values for certain expensive functions, given certain inputs, thus replacing the expensive operation with a lookup inside a certain data structure (which is usually really fast).

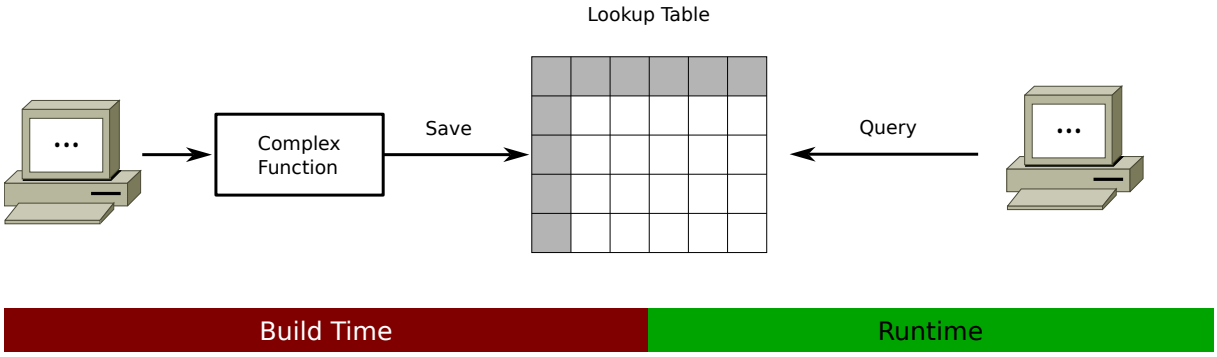


Figure 456: How a lookup table works

This has a tradeoff: you're trading CPU time for Memory space, since the lookup tables are meant to stay into RAM.

In modern games instances of lookup tables are as rare as hens' teeth, but it's an interesting historical view over some older forms of optimization.

25.2.8 Memoization

Memoization (sometimes known as "tabling") is an optimization technique that consists in saving the result of an expensive function, as well as the function's arguments for later calls: this way when the same arguments are passed to the function, you can return the stored value instead of performing the calculation again.

This is due to the fact that functions are deterministic, so if you have the same inputs you will always receive the same outputs: this allows to minimize expensive computations at the expense of memory.

Obviously this technique can't really be applied to functions that make use of pseudo-random numbers and connected functionalities, because memoization would completely void such randomization.

Memoization is usually implemented via decorators that check if the arguments passed are inside a defined data structure (usually a hash table): if there is a hit, the result is returned immediately, if not the original (expensive) function is run and its result is memorized in said structure.

A simple memoization system could work like the following UML diagram:

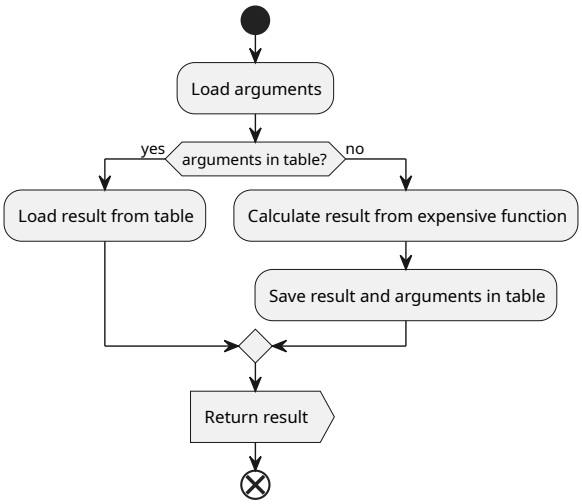


Figure 457: How a simple memoization pattern works

The biggest problem with this simple approach is that every different call to the function would be memorized, this would end up eating more and more memory, without any form of control.

The solution is deciding the "table size" of the results we want to keep: sometimes keeping the 10 most recently used calls is enough, sometimes we need more. Being able to control this will allow us to fine-tune the CPU vs. memory balance.

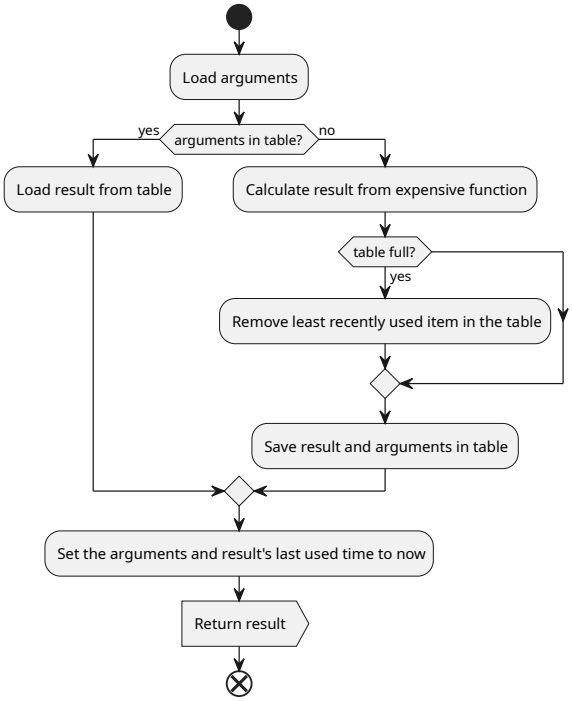
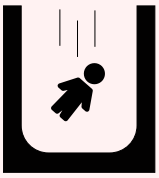


Figure 458: How a more complex memoization pattern works

Considering what we've seen so far, we can say that memoization should be used only on functions that are:

- Expensive
- Called often with the same arguments

Pitfall Warning!



If we start using memoization techniques on all functions, we may end up with a software that occupies a lot of memory without any significant speedup. Moderation is key.

Here's how a memoization pattern could be implemented:

Listing 143: An example of memoization

```
class MyMemoizedObject{
    let memory = {};

    memoizedfunction(parameter){
        if (parameter in this.memory){
            // If the result was calculated earlier, we can just return it
            return this.memory[parameter];
        }
        // If the result has never been calculated we do so.
        // ...
```

```
11          // Very complex and heavy calculations here
12          // ...
13          result = something_complex;
14          // Now we save the result in our memory, so other calls with the same parameter will be
        faster
15          this.memory[parameter] = result;
16          return result;
17      }
18  }
```

25.2.9 Approximations

Many times when developing games we don't need to have a value that is precise to the 10th decimal digit, that's where approximation comes into hand.

A prime example of approximation was used in Quake III Arena, via the algorithm known as "Fast Inverse Square Root". Back in 1999 calculating the inverse square root of a number was an expensive calculation for the CPU, so the developers decided to create an algorithm that would calculate an approximation quickly.

This was done by playing around with the floating point low-level structure and using a "magic constant" (`0x5f3759df`) to create a good "first guess", after that a single iteration of the Newton-Raphson Method is applied to refine the guess.

This proved to be faster than directly calculating a normalized vector (which uses a square root and a division, expensive at the time) and also faster than using a lookup table. The algorithm proved to be slower (and less precise) than the dedicated SSE instruction in the newer x86 CPUs.

25.2.10 Eager vs. Lazy Evaluation

Lazy objects are yet another possibility when it comes to optimization, with some drawbacks: you create an object but the calculations related to its state are performed when the object is first used, instead of when it is constructed.

This can be really useful when you have a great quantity of items that you are iterating through, one at a time, but don't need the whole collection at hand at once. When it comes to collections, lazy objects help saving memory at the cost of more CPU cycles while the game is running.

In some languages, this concept is abstracted in a language feature (like "generator expressions" in Python), while in others you'll have to work a little bit harder to get them.

Let's take an example, we have a custom object that contains a reference to a list of numbers: when we iterate through this object, we want it to return the numbers saved, halved.

Note!

What follows is just a didactic example, but should be simple enough to understand the difference between “eager” and “lazy” objects.

25.2.10.1 Eager approach

The eager approach is to take the list of numbers, create a second list inside our object with the numbers halved: this will make sure that the values are always ready and readily available, but will consume more memory. Here’s the example:

Listing 144: An eager object

```
class EagerObject{
    constructor(numbers){
        this.halved_numbers = [];
        // Prepares the halved numbers list
        for (number in numbers){
            this.halved_numbers.push(number / 2);
        }
    }

    getObject(index){
        // Returns the pre-calculated object at the requested index
        return halved_numbers[index];
    }
}
```

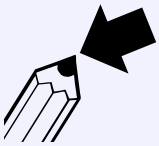
25.2.10.2 Lazy approach

If we know that we are working with millions of values, and we are going through them kind of rarely, saving all the halved values in RAM may not be a good idea. This is where lazy evaluation comes into play: instead of memorizing the value in RAM, we calculate it on-demand. Here’s the example:

Listing 145: A lazy object

```
class LazyObject{
    constructor(numbers){
        this.numbers_reference = numbers;
    }

    getObject(index){
        // Calculates the halved number on-demand
        return this.numbers_reference[index] / 2;
    }
}
```

Tip!



Lazy objects are great when you're working with bigger-than-RAM lists: each single value is in memory when needed, instead of the whole list. This comes at a cost: when you need to iterate on such list multiple times, the cost or re-evaluating the result may become a problem.

25.2.11 Detach your updating from drawing

An interesting approach to squeeze a bit more performance from your game could be detaching your updates from the drawing routine. For instance, your game could be refreshing its screen at 60fps, but its internal state is updated only 20 times a second.

This will obviously introduce some complications, since you may need some interpolation to make things work smoothly. This approach will get rid of a lot of heavy work (in the case above, you will get rid of $\frac{2}{3}$ of the updates), freeing resources for new things.

This means that you can process AI less times (thus "check for player's presence" once every 50ms instead of once every 16), physics can be processed less too (we don't care if a block starts falling after 15 or 50ms, the time is still too short for us to notice).

Detaching your updates from the drawing routines usually entails a change of language too: when you're talking about drawing cycles, we talk about "frames", while when you're talking about updates, you should be talking about "game ticks".

25.2.12 Be mindful of how you query your data structures

Sometimes small precautions can avoid a great deal of pain when it comes to optimizing our game.

Let's imagine that our game makes use of a single-file database (for instance SQLite) to save and load settings.

It is a lot harder for a database to deal with multiple small queries than dealing with one big query instead. So if you need to extract some data, try to avoid querying a database inside of a loop: instead focus on what data you need, extract all the data you need at once and then loop on the result of the extraction.

This doesn't apply only to databases, but also other data structures that allow for filtering.

[Do you know more about this? You can contribute, this book is open source!]

25.3 Tips and tricks

25.3.1 Be mindful of your "updates"

It is a common mistake among new game developers of putting the whole game logic inside the engine's `update()` method: this will eventually bog down the game and create inconsistencies when the framerate varies.

Input should be handled in your engine's event-based input system (very rarely you will need to check the keyboard

status inside the `update()` method), also you should absolutely take advantage of your engine's facilities when it comes to managing how the game updates.

For instance, Unity offers 3 update functions:

- `FixedUpdate()`
- `Update()`
- `LateUpdate()`

`FixedUpdate()` is executed with the Physics engine, so here is where you should apply forces, torques and any other physics-related function. Being run with the physics engine, this function may be called zero, one or more times per frame.

`Update()` is your run of the mill update function, it is always executed once per frame, without fail. This is used for other kinds of updates, if you do physics operations here the results may be inconsistent (since it doesn't run in sync with the physics engine). You can still move objects that are not tied to physics.

`LateUpdate()` is a utility function that is run once per frame, after the `Update()` function. This is useful for all kinds of operations that would require the `update()` calculations to be completed.

25.3.2 Use the right data structures for the job

Choosing the appropriate data structure for a task can have a lot more impact on performance that we may expect, and choosing the wrong one can have an even bigger impact.

So here are some small tips that work with the majority of programming languages.

Arrays are contiguous memory sections, thus indexing (finding an element at a certain position) is fast, as is scanning through the entire array itself. The limitation is that "pure arrays" have a well-defined size and cannot be resized: if you need a bigger array, you need to allocate memory for it and copy over the data. Inserting an item at the end of the array (if not full) is fast, but inserting an item at the head or in the middle of the array can be quite slow (since the items would need to be moved over).

Dynamic Arrays (sometimes called "Vectors") try to solve the "frozen size" of Arrays, while keeping the advantages. Pushing to the end of a dynamic array is usually fast (with the exception of the times the array is automatically resized to hold more items), but pushing items at the beginning (or in the middle) of the array is usually quite slow (because all items would need to be moved). Dynamic arrays tend to "overcommit memory", so they may be bigger than necessary (to save on the computationally heavy task of resizing and copying over items).

Linked lists are good if you need fast insertion anywhere, but they tend to lack in the iteration department: since the nodes are not contiguously packed in memory, iteration can be slow.

Hash tables are good if you need to memorize items in a "key-value" fashion and retrieve them very quickly, but they use more memory and may fall short in terms of performance if a bad hashing function is involved.

There is no "silver bullet" when it comes to data structures, but knowing the basics can make your code a lot better:

there are more advanced data structures, like heaps, that are discussed in this book, check them out!

25.3.3 Dirty Bit

Not all entities in your game need to have their state updated all the time. Continuously updating all entities' internal state can be really costly in terms of game performance.

A quick way to make your game lighter on resources (and thus more performing) can be putting a boolean check at the beginning of the update function, checking if the object really needs to have its internal state updated.

A possible example could be the following:

Listing 146: Example on how to optimize entities with a dirty bit

```
class Player{
    constructor(){
        this.speed = [0, 0];
        this.needs_update = false;
    }
    // ...
    input(){
        // ...
        if (right_key.is_Pressed){
            this.speed[0] = this.speed[0] + 1;  // Move right
            this.needs_update = true;
            // ...
        }
        if (up_key.is_Pressed){
            this.needs_update[1] = this.speed[1]  -100;  // Move up (jump)
            this.needs_update = true;
        }
        // ...
    }

    update(dt){
        if (this.needs_update){
            // Do Update instructions
            // ...
        }
    }
}
```

If your code is well-done, you won't have issues like animations freezing, because those will be separated from the "update routine", since the animator will chug along its frames when requested by the `draw` function.

25.3.4 Far-Away entities (Dirty Rectangles)

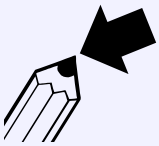
Another way to optimize your game performance is not updating entities way off screen: this is also a technique used in the game Minecraft, where entities are frozen when you are far away from them, to save on resources.

A possible idea would be having an “updatable rectangle” (sometimes called “Dirty Rectangle”), bigger than the screen, and only the entities inside such rectangle will be updated.

This could create some issues when it comes to games that have their challenge deriving from entities updating in sync with each other, thus if we implement this “updatable rectangle” one or more entities would fall “out of sync”, possibly making beating the level impossible.

In that case we may just put out an exception (where certain entities are updated no matter what) or divide our level into smaller “rooms” that are instead entirely updated all the time. Another way could be updating the internal state of the objects, but not drawing them at all, which would still lighten the workload.

Tip!



If you want to keep updating far away entities (for instance to avoid seeing all entities start updating as soon as they enter the screen), you can update entities every other frame, or just update one half of the entities on one frame and the other half on the next.

25.3.5 Tweening is better than animating

Animators and animation frames are performance-hungry and should absolutely not be used in all those situations where you can instead use inbetweening techniques.

This means that frame-by-frame animations should not be used when taking care of moving UI parts: if you want to slide a piece of UI (take for instance a drop-down terminal from a “computer hacking” game) from the top, you can just tween its position and save a lot of memory.

Remember that Tweening doesn’t apply only to positions, you can tween any property of a game object.

So a quick way you can optimize your game, is removing all the unnecessary animations and replace them with tweening, your game will surely benefit from that.

25.3.6 Remove dead code

There are many definitions for “dead code”, some use the “unreachable code” definition (for instance code placed after a “return statement”) some use a more extensive definition.

I like to think of dead code as “wasted code”, which is:

- Anything that happens to be written after a “return statement” in a function: return statements are used to give control of the program back to the caller of a function, so this code will never be executed;
- Unused variables: variables are allocated in memory, require calculations and CPU cycles, if not used that’s just a waste;
- Unused code: complete functions that are never called are a waste of memory (because they may be loaded in RAM) and of disk space, making the executables bigger;

- Debug code: sometimes we need to write code to debug other code, this code may end up being part of a "release version" and weigh it down, this may also make the game more sensitive to cheating and hacking.

You should be careful when optimizing out dead code, even more when you are dealing with functions which result is not used: those functions may change some global state (or change stuff by usage of *side effects*[g]).

25.4 Non-Optimizations

In this small section we take a look at some alleged "optimizations" that actually do nothing (or close to nothing) for our game's performance.

25.4.1 "Switches are faster than IFs"

Some people allege that using "switch" statements instead of "if" statements is bound to optimize the game. This an overstatement, and we can prove it with a simple test.

Let's create two C++ listings, like follows:

Listing 147: IFs vs Switch - IF Statements

```
#include <iostream>
using namespace std;
int main(){
  for (int i = 0; i < 10000000; i++){
    int x = rand() % 5;
    if (x==1){
      cout << "One" << endl;
    }else if (x==2){
      cout << "Two" << endl;
    }else if(x==3){
      cout << "Three" << endl;
    }else if(x==4){
      cout << "Four" << endl;
    }else if(x==5){
      cout << "Five" << endl;
    }
  }
  return 0;
}
```

Listing 148: IFs vs Switch - Switch Statements

```
#include <iostream>
using namespace std;
int main(){
  for (int i = 0; i < 10000000; i++){
    int x = rand() % 5;
    switch(x){
      case 1:
```

```
8           cout << "One" << endl;
9           break;
10        case 2:
11          cout << "Two" << endl;
12          break;
13        case 3:
14          cout << "Three" << endl;
15          break;
16        case 4:
17          cout << "Four" << endl;
18          break;
19        case 5:
20          cout << "Five" << endl;
21          break;
22      }
23    }
24    return 0;
25  }
```

These pieces of code will be compiled without any optimization, using G++, using the following command:

```
1 g++ -Wall -Wextra -Werror -O0 filename.cpp -o filename.bin
```

Where “filename” is replaced by the source name, then each file will be executed using the “time” Linux command, like follows:

```
1 time ./filename.bin
```

Below we can see the results for both the codes:

```
Two
One
./if.bin  3.94s user 14.39s system 95% cpu 19.166 total
```

Figure 459: Time taken by the IF code

```
Four
Two
One
./switch.bin  3.84s user 14.22s system 95% cpu 18.915 total
```

Figure 460: Time taken by the Switch code

We can see a difference of just around 0.25 seconds, over 10 Million iterations. If you changed an equivalent IF statement for a Switch statement, you would earn a quarter of a second every 46 hours of gameplay at 60fps.

The right choice is the simply choose the structure that lets you have the most readable code: the more your code is readable, the easier it is to understand; the easier to understand, the lower the probability that there is a bug in there (or a performance hog of some sort).

25.4.2 Blindly Applying Optimizations

There rarely is something more wrong you can do when optimizing than blindly applying optimizations without considering your application context.

Using resource pooling in an environment with limited memory (but plenty of CPU power) can prove a disaster: it's better to instantiate and destroy objects in such case.

Sometimes animators can be faster than LERPing/Tweening, mostly when you have to tween objects with multiple children: tweening would create a lot of CPU-bound calculations for the new position and size that will make the whole thing bog down and get choppy.

The only thing you can do is think first and try later: this book can give you some suggestions, but nothing should be taken at face value. Remember the context your game is working in and **do not treat all platforms like they're the same**: WebGL is different than Console which is different than Mobile.

Part 7: Marketing and Communities

26 Marketing your game

> Here's my whole marketing idea: treat people the way you want to be treated.
>
> *Garth Brooks*

26.1 An Important Note: Keep your feet on the ground

Let's be honest: the "indie success stories" you see everywhere are going to give you a false sense of hope, that "there is space for everyone in the industry".

I'm going to break it to you: if you're in for the money, you already lost.

The reality is that many game developers are horribly underpaid and many projects failed and many others are going to fail.

If you're in for the sake of creativity, to make something you want to make, seeing your project perform "under par" will hurts less, if at all. Your first project won't be a best-seller, but doing something you enjoy will make it better.

This book is meant to teach the basics of game development as a way to channel your creativity, and to help you understand the pitfalls and mistakes that can keep you from showing your best product, in the best possible way.

26.2 The importance of being consumer-friendly

We live in a "Money-Driven" world. This is a fact, not an opinion. So, leaving morality out of the discussion, you can argue that the general game publisher mentality of "getting all the money" is not that wrong. But you don't have to be an idealist to realize that we should see the world for what it could be, and not for what it is at the moment. We should apply this mentality to every aspect of our world, game industry included.

We are NOT here to enlighten you about how game industry has to change, but every game developer should realize that the true success of a game is **not** based on sales, it is based on customer satisfaction. So, even if this cursed "publisher mentality" could be applied to small indie developers (spoiler: it is not), we have to fight it back, and restore the "customer satisfaction above all" philosophy. So, fun fact: the only thing we (small indie developers) can do is *doing the right thing*.

Focus your effort on customer satisfaction, you have to be consumer-friendly!

My first advice is: instead of implementing a thousand features, make one single thing a thousand times better (Bruce Lee style). If you promise the client a ton of features to do you can generate hype (we will discuss it later), but if you are an indie developer (and most of times even if you're not) making one thing extremely enjoyable is way better.

Why? Because your goal is to create something (even if it's only one thing) that will make the customer remember your game: Quality over Quantity.

Satisfying customers is no easy feat, we all know this. So one question you may ask is: How in hell can I be original?

Answer is: You don't have to.

So, my second advice: There are million of games out there, so creating something never seen before is very, very, very, very difficult. So, try to innovate what already exists, make usual features in your own way. This is how sub-genres come to life.

You think this is not the right way? Then go and tell the people from `From Software` that the "Souls-like" sub-genre was not innovating.

Last advice: Gameplay over Graphics.

We're not Activision, we can't afford to spend 10 million bucks for a cutscene. Aim for the "old fashion gamers", the customers that play for fun, not because some game is gorgeous to see. Whatever your working on, whatever your game genre is, focus on making the gameplay fun to play. Things are really this simple. Work 2 hours less on a model or sprite and 2 hours more on thinking about how to not get your customer bored.

Random Trivia!

Around July 2015 a game called "Brutal Force" (a clone of the more successful "Nuclear Throne") was hit with a wave of criticism after they raised their base price during a sale. The game was 75% off, but its sale price quickly went from 99¢ to $1.99 until it reached $3.24. The moral of the story is "never lie to your customers".

26.3 Pricing

If you want to sell your game, *Pricing* is one of the 4 "P"s of marketing you should pay attention to (along with "Place", "Product" and "Promotion"), here you will find some tips on how to price your game.

26.3.1 Penetrating the market with a low price

The video games market is quite large but also quite saturated, and it is really hard to stand out without a competitive visuals, mechanics and most of all: price.

Penetrating the market is easier when done with a low price, even more when your game does not boast "innovation" as one of its own strong points. Usually "retro" titles benefit from this pricing model.

After the initial "cheap" phase, you should be able to recover your investment by selling "premium" products later (like higher-budget games). The main objective of this strategy is to create the so-called "brand awareness", in short: letting yourself get known.

26.3.2 Giving off a "premium" vibe with a higher price

When instead your game has something more (like cutting-edge, never seen before graphics capabilities), you may be able to keep a high price, giving off the vibe that your product is "superior" to its competition.

This can also work with products that offer really innovating concepts, and rarely works with "run of the mill" games that you see almost anywhere.

26.3.3 The magic of "9"

There is something psychologically magic about the number 9. When you price something at $0.99 instead of $1.00, it looks like you're giving away the product for almost-free. Same goes for pricing your game $39.99 instead of $40.00, the price will look a lot closer to $30.00 than $40.00, even though the difference to $40.00 is just 1 cent.

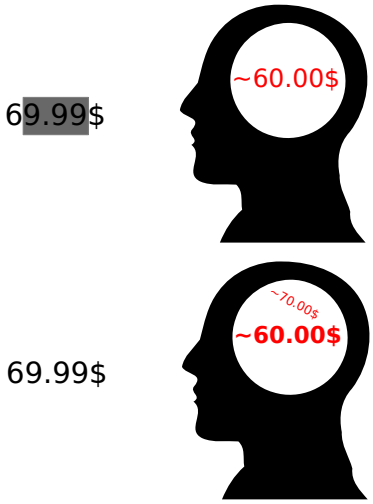


Figure 461: How first impressions leave a mark, even when it comes to price

This effect is due to how our brains work: we're always trying to "be lazy", so our brain does a "first pass" which reads the price as being in the 60$ range (for the reference image), and only then we realize it's actually much closer to 70. By that time, we already had a "first impression" of the price, which will be hard to remove.

26.3.4 Launch Prices

Another way to create product awareness is setting a "launch price": a discounted price for the period the game is launching (for example a week), after that the game's price goes back to its "listing price".

This will ensure a quick influx of players (even more if you built hype well) that want to get your game for a bargain.

A variation could be having a low-priced base game, which then can be monetized with Downloadable Content (DLC) that add more to the base game.

26.3.5 Bundling

Another way to entice players to buy your product is "bundling", that is "offering more for the same price". If your game has a really notable soundtrack, a good idea could be selling such soundtrack along with the game (which could help you pay your composers), or maybe a digital version of an artwork book, or sketches that can give an insight on the development.

Other types of bundling can include other games that you produced, or being part of a bundle of games of the same genre. An instance of "bundling" could be the famous "Humble Bundle", which bundles games around a certain theme or for a defined occasion (like Christmas Holidays).

It could be interesting to offer flexible prices on your bundles, like the following:

- Base Game: 9.99$;
- Base Game + Soundtrack: 14.99$;
- Base Game + Artwork Book: 15.99$;
- Base Game + Soundtrack + Artwork Book: 18.99$.

This will allow your players to choose which "extras" they want to buy, making them more comfortable in the process.

26.3.6 Nothing beats the price of "Free" (Kinda)

Free to play games are all the rage these days, you can play a great game for free, if you're willing to resist all kinds of microtransactions and premium DLCs.

That is actually one of the problems with Free to Play games: you must not get "too naggy" about microtransactions, or your player will get annoyed and stop playing. Also your game must still have enough qualities to be played: if your game is evidently low-quality, no one will play it, not even if it's free.

26.4 Managing Hype

It's time to discuss about promoting our works.

Let's start with an obvious (sadly not that much obvious) statement: We are **not** a big publisher, we cannot afford to make big mistakes.

Truth is that companies like EA, Ubisoft, Square Enix, ActiVision (and many more) can do pretty much everything they want. Why? Because they know that a massive part of their audience will buy the product "sight unseen", no matter what.

We are not in this situation, so we have to be very cautious. We must be aware that bad moves (not mentioning illegal ones) lead to major consequences, and most of the time indie developers cannot pay this price.

Most important thing to keep in mind is: never promise without keeping your word. This is very important, especially if you're making people pay and the game is not complete yet (so-called "Early Access").

You're not a big publisher, people don't trust you and will not pay you for a promise. You have to give them something worth the price, immediately enjoyable. Overpricing your game and justifying that because of "future release/contents" is NOT going to work either.

Patreon (or crowdfunding websites in general) could be an interesting compromise for such situation, but you have to use it in the right way, and never abuse your supporters trust. Keep in mind all the basic tips for managing hype we just discussed, and make sure to respect promised releases/updates when related to a certain amount of income from your crowdfunding (so-called crowdfunding "goals").

26.5 Downloadable Content

DLC changed the game industry, this is a fact. How? You may ask.

The answer is very simple: it's an excellent way to extend a game's lifetime, especially for a single-player games. They are both a way to keep people playing a game (or make people come back) and, at the same time, earning money.

As always, there is a good way to do things, and a very bad way.

26.5.1 DLC: what to avoid

In my opinion there are two major behaviors to avoid, besides the obvious "Do NOT overprice". I'll illustrate them, showing some examples.

1) Putting the "true" ending in a DLC This behavior is almost insulting for the customer. Creating a game and put its ending in a DLC is equal to sell an unfinished product, this is the truth. For a "casual fan" it may not matter much, but it's very annoying for a true fan of the game/saga. Why? Because you are implicitly forcing him to pay twice: The game and its DLC. Some major game companies exploit this mechanic to simply cut off some content of the original game and put it in a DLC: same effort and more income. I found the pair EA/BioWare to be the major example for this behavior with Dragon Age Inquisition and its DLC "The Trespasser" comes immediately to mind.

2) Selling a Season Pass "on trust" With this behavior you're demanding a lot from your customers, things can go really wrong. To put it simple you're selling a Season Pass without presenting the DLC. Ideally if you're selling a season pass from day 1 than you should, at least, let the audience know about the number of future DLC and their basic content/quality/quantity. So what happens if an over-hyped game is released alongside its season pass and the whole thing is quite a flop? That's the case of FFXV. Square Enix made quite the gamble, and it really didn't pay off. But Square Enix is Square Enix, so it won't go bankrupt any time soon.

26.5.2 DLC: what to do

The Golden Rules are:

- Limit DLC to **additional content**. Related to the main story, but not concerning major twists or the finale itself;
- Create the DLC with an adequate playtime. If your game is an RPG with over 60 hours of content you simply cannot make a 5-hour long DLC;
- Sell a Season pass only if you're going to publish at least 3 distinct DLCs, or two very extensive ones;
- Obviously make the customer pay a fair price.

I find CD Projekt RED the absolute best company when it comes to DLCs. In particular "The Witcher 3 Wild Hunt" 's two DLCs have all the good qualities that we want.

The adequate content, a fair price and we're talking about a combined 50 hours of extra game time.

26.6 Digital Rights Management (DRM)

Another thing that can make or break a game's enjoyability is the presence of a Digital Rights Management System (shortly referred to as "DRM").

A DRM is a piece of software that silently sits in the background of a game to verify its authenticity, thus limiting piracy for the first months of the game being released.

A question comes up to mind: Why would an invisible piece of software influence a game?

There are some major reasons why a DRM System could drive away customers:

- The DRM interacts with the game in an annoying way;
- The DRM behaves outside of its scope;
- (An extension of the previous point) The DRM is known to spy on its users.

We have cases of "DRMs interacting with the games badly" left and right, for instance the "always online" DRMs that stop you from playing the game if your connection drops for a second, or some famous instances where more DRM Systems were stacked on top of each other, slowing down the game's performance.

This brings to something that many find annoying and unfair: people pay to play an original game and are hindered by DRM, while pirates are free from such hindrance for the fact that they pirated the game. You paid for the game, and you pay the price for other people pirating it, in the form of frame drops, slowdowns and sometimes flat out unplayability.

Another thing players hate is having a piece of software that acts out of its scope or, even worse, similarly to a *malware*[g].

There have been cases of DRM systems installing themselves as *rootkits*[g], making it impossible for the player to uninstall them, even after removing the game said DRM was tied to.

For instance that's what happened with the free game "Sonic Gather Battle", a tournament fighter fan game starring the beloved Blue Hedgehog. This game contained a piece of software that would track your browser tabs: if you searched for cheats/mods/hacks, it would close your browser and enable a "protection stage", making the game impossible to play, by adding immortal enemies that would one-shot you.

At the same time such DRM would have raw disk access and would also search for Cheat Engine installations. To top it off, the game would send the computer IP to a server to be saved in a list, so that the "protection stage" would still engage, even after a fresh install.

Some research found out that the game would sometimes require administrative access to the system, which shouldn't usually happen.

The bottom line is: a DRM doesn't always guarantee more people will play your game, so if you have to implement a DRM system, **do it well**, do some tests, see if there are framerate drops, stutters, see how the game behaves when the internet goes down while playing, etcetera…

Some games even decided to go DRM-less, such as any game sold on the GOG (Good Old Games) marketplace. This is another important decision you have to think through, to ensure your game has the highest chance of succeeding as possible.

26.6.1 How DRM can break a game down the line

In November 2019, Disney's game "TRON: Evolution" has been reported as "unplayable" due to the Disney's SecuROM license expiring. This had serious ramifications.

The game cannot be played, since SecuROM checks periodically the activation status and reports a "serial deactivated" error and the people who bought a legit copy of the game found themselves unable to enjoy their rightfully acquired product.

This is another situation where piracy wins, since people who pirate games have a way better experience compared to the people who legally bought the game. Players are in the hands of the game publisher and their intention of maintaining the game; as well as in the hands of the DRM manufacturer, since no software is eternal, the DRM servers may close down, blocking the authentication.

26.7 Free-to-Play Economies and LootBoxes

Let's face the most controversial topic, here we go.

26.7.1 Microtransactions

This is quite the hot topic at the moment: the infamous, devilish microtransactions. Let's put this clear from the start, microtransactions are NOT intrinsically bad. As always there's a good way and a bad way to make things. In that case the bad way can bring really backfire nastily.

Microtransactions should be limited to fashion accessories and avatar customization. Nothing related to game mechanics or gameplay-related elements. Everyone **hates** pay-to-win.

We must acknowledge that there are people willing to pay in order to have some bling for their avatar. Microtransactions, and lootboxes as well, should be used for this kind of audience.

I found Blizzard to be the best example for a good implementation for Microtransactions and lootboxes. Overwatch, for example, presents the best example of lootboxes, giving the audience access only to fashion/customization features for the heroes, like: different skins, animations, vocal sounds.

On the other hand EA, with the Fifa series, is the perfect example of how to NOT DO microtransaction/lootboxes. The reason is very simple: Fifa Ultimate Team is the biggest, most greedy, pay-to-win mechanic ever.

You could question me, stating that lootboxes are gambling microtransactions. You're right. It's a tricky topic, so it has to be done right, but it's not bad for the sake of existing. Otherwise we wound end up making an inquisition about how people spend their money.

Now many countries are making an effort to regulate lootboxes (some even making them straight up illegal), so you should be always informed about local laws, and if you decide to implement them, you should know the duties that come with it.

26.7.1.1 The human and social consequences of Lootboxes

Among all monetization schemes, lootboxes are probably among the most controversial. At least the ones that are paid with real-life money.

The objective of lootboxes and microtransactions is to generate a secondary revenue that some experts call “recurrent user spending”.

The idea behind lootboxes is having a prize pool from where a random prize is selected, the user buys a lootbox and gets a randomized item (or set of items), usually accompanied by an epic set of animations, colors and music.

This mechanic can remind someone of slot machines, which share the same psychological principle behind their idea.

Flashy lights and music are tailored to trigger certain chemical reactions in our brains, which will make them seem fun. This adds to all the “off by one” psychology, that will hook you to have “one more go” (how many times you heard of people losing the lottery because of numbers “off by one digit”?).

The situation gets even more serious when you want a very specific item in the prize pool, which lowers the chances of a win in the player’s eyes and consolidates the “one more go” way of thinking.

Things become horrible when kids are involved, if they see their father buying them a lootbox and learn how to do it, sooner or later the credit card will be rejected (and it happened - internet archive link to a BBC article).

The situation becomes catastrophic when you start putting people with compulsive disorders, or people with gaming addiction, gambling addiction and the so-called “whales” (people with high amounts of money and high willingness to spend such money in video games). Some people use videogames as a mean to escape their gambling addiction and lootboxes can trigger the same mechanisms in the peoples’ brains.

This obviously has serious consequences when it comes to consumer trust and the image of your game development studio (or even yourself) that can culminate into a full-blown boycott of your products.

Remember: you are not a big publisher, you cannot take the hit of a really bad decision.

26.7.2 Free-to-Play gaming and Mobile Games

Let’s talk a bit about free to play games.

We cannot approach that topic without speaking a little about mobile games, the very kingdom of free to play gaming.

First of all: **What NOT to do**:

- Make people wait/pay in order to play (The so-called “energy system”, very popular at the moment). Pay-to-Win mechanics are bad, but almost understandable, but Pay-to-Play ones are simply disgusting;
- Make intrusive banners/advertisements, that stop gameplay or reduce the available screen size;
- Avoid Pay-to-Win, especially in a multiplayer game.

I also want to give you a major advice about how to use banners/advertising in a smart way:

Let people decide if they want to view an advertisement or not. Especially in a single player game, if you give them something big in return (for example double points/coins/gold for next game, or the chance to continue after a death) the will accept the offer.

This mechanism has two major positives:

- Gamers are NOT annoyed by the use of advertising, they usually see this mechanism as a trade deal: "Do ut des" (I give you, you give me in exchange);
- If they do it once, they're probably going to do it again. At the end they willingly watch advertisement for you.

It is obvious that you shouldn't kill your player in-game over and over so you can offer them the chance of watching an advertisement to continue playing, that will be seen as a form of "pay to play", from the player's perspective.

26.8 Assets and asset Flips

Pre-made assets are both a resource and a liability when it comes to video game development, depending on how they are used.

Assets can be purchased by developers in a so-called "asset store" **as a basis**, a foundation upon which they can build their own game, that's how they are intended to be used and how they should indeed be used.

Sadly for every good thing, there must be a bad thing too: **asset flippers**.

Like "flipping a house" (minus the work), a group of assets can be put together (a bit like building blocks) into a somewhat working final product without much (or any) work, making a so-called **asset flip**.

These products are usually sold to adventurous or unknowing consumers for a quick buck or people who hope to make some money out of "out-of-game economies" (like Steam Cards or Tradeable Items), while the quality of the final product is lacking at best or terrible at worst.

This circles back to "being consumer-friendly": you're not selling to a single person, but to a lot of people with different backgrounds and expertise, and one of those **will** find out if your game is an "asset flip".

Flipping assets is the equivalent of betraying your customers, as well as living entirely on the work of who made said assets.

In short: assets are a good basis to work on your game (or make placeholders), but they need to be refined to elevate the quality of your game.

26.9 Crowdfunding

Crowdfunding has been a small revolution in the world of the "idea to reality" business. You publish an idea, with a working sample and some explanation and people who are interested in your idea invest money.

This makes transforming an idea into reality a lot easier and removes a lot of economic worries from the developers.

Although for every good thing, there is always a pitfall: be it scams, things that utterly impossible to create or mismanagement, not everything that gets “funded” is going to go well.

Here are some tips that will help you in this confusing world.

26.9.1 Communication Is Key

There is an Italian saying that can be translated to something like the following:

> A good worker must work, know, know how to work and let know about their work

This underlines that you may have knowledge and practice, but communication is a key factor in the success of any kind of work.

Remember to keep your backers updated at least weekly, avoid any language that could create misunderstandings (someone could misunderstand a “cutesy language” for a “mockery of the backers”) and always justify your decisions. After all you’re talking to your own investors.

26.9.2 Do not betray your backers

Your backers are your first and most precious customers: they’re the ones that bought your product “sight unseen”, and invested money without knowing what the result will be.

Betraying your backers by taking unpopular decisions is a surefire way to see chargebacks and retaliation. If you suddenly decide to change the store where your game will be published to something different than promised, you better bring some **very strong** arguments for it.

Don’t lie, if you receive a cash infusion to publish your product on a certain platform (even as a “temporary exclusive”), tell your backers about your decision and the reason being said cash infusion: some will understand, some won’t like it, some will get angry. It’s still better than everyone getting angry when they find out you lied to them.

26.9.3 Don’t be greedy

Greed is never well seen in the world of crowdfunding, having many projects asking for funds at the same time does not look good and opening a new crowdfunding campaign while a funded project has not yet been delivered looks even worse.

Having many ways to fund the same project may seem a good idea, but it usually isn’t and gives out the impression of greed, not even to give the chance of “people who are late” to fund your project.

Another mistake are the so-called “stretch goals”: encouragements to fund the project over its goal, in exchange for more features. This is a serious mistake that can break an entire project if you are not able to manage it correctly, because you’re trading money for more work, and more work usually means a delayed release.

Stretch goals should be very few, already planned in the time table and well thought out. If they are not, you will pay for that mistake.

26.9.4 Stay on the “safe side” of planning

A common mistake made by people who are getting started in development, is getting the times wrong. It happens. It’s normal.

Planning a project is hard, and it’s way too easy to get it wrong and find yourself with a deadline knocking at your door and having nothing functional in your hands.

> When you plan, you should always double your project duration - in case of unforeseen events. Then you should double that time again - in case of events you really cannot foresee.

26.9.5 Keep your promises, or exceed them

When you present your idea, with a working sample (usually requested by many crowdfunding websites), you should keep in mind that people will use that sample as an “anchor”: a fixed reference for the minimum quality expected for the product.

There is no “not game footage” that will save you. You promised, you have to deliver.

The only way to deliver a product that is different from the promised one, is to deliver something that is objectively better than what you promised (better graphics, gameplay, sound, story, features, ...)

26.9.6 A striking case: Mighty No. 9

I want to spend a few words on a crowdfunding campaign that failed to meet expectations and broke all the rules we just saw.

The game was asking for a pledge of 900000$, clearing that objective in just 2 days, and managed to raise over 4 Million $ between kickstarter and PayPal. The game trailers showed great graphics, amazing soundtrack and fast-paced gameplay.

Sadly the crowdfunding was really mismanaged, the campaign updates happened often but were kind of disconnected with the reality of development and the campaign was littered with **SIXTEEN** stretch goals (with more foreseen, given the graphics in the kickstarter page).

The product delivered was under the expectations, the graphics were nothing like the trailers, they looked poor and cheap, the game was set to release in April 2015 but was delayed until June 2016.

The creator of the campaign was dreaming for a whole “Mighty No.9” franchise, with physical rewards and an animated series, but didn’t make the cut, even after receiving almost 5 times the asked money and a partnership with the publisher Deep Silver.

What did this game get wrong?

- Too many stretch goals;
- Not delivered in the time promised (the game was delayed a lot of times);
- The final product was way under the expectations;

- Too many platforms to release on (10 of them!);
- The official trailer's "Make them cry like an Anime fan on prom night" - given that the game is very anime-like, this sounds a lot like a mockery to the fans and backers;
- Before delivering the game, the kickstarter account used for Mighty No. 9 was used to create a new kickstarter (Red Ash -Magicicada-), which does not look good when you're late in delivering a product;
- Too much to do for a project in its beginnings: there were physical releases and rewards planned, as well as an animated series and a whole franchise. This is too much of a gamble for a product that has still to get its first sale.

If you want to know more about what happened, there are a lot of YouTube videos and articles talking about the matter, it really makes you think about how you should behave when opening a crowdfunding campaign.

26.10 Engagement vs Fun

In the games industry we can sometimes come across the terms "fun" and "engagement" in a way that makes them seem like synonyms, but they definitely aren't.

Let's start with the hardest to define: fun. Fun is really subjective, as humans each one of us has a radically different concept of "fun". The Oxford English dictionary comes to help us, in a way:

> Light-hearted pleasure, enjoyment, or amusement; boisterous joviality or merrymaking; entertainment.

This is a really broad definition, but the concept of "fun" itself is really broad and complex, and cannot become more than a few common points that we can define across games, thus making "designing fun" a form of art, instead of just an act of engineering.

"Player engagement" is completely different, it has a well-defined metric: play time.

It is possible to artificially extend the players' play time by tweaking the difficulty so that a player becomes frustrated enough to keep trying, but not enough to drop the game, occasionally adding a prize to keep the player gaming on (no matter how well-tweaked your experience is, without payoff players will leave sooner or later).

Is the player having fun when the experience is "tweaked" as such? Probably not, but the game is tailored so that the play time is longer.

The fun goes down, but the play time goes up. Fun and engagement are not the same thing.

"Player engagement" (and thus "play time") is one of the metrics that is used by big publishers to analyze and hook players into in-game economies and lootboxes, the more the player engages with your game, the more you can "advertise" lootboxes and in-game economies. These metrics take a bigger role when the game takes the form of a "live service".

26.11 Streamers and Content Creators

When you're developing or publishing a video game, you should not underestimate the power of so-called "content creators": streamers and YouTube stars, small or big, can turn your game from a niche project right into a cash buster.

Let's take some examples of games that, even though great on their own, would not be the same without streamers:

- **Minecraft:** this game is great on its own, but having a big number of streamers covering it definitely helped its growth, so much that Mojang got bought by Microsoft Corporation;
- **Fortnite:** even though pushed by a big company like Epic Games, this game benefited from the streaming community a lot;
- **Fall Guys:** published by Devolver Digital, this game cashes in on the "free for all/battle royale" genre, while giving a quirky and funny twist to it. Being a quirky game on its own right makes it an ideal game to stream, which definitely helped its sales.

I want to bring up one last example before delving into the nitty-gritty of the Streamers/Developers symbiotic relationship: **Among Us**.

"Among us" is a game that went from "unknown" to "cash buster" in a few weeks thanks to streaming. It got its "prime time" in 2020, after streamers brought it to light: it's a multiplayer, "Town of Salem"-like experience based on deception.

The thing is, this game was released in 2018, and it was not doing too well on its own, until streaming came in the picture. Being a very interactive game, this made it ideal for collaborations between streamer friends, striking a nice balance between suspect, suspense and strategy. After streamers came into play, the sales exploded.

The Content Creator/Developer symbiotic relationship is one of the best win/win situations that can happen in the video game development, let's see how.

26.11.1 The game developers' side

For a game developer, having a game covered by a streamer has one big advantage: **free advertisement**.

Some big streamers can gather between 15.000 and 35.000 viewers *alone*, giving your game a huge boost in coverage. Most of the times the game is bought by the streamer themselves, with their own money, so you already sold a copy, but its worth is a lot more than meets the eye.

This boost in advertisement has consequences, if the game looks fun people will be encouraged to buy it, resulting in a surge in sales. The more the people who know about the game, the more it spreads via "word of mouth", furthering your sales.

There is no "traditional advertising" that can bring that much attention to your game.

26.11.2 The Streamers' side

For a streamer, having a new game is definitely advantageous: it's fresh new content.

Fresh content keeps a streaming channel interesting, engaging the public, which in turn generates advertising revenue.

Such revenue can be spent by the streamer for better gear, but also new games which can bring even more fresh content to the channel, thus keeping viewer engagement high.

26.11.3 Other entities and conclusions

This “symbiosis” doesn’t limit itself to game developers and streamers: advertisers earn because their products are exposed, other game developers earn money by having their products bought to bring new content, gear manufacturers (keyboard, mice, monitors, cameras, lights, green screens, …) get revenue from streamers looking to improve the quality of their streams. This is sector economic growth at its finest.

On October 22nd, 2020 the Creative Director at Google Stadia threw out the idea of making streamers pay a fee (or a revenue share) to game developers (or publishers) for the content they stream. Given what we just saw, and the great advantages it brings, having a “streamer tax” would remove content creators from the video game environment, nullifying all the good that comes from them.

Small streamers would not be able to pay such “tax”, bigger ones would be willing to change their content (moving away from videogaming) to not pay such “tax” and not suffer many losses: people are watching streams because they like the streamer’s behaviour, reactions and jokes, as well as gameplay style and technique. If people watched streamers and content creators for the games they play, views and subscriptions would not be as stable as they actually are.

It would be changing a system that works for one that will, in the most absolute way, not work.

One of the best things you can do as a game developer is harnessing the power of the streaming community: reach out to some small streamers, give out free keys, offer review copies, give out preview copies of DLCs, be friendly and supportive. Small streamers may become big ones one day, and supporting mutual growth is a really satisfying experience.

27 Keeping your players engaged

> There is no power for change greater than a community discovering what it cares about.
>
> *Margaret J. Wheatley*

Having a solid story and great game mechanics is only part of your objective: you want your players to talk about your game or play it for as long as possible, this can be done in two main ways: **communities** and **replayability**.

27.1 Communities

Coding a game is just part of the whole game development ecosystem, engaging your community and having a fruitful exchange of ideas and knowledge can be an amazing resource to keep your game inside players' minds.

27.1.1 Forums

Forums (also known as "boards") are websites where people (in our case players) engage in discussions on certain matters (fittingly called "topics").

Forums can be places where our players can share their fan work, discuss about the game and give suggestions.

Forums can be really useful a makeshift "service desk", where players can report bugs and any kind of weirdness your games may present to them.

27.1.2 Wikis

Wikis are a collaborative knowledge source where people write different pages about a certain matter.

In our case having a wiki about our game can contain various pages talking about:

- Characters
- Levels
- Bosses
- Enemies
- Walkthroughs

The great thing having many players collaborating on the same matter is that someone may be able find something someone else missed, helping to build a better understanding of the game.

Searching for more information about a game can be a kind of game itself, building and strengthening a community.

27.1.3 Update Previews

Beta channels and update previews can be an incredible way to engage your most faithful fans as well as:

- **Give you more testing ground:** having many players test your game can give you further testing than any small team and also let you test server load;

- **Give you some market insight:** having some different point of view can give you some insight on whether your players will like the new features you're preparing or not.

27.1.4 Speedrunning

When your game has a decent success (or even if it doesn't!), there may be someone who wishes to challenge themselves into completing it as fast as they possibly can, sometimes with the aid of glitches.

The speedrunning community can really change your game from "obscure" to "well known", showing it at events and gatherings, making a fun show out of it, for better or for worse (hey, not all donuts come with a hole!): even when your game is "not so good" it can still be a fun oddity that people may want to speedrun.

Giving your own game a "speedrun mode" can prove beneficial, lowering the entrance bar to speedrunners and letting them know that you are thinking about them and would like them to speedrun your game.

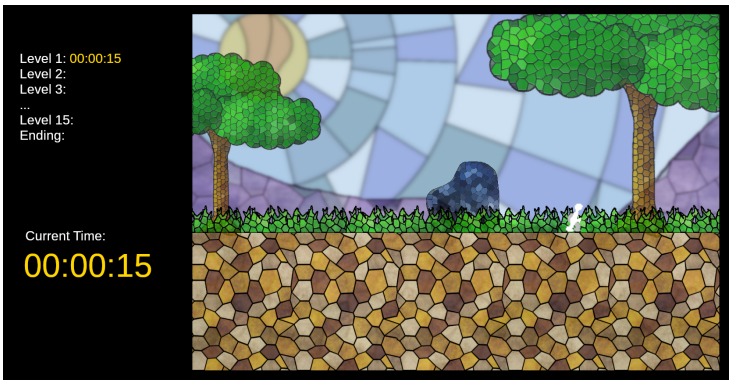


Figure 462: What a simple "speedrun mode" may look like

What could such "speedrun mode" contain? More features than you may think about:

- **A game-wide chronometer:** this will allow to register how long beating the game takes;
- **A time for each section:** usually called "splits", they can feature the difference with older recorded times for each single level or section of the game;
- **Automatic Cutscene Skip:** cutscenes are usually just a waste of time in speedrunning, some may appreciate the fact that in this mode cutscenes don't play;
- **Automatic chronometer stop:** the chronometer can automatically stop at the end of the game, and the game itself can identify when a section is finished, giving a consistent way of evaluating speedruns;
- **Automatic save of best times:** this will help speedrunners with the management of their times and remember their "PB" (personal best);
- **An input monitor:** this will show the buttons pressed by the speedrunner, it may prove to be an invaluable tool if a new strategy is discovered.

Here's how a more complex and advanced "speedrun mode" screen could look like:

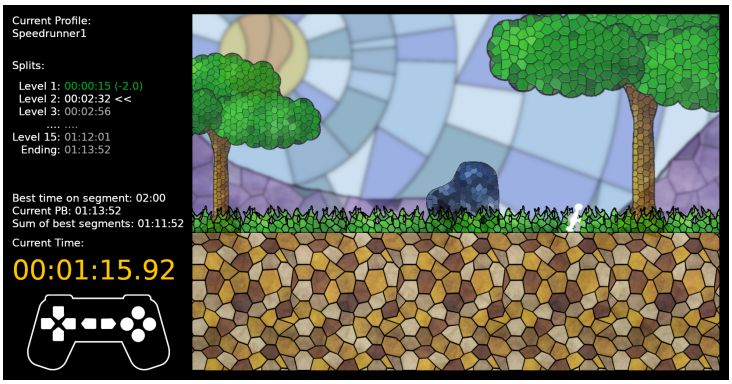


Figure 463: What a mode advanced “speedrun mode” may look like

Some game developers went to the extent of not fixing some glitches for the sake of speedrunning.

27.1.5 Streaming

If your game is going to be free or open source, it may be a good idea to stream its building process! This could also help attract some people who can give a hand in the future.

This is also a nice way to build a community around your game, get feedback and suggestions that could give your game that edge that it might be missing.

27.2 Replayability

27.2.1 Modding

A great way to gather some more tech-savvy fans of your game is opening your game’s structure to editing, which means enabling people to “mod” your game.

Mods are a great way to engage your community, and the are just so many types of mods that can be made:

- **Data Packs** - These add new user-created content to the game, such as:
  - New Maps/Levels;
  - Upgraded Textures;
  - Upgraded Models (if the game is 3D);
  - Upgraded Sounds;
- **Feature Mods** - These add new features to the game, such as:
  - New game modes;
  - HUD Modifications or upgrades (mini maps for instance);
  - New options and settings;
  - Additional/tweaked difficulty levels;
- **Total Conversions:** These take the engine of a game (like the first DOOM) and make a completely new game out of it, examples are “Castlevania: Simon’s Destiny” and “Sonic Robo Blast 2”.

Refining and giving out your own map editor can be a really great way to make your game more “open to editing”

and allow modders to enjoy working on it, after that, if your game structure allows it, you could document the engine and allow people to create their own masterpieces, maps and fan-made stories.

Random Trivia!

Even after over 30 years since its release, the original DOOM is still receiving a lot of support from modders, and not only modders! John Romero has released "Sigil" in 2019, a megawad that is an "officially unofficial" 5th episode. And in 2023 "Sigil II" was released. That's quite a lifespan!

27.2.2 Fan games

If your game becomes so famous that you manage to get a full franchise out of it, fan games are bound to happen: small games that take inspiration from your storyline and take different directions, according to the creators' will. This is a great opportunity to test how your current game is liked, and how people would like for its story to develop.

Some developers prefer to squash fan games under the Digital Millennium Copyright Act (DMCA) and get them removed, to avoid "tarnishing the franchise", some others instead prefer turning a blind eye, and some even go to hire the fan game creators to make a fully fledged official game.

In my humble opinion, fan games are good, they signal the community's desire for more, and you should probably play them too, new ideas come from comparing your own thoughts with other peoples'.

27.2.3 Mutators

Your game has a well-defined gameplay, everything seems balanced and all is well, until you start feeling that the gameplay has become stale and needs a shakeup.

This is an absolutely normal feeling that you'll get after testing your game for a while, and it will be the feeling that some players will get after playing your game for a long time.

This shakeup can come in the form of "mutators", small options that change some rules of the base game, making it easier, harder or just adding some cosmetic effect. This gives a layer of customization that players will appreciate, as well as either a challenge or a relaxing (or silly) time.

Some examples of mutators can be:

- **Instant Death:** There is no energy bar, you die as soon as you're hit. This is a good idea for players who want a challenge;
- **More enemies:** This mutator changes the enemy spawn points to spawn more enemies (can be double the amount, for instance), another way to give a challenge to players;
- **Forced weapon/build:** This mutator forces the player to start with a certain weapon or build, thus limiting their choices and forcing them to play differently from the way they're used to;
- **Draining Health:** The energy bar drains as time passes, making the player search for health pickups to be able to survive;
- **Limited powerups:** Only certain powerups can spawn;

- **Changed assets:** Some games like putting a "big head mode" (where characters have comically huge heads), or change the normal game sounds to silly ones, this could be a way to change up the game without changing its gameplay;
- **Power Fantasy:** Make the player hilariously overpowered, many players enjoy the feeling of power as they mow down dozens of enemies effortlessly, couples well with the "more enemies" mutator to counterbalance the difficulty (thus having more enemies, but they are easier to beat);
- ...

These mutators are usually compatible with each other, so you can have an overpowered player, but instant death, for instance.

27.2.4 Randomizing

In recent years, a certain kind of hacks for old games (mostly built around the Legend Of Zelda and Metroid series) that mix up the gameplay by mixing up the item locations, with a certain logic to avoid unbeatable games.

If your game has a somewhat linear story and no "roguelike" elements, you may be able to extend it replay value by adding a "randomized mode". This new mode would have some logic implemented to avoid *soft locking*$_{[g]}$ a save file and mixes up the items and powerups, usually making the game a bit more challenging.

27.2.5 New Game+

A good way to lengthen the life span of your game is adding a "New Game+" mode. Let's see what it is.

New Game+ is a mode that gets unlocked when you finish a game on a certain save file, this mode allows you to re-play the same story and game with some additions (hence the "plus"), those additions may be multiple, like:

- **Keep your inventory/equipment:** this is especially handy in story-driven, multiple-path/multiple-ending games. This allows the player to gloss over the fighting and instead concentrate more on "taking a different path" in the story. Let's not understate the willingness of a player to play a "power fantasy" by mowing through enemies in a fighting-heavy game, though;
- **Keep your statistics:** this is very useful, in a similar way to the "keep your equipment" bullet point, just more specific to RPG-style games;
- **Unlock new items:** this makes the game a bit different by allowing to use brand new, previously locked, equipment in the new "run";
- **Make the game more challenging:** this makes every new run a harder challenge to the player, by making enemies stronger, more resistant or just by adding more enemies in the various areas. A good idea is also making making enemies from mid and end-game appear earlier.

27.2.6 Transmogrification

We all know that looks matter, and there are people who are willing to "pay extra" to look how they want. This is especially true in MMO games, where (sadly) peer pressure tends to be quite high.

Say hello to "Transmogrification" (sometimes called "transmog", "tmog", "xmog" or just "mog"): it's a system that

allows players to change the look of a weapon, armor or anything to make it look like another weapon, armor or whatever.

In short, transmog solves the issue that is represented by the following sentence:

> This thing looks so cool! But its stats are horrible…

Using transmog you can have both the power of your best armor (or weapon, or mighty stick) with the looks of the (less powerful) armor (or weapon, and don't forget the stick) that just looks so cool.

Transmogrification can be implemented quite easily by allowing each instance of an object have a customizable sprite, instead of locking it behind the item itself. You just have to remember to save the skin used by each object's skin in your inventory/save file.

Design-wise, you should make transmogrification quite challenging to get, since it's a "quality of life" enhancement, but at the same time it shouldn't be so frustrating that the players just give up trying.

Any decision that will be frustrating to the player (maybe to "encourage" them to make use of microtransactions) will always come back to bite you.

Remember: you are not a AAA developer or publisher, **you can't afford to take the hit of these decisions**: being too greedy can completely destroy a good game's performance.

28 When the time for retirement comes

In all the computer science texts, you will find that the software life cycle is roughly divided in 6 steps:

1. Analysis;
2. Design;
3. Development;
4. Testing;
5. Deployment;
6. Maintenance;

But nothing is eternal, and thus sooner or later, we will need to talk about “software retirement”.

Software retirement happens when a software reaches the so-called “end of support” stage (sometimes called “end-of-life”): the software doesn’t get updated anymore, security patches are no applied and development stops. Sometimes even distribution stops, as well.

End of life can happen for different reasons:

- The platform your game is based on is not supported anymore, that includes: console, operating system, runtime libraries...
- Financial reasons: this usually involves multiplayer games, when the cost of server upkeep makes the maintenance of the game “antieconomic”;
- Loss of interest from the community: the game may not be played as much, and thus maintenance may be a draining task (either financially or mentally);
- Simple lack of time: people have their own lives and sometimes they have to decide that something needs to be cut to make space for something else (newer projects or just personal time);

So what can we do when a game reaches its “end of support” phase? Let’s see.

Note!

This section assumes that you’re publishing the game yourself, or you have a deal with your publisher that allows what I’m about to show.

28.1 Remonetization

This is usually done by big companies when a console reaches its end of support phase: they update the game (or build it anew, if necessary) and release it on the new platform.

This has some advantages, such as:

- The game is preserved (at least for the foreseeable future);
- The game is updated for new platforms, thus extending its lifespan;

- The game can be “remastered”, thus making it more appealing to newcomers (usually in the form of a graphical upgrade);
- The developer and publisher can re-sell the game, thus getting more funds for new projects.

There are also some disadvantages, as well:

- The developers (and publisher) will have to face an upfront cost for the upgrade (not knowing if the new sales will be worth the risk);
- The P.R. hit coming from “reselling the same thing to the same people”;

Usually the second issue is addressed with substantial discounts for owners of the “older version” or sometimes even just and outright “free update” option.

28.2 Free release

If the product is paid, the developers/publisher may decide to release the game for free if the situation allows it, sometimes even by removing DRM entirely.

This means that the game will continue to live past its “official life cycle” and new players may decide to try the game themselves.

When technical problems arise, there is a possibility that a community of fans will unite in the effort of building compatibility layers for the game and allow it to continue existing without the intervention of the original development team.

Sometimes those projects may even take the form of complete re-implementations of parts of the game, allowing for very old games to be experienced on modern systems. An example of this is “SCUMMVM” for adventure games: with it, you can play adventures as old as Maniac Mansion (1987) on modern PCs in 2024 (the time of writing this section, that’s 37 years!).

28.3 Open Sourcing

A more radical version of releasing a game for free, is taking the source code and publish it on a versioning platform with a permissive open-source license.

This allows the community to take over the maintenance of the game they love, without the effort of having to “reverse engineer” parts of the engine to make things work.

This is essentially giving the game away for free and allow people to look and edit the source code: this way any person who wants the game to be preserved can tweak it until it works natively on the device they want it to work on, or just analyze the source code and build the necessary “compatibility layers” to make it work.

In case of multiplayer games, it is ideal to release the server code too, allowing players to build their own servers and continue enjoying what you created for years to come (eventually by doing a final code commit on your game, allowing for custom server IPs to be used).

Note!

Author's note: this is the way I prefer things to go, personally. I think it's just ideal.

28.3.1 When "Open Sourcing" is not enough

I want to mention a game where sadly open-sourcing didn't do much to save it: the name is "Mother ZerOS: An open source hacking simulation" by Massimo "V4ldemar" Pinzaglia (which itself is the spiritual successor to "Mother: A computer hacking simulation" by the same author).

It was supposed to be an open source "Hacking simulation" game, much in the style of "Uplink". The game and source code (made using IRRlicht) were downloadable from V4ldemar's website: sadly (from the news I could find) V4ldemar left us in late 2018, his domain has expired and the game has disappeared from the face of the internet.

That is because the source code (as well as the installers) was available exclusively from his website and the website itself used a mix of PHP and AJAX requests (probably to save on bandwidth), thus stopping scrapers (Like Internet Archive's "Wayback machine") from processing the website correctly.

The game was in alpha state, but it had a lot of charm and it is painful to know that such a piece of my childhood is lost forever.

28.4 Hybrid Approaches

A possible hybrid approach could be releasing the engine as open source but keeping the art and assets under a "All copyrights reserved" license instead.

This is a way to protect your intellectual property (in case you want to remonetize it down the road), while allowing the current user base to at least make things work and do their own ports, but the graphics, setting, sound, levels and characters will remain under your control.

This is the solution that was adopted by DOOM developer "id Software", when they released the DOOM engine source code under a "not-for-profit license" (later changed to GPL2) but kept the assets proprietary.

A hybrid approach is the ideal way to allow the community to have a game supported by other (more tech-savvy) members of the community, while avoiding "diluting" the value of the brand you created.

28.5 What not to do

And here we are, the event that sparked this whole section of the book and what not to do when dealing with the "end of support" phase of your game.

On December 14th 2023, Ubisoft declared that their online racing game "The Crew" was going to be retired from market "due to upcoming server infrastructures and licensing constraints".

The game was retired from storefronts and could not be sold anymore.

Sadly this is not the end of it, because the game stopped being playable in April 1st 2024: the servers were taken down and the game (being “always online” even in its “single player form”) stopped working.

But that is not all, on or around April 14th 2024, people started noticing that the game’s license has been revoked. This is the equivalent of a publisher entering your home and taking away your game’s disc.

This is a textbook example of what **not to do** when your game reaches its end of support phase.

Note!

What follows **is not legal advice**. I am not a lawyer.
If you want to know more (as in quantity and quality of information), contact your favourite lawyer. What follows is just my somewhat objective evaluation of the situation, as well as an opinion.

First of all, the “server infrastructure limitations” issue could have been solved by allowing players to host their own servers. Releasing the source code for the server was not necessary, since an executable form of the server would have been enough for players to be able to enjoy the game they have spent their hard-earned cash on.

The only way I think these limitations could come to be, without being able to give the community a self-hostable server, would be that the current server infrastructure is shared between games, which is somewhat worrisome.

The licensing issue, I assume is due to in-game music, which I don’t think should be a problem, because the publisher decided to stop selling the game. That means that they should not infringe on any license, since the game is not sold anymore.

Again, the only way I think there could be licensing issues, is that the servers are themselves using pieces of software that are subject to a license agreement. That would be a reason to not give out a self-hostable server, because doing so would infringe on the software license of some components of the server itself (which may have expired).

But even accounting for all these motivations, there is absolutely **no excuse** (ethically speaking, I’m sure the EULA has something that allows them to do that) for Ubisoft to just decide to revoke the licenses for a product that people paid for.

There have been some amazing projects from the fans who managed to bring games “back to life” from seemingly desperate situations, just because the fans were passionate enough analyze every last bit in the effort to reverse engineer a way to play a beloved game again.

Random Trivia!

An example of a game that was “brought back from the dead” is “Sonic Runners”, a game that was released in June 2015 and retired July 2017, when the servers were shut down, making the game unusable. Through reverse engineering, a team of fans managed to reverse engineer a client and server, making it playable again.

Part 8: Learning from others and putting yourself out there

29 Dissecting games: three study cases

> I'm a writer, so I like dissecting things.
>
> *Hal Sparks*

In this section we will talk about three games, one bad and two good, and study their decisions closely, how a bad game inconveniences the player or how a good one reduces the space between the player's thought and the action on the screen.

This is not intended as a section to diss on a bad game or sing the praises of a good one, it is meant to be a study on how bad decisions pile up into what is universally recognized as a "bad game", while there are so many good decisions that need to be taken to make a "good game".

Note!

This section will contain spoilers of the games. This is necessary to understand them completely.

29.1 A bad game: Hoshi wo miru hito

29.1.1 Introduction

"Hoshi wo miru hito" (roughly translated to "stargazers") is a Japanese turn-based RPG for the NES/Famicom, set in a future where everyone has "Extra Sensory Perception" (ESP) and where a supercomputer has enslaved humanity via brainwashing.

The story may sound thrilling but the game is not. Let's see why this game is also known as "Densetsu no Kusoge" (I will leave finding the meaning to the reader).

29.1.2 Balance Issues

29.1.2.1 You can't beat starter enemies

At the beginning of the game, you deal only 1 point of damage per attack, way less than the health of the common starter enemy. This, together with your starting health points, make the starter enemies unbeatable most of the times. Speedrunners usually end up resetting in case they get an enemy encounter at the beginning of their run.

29.1.2.2 The Damage Sponge

One of the characters, called "Misa", is the only character that is able to walk on damage tiles without getting hurt. There is no explanation on the reason behind it.

This means that you have to die multiple times before finding out that only one of the four characters in your party is able to cross certain floor tiles, that may be no different than the other tiles.

29.1.2.3 You can't run away from battles, but enemies can

In this RPG you lack the option to run away from battles.

Enemies instead have a chance to run away from battle when their health points drop below 25% of their original health. Talk about fairness.

The "escape" option is instead hidden behind the "teleport" spell that you acquire after leveling up, in addition such spell is really weird in its way of working.

After selecting the "teleport" spell, you select a team mate to target such spell to, the spell can either succeed or fail:

- If the spell succeeds, the selected team member escapes the battle, while the others continue fighting for the turns that follow;
- If the spell fails, the whole team gets ejected (read "escape") from the battle.

This means that the teleport spell is more beneficial (4 times faster) when it fails than when it succeeds.

Note!

To be more precise, if the teleport spell succeeds and you manage to get each character (one by one) out of the battle, you keep your position on the current map. If instead it fails, your entire party will be kicked from the battle and you will find yourself at the beginning of the map. In practical terms, that doesn't make much of a difference, because you will probably need to get to the nearest healer anyway.

29.1.2.4 Statistics

There are some statistics that make sense in other RPGs, but do not in this game.

For instance the "defense" statistic scales so poorly that you barely notice its effect in this game.

In other games the "speed" statistic is tied to the order of attack (from the quickest to slowest character), but in this game the order is always "player's team" first, and "enemy team" after.

In conclusion, in "Hoshi wo miru hito", defense is effectively useless while speed is not even implemented.

29.1.3 Bad design choices

29.1.3.1 You get dropped in the middle of nowhere

In the NES era, it was common thing to have the story written in the manual. To save space on the cartridge, the beginning story elements were reduced to a minimum or completely removed, but in most games you still had a sense of where to go.

In this game, you just get dropped in the middle of nowhere, with no direction whatsoever. And you don't have the "Legend Of Zelda" style of exploration, since any enemy can make minced meat of you.

As a comparison, Dragon Quest, a game from the same period, had at least a hearing with the king to still introduce you into the story.

29.1.3.2 The starting town is invisible

The previous point is not really true, you actually start near a town, but such town is invisible.

The game makes a really lousy attempt to justify the fact that the town is invisible, but such explanation falls absolutely flat.

This just adds to the confusion of the story, as well as the lack of direction given to the player which can result in frustration and abandoning the game.

29.1.3.3 The Jump Ability

At level 1, you acquire a “jump ability”, that allows you to jump over certain tiles, like rivers. The issue is that such tiles are not distinguishable in any way from all the other tiles.

So you will find yourself mashing your main character’s body against various tiles, trying to find which ones you can skip with your jump ability, and probably die in the process by finding an unrecognizable damage tile.

29.1.3.4 Items are invisible

All items in the game are invisible, including all plot-crucial and revive items. The only thing telling you that you found an item, is a “beep” sound when you collect them.

This further piles up with the lack of direction the player faces in this game since the beginning. While it’s understandable that the limited size (and therefore duration) of NES/Famicom games kind of forced the developers’ hands into making harder games (to make them last longer), but introducing confusing of flat-out unfair mechanics is just bad design.

29.1.3.5 Item management

Usually when you buy a new weapon inside an RPG, you get to un-equip the old weapon and substitute it with the new one, then eventually sell the old one to recover some currency. This gives the game’s challenge new dimensions: item management and a simple economy system.

Well, this game instead lacks any kind of item management: every time you buy a new weapon, the old one will be automatically discarded. You cannot sell old weapons, and the auto-discard removes the possibility of trying a new weapon and in case go back to the old one.

And you cannot un-equip items and weapons.

29.1.3.6 Buying Weapons makes you weaker

When unarmed, from level 1 onward, the fight option lets you deal a damage equal to a random number between 0 and 4 (bounds included), regardless of the enemy defense stat, which is a real low amount of attack power.

When armed, the enemies defense values are taken into account instead, which means that most of the time, the boosted attack power given by the weapon doesn't overcome the enemies defense enough to make using weapons an advantage.

In few words: buying weapons makes you weaker.

And, as stated before, you cannot un-equip weapons, so your game session is probably ruined.

29.1.3.7 Enemy Abilities

Many enemies have an ability which is essentially a permanent, non curable in battle, paralysis + poison combo that will make your battle really hard and frustrating. That means that you will lose all the turns of the character that has been hit with such status effect.

And in case all your party members are hit with such status effect, you don't game over immediately, instead you will keep losing turns while the enemies slowly chip away at your party's health until you eventually game over.

Such effect lasts outside of battle too, so every step you take the affected party members will lose health until you see a healer.

29.1.3.8 You can soft lock yourself

In the vast majority of games, keycards are usually a permanent item that can be reused after finding it. In other games instead doors opened with keycards stay open for the rest of the game.

In this game, keycards have to be bought for quite the price, and disappear on use, and there is a serious chance that you softlock yourself somewhere if you don't buy enough.

29.1.4 Confusing Choices

29.1.4.1 Starting level for characters

This may be minor, but your characters start at level 0. Simply confusing and weird from a player standpoint.

As a programmer I find it quite amusing, though.

29.1.4.2 Slow overworld movement

The movement in the overworld is really slow, at around 1 tile every 2 seconds. This is really confusing, since the NES/Famicom is definitely capable of higher performance.

This is not due to lag or performance issues, it is a conscious decision to make the characters walk so slowly.

29.1.4.3 Exiting a dungeon or a town

Every time you exit a town or a dungeon, you won't find yourself at the entrance or exit of such place (like you'd expect), but instead you will find yourself at the default spawn of the world you're in.

So you may find yourself at the beginning of the game or in some not easily recognizable point of the map.

29.1.4.4 The Health Points UI

In the battle UI, the health of your team members is shown on top of their pictures, as an overlay.

Given the size of the font and the size of the pictures, only 4 digits fit. Given the game's health scaling, there is a serious chance that you get your health points to 5 digits.

The solution adopted was to drop the last digit of the health counter in all cases (even if your maximum health has less than 5 digits): so if you see "15" your health is actually between "150" and "159".

Also for some reason, if your health is lower than 10 points, your health shows as 0 (my speculation is that is would be written as "00" to "09").

29.1.5 Inconveniencing the player

29.1.5.1 The battle menu order

In the great majority of turn-based RPGs, the options are shown in the following order:

1. Fight
2. Magic (ESP)
3. Items
4. Escape

This is done in order, from most used (fight) to least used (escape).

In "Hoshi wo miru hito", the menu order presents the ESP option as the first option, selected by default, so most of the time you will have to move your cursor to the "fight" option and select it. This compounds with another problem exposed below.

29.1.5.2 Every menu is a committal

There is no "back" option in any menu, this means that every menu is a committal and you can't back off from any decision, even accidental ones.

That means that if you accidentally select the ESP option in battle and you don't have enough energy/mana to execute any attack, you will end up losing a turn.

If you select the wrong ingredient to make a potion, you most probably will have to waste that ingredient.

29.1.5.3 Password saves

In the NES/Famicom era, games that made use of battery-backed RAM modules to save game progress were rare. This means that the most used save method was using "passwords": a jumble of letters (and eventually numbers and symbols) that needed to be written down precisely, or you wouldn't be able to restore your progress.

This game's passwords are **massive** and use a mix of katakana Japanese characters and English alphabet, (while the rest of the game uses hiragana characters), which can be confusing.

Also passwords don't really save your progress "as is": your levels are saved in multiples of 4 (so if you're level 6, you will restore your progress but be level 4) and money is saved in multiples of 255 (if you have 3000 gold, you will restore your progress but have $255 \cdot 11 = 2805$ gold)

29.1.5.4 Each character has their own money stash

In most RPGs that feature a party, there is a shared pool of money that is used for all expenses, this may not be "realistic" but it's a good enough approximation that has a major upside: it is practical.

This game instead inconveniences the player further by giving each party member their own separated money stash. This is realistic and sometimes used in more modern RPGs, but it is not practical: every time you need to purchase potions used by the whole party (remember: there is no item management) you will have to switch characters or you'll find yourself running out of money.

29.1.6 Bugs and glitches

29.1.6.1 Moonwalking and save warping

This game doesn't interpret inputs as well as it should, so if you press the up and down buttons at the same time, you will find yourself "moonwalking".

Besides the perceived coolness of such move, moonwalking will allow you to go through obstacles, and eventually corrupt the graphics of the tilemap (like loading the right side of the map on the left side of the screen).

This is due to the game checking one direction for wall collisions, but moving the character in the opposite direction.

Pressing up and down at the same time on a controller is not possible, due to the fact that the NES/Famicom D-Pad does not have separated buttons, but if you connect any accessory that allows you to connect up to 4 controllers, the game won't be able to distinguish between the inputs from Controller 1 and the ones Controller 3.

A side effect of moonwalking, used in speedrunning is "Save Warping", you are able to manipulate the tilemap and your position via moonwalking, then save and *voilà* you will be warped to another point of the map.

29.1.6.2 The final maze

The final maze is divided in multiple floors, and is the greatest proof of how rushed this game was.

In the first floor of this maze, which is supposed to be really hard, no encounter tiles have been programmed: this means you won't have to fight anything on this floor. Also no "wall collision" was programmed either, so you can go through the maze walls with the same ease you walk on the floor.

In the other maze floors, encounter tiles were programmed, but still no wall collision was implemented, and since you can't encounter anything on walls, you can just minimize your encounter chance by taking a stroll inside the maze's walls.

29.1.6.3 The endings

This game, very ambitiously I shall say, features multiple endings. Towards the end you have to take a very hard decision:

- Join your enemy and leave humanity and live peacefully
- Leave the disputed territory and let the enemy live in peace
- Fight to gain control over the disputed territory

This can result in four different endings, which is really ambitious for a NES/Famicom game. If only the final boss fight was implemented...

If you choose to fight, you will automatically lose the battle and the game will end with a "bad ending".

29.1.7 Conclusions

"Hoshi wo miru hito" is a product of its situation, rumors state that this game was programmed by only one person, and rushed beyond belief so it could compete with *Dragon Quest* in some way. For the sake of fairness, I will assume that this game was made by a team.

The game has interesting ideas for its time: a cyberpunk theme, Extra-sensory powers, the character sprites "grow up" as they gain levels, the enemy sprites are artistically well-done, ... but the execution is littered with problems, obstacles to the player, bad design decisions and bugs.

It seems that the developers were not familiar with the NES/Famicom architecture, game designers weren't really familiar with game design concepts and play testers were completely nonexistent.

Even though this game has earned the status of "legendary bad game" (not a literal translation of "Densetsu no Kusoge"), "Hoshi wo miru hito" has gained a cult following that is really devoted, to the point that a hacker took upon themselves the daunting task of patching the game and redraw the graphics, as well as rebalancing the weapons and fix the walking speed.

There is even a "remake" called "STARGAZER" for windows.

29.2 The first good game - VVVVVV: Slim story and essential gameplay

VVVVVV is a 2D platformer created by Terry Cavanagh that features essential gameplay mechanics, a slim story that gives the player a reason to explore the game world and get to the end, as well as a satisfying level of challenge. Let's see what makes this a good game.

29.2.1 A Slim story that holds up great

VVVVVV's story is as essential as it gets: you're in a space ship, you run into some trouble and try to teleport out of the ship. Now you and your crew are scattered in a new dimension. Your mission is to rescue your crew and explore the new dimension you're in.

This is the story, it gives you enough of a reason to move from one level to the other, without being too burdensome. After you finish the game, you are free to explore the entire "dimension VVVVVV", searching shiny trinkets to unlock a final secret and learn more about the dimension you're in.

29.2.2 Essential gameplay: easy to learn, hard to master

In VVVVVV you can move left and right, but you cannot jump: you can only flip gravity. This, combined with the plentiful quantity of spikes, makes for a game that is easy to learn (it's just 3 buttons) but hard to master (you will find yourself dying over and over on the same screen).

The gameplay is so simple and well-implemented that it's really hard to get mad at the game: if you die and respawn, you know it's your fault. The great majority of time, it is evident. When you learn what "not to do", the rest becomes a lot easier. Enough so that you don't get stuck in a screen for more than a few minutes.

Checkpoints are plentiful and well distributed, so you won't end up going back too far if you die.

29.2.3 Diversified challenges

Each zone of the game (there are 6 in total, plus 2 intermissions and the finale) features the tight gameplay explained above, and sometimes add a new gimmick to the mix.

Let's explore those new gimmicks quickly:

- In the "Warp Zone", when you exit the screen on one side, you will re-enter on the opposite side: this means that if you fall on the bottom, you'll come out the top. This goes on until you reach the exit of the screen;
- In "the lab", there are lines that make you invert your gravity when touched. This can end up being a bit confusing at first, until you get used to the mechanic. After that you'll be bouncing around with no issue;
- In "the tower" the level automatically scrolls vertically: you can't go too slow or too fast, or spikes will come out of the top/bottom of the screen to kill you;
- In "intermission 1" you are followed by a crew mate: if you're standing on the ground, they will walk towards you. If you're on the ceiling, they will stay still. This introduces a new layer of difficulty and management;
- In "intermission 2" (also known as "the gravitron") you are kind-of-followed by a crew mate, but in reality you're playing alone. You will have to survive 60 seconds (with a checkpoint every 5 seconds) between two gravity-inverting lines, while items are shot at you.

In the finale, you'll have to put everything you have learned to the test ("warping", "bouncing", dealing with auto-scrolling levels, ...) to save yourself from being a prisoner of dimension VVVVVV.

29.2.4 Graphics

The graphics try to imitate the Commodore 64 (there is even a fake C64-style loading screen!); but they don't give up special effects like flashes, animated sprites, animated tiles, screen shaking, and fully moving backgrounds.

Even though the graphics are superior to what a Commodore 64 would be able to output, the special effects used still fit the chosen style and never really feel "out of place".

29.2.5 Amazing soundtrack

The soundtrack in VVVVVV is definitely one of the game's highlights: the chiptune-style songs that characterize every zone are catchy and so memorable that you will find yourself humming the tunes from time to time.

Each tune fits the zone it's used in, and kind of "tells a story" of its own: from "Passion for exploring" (the overworld theme) to "Predestined Fate" (used in intermissions and, in a remixed fashion, in the finale).

29.2.6 Accessibility Settings

VVVVVV is as accessible as it gets: there are a ton of accessibility features crammed in such a small game. Let's take a look at them.

- **Invulnerability** in case the challenge is too much, or you don't want to feel challenged all the time. This also helps people who want to enjoy the game, but can't due to mobility problems;
- **Slowdown mode** some people with mobility issues or slower reflexes may benefit from playing the game at 75, 50 or even 25% speed;
- **No screen flashing or shaking** some people with photosensitivity may have huge issues with the flashing and screen shaking, there are options to disable these effects and make the game much safer for those people. It also helps if flashing and screen shaking just annoy you or simply give you headaches;
- **No animated backgrounds** this may help with visual clarity or if the movement in the background gives you issues, like headaches.

29.2.7 Post-endgame Modes

After finishing the game, VVVVVV still offers some challenges. If you haven't collected all the trinkets, that is a good start, since they're hard to get and will reveal a secret back at the ship.

VVVVVV offers other game modes too, after you finish the main story:

- Time trials: you can replay any level, but you have to finish it under a certain time limit.
- Flip mode: the whole game is flipped vertically;
- No death mode: the entire game must be played without dying, you cannot save, there are no checkpoints and your companions in intermissions cannot die either;
- Intermissions mode: replay all completed intermissions, you can choose your companion too.

29.2.8 User-generated content

If the main story didn't satisfy you enough, VVVVVV features a level editor. The game already includes some selected user levels you can play, each with its own story and mechanics.

This also means that you can access the level editor yourself and create your own adventure from scratch, featuring the mechanics and characters of the original game.

29.2.9 “Speedrunnability”

Being an exceptionally difficult game, as well as the simple controls, the game has attracted an active and passionate speedrun community. The first playthroughs can take up to a couple hours, but an average speedrun can take less than 20 minutes!

The barrier of entry (usually for “glitchless” speedruns) is very low: you just need to go fast and die as little as possible. There aren’t too many tricks to be learned, and those are usually quite simple, with very few exceptions.

29.2.10 Characters are memorable, even if you don’t see them a lot

Even though the characters are not a continuous presence in the game, each of them is memorable: they all have very different colors, and different characters.

For instance: Vermillion is adventurous and always excited about exploring, after rescuing him, you will find him here and there in the overworld, sometimes even in the zones where other crew mates are stuck.

Victoria is a bit of a crybaby, very emotional and gets depressed very easily, she’s always feeling blue (and blue is her color too) and sniffles a lot when talking.

Verdigris is technical, being the ship’s technician (you will find him working on the ship’s antenna after rescue). Professor Vitellary is analytical and curious (if you bring him into an intermission, he will express marvel at what’s happening).

29.2.11 Conclusion

VVVVVV is a small game (as I said, it can be completed in less than 20 minutes if you’re quick), but it gives a lot of options for everyone. Lots of accessibility, replayability and “speedrunnability”. Custom levels and the level editor are the cherry on top of a game that is feature-complete and fun to play.

29.3 Another good game - Undertale: A masterclass in storytelling

Undertale is an RPG game created by Toby Fox that features some unique mechanics and masterful story telling.

We’ll take a deeper dive into the game immediately!

29.3.1 The power of choice

The game features a huge innovation in the field of RPGs: you can run the entire game without killing anything. In fact you are encouraged from the beginning to do so.

This innovation is not forced onto you though, you can play it as any other RPG out there (but you will miss a lot of the game, more on that later).

29.3.2 The game doesn’t take itself very seriously (sometimes)

Undertale is a very unique game that doesn’t take itself very seriously, there are 4th wall breaks, bad puns, jokes, worse puns and more. This gives the game a very lighthearted tone. That is if you’re doing a “pacifist run”.

The game becomes more somber the more you lean into a “neutral” or even “genocide” run.

This gives the game a lot of layers and depth, making each “type of playthrough” a different experience.

29.3.3 All the major characters are very memorable

Each major character is extremely memorable, and will be etched into your memory for the entire game (and probably part of your life too), I can recall practically all of them on the top of my head:

- Toriel, a gentle and motherly monster that loves puns and jokes;
- Sans, a skeleton that loves bad puns and “knows shortcuts” to every place in the game, somewhat lazy (that is explained in a lot more detail in genocide runs);
- Papyrus, Sans’s brother. Hates bad puns, loves spaghetti and wants to enter the royal guard. Has high self-esteem;
- Undyne, a brave, brash muscle-for-brains fish girl that has actually a very kind heart;
- Alphys, a pessimist doctor with very low self-esteem, even though she’s essentially a genius. Somewhat a nerd too;
- Mettaton, the robot that wants to become a TV superstar,
- Muffet, the leader of the spider bakery,
- Asgore, the “final boss” that you have to face. Regretful of his actions and past;
- Flowey, a yellow flower that can’t have feelings (this is explained well in the pacifist run).

29.3.4 The game continuously surprises the player

The game is able to surprise the player continuously. At a certain point the player realizes that the game is playing by different rules than expected: the game (as a piece of software) and the world become blurred when Asgore breaks the “mercy” option in a pacifist run.

From that point on, the player realizes that the UI, saving, loading are all characteristics of the world itself, and not of the game: each playthrough is treated like a timeline, and each reset is a new timeline (although some characters may have memories from previous playthroughs).

It feels like the world inside the game actually exists.

29.3.5 Player choices influence the game

Each run can be a bit different than any previous run, but they can all be classified into 4 categories.

- **Neutral runs** these are the runs that entail killing some enemies, but not everyone. The ending of this kind of runs is not satisfactory, and the game loses its meaning. The game will suggest (via its characters) to try a different approach;
- **Pacifist runs** these runs entail not killing anyone, this will bring an extended playthrough and ending, explaining a lot more about the world (and the “meta” nature of the game);
- **Genocide runs** these runs entail killing everything, even the “random encounters”, this will bring a different ending and will permanently change the game, even in successive runs (unless you physically delete your

save file from the game folder).

- **Partial/Aborted Genocide** these runs entail killing everything, besides one of the main characters. This will bring the player a different ending each time, but none of them will be “good”.

29.3.6 Great (and extensive!!) soundtrack

The soundtrack features over 100 (a hundred!!) tracks, each of them is unique in itself and memorable. Each zone has its own fitting theme, as does each boss (so much so that many of them have different themes for normal and genocide).

It’s hard to describe the soundtrack here, so I suggest you to try the game or at least listen to some of the most famous tracks.

29.3.7 Conclusion

I tried to keep this analysis vague so not to ruin the game too much to the people who didn’t play it. The pacifist story is extremely well-written and ties extremely well with the genocide one, together giving two sides of the same world.

The fact that such world is working with the mechanics of a video game is a surprise to the player, who will be a bit confused at the beginning but will soon understand things that may have felt weird before.

30 Project Ideas

> In theory, theory and practice are the same. In practice, they are not.
>
> *Albert Einstein*

This section tries to give you some ideas for some small projects you can try to do by yourself.

The projects will be put in order of (perceived) difficulty and each one will use a larger set of skills you have learned. Each project will have three levels of completion:

- **Basic:** This is the baseline to consider a project "complete", this is also the set of requirements the projects are sorted by;
- **Advanced:** This is more of a challenge, requiring more advanced skills and techniques. If you complete this level, you can consider yourself comfortable with most of the matters treated in the project;
- **Master:** This can be a real challenge, requiring skills and techniques that may not be taught in this book, but such problems can be solved with a bit of planning. If you complete this level successfully, pat yourself on the back and keep up the great work!

A title screen for each game is not required or necessary, but if you want to make one, feel free to do so.

30.1 Life

Also known as "Conway's Game of Life", this is a cellular automaton and it's also called a "zero-player" game. Every step of its evolution is determined by the previous step.

Each cell has 8 neighbours, and its new state is defined by 4 simple rules:

- If an alive cell has less than two live neighbours, it dies due to loneliness;
- If an alive cell has two or three live neighbours, it continues living;
- If an alive cell has more than three live neighbours, it dies by overpopulation;
- If a dead cell has exactly three neighbours, it becomes alive by reproduction.

30.1.1 Basic Level

Make a simple 50x50 grid that implements the rules of Life, you can set the initial state inside the source code and make it run as soon as the game begins. Make sure that the normal patterns work as they should, here's a couple of those:

- Block: a 2x2 block of alive cells, it's a so-called still life, it shouldn't change;
- Blinker, a 1x3 row of alive cells, it should switch between two states;

If these work well, check Wikipedia for some more patterns.

Skills Required:

- Drawing to a screen;
- Manipulating the game's state.

30.1.2 Advanced Level

Instead of setting the initial state from the source code and making it start at the opening of the game, start the game with an empty grid and the simulation paused.

Allow the player to enable/disable cells using the mouse and then start the simulation with the press of a button. Also allow the player to reset the simulation with another button.

Further skills required:

- Keyboard handling;
- Mouse handling;
- Event handling.

30.1.3 Master Level

Make a new version of Life where coordinates "wrap around", so a cell on the very right side of the screen is influenced by the cells at the very left of the screen. How do some of the more advanced patterns (like the gliders, glider guns, etc...) behave?

Further skills required:

- Some modulo maths;

30.2 Tic-Tac-Toe

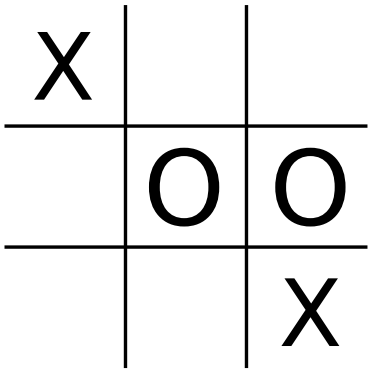


Figure 464: Example picture of Tic-Tac-Toe

Tic-Tac-Toe is a "finite" game, there are a finite number of choices and strategies but that makes it ideal for a simple project.

The objective of the game is to score 3 of your own symbol (either an `X` or an `O`) in a row or in diagonal, in a 3x3 grid; the players take turns and put their symbol on the game board in an empty spot.

30.2.1 Basic Level

Make a simple Tic-Tac-Toe style of game, where the mouse commands both players (so it alternatively switches between `X` and `O`), the game should be able to detect winning conditions for a certain symbol or a draw.

Skills required:

- Drawing to a screen;
- Event handling;
- Mouse Events;
- Winning/Losing Conditions.

30.2.2 Advanced Level

Make a computer-controlled player, by making the mouse write only the `X` symbol, while the computer will randomly put the `O` symbol onto a blank spot. The game should still be able to detect winning conditions for each player, as well as a draw.

Furthermore, the game should enforce turns, so that the human player is not able to put their own symbol when it is the computer's turn.

Further skills required:

- Random number generation;
- Coding basic game logic.

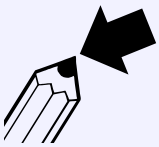
30.2.3 Master Level

Improve your computer-controlled player by making it actually seek for a way to win against the human player. The AI should do these checks, in order:

1. "Can I make a 3-in-a-row in a single move?": this means that the AI has 2-in-a-row with an empty space available. The AI should put their symbol in the empty spot to win.
2. "Is my adversary about to make a 3-in-a-row?": this means the human player has 2-in-a-row with an empty space available. The AI reaction should be to put their symbol in the empty spot.
3. If none of those cases happen, then make the AI fall back to a "random choice".

The AI does not need to be perfect, just mildly challenging.

Tip!



To some people, points 1 and 2 may be problematic, in the case that two among rows, columns or diagonals satisfy one of those conditions.
If both satisfy condition 1, there is actually no problem, the AI would win anyway: the AI can choose randomly.
In case 2, the AI should check if there is a way to block both "potential wins" in one move, but it is a perfectly acceptable solution to consider the case a loss for the AI and just choose randomly.

30.3 Space Invaders

Manufactured by Taito in 1978, the arcade Space Invaders is probably one of the most known games around for its historical value.

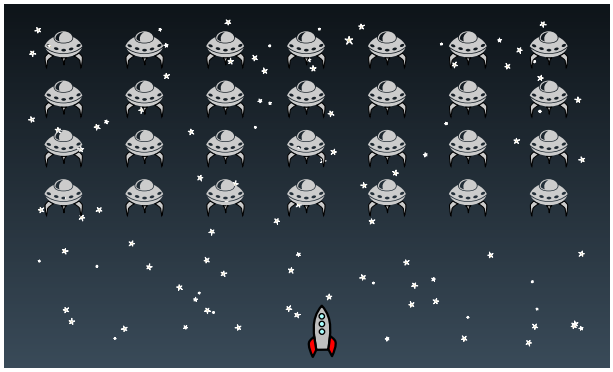


Figure 465: Example of a "Space Invaders"-style game

Your objective is to prevent a horde of aliens from landing, by shooting them with your monochrome laser cannon. The concept is deceptively simple, but the implementation can be really complex if you look more thoroughly. The more aliens are killed, the more the remaining ones descend faster, there is a bonus ship that pops out at random intervals, the aliens shoot back at you, there are destructible "shields" to give you some defense from the alien bullets, the win and lose mechanics...

Random Trivia!

The aliens getting faster wasn't initially intended in Space Invaders, the position of the aliens gets updated every frame, but the hardware couldn't process all the entities fast enough. The more aliens got killed, the less entities had to be processed per frame, making the aliens move faster.
Instead of coming up with a solution, this "bug" was kept as a challenging "feature"

30.3.1 Basic Level

In the basic level, we will just create a static, unarmed horde of aliens menacing our base (although being static it won't be much of a threat), we can shoot out projectiles at them and they disappear when they're hit, adding some

points to our score.

Since our unwanted alien guests are not much of a threat, we won't need to put up any shields to defend against bullets. When all the aliens are dead, we win the game.

Skills Required:

- Drawing on screen;
- Collision Detection;
- Keyboard Input;
- Bullets;
- Score management;
- Winning conditions;
- Managing entities.

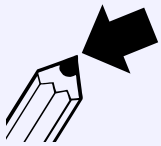
30.3.2 Advanced Level

For the advanced level, we will make our alien foes more menacing by introducing movement and making the shoot at random intervals towards our laser cannon. The aliens do not need to accelerate as in the original game (but it's not really difficult to implement).

When an alien touches the ground, we will lose the game. Thus making the game a bit more difficult to manage (as well as to play).

We will also introduce a bonus ship that appears on top of the alien horde, floating from left to right, awarding a bonus when successfully hit.

Tip!



To make the bonus ship, you may be tempted to create an entirely new object, but at the same time a "bonus ship" IS-AN "alien ship", which would call for subclassing.
If you separate behaviour from the objects well, with methods, you can use subclassing to "change the ship's behaviour" by overriding the `update()` method.

Further Skills Required:

- Timers (for the ship movement, and if you want, the bonus ship);
- Random number generation (for the bonus ship and the alien bullets);
- Subclassing;
- Losing conditions.

30.3.3 Master level

For the master level we will make the shields, which are complex to code in the way of the original game, so we will "cheat".

Our shields will be “force fields” which can withstand 3 shots before getting disabled: such shots can come from the aliens or our ship. After 5 to 10 seconds, the shields will come back online and ready to stop bullets again.

Like the original game, the shields can block both our and our enemy’s bullets, making for a more challenging gameplay.

For an even more challenging gameplay, we can (rarely) make a random ship detach from the horde to try and attack us directly: on such event, the ship will start shooting more often and move towards our laser cannon, before going back into position (a bit like Galaxian).

Further Skills Required:

- Timers (this time for the shields);
- Minimal AI (for the “ship attack”);
- Managing object states.

30.4 Breakout

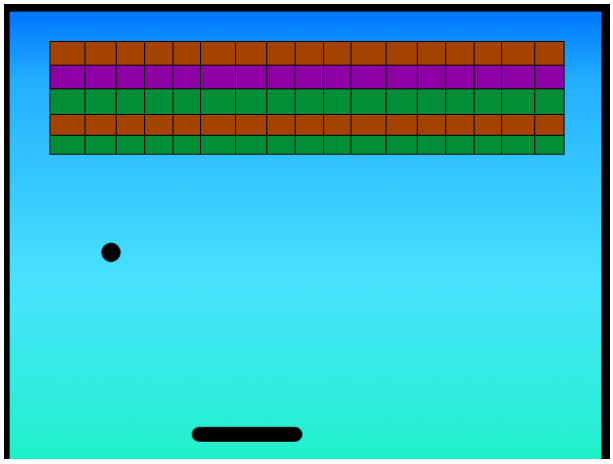


Figure 466: Example picture of a breakout-style game

Breakout is a well-known brick-breaker style of game, where the player drives a paddle trying to keep a ball from getting to the bottom of the screen, while breaking all the blocks on the screen.

30.4.1 Basic Level

The basic level is just making the basic game: make a single level that works. If the player loses their ball, they lose, if they manage to clear the screen they win.

The paddle should be moved with mouse or keyboard (you can choose to implement either or both) with sufficient speed to avoid useless deaths but slow enough to be usable. The ball should bounce like it bounces on a wall: keeping the angle with the wall without changing it.

Skills Required:

- Drawing to a screen;
- Vectors;
- Moving elements of the game;
- Event handling;
- Winning/Lose Conditions;
- Mouse/Keyboard Controls;
- Collision detection and reaction;
- Object management (creation and deletion);
- Score keeping.

30.4.2 Advanced Level

In the advanced level you should implement a basic life system: each time the ball crosses the lower side of the screen, you lose a life; when all lives are lost, you lose the game.

To make the game a bit more interesting, it could be an idea to implement different block types:

- **Explosive Blocks:** When destroyed, these blocks explode, destroying the surrounding blocks;
- **Multiple-hit Blocks:** They simply require more than one hit to destroy;
- **Score Blocks:** When destroyed, these blocks drop score items that slowly descend toward the bottom of the screen. If you catch these items with the paddle, you get extra points.

Furthermore, the ball should progressively get faster with gameplay, you can either do it every few bounces or just every bounce with the paddle. The choice is yours.

Further Skills Required:

- Managing Object's states;
- Subclassing;
- Managing Game State.

30.4.3 Master Level

With the "master level" we are going to complete this game by adding powerups: they work in a similar way to the advanced level's score blocks with a descending item that grants a certain status to either the ball or the paddle.

You can make it a random chance for each block destruction or make a dedicated block. Here are some suggestions for some powerups:

- Larger/Smaller paddle;
- Paddle that shoots to destroy blocks;
- Faster/Slower ball;
- Ball that goes through blocks, destroying them;
- "Sticky Paddle" that allows to stop and then release the ball;
- Multiball.

Furthermore, you can implement a sort of “biased bouncing” for the paddle: the further left or right on the paddle the ball touches, the more it will take a horizontal bias towards that direction.

This way the center of the paddle is “neutral”, keeping the normal bouncing mechanics, while the leftmost and rightmost sides allow the player to direct the ball the way they want.

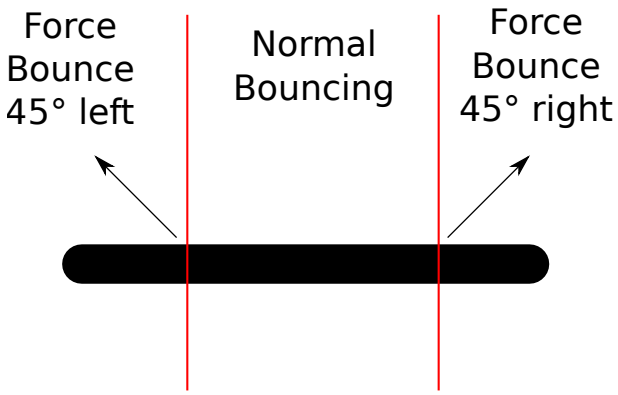


Figure 467: Biased Bouncing for breakout

30.5 Shooter Arena

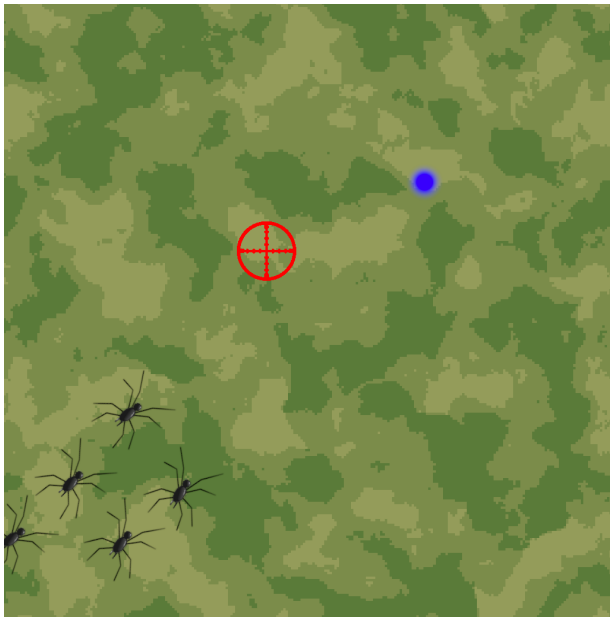


Figure 468: Possible Shooter Arena Game

This is a simple keyboard and mouse-controlled arena shooting game (technically the genre is called “twin-stick shooter”), much in the style of “Crimsonland” and “R.I.P.”: the objective of the game is surviving for as long as possible.

30.5.1 Basic Level

This level entails making the basic game: being a horde-based game, the AI can be really basic, following your movements and trying to touch you.

A single type of monsters will spawn from outside the screen and slowly make its way towards you, while the playable character will shoot bullets towards the mouse cursor.

If a bullet hits an enemy it will die and your score will be incremented, if an enemy hits you you will die.

Skills required:

- Drawing to a screen;
- Vectors;
- Basic AI;
- Collision detection and reaction;
- Projectiles;
- Lose Conditions;
- Mouse/Keyboard Controls;
- Spawning entities;
- Score Keeping.

30.5.2 Advanced level

In the advanced level you should implement a life system for our playable character, so instead of dying our player will get hurt and will have its life reduced. A health bar should be shown on screen.

The same should be done for each enemy, so you will need to be able to manage the state of each object (enemy) separately. Extra points if you show a small health bar on top of a hit enemy, showing its current health.

You should also implement a way to easily code in new enemy types, this will require refactoring your code to support importing entities from data. Each enemy should at least have different speeds and health.

On each death, the enemy should have a random chance of dropping e medkit that will heal you when touched, such item should stay on screen for a limited amount of time, then disappear if not used.

Further Skills Required:

- Drawing an HUD;
- Managing an object’s state;
- Coding entities as data;
- Random number generation;
- Timers.

30.5.3 Master Level

In the most difficult level, you should start coding powerups, like new weapon types, temporarily increased walking speed, higher damage projectiles, etc... Similarly to medkits, these powerups should disappear after a certain amount of time.

An interesting weapon to implement would be a “railgun”, with bullets that can go through enemies, this will be easier when you use ray casting (and some tricks for drawing), if you didn’t use it already.

You should also animate the characters, thus getting used to your favourite engine’s animator nodes/classes: this will make the game feel more complete.

On each death, the enemy should leave a blood (or if you prefer, goo) splatter that will disappear after a few seconds: this will make the game more messy and in its own way fun.

Further Skills Required:

- Animators;
- Ray Casting.

31 Game Jams

> You miss 100% of the shots you don't take.
>
> *Wayne Gretzky*

Game jams are a great opportunity to test your game making skills in a tight situation, where time is at a premium and you need to "hack your way" through a fully-fledged prototype of a video game.

In this section there will be some suggestions on how to survive a game jam.

31.1 Have Fun

The biggest prize you can get from a Game Jam is experience and comparing yourself to other participants constructively. You shouldn't take part to a game jam just for the prize (although aiming for it could make you strive to do better).

If you don't have fun, then it's probably not worth it.

31.2 Stay Healthy

Don't forget to eat, take regular breaks, go to bed early and just keep some healthy working habits when you're participating a Game Jam.

If you don't keep a healthy work style, your productivity is going to take a dive: you'll find yourself having trouble solving the simplest problems, your creativity will be nonexistent and you'll get extremely frustrated.

31.3 Stick to what you know

A Game Jam event is not a good place where you learn a new language or game engine, you barely have time to make a game, let alone learn a completely new language and even a new engine!

If you find yourself having to choose between the newest game engine and something you already used two or three times: **go with what you know**.

31.4 Hacking is better than planning (But still plan ahead!)

During Game Jams time is at a premium: you shouldn't use complex data structures or be concerned too much about "best practices".

Sometimes "a hack" is better than "a solution": you're building what is essentially a prototype, if the game works and is playable, you have already reached your objective.

You should still plan ahead for your Game Jam experience:

- Make sure your PC is well set up;
- Make sure your development environment works correctly;
- Ensure you can compile some test programs correctly;

- Have a good *IDE*[gj] ready;
- Plan your meals well, the less time you cook, the more time you can rest and think;
- Have a generic roadmap that tells you how much time you want to dedicate to each phase;
- Have some ready-to-use resources to use as placeholders;
- If the game jam allows it, have a basic game structure ready (Like a title screen with a "Play" button).

31.5 Graphics? Sounds? Music? FOCUS!

When it comes to Game Jams you can't afford to waste much time on graphics or sounds: having a library of ready-to-use resources can prove vital, as well as a good way to test game mechanics while someone else is drawing (if you're participating as a group).

Sometimes it could be better having no graphics at all, to an extent (just see the game: "Thomas was alone", where graphics are rectangles), as long as it doesn't stop the game from being enjoyable. This is even more important for more "extreme" Game Jams, like the "0 Hour Game Jam" (where you make a game in the hour that is affected by the DST time change, usually from 2am to 2am DST).

Be essential, if you want to make a fully-fledged RPG with procedurally generated weapons, a Game Jam is not the right place to create it.

Again, time is at a premium, cut everything that can be cut:

- Menus can be replaced with a simple title screen, no options;
- Don't have a dedicated credits screen if the creator names fit in the title screen;
- Keep the HUD minimal or just remove it entirely;

Focus on one or a small bunch of game mechanics and do them well, if you try to do a lot of stuff nothing will come out good enough to make an enjoyable game.

31.6 Find creativity in limitations

With experience, you'll find out that it's much harder to find inspiration when you have full freedom on a project.

Some Game Jams define a "subject" the games have to stick to, this is a nice way to boost creativity: through limitations. Usually these subjects are conveyed through a single word or by expressing a game mechanic.

An example of "game mechanic" could be the following: "2 button controls" (Which, by the way, was one of the themes of Ludum Dare 34).

A "theme" or "subject" could be "gravity", which could mean any of the following:

- Newton's Gravity
- Solemness, seriousness
- Weight/Pressure (Usually emotional)

Another one could be "growth" (Ludum Dare 34's other Theme) as in:

- Planting and growing plants
- Personal Growth (getting experience in life)
- Eating, Surviving and Growing to adult

Many times you'll find a hidden meaning behind a theme that could give you something really unique and eventually an edge over the other participants in the Game Jam.

31.7 Involve Your Friends!

A Game Jam doesn't have to be a quest for "lone wolves", you can involve your friends in the experience, even if they're not coders or game developers: talk about your ideas, get new points of view, suggestions and help.

Game Jams can be a really strong bonding experience for friends and colleagues.

31.8 Write a Post-Mortem (and read some too!)

One of the most useful things you can do after a Game Jam, both for yourself and others, is writing a so-called "Post-Mortem" of your game, where you state your experience with the Game Jam itself, what went right, what went wrong and what you would do differently in a following event.

A Post-Mortem is a reflection on what happened and how to make yourself and your work better while leaving a document that allows other people to learn from your mistakes and learn from a "more experienced you".

Obviously a smart move would be reading a bunch of Post-Mortems yourself, so to learn more about Game Jams and avoid some common pitfalls of such an experience.

An interesting take on Post-Mortems could be making a time-lapse video of your work, there are applications ready to use that will take screenshots at regular intervals of your desktop and your webcam (with a nice picture-in-picture) if enabled. At the end of everything, the program will take care of composing a time-lapse video for you. It's interesting to see a weekend go fast forward and say "oh, I remember that".

31.9 Most common pitfalls in Game Jams

To conclude this section, we'll see some of the most common problems in game jams:

- **Bite more than you can chew:** Aiming too high and being victims of "feature creep" is a real problem, the only solution is staying focused and keep everything as simple as possible and be really choosy on the features to add to the game: take time to refine what you have, instead of adding features;
- **Limitations related to tools:** If your tools have issues importing a certain asset or you are not able to create your art for the game then you're in real trouble. You can't afford to waste time troubleshooting something that is not even related to your game. You need to be prepared when the jam starts, test your tools and gather a toolbox you can rely on, something that will guarantee you stability and productivity.
- **Packaging:** This is a hard one - sometimes the people who want to try your game won't be able to play it, either because of installer issues or missing DDLs. Make sure to package your game with everything needed

and eventually to link to the possible missing components (like the Visual C++ Redist from Microsoft). The easier it is for the evaluators to try your game, the better.

- **Running out of time or motivation:** plan well and be optimistic, you can do it!

32 Where To Go From Here

> Be grateful for all the obstacles in your life. They have strengthened you as you continue with your journey.
>
> *Anonymous*

It has been a very long trip, hopefully this book has satisfied great part of your curiosity about 2D Game Development and Design, but the adventure is far from over. There are infinite resources on- and off-line to further your experience in game development and design.

This book has been mainly about getting you to touch the nitty-gritty of game development, giving you the tools to be able to create your game without overly relying on your toolkit (be it Unity, UDK or any other fully-fledged game engine), or even be able to modify, at a low level, the behaviour of the tools given to you to reach your objective.

There are still countless questions remaining, which we can condense in one big question:

> "Where do I go from here?"

You can try and take on a game jam, Itch hosts a myriad of game jams at https://itch.io/jams and many of them are beginner friendly, then there is the famous "Ludum Dare" game jam that you can find at https://ldjam.com/.

If you want to know more about the game development process and how it works, Here are some resources you can check out to become a better game developer and designer. Everything is divided by category for convenience.

But first, here's a small legend to distinguish paid products from free products.

- **Free Product: [F]**
- **Accepts donations or Partially Free: [D]**
- **Paid Product: [P]**

32.1 Collections of different topics

32.1.1 Books

- **Apress-Open Ebooks:** https://www.apress.com/it/apress-open/apressopen-titles **[F]**
- **"OpenBook" offers from O'Reilly:** https://www.oreilly.com/openbook/ **[F]**

32.1.2 Videos

- **freeCodeCamp.org's YT:** https://www.youtube.com/channel/UC8butISFwT-Wl7EV0hUK0BQ **[F]**
- **3DBuzz (Archived on the Internet Archive):** https://archive.org/details/3dbuzz-archive **[F]**

32.1.3 Multiple formats

- **Daily "Free Learning" offer from PacktPub:** https://www.packtpub.com/free-learning **[F]**
- **"Awesome lists":** https://github.com/sindresorhus/awesome **[F]**
- **Game Dev Nexus:** https://gamedevnexus.com/ **[F]**

32.2 Pixel Art

32.2.1 Multiple Formats

- **Pedro Medeiros's Patreon Page:** https://www.patreon.com/saint11/ **[D]**

32.3 Sound Design

32.3.1 Multiple Formats

- **EpicSound's Sound Effects Tutorials:** https://www.epicsound.com/sfx/ **[F]**

32.4 Game Design

32.4.1 Books

- **100 Game Design Tips and Tricks** *(Wlad Marhulets)* https://archive.org/details/100-design-tips-and-tricks/ **[F]**
- **A theory of fun for game design** *(Raph Koster)* **[P]**
- **The Art of Game Design: A Book of Lenses** *(Jesse Schell)* **[P]**

32.5 Game Development

32.5.1 Web Resources

- **Amit's Game Programming Information:** http://www-cs-students.stanford.edu/~amitp/gameprog.html **[F]**

32.5.2 Videos

- **OneLoneCoder's YT:** https://www.youtube.com/channel/UC-yuWVUplUJZvieEligKBkA **[F]**

32.6 References and Cheat Sheets

- **Easing functions Cheat Sheet:** https://easings.net/ **[F]**

Appendices

A Glossary

A

A.L.U. Short for "Arithmetic Logic Unit", it's a component in modern CPUs that does arithmetic and bitwise operations[g] on integer binary numbers.

API Short for "Application Programming Interface", it's a set of definitions, tools and clearly defined methods of communication between various software components.

B

Bitwise Operation Bitwise operations are operations that operate on individual bits. For instance, if bitwise operations are done between bytes, a bitwise operation will still operate bit by bit. (Example: `01001100 AND 10001100 = 00001100`)

C

Call by reference Evaluation strategy where a function parameters are bound to a function by passing a reference to the arguments, this could cause *side effects*[g] since the function would be able to change variables outside its local scope.

Call by value Evaluation strategy where a function parameters are bound to a function by making a copy of the values used as an argument.

D

Dangling Pointer A dangling pointer is a memory pointer that references an area of memory that doesn't contain a valid object. A dangling pointer usually happens when an object is deleted from memory forcibly, but the pointers referencing said object are not invalidated (usually by setting such pointers to a null value).

Dynamic Execution See *out of order execution*

E

EULA Short of "End User License Agreement", is essentially a contract that establishes the purchaser's right to use the software, usually with some limitations on how the copy can be used.

F

FOV Short of "Field of View", it denotes what can be visible by something, it is measured in degrees. Common used values are 90 (in the style of the original DOOM), but it can go up to 120 (standard binocular vision). Higher values may make the environment look distorted.

G

Greedy Algorithms Class of algorithms that try to solve a problem by making the locally optimal choice **at each stage**, approximating the global solution, without any global planning.

H

Hash Function A hash function is a special function that is used to map data of arbitrary size to fixed-size values. This function has some features like being able to spread values in an uniform way (minimizing the different values that have the same hash, called “hash collisions”), is fast and deterministic (given the same input will generate the same hash).

HUD Short of “Heads Up Display”, in games it usually shows your health, ammunition, minimap and other information.

I

IDE Short for “Integrated Development Environment”, it is a program that integrates a text editor with syntax highlighting, a compiler, a code checker, a project explorer and other features (like a tag explorer, for instance).

Information Hiding Information hiding is one of the basic principles of programming: each part of a program (a “module”) should not expose its inner workings, but rather expose a stable “interface” to the outside world. This will help separating modules from each other and avoid “snowball effects” when modifying the inner workings of one of them.

K

Kanboard Short for “Kanban Board”, are boards used to manage work. The board is usually divided into swimlanes and “cards” that represent the work to do are moved from left to right, to represent the progress of the work itself.

L

Letter Notation Also known as “letter music notation”, it’s a music notation system that uses letters A through G to write music.

M

Malware Short for “malicious software”, it’s a “catchall term” for viruses, trojan horses and any kind of software that is programmed to behave maliciously. Such software can steal information (passwords, key presses, habits, etc…) or flat out try to make your computer unusable (deleting system files, encrypting your documents and asking for a ransom, etc…)

Memory Leak A memory leak is usually the result of a programming error, where the memory is not correctly managed. This usually entails allocating and using memory without releasing it, thus the program will eat more and more memory as it keeps running.

Meta-Gaming Meta-Gaming consists in the act of playing a game by applying rules that are outside of the game itself to gain an advantage. This includes thoughts like “It’s the first level, the game wouldn’t kill my character so early”. This mostly consists in applying real-world logic to a game as “a piece of the real world”, instead of being fully immersed in the game itself and its world.

Modern music notation The most common way to write music, using symbols to indicate the duration and type of note, while the symbol's positioning in a 5-line staff defines its pitch.

O

Oscillator An oscillator is a device (usually hardware) used to create an alternating current. Oscillators can be used in audio synthesis to create pitches.

Out of order execution Paradigm used in high performance CPUs to reduce the wasted instruction cycles. The instructions are performed in an order given by the availability of the input data and execution units instead than the original order defined by the programmer.

P

Pre-emption See *preemptive multitasking*

Preemptive multitasking A multitasking environment where the operating system forcibly initiates context switches (saves the state and interrupts temporarily) between running processes, regardless whether their job is finished or not.

Process Starvation See *starvation*

R

Race Condition A condition where two or more threads are competing for a resource (variable, screen buffer, ...) and the order that the threads happen to take hold of such resource changes the result.

REPL Short for "Read-Eval-Print Loop", sometimes called "language shell". It's a simple computer program that presents a prompt where the user can write code in real time, line by line, and see each piece being executed immediately.

Rootkit Usually associated with malware, a rootkit is a software that manages to get access to reserved areas of an operating system, usually to hide its own, or other softwares' presence.

S

Side Effect In computer science a function is said to have a "side effect" when it changes variables outside its local environment, this can happen in languages which use *call by reference*$_{[g]}$ evaluation strategies.

Single Point of Failure This usually defines a part of a system that, if it fails, will stop the entire system from working. This is often used to indicate objects that have too many responsibilities.

Soft Locking This is usually referred as an anomalous situation where a game is not "frozen" and still working correctly, but the player is unable to continue playing. An example of soft-locking would be the player getting trapped inside level geometry, unable to move: the game is not frozen (the level can be restarted), but the player cannot finish the level.

Stack Overflow A stack overflow is a situation where too much data is pushed into a data structure called a "stack". One of the most common cases of "stack overflow" happens during recursion: when a function is called all the current work variables are saved and pushed into a stack data structure in memory, along with a "return address" that will allow us to come back to this point of the program. When a recursion is too deep (the

recursive function calls itself too many times), the call stack gets filled up and it's not able to continue the execution, leading to an aborted operation.

Starvation Also known as "process starvation", it's a phenomenon where a certain process (or group of processes) has a lower priority than others, and is not able to access resources (like the CPU) because it's always "overtaken" by higher priority tasks. This leads to the process itself never being executed. When this happens, a process is labeled as "in starvation".

Static Typing Languages characterized by *static typing* are the ones where the type of a certain variable (integer, string, class, ...) is known at compile time.

U

UI Short of *User Interface*, defines the elements shown to and interacted by the user in a software or, in this case, a video game menu.

Unreachable Memory This is a phenomenon where some dynamically allocated memory has no more references pointing to it. This is a common cause of memory leaks in programming languages without automatic garbage collection (like C and C++).

W

Wiki A wiki usually refers to a knowledge base website where users collaboratively modify content and structure by using their own web browsers.

B Engines, Libraries And Frameworks

Here follows a list of game engines, libraries and frameworks, so you can make an informed choice about your platform, or just try one!

For each proposed engine or framework, along with a short description, you will find the following sections:

- **Website:** This contains a link to the official website of the engine/framework/library;
- **License:** Here you will see if the product is free to use or if you need to pay a price for it, and anything in between;
- **Bindings:** In this table you will find if the product supports one of the many famous programming languages available, as well as the following:
  - **Dedicated Language:** This engine/framework makes use of its own scripting language, usually easier to learn than general-purpose languages. You may need to learn it;
  - **Visual Programming:** This product makes use of a "Visual Scripting" (codeless) paradigm, where you can program your own game without writing code, usually by using directed graphs and nodes.
- **Platform compatibility:** This will tell you if the product is good for your objective, by telling you which platforms you can deploy on, like Linux, Windows or even on the web via HTML5;

General Purpose

General purpose engines are the basic frameworks that allow you to build any game you want, without being tied to a specific genre. These are the tools that give you the most freedom, but also will be a bit harder to master.

Axys

Website: https://github.com/axys1/axys

License: Open Source

Axys aims to be a radical fork and refactoring of the Cocos2d-x engine, after the producer decided to work on its newest product. It aims to support OpenAL on all platforms and use more modern C++ features.

Bindings: C++, Lua.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

Cocos2d-x/Cocos Creator

Website: https://www.cocos.com/

License: Mixed: The code is Open Source, but has many different licenses

Cocos is a set of tools and engines that allow to code your games. It supports both 2D and 3D games and it provides all the basic functions needed to create your own game, like rendering, audio, physics and user input.

The producer is now working on the new iteration: Cocos Creator, but Cocos2d-x is still available and considered a very mature project. If you fear it is going to be abandoned, you may want to check Axys

Bindings: C++, Lua, Javascript.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

ENIGMA

Website: https://enigma-dev.org/docs/Wiki/ENIGMA

License: Open Source

ENIGMA (Extensible Non-Interpreted Game Maker Augmentation) is an environment that is derived from Game Maker, but with time has taken a different approach than its counterpart: it compiles its own language (EDL) to C++ and it also allows to tie into C++'s templates and functions.

ENIGMA's license is GPL3, with a special exemption that allows you to sell the products made with the software.

Bindings: C++, Dedicated Language (EDL), Visual Programming

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

Game Maker Studio

Website: https://www.yoyogames.com/gamemaker

License: Paid

Game Maker Studio is one of the simplest game-making toolkits available on the market, but that doesn't mean it's not powerful. In fact, one of the most famous games of recent history was made with it: Undertale.

It makes use of its own scripting language, and some visual toolkits as well.

Game Maker Studio is commercial software, regulated by its own EULA, but it was added here for sake of completeness.

Bindings: Dedicated Language (GML), Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

GDevelop

Website: https://gdevelop-app.com/

License: Open Source

GDevelop is an open-source toolkit to make games, mostly based on visual programming, GDevelop supports the use of JavaScript to code parts of the game, as well as JSON and support for HTTP Requests. GDevelop also supports exporting to Android and Facebook Instant Games, as well as exporting to iOS and web-based platforms.

GDevelop is distributed under the MIT license (although the name and logo are copyright), although the main license file refers to other license files inside each folder. So you may want to check the GitHub repository for more information.

Bindings: JavaScript, Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

GLFW

Website: https://www.glfw.org/

License: Open Source

GLFW is an Open-Source library for OpenGL, OpenGL ES and Vulkan, that allows to create windows, context and surfaces, as well as receiving input and events. It is written in C.

GLFW is distributed under the ZLib/png license, which allows for both commercial and personal use, both in proprietary and open-source situations.

Bindings: C++, C, C#, Go, Java, Python, Ruby, Lua, Rust.

Platform Compatibility: Windows, Linux, Mac OS.

Godot

Website: https://godotengine.org/

License: Open Source

Godot is a fully-fledged engine that makes use of a node-based approach and supports many ways of programming your own game, both in 2D and 3D, including its own language (GDScript) and visual scripting.

Godot is currently distributed under the MIT license, you should check the Legal section of the Godot Documentation for all the additional licenses that you may need to know about.

Bindings: C++, C#, Dedicated Language (GDScript), Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

Gosu

Website: https://www.libgosu.org/

License: Open Source

Gosu is a 2D game development library that focuses on short game development competitions (game jams) and prototyping. It is lightweight and have few dependencies.

The library is distributed under the MIT license, which is a very permissive one.

Bindings: Ruby, C++.

Platform Compatibility: Windows, Linux, Mac OS, iOS.

Löve

Website: http://love2d.org/

License: Open Source

Löve is a lua-based framework for creating games, features an extensive documentation and features some levels of abstraction that help with game development.

Löve is distributed under the ZLib/png license, which allows for both commercial and personal use, both in proprietary and open-source situations.

Bindings: Lua.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android.

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microStudio

Website: http://microstudio.dev/

License: Open Source

microStudio is a free, open source online game engine, and a platform to learn and practice programming.

Bindings: Microscript, Lua, Javascript, Python.

Platform Compatibility: Windows, Linux, Mac OS, Android, Web.

MonoGame

Website: http://www.monogame.net/

License: Open Source

MonoGame is an open-source porting of XNA 4, it allows for people used to Microsoft's XNA framework to port their games to other platforms, as well as creating new games from scratch. Many games make use of this framework, one above all: Stardew Valley.

MonoGame is distributed under a mixed license: Microsoft Public License + MIT License. You may want to check the license yourself on the project's GitHub page.

Bindings: C#.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

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olcPixelGameEngine

Website: https://github.com/OneLoneCoder/olcPixelGameEngine

License : Open Source

A small framework for pixel drawing and user interface framework coded by javidx9 (also known as OneLoneCoder), it is made up of a single header file, making it extremely lightweight but also rich in features.

Bindings: C++, C, C#, Java, Lua, Rust.

Platform Compatibility: Windows, Linux, Mac OS.

SDL

Website: https://www.libsdl.org/

License: Open Source

SDL (Simple DirectMedia Layer) is one of the most famous libraries around to make multimedia applications as well as video games.

SDL is distributed under the ZLib license, which allows for both commercial and personal use, both in proprietary and open-source situations. Many of the languages listed as "usable" are compatible via extensions.

The versions of SDL up to version 1.2 are instead distributed under the GNU LGPL license, which is more complex and may need to be analyzed by legal experts.

Bindings: C++, C, C#, Go, Python, Lua, Rust.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

SFML

Website: https://www.sfml-dev.org/

License: Open Source

SFML (Simple Fast Multimedia Library) is a library dedicated to creation of multimedia applications (not limited to video games), providing a simple interface to the system's components.

SFML is distributed under the ZLib/png license, which allows for both commercial and personal use, both in proprietary and open-source situations.

Bindings: C++, C, C#, Go, Java, Python, Ruby, Rust.

Platform Compatibility: Windows, Linux, Mac OS, iOS*, Android*.

*: Currently in development

Genre-Specific

Sometimes you don't need all the tools available to make your own game, sometimes you want your adventure to fit some pretty common standards and not deviate too much from them. Genre-specific engines allow you to ditch both the complexity and freedom of the general-purpose engines to have an easier time making your own adventure.

These engines allow you to focus more on the content of your game, instead of wasting time crunching code.

EasyRPG

Website: https://easyrpg.org/

License: Open Source

EasyRPG is a development environment focused on creating RPG games, trying to maintain compatibility with the older RPG Maker 2000/2003 software.

Bindings: Dedicated Language, Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

O.H.R.RPG.C.E.

Website: http://rpg.hamsterrepublic.com/ohrrpgce/About

License: Open Source

The “Official Hamster Republic Role Playing Game Construction Engine” is an utility that allows you to create RPGs in the style of Final Fantasy on the NES/SNES/GBA. It focuses primarily on visual programming, but it also features a custom scripting language for more advanced usage.

Bindings: Dedicated Language, Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS, Android.

OpenBOR

Website: https://github.com/DCurrent/openbor

License: Open Source

OpenBOR is an open source implementation of a side-scroller engine (mostly famous for beat-em-ups in the style of Streets of Rage or Final Fight). It has an active community, even if releases are infrequent.

Bindings: C

Platform Compatibility: Windows, Linux, Android.

Ren’Py

Website: https://www.renpy.org/

License: Open Source

Ren’Py is an engine that allows you to build visual novels and life simulation games. It features its own dedicated scripting language as well as Python for more advanced usage.

This engine is used to create over 4000 visual novels, games and other works.

Bindings: Dedicated Language, Python.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

RPG Maker

Website: https://www.rpgmakerweb.com/

License: Paid

RPG Maker is probably the most well-known engine for building top-down RPG games that usually feature story-heavy elements.

Bindings: Ruby (from RPG Maker XP to RPG Maker Vx Ace), JavaScript (From RPG Maker MV onward), Dedicated Language, Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

RPG Paper Maker

Website: http://rpg-paper-maker.com/

License: Open Source. Free for non-commercial projects, requires a one-time fee for commercial use.

RPG Paper Maker is an engine that allows you to create 3D RPG games with 2D sprites, in the style of the famous "Paper Mario" series.

It allows for creation of jRPGs, action RPGs and Tactical RPGs alike, as well as other kinds, it supports visual programming and makes use of JavaScript for more advanced features.

Bindings: Visual Programming, JavaScript.

Platform Compatibility: Windows, Linux, Mac OS, and Web.

Stratagus

Website: https://github.com/Wargus/stratagus

License: Open Source

Stratagus is an open source real-time strategy engine, it's the engine behind the Wargus mod for Warcraft II. It makes use of the Lua language for scripting.

Bindings: Lua.

Platform Compatibility: Windows, Linux, Mac OS.

3D Engines

Sometimes you may want to go premium, and make use of a 3D engine for your 2D game ("Castlevania: Symphony of the Night" made use of some 3D capabilities of the PlayStation, for instance). Be it either to have 3D backgrounds, make a 2.5D game or particle effects, you can use a 3D engine to make 2D games without any issue.

IRRlicht

Website: http://irrlicht.sourceforge.net/

License: Open Source

IRRlicht is a 3D engine (as in does only 3D rendering) made in C++ that aims to be high-performance.

IRRlicht distributed under the ZLib/png license, which allows for both commercial and personal use, both in proprietary and open-source situations.

Bindings: C++, C#, Java, Python, Ruby.

Platform Compatibility: Windows, Linux, Mac OS.

Ogre3D

Website: https://www.ogre3d.org/

License: Open Source

Ogre3D is an open source 3D graphics engine (it's used to render 3D graphics only).

Ogre3D comes in 2 versions: version 1.x is distributed under the GNU LGPL license, while the more recent 2.x version is distributed under the more permissive MIT license.

Bindings: C++, C#, Java, Python, Ruby.

Platform Compatibility: Windows, Linux, Mac OS.

Panda3D

Website: https://www.panda3d.org/

License: Open Source

Panda3D is a complete and open source 3D game engine.

Panda3D itself is distributed under the modified BSD license, which is very permissive, but it brings together many third-party libraries that are released under different licenses. It is suggested to check the license section of the manual for more information.

Bindings: C++, Python.

Platform Compatibility: Windows, Linux, Mac OS.

Unity 3D

Website: https://unity.com/

License: Free for Personal Use + Paid Version for commercial projects

Unity is probably among the most famous 3D engines used to create video games, as well as other projects that make use of its 3D capabilities (like VR/AR-based projects). It uses the C# programming language.

Unity is a proprietary engine, distributed under a proprietary license, with a Free edition available.

Relevant Bindings: C#.

Platform Compatibility: Windows, Linux, Mac OS, iOS, Android and Web.

UPBGE

Website: https://upbge.org/

License: Open Source

Since 2018, Blender doesn't ship with its own game engine anymore. UPBGE is a project to bring back such feature, built on top of the newer Blender versions, and then some.

Bindings: Python, Dedicated Language, Visual Programming.

Platform Compatibility: Windows, Linux, Mac OS.

Fantasy Consoles

If you want to challenge yourself, or feel "burdened by choice" (which can be a real problem, having "too much power" can hinder creativity), you can make some small (or not-so-small) games on a Fantasy Console: a software that pretends to be a console from a certain era, adhering to arbitrary limitations.

MEG4

Website: https://gitlab.com/bztsrc/meg4

License: Open Source

The "Amiga" of fantasy consoles. Has more features than other fantasy consoles, with a 320x200 (DOS) resolution, with 256 colors, 4 channel tracker music and 1024 8x8 pixel sprites.

Its programs are stored in "floppies" which can be just PNG images with additional data.

Bindings: ANSI C, Basic, Assembly, Lua.

Platform Compatibility: Windows, Linux.

Pico-8

Website: https://lexaloffle.itch.io/pico-8

License: Commercial

Probably the most famous fantasy console, features a 128x128 pixel resolution with 16 colours taken from a pre-defined palette, 4 channel sound, and 8x8 pixel sprites.

Contains its own code editor, sprite editor, tracker and even a cartridge browser (called "Splore") to download and play other people's carts.

Bindings: Lua.

Platform Compatibility: Windows, Linux, Mac OS.

Pyxel

Website: https://github.com/kitao/pyxel

License: Open Source

A slightly more powerful Pico-8, programmed in Python. Features a 256x256 pixel resolution with 16 colors, taken from a pre-defined palette and 4 channel soundtracks.

Features its own sprite and music editor.

Bindings: Python.

Platform Compatibility: Windows, Linux, Mac OS.

TIC-80

Website: http://tic80.com/

License: Open Source

Similar to the Pico-8 in specifications, but features a widescreen resolution of 240x136 pixels, has its own code editor, music editor/tracker as well as sprite and map editor.

Bindings: Lua.

Platform Compatibility: Windows, Linux, Mac OS.

WASM4

Website: https://wasm4.org/

License: Open Source

What if there was a console with features very close to a famous portable console of the past, but with modern multiplayer netcode? WASM4 would probably be that console.

Features a 160x160 pixel resolution and 4 colors a 4 channel audio (2 pulse waves, 1 triangle, 1 noise) and **huge** programming language support.

Bindings: Any programming language that compiles to WebAssembly (C, C++, Go, Odin, Zig, Rust...).

Platform Compatibility: Windows, Linux, Mac OS.

C Some other useful tools

Graphics

Aseprite

Aseprite is an open-source and commercial tool for creating pixel-art graphics. At the time of writing, it can be either bought or compiled from source code, but cannot be redistributed according to its *EULA*[g].

https://www.aseprite.org/

Blender

Blender is an open-source surface modeling program, used in the movie industry and in many other 3D projects, it allows you to create your models and worlds for free.

https://www.blender.org/

Gimp

Gimp is an extensible drawing and photo-manipulation tool, it can be used to draw, edit, filter, balance or compress your graphic resources.

https://www.gimp.org/

Inkscape

Inkscape is an open-source software to work with vector graphics, if you want to give a really clean look to your graphics, you should probably take a look at this software.

https://inkscape.org/

Krita

Krita is a drawing program principally aimed towards artists, with all kinds of brushes and tools it's a real treat to any graphical artist and whoever wants to give a "painted feeling" to their game.

https://krita.org/en/

Laighter

This is a great "name your own price" tool that allows you to create normal maps for 2D textures, to give your game a more "professional" feeling.

https://azagaya.itch.io/laigter

LibreSprite

LibreSprite was created as a fork of Aseprite, but it keeps the previous GNU GPL 2 license.

https://libresprite.github.io/

NormalMap-Online

This is a very nice tool that allows you to create normal maps automatically, for free. Good for a quick game jam or for starters.

https://cpetry.github.io/NormalMap-Online/

Piskel

Piskel is an open-source web-based tool for creating pixel-art graphics and animations. On the website there is a downloadable version too, but usually the web-based one is a bit more performing.

https://www.piskelapp.com/

Pixelorama

Pixelorama is a really interesting beast, since it is made using the Godot game engine and GDScript, it features tools, animation timeline, tool options, pattern filling, and many many more interesting features.

https://orama-interactive.itch.io/pixelorama

Shaders

SHADERed

SHADERed is a fully-featured IDE for coding GLSL and HLSL shaders, it supports debugging as well as live preview of the shader you're coding. It's cross-platform and open source.

https://shadered.org/

Sounds and Music

Audacity

Audacity is an open-source tool for audio editing and recording, extensible with plugins. In the hands of an expert, this seemingly simple program can perform great feats.

This program has been object of controversy due to licensing and telemetry, thus a fork has been made, called "Tenacity", which is described below

https://www.audacityteam.org/

Bosca Ceoil

Created by Terry Cavanagh (creator of VVVVVV, Super Hexagon and Dicey Dungeons), this tool is geared towards beginners, containing chords and scales as well as pre-made instruments. This allows to get something good going on straight away.

This tool is open source, made in Adobe AIR, available for Windows, Linux and Mac

https://terrycavanagh.itch.io/bosca-ceoil

Bosca Ceoil: The Blue Album

A Godot port of the original Bosca Ceoil, created by Yuri Sizov (with Terry Cavanagh's permission). This allows the tool to be ported to new platforms and be maintained.

https://yurisizov.itch.io/boscaceoil-blue

Chiptone

An alternative to SFXR, Chiptone is an online tool (made with HTML5, Haxe and OpenFL) that can be used to create chiptune-like sounds for your games.

It also features downloadable versions for Windows and Mac OS.

https://sfbgames.itch.io/chiptone/

FamiStudio

FamiStudio is a DAW (Digital Audio Workstation) software that allows you to edit and create tracker files for the Nintendo Famicom/NES. Its interface is very intuitive and resembles a lot of DAW software you can find.

The software is available for Windows, MacOS and Linux.

https://famistudio.org/

Furnace

Furnace defines itself as "the ultimate music tracker", supporting almost all 8 and 16 bit systems, including the Atari 2600, Commodore VIC-20, Sega Genesis/Master System and more. It counts over 60 supported chips, it has a really slick interface and supports DefleMask (*.dmf and* .dmw).

The software is available for Windows, MacOS and Linux.

https://tildearrow.org/furnace/

Helio

The Helio project poses itself as "an attempt to rethink a music sequencer to create a tool that feels right". It strives for an extremely clean interface, throwing the idea of it being used as a tool to "grow as a composer".

It is still in its early stages, but what's there seems really good.

This software is available for Windows, Linux, MacOS, Android and iOS.

https://helio.fm/

JFXR

Another alternative to SFXR, JFXR aims to be a browser-based alternative that is more powerful than the original.

Being completely browser-based, this software should work on all systems, including modern smartphones.

https://github.com/ttencate/jfxr

LMMS

LMMS (Linux Multimedia Studio) is a software used to create digital music, it works in a similar fashion to the commercially available FruityLoops.

https://lmms.io/

MilkyTracker

MilkyTracker is an editor for tracker files that takes a lot of inspiration from FastTracker II, it has a lot of functionality, it's well-documented and the community is active.

https://milkytracker.titandemo.org/

OpenMPT

OpenMPT (Open Modplug Tracker) is a tracker software that supports VST plugins and the ASIO protocol. It also hosts libopenmpt, a cross-platform library to play module files; this allows you to implement native tracker music (without converting it to PCM files) in your games.

The software is available for Windows, but it works on Linux via Wine.

https://openmpt.org/

rFXGen

This is another web-based alternative to SFXR. It aims to be simpler to use, has a less dated interface and lets you save files that contain the generation parameters, thus extremely lightweight.

This software is available both on browsers and as a downloadable. It's open source and downloadable from itch.io in a "name your own price" fashion.

https://raylibtech.itch.io/rfxgen

Rimshot

Rimshot is a simple but effective drum machine that can be used to lay down the rhythm of your next jam. Useful to help in the creation of background music for your games.

http://stabyourself.net/rimshot/

SFXR/BFXR/CFXR

SFXR (and its other iterations) is a small software that can help you create 8-bit style sound effects for your games, easily. There are versions for Windows, Mac and even online.

http://www.drpetter.se/project_sfxr.html

Tenacity

Tenacity is a fork of the Audacity audio editor, mostly due to licensing woes and complaints about new telemetry functionalities and data collection inserted by the company that bought Audacity.

https://tenacityaudio.org/

Wafxr

This is another SFXR-inspired sound generator, but instead of generating static WAV files, it generates audio dynamically via Web Audio. This, coupled with the `wasgen` library allows you to save the "playback code" of a certain sound to include it in your games.

https://github.com/fenomas/wafxr

Zrythm

Zrythm is an open-source Digital Audio Workstation (DAW) that aims to be intuitive and highly automated. It features chord assistance, mixing, effects, support for VST plugins (among others), automatic project backups.

The basic release is free with some limitations (25 tracks maximum).

The software is available for Windows, MacOS and Linux.

https://www.zrythm.org/en/index.html

Maps

LDtk

LDtk is a level editor focused on user-friendliness, it also has some interesting features like auto-rendering where it takes care of placing decorations automatically. It has a pay-what-you-want policy, but you can also get it for free (if you like it, support the creator!).

https://ldtk.io/

Tiled

Tiled is a map editor tool that can be used to draw your maps using a tilemap, supports orthogonal, isometric and even hexagonal maps.

https://www.mapeditor.org/

Libraries

Open Adaptive Music Library (oaml)

Oaml is a library dedicated to the development of adaptive music in video games, it supports Unity and Godot as of now. Last development at the time of writing was in 2018, though.

https://oamldev.github.io/

Misc

Chronolapse

Chronolapse is a software specialized in timelapse creation. This can be a really nice addition to a Game Jam's post-mortem post, which will show a timelapse of the entire game development process, thus making it more interesting.

https://github.com/collingreen/chronolapse

Open Broadcaster Software (OBS Studio)

While not strictly a coding-related software, OBS is an amazing software for broadcasting your streams, it could be a good idea to show your progress by programming while streaming.

https://obsproject.com/

D Free assets and resources

In this appendix we'll take a look at a small list of websites that contain resources to help with the development of your game, like assets, royalty-free music and the like.

For each website we will have, along with a short description:

- **Website:** containing a link to the website where you can find the resources listed;
- **Resource Types:** a small table describing the kind of resources you will find on the website.

The licenses vary from website to website and even from single resource to another, so you should pay close attention to what you use and how you use it.

Openclipart.org

Website: https://openclipart.org/

Types of resources available:

Graphics	Music	Sound Effects
✓		

Openclipart is a website that contains lots of freely available cliparts , in vector format, that can be a good starting point for your own art: buttons, characters, symbols, ...

The website is currently (as of December 18th, 2020) undergoing a phase of transition, after switching backend and what seems to have been an attack by cyber-criminals.

It now features quite a confusing interface but now features a search function, but it's a precious source of art if you know what you're looking for.

Opengameart.org

Website: https://opengameart.org/

Types of resources available:

Graphics	Music	Sound Effects
✓	✓	✓

Opengameart is a website specifically dedicated to game development, with all kinds of resources to use in your game, either as placeholders or good-quality assets that are meant to stay in your product.

The interface is quite essential and sometimes the website can be slow, but this is a great source for anything you need to make your own video game.

Freesound

Website: https://freesound.org/

Types of resources available:

Graphics	Music	Sound Effects
	✓	✓

The Freesound project is a collaborative database of sounds licensed under the "Creative Commons" license, it also features a very interesting blog that will teach you about sounds and soundscapes.

PublicDomainFiles

Website: http://www.publicdomainfiles.com/

Types of resources available:

Graphics	Music	Sound Effects
✓		

Similarly to OpenClipart, PublicDomainFiles contains a ton of graphics that can be a really good starting point for all your projects. It has a nice search function with filters and a good interface, even though it does not look as modern as other dedicated websites.

CCMixter

Website: http://ccmixter.org/ and http://dig.ccmixter.org

Types of resources available:

Graphics	Music	Sound Effects
	✓	

CCMixer is a gold mine when it comes to music you can use in your games: its interface is simple and easy to understand, features a tag-based search and the quality is great.

Definitely a website to check when you want to add some beat to your game.

SoundBible.com

Website: http://soundbible.com

Types of resources available:

Graphics	Music	Sound Effects
		✓

SoundBible.com is another great website where you can look for sound effects for your games: the interface is quick and all sound effects are listed along with their license, which is definitely a plus.

Incompetech

Website: https://incompetech.com/

Types of resources available:

Graphics	Music	Sound Effects
	✓	

One of the most known websites in the field, and sometimes a bit overused, Incompetech contains lots of music for your game, but you need to check the license yourself. It surely gives a great set of placeholders for your project, while waiting for a more fitting soundtrack.

E Contributors

This e-book is the collective work of many game developers, critics and enthusiasts. Here is a list of who contributed to the making of this work.

- **Daniele Penazzo (Penaz)** - Project creator, maintainer and main writer
- **Luca Violato (Rei)** - Marketing section
- **Sjofin (Fin)** - Art Tips
- **DashingRaven** - Art (https://linktr.ee/DashingRaven)

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List of Tables

1 Some rules that would help us calculating logarithms . . . 12
2 Some simple derivation rules (k is any constant number and e is Euler's number) . . . 14
3 Some derivation rules for combined functions (a and b are constants) . . . 14
4 Conversion between degrees and Radians . . . 30
5 Some reflection formulas for trigonometry . . . 31
6 Some Shift Formulas for Trigonometry . . . 32
7 Some addition and difference identities in trigonometry . . . 32
8 Some double-angle formulae used in trigonometry . . . 32
9 Counting the possible outcomes of two coin tosses . . . 40
10 Comparison between decimal and binary representations . . . 50
11 Comparison between decimal and octal representations . . . 50
12 Comparison between decimal and hexadecimal representations . . . 51
23 The first truth table we'll simplify with Karnaugh Maps . . . 81
24 Truth table with a "don't care" value . . . 82
27 A simple adjacency list for our reference image . . . 115
28 How to read an adjacency matrix . . . 116
29 Performance table for Dynamic Arrays . . . 125
30 Summary Table for Dynamic Arrays . . . 125
31 Performance table for Linked Lists . . . 127
32 Summary Table for Linked Lists . . . 127
33 Performance table for Doubly-Linked Lists . . . 128
34 Summary Table for Linked Lists . . . 128
35 Performance table for Hash Tables . . . 129
36 Summary Table for Hash Tables . . . 129
37 Performance table for Binary Search Trees . . . 130
38 Summary Table for Binary Search Trees . . . 130
39 Performance table for Heaps . . . 131
40 Summary Table for Heaps . . . 131
42 Summary of linear gameplay . . . 290
48 An example of exponential level curve . . . 302
49 An example of "level-based" experience rewards . . . 302
52 Summary table for the Singleton Pattern . . . 375
53 Summary table for the Dependency Injection Pattern . . . 376
54 Summary table for the Prototype design pattern . . . 377
55 Summary table for the Flyweight Pattern . . . 379
56 Summary table for the Component/Composite design pattern . . . 381
57 Summary table for the Decorator design pattern . . . 382
58 Summary table for the Adapter design pattern . . . 382

59 Summary table for the Facade design pattern 384
61 Summary table for the Command Pattern 388
62 Summary table for the Observer Pattern 390
63 Summary table for the Strategy Pattern 392
64 Summary table for the Chain of Responsibility Pattern 393

List of Figures

1 Example of a Cartesian plane 14
2 Image of a vector 15
3 Graphical representation of a sum of vectors 16
4 Example of a vector multiplied by a value of 3 16
5 Example of a vector multiplied by a value of 0.5 17
6 Example of a vector multiplied by a value of -2 17
7 Example of a convex shape 19
8 Example of a concave shape 20
9 Example of a self-intersecting polygon 20
10 Projecting the point P onto the line r 23
11 Projecting a line onto the axes 25
12 Unit Circle definition of sine and cosine 30
13 Graphical plotting of the angle of a vector 33
14 Image of a coordinate plane 34
15 Image of a screen coordinate plane 34
16 Reference image for transformation matrices 35
17 Stretching along the x and y axes 36
18 The result of applying a rotation matrix 37
19 Shearing along the x and y axes 38
20 Running the probability_20 example shows the probability floating around 20% 44
21 Running the probability_le_13 example shows the probability floating around 13% 45
22 Intuitive representation of our prize pool 46
23 How we can pack wall information with a 4-bit integer 58
24 Example of a compiler output (G++) 62
25 Python's REPL Shell 63
26 Results of the simple float precision test 68
27 Python 2 has the same issues with precision as C++ 68
28 Python 3 doesn't fare much better when it comes to precision 68
29 Running a random number generator with the same seed will always output the same numbers . . . 71
30 Using the system time as RNG seed guarantees a degree of randomness 72
31 O(n) growth rate, compared to $O(n^2)$ 78
32 When coefficients have important values, asymptotic complexity may trick us 79

33 Big-O Estimates, plotted . . . 80
34 How O(2^n) overpowers lower complexities . . . 80
35 Karnaugh Map for A XOR B . . . 81
36 Karnaugh Map where the elements of the two “rectangles” have been marked green and red . . . 81
37 Karnaugh Map with a “don’t care” value . . . 82
38 Karnaugh Map where we pretend the “don’t care” value is equal to 1 . . . 82
39 First Rectangle in the Karnaugh map . . . 82
40 Second Rectangle in the Karnaugh map . . . 83
41 A more complex Karnaugh map . . . 84
42 First rectangle of the more complex Karnaugh map . . . 84
43 Second rectangle of the more complex Karnaugh map . . . 84
44 Guided Exercise: Karnaugh Map (1/4) . . . 85
45 Guided Exercise: Karnaugh Map (2/4) . . . 85
46 Guided Exercise: Karnaugh Map (3/4) . . . 86
47 Guided Exercise: Karnaugh Map (4/4) . . . 86
48 Example of a diamond problem . . . 88
49 How an object may look using inheritance . . . 91
50 How inheritance can get complicated quickly . . . 91
51 How components make things a bit simpler . . . 92
52 Example of a use case diagram . . . 96
53 Example of an actor hierarchy . . . 96
54 Example of a use case . . . 97
55 Example of a use case hierarchy . . . 97
56 Example of a use case extension . . . 97
57 Example of a use case inclusion . . . 98
58 Example of a sub-use case . . . 98
59 Example of classes in UML . . . 99
60 Defining an interface in UML . . . 99
61 Interface Realization in UML . . . 99
62 Relationships between classes in an UML Diagram . . . 100
63 Example of inheritance in UML class diagrams . . . 100
64 Example of interface realization in UML class diagram . . . 100
65 Example of association in UML class diagrams . . . 101
66 Example of aggregation and composition in UML class diagrams . . . 101
67 Example of dependency in UML class diagrams . . . 102
68 Example of an activity diagram . . . 103
69 Example of activity diagrams start and end nodes . . . 103
70 Example of Action in activity diagrams . . . 103
71 Example of decision, using hexagons to represent the condition . . . 104
72 Example of loops, using guards to represent the condition . . . 104

73 Example of how nested loops and conditions are performed . . . 105
74 Example of concurrent processes in activity diagrams . . . 105
75 Example of swimlanes in activity diagrams . . . 106
76 Example of signals in activity diagrams . . . 107
77 Example of a note inside of an activity diagram . . . 107
78 Example of a sequence diagram lifeline . . . 108
79 Some alternative shapes for participants . . . 109
80 Messages in a sequence diagram . . . 109
81 Object instantiation and destruction in a sequence diagram . . . 110
82 A loop grouping in a sequence diagram . . . 110
83 Example of notes in a sequence diagram . . . 111
84 Example of a 4-element integer array . . . 112
85 Graphical representation of a simple graph . . . 115
86 Example of a tree structure . . . 117
87 Order in which the nodes are visited during DFS . . . 118
88 Example tree that will be traversed by DFS . . . 119
89 Order in which the nodes are visited during BFS . . . 122
90 Dynamic Arrays Reference Image . . . 123
91 Adding an element at the beginning of a Dynamic Array . . . 124
92 Adding an element at the end of a Dynamic Array . . . 124
93 Adding an element at an arbitrary position of a Dynamic Array . . . 124
94 Linked List Reference Image . . . 125
95 Double-Ended Linked List Reference Image . . . 126
96 Inserting a new node at the beginning of a linked list . . . 126
97 Inserting a new node at the end of a (double-ended) linked list . . . 126
98 Inserting a new node at an arbitrary position in a (double-ended) linked list . . . 127
99 Doubly Linked List Reference Image . . . 127
100 Hash Table Reference Image (Hash Table with Buckets) . . . 128
101 Binary Search Tree Reference . . . 129
102 Heap Reference Image (Min-Heap) . . . 131
103 How a stack works . . . 132
104 Array and linked list implementations of a stack . . . 132
105 How a queue works . . . 132
106 Array and linked list implementation of a queue . . . 133
107 How a circular queue works . . . 133
108 Array and linked list implementation of a circular queue . . . 134
109 The result of running the identity vs equality code . . . 135
110 The “easy way” of dealing with frames . . . 138
111 A sample sprite sheet with the same frames as before . . . 138
112 Singly-Linked List has no redundancy . . . 141

113 A doubly linked list is an example of redundancy 141
114 In a multi-processing environment, each CPU takes care of a task 142
115 In multi-threading, the CPU uses I/O wait time to take care of another task 142
116 Two threads and a shared variable 144
117 Thread 1 reads the variable 144
118 While Thread 1 is working, Thread 2 reads the variable 145
119 Thread 1 writes the variable 145
120 Thread 2 writes the variable 145
121 Both Threads Terminated 146
122 How mutex works (1/8) 147
123 How mutex works (2/8) 147
124 How mutex works (3/8) 147
125 How mutex works (4/8) 148
126 How mutex works (5/8) 148
127 How mutex works (6/8) 148
128 How mutex works (7/8) 149
129 How mutex works (8/8) 149
130 How an arcade machine usually looks like 154
131 A portable console 155
132 A personal computer 156
133 How many abstraction layers are used just for a game to be able to play sounds 156
134 A smartphone 157
135 Fully fledged games can run in your browser nowadays 157
136 When there is no console people just invent theirs, in software form 158
137 How to approach improvements on your game 176
138 Diagram of the waterfall life cycle model 177
139 Diagram of the incremental life cycle model 178
140 High-level diagram of the evolutionary life cycle model 178
141 Diagram of the evolutionary life cycle model 179
142 Example of a Kanban Board 181
143 An example screen from Git, a version control system 182
144 UML of the program which we'll calculate the cyclomatic complexity of 184
145 Flow diagram of the program we'll calculate the cyclomatic complexity of 185
146 UML Diagram of the input-update-draw abstraction 200
147 An example of screen tearing 206
148 A small example of the "painter's algorithm" 207
149 How not clearing the screen can create glitches 208
150 Another type of glitch created by not clearing the screen 208
151 Reference image for Point-Circle Collision detection 211
152 Reference image for Circle-Circle collision detection 212

153 Example used in the AABB collision detection . . . 213
154 Top-Bottom Check . . . 214
155 Top-Bottom Check is not enough . . . 214
156 An example of a left-right check . . . 214
157 Example of the triangle inequality theorem . . . 216
158 Example of a degenerate triangle . . . 216
159 Point/Triangle Collision Detection: division into sub-triangles . . . 221
160 Example image for line/line collision . . . 224
161 Example of a Jordan Curve . . . 227
162 A simple case where a point is outside the polygon . . . 228
163 A simple case where a point is inside the polygon . . . 228
164 How a non-convex polygon makes everything harder . . . 229
165 Decomposing a polygon into triangles . . . 229
166 Example of a polygon with its bounding box . . . 230
167 Example image used for circle/polygon collision detection . . . 235
168 An edge case of the circle/polygon check . . . 236
169 Example image used for line/polygon collision detection . . . 237
170 Example image used for polygon/polygon collision detection . . . 239
171 How a non-convex polygon still makes everything harder . . . 240
172 Counting how many times we hit the perimeter gives us the result . . . 240
173 Issues with vertices make everything even harder . . . 241
174 Triangulating a non-convex polygon . . . 242
175 Two Bitmasks that will be used to explain pixel-perfect collision . . . 242
176 Two Bitmasks colliding, the ‘AND’ operations returning true are highlighted in white . . . 243
177 Example for collision detection . . . 244
178 Graphical example of a quad tree, overlaid on the reference image . . . 246
179 A quad tree . . . 246
180 Quad trees as spacial acceleration structures . . . 247
181 Redundancy in quad-tree pointers . . . 247
182 How an AABB-tree would process our example image . . . 248
183 How a possible AABB-tree structure would look like . . . 249
184 Example of a search in an AABB-Tree . . . 249
185 Querying an AABB-tree (1/3) . . . 250
186 Querying an AABB-tree (2/3) . . . 250
187 Querying an AABB-tree (3/3) . . . 251
188 Example tile-based level . . . 252
189 Tile-based example: falling . . . 252
190 Example tile-based level with a bigger object . . . 254
191 Tile-based example with a bigger object: falling . . . 254
192 Example of a hitbox (red) and a hurtbox (blue) . . . 256

193 Images used as a reference for collision reaction 257
194 How the naive method reacts to collisions against a wall 258
195 How the naive method reacts to collisions against the ground 259
196 Example of shallow-axis based reaction on a horizontal plane 259
197 Example of shallow-axis based reaction on a vertical plane 260
198 How the the interleaving method reacts to collisions on a horizontal plane 262
199 Example of the “Bullet through paper” problem 265
200 How velocity changing direction can teleport you 266
201 Example of how you can draw a line between two convex non-colliding polygons 267
202 Why the SAT doesn’t work with concave polygons 267
203 How the SAT algorithm works 268
204 Finding the axes for the SAT (1/2) 269
205 Finding the axes for the SAT (2/2) 269
206 Projecting the polygons onto the axes 270
207 Projecting our projections onto the x and y axes 271
208 How Ray Casting Works: Gun (1/2) 272
209 How Ray Casting Works: Gun (2/2) 272
210 How a scene tree looks (specifically in Godot) 273
211 Example of a ship attack formation 274
212 What happens when the ship attack formation rotates 274
213 Reference Image for Screen Space and Game Space 275
214 How a person sees things 276
215 How videogame cameras see things 277
216 Example of an horizontally-tracking camera 278
217 Example of a full-tracking camera 278
218 Example of camera trap-based system 279
219 Example of look-ahead camera 279
220 How the camera may end up showing off-map areas 280
221 Example of how to induce lateral thinking with environmental damage 284
222 Example of how to induce lateral thinking by “breaking the fourth wall” 284
223 Example of secret-in-secret 285
224 Example Scheme of linear gameplay 290
225 Example Scheme of branching gameplay 290
226 Example Scheme of parallel gameplay 291
227 Example Scheme of threaded gameplay 291
228 Example Scheme of episodic gameplay 292
229 Example Scheme of looping gameplay with a overarching story 293
230 Example of a telegraphed screen-filling attack in a shooter 298
231 A pixel perfect detection would trigger in this case 304
232 A smaller hitbox may save the player some frustration 304

233 A bomb spawned behind a chest, drawn in a realistic order 305
234 Moving the bomb in front of the chest may ruin immersion 305
235 Highlighting the hidden part of a danger can be useful 305
236 Making objects transparent is a solution too 306
237 The “color wheel” for screens 313
238 An example of an RGB picker 314
239 A simplified “slice” of an HSV representation 315
240 More slices of the HSV representation show how value changes 315
241 The color star shows how complementary colors are on opposite sides 317
242 Reference image that we will for bit depth comparison 318
243 Reference image, quickly converted to 1-bit color depth 318
244 Reference image, converted to 2-bit color depth in CGA style 318
245 Reference image, converted to a 4-bit color depth in EGA style 319
246 Reference image, converted to an 8-bit color depth 319
247 Indexed transparency takes a color and “marks it” as transparent 321
248 Example of using smearing for animating a sword slash 323
249 Example sprite that gets padded to match hardware constraints 324
250 Example spritesheet that gets padded to match hardware constraints 325
251 Two platforms in a spritesheet 325
252 Two platforms in a spritesheet after heavy compression 326
253 Windowed Game Example - A 640x480 game in a 1920x1080 Window 326
254 Fullscreen Game Example - Recalculating items positions according to the window size 327
255 Fullscreen Game Example - Virtual Resolution 327
256 Dithering example 328
257 Dithering Table Example 328
258 Some more dithering examples 329
259 Which spaceship is easier to spot at a glance? 331
260 How contrast and detail can help distinguishing foreground and background 332
261 Breaking down the image allows us to see the differences between the layers 332
262 A diagram to show how each section affects our perception of a layer 333
263 A texture (on the left), with a possible normal map (on the right) 335
264 Aseprite’s normal mapping color picker (both in its normal and discrete versions) 335
265 A box that will be used to show how normal maps influence light 336
266 How the lack of normal mapping makes lighting look artificial 336
267 How normal mapping changes lighting 336
268 A more detailed normal map results in better lighting 336
269 Surface blending - addition (1/2) 337
270 Surface blending - addition (2/2) 338
271 Simple lighting effect using 2 surfaces and blending 338
272 Surface blending - subtraction (1/2) 338

273 Surface blending - difference (2/2) . . . 339
274 Simple shadow effect using 2 surfaces and blending . . . 339
275 Example of tile “alternatives” . . . 340
276 Tile “Rotation Trick” (1/3) . . . 340
277 Tile “Rotation Trick” (2/3) . . . 341
278 Tile “Rotation Trick” (3/3) . . . 341
279 Example of black and transparent tileset used in “inside rooms” . . . 342
280 Example of incomplete “inside room” . . . 342
281 Example of “inside room” with the black/transparent overlay . . . 342
282 How color can completely change an object . . . 343
283 Graphical Representation of Sample Rate (44.1KHz) . . . 344
284 Graphical Representation of Sample Rate (8KHz) . . . 344
285 Example of audio clipping . . . 346
286 Example of AM Synthesis . . . 347
287 Example of FM Synthesis . . . 347
288 How a sine wave looks . . . 349
289 How a square wave looks . . . 349
290 How a triangle wave looks . . . 349
291 How a sawtooth wave looks . . . 350
292 How a noise wave looks . . . 350
293 Representation of an ADSR Envelope . . . 350
294 Attack on ADSR Envelope . . . 351
295 Decay on ADSR Envelope . . . 351
296 Sustain on ADSR Envelope . . . 352
297 Release on ADSR Envelope . . . 352
298 A freeze frame of a C64 song, you can see the instruments changing . . . 355
299 A source of sound right in front of you sounds the same in both ears . . . 359
300 A source of sound out of center sounds different in each ear . . . 359
301 Example of a piano roll in LMMS . . . 360
302 Example of a piano roll in FamiStudio . . . 361
303 A screen from MilkyTracker . . . 361
304 Simple overview of a tracker . . . 362
305 How each tracker channel is divided . . . 362
306 A single bar in our basic rhythm . . . 364
307 A basic four on the floor rhythm . . . 364
308 Four on the floor with off-beat hi-hats . . . 364
309 A simple rock beat . . . 364
310 Example of a serif font (DejaVu Serif) . . . 365
311 Example of a sans-serif font (DejaVu Sans) . . . 365
312 Example of a proportional font (DejaVu Serif) . . . 366

313 Example of a monospaced font (Inconsolata) . . . 366
314 A simple spritesheet for rendering text using textures . . . 366
315 Indexing our spritesheet for rendering . . . 366
316 Godot's "Visual Shader" Editor . . . 370
317 The UML diagram for a singleton pattern . . . 374
318 A naive implementation of a local file upload system . . . 375
319 A naive implementation of a file upload system on S3 . . . 375
320 Using Interfaces and DI to build a flexible file upload . . . 376
321 Possible class structure for a DI file upload . . . 376
322 Diagram of the Prototype Pattern . . . 377
323 UML Diagram of the Flyweight pattern . . . 378
324 Diagram of the Component Design Pattern . . . 379
325 Diagram of the Decorator Pattern . . . 382
326 Diagram of the Object Adapter Pattern . . . 383
327 Diagram of the Class Adapter Pattern . . . 383
328 Diagram of the Facade Pattern . . . 384
329 Diagram of the Proxy Pattern . . . 386
330 UML diagram for the Command Pattern . . . 388
331 The UML diagram of the observer pattern . . . 389
332 The UML diagram of the strategy pattern . . . 391
333 UML Diagram of the Chain of Responsibility Pattern . . . 392
334 Diagram of a character's state machine . . . 396
335 Diagram of a menu system's state machine . . . 396
336 Example of a simple menu stack . . . 399
337 Some examples of particles . . . 399
338 Map we will create a navigation mesh on . . . 415
339 Dividing the map into many convex polygons and labelling them . . . 416
340 Creating the graph . . . 416
341 The final data structure . . . 417
342 Example of Manhattan distance . . . 417
343 Example of Euclidean Distance . . . 418
344 Pathfinding Algorithms Reference Image . . . 419
345 Pathfinding Algorithms Heuristics Reference Image . . . 419
346 Simple wandering algorithm 1/2 . . . 420
347 Simple wandering algorithm 2/2 . . . 420
348 This maze breaks our wandering algorithm . . . 421
349 The path taken by the greedy "Best First" algorithm . . . 426
350 The path taken by the Dijkstra Algorithm . . . 428
351 Finite state machine representing an enemy AI . . . 430
352 Example of a decision tree . . . 430

353 Example of a behaviour tree . . . 431
354 How the Midpoint Displacement Algorithm Works (1/4) . . . 436
355 How the Midpoint Displacement Algorithm Works (2/4) . . . 436
356 How the Midpoint Displacement Algorithm Works (3/4) . . . 437
357 How the Midpoint Displacement Algorithm Works (4/4) . . . 437
358 How the diamond-square algorithm works (1/5) . . . 439
359 How the diamond-square algorithm works (2/5) . . . 440
360 How the diamond-square algorithm works (3/5) . . . 440
361 How the diamond-square algorithm works (4/5) . . . 441
362 How the diamond-square algorithm works (5/5) . . . 441
363 How the recursive backtracker algorithm works (1) . . . 442
364 How the recursive backtracker algorithm works (2) . . . 442
365 How the recursive backtracker algorithm works (3) . . . 443
366 How the recursive backtracker algorithm works (4) . . . 443
367 How the Randomized Kruskal's Algorithm Works (1/6) . . . 448
368 How the Randomized Kruskal's Algorithm Works (2/6) . . . 448
369 How the Randomized Kruskal's Algorithm Works (3/6) . . . 448
370 How the Randomized Kruskal's Algorithm Works (4/6) . . . 449
371 How the Randomized Kruskal's Algorithm Works (5/6) . . . 449
372 How the Randomized Kruskal's Algorithm Works (6/6) . . . 449
373 How the Recursive Division Algorithm Works (1/6) . . . 450
374 How the Recursive Division Algorithm Works (2/6) . . . 450
375 How the Recursive Division Algorithm Works (3/6) . . . 451
376 How the Recursive Division Algorithm Works (4/6) . . . 451
377 How the Recursive Division Algorithm Works (5/6) . . . 451
378 How the Recursive Division Algorithm Works (6/6) . . . 452
379 The bias of Recursive Division Algorithm . . . 452
380 How the Binary Tree Maze generation works (1/6) . . . 453
381 How the Binary Tree Maze generation works (2/6) . . . 453
382 How the Binary Tree Maze generation works (3/6) . . . 453
383 How the Binary Tree Maze generation works (4/6) . . . 454
384 How the Binary Tree Maze generation works (5/6) . . . 454
385 How the Binary Tree Maze generation works (6/6) . . . 454
386 How Eller's Maze Generation Algorithm Works (1/7) . . . 455
387 How Eller's Maze Generation Algorithm Works (2/7) . . . 455
388 How Eller's Maze Generation Algorithm Works (3/7) . . . 455
389 How Eller's Maze Generation Algorithm Works (4/7) . . . 455
390 How Eller's Maze Generation Algorithm Works (5/7) . . . 455
391 How Eller's Maze Generation Algorithm Works (6/7) . . . 456
392 How Eller's Maze Generation Algorithm Works (7/7) . . . 456

393 Example of Random Noise 457
394 Example of a tileset and a tilemap drawn with it 466
395 Simple structure of a hexmap 467
396 The outer circle or an hexagon 467
397 The size of an hexagon, calculated 468
398 Making a hexmap 469
399 A simple isometric tiles and a tilemap 469
400 Demonstration of an image with loop points 470
401 How we can split our game into layers 472
402 Rough UML diagram of a multi-threaded loading screen 475
403 Example chart of how movement without inertia looks 476
404 Example chart of how movement without inertia looks: reversing directions 476
405 Example chart of how movement with inertia looks 477
406 Example of character running 477
407 Applying an acceleration to a character running 478
408 Applying an acceleration frame by frame leads to the feeling of inertia 478
409 What would be a good collision response for this situation? 480
410 Corner correction makes for a more fluid experience 480
411 Plotting a physics-accurate jump 483
412 Plotting a jump with enhanced gravity 483
413 Plotting a jump with multiple gravity changes 484
414 Example of how jump buffering would work 486
415 Example of how coyote time would work 487
416 Example of how timed jumps would work 488
417 A simple example of fake height in RPG 489
418 How few collisions may “sell” the effect of height 489
419 A more complex example of fake height 490
420 Even with complex tilemaps, the texture sells the height effect 490
421 Reference image for video lag 491
422 Reference image for audio lag 491
423 Plotting amplitude against time 493
424 Plotting frequency domain 493
425 Example of how to better “highlight” bullets 494
426 An example of a Bullet Hell ship hitbox 495
427 Example of a matrix, saved as “array of arrays” 498
428 What happens when deleting a match immediately 501
429 A Flat line difficulty curve 513
430 A linearly increasing difficulty curve 514
431 As the player learns to predict, the difficulty curve changes from our design 514
432 A Logarithmic difficulty curve 515

433 An exponential difficulty curve . . . 515
434 A linearly increasing wavy difficulty curve . . . 516
435 A Logarithmically increasing wavy difficulty curve . . . 517
436 A simple wavy difficulty interval . . . 518
437 A widening and wavy difficulty interval . . . 518
438 A widening wavy difficulty interval with a logarithmic trend . . . 519
439 A sawtooth difficulty curve . . . 520
440 Difficulty spikes are not good . . . 520
441 A simplified vision of supply and demand . . . 522
442 Example of a P2P connection . . . 526
443 Two cheaters meet in P2P . . . 526
444 What would happen if one of the Peers had the authoritative game state . . . 527
445 Example of a dedicated server . . . 527
446 Two cheaters vs. Dedicated server . . . 528
447 How the effort put in automated testing can give good returns . . . 538
448 Using Valgrind's Callgrind tool and Kcachegrind we can see what is bogging down our game . . . 542
449 Not putting off-screen objects in the drawing queue can be a good optimization . . . 544
450 An example screen from LLVM's scan-build . . . 546
451 A screenshot from Valgrind, looks like we have a memory leak here . . . 546
452 A screenshot from Godot's profiler . . . 546
453 A resource pool instantiates objects and "keeps them" ready when needed . . . 547
454 Pulling an object from a resource pool . . . 547
455 Returning an object from a resource pool . . . 548
456 How a lookup table works . . . 548
457 How a simple memoization pattern works . . . 549
458 How a more complex memoization pattern works . . . 550
459 Time taken by the IF code . . . 558
460 Time taken by the Switch code . . . 558
461 How first impressions leave a mark, even when it comes to price . . . 563
462 What a simple "speedrun mode" may look like . . . 576
463 What a mode advanced "speedrun mode" may look like . . . 577
464 Example picture of Tic-Tac-Toe . . . 599
465 Example of a "Space Invaders"-style game . . . 601
466 Example picture of a breakout-style game . . . 603
467 Biased Bouncing for breakout . . . 605
468 Possible Shooter Arena Game . . . 605

List of Code Listings

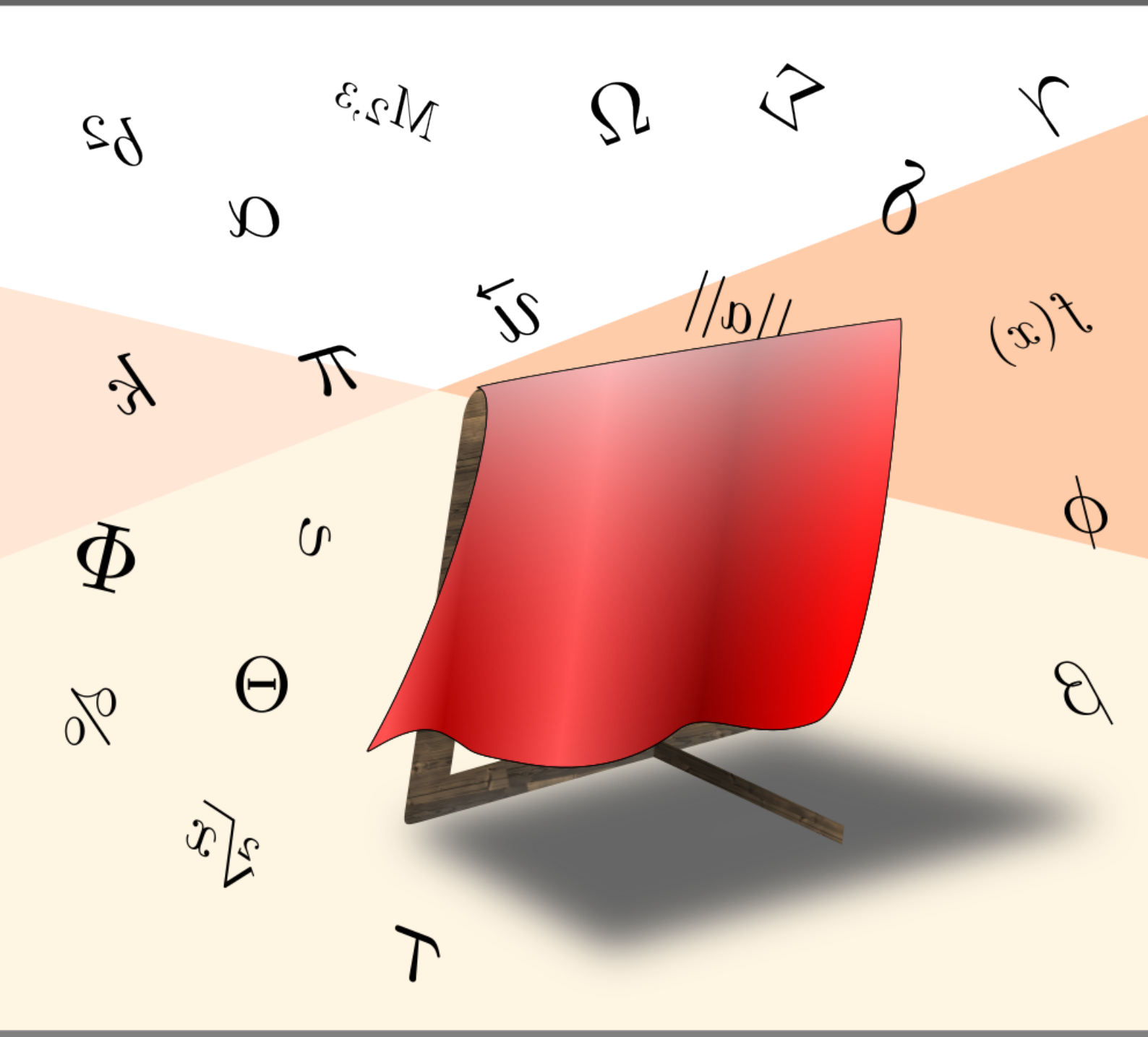
1 Example code listing . . . 3

2 1 (out of 5) will be extracted with about 20% probability . . . 44
3 A number less or equal than 13 (out of 100) has 13% probability of appearing . . . 45
4 How to implement a tiered prize pool selector . . . 46
5 A possible implementation of a luck stat . . . 47
6 Counting from n to 0 using a loop . . . 60
7 Counting from n to 0 using recursion . . . 60
8 Counting from 0 to n using head recursion . . . 61
9 A simple float precision test . . . 67
10 A simple random number generation program . . . 70
11 A random number generation program that uses system time as seed . . . 72
12 Example of an O(1) algorithm . . . 73
13 Example of an O(log(n)) algorithm (Binary Search) . . . 74
14 Example of an O(n) algorithm (printing of a list) . . . 74
15 Example of an $O(n^2)$ algorithm (bubble sort) . . . 75
16 A simple O(1) algorithm . . . 76
17 A simple o(n) algorithm . . . 76
18 The bubble sort algorithm, an $O(n^2)$ algorithm . . . 77
19 A more complex algorithm to estimate . . . 77
20 An example of inheritance: Shapes . . . 89
21 An example of inheritance: A coffee machine . . . 90
22 Example of an entity declared as YAML data . . . 94
23 Example of an entity declared as JSON data . . . 94
24 A code block featuring some magic numbers . . . 113
25 A code block featuring enums for readability . . . 113
26 A possible implementation of a tree class . . . 117
27 Pre-order traversal of a tree using DFS . . . 119
28 In-order traversal of a tree using DFS . . . 120
29 Post-order traversal of a tree using DFS . . . 121
30 Traversal of a tree using BFS . . . 122
31 Checking identity vs. checking equality . . . 134
32 Using operators as functions . . . 136
33 Counting the elements in a list . . . 139
34 Counting the elements in a list with data redundancy . . . 140
35 Finding the previous element in a singly linked list . . . 141
36 A naive array initialization . . . 150
37 A smarter array initialization . . . 150
38 A piece of code featuring some magic numbers . . . 151
39 A piece of code after magic numbers are removed . . . 152
40 Game Loop example . . . 201
41 Game loop with fixed timesteps . . . 202

42 Game loop with variable time steps . . . 203
43 Game loop with Semi-Fixed time steps . . . 203
44 Game loop with Frame Limiting . . . 204
45 Point to point collision detection . . . 210
46 Point to point collision detection with epsilon values . . . 210
47 Point to circle collision detection . . . 211
48 Circle to Circle Collision Detection . . . 213
49 Axis-Aligned Bounding Box Collision Detection . . . 214
50 Line to Point Collision detection . . . 216
51 Partial Implementation of a Line to Circle Collision Detection . . . 217
52 Line to circle collision detection . . . 219
53 Point/Rectangle collision detection . . . 220
54 Point/Triangle Collision Detection . . . 221
55 Point/Triangle Collision Detection with epsilon . . . 222
56 Rectangle to Circle Collision Detection . . . 222
57 Implementation of the line/line collision detection . . . 225
58 Implementation of the line/rectangle collision detection . . . 226
59 How to find the bounding box of a polygon . . . 231
60 A (not so) simple polygon class . . . 233
61 Polygon vs Point collision detection . . . 234
62 Polygon vs Circle collision detection . . . 236
63 Polygon vs Line collision detection . . . 238
64 Polygon vs Polygon collision detection . . . 239
65 Example of a possibile implementation of pixel perfect collision detection . . . 243
66 Brute Force Method of collision search . . . 245
67 Converting player coordinates into tile coordinates . . . 252
68 Tile-based collision detection . . . 253
69 Tile + Offset collision detection . . . 255
70 Code for the naive collision reaction . . . 257
71 Possible implementation for a shallow axis collision reaction . . . 260
72 Code for interleaving movement and collision reaction . . . 261
73 Example of the ”snapshot” collision reaction method . . . 262
74 Code for the collision reaction between moving objects . . . 264
75 How FM music may be saved on an old console . . . 348
76 Example of swappable sound effects . . . 358
77 A simple algorithm to create a text using a texture . . . 367
78 Simple GLSL Fragment shader . . . 369
79 Example of a singleton pattern with lazy loading . . . 374
80 Code for a flyweight pattern . . . 378
81 Example Implementation Of the Component Pattern . . . 380

82 Example Implementation Of the Facade Pattern . . . 384
83 Example Implementation Of the Proxy Pattern . . . 386
84 Example code for the Command Pattern . . . 388
85 Code for an observer pattern . . . 389
86 Code for a strategy pattern . . . 391
87 Code for a chain of responsibility pattern . . . 392
88 A simple finite state machine . . . 397
89 An example of usage of a FSM . . . 397
90 A simple particle class . . . 400
91 A more complex particle class . . . 401
92 A simple particle emitter class . . . 402
93 A more complex (and complete) particle emitter class . . . 403
94 A particle with mass and force application . . . 404
95 A simple timer class . . . 405
96 A naive approach to account for leftover time . . . 407
97 A possible solution to account for leftover time . . . 408
98 Another possible solution to account for leftover time . . . 408
99 Linear Tweening . . . 409
100 Example of a simple easing function . . . 410
101 Ease-in . . . 410
102 Ease-out . . . 411
103 Ease-in-out . . . 411
104 Clamping values . . . 411
105 A bouncy tween function . . . 412
106 Possible representation of a 2D grid . . . 413
107 Example code calculating the Manhattan distance on a 2D grid . . . 418
108 Example code calculating the Euclidean distance on a 2D grid . . . 418
109 Implementation of a simple wandering algorithm . . . 421
110 Implementation of a better wandering algorithm . . . 423
111 The node structure used in the greedy ”Best First” algorithm . . . 425
112 The greedy ”Best First” algorithm . . . 425
113 The node structure used in the Dijkstra Algorithm . . . 427
114 The Dijkstra Algorithm . . . 427
115 The A* Algorithm . . . 428
116 Example code for ”jump when player shoots” AI pattern . . . 432
117 Example code for a ”ranged” AI pattern . . . 433
118 Example code for a ”melee” AI pattern . . . 434
119 Example implementation of the midpoint displacement algorithm . . . 437
120 Example implementation of recursive backtracker maze generation . . . 443
121 Example implementation of recursive backtracker with an explicit stack . . . 445

122 Example implementation of randomized noise 457
123 Example procedural weapon creation 459
124 Example Randomized weapon creation 459
125 Example of I-Frames Implementation 465
126 Example of an infinitely scrolling background implementation 470
127 Example of parallax scrolling implementation 472
128 Code for simulating inertia 479
129 Possible implementation of a simple corner correction 481
130 Code for applying gravity to an object 482
131 Code for jump with enhanced gravity while falling 483
132 Code for jump with more gravity while falling and less when peaking 484
133 Code for jumping without buffering 485
134 Jump buffering example 486
135 Coyote time code example 487
136 Example code of how timed jumps work 488
137 Finding horizontal matches in a match-3 game 499
138 Finding vertical matches in a match-3 game 500
139 Eliminating matches and preparing the tween table 502
140 Creating new tiles and preparing another tween table 503
141 Double Engine Movement Call 544
142 Single Engine Movement Call 545
143 An example of memoization 550
144 An eager object 552
145 A lazy object 552
146 Example on how to optimize entities with a dirty bit 555
147 IFs vs Switch - IF Statements 557
148 IFs vs Switch - Switch Statements 557



There are a lot of things that come to mind when you want to create your very own videogame: graphics, sound, gameplay... But the biggest question of them all is definitely

"Where do I start?"

This book wants to be a starting point for game developers: in depth, accessible and easy to understand.
It will show everything behind the scenes of videogames and how they work: From Zero to Hero!

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